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* Electrostatics :-

YT :- Indian School of Physics
(INSP)

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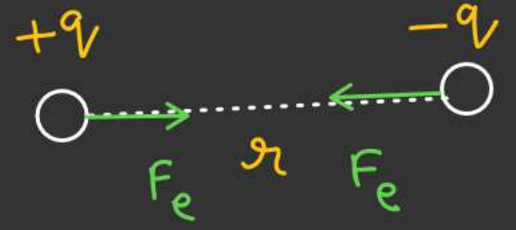
Electric charge :

→ property of fundamental particles (e^-, p) by which Attraction or repulsion b/w the particles can be explained.

like charges



Unlike charges

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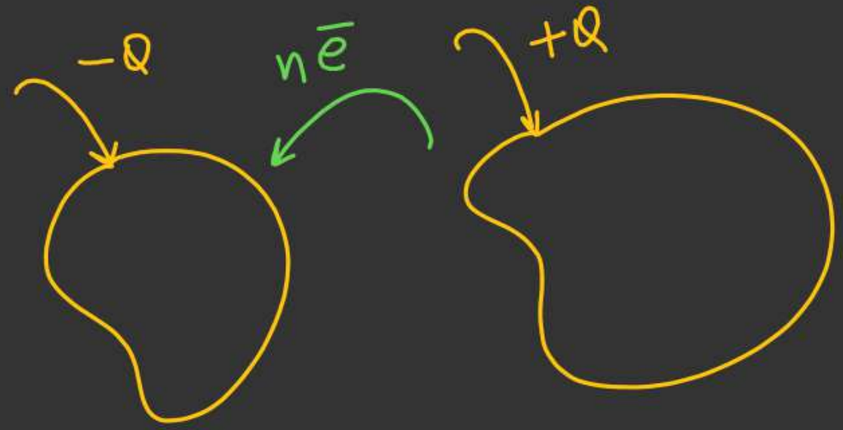
* Charge is invariant $(v \ll c)$

* Charge is quantised

$$q = ne$$

↓

Integer



$q = n\bar{e}$

electric Current

$$i = \frac{dq}{dt} = \left(\frac{n}{t}\right) e$$

$$= \left(\frac{dn}{dt}\right) \cdot e$$

quarks

↳ $\pm e/2$

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Coulomb's Law :- (Valid for point charges only)

$$F_e \propto q_1 \cdot q_2$$

$$F_e \propto \frac{1}{r^2}$$

$$F_e \propto \frac{q_1 q_2}{r^2}$$

$$F_e = \frac{k q_1 q_2}{r^2}$$

k → depends on the medium.

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \ N\text{-}m^2/C^2$$

ϵ_0 → permittivity of free space

$$F_g \propto m_1 m_2$$

$$F_g \propto \frac{1}{r^2}$$

Always Attractive

$$F_g \propto \frac{m_1 m_2}{r^2}$$

$$F_g = \frac{G m_1 m_2}{r^2}$$

$G \rightarrow N\text{-}m^2/kg^2$

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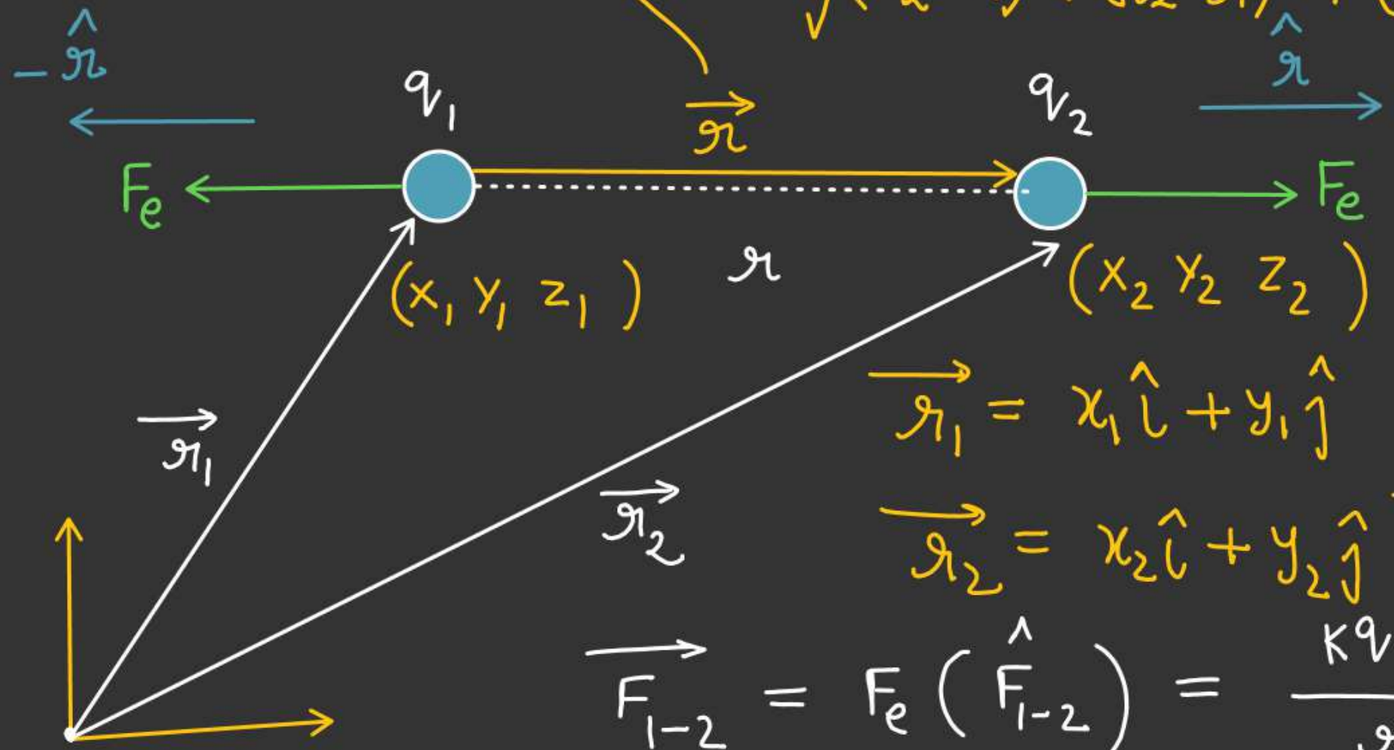
Vector form :-

$$\vec{r} = \vec{r_2} - \vec{r_1} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$\hat{r} = \frac{\vec{r}}{r}$$



$$F_e = \frac{k q_1 q_2}{r^2}$$

F_{A-B} → Force on A by B

$$\vec{r_1} = x_1\hat{i} + y_1\hat{j}$$

$$\vec{r_2} = x_2\hat{i} + y_2\hat{j}$$

$$\vec{F_{1-2}} = F_e(\hat{F}_{1-2}) = \frac{k q_1 q_2}{r^2}(-\hat{r}) \Rightarrow -\frac{k q_1 q_2}{r^3}(\vec{r})$$

$$\vec{F_{2-1}} = \frac{k q_1 q_2}{r^2}(+\hat{r}) = \frac{k q_1 q_2}{r^3}\cdot\vec{r} = \frac{k q_1 q_2\left[(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}\right]}{\left[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2\right]^{3/2}}$$

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* limitations of Coulomb's Law :-

(i) Charges should be at rest otherwise magnetic interaction may come in to Force Calculation

(ii) only for point charges

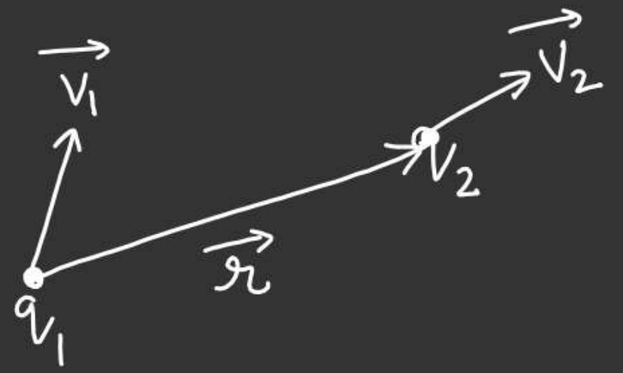
(iii) for moving charges

$$\vec{F}_{net} = \vec{F}_e + \vec{F}_m$$

↓

Lorentz force

$$\frac{kq_1q_2}{r^2}(\hat{r}) + q_2(\vec{v}_2 \times \vec{B}_1) = \vec{F}_{2-1}$$



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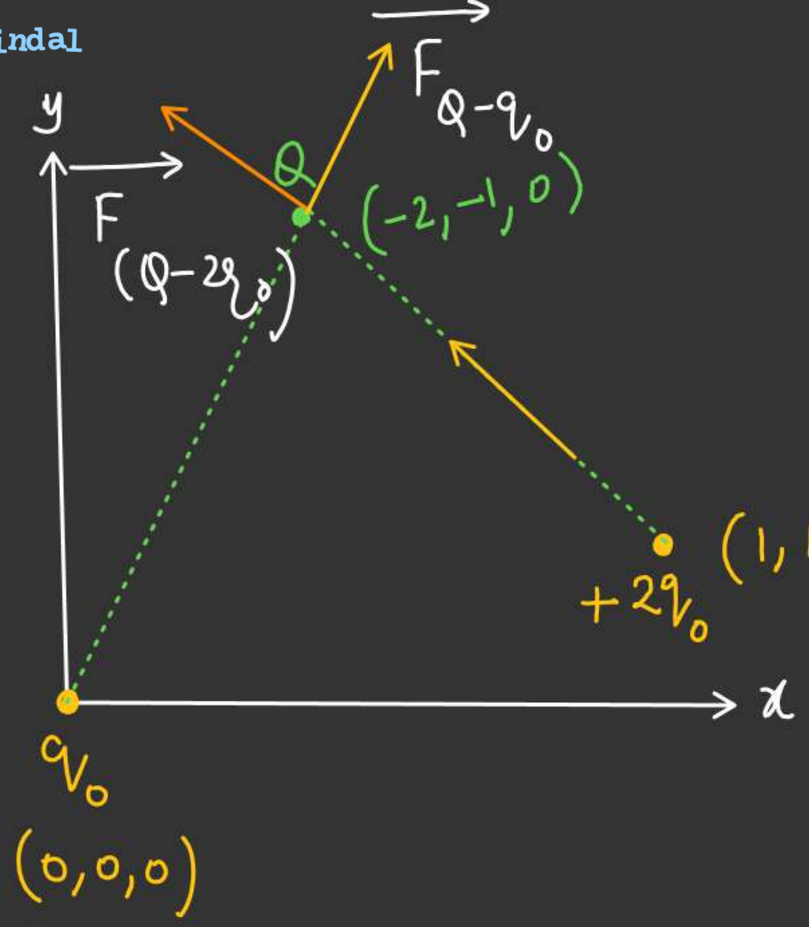

$\vec{F_Q} = ?$

$$\vec{F}_{Q-2q_0} = \frac{KQ\cdot 2q_0\left(-3\hat{i}-2\hat{j}-\hat{k}\right)}{\left(3^2+2^2+1^2\right)^{3/2}}$$

$$\vec{F}_{Q-q_0} = \frac{KQ\cdot q_0\left(-2\hat{i}-\hat{j}\right)}{\left(2^2+1^2\right)^{3/2}}$$

$$\vec{F}_{Q\,net} = \vec{F}_{Q-q_0} + \vec{F}_{(Q-2q_0)} = (\quad)\hat{i} + (\quad)\hat{j} + (\quad)\hat{k}$$

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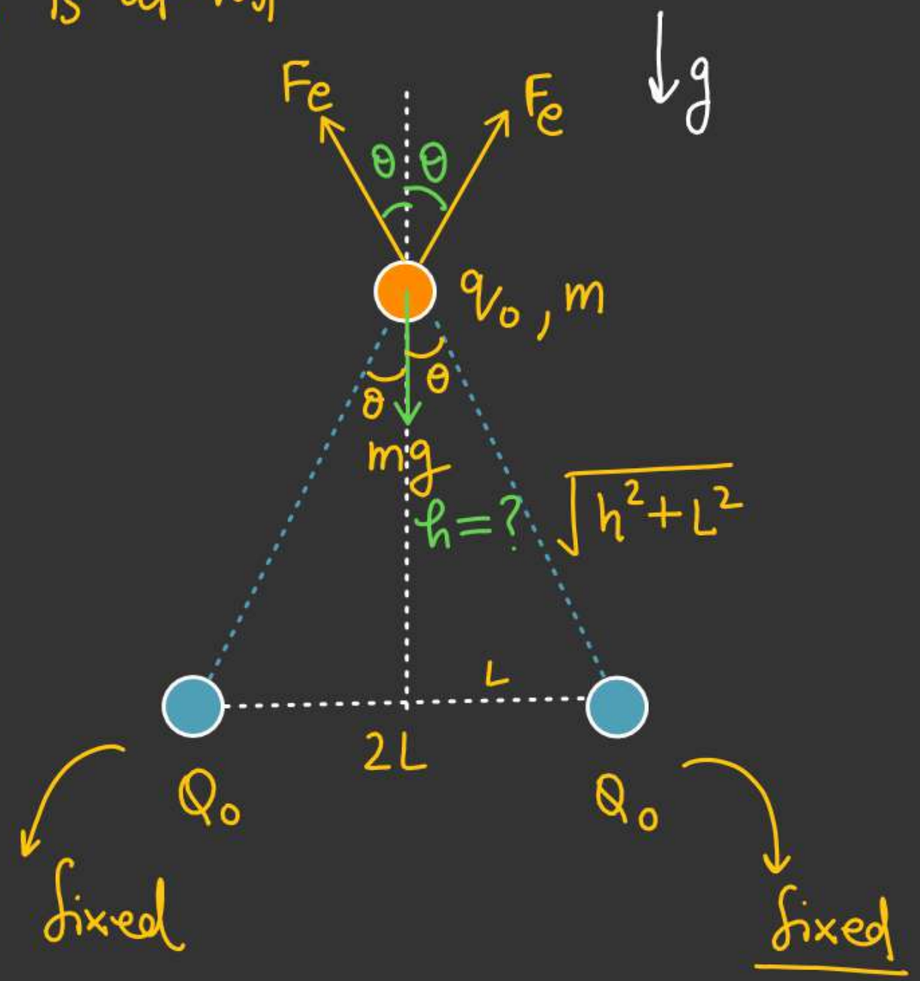
* q_0 is at rest



$$2F_e \cos\theta = mg$$

$$2\left(\frac{K q_0 Q_0}{h^2+L^2}\right)\cdot\frac{h}{\sqrt{h^2+L^2}} = mg$$

$$\boxed{\frac{2kQq_0 h}{(h^2+L^2)^{3/2}} = mg}$$

$h = L$

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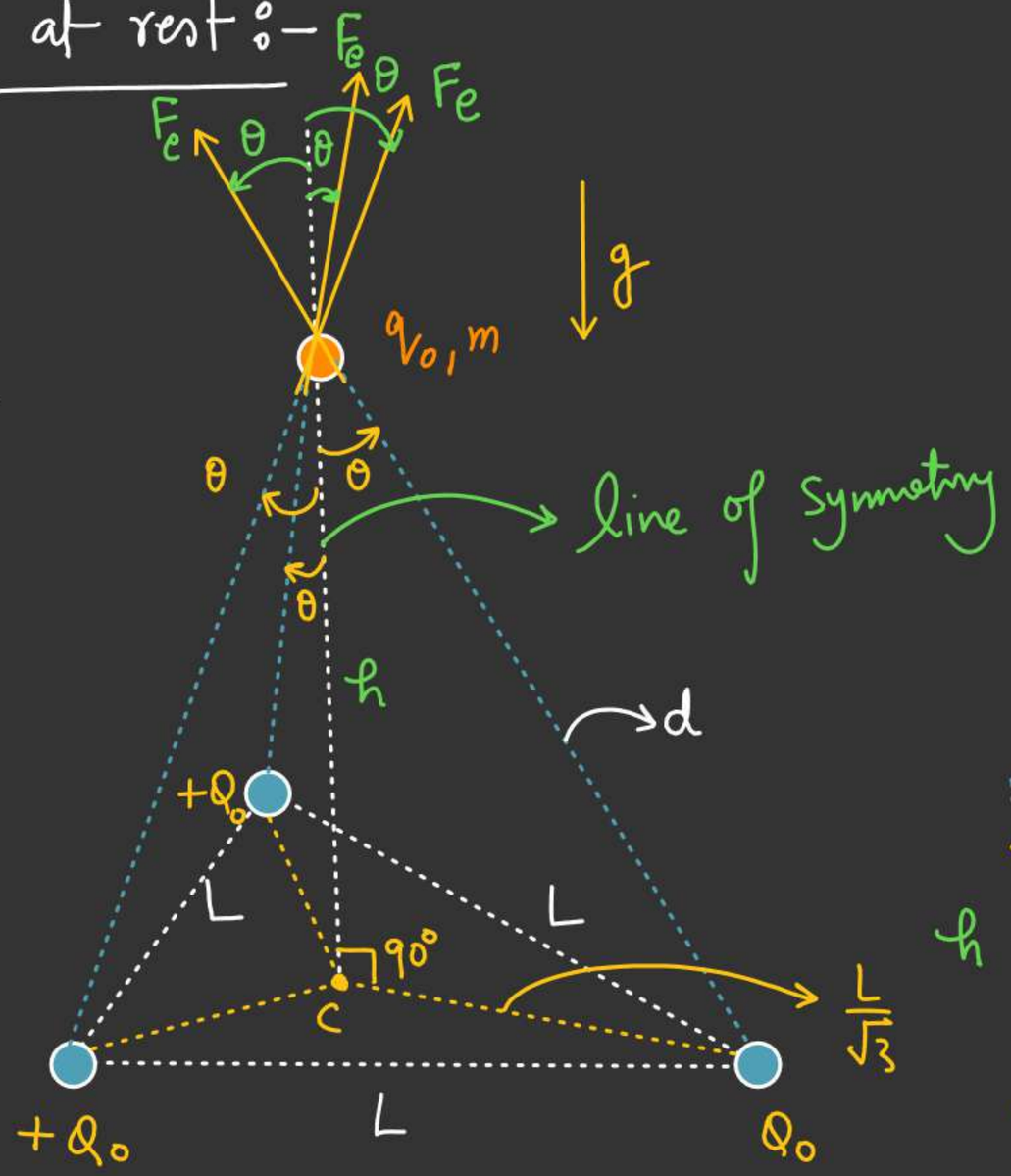
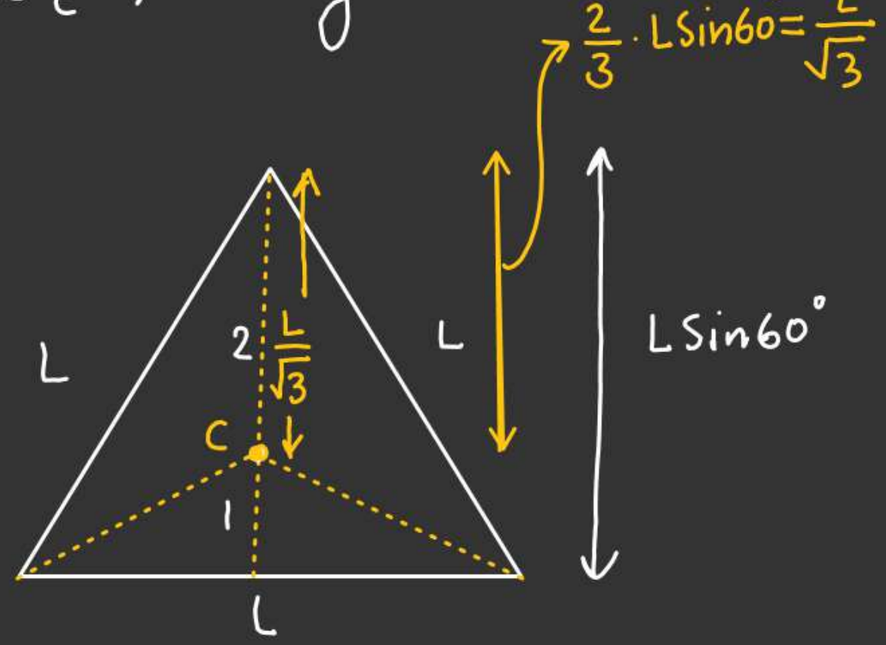
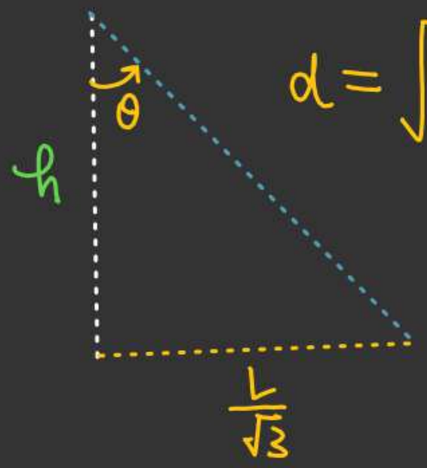
* q_0 is at rest :-



$$F_e = \frac{KQq_0}{d^2}$$

$$3F_e\cos\theta = mg$$





$$d = \sqrt{\left(\frac{L}{\sqrt{3}}\right)^2 + h^2}$$

$$\cos\theta = \frac{h}{d} = \frac{h}{\sqrt{\left(\frac{L}{\sqrt{3}}\right)^2 + h^2}}$$

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$$3 \cdot \left(\frac{kQq_0}{\left(h^2 + \frac{L^2}{3}\right)} \right) \cdot \frac{h}{\sqrt{h^2 + \frac{L^2}{3}}} = mg$$

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*

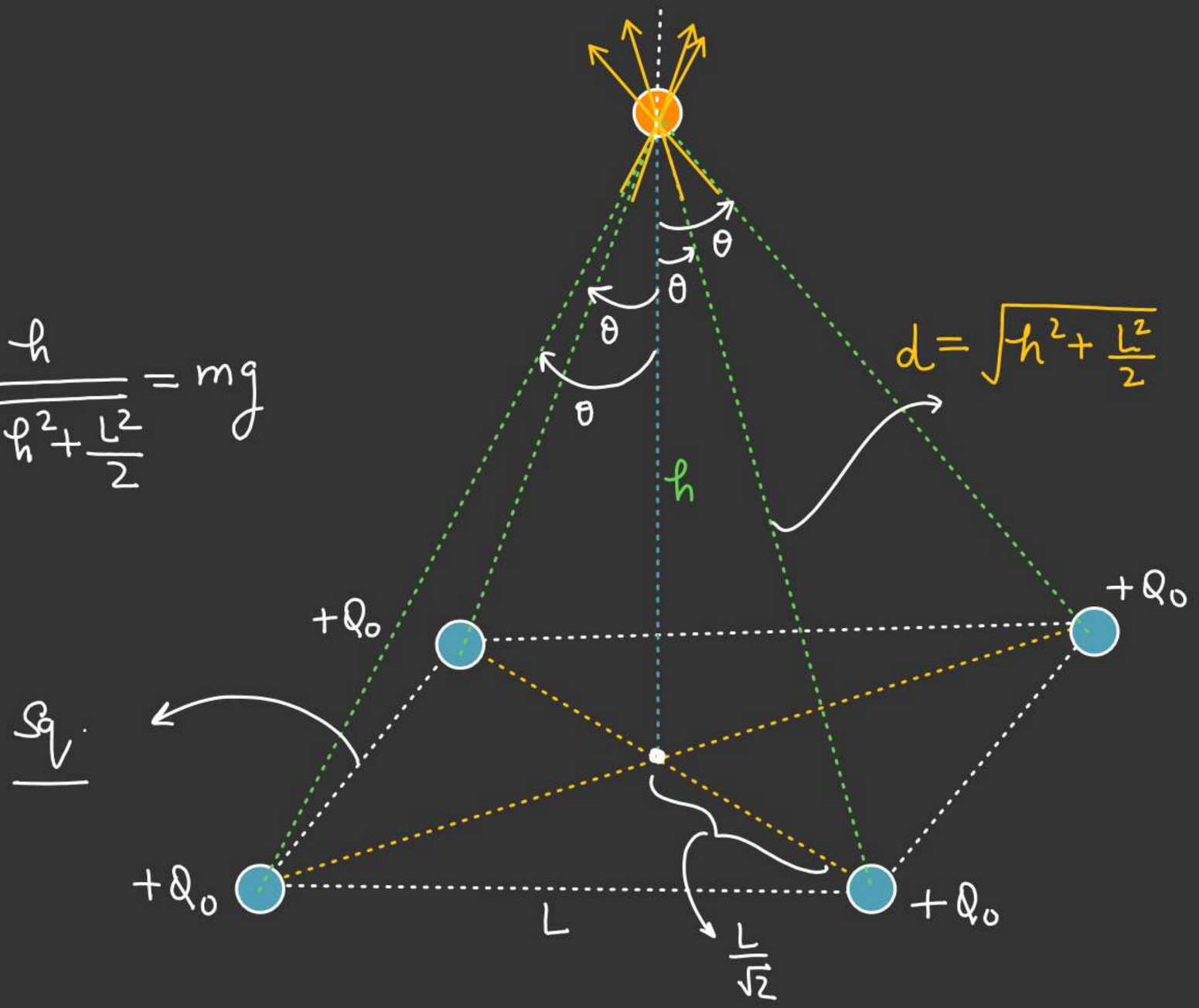
$$4F_e\cos\theta = mg$$

$$4\cdot\frac{kQ_0q_0}{\left(h^2+\frac{L^2}{2}\right)}\cdot\frac{h}{\sqrt{h^2+\frac{L^2}{2}}} = mg$$

$$F_e = \frac{kQ_0q_0}{\left(h^2+\frac{L^2}{2}\right)}$$

$$d = \sqrt{h^2+\frac{L^2}{2}}$$

$$\cos\theta = \frac{h}{\sqrt{h^2+\frac{L^2}{2}}}$$



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H.C. Repeat the previous Case for N-Sided regular polygon.

* find max m_0 for a given value of h.

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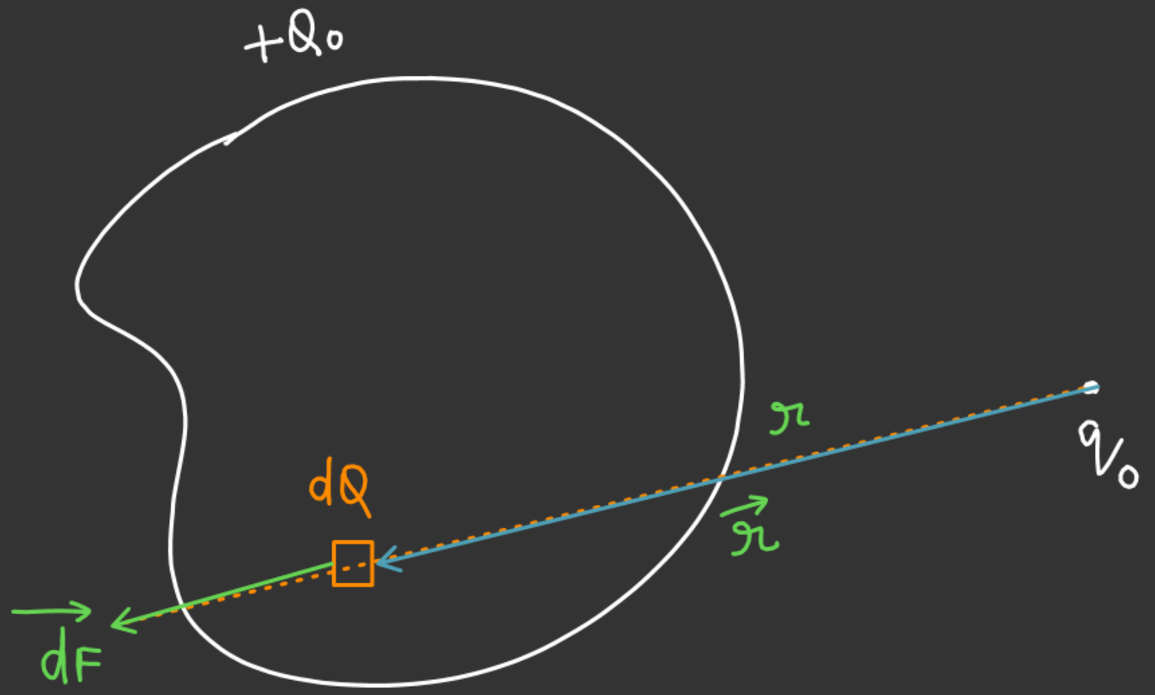
Continuous charged system



$$\vec{dF} = \frac{K \cdot q_0 \cdot dQ}{r^2} \cdot (\hat{r})$$

$$\vec{F_{net}} = \int \vec{dF}$$

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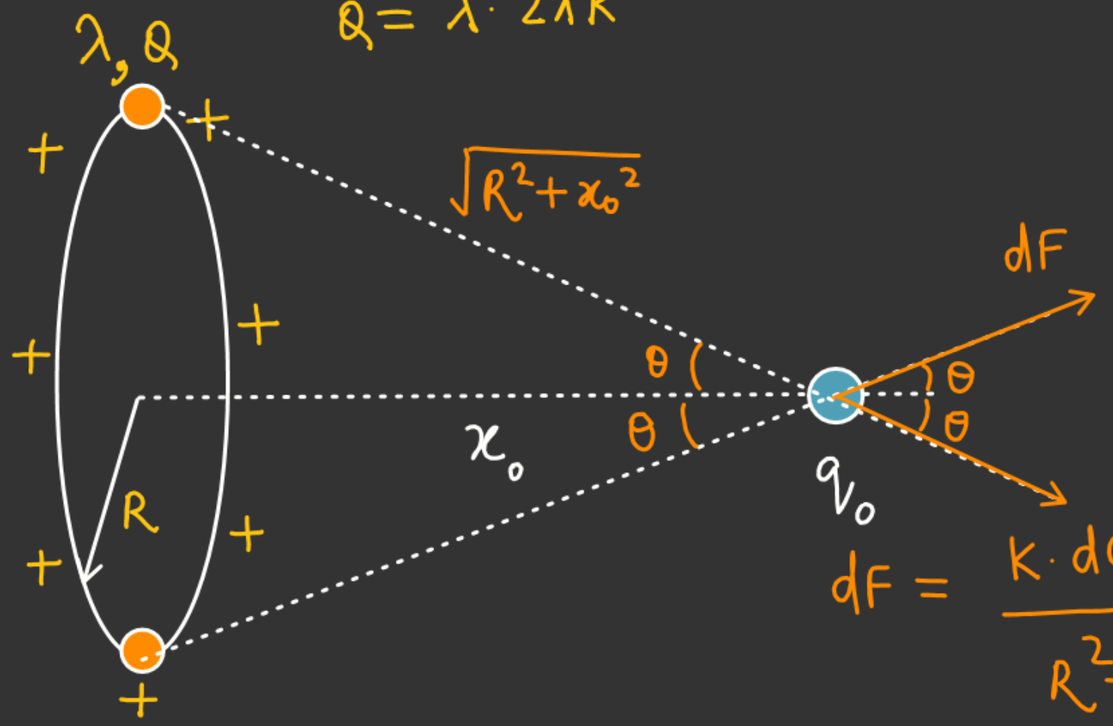
* Force b/w Uniform Charged Ring & point charge :-



$Q = \lambda \cdot 2\pi R$

$$dF = \frac{K \cdot dQ \cdot q_0}{R^2+x_0^2}$$

$$\int dF \sin\theta = 0 \qquad F_{net} = \int dF \cos\theta$$

$$F_{net} = \int \frac{K \cdot dQ \cdot q_0}{R^2+x_0^2} \cdot \frac{x_0}{\sqrt{R^2+x_0^2}}$$

$$\Rightarrow \frac{K q_0 x_0}{(R^2+x_0^2)^{3/2}} \cdot \int dQ$$

$$\boxed{F_{net} = \frac{K Q q_0}{(R^2+x^2)^{3/2}}}$$

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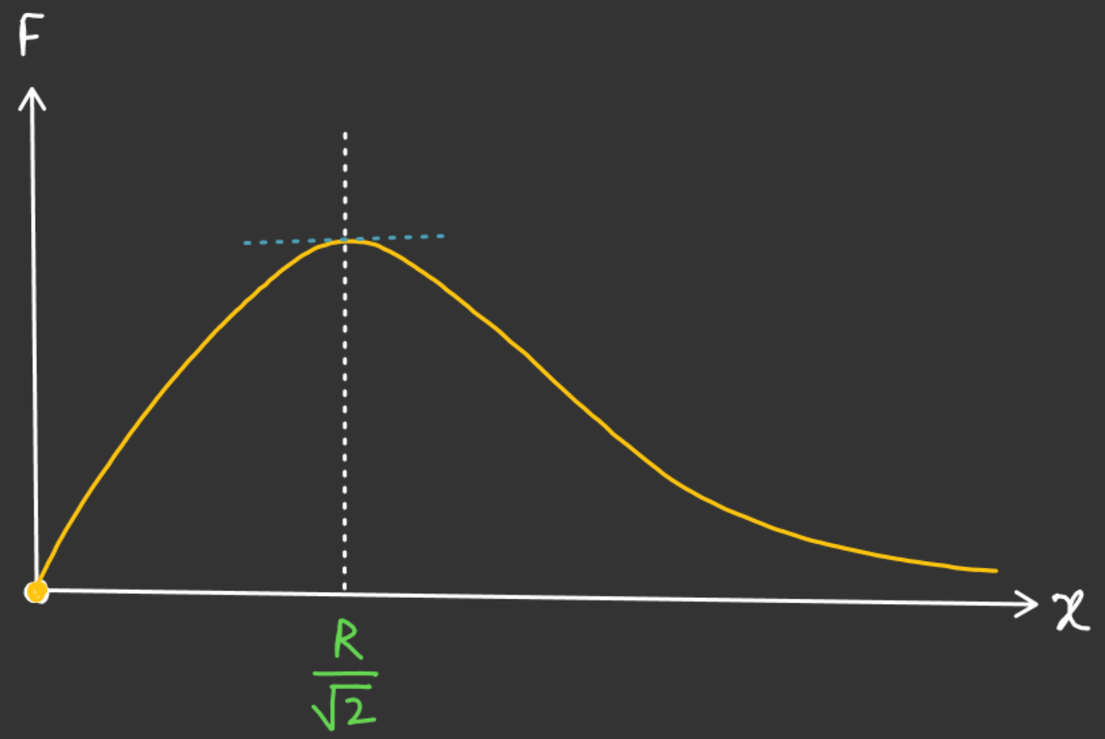

$x = 0 \Rightarrow F = 0$

$x \to \infty \quad R^2 + x^2 \approx x^2$

$$F = \frac{KQq_0 x}{(x^2)^{3/2}} = \frac{KQq_0}{x^2} \to 0$$

$$F_{net} = \frac{KQq_0 x}{(R^2+x^2)^{3/2}}$$

$\hookrightarrow$ odd f^n.

$$\frac{dF}{dx} = 0 \Rightarrow x = \pm \frac{R}{\sqrt{2}} \quad (\text{N.C.})$$

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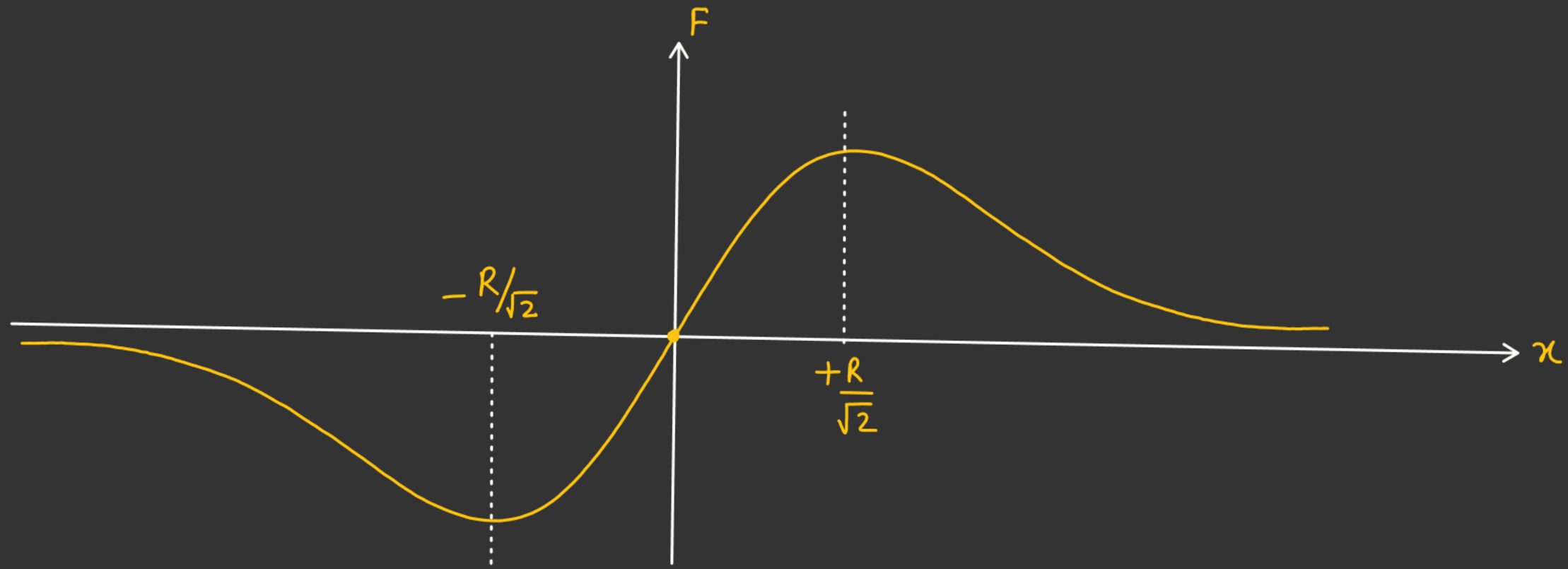

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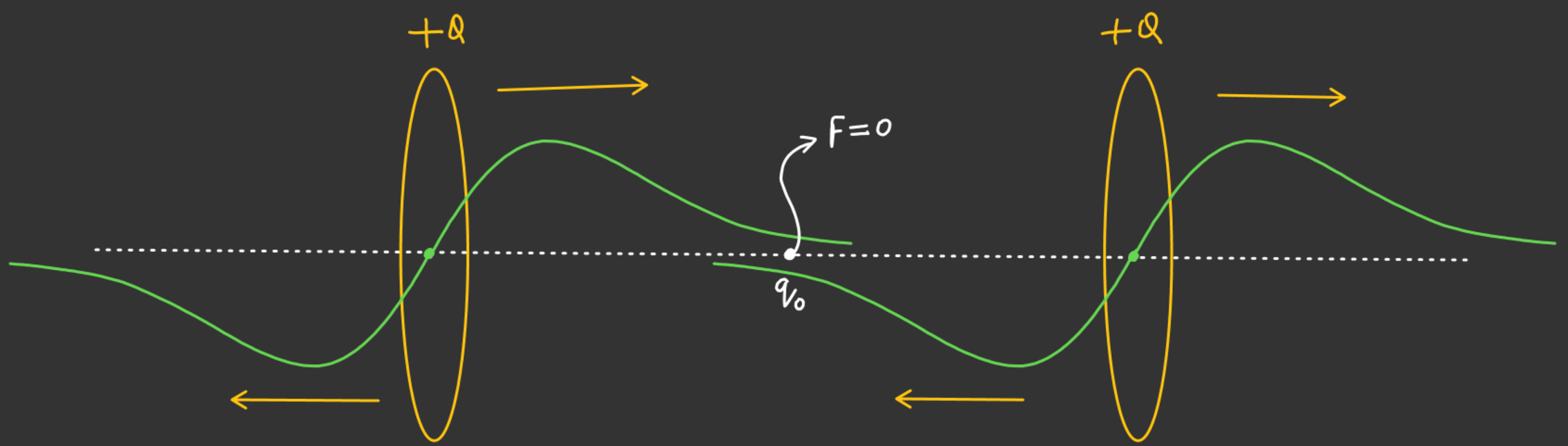

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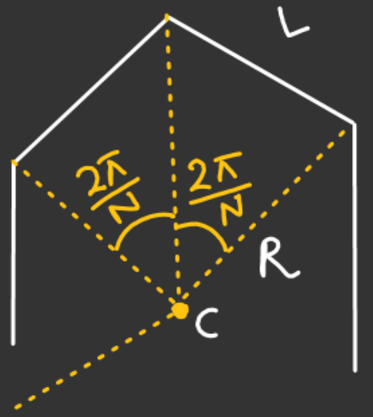
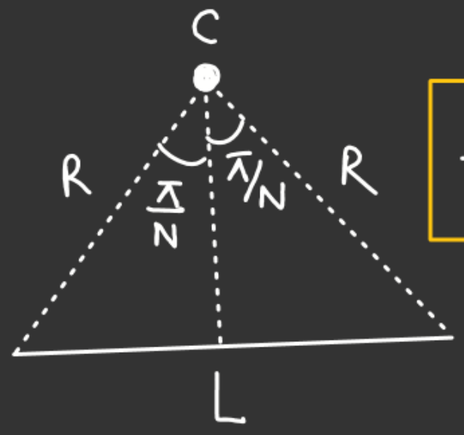
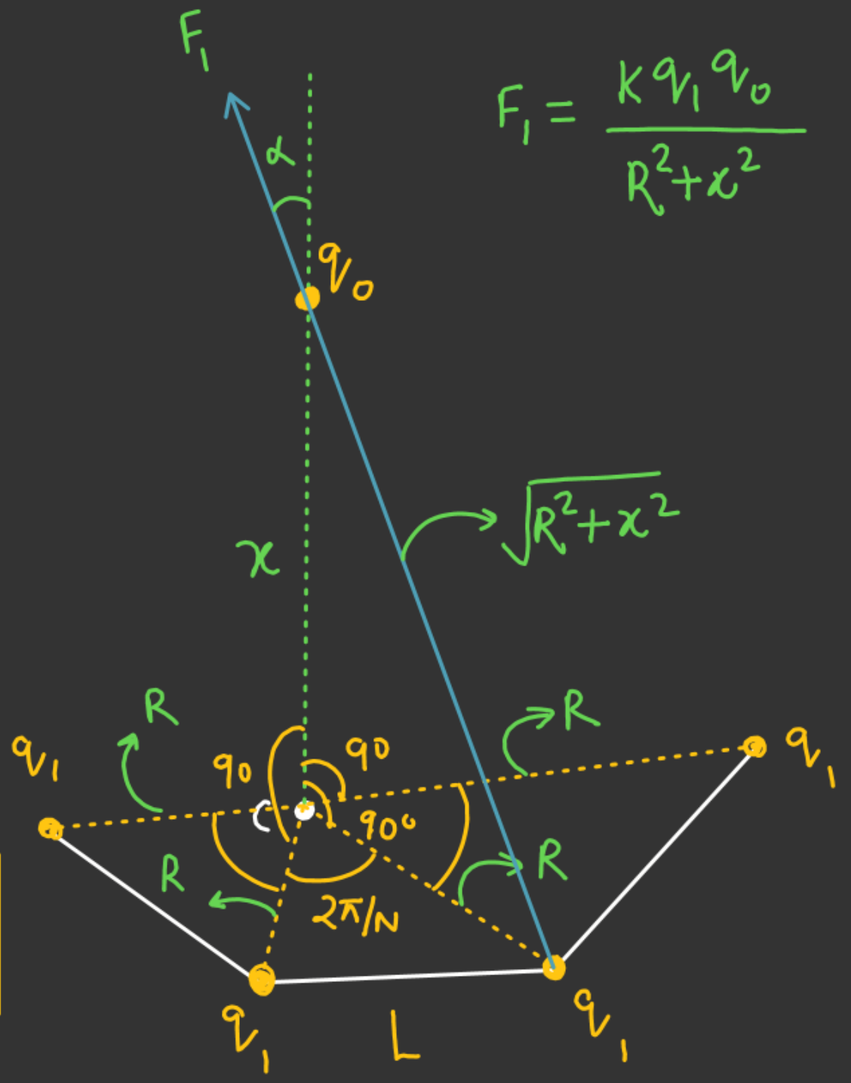
* N - Sided Polygon :-





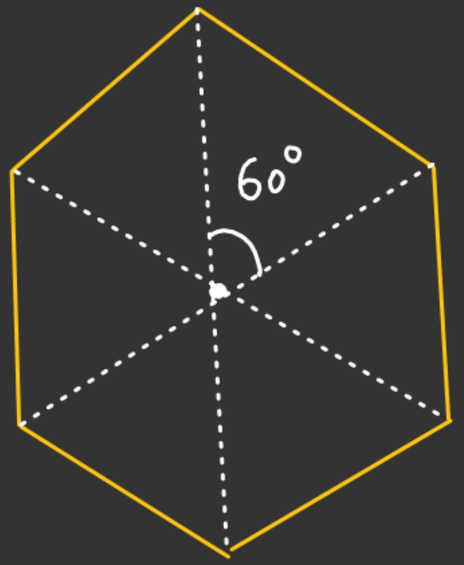
$$\frac{L}{2} = R \sin \pi/N$$



$$F_1 = \frac{k q_1 q_0}{R^2 + x^2}$$

$N \to \infty$

$N q_1 = Q$



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$$F_{net} = N \cdot F_1 \cos\alpha$$

$$F_{net} = N \cdot \frac{k q_1 q_0}{(R^2+x^2)} \cdot \frac{x}{\sqrt{R^2+x^2}}$$

$$\underbrace{N q_1 = Q}$$

$$\boxed{F_{net} \Rightarrow \frac{k Q q_0 x}{(R^2+x^2)^{3/2}}}$$

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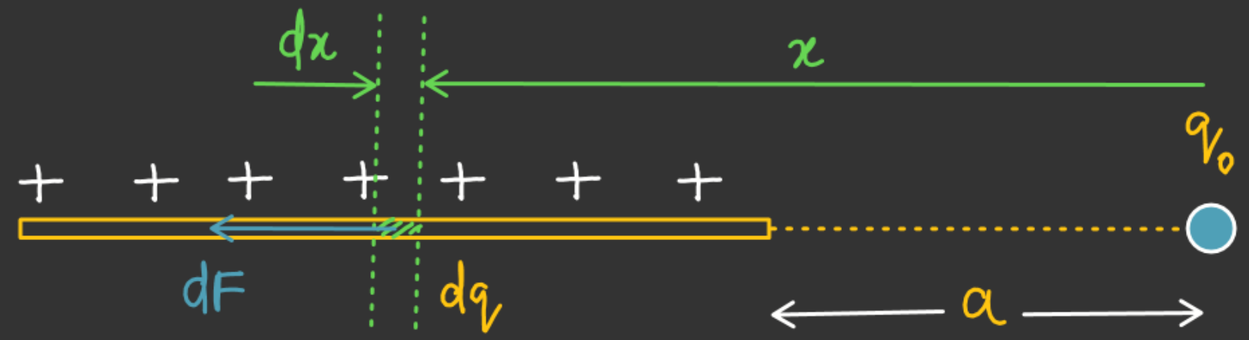
* Force b/w uniform rod and point charge :

$\lambda L = Q_0$



$$dq = \lambda \cdot dx = \frac{Q_0}{L} \cdot dx$$

$$dF = \frac{k \cdot dq \cdot q_0}{x^2} = \frac{k q_0}{x^2} \frac{Q_0}{L} \cdot dx$$

$$F_{net} = \int dF = \frac{k Q_0 q_0}{L} \cdot \int_a^{a+L} \frac{dx}{x^2}$$

$$\boxed{F_{net} = \frac{k Q_0 q_0}{L} \left(\frac{1}{a} - \frac{1}{a+L} \right)}$$

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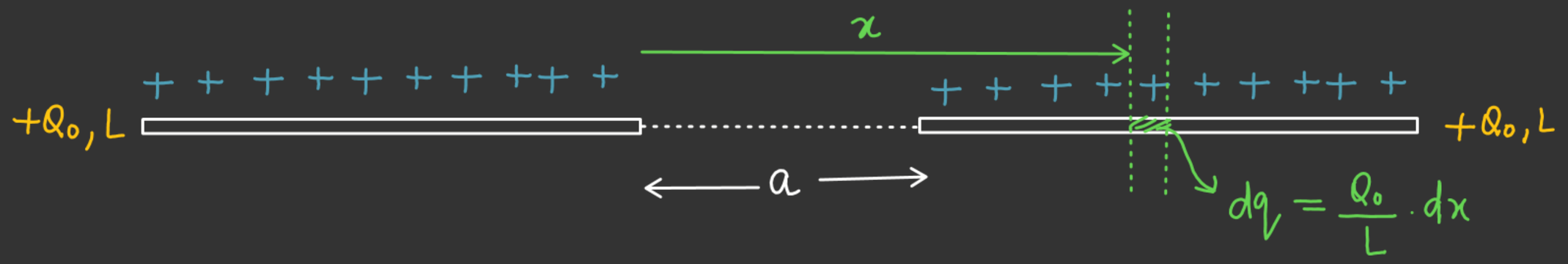
* F = ?



$$dF = \frac{K \cdot Q_0 \cdot dq}{L}\left[\frac{1}{x} - \frac{1}{L+x}\right] = \frac{K \cdot Q_0 \cdot Q_0}{L \cdot L}\left[\frac{1}{x} - \frac{1}{L+x}\right] \cdot dx$$

$$F_{net} = \int dF = \frac{KQ_0^2}{L^2}\left[\int_a^{a+L} \frac{dx}{x} - \int_a^{a+L} \frac{dx}{L+x}\right] = \frac{KQ_0^2}{L^2}\left[\ln\left(\frac{a+L}{a}\right) - \ln\left(\frac{a+2L}{a+L}\right)\right]$$

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*

$$\int \vec{r} \cdot dm = \vec{r}_{cm} \cdot M$$

$$\int \vec{r} \cdot dQ = \vec{r}_{cm} \cdot Q$$

in Coumbombs law $\int \frac{dQ}{r^2}$ ✗ CM logic will not work in force Calculation

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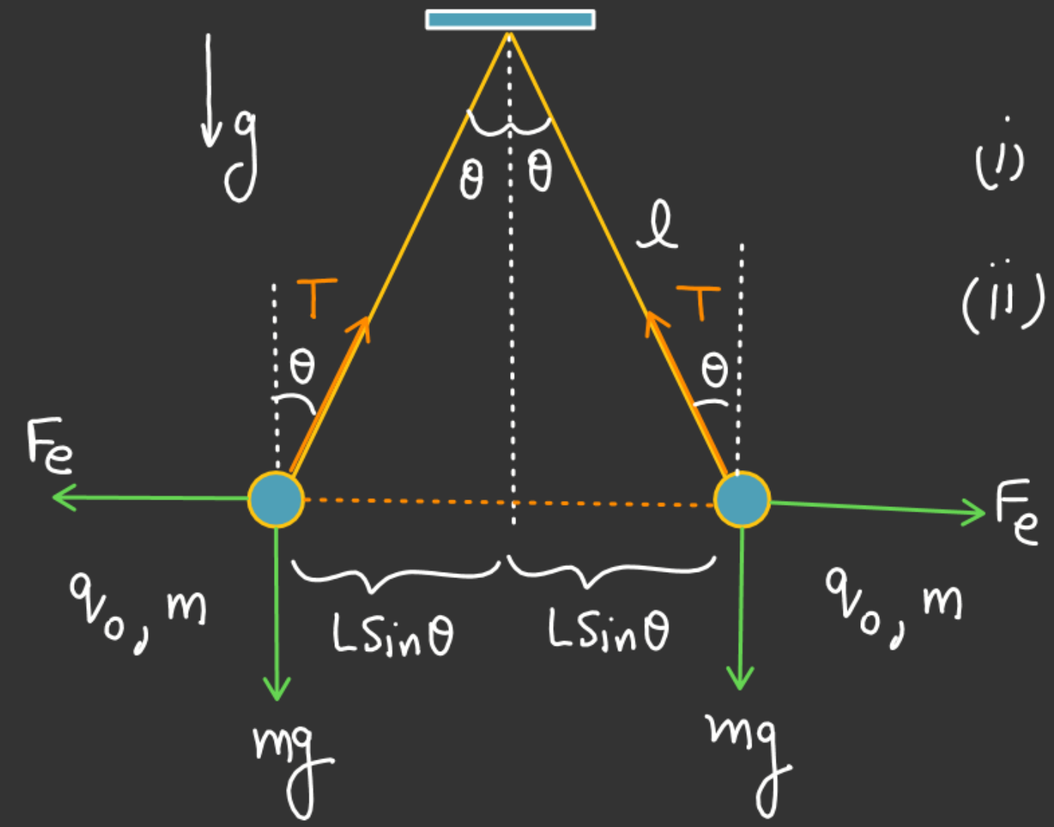
* if equilibrium is at $\theta = 60^\circ$, then find $q_0 = ?$



(i) $\leftarrow F_e = T\sin\theta = \dfrac{K \cdot q_0^2}{4L^2\sin^2\theta}$

(ii) $\leftarrow mg = T\cos\theta$

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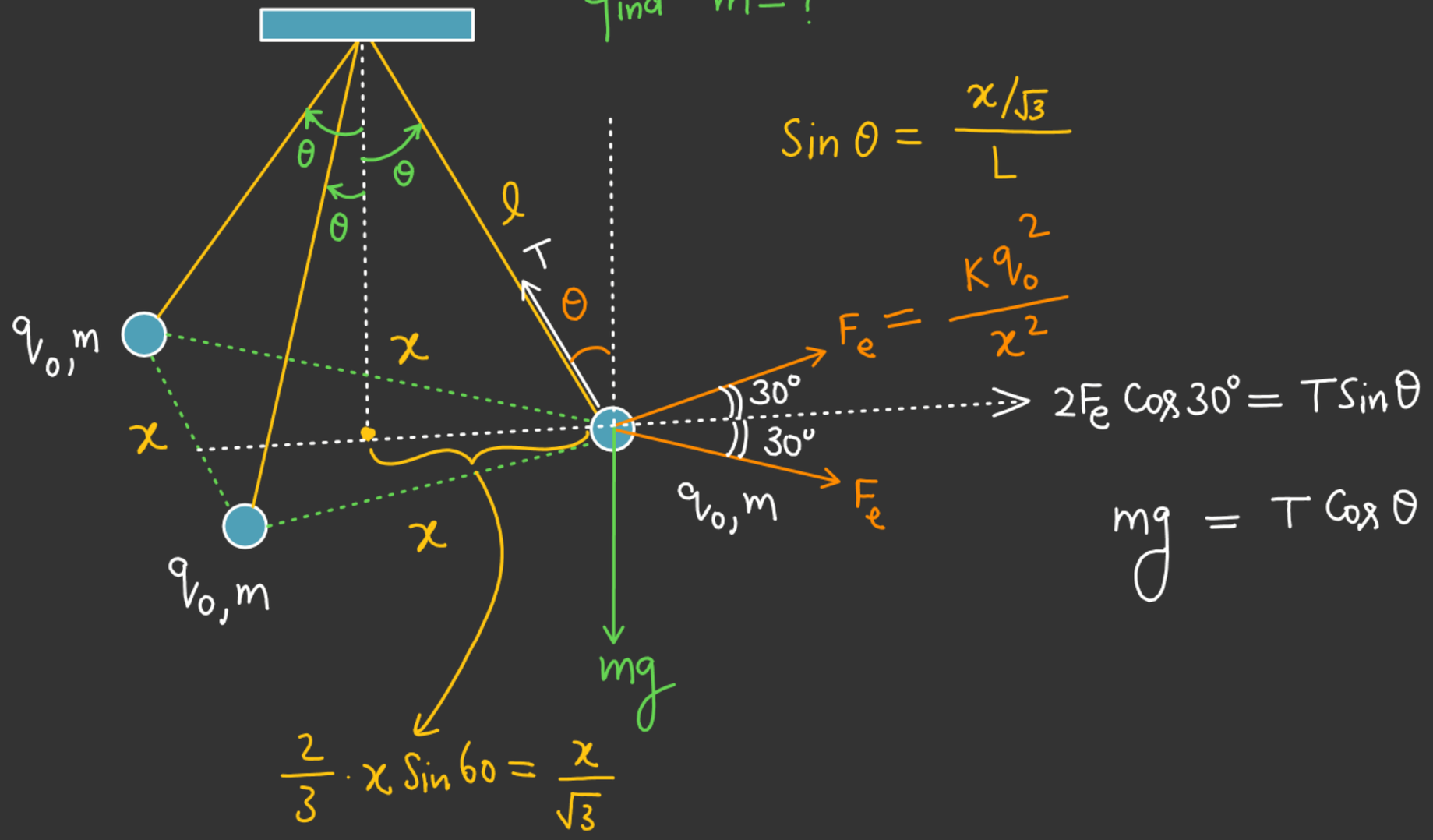
* 3 Pendulums

$q, \theta \rightarrow$ given

find $m = ?$



$$\sin\theta = \frac{x/\sqrt{3}}{L}$$

$$F_e = \frac{K q_0^2}{x^2}$$

$$2F_e \cos 30^\circ = T \sin\theta$$

$$mg = T\cos\theta$$

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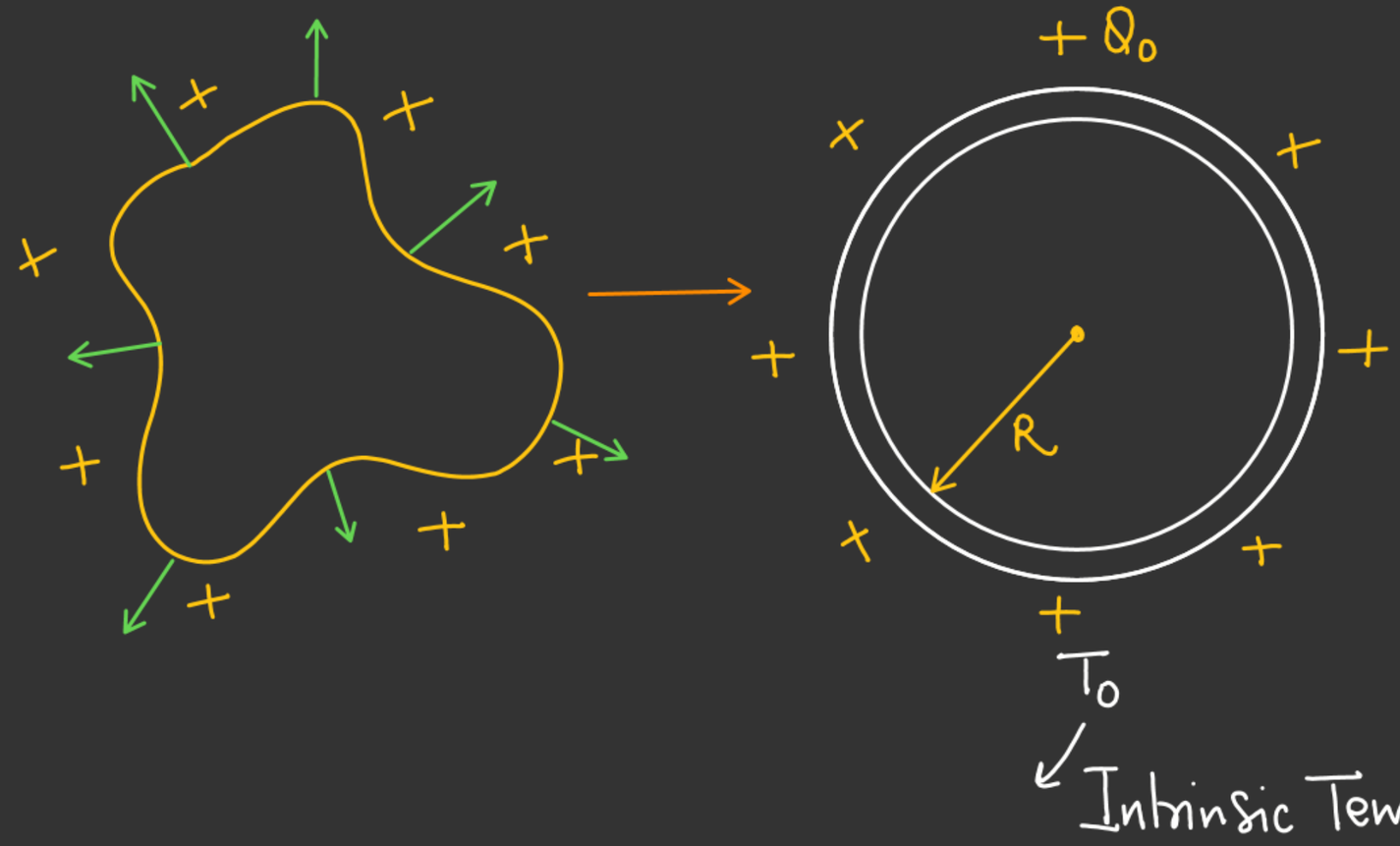
* Increment in Tension of Uniform Charged Ring :-



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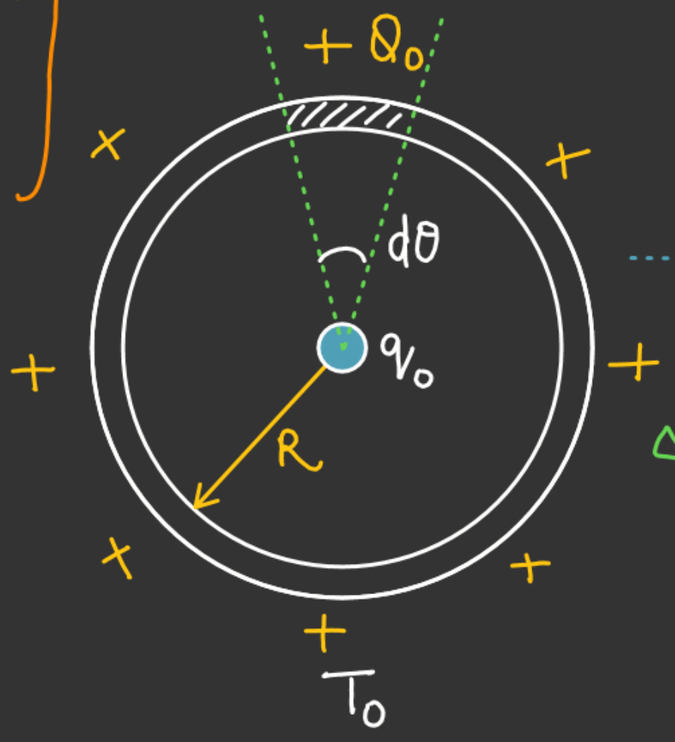
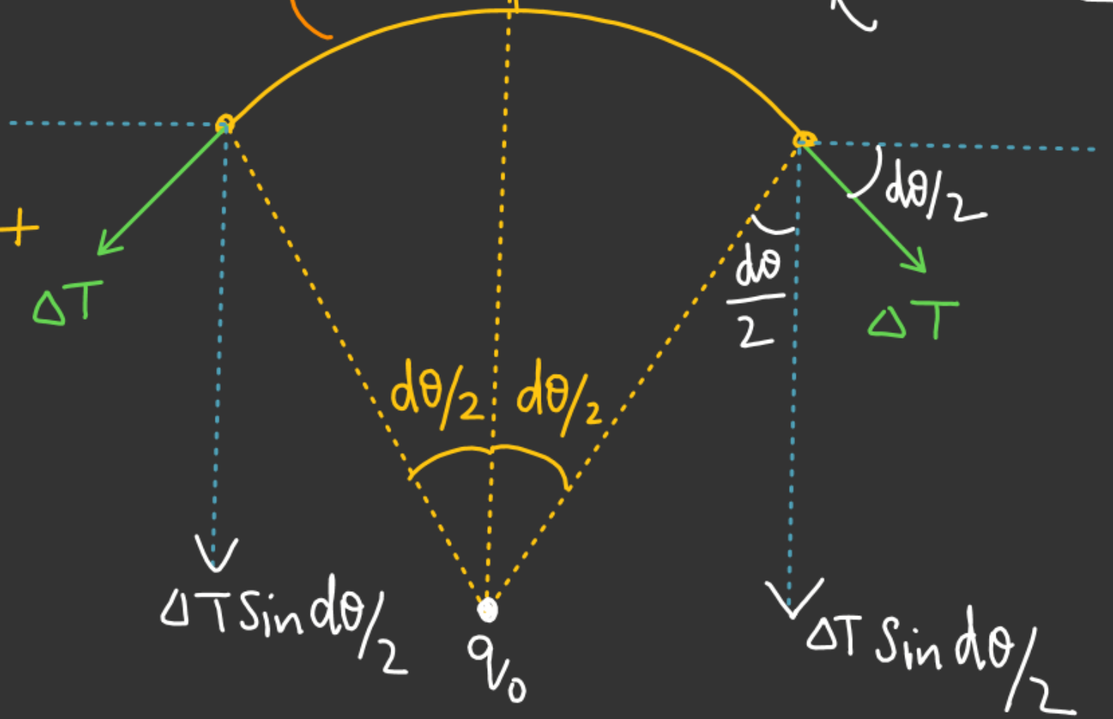
* Increment in Tension of Uniform Charged Ring :-

[for Time period Sol^{n}. is at INSP YT]



$$dQ = \frac{Q_0}{2\pi} \cdot d\theta$$

$$dF_e = \frac{K q_0 \cdot dQ}{R^2} = \frac{K q_0 \cdot \frac{Q_0}{2\pi} \cdot d\theta}{R^2}$$



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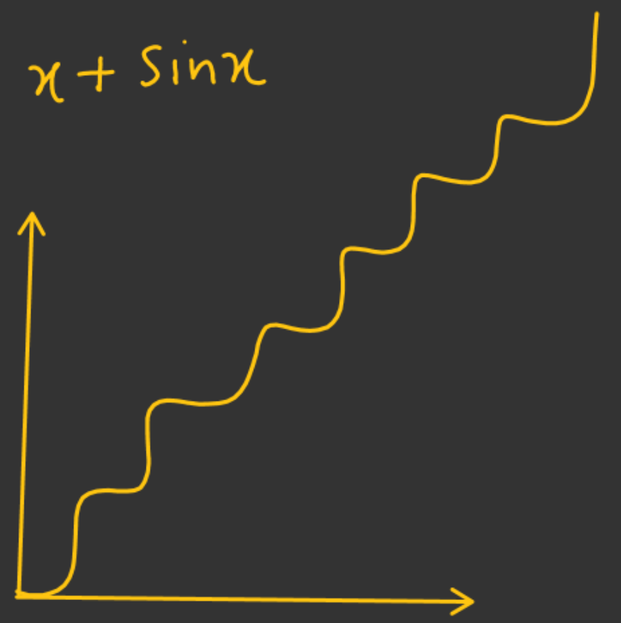
$\sin\theta \approx \theta$

$$2\,\Delta T \sin\frac{d\theta}{2} = dF_e$$

$$2\,\Delta T \cdot \frac{d\theta}{2} = \frac{K \cdot q_0 \cdot \frac{Q_0}{2\pi} \cdot d\theta}{R^2}$$

$$\boxed{\Delta T = \frac{K Q_0 q_0}{2\pi R^2}} \quad \star$$

$y = x + \sin x$



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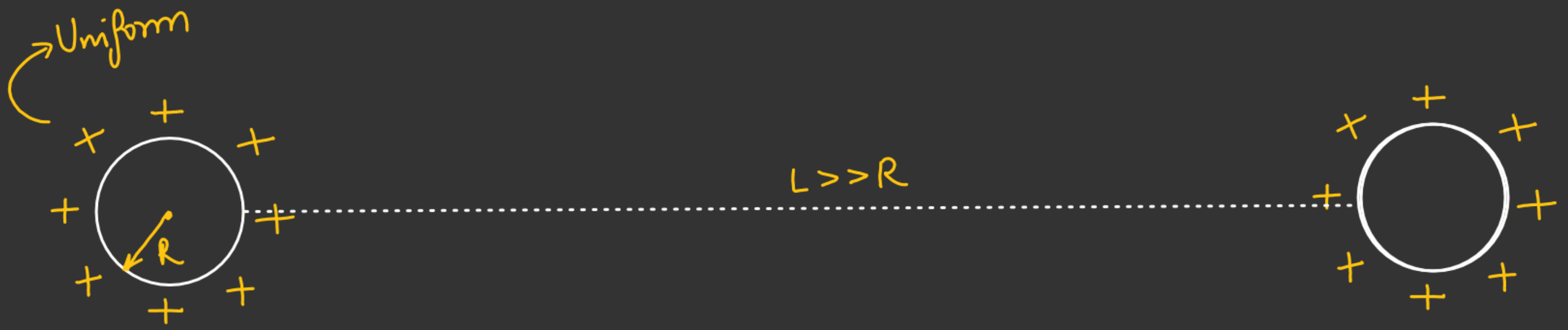
*

Electric field :-



$$F < \frac{kQ_1Q_2}{(L_1+2R)^2}$$

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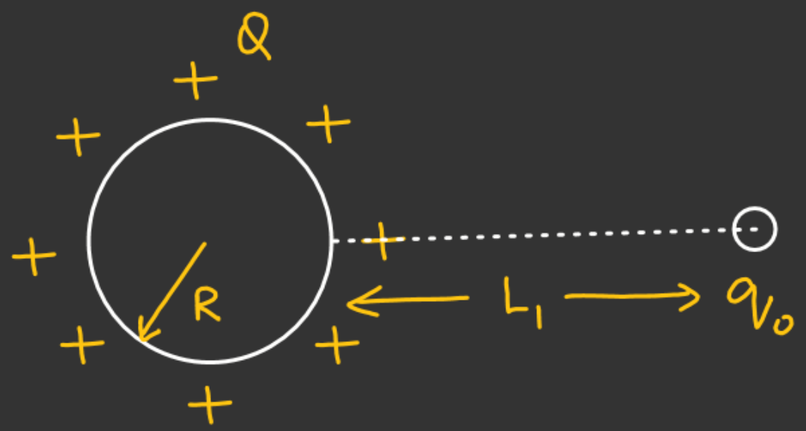
* Test charge :



$$F = \frac{k Q q_0}{(L_1 + R)^2}$$

$q_0 << 1C$

$\lim_{q_0 \to 0}$

$$F = \frac{9 \times 10^9 \times 1 \times 1}{(10^3)^2}$$

$$= 9 \times 10^3$$

$\approx$ 1000 kg of wt.

$$F = \frac{9 \times 10^9 \times 1 \times 1}{1^2}$$

$$F = 9 \times 10^9 \text{ N}$$

$\approx 10^9$ kg of wt.

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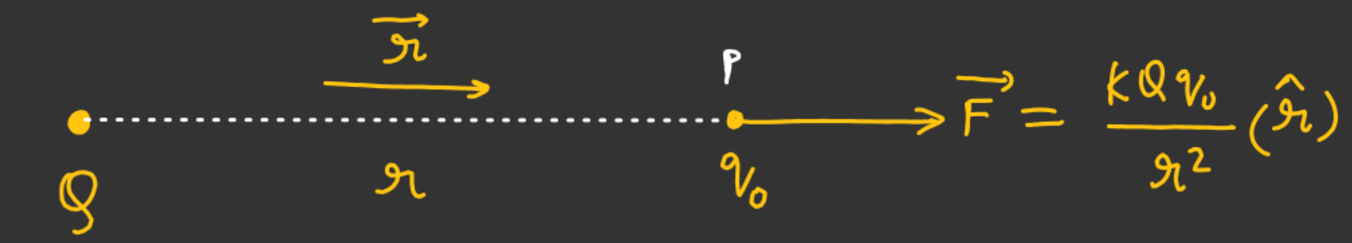

fixed

$$\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0} = \frac{kQ}{r^2}(\hat{r})$$

$$\vec{F} = q_0 \cdot \vec{E}$$

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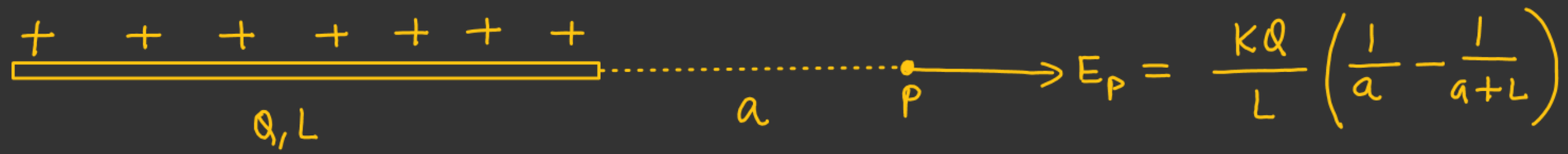

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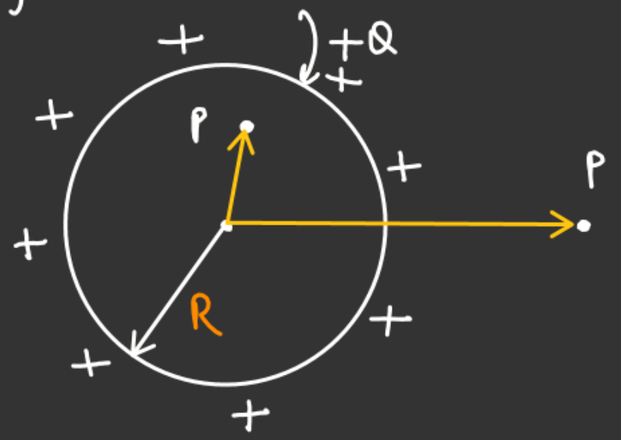
* Shell Theorem :-

* for a uniform charged shell the field inside is zero and for outside points shell will behave like a point charge at the centre.

Uniform charged shell



$r \leq R$ $\quad E = 0$

$r > R$ $\quad E = \frac{kQ}{r^2}$

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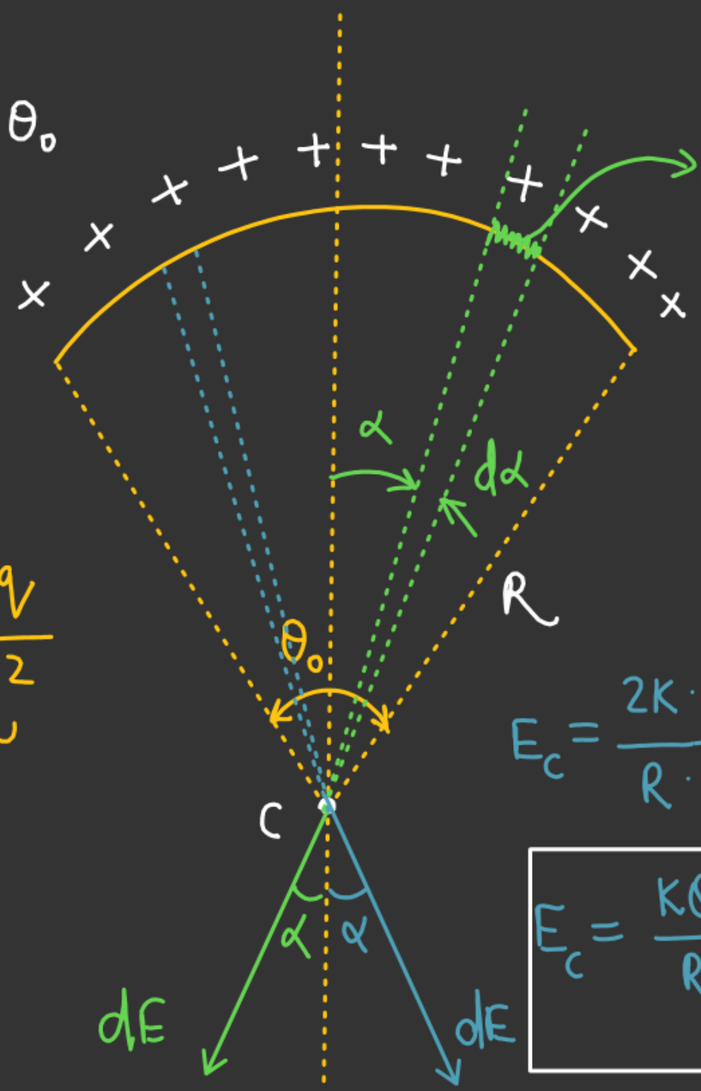
* Field due to Circular Arc



$Q_0 = \lambda \cdot R\theta_0$

$dE = \dfrac{k \cdot dq}{R^2}$

$E_C = \int_0^{\theta_0/2} 2\,dE\cos\alpha$

$E_C = \int dE\cos\alpha$

$\Rightarrow \int \dfrac{k \cdot \lambda R d\alpha}{R^2} \cdot \cos\alpha = \dfrac{k\lambda}{R} \int_{-\theta_0/2}^{+\theta_0/2} \cos\alpha \cdot d\alpha$

$E_C = \dfrac{2K \cdot Q_0}{R \cdot R\theta_0} \cdot \sin\dfrac{\theta_0}{2}$

$E_C = \dfrac{2k\lambda}{R} \cdot \sin\theta_0/2$

$E_C = \dfrac{kQ_0}{R^2}\left(\dfrac{\sin\theta_0/2}{\theta_0/2}\right)$

H.C (only for that bhaiya)

$Q_0 = 1\ mC$
$r = 2\ cm$
$\theta_0 = 60°$

find E upto 77 decimal place

Along the bisector

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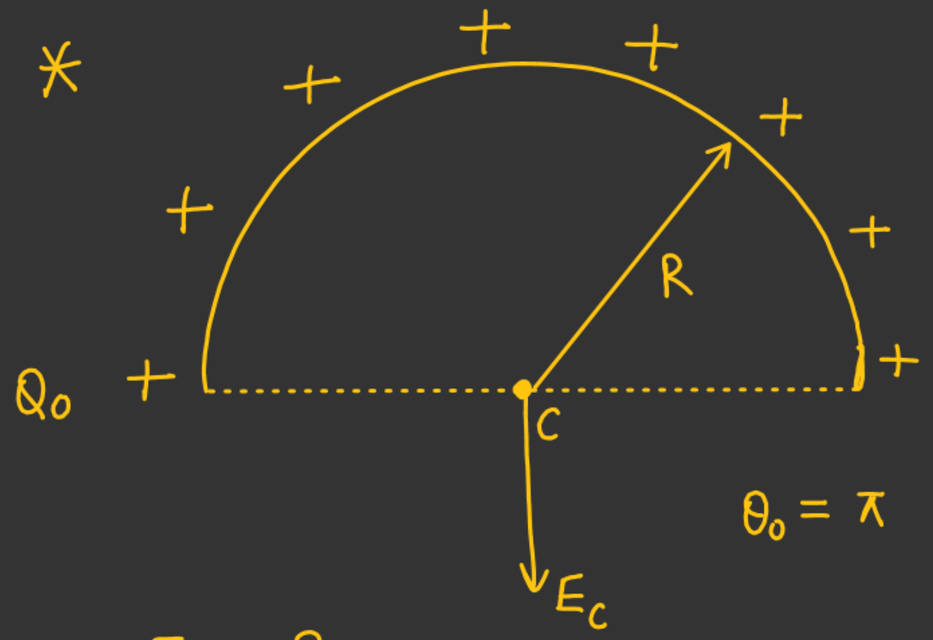


$Q = \lambda \cdot \pi R$

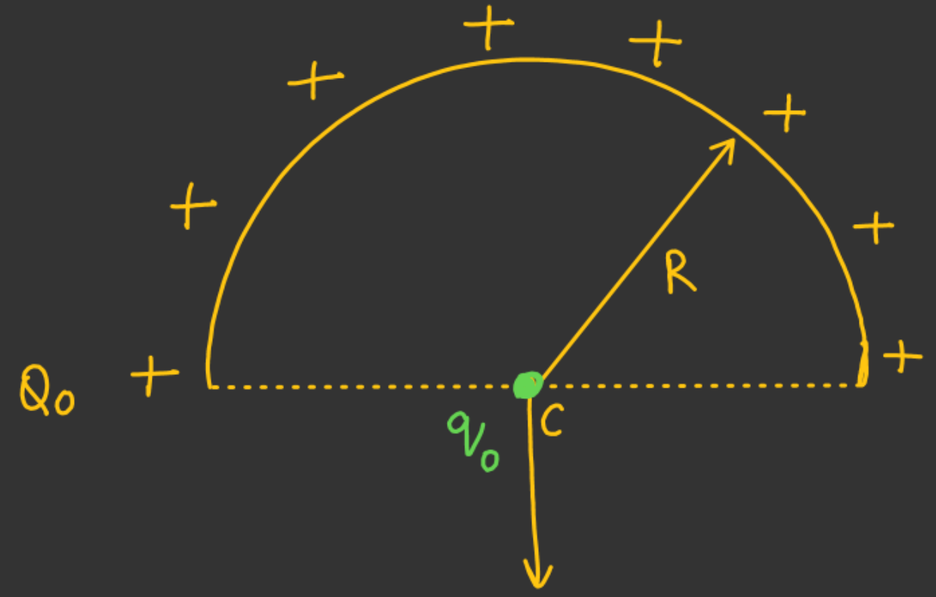
$\theta_0 = \pi$

$E_C = ?$

$$E_C = \frac{2k\lambda}{R} \cdot \sin\frac{\theta_0}{2}$$

$$E_C = \frac{2 \cdot k \cdot \left(\frac{Q}{\pi R}\right)}{R} \cdot \sin\frac{\pi}{2}$$

$$\boxed{E_C = \frac{2kQ}{\pi R^2}}$$



$$F = \frac{kQq_0}{\pi R^2}$$

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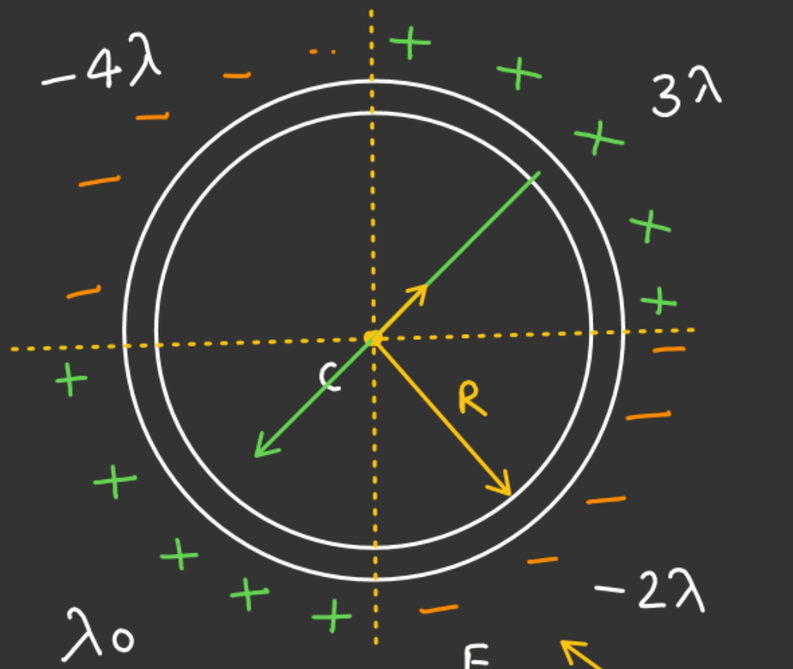

E_0

$\vec{E_C} = ?$

$\frac{2k\lambda}{R} \cdot \sin\theta_0/2$

$\frac{2k \cdot 2\lambda}{R} \cdot \sin\pi/4 = E_0$

$\vec{E_C} = -\sqrt{2}\, E_0\, \hat{i}$

$\Rightarrow \sqrt{2} \cdot \frac{2k \cdot 2\lambda}{R} \cdot \frac{1}{\sqrt{2}} (-\hat{i})$

$\Rightarrow -\frac{4k\lambda}{R} (\hat{i})$

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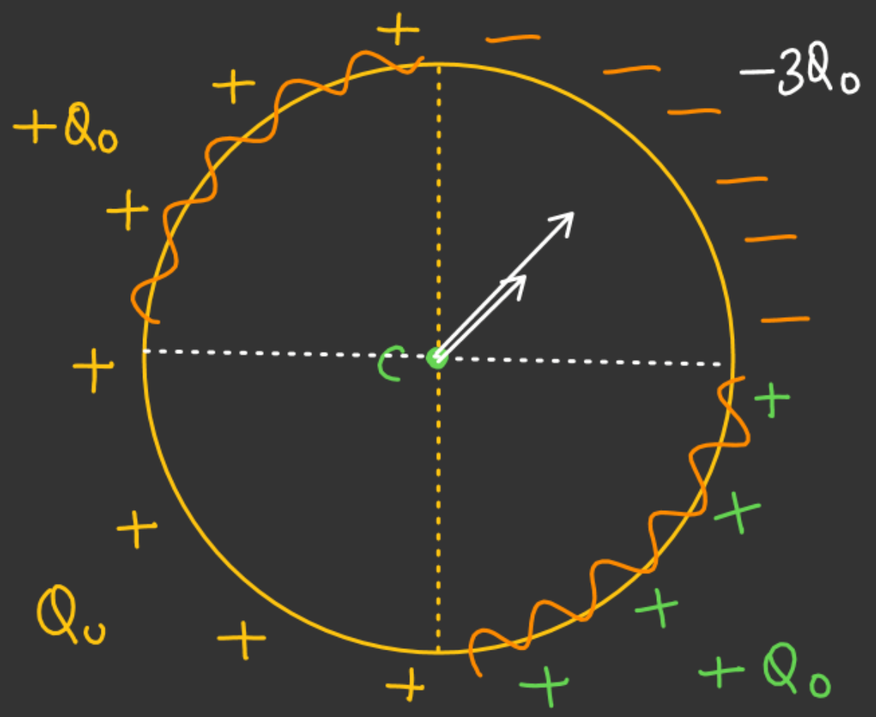
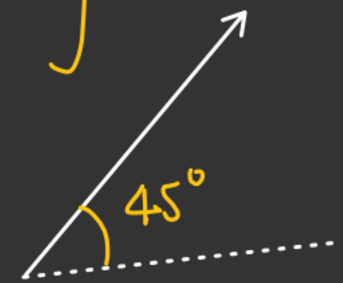


$\theta_0 = \pi/2$

$\vec{E_C} = ?$

$$\frac{K \cdot 4Q_0}{R^2} \cdot \left(\frac{\sin \pi/4}{\pi/4} \right)$$

$$\Rightarrow \frac{8\sqrt{2} K Q_0}{\pi R^2}$$



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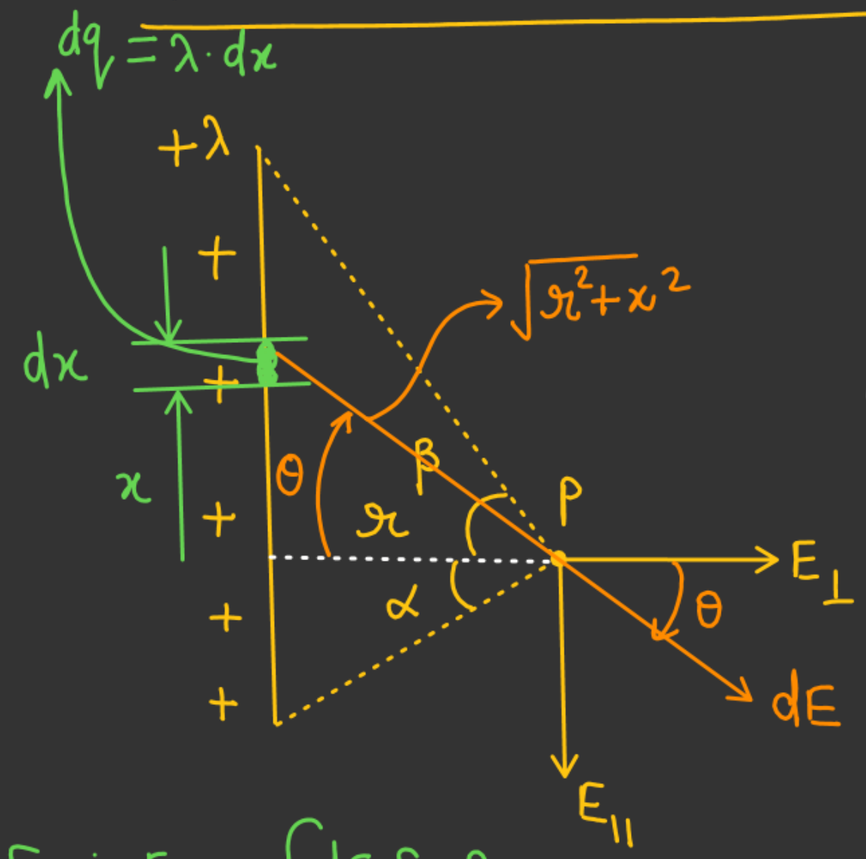
* field due to finite wire :-



$$* \quad dE = \frac{K\cdot(\lambda\cdot dx)}{\left(\sqrt{r^2+x^2}\right)^2}$$

$$r\,Tan\theta = x$$

$$r\,Sec^2\theta\cdot d\theta = dx$$

$$dE = \frac{K\cdot\lambda\cdot r\,Sec^2\theta\cdot d\theta}{r^2\,Sec^2\theta} = \frac{K\lambda}{r}\cdot d\theta$$

$$E_\perp = \int dE\,Cos\theta$$

$$E_{||} = \int dE\,Sin\theta$$

$E_\perp$:-

$$E_\perp = \int dE\,Cos\theta = \int_{-\alpha}^{+\beta} \frac{K\lambda}{r}\cdot Cos\theta\cdot d\theta$$

$$\boxed{E_\perp = \frac{K\lambda}{r}\left[Sin\alpha + Sin\beta\right]}$$

$E_{||}$: $E_{||} = \int dE\,Sin\theta$

$$= \frac{K\lambda}{\gamma}\int_{-\alpha}^{+\beta} Sin\theta\cdot d\theta \Rightarrow \frac{K\lambda}{r}\left[Cos\alpha - Cos\beta\right]$$

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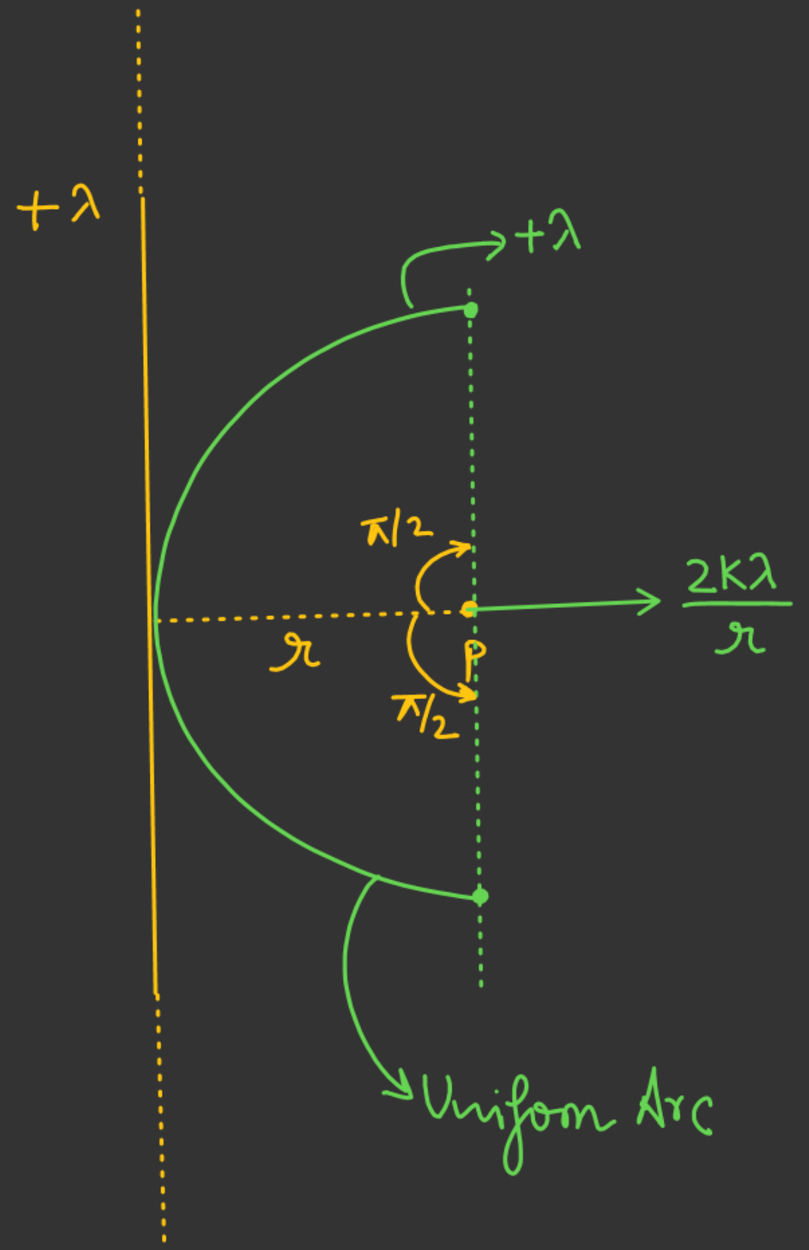

$$\left\{\begin{array}{l}\alpha = \pi/2 \\ \beta = \pi/2\end{array}\right\}$$

$$E_{\perp} = \frac{K\lambda}{r}\cdot\left(\sin\pi/2 + \sin\pi/2\right) = \frac{2K\lambda}{r}$$

$$E_{\parallel} = \frac{K\lambda}{r}\left[\cos\pi/2 - \cos\pi/2\right] = 0$$

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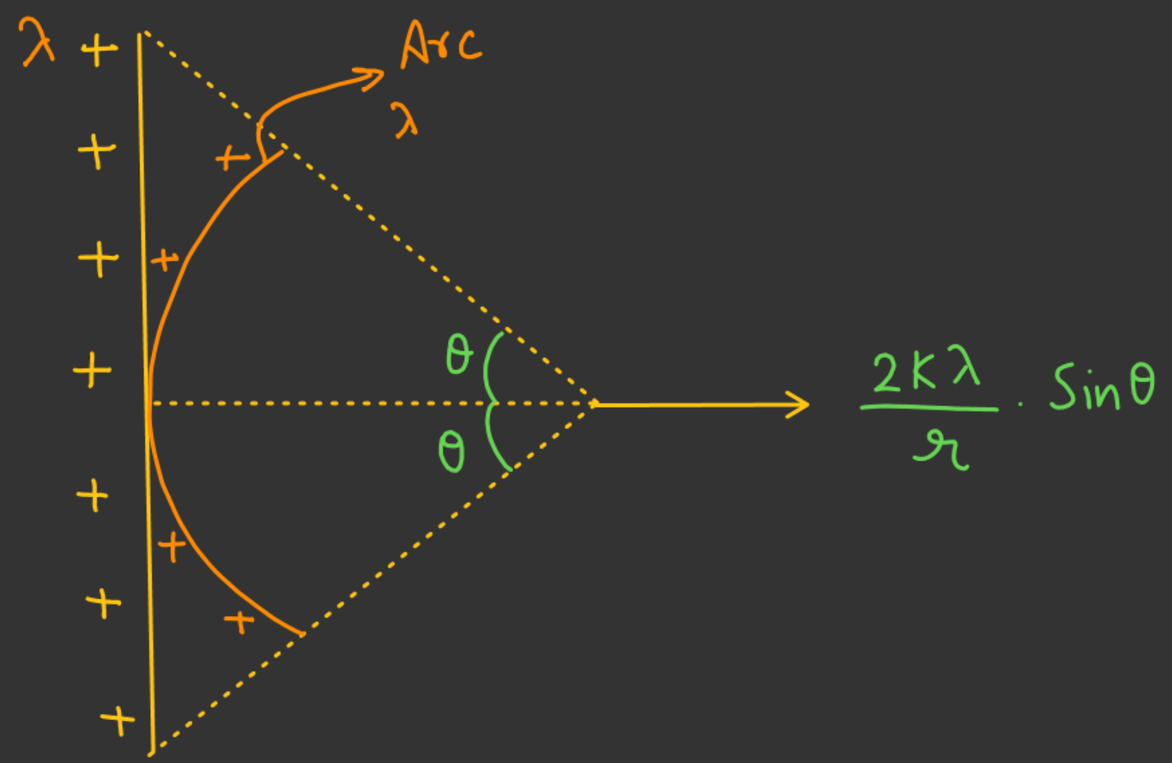
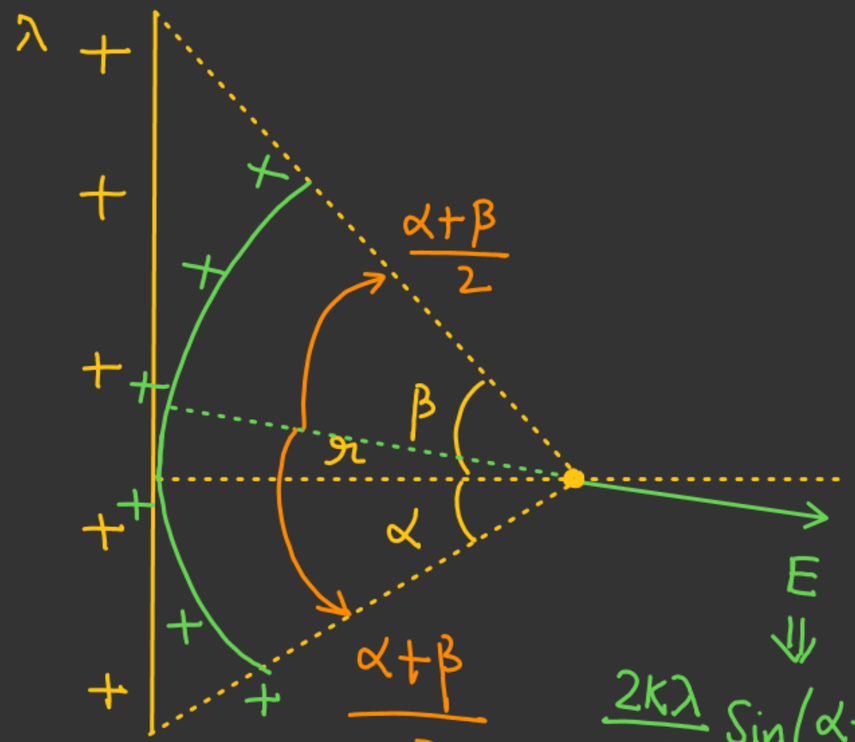


$$\frac{2K\lambda}{r}\cdot \sin\theta$$

$$E_{\perp} = \frac{K\lambda}{r}\left(\sin\theta + \sin\theta\right) = \frac{2K\lambda}{r}\sin\theta$$



$$E \Downarrow \quad \frac{2K\lambda}{r}\sin\left(\frac{\alpha+\beta}{2}\right)$$

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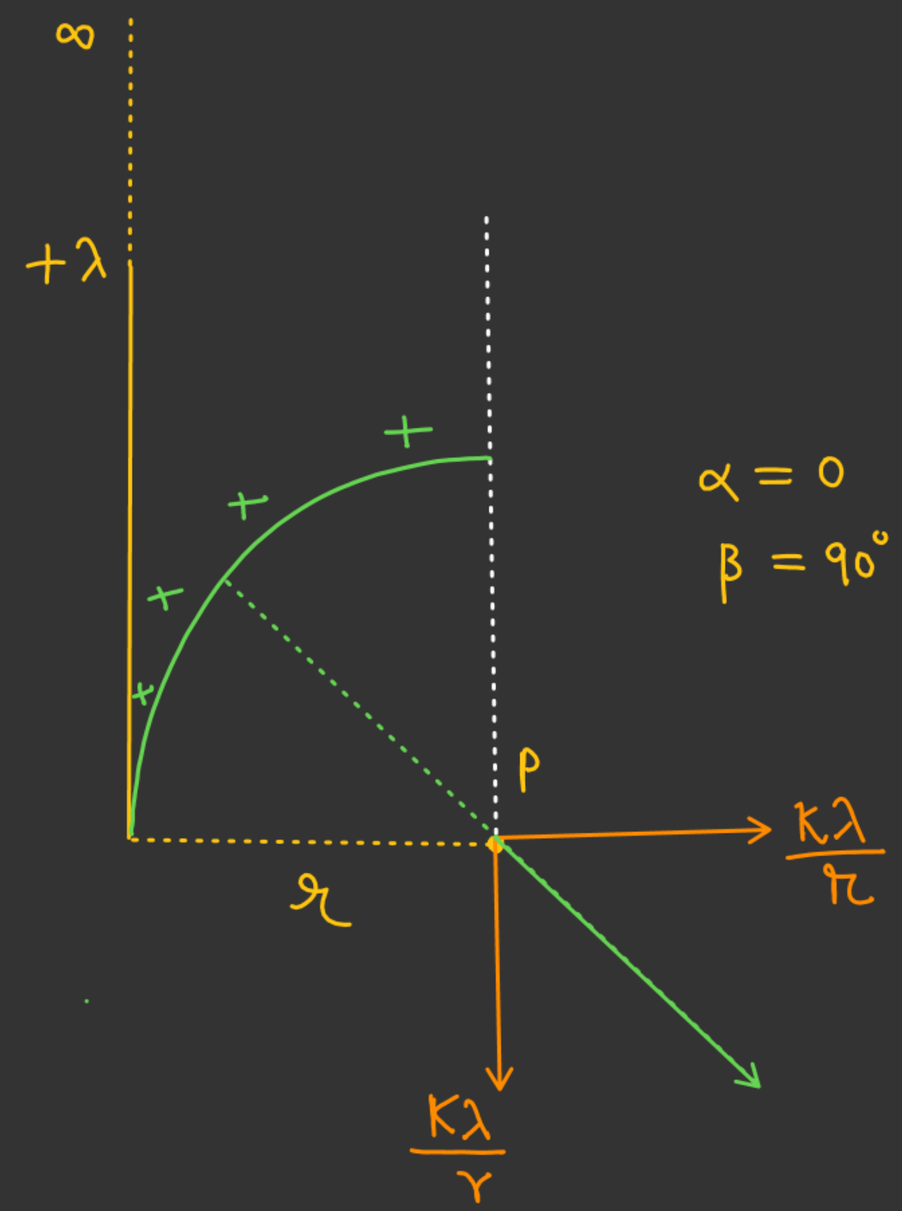
Semi infinite (At one end of wire)



$\alpha = 0$

$\beta = 90^\circ$

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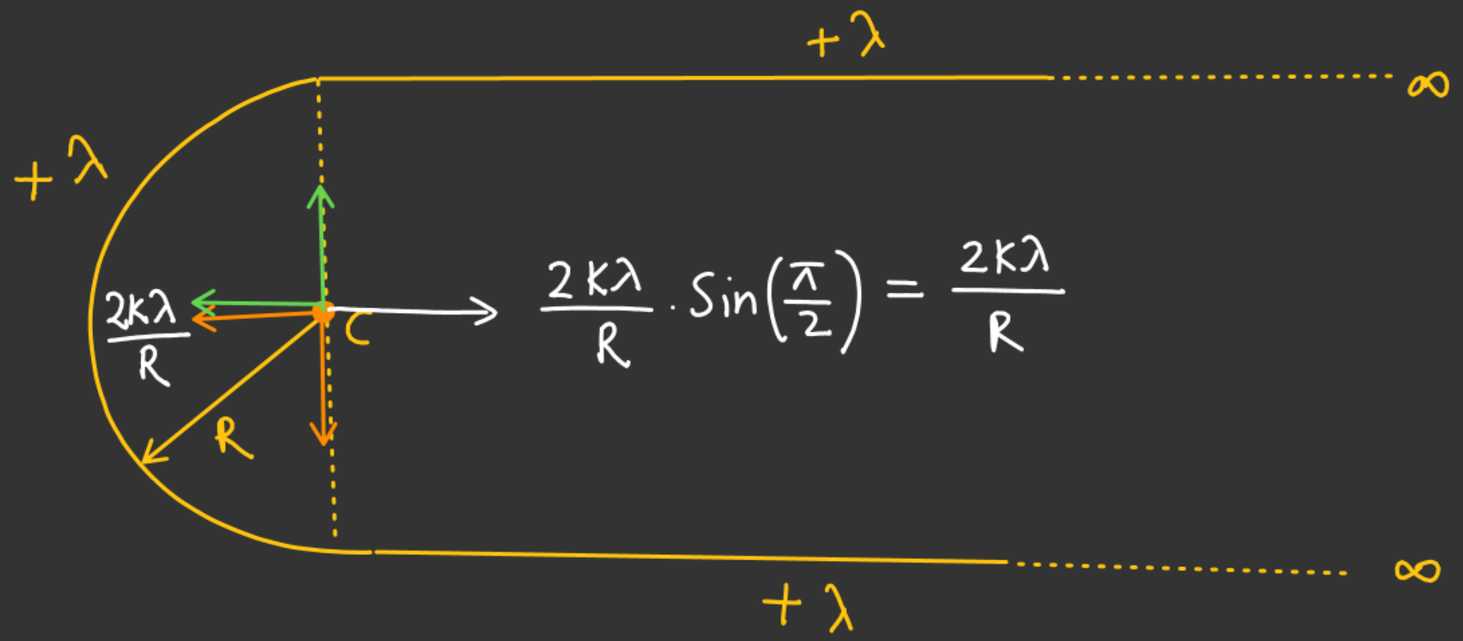
*



$$\frac{2k\lambda}{R} \cdot \sin\left(\frac{\pi}{2}\right) = \frac{2k\lambda}{R}$$

$E_C = ?$

$$\boxed{E_C = 0}$$

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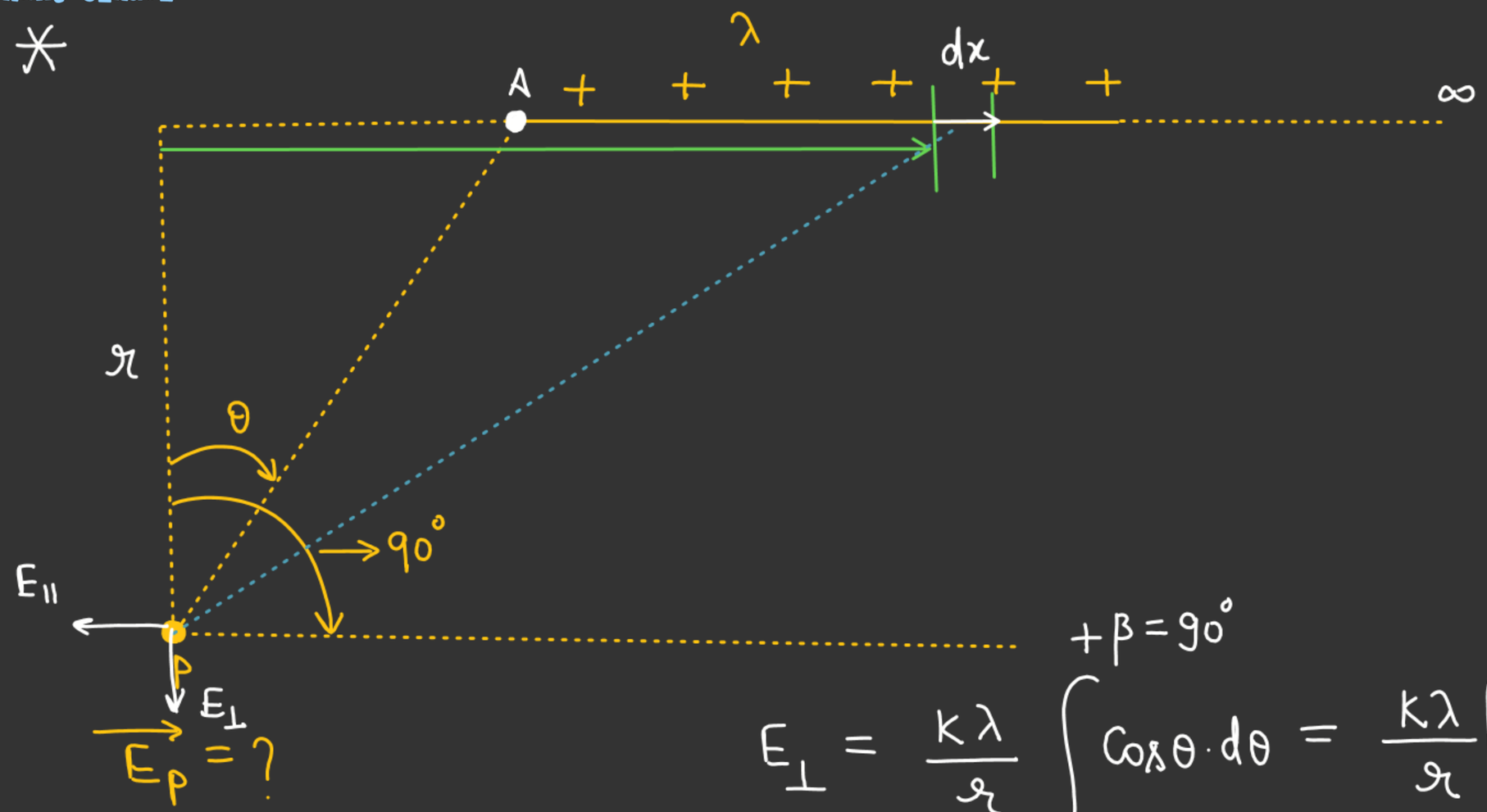

$$+\beta = 90^\circ$$

$$E_\perp = \frac{k\lambda}{r}\int \cos\theta \cdot d\theta = \frac{k\lambda}{r}\left[1 - \sin\theta\right]$$

$$-\alpha = \theta$$

$$E_\parallel = \frac{k\lambda}{r}\left[\cos\theta - \cos 90^\circ\right]$$

$$E_\parallel = \frac{k\lambda}{r}\cos\theta$$

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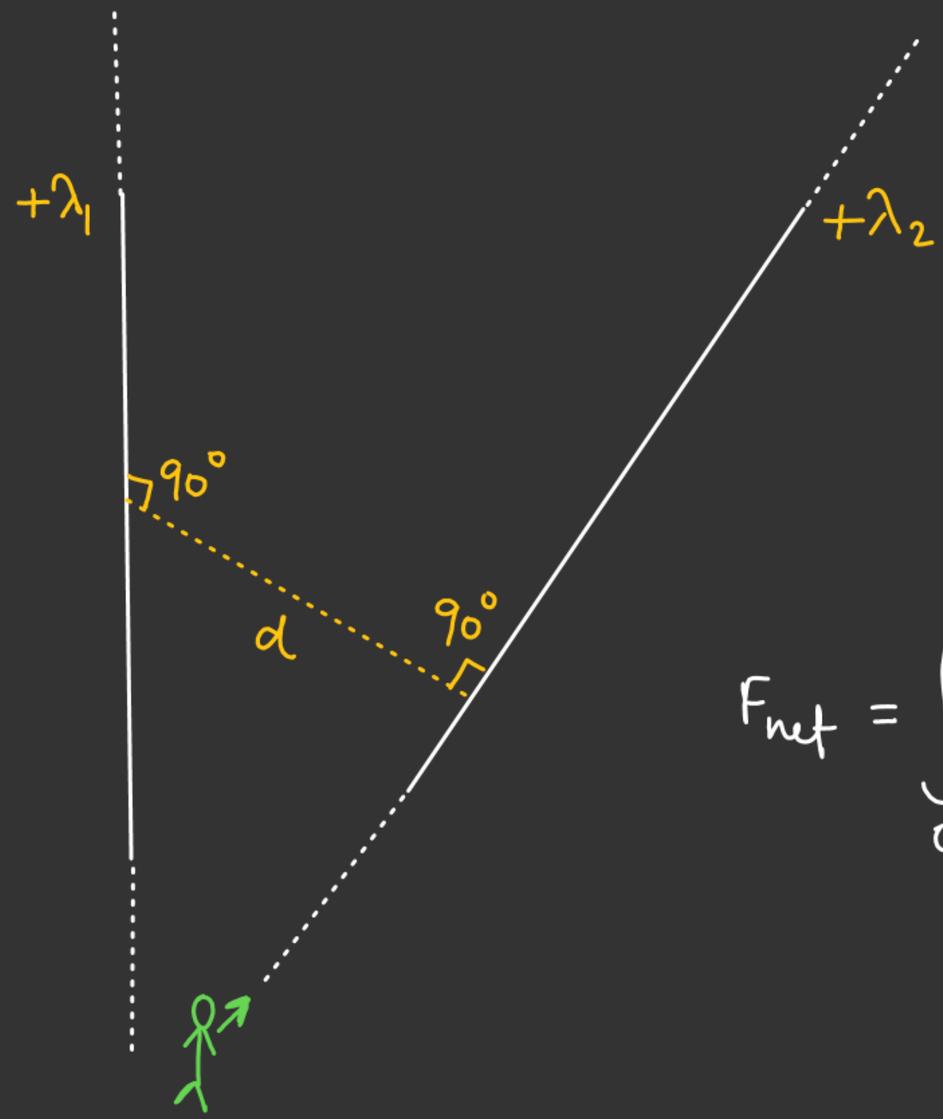
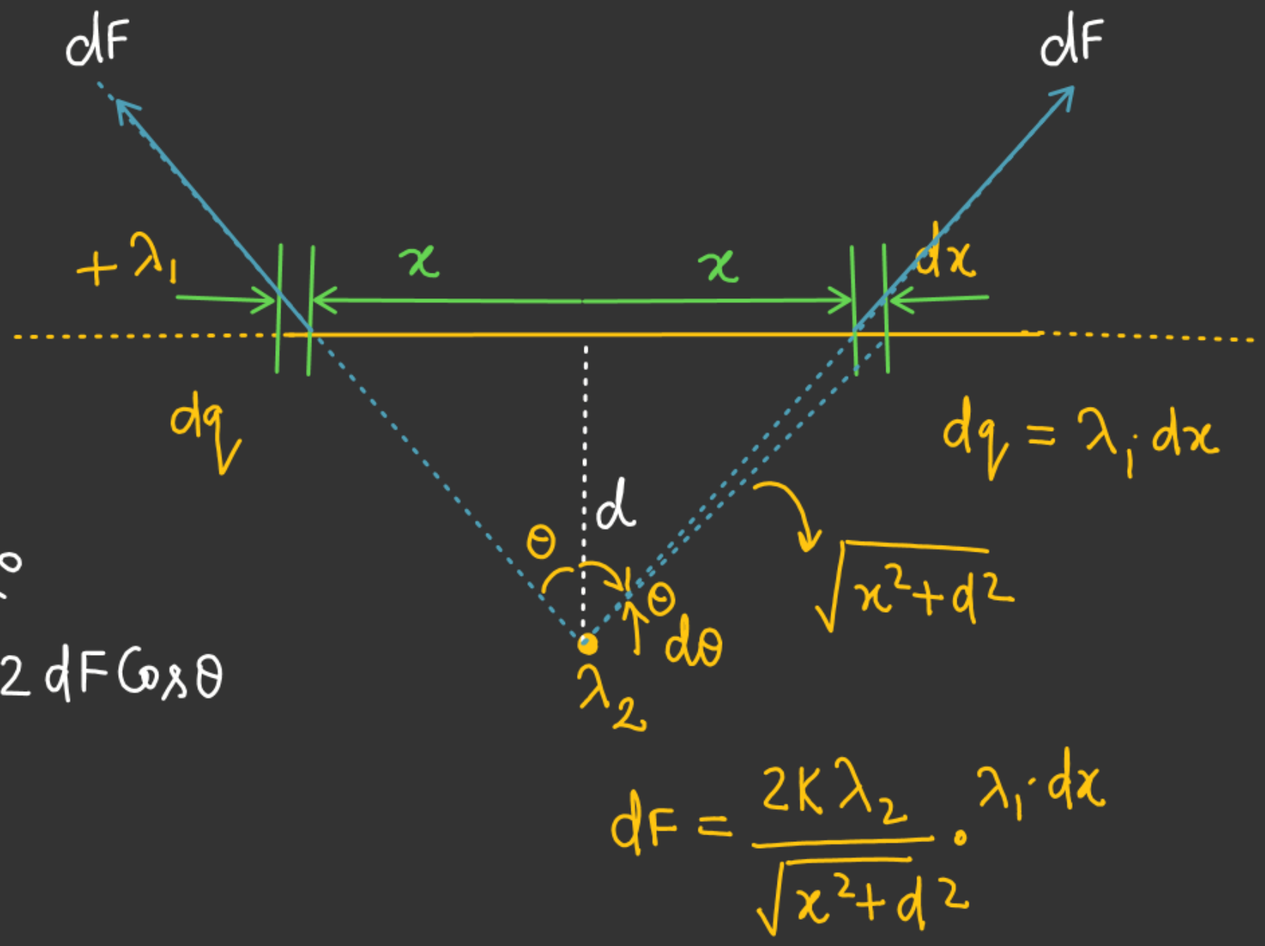
* Force b/w Skew line Charges



$$F_{net} = \int_0^{\infty} 2\,dF\cos\theta$$



$$dF = \frac{2K\lambda_2}{\sqrt{x^2+d^2}} \cdot \lambda_1 \cdot dx$$

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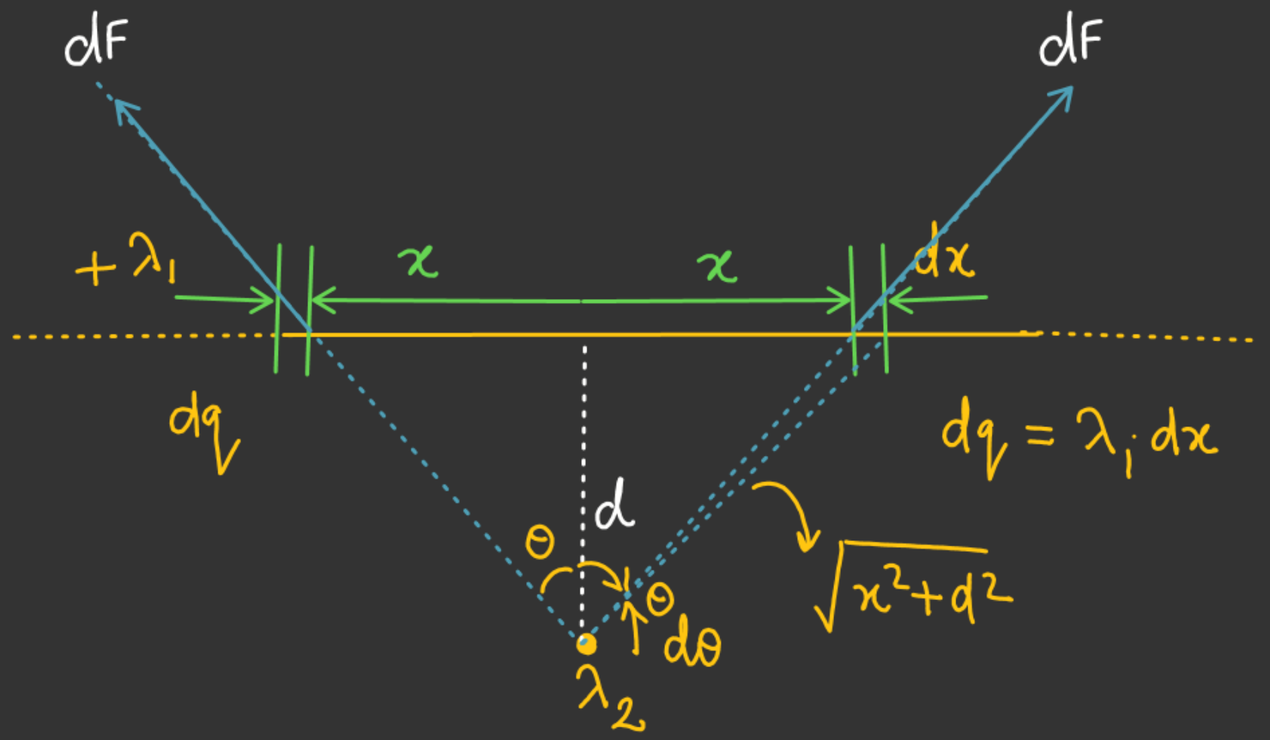
$$F_{net} = \int_0^{\infty} 2\,dF\cos\theta$$

$$dF = 2k\lambda_1\lambda_2 \sec\theta \cdot d\theta$$

$$F_{net} = \int_0^{\pi/2} 2 \cdot 2k\lambda_1\lambda_2 \cdot d\theta$$

$$\Rightarrow 2\pi k\lambda_1\lambda_2$$

$$\boxed{F \Rightarrow \frac{\lambda_1\lambda_2}{2\epsilon_0}}$$



$$dF = \frac{2k\lambda_2}{\sqrt{x^2+d^2}} \cdot \lambda_1 \cdot dx \Rightarrow \frac{2k\lambda_1\lambda_2 \cdot d\sec^2\theta \cdot d\theta}{d\sec\theta}$$

$$x = d\tan\theta$$

$$dx = d\sec^2\theta \cdot d\theta$$

$$d^2 + x^2 = d^2(1+\tan^2\theta)$$

$$= d^2\sec^2\theta$$

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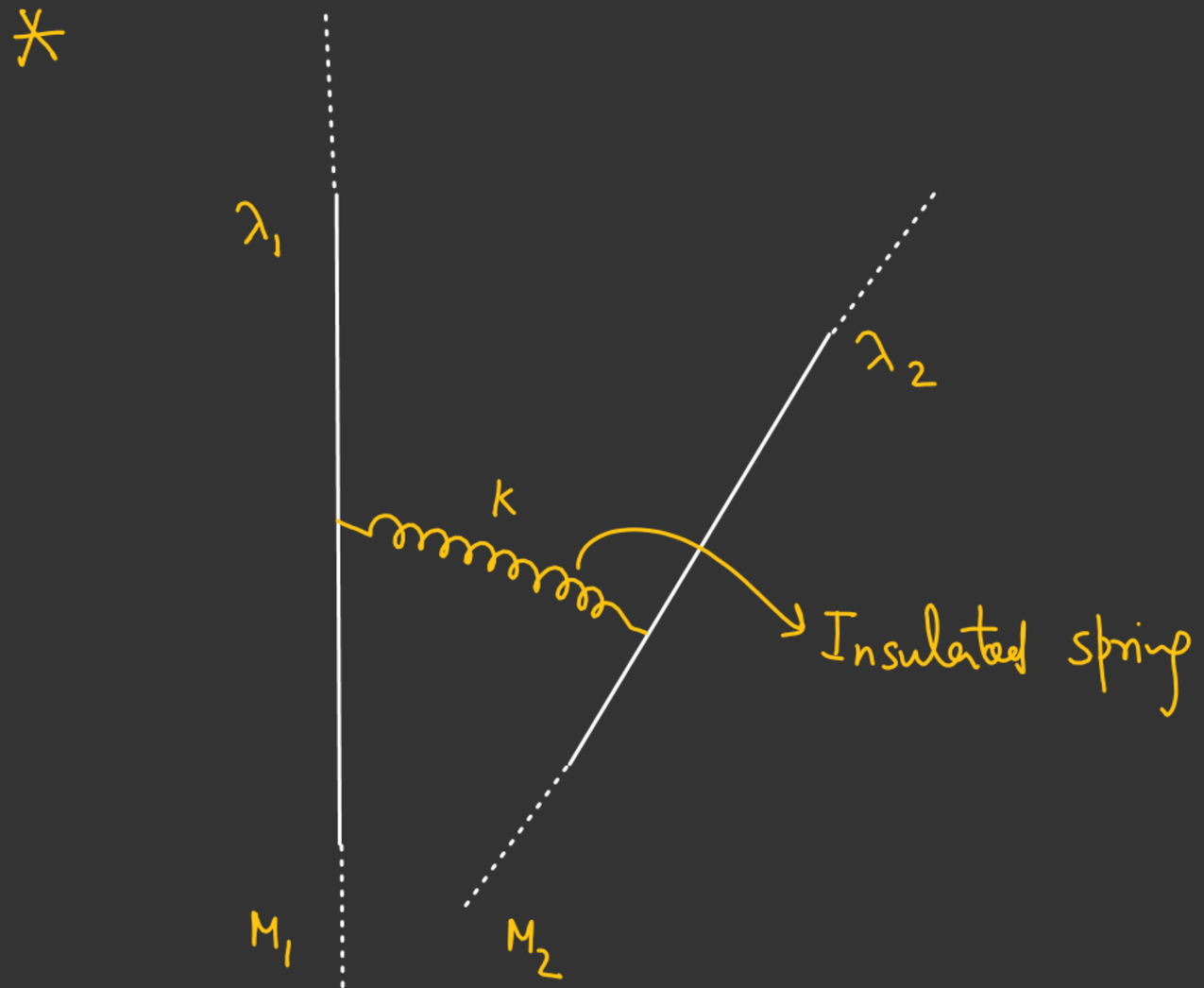

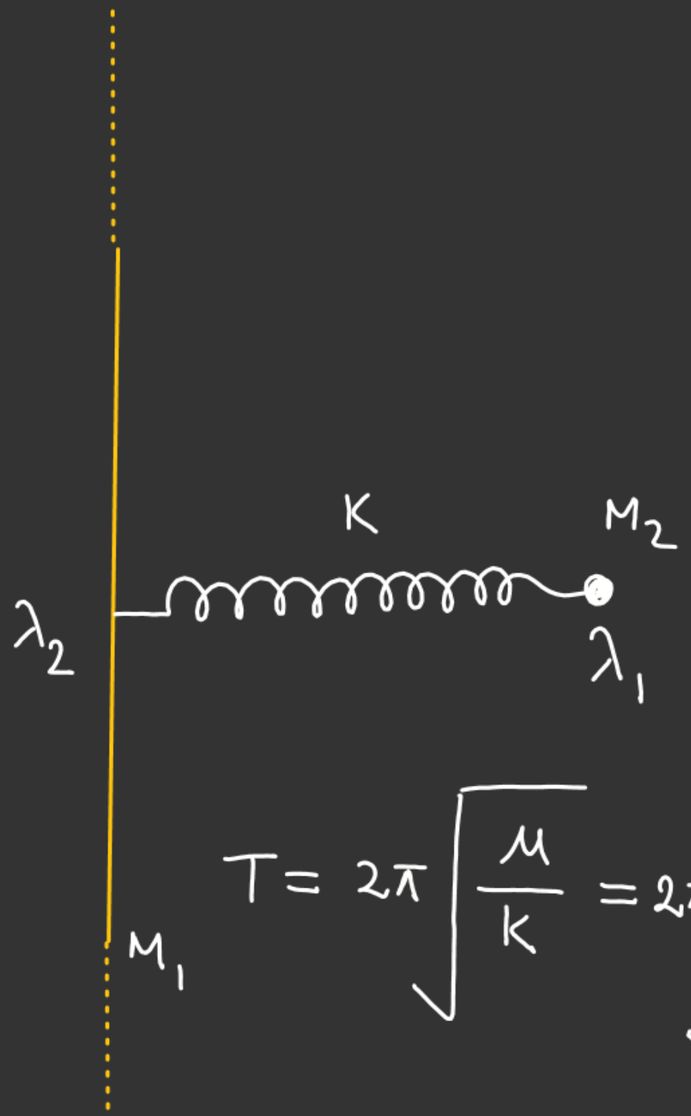

$$T = 2\pi\sqrt{\frac{\mu}{k}} = 2\pi\sqrt{\frac{m_1 m_2}{(m_1+m_2)k}}$$

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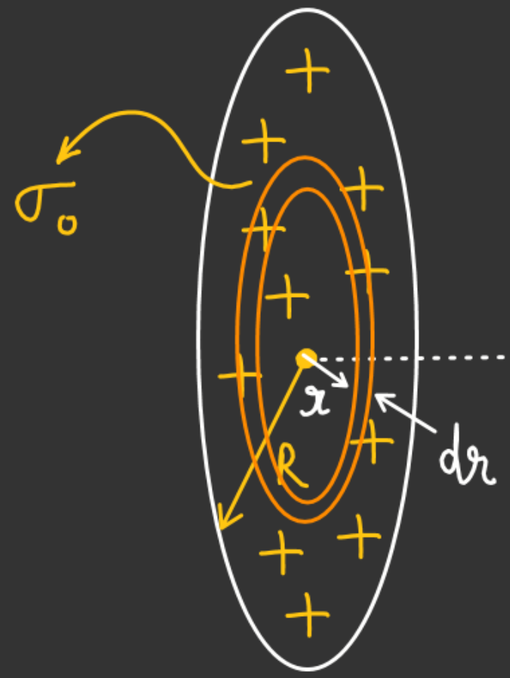
* Field due to Disc :-



$$dE = \frac{K \cdot dq \cdot x}{(r^2+x^2)^{3/2}}$$

$$dq = \sigma_0 \cdot dA = \sigma_0 (2\pi r \cdot dr)$$

$$r^2 + x^2 = v^2$$

$$r\,dr = v \cdot dv$$

$$dE = \frac{K \cdot \sigma_0 \cdot 2\pi r\, dr \cdot x}{(r^2+x^2)^{3/2}}$$

$$dE = \frac{K \cdot \sigma_0 \cdot 2\pi\, v \cdot dv \cdot x}{v^3}$$

$$dE = 2\pi K \sigma_0 x \frac{dv}{v^2}$$

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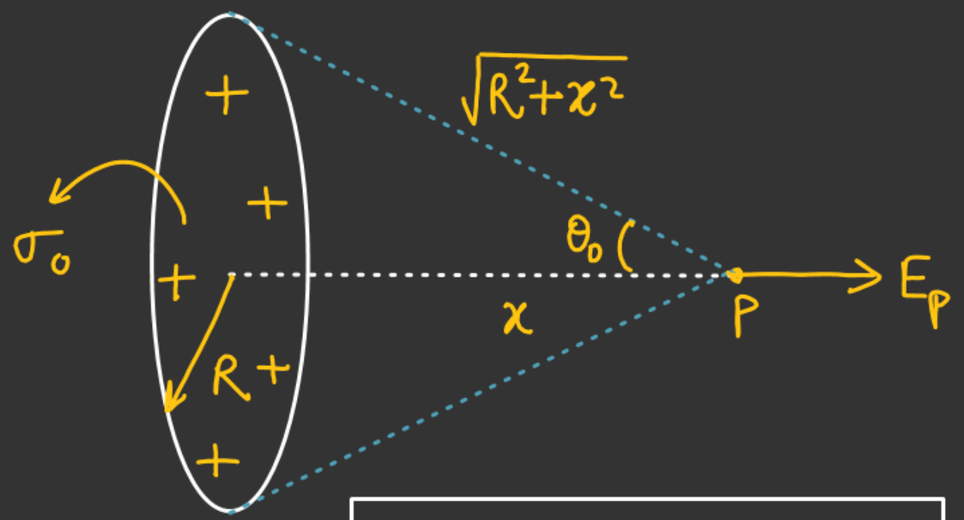
$$\int dE = 2\pi k \sigma_0 x \int_{x}^{\sqrt{R^2+x^2}} \frac{du}{u^2}$$

$$E_p = \frac{\sigma_0}{2\epsilon_0} \cdot x \left(-\frac{1}{u} \Big|_{x}^{\sqrt{R^2+x^2}} \right)$$

$$E_p = \frac{\sigma_0}{2\epsilon_0} x \left(\frac{1}{x} - \frac{1}{\sqrt{R^2+x^2}} \right)$$

$$\boxed{E_p = \frac{\sigma_0}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{R^2+x^2}} \right)}$$



$$\cos\theta_0 = \frac{x}{\sqrt{R^2+x^2}}$$

$$\boxed{E_p = \frac{\sigma}{2\epsilon_0} \left(1 - \cos\theta_0 \right)}$$

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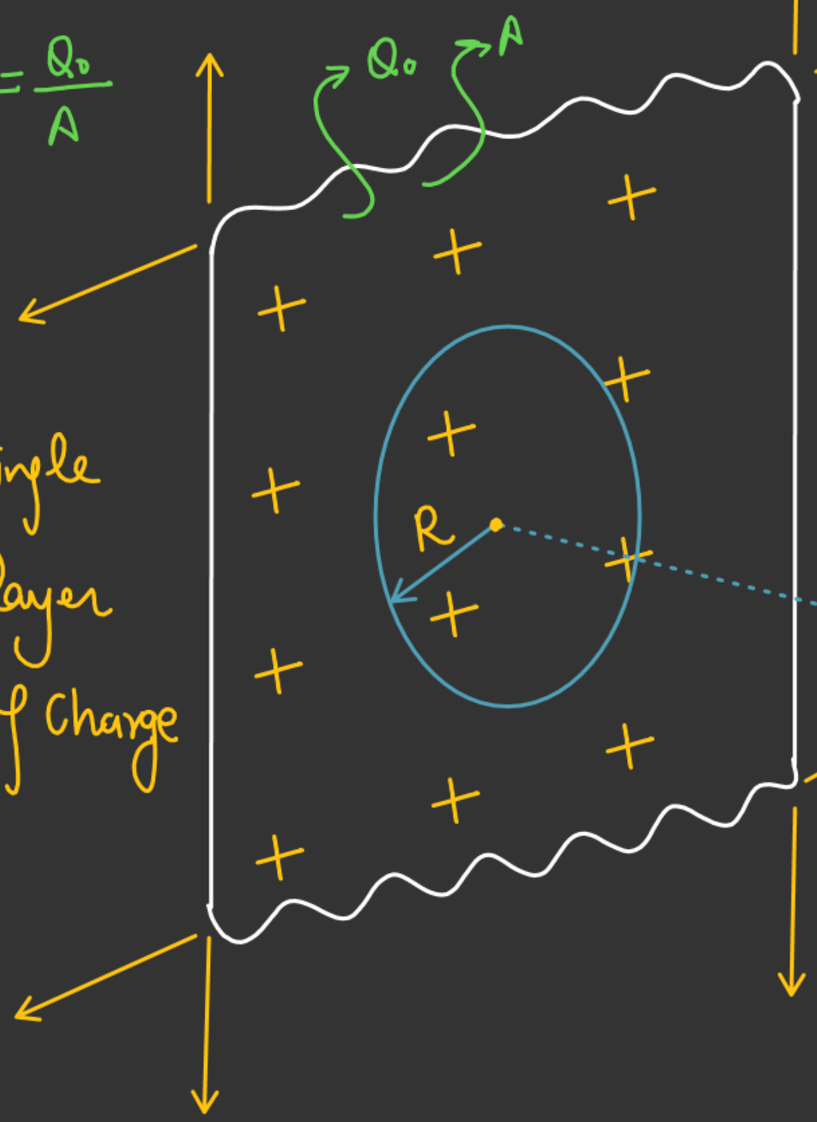
* Infinite sheet :-

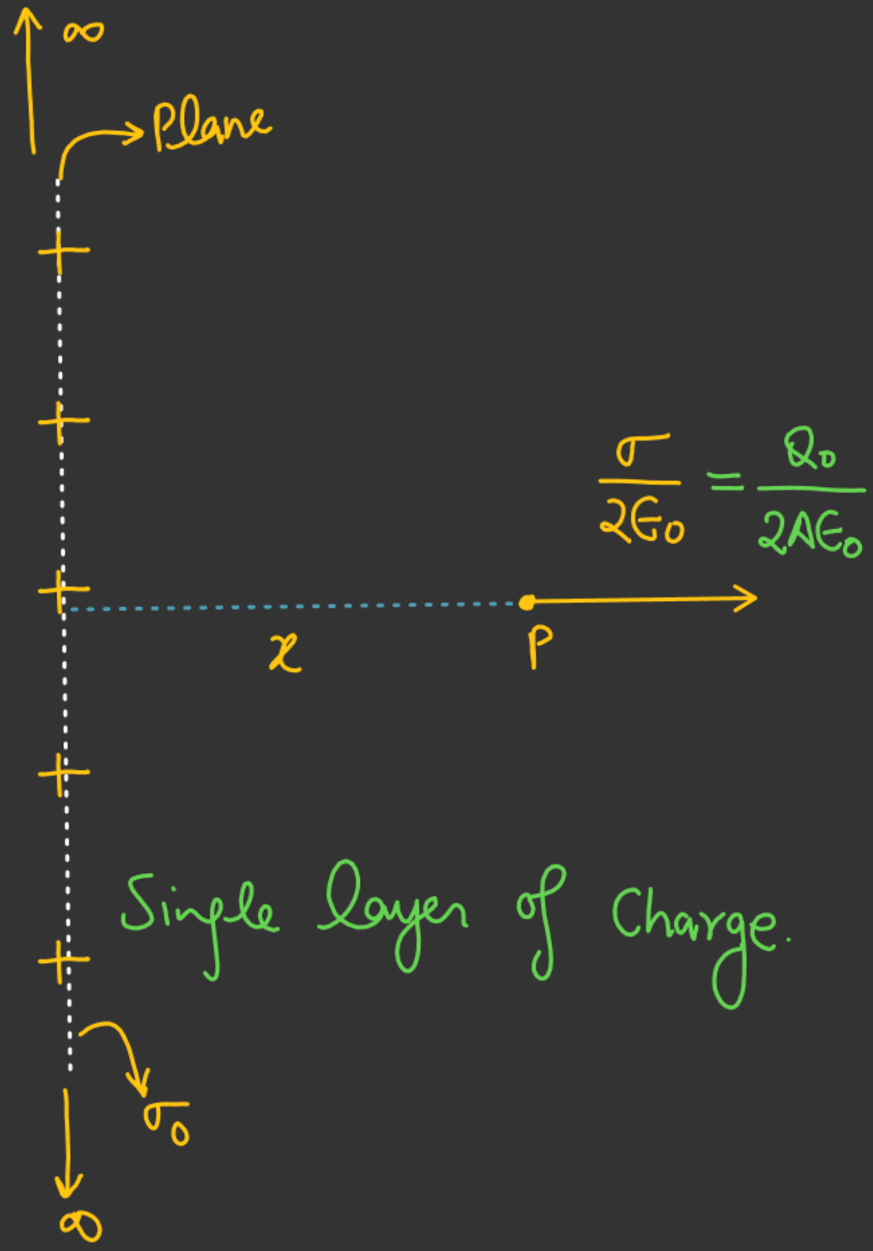
$\sigma = \frac{Q_0}{A}$



$$E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$$

$R \longrightarrow \infty$

$E \longrightarrow \frac{\sigma}{2\epsilon_0}$



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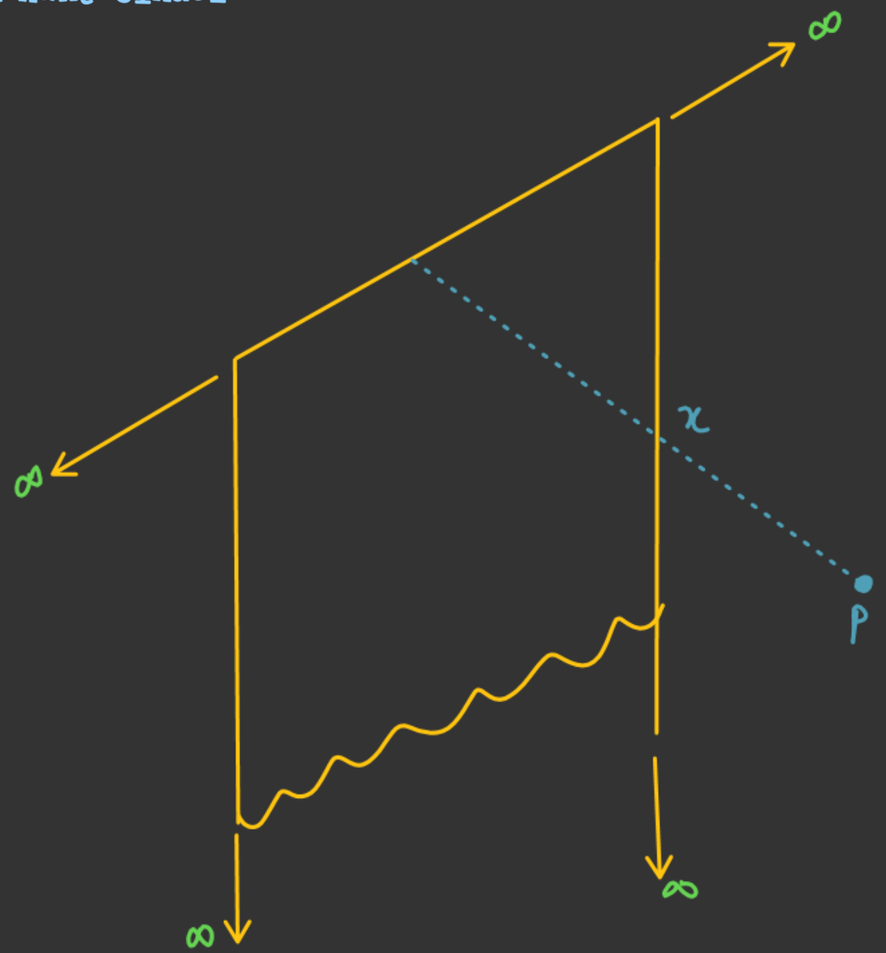
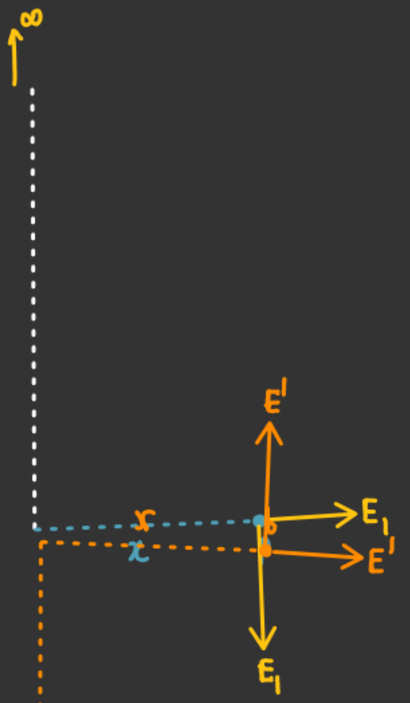
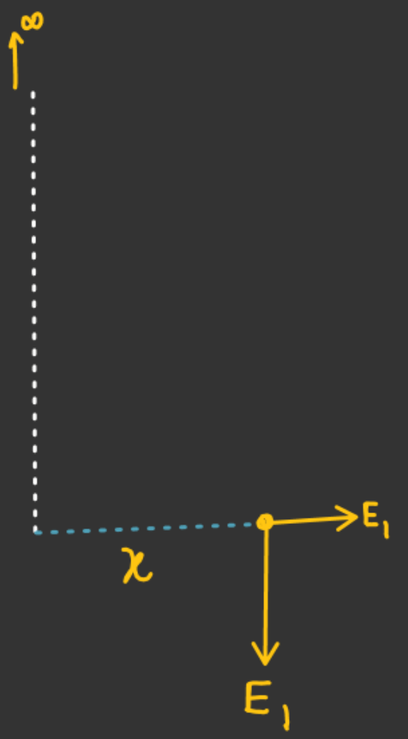




$$2E' = \frac{\sigma}{2\epsilon_0}$$

$$E' = \frac{\sigma}{4\epsilon_0}$$



$$\sqrt{2}E_1 = \frac{\sigma}{2\sqrt{2}\epsilon_0}$$

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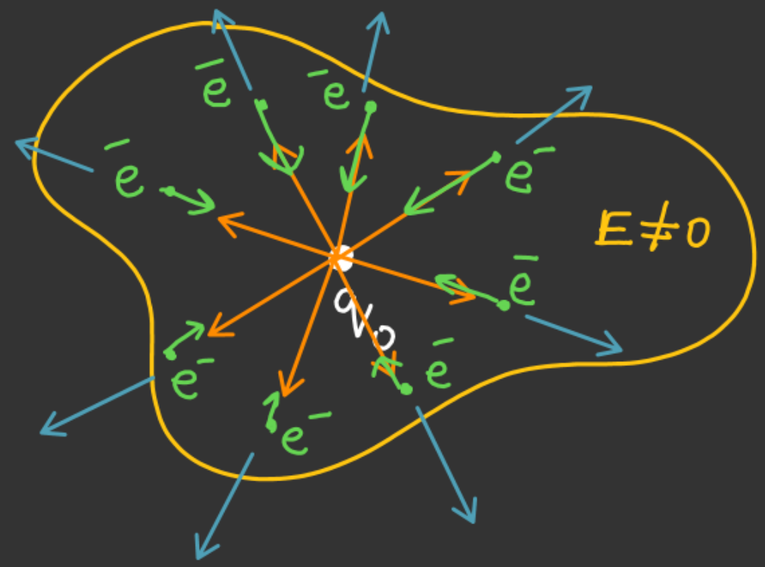
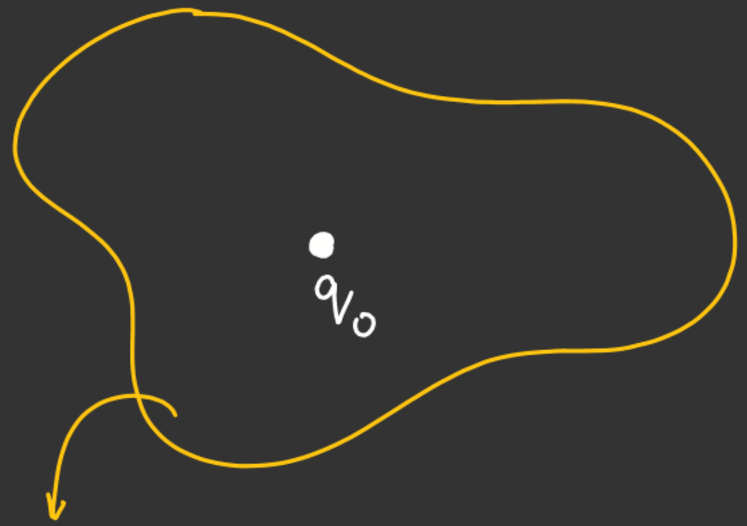
* Electrostatic Condition of a Conductor :-





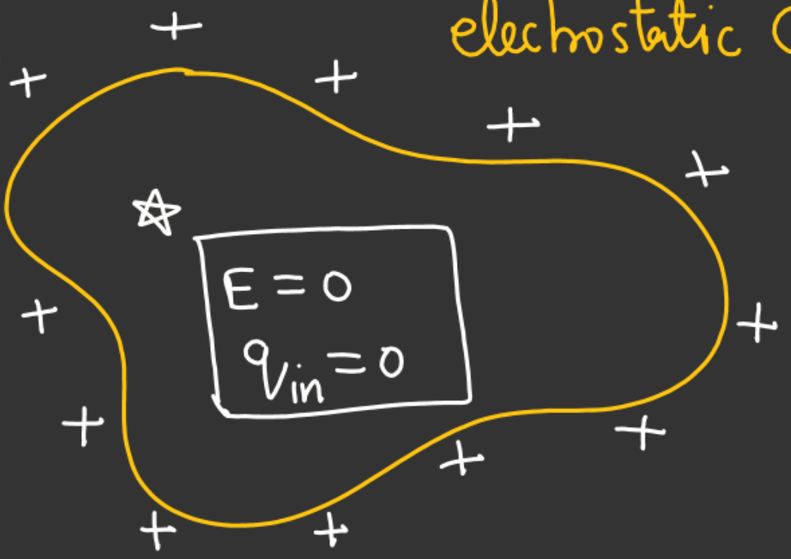
Conductor

↳ large no. of free $\bar{e}$

non Uniform

$\int \sigma \cdot dA = q_0$

electrostatic Condition



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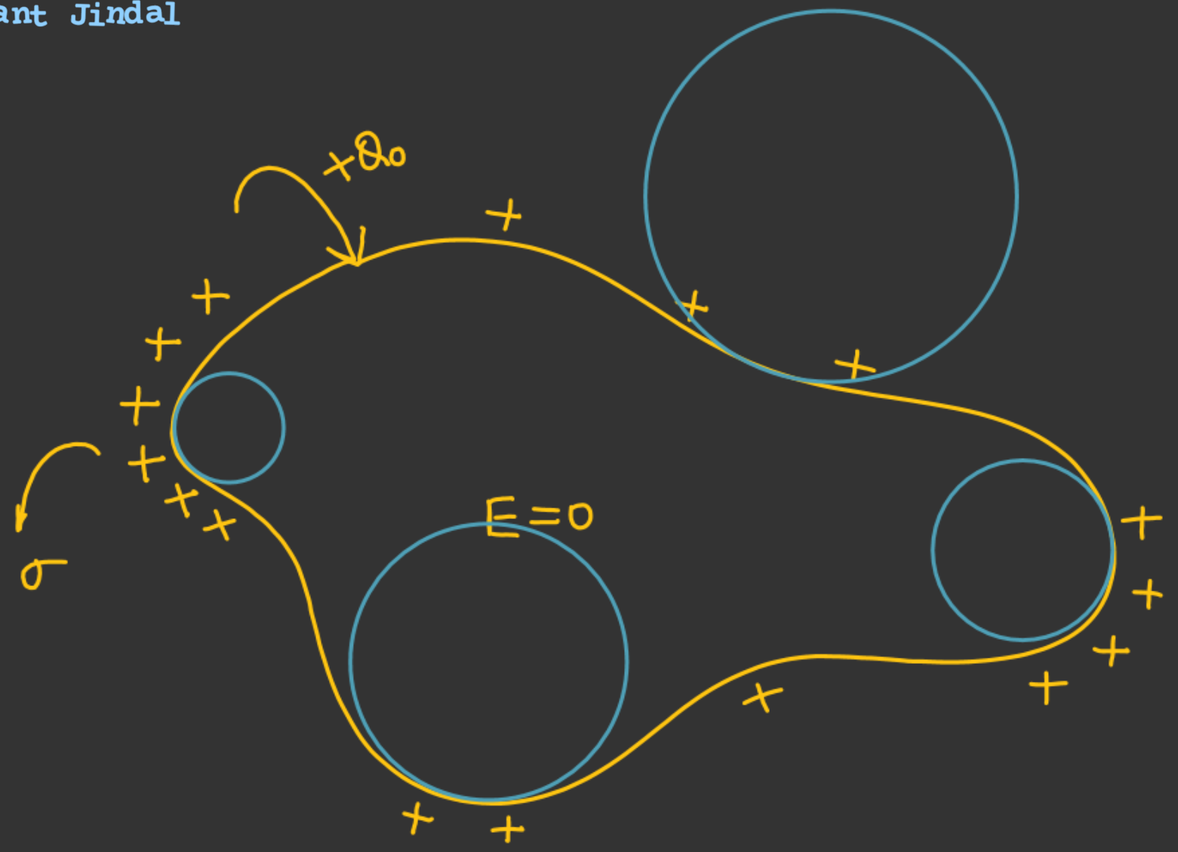
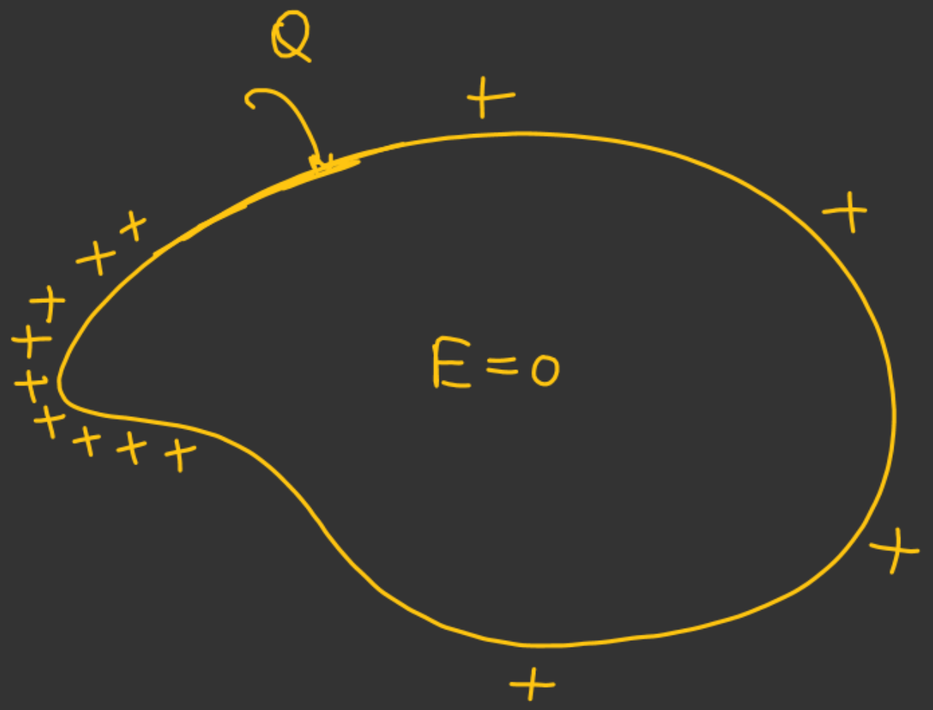


$$\sigma \propto \frac{1}{R_c} \quad (\text{Approx})$$



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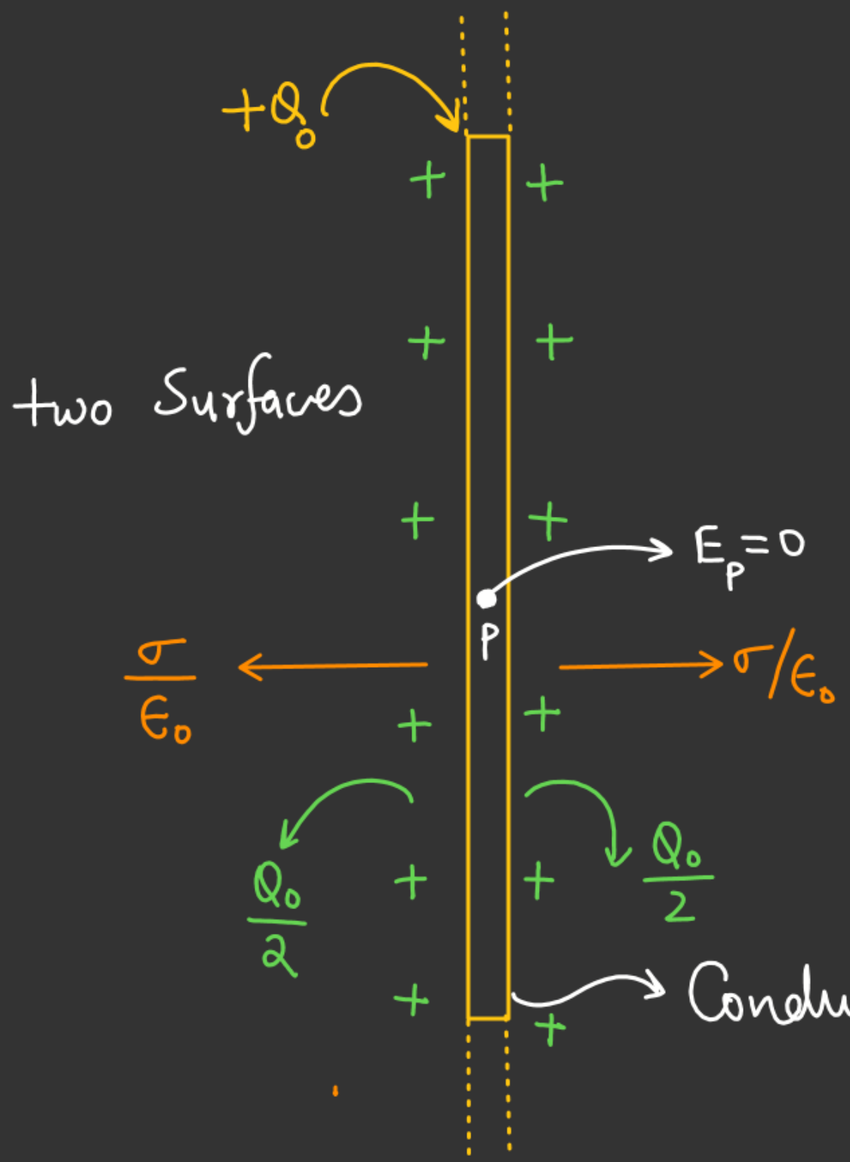
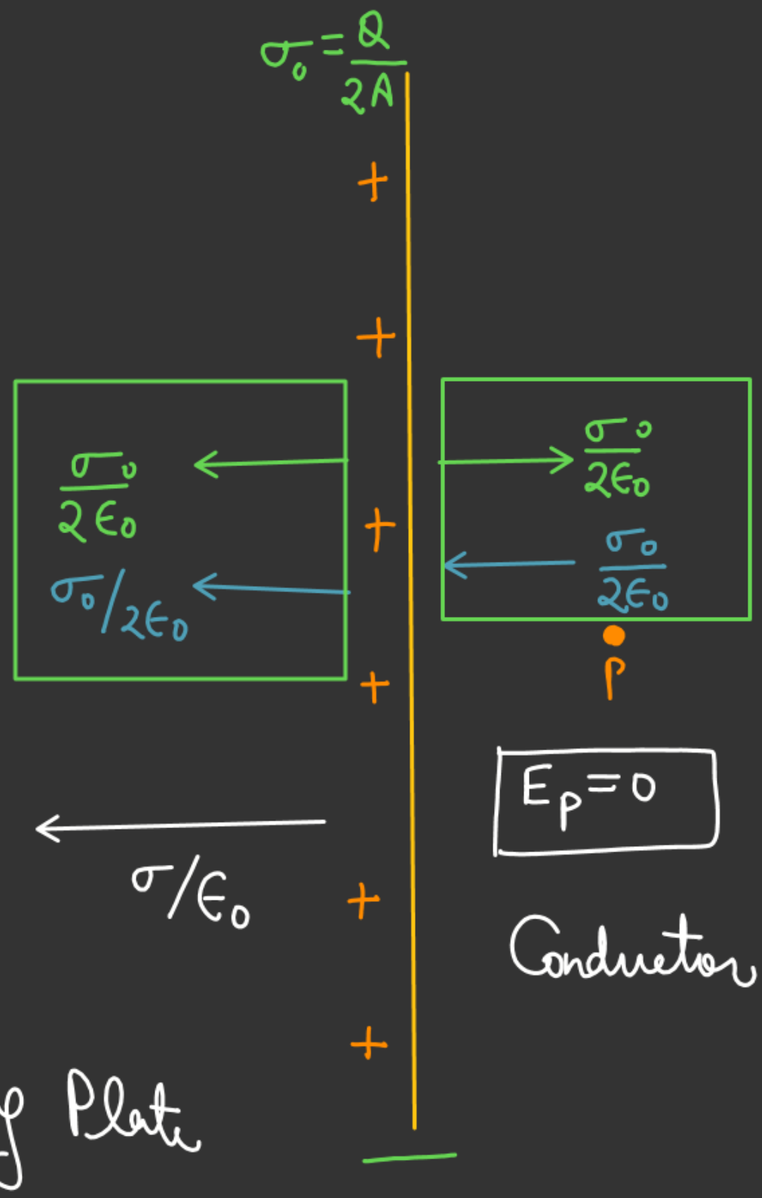
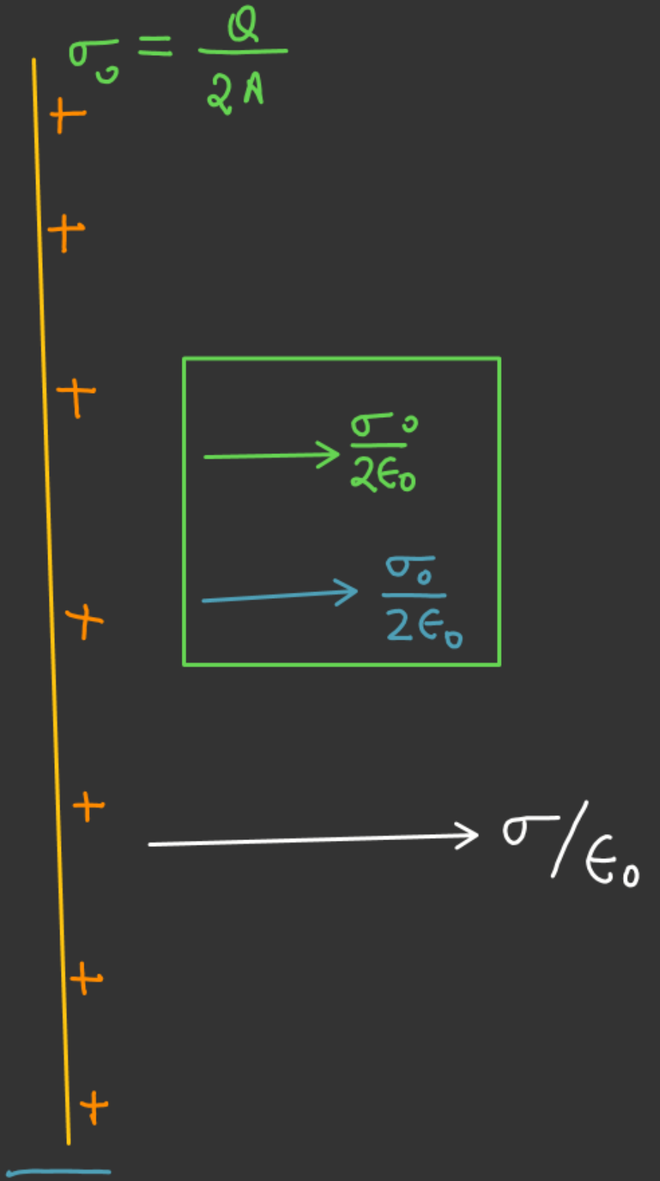
Conducting Plate :-







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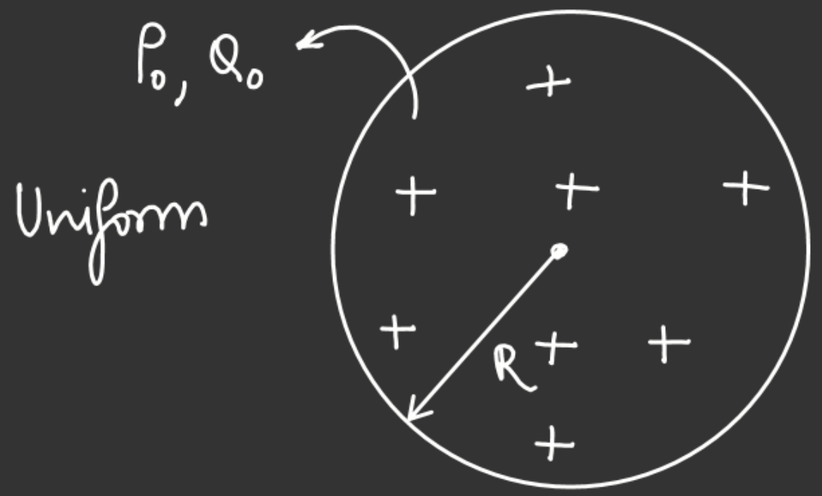
*

Solid Non Conducting



$$Q_0 = \rho_0 \cdot \frac{4}{3}\pi R^3$$

E Vs r graph

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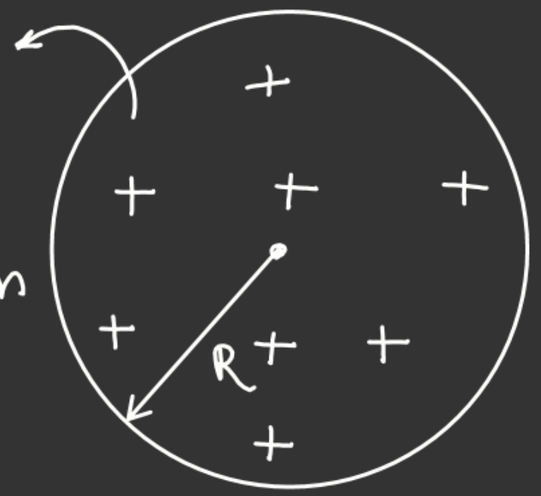
*

Solid Non Conducting



Non Uniform

(i) $\rho_r = \rho_0 \frac{r}{R}$

$\rho_r = \rho_0 \left(1 + \frac{r}{R}\right)$

$\rho_r = \rho_0 \left(1 - \frac{r}{R}\right)$

$\rho_r = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$

(i) E Vs r graph

(ii) Total Charge.

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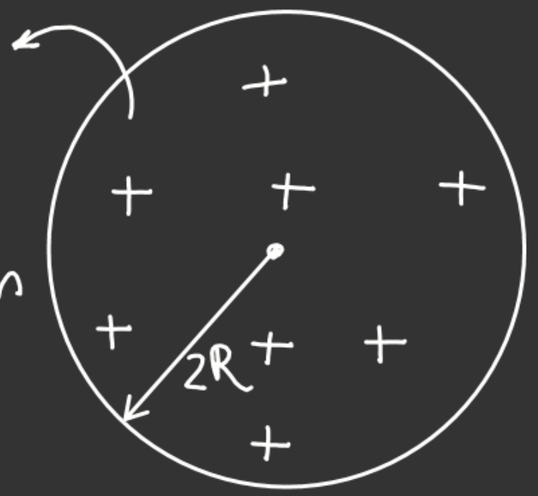
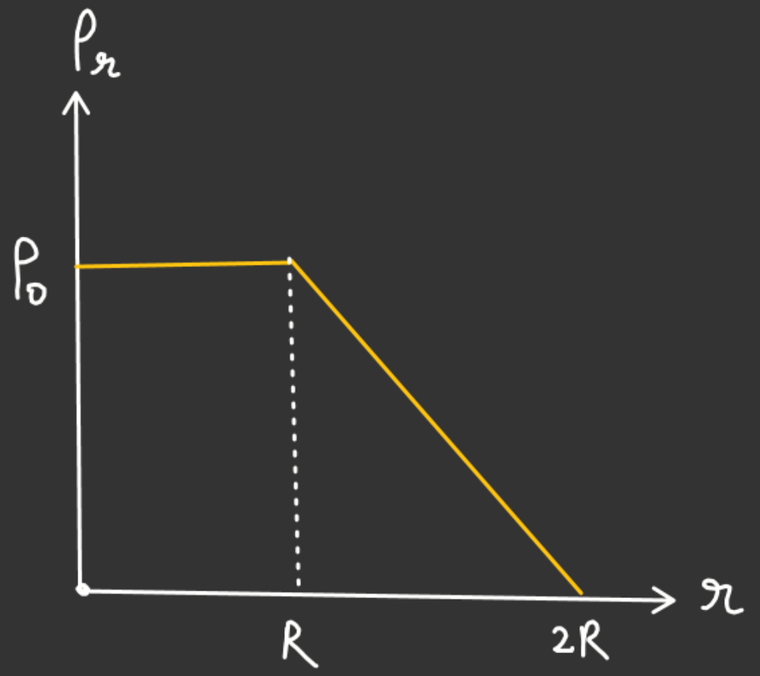
*

Solid Non Conducting



Non Uniform



(i) Total Charge = ?

Draw E Vs r graph

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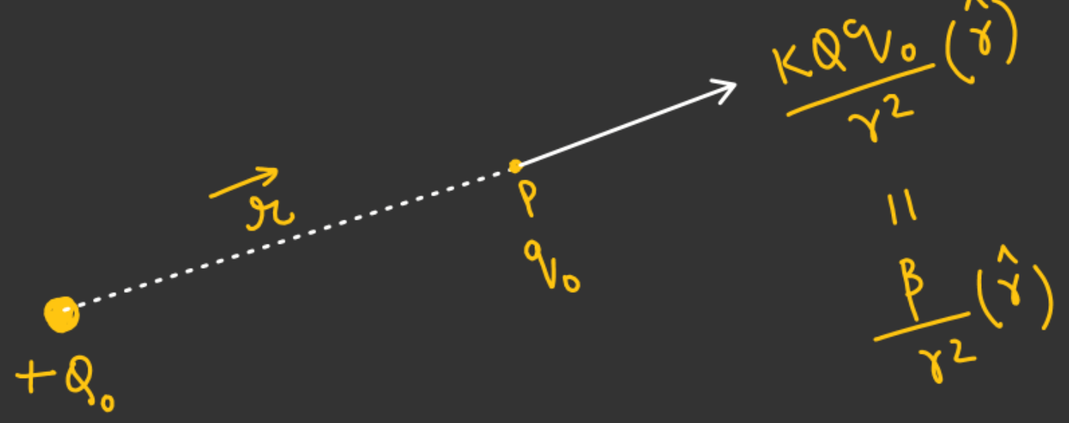
* Electric potential & Potential Energy :-

electrostatic force ⟶ Conservative



$$= \frac{\beta}{r^2}(\hat{r})$$

Curl of Force $= 0$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{k} \\ \frac{\partial}{\partial r} & \frac{\partial}{r\partial\theta} & \frac{\partial}{\partial z} \\ \frac{\beta}{r^2}(\hat{r}) & 0 & 0 \end{vmatrix} = 0$$

Always Correct →

$$W_c = W_e = U_i - U_f = -\Delta U \quad (\text{by def}^n)$$

$$W_{agent} + W_e = K_f - K_i$$

$W_e \to -\Delta U$, $K_f - K_i \to 0$ (very slowly)

$$W_{agent} = \Delta U = U_f - U_i$$

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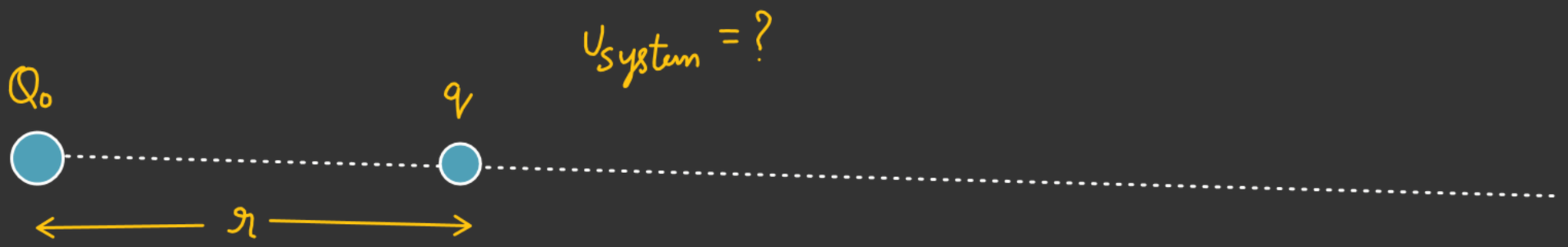

* P.E. is defined as the work done in Constructing the System (work done in bringing a charge q from ∞ to point P)

$$W_{agent} = U_f - U_i = U_P - U_\infty$$

ref $U_\infty = 0$

$$U_P = W_{agent}{}_{\infty \to P}$$

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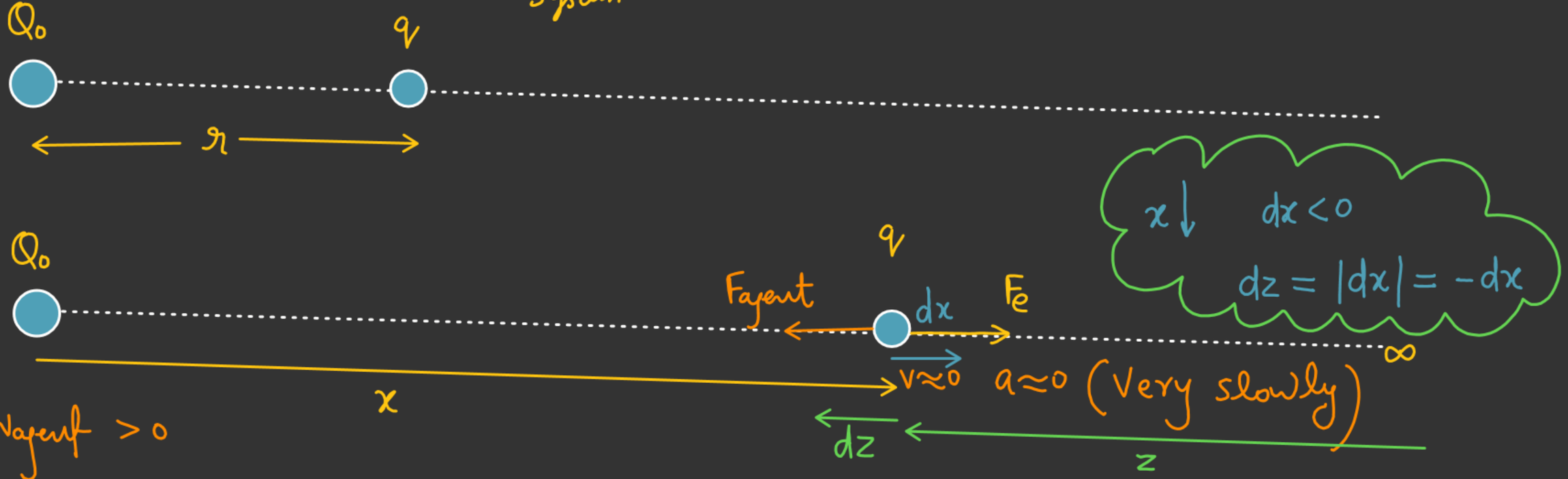
$U_{system} = ?$



$dW_{agent} > 0$

$dW_{agent} = F_{agent} \cdot dz \cdot \cos 0^\circ$

$= \dfrac{KQq_0}{x^2}(-dx)$

$F_{agent} = \dfrac{KQq_0}{x^2} = F_e$

$z \uparrow \quad dz > 0$

$z + x = \text{Const}$

$dz = -dx$

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$$dW_{agent} = -\frac{kQq_0}{x^2}dx$$

Note: Charges will be taken with sign in the expression of P.E.

$$W_{agent} = \int dW_{agent} = -kQq_0\int_{\infty}^{r}\frac{dx}{x^2} = -kQq_0\left(-\frac{1}{x}\Big|_{\infty}^{r}\right) = \frac{kQq_0}{r}$$

$$W_{agent} = \Delta U = U_r - U_\infty$$

U_∞: ref at ∞

$$U_r = W_{agent} = \frac{kQq_0}{r}$$

* It is the potential Energy stored in the electric field of Q_0 & q_0

Work done by the agent in moving q_0 in the electric field of Q_0.

* $$W_e = -\frac{kQq_0}{r}$$

Very Slowly

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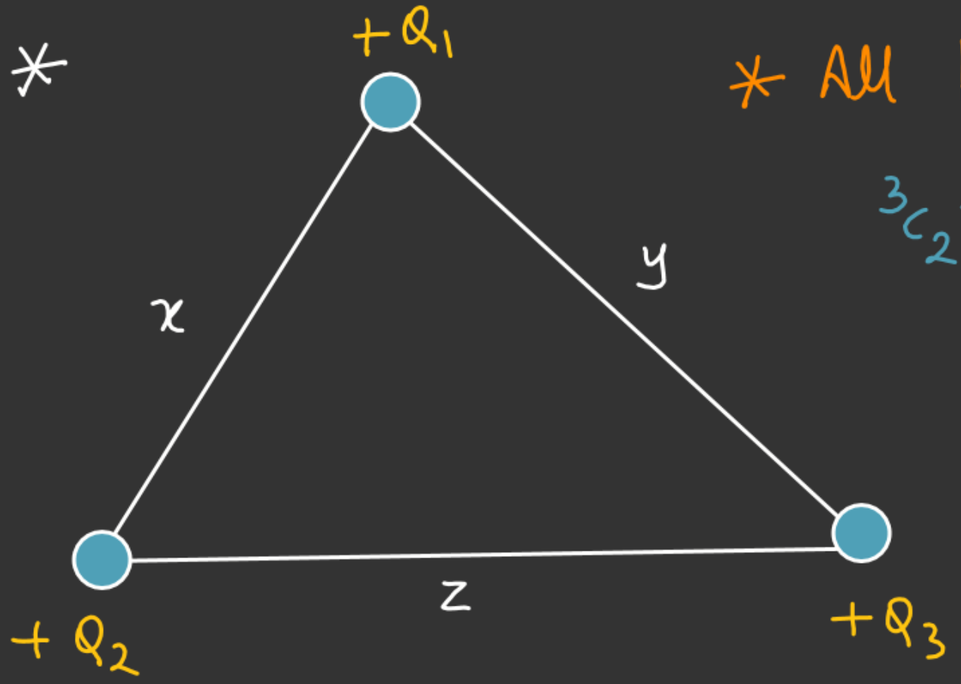
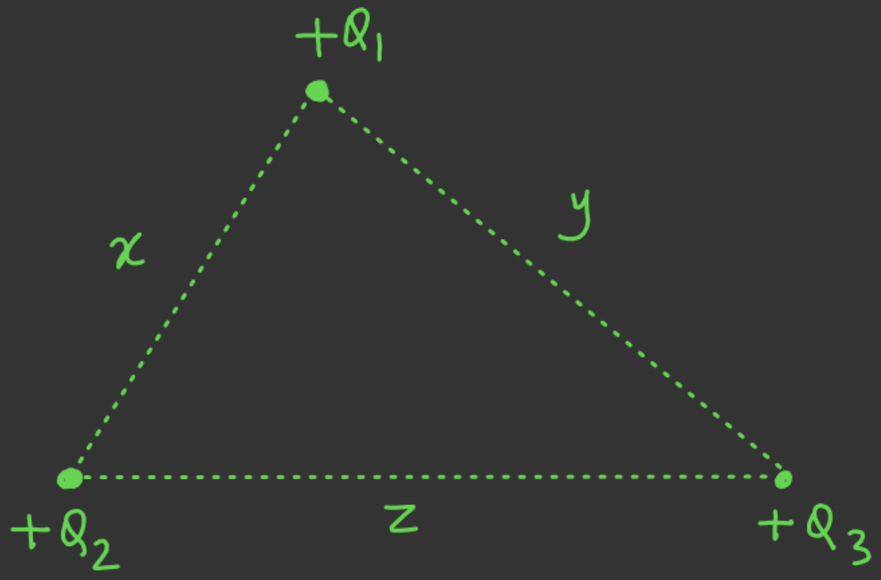
*



* All possible pairs

$^3C_2 \longrightarrow 3$



$$W_{agent} = W_{construction} = -W_{destruction} = U_f - U_{\infty} = 0 + \left(\frac{KQ_1Q_2}{x}\right) + \left(\frac{KQ_1Q_3}{y} + \frac{KQ_2Q_3}{z}\right)$$

($U_{\infty} \to 0$)

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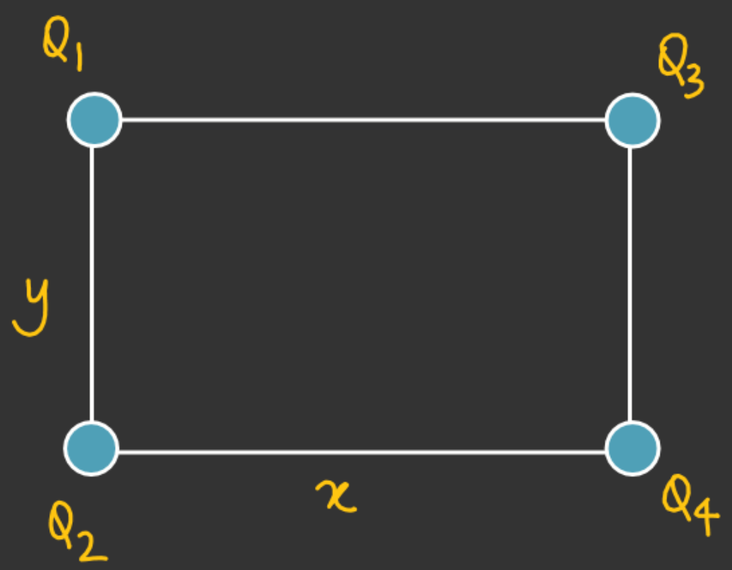
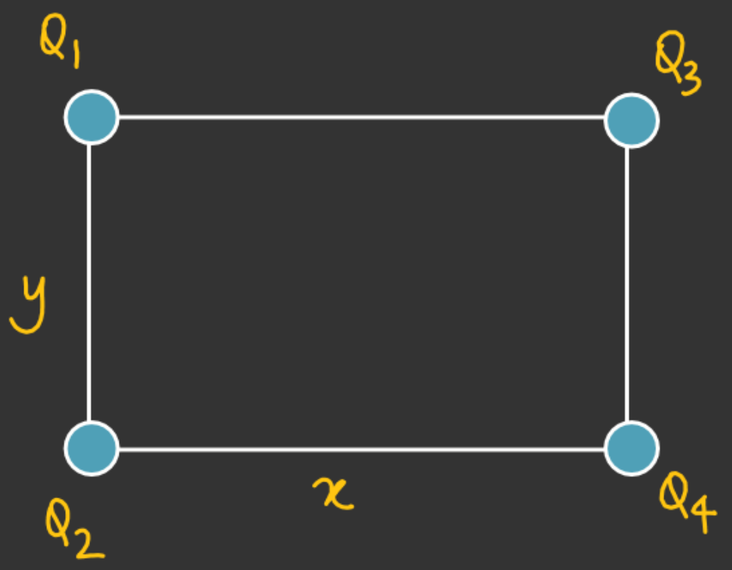
* All possible pairs



$^4C_2 \rightarrow 6$



$$U = (0) + \left(\frac{kQ_1Q_2}{y}\right) + \left(\frac{kQ_1Q_3}{x} + \frac{kQ_2Q_3}{\sqrt{x^2+y^2}}\right) + \left(\frac{kQ_1Q_4}{\sqrt{x^2+y^2}} + \frac{kQ_2Q_4}{x} + \frac{kQ_3Q_4}{y}\right)$$

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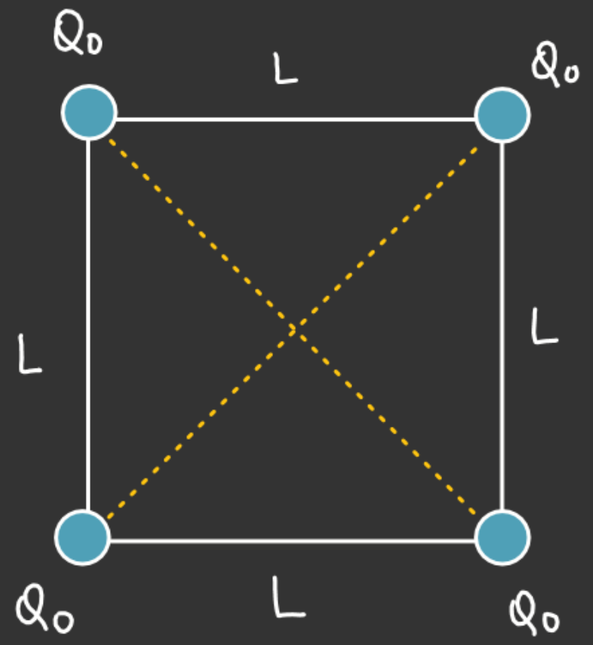

$$U = 4 \cdot \frac{kQ_0^2}{L} + 2\frac{kQ_0^2}{\sqrt{2}L}$$

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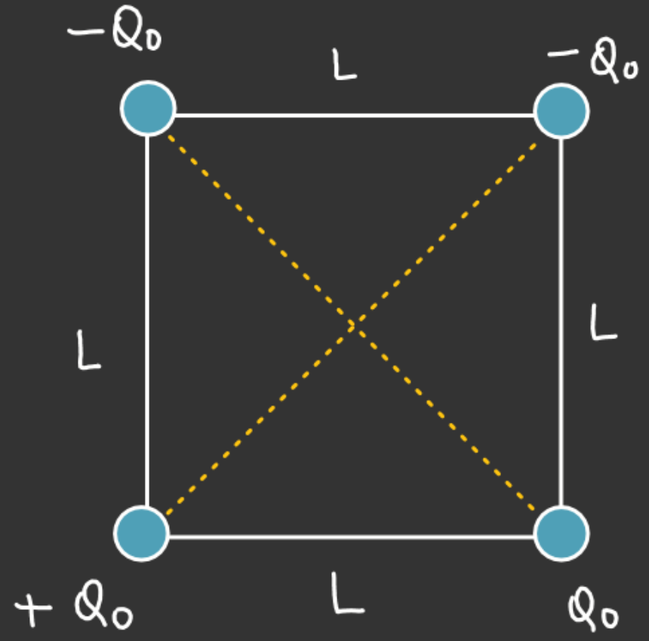
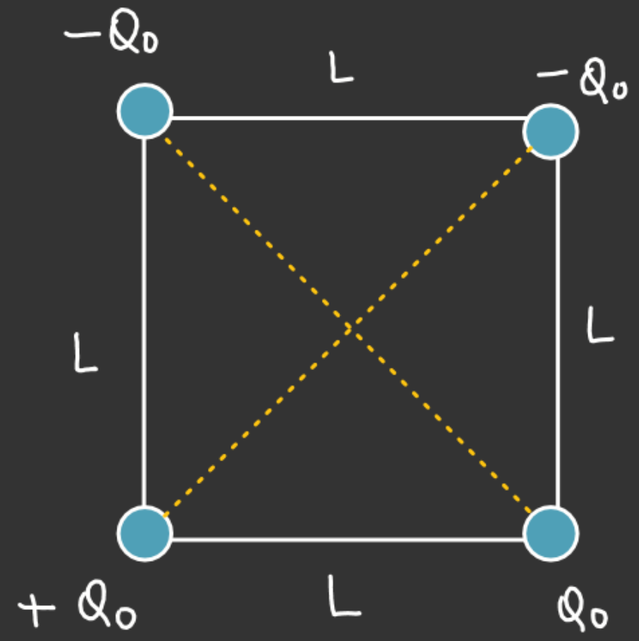
*





$$U = (0) + \left(-\frac{kQ_0^2}{L}\right) + \left(-\frac{kQ_0^2}{\sqrt{2}L} + \frac{kQ_0^2}{L}\right) + \left(+\frac{kQ_0^2}{L} - \frac{kQ_0^2}{\sqrt{2}L} - \frac{kQ_0^2}{L}\right)$$

$$U = -\frac{2 \cdot kQ_0^2}{\sqrt{2}L} = -\frac{\sqrt{2}kQ_0^2}{L}$$

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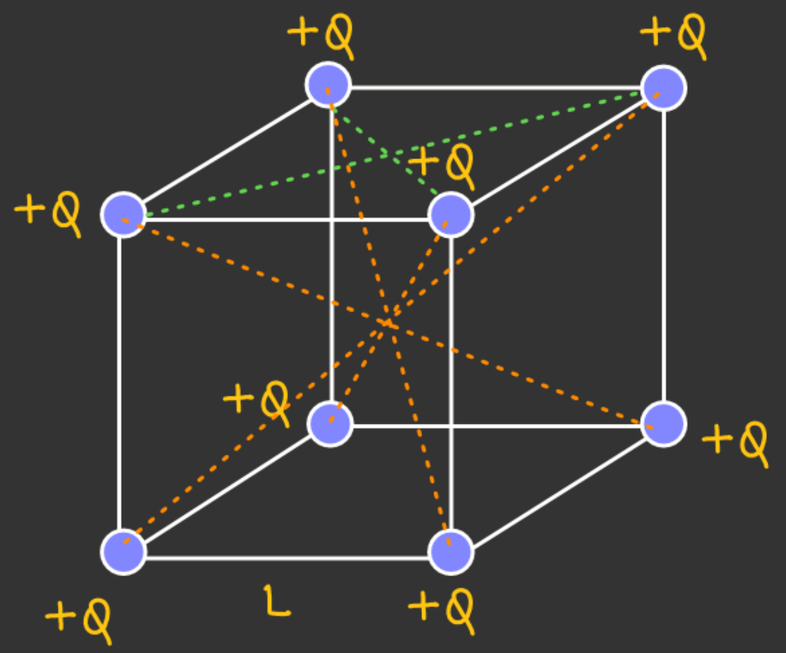
*



$L \longrightarrow 12$

$\sqrt{2}L \longrightarrow 12$

$\sqrt{3}L \longrightarrow 4$

$${}^{8}C_{2} \longrightarrow \frac{8!}{2!\,6!} = 28$$

$$U = 12\,\frac{kQ_0^2}{L} + 12\cdot\frac{kQ_0^2}{\sqrt{2}L} + 4\cdot\frac{kQ_0^2}{\sqrt{3}L}$$

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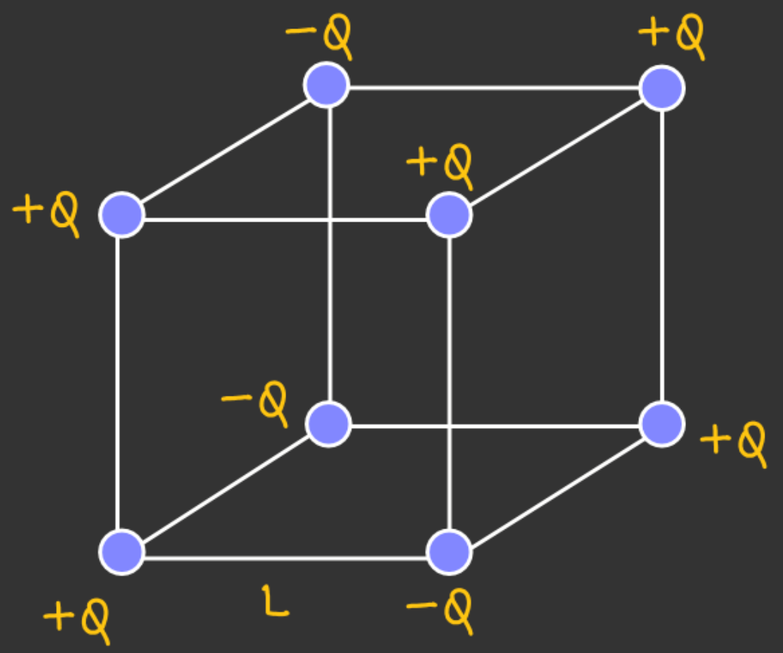
* H.C.



U = ?

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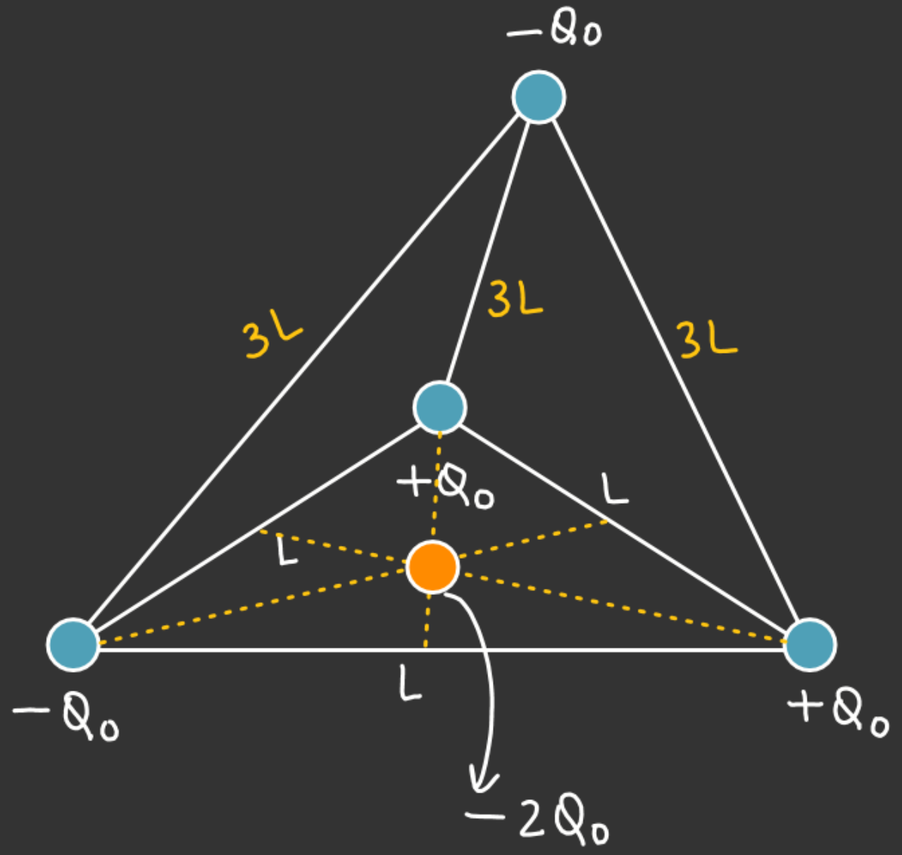
*

H.C.



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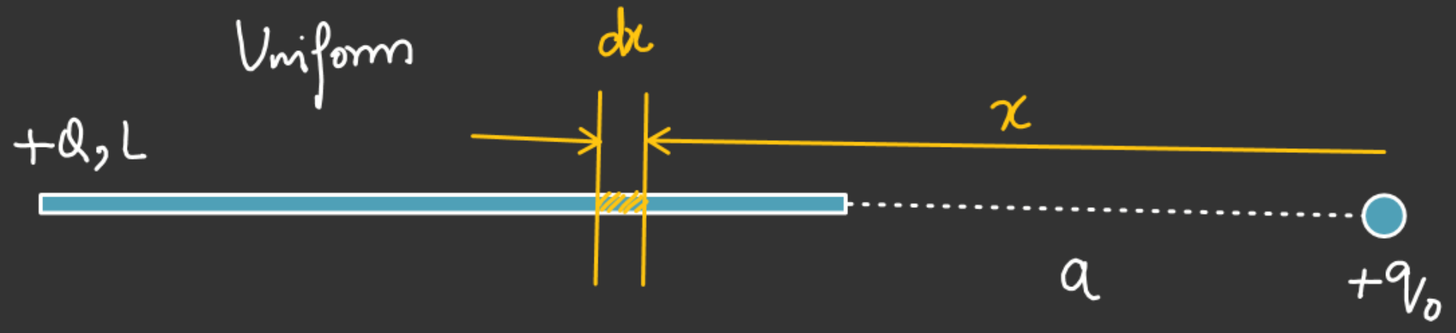

$$dq = \frac{Q}{L} \cdot dx$$

$$dU = \frac{K \cdot q_0 \cdot dq}{x} = \frac{Kq_0}{x} \cdot \frac{Q}{L} \cdot dx = \frac{KQq_0}{L} \cdot \frac{dx}{x}$$

$$U = \int dU = \frac{KQq_0}{L} \int_{a}^{a+L} \frac{dx}{x} = \boxed{\frac{KQq_0}{L} \ln\left(\frac{a+L}{a}\right) = U}$$

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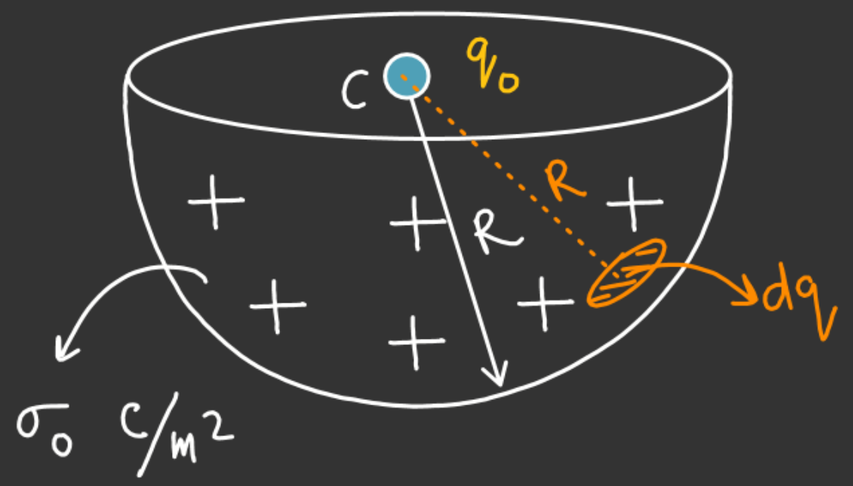
*



$$W_{agent} = U_f - U_i = \cancel{U_\infty}^{0} - U_c = -U_c$$

$$U_c = \int \frac{k \cdot dq \cdot q_0}{R} = \frac{k q_0}{R} \int dq$$

$$\Rightarrow \frac{k q_0}{R} \left[\sigma_0 \cdot 2\pi R^2 \right]$$

* find work done in shifting the charge to Infinity

$$W_{agent} = -\frac{1 \cdot q_0 \cdot \sigma_0 \cdot \cancel{2}\pi R}{\cancel{4}\pi\epsilon_0 \cdot \cancel{R}_{2}} = -\left(\frac{\sigma R}{2\epsilon_0} \cdot q_0 \right)$$

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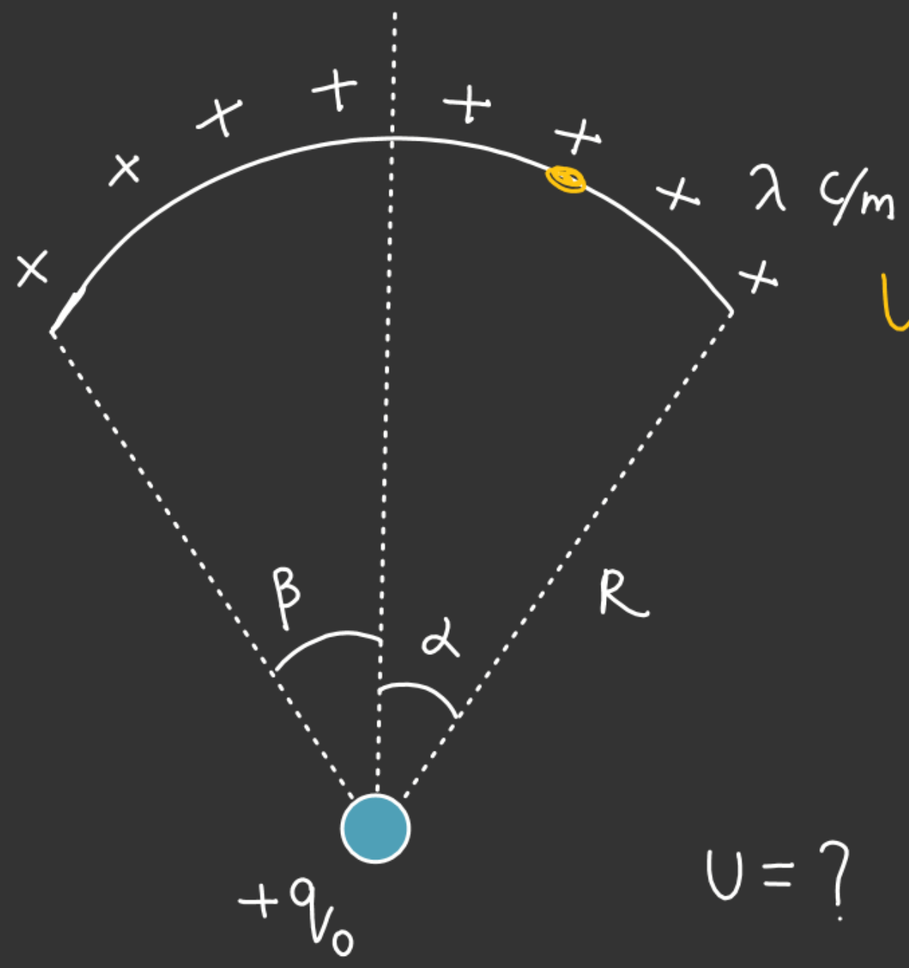

Uniform Arc of Charge

$U = ?$

$$dU = \frac{Kq_0 \cdot dq}{R}$$

$$U = \frac{Kq_0}{R}\int dq = \frac{Kq_0}{R} \cdot \lambda R(\alpha+\beta)$$

$$\star \quad \boxed{U \Rightarrow Kq_0\lambda(\alpha+\beta)}$$

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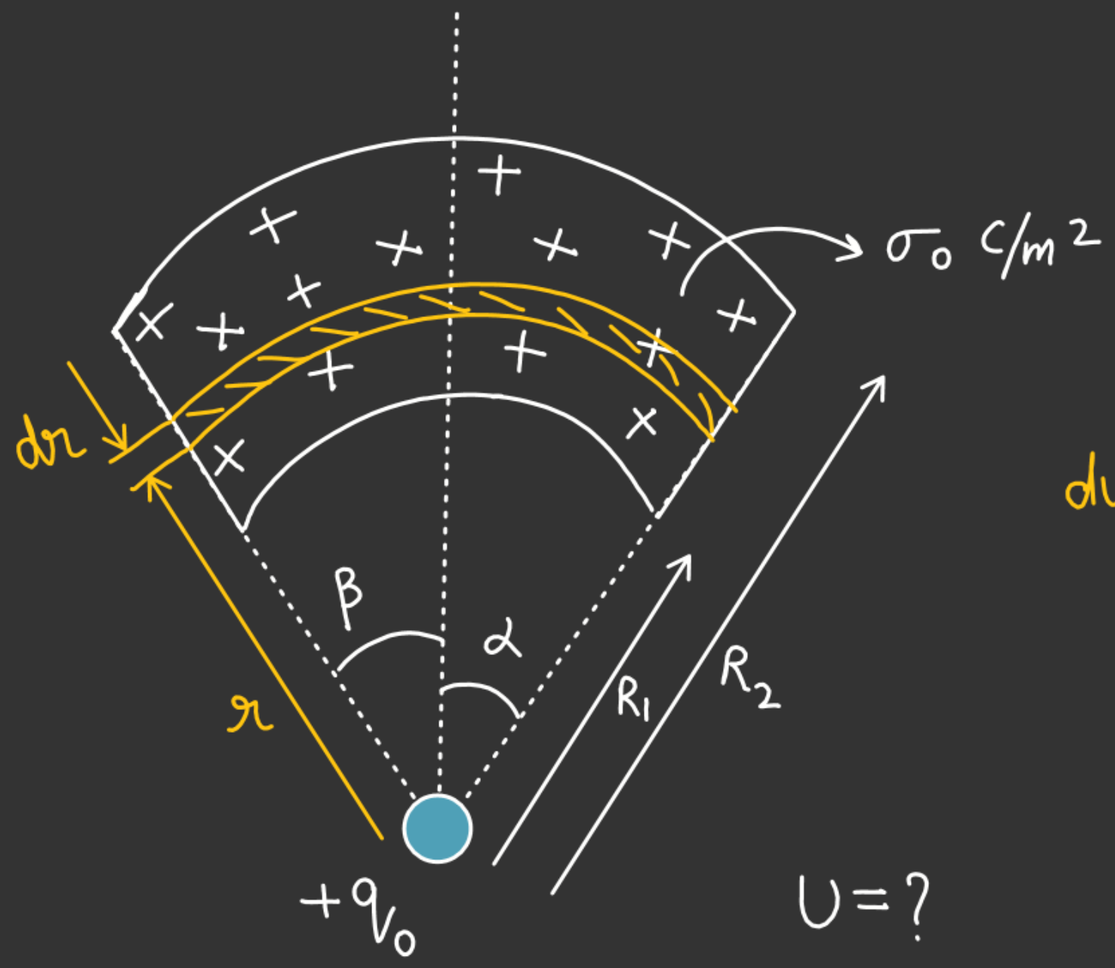
*



U = ?

$$dq = \sigma_0 \cdot r(\alpha+\beta) \cdot dr$$

$$dU = \frac{K \cdot q_0 \cdot dq}{r} = \frac{K q_0 \cdot \sigma_0 \cancel{r}(\alpha+\beta) \cdot dr}{\cancel{r}}$$

$$U = K q_0 \sigma_0 (\alpha+\beta) \cdot \int_{R_1}^{R_2} dr$$

$$\boxed{U = K q_0 \sigma_0 (\alpha+\beta)(R_2 - R_1)}$$

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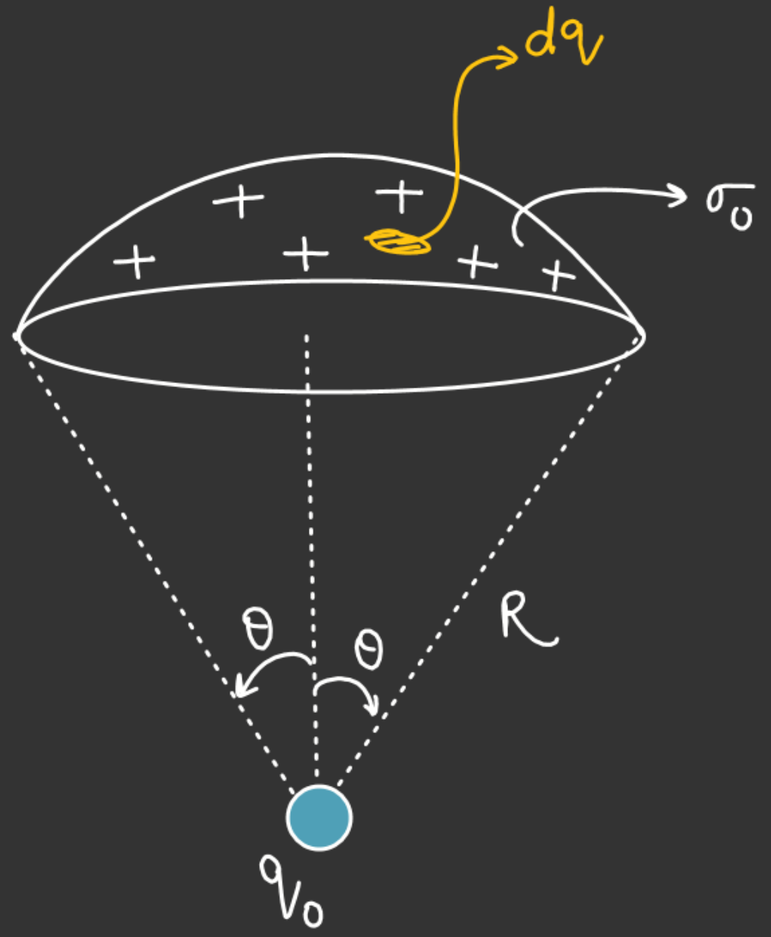
* part of Uniform shell (spherical cap)



$$dU = \frac{Kq_0 \cdot dq}{R}$$

$$U = \frac{Kq_0}{R}\int dq$$

$$U \Rightarrow \frac{Kq_0}{R} \cdot \sigma_0 \cdot 2\pi R^2 (1-\cos\theta)$$

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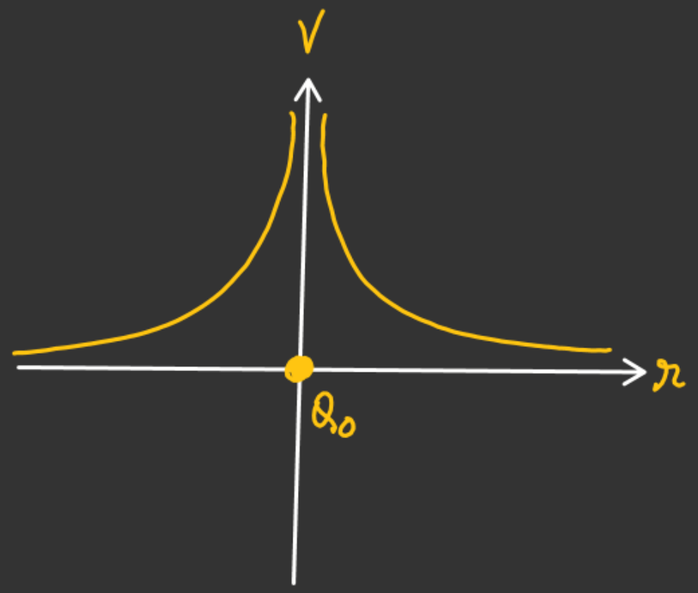
Electric potential

$$W_{elec} = -\Delta U = U_i - U_f = q_0 (V_i - V_f)$$

↓

Conservative



$$\boxed{V_i - V_f = \frac{W_e}{q_0}}$$

$$V_p - \cancel{V_\infty}^{0} = \left(\frac{kQ_0 q_0}{r \cdot q_0}\right) = \frac{kQ_0}{r}$$

★

$$\boxed{V_p = \frac{kQ_0}{r}}$$

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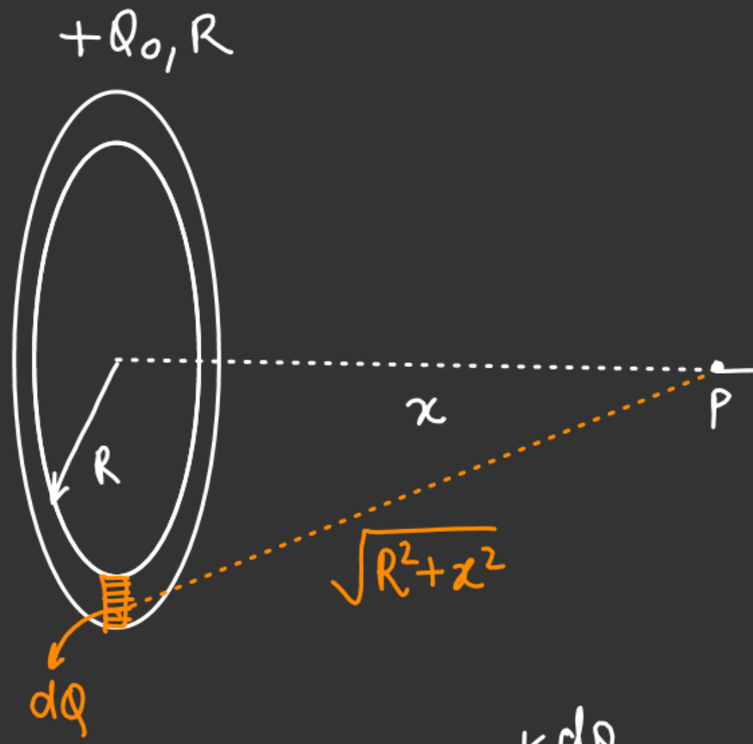
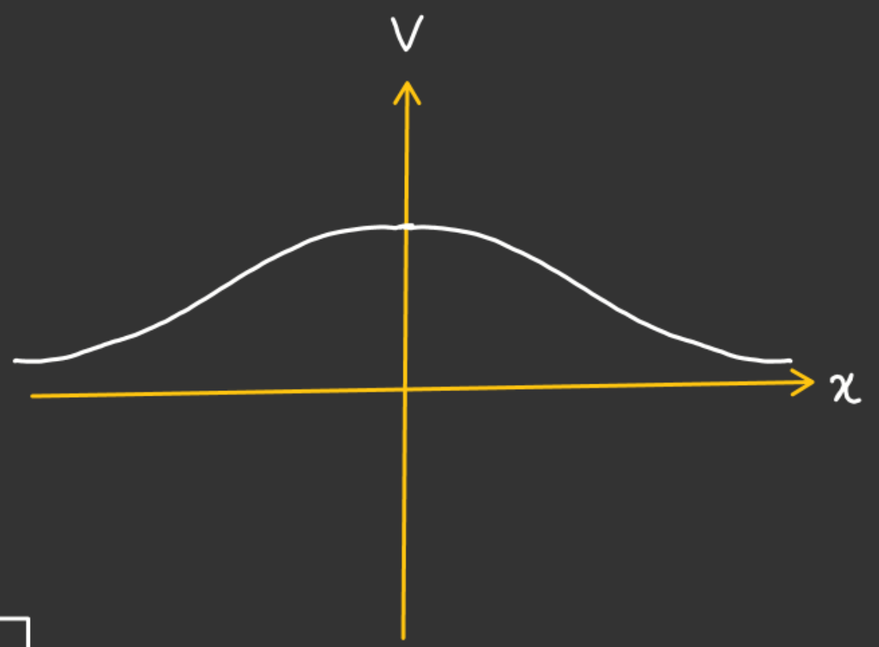
* Uniform Ring.



$$E_p = \frac{kQ_0 x}{(R^2+x^2)^{3/2}}$$



$$dV_p = \frac{k\,dQ}{\sqrt{R^2+x^2}}$$

$$V_p = \int dV_p = \frac{k}{\sqrt{R^2+x^2}} \int dQ = \frac{kQ_0}{\sqrt{R^2+x^2}}$$

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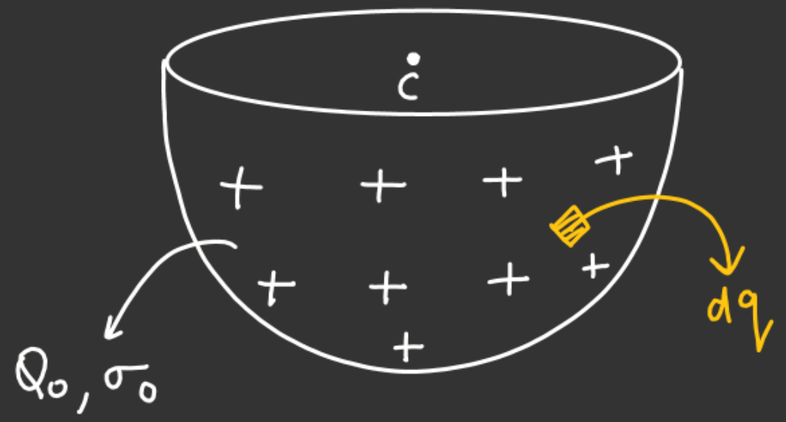
* Uniform Hemispherical Shell

$V_c = ?$



$$Q_0 = \sigma_0 \cdot 2\pi R^2$$

$$dV_c = \frac{k \cdot dq}{R}$$

$$V_c = \int dV_c = \frac{kQ}{R} = \frac{1}{4\pi\epsilon_0} \frac{\sigma_0 \cdot 2\pi R^2}{R} \Rightarrow \frac{\sigma_0 R}{2\epsilon_0}$$

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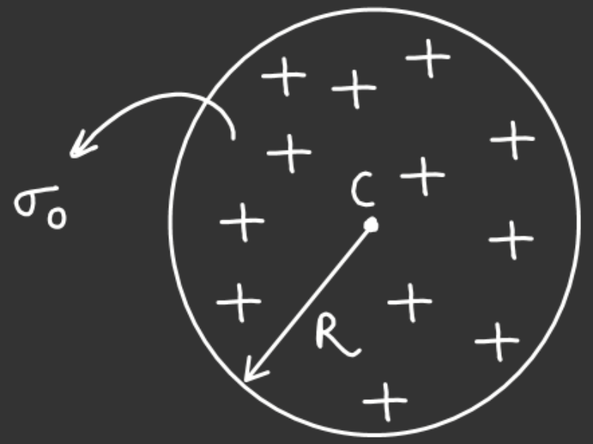
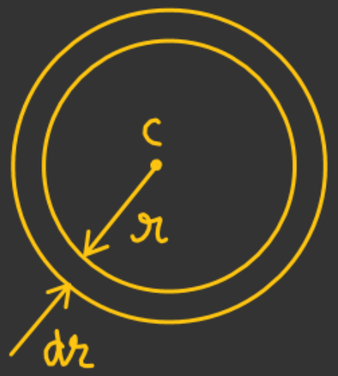
* Uniform Disc :-





$$V_C = \int dV_C = \frac{\sigma_0}{2\epsilon_0}\int dR$$

$$\boxed{V_C = \frac{\sigma_0 R}{2\epsilon_0}}$$

$$dq = \sigma_0 \cdot 2\pi r\, dr$$

$$dV_C = \frac{k\cdot dq}{r} \Rightarrow \frac{1\cdot \sigma_0\, 2\pi r\, dr}{4\pi\epsilon_0\, r} = \frac{\sigma_0}{2\epsilon_0}\cdot dr$$

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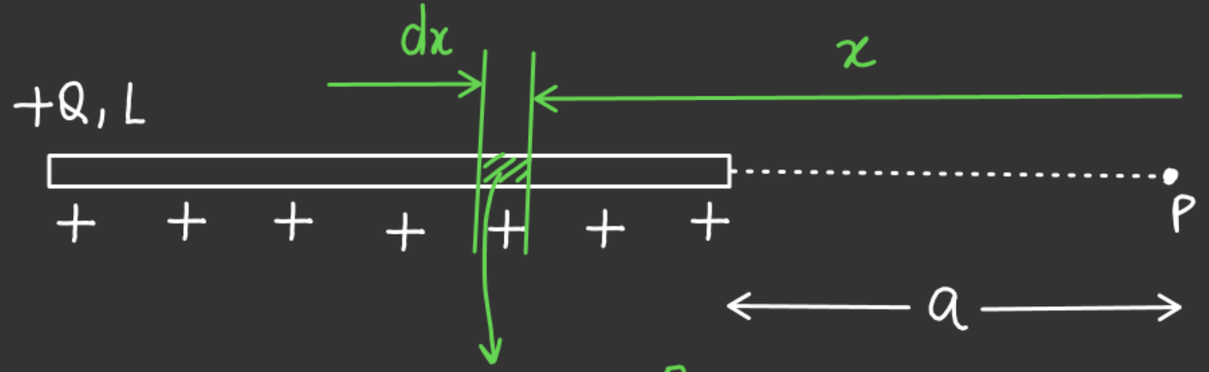
* Uniform Rod :-



$$dq = \frac{Q}{L} \cdot dx$$

$V_p = ?$

$$dV_p = \frac{k \cdot dQ}{x} = \frac{kQ}{L} \cdot \frac{dx}{x}$$

$$V_p = \frac{kQ}{L} \int_{a}^{a+L} \frac{dx}{x} = \frac{kQ}{L} \ln\left(\frac{a+L}{a}\right)$$

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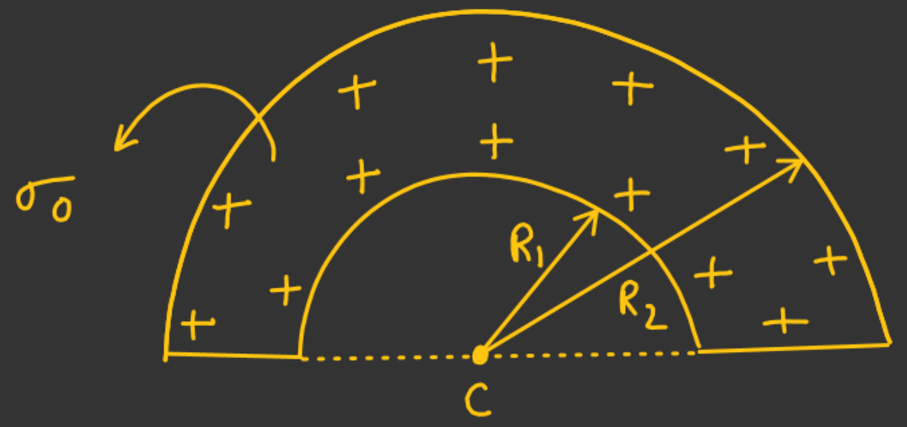
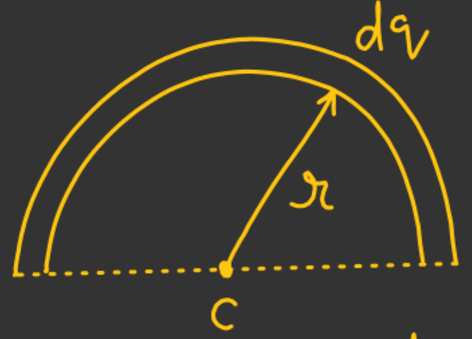
*



$$V_C = \int dV_C = \frac{\sigma_0}{4\epsilon_0}\left(R_2 - R_1\right)$$

$V_C = ?$



$dq = \sigma_0 \cdot \pi r\, dr$

$$dV_C = \frac{k \cdot dq}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma_0 \cdot \pi r\, dr}{r} = \frac{\sigma_0}{4\epsilon_0}\, dr$$

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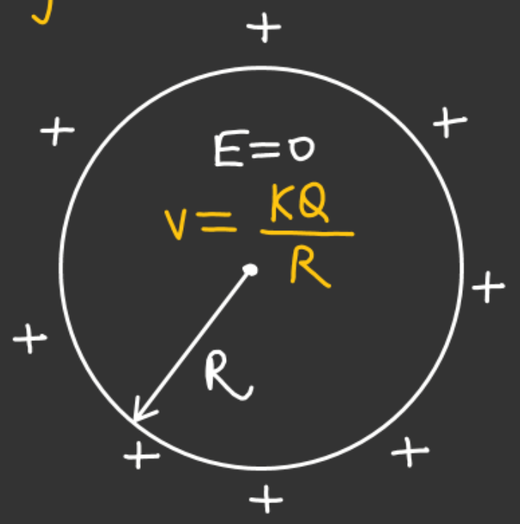
* Uniform Shell :-

Conducting shell

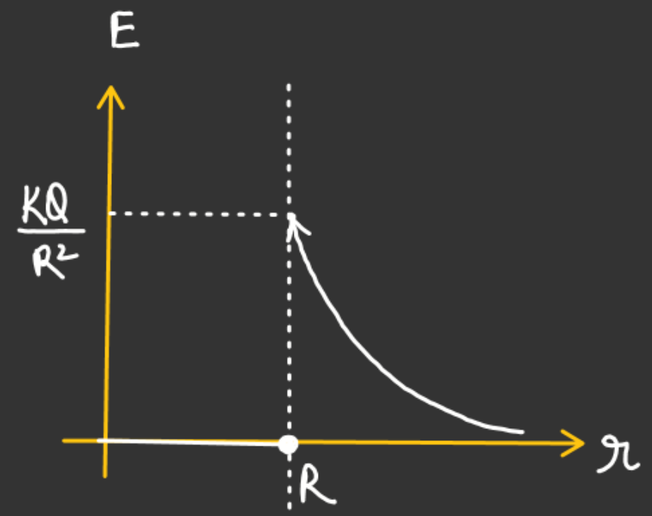


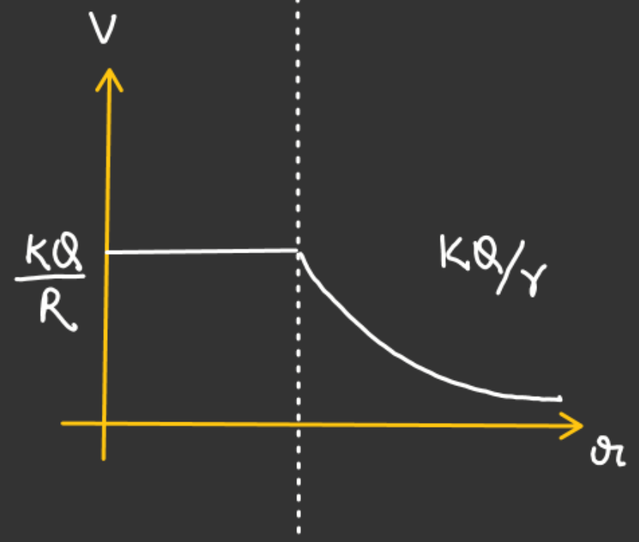
$\frac{KQ}{R}$

$S \leftarrow \infty$

$$E = \begin{cases} 0 & r \le R \\ \frac{KQ}{r^2} & r > R \end{cases}$$

$$V = \begin{cases} \frac{KQ}{R} & r \le R \\ \frac{KQ}{r} & r > R \end{cases}$$





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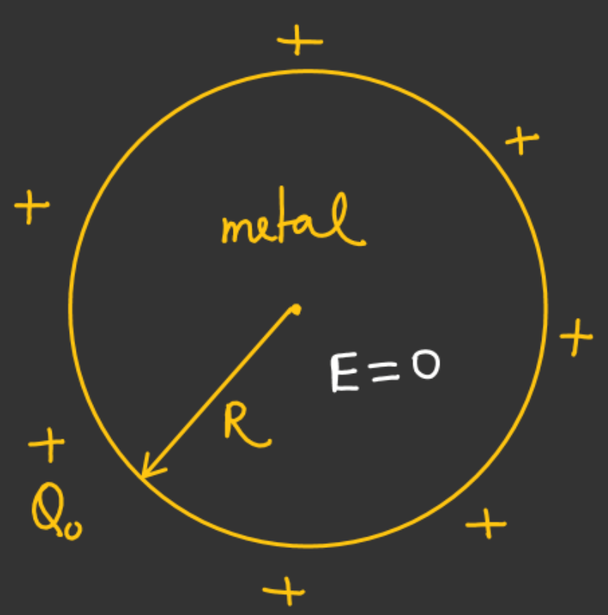
Solid Conducting sphere :-



* E & V will be same as Hollow sphere (Shell)

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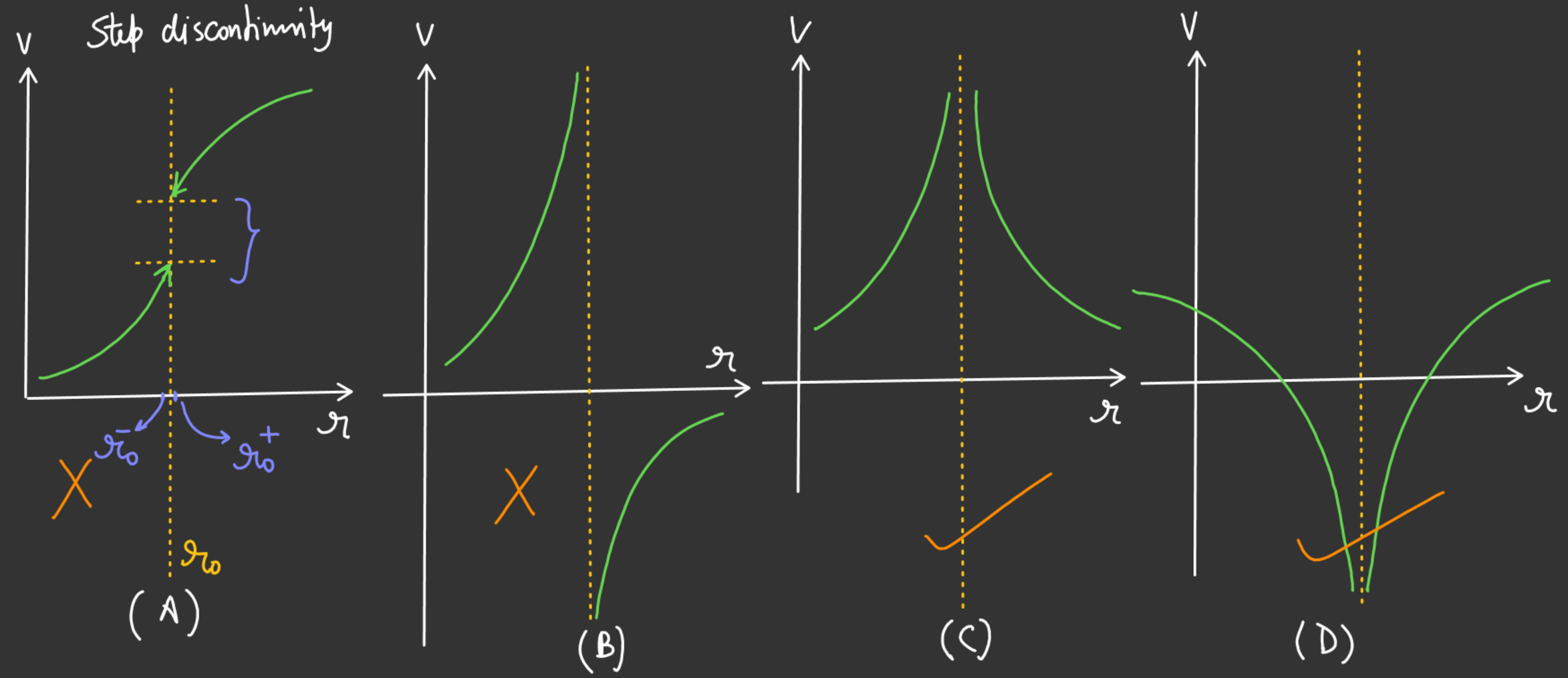
* Continuity and discontinuity of Potential function :-



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*

$$U_f - U_i = -\int \vec{F_c} \cdot d\vec{r}$$

$$\Delta U = -W_c$$

$$q_0(V_f - V_i) = -\int \vec{F_c} \cdot d\vec{r}$$

$$V_f - V_i = -\int \frac{\vec{F_c}}{q_0} \cdot d\vec{r}$$

$$\boxed{V_f - V_i = -\int \vec{E} \cdot d\vec{r}}$$

*

$$-dU = \vec{E} \cdot d\vec{r}$$

In 1D (along x)

$$E_x = -\frac{dV}{dx}, \quad E_y = E_z = 0$$

In 1D (along y)

$$E_y = -\frac{dV}{dy}; \quad E_x = E_z = 0$$

In 1D (along z)

$$E_z = -\frac{dV}{dz}; \quad E_x = E_y = 0$$

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* $V \rightarrow f(x, y, z)$

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

$$E_x = -\frac{\partial V}{\partial x}$$

$$E_y = -\frac{\partial V}{\partial y}$$

$$E_z = -\frac{\partial V}{\partial z}$$

$$-dV = E \cdot dr$$

$\Rightarrow$ In the direction of $\vec{E}$ potential decreases

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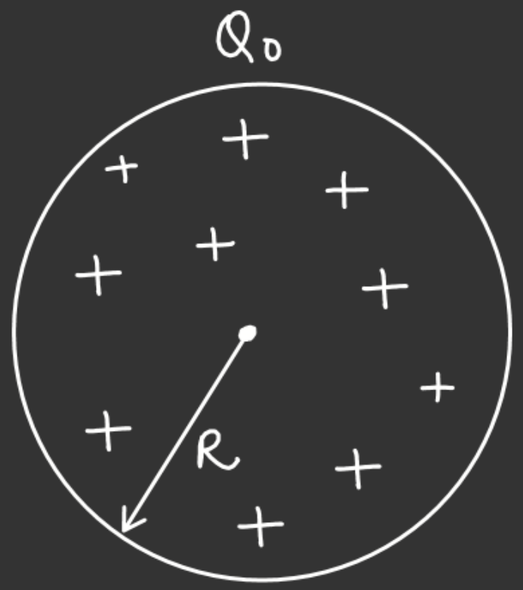
* Uniform Solid Sphere :-



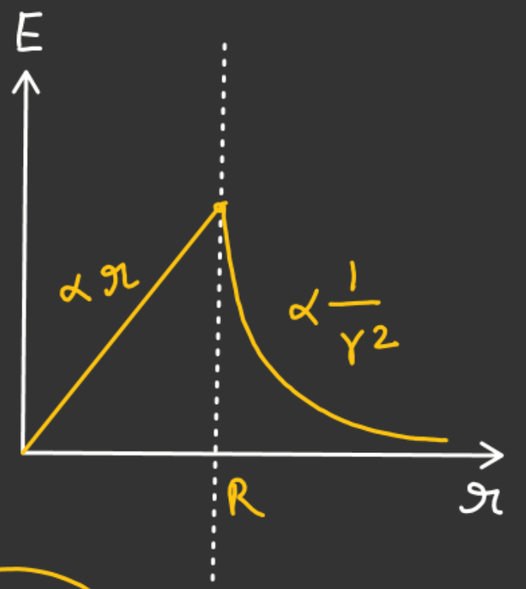
$$\rho_0 = \frac{Q_0}{\frac{4}{3}\pi R^3}$$

Solid Sphere (Non Conducting)

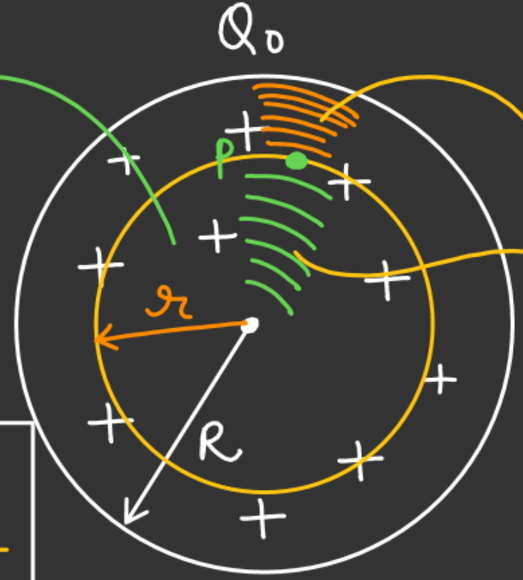
$$q = \rho_0 \cdot \frac{4}{3}\pi r^3 = \frac{Q_0 r^3}{R^3}$$

$$E = \frac{KQr}{R^3} = \frac{\rho r}{3\epsilon_0} \qquad r \le R$$

$$\frac{KQ}{r^2} \qquad r > R$$



r < R :-



$$E_p = \frac{Kq}{r^2}$$

$E_p = 0$

$E_p = \frac{Kq}{r^2}$

$$E_p = \frac{KQr}{R^3} = \frac{\rho r}{3\epsilon_0}$$

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HC * find potential due to Solid Non Conducting Uniform Sphere.

Draw V-r graph.

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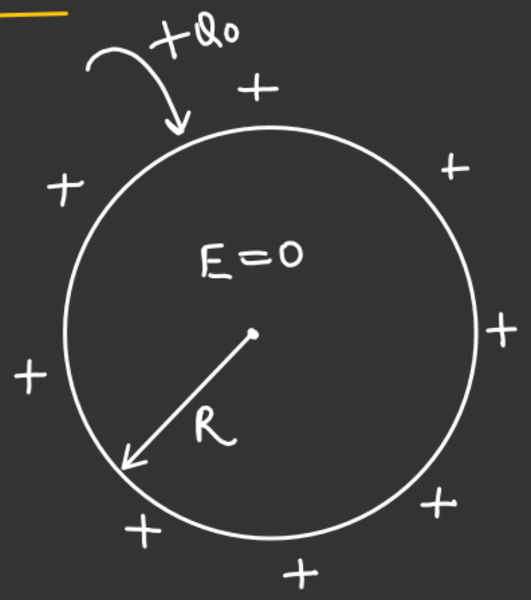
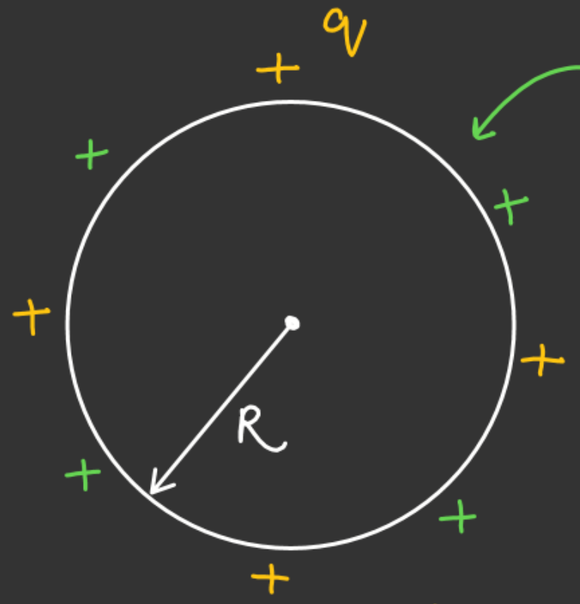
* Self Energy :-

work done by agent in constructing the system very slowly.

Uniform Shell :-





$$dW_{agent} = V_s \cdot dq = \frac{Kq}{R} \cdot dq$$

$$U_{self} = W_{agent} = \frac{K}{R}\int_0^{Q_0} q \cdot dq = \frac{KQ_0^2}{2R} = \frac{Q_0^2}{8\pi\epsilon_0 R}$$

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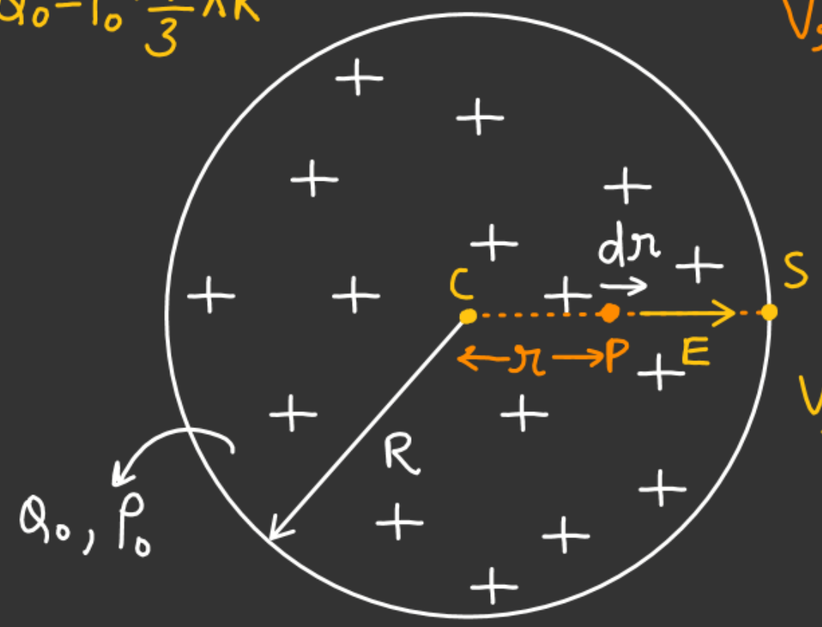
✳ Potential due to Solid Non Conducting sphere :-

$Q_0 = \rho_0 \cdot \frac{4}{3}\pi R^3$

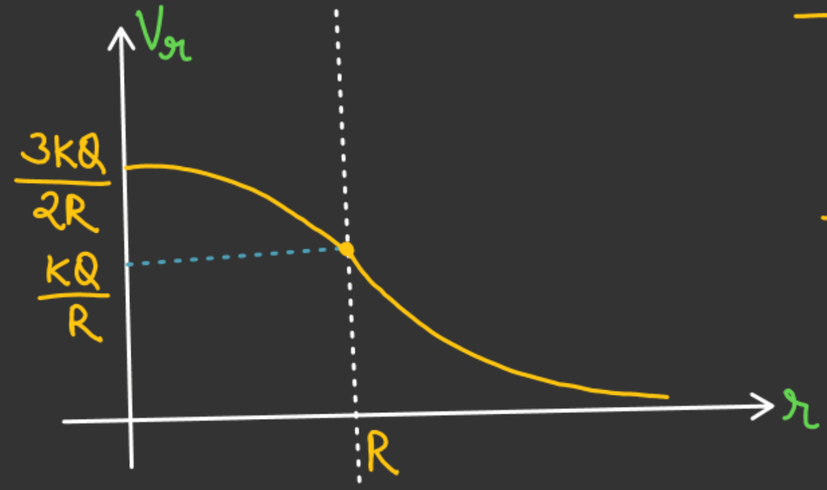
$V_r = ?$



$V_S = \frac{KQ_0}{R}$

$E = \frac{KQ_0 r}{R^3} = \frac{\rho_0 r}{3\epsilon_0}$ $\quad r \le R$

$\frac{KQ_0}{r^2}$ $\quad r > R$



$r > R$ $\quad V_r = \frac{KQ_0}{r}$

$r < R$:- $\quad V_f - V_i = -\int \vec{E} \cdot d\vec{r}$

$$V_S - V_r = -\int_r^R \frac{KQ_0 r}{R^3} \cdot dr$$

$$\frac{KQ_0}{R} - V_r = -\frac{KQ_0}{R^3} \left.\frac{r^2}{2}\right|_r^R$$

$$\frac{KQ_0}{R} - V_r = -\frac{KQ_0}{2R^3}\left[R^2 - r^2\right]$$

$$\boxed{V_r = \frac{KQ_0}{2R^3}\left[3R^2 - r^2\right]}$$

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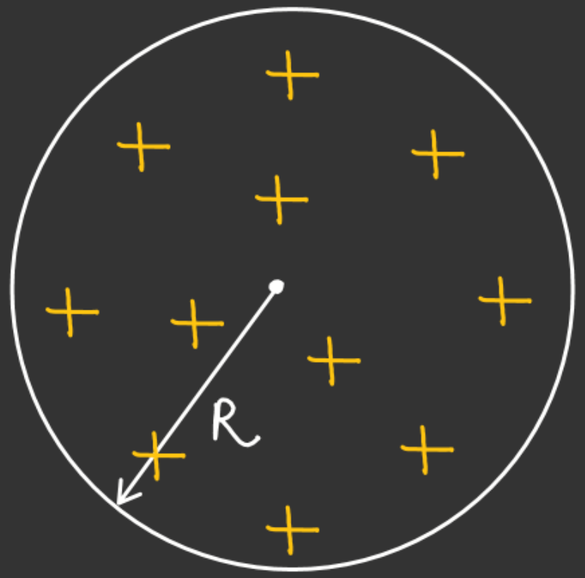
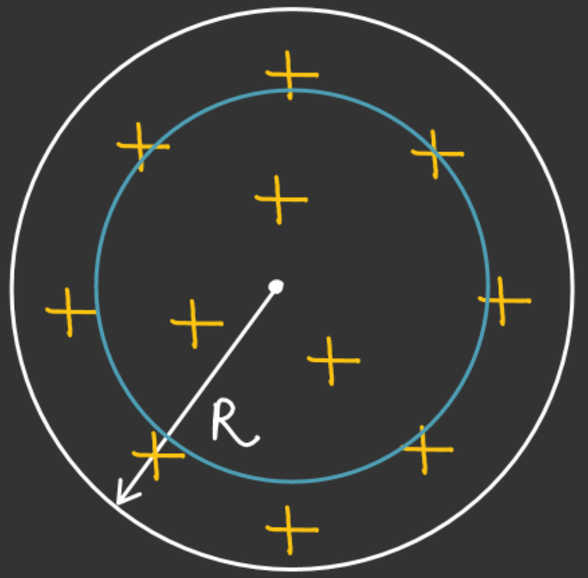
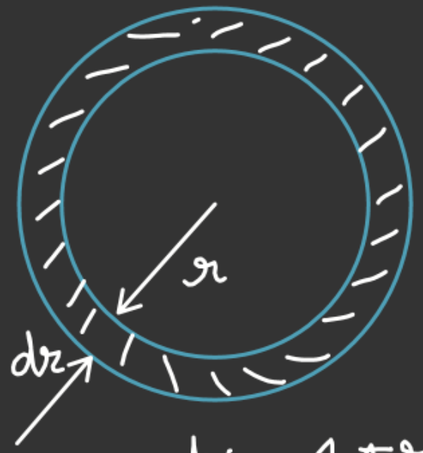
* Non Uniform Charged Non Conducting sphere :-

$\rho_r = f(r)$

$Q_0 = \int \rho_r \cdot dV = \int \rho_r \cdot 4\pi r^2 dr$







$dV = 4\pi r^2 dr$

(i) $\rho_r = \rho_0 \left(1 + \frac{r}{R}\right)$

a) Draw E-r graph

b) total Charge of the Sphere.

$r > R$ $\quad E = \frac{kQ_0}{r^2}$

$r < R$ $\quad E = \frac{kQ_r}{r^2}$

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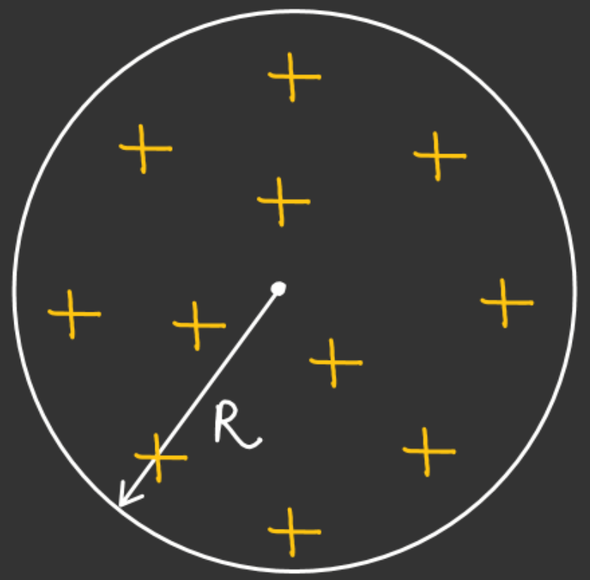
* Non Uniform Charged Non Conducting sphere :-

$\rho_r = f(r)$

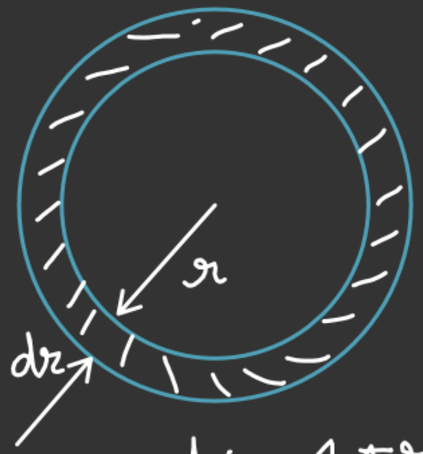


(i) $\rho_r = \rho_0\left(1+\frac{r}{R}\right)$

a) Draw E-r graph

b) total Charge of the Sphere.

$$Q_0 = \int \rho_r \cdot dv = \int_0^R \rho_r \cdot 4\pi r^2 dr$$



$dV = 4\pi r^2 dr$

$$Q_r = \int_0^r \rho_0\left(1+\frac{r}{R}\right)\cdot 4\pi r^2 \cdot dr$$

$$Q_r = 4\pi\rho_0 \int_0^r \left(r^2+\frac{r^3}{R}\right)\cdot dr = 4\pi\rho_0\left(\frac{r^3}{3}+\frac{r^4}{4R}\right)$$

$$Q_0 = \frac{7\pi\rho_0 R^3}{3}$$

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$$* \quad \rho_r = \rho_0 \frac{r}{R}$$

$$\rho_r = \rho_0 \left(1 - \frac{r}{R}\right)$$

$$\rho_r = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$$

$$\rho_r = \rho_0 \left(1 + \frac{r^2}{R^2}\right)$$

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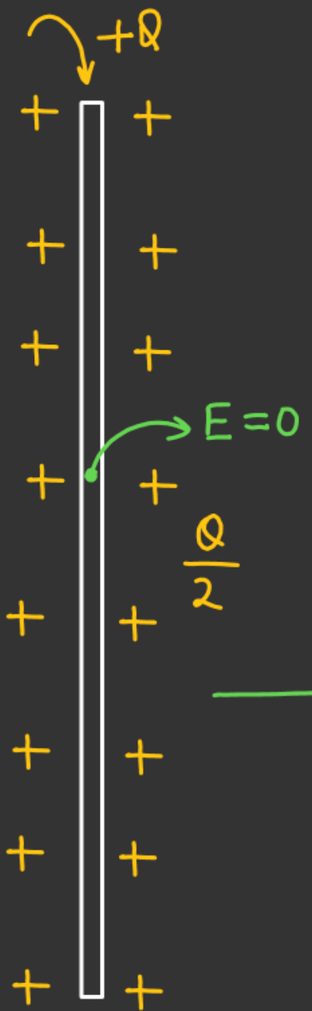
*



$\sigma = \frac{Q}{2A}$

$V_A > V_B$

$V_A - V_B = E \cdot L = \frac{\sigma}{\epsilon_0} \cdot L$

$\rightarrow \frac{\sigma}{\epsilon_0} = \frac{Q}{2A\epsilon_0}$

Conducting Plate

$\sigma = \frac{Q}{A}$

$\rightarrow \frac{\sigma}{2\epsilon_0} = \frac{Q}{2A\epsilon_0}$

$V_A > V_B$

$V_A - V_B = E \cdot L = \frac{\sigma}{2\epsilon_0} \cdot L$

Sheet of Charge

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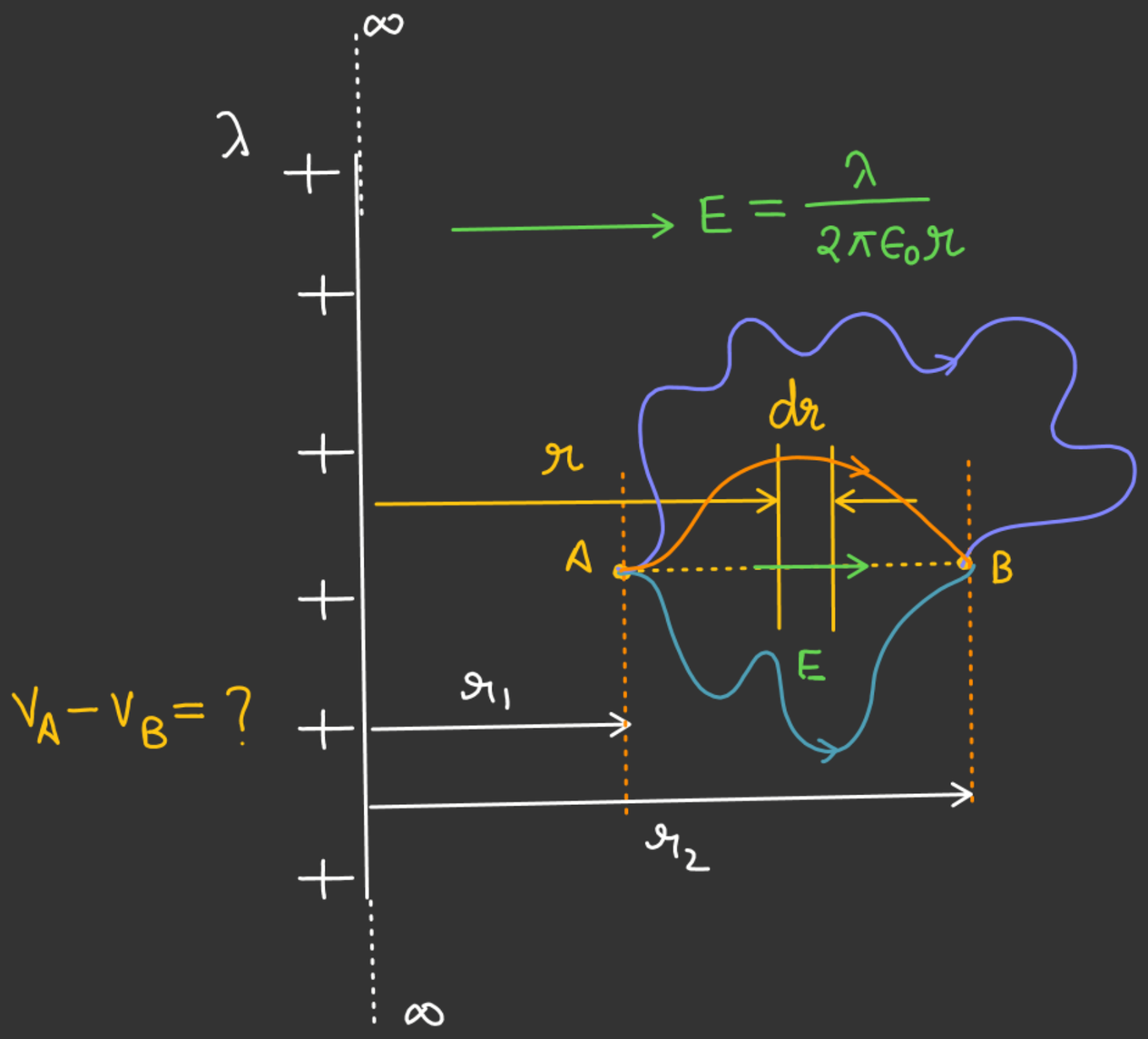
Infinite wire :-



$V_A > V_B$

$$V_B - V_A = -\int_{r_1}^{r_2} E \cdot dr = -\int_{r_1}^{r_2} \frac{\lambda}{2\pi\epsilon_0 r} \cdot dr$$

$$V_B - V_A = -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_2}{r_1}$$

$$\boxed{V_A - V_B = V_1 - V_2 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_2}{r_1}\right)}$$

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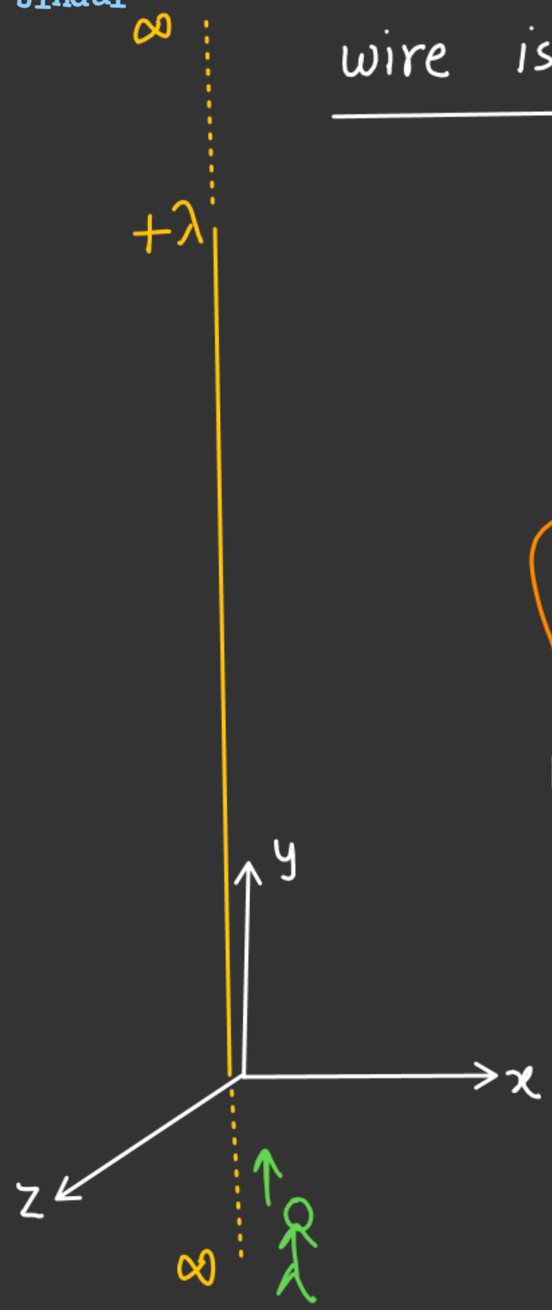
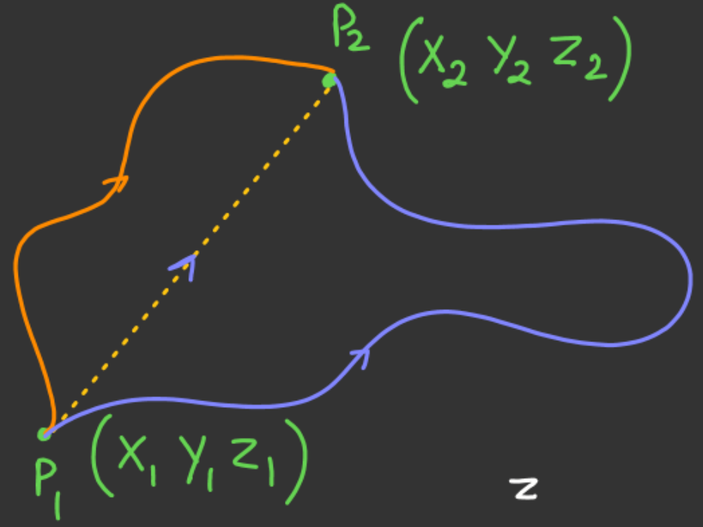
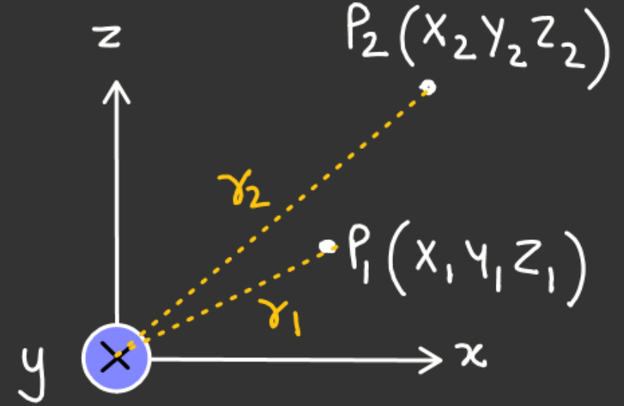
wire is along y axis :-





$$V_1 - V_2 = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_2}{r_1}$$

$$\boxed{V_1 - V_2 = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{\sqrt{x_2^2 + z_2^2}}{\sqrt{x_1^2 + z_1^2}} \right)}$$



$$r_1 = \sqrt{x_1^2 + z_1^2} \quad , \quad r_2 = \sqrt{x_2^2 + z_2^2}$$

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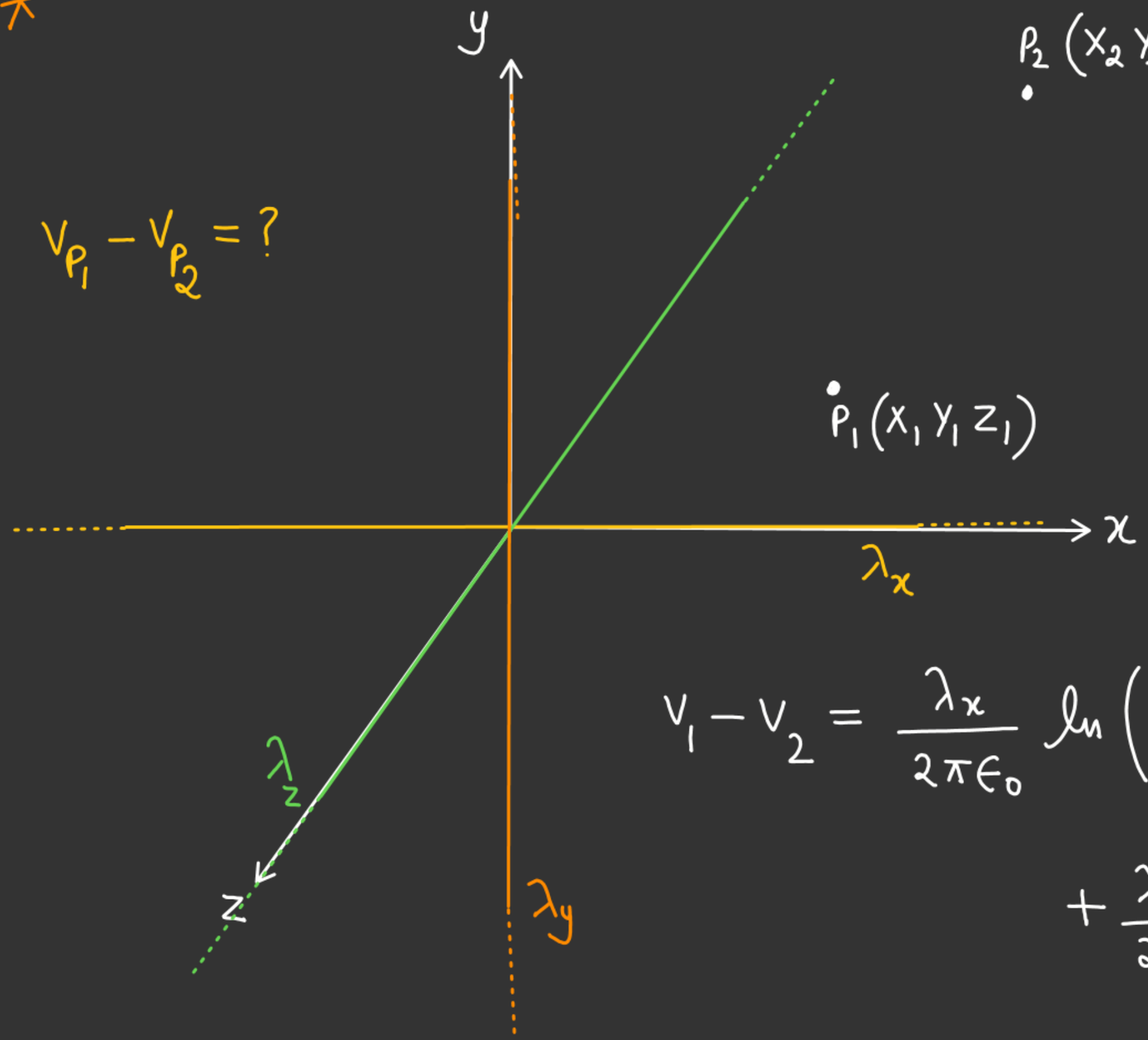

$P_2 (x_2\ y_2\ z_2)$

$P_1 (x_1\ y_1\ z_1)$

$V_{P_1} - V_{P_2} = ?$

$$V_1 - V_2 = \frac{\lambda_x}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{y_2^2+z_2^2}}{\sqrt{y_1^2+z_1^2}}\right) + \frac{\lambda_y}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{x_2^2+z_2^2}}{\sqrt{x_1^2+z_1^2}}\right) + \frac{\lambda_z}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{x_2^2+y_2^2}}{\sqrt{x_1^2+y_1^2}}\right)$$

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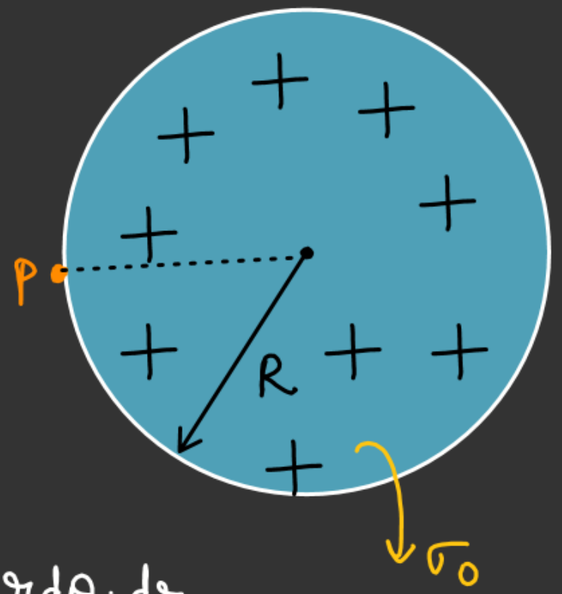
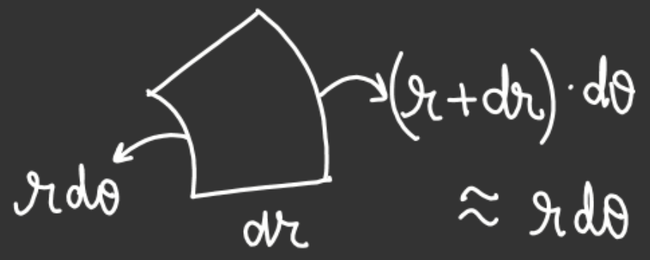
* Potential due to a Uniform Disc at the edge :

Jmb



$dA = r\,d\theta \cdot dr$

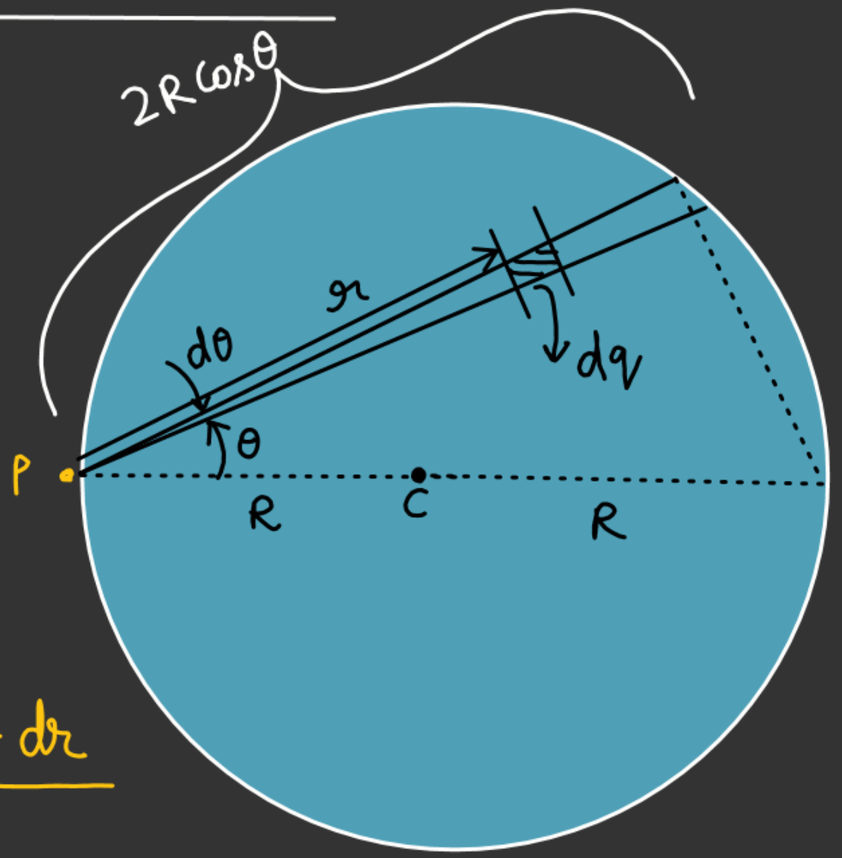


$\approx r\,d\theta$

$$dV_P = \frac{K\,dq}{r}$$

$$= \frac{K \cdot \sigma_0 \cdot dA}{r}$$

$$\Rightarrow \frac{K\sigma_0 \cdot r\,d\theta \cdot dr}{r}$$



$$dV_P = K\sigma_0\,d\theta\,dr$$

$$= K\sigma_0 \cdot d\theta \int_0^{2R\cos\theta} dr \Rightarrow K\sigma_0 \cdot 2R\,\cos\theta \cdot d\theta$$

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$$dV_p = K\sigma_0 \, d\theta \, dr$$

$$= K\sigma_0 \cdot d\theta \int_0^{2R\cos\theta} dr \Rightarrow K\sigma_0 \cdot 2R \cos\theta \cdot d\theta$$

$$V_p \int dV_p = \frac{1}{4\pi\epsilon_0} \cdot \sigma_0 \cdot 2R \int_{-\pi/2}^{+\pi/2} \cos\theta \cdot d\theta$$

$$\downarrow$$

$$2$$

$$V_p = \frac{\sigma R}{\pi \epsilon_0}$$

$$V_C = \frac{\sigma R}{2\epsilon_0}$$

$$\frac{V_p}{V_C} = \frac{2}{\pi}$$

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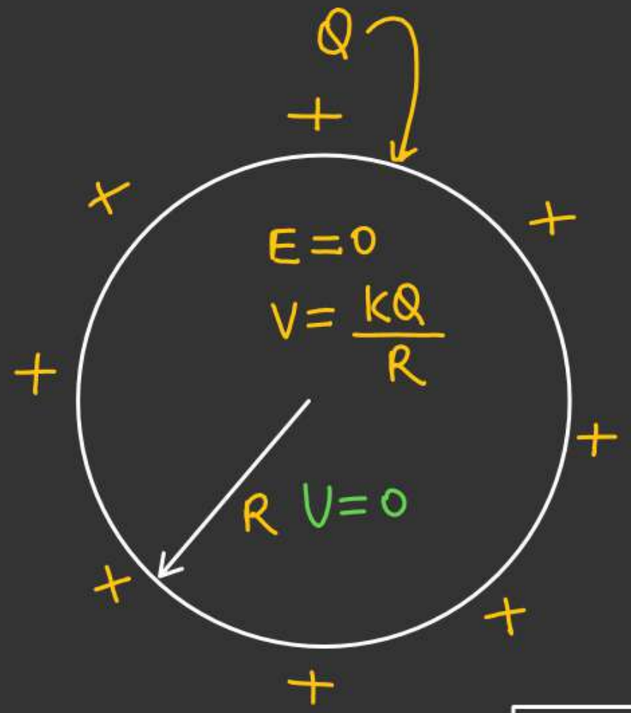
* Self Energy of Solid Conducting sphere :-



E - r
V - r
→ Same as shell

$$U_{self} = \frac{kQ^2}{2R}$$

$$U_{self} = \underbrace{U_{in}}_{0} + U_{out} = \frac{kQ^2}{2R}$$

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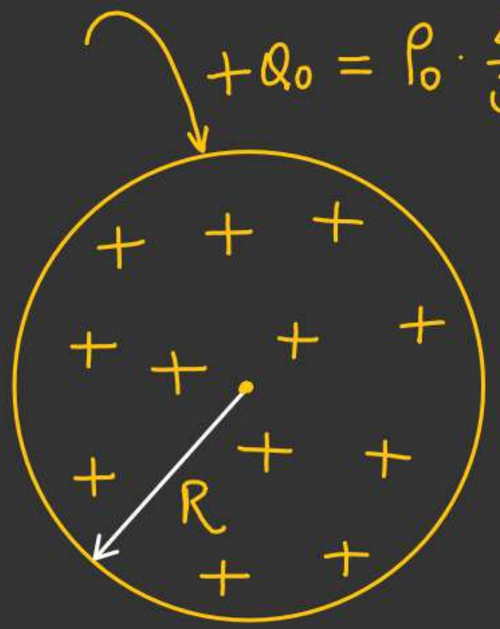
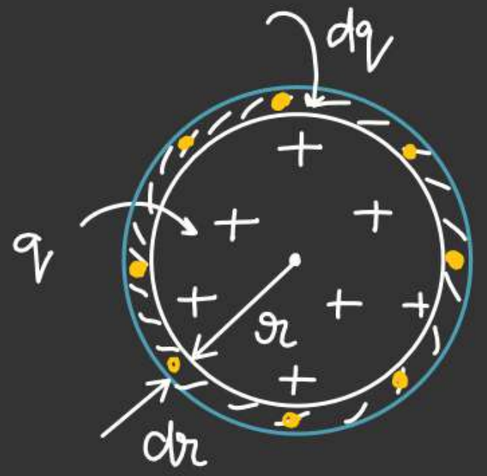
* Self Energy of Solid Non Conducting sphere :-



$$q = \rho_0 \cdot \frac{4}{3}\pi r^3$$

$$q \Rightarrow \frac{Q \cdot r^3}{R^3} \Rightarrow dq = \frac{3Qr^2}{R^3} \cdot dr$$



$$dW = V_s \cdot dq$$

$$dW = \frac{kq}{r} \cdot dq$$

$$U_{self} = W = \int dW = k \int \frac{q \cdot dq}{r}$$

$$U_{self} = k \int \frac{Q \cdot r^3}{R^3} \cdot \frac{3Qr^2}{R^3} \cdot \frac{dr}{r} = \frac{3kQ^2}{R^6} \int_0^R r^4 \cdot dr$$

$$\star \quad \boxed{U_{self} = \frac{3kQ^2}{5R}}$$

∞
dq

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$$U_{self} = U_{in} + U_{out}$$

$$\frac{3kQ^2}{5R} = U_{in} + \frac{kQ^2}{2R}$$

$$U_{in} = \frac{kQ^2}{R}\left[\frac{3}{5} - \frac{1}{2}\right]$$

$$\boxed{U_{in} = \frac{kQ^2}{10R}}$$

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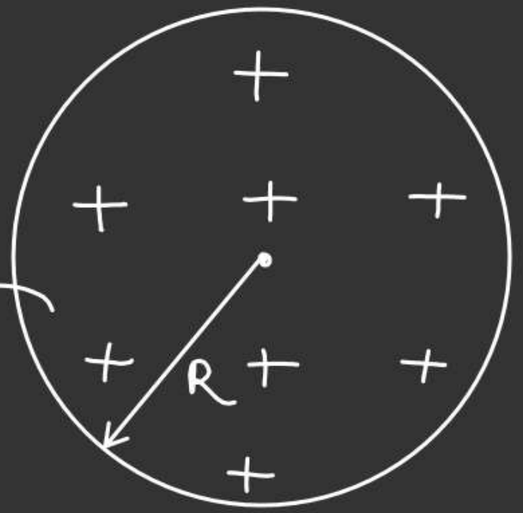
* Solid Non Conducting



$$\rho_r = \frac{\rho_0 r}{R}$$

* $U_{self} = ?$

* $\frac{U_{in}}{U_{out}} = ?$

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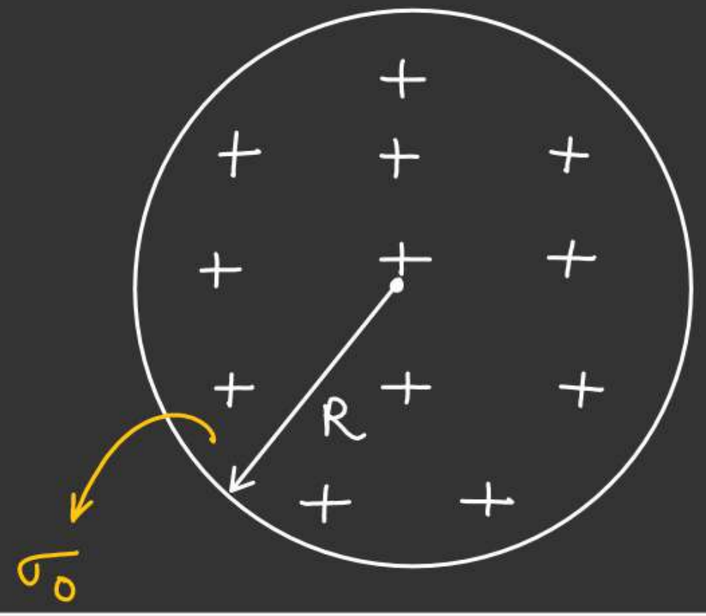
* Self Energy of a Disc :-

$Q = \sigma \cdot \pi R^2$

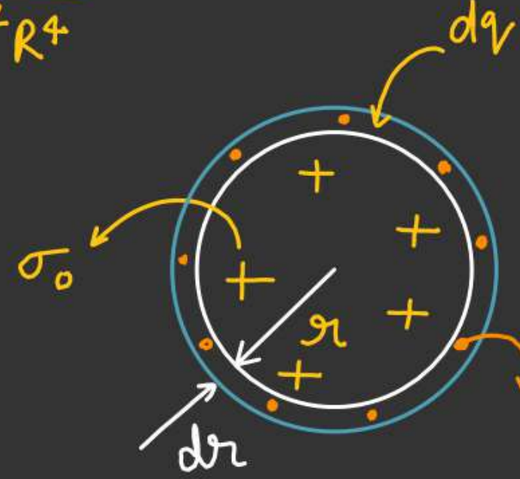


$$U = \frac{2R^3}{3\epsilon_0} \cdot \frac{Q^2}{\pi^2 R^4}$$

$$= \frac{2}{3\pi^2 \epsilon_0} \cdot \frac{Q^2}{R}$$

$$= \frac{8KQ^2}{3\pi R}$$

$dq = \sigma_0 \cdot 2\pi r dr$



$dw = V_{edge} \cdot dq$

$$dw = \frac{\sigma_0 r}{\pi \epsilon_0} \cdot \sigma_0 \cdot 2\pi r dr$$

$$dw \Rightarrow \frac{2\sigma_0^2}{\epsilon_0} \cdot r^2 dr$$

$$U_{self} = \int dw = \frac{2\sigma_0^2}{\epsilon_0} \int_0^R r^2 dr = \frac{2\sigma_0^2 R^3}{3\epsilon_0}$$

$\int_0^{\infty} dq$

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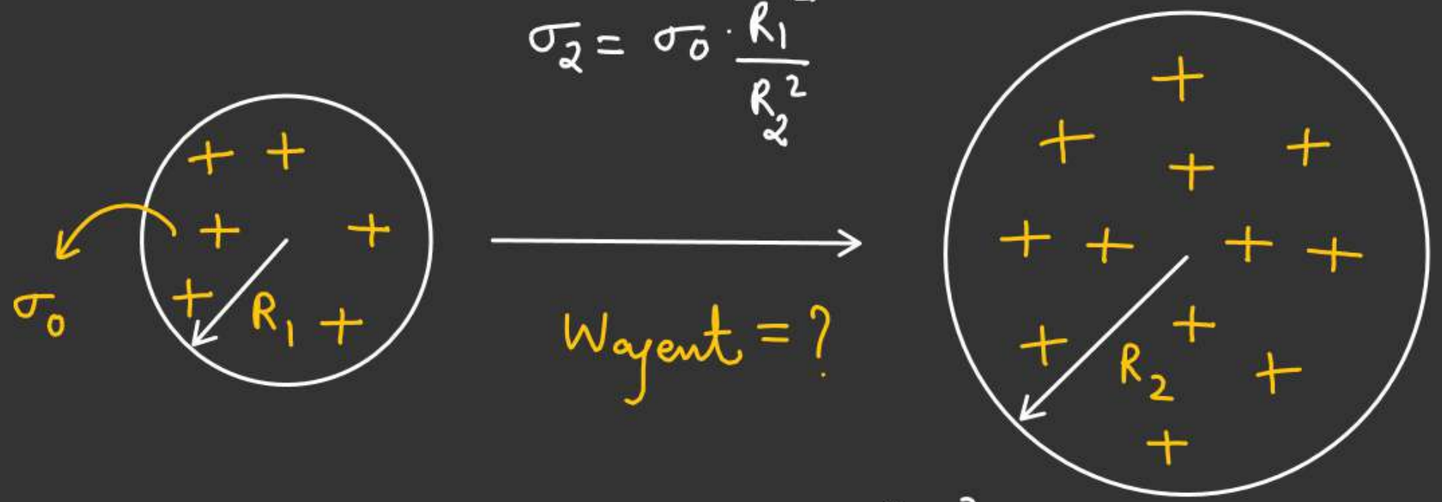
Disc

$$Q = \sigma_0 \cdot \pi R_1^2 = \sigma_2 \cdot \pi R_2^2 \Rightarrow$$

$$\sigma_2 = \sigma_0 \cdot \frac{R_1^2}{R_2^2}$$



M_1) $$W_{agent} = U_f - U_i = \frac{2\sigma_2^2 R_2^3}{3\epsilon_0} - \frac{2\sigma_1^2 R_1^3}{3\epsilon_0}$$

$$= \frac{2}{3\epsilon_0}\left[\sigma_0^2 \cdot \frac{R_1^4}{R_2^4} \cdot R_2^3 - \sigma_0^2 \cdot R_1^3\right] \Rightarrow \frac{2\sigma_0^2 R_1^3}{3\epsilon_0}\left[\frac{R_1}{R_2} - 1\right] = -W_e$$

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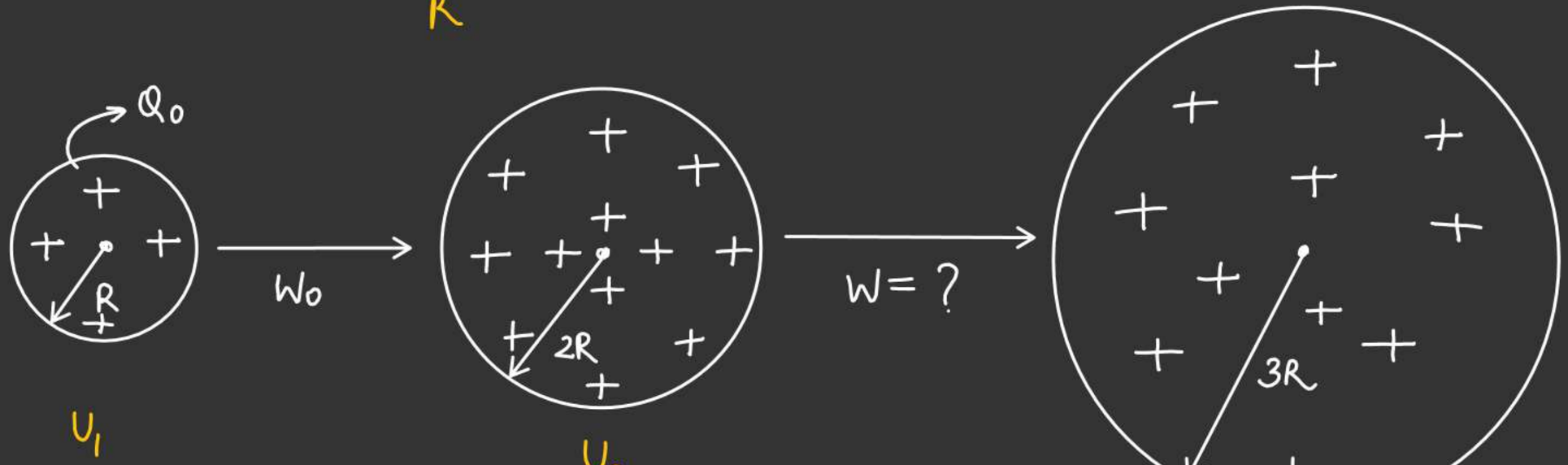
Disc :

$$U = C\cdot\frac{Q^2}{R}$$



U_1 U_2 U_3

$$W_0 = U_2 - U_1 = CQ^2\left(\frac{1}{2R} - \frac{1}{R}\right) = -\frac{CQ^2}{2R}$$

$$W = U_3 - U_2 = CQ^2\left[\frac{1}{3R} - \frac{1}{2R}\right] = -\frac{CQ^2}{6R} \Rightarrow \frac{W_0}{3}$$ Ans

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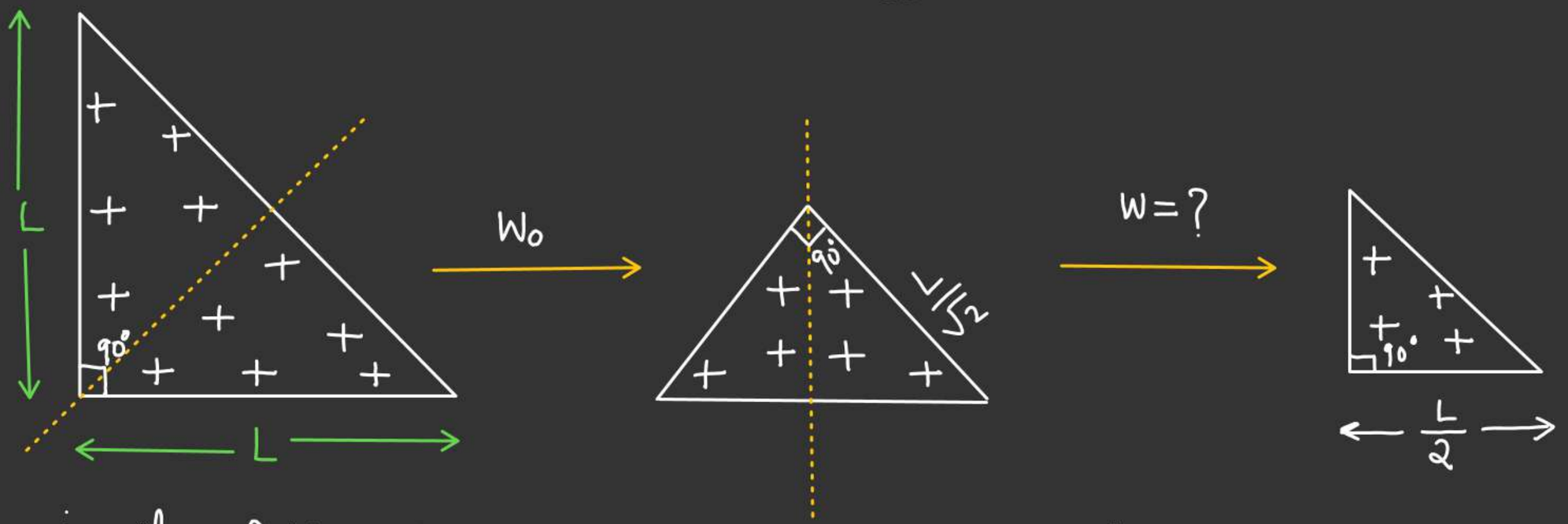
$M_1)\ V_1 = \frac{C \cdot Q^2}{L}$

$V_2 = \frac{C Q^2}{L/\sqrt{2}}$

$V_3 = \frac{C \cdot Q^2}{L/2}$



isosceles Right angled

$W_0 = V_2 - V_1 = \frac{CQ^2}{L}(\sqrt{2}-1)$

$W = V_3 - V_2 = \frac{CQ^2}{L}(2-\sqrt{2}) = \sqrt{2}W_0$

$\boxed{W = \sqrt{2}W_0}$

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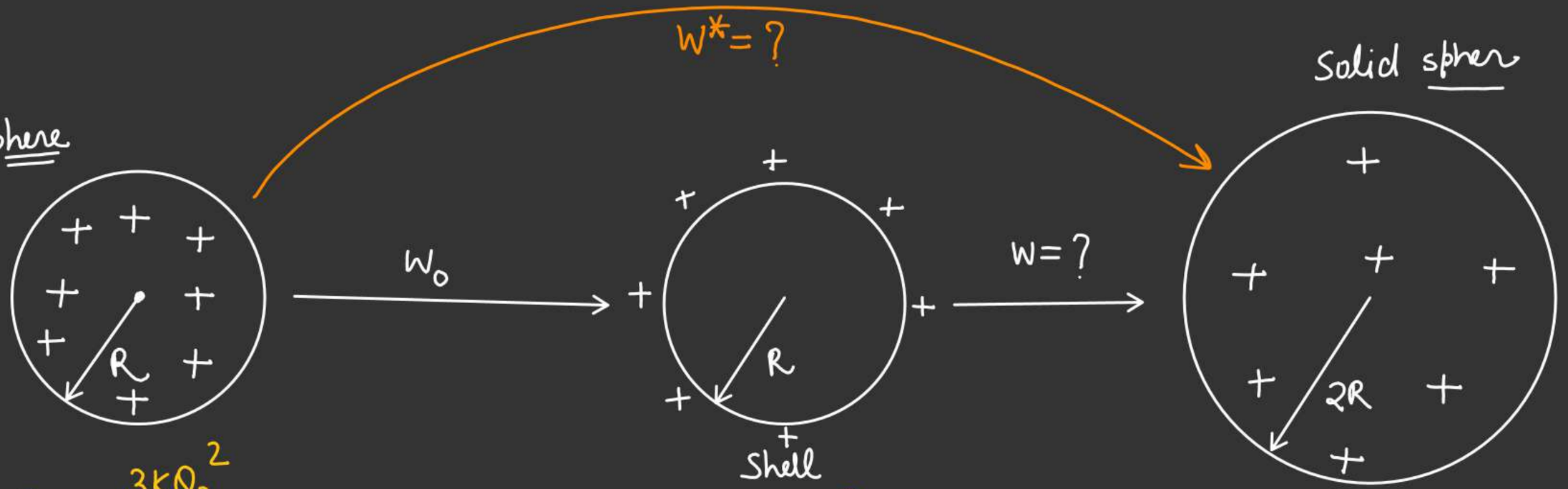

$U_1 = \frac{3KQ_0^2}{5R}$

$U_2 = \frac{KQ^2}{2R}$

$U_3 = \frac{3KQ^2}{5 \cdot 2R}$

$W_0 = U_2 - U_1 = \frac{KQ^2}{R}\left(-\frac{3}{5} + \frac{1}{2}\right) \Rightarrow -\frac{KQ^2}{10R}$

$W = U_3 - U_2 = \frac{KQ^2}{R}\left(\frac{3}{10} - \frac{1}{2}\right) = \frac{-KQ^2}{5R} = +2W_0$

$W^* = U_3 - U_1 = -\frac{3KQ^2}{10R}$

$\Rightarrow 3W_0$

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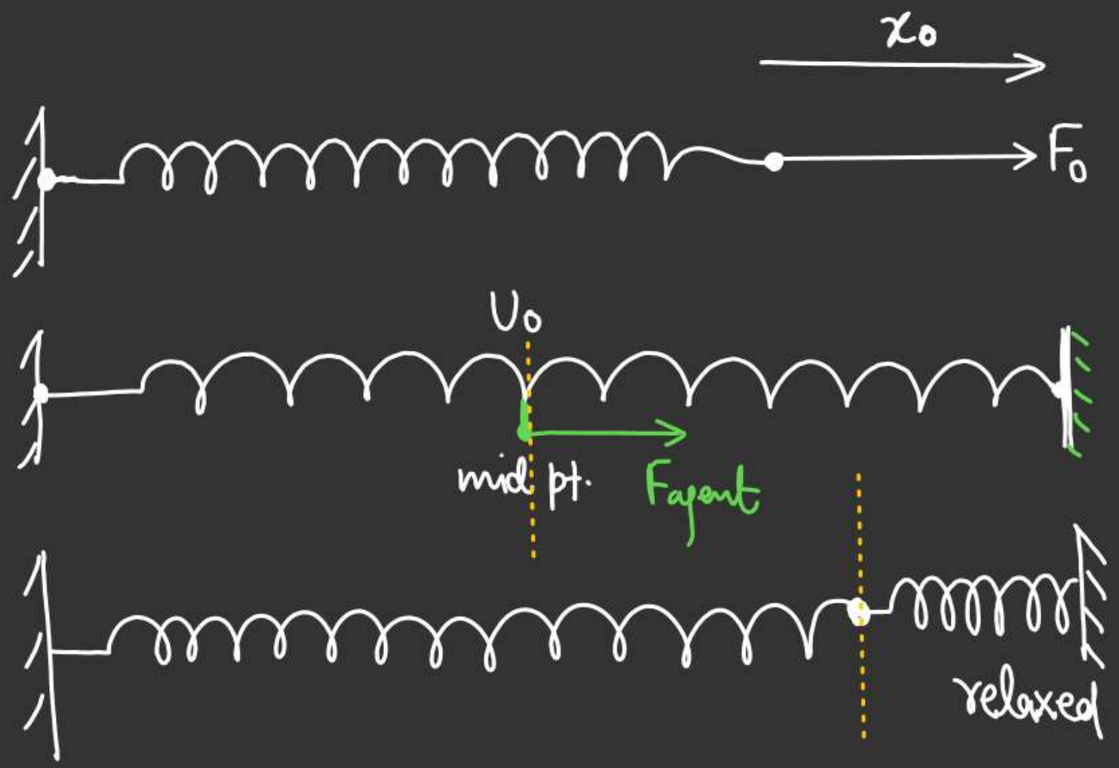
*

H.C.



$[W_{agent} = ?]$

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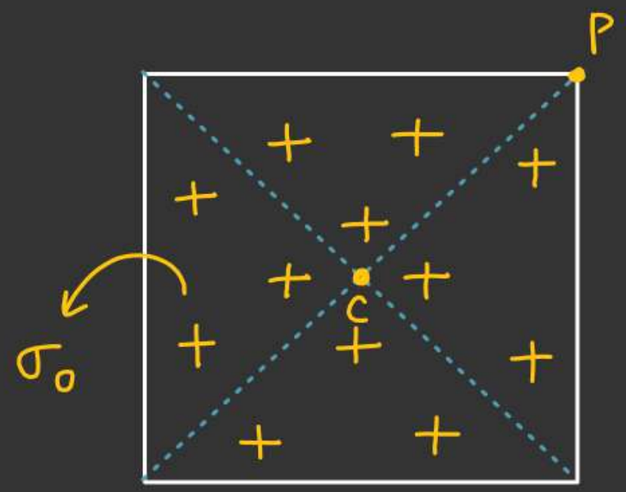
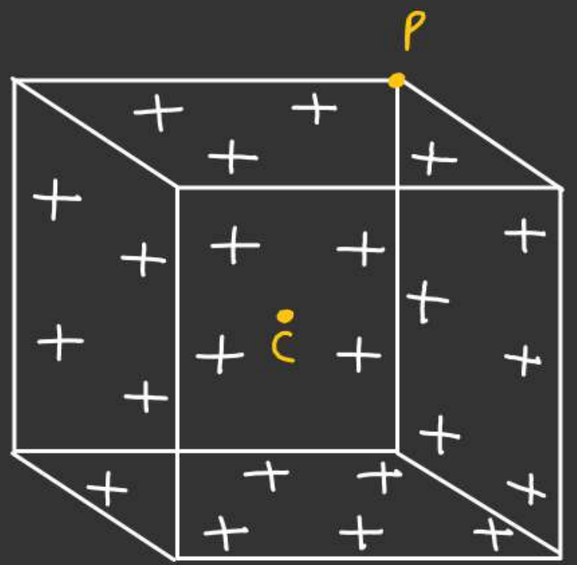


Sq. plate

$$\frac{V_C}{V_P} = ?$$



$$\frac{V_C}{V_P} = ?$$

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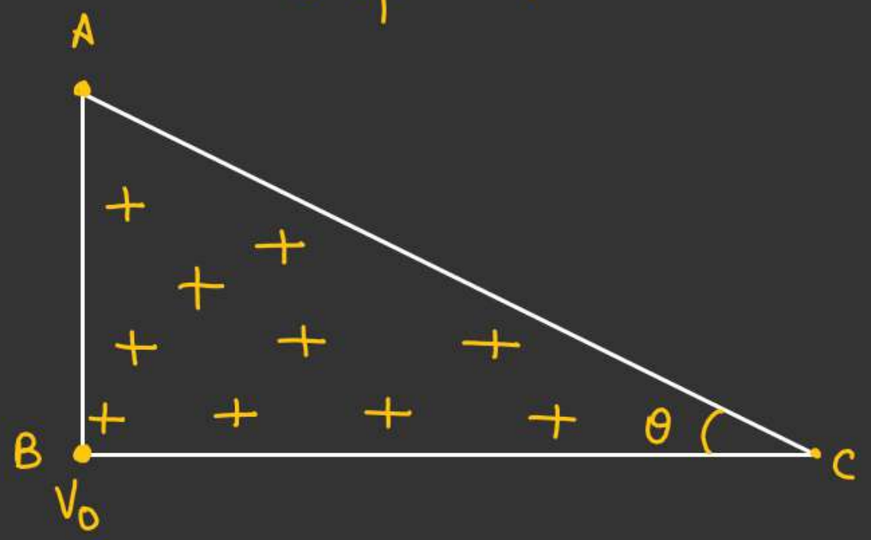
Uniform plate



$V_A = ?$

$V_C = ?$

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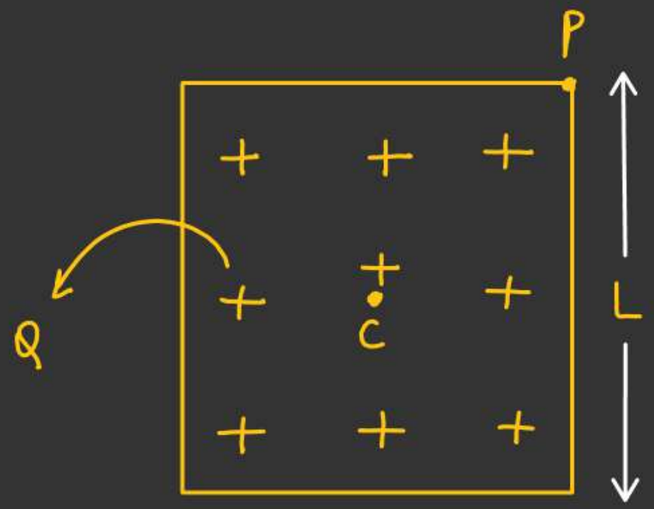
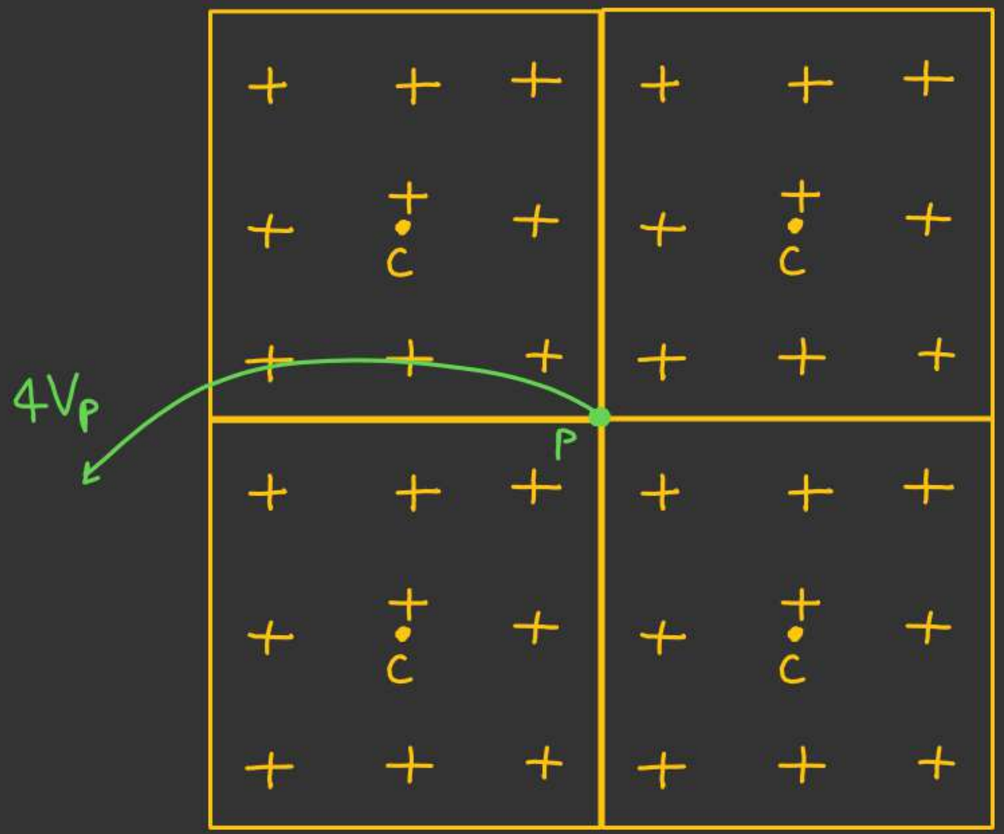


$$V_C = \alpha \cdot \frac{Q}{L}$$

$\alpha \rightarrow$ Const



$$4V_P = \alpha \cdot \frac{4Q}{2L} \Rightarrow V_P = \frac{\alpha Q}{2L}$$

$$\frac{V_C}{V_P} = 2$$

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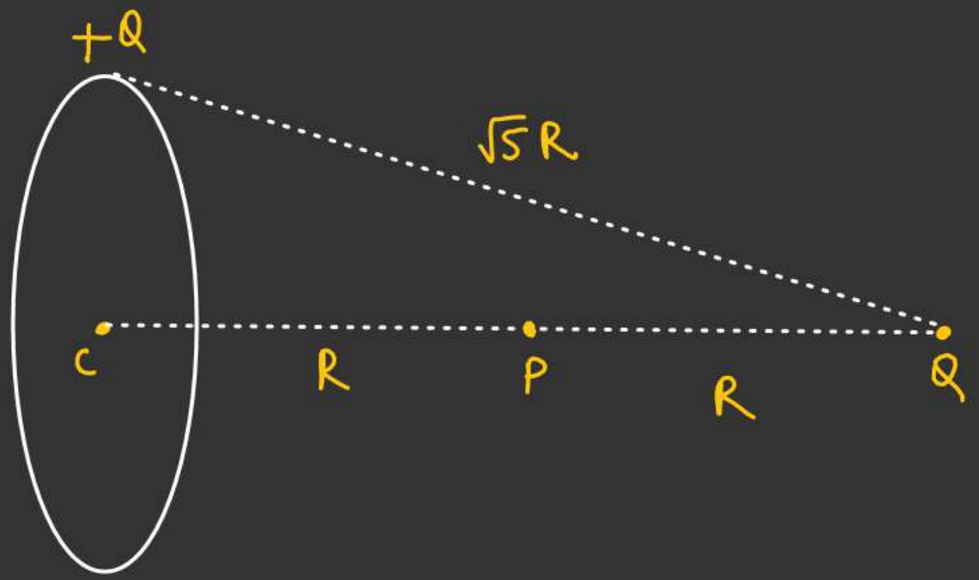
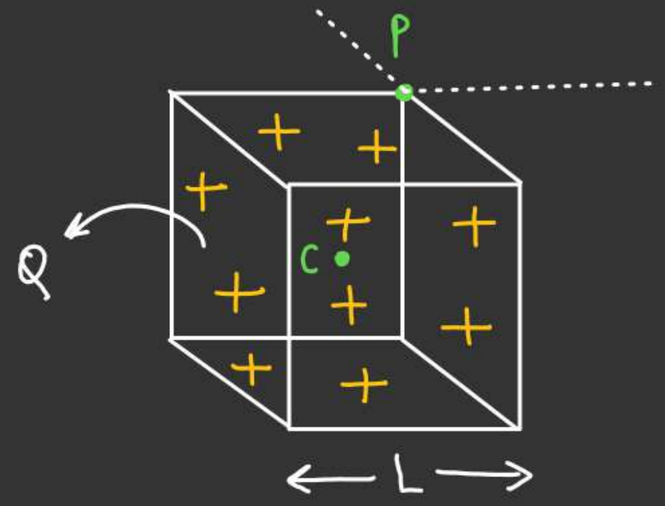


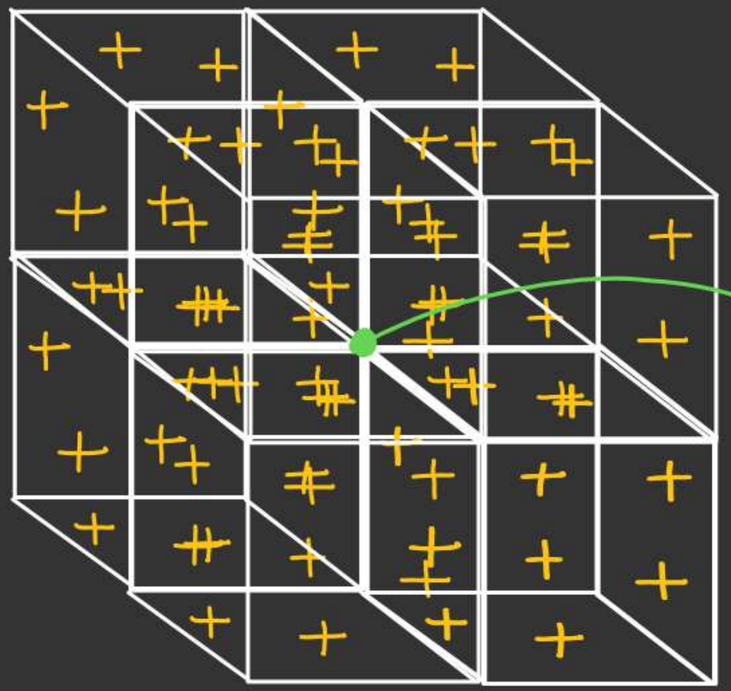
$$V_C = \frac{KQ}{R} \quad , \quad V_P = \frac{KQ}{\sqrt{2}R} \quad , \quad V_Q = \frac{KQ}{\sqrt{5}R}$$

$$V_P = \frac{V_C}{\sqrt{2}} \quad , \quad V_Q = \frac{V_C}{\sqrt{5}}$$

Nishant Jindal



$$V_C = \alpha \cdot \frac{Q}{L}$$



$$8V_P = \alpha \cdot \frac{8Q}{2L}$$

$$V_P = \frac{\alpha Q}{2L}$$

$$\frac{V_C}{V_P} = 2$$

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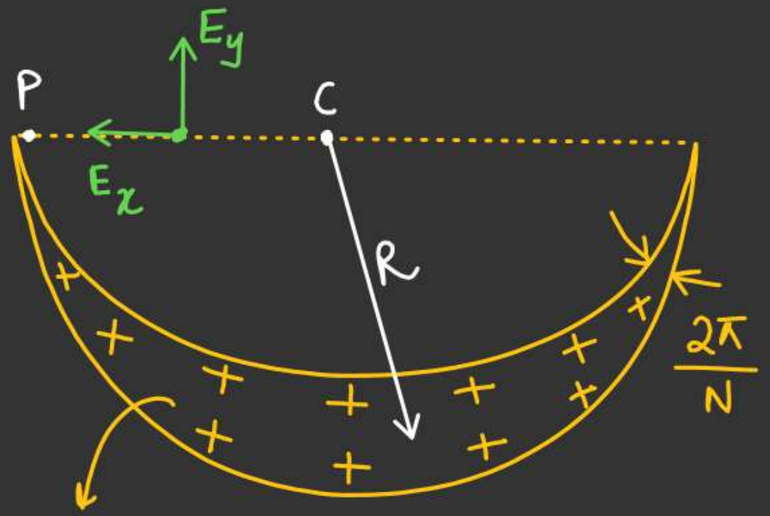
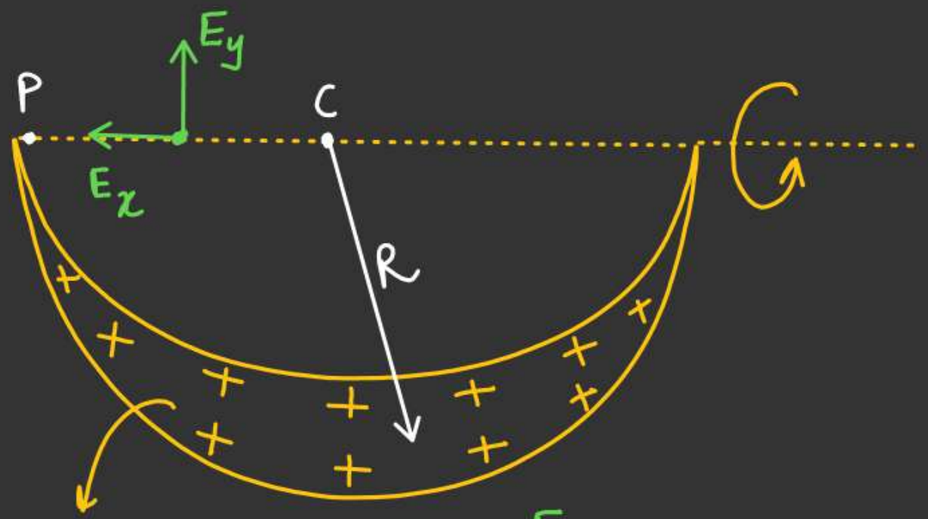
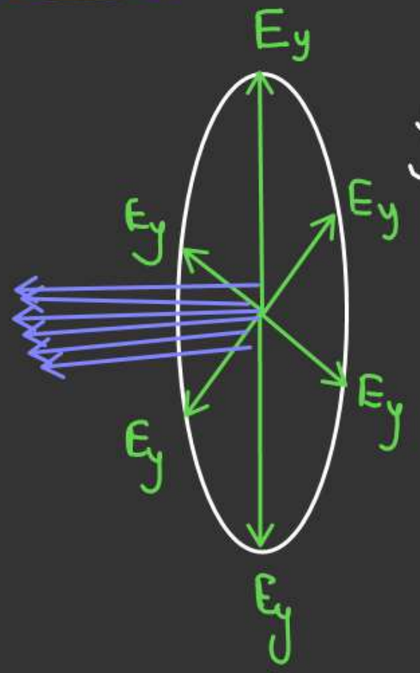
* Watermelon Problem :-

Non Conducting



σ_0 Const





for Complete shell

NE_x

$$\frac{V_C}{V_P} = ?$$

$$V_P - V_C = -\int E_x \cdot dx =$$

$$V_f - V_i = -\int \vec{E} \cdot \vec{dr}$$

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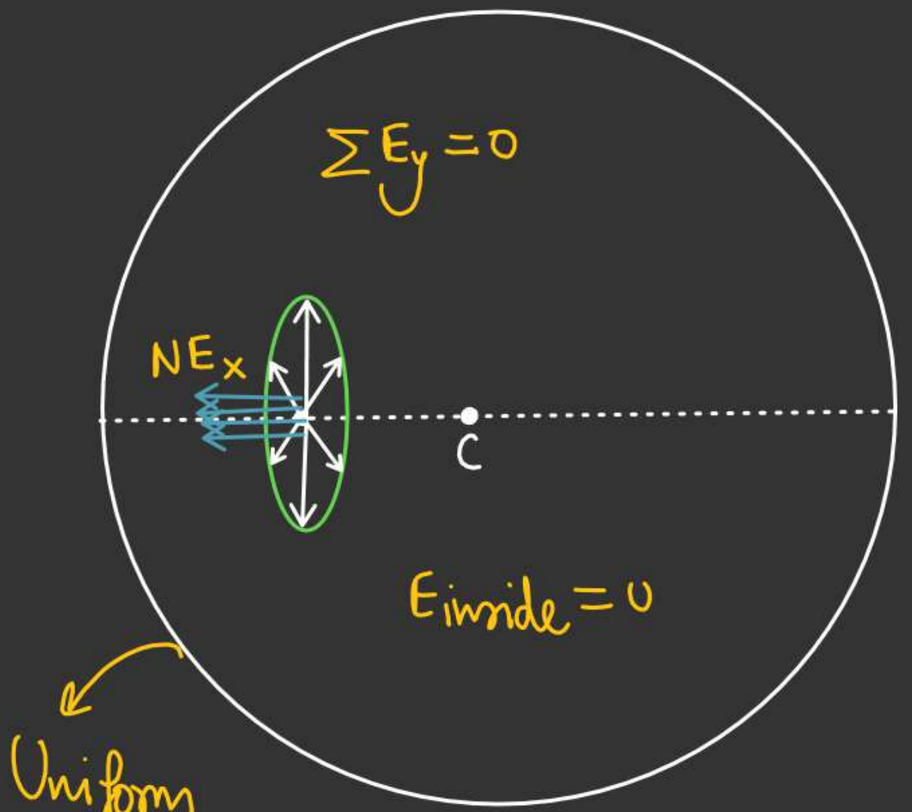
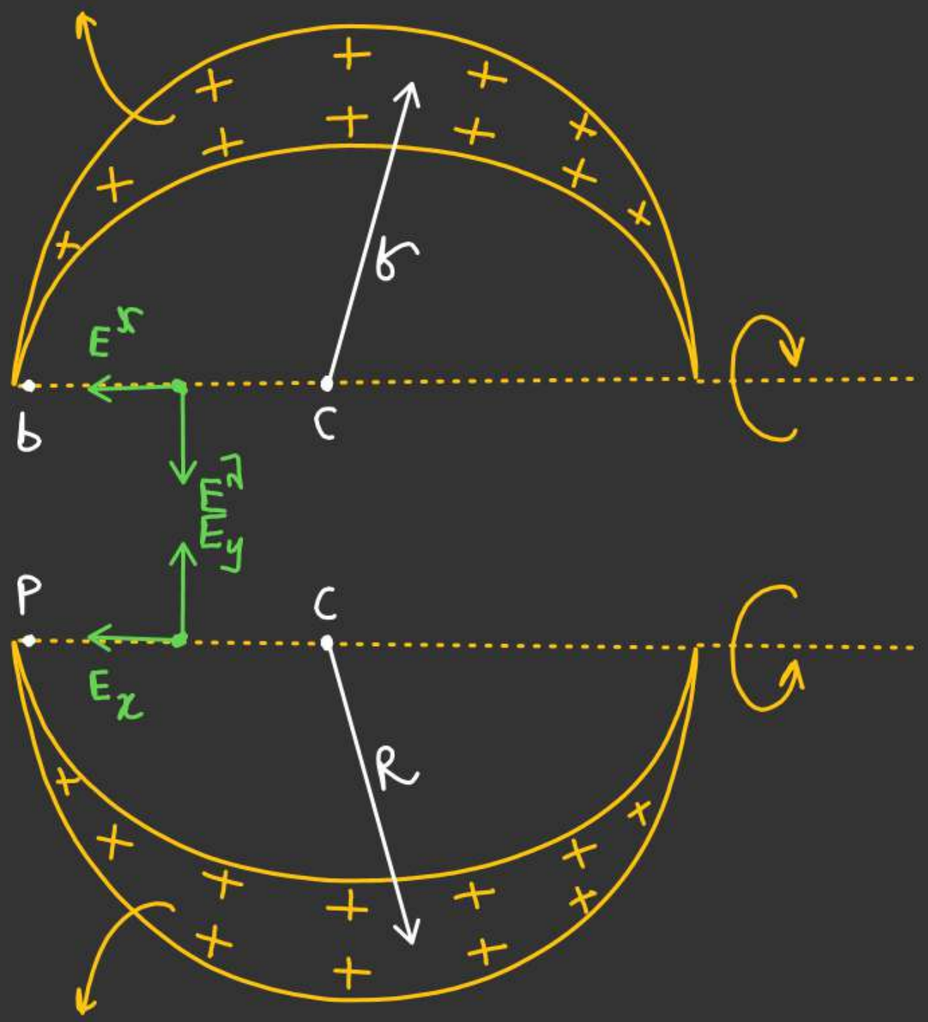


$$NE_x = 0$$

$$\boxed{E_x = 0}$$

$$V_C - V_P = 0 \Rightarrow \frac{V_C}{V_P} = 1$$



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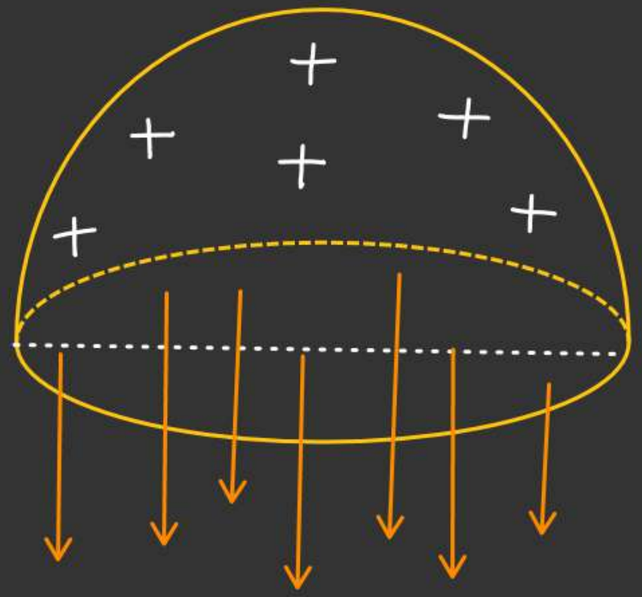
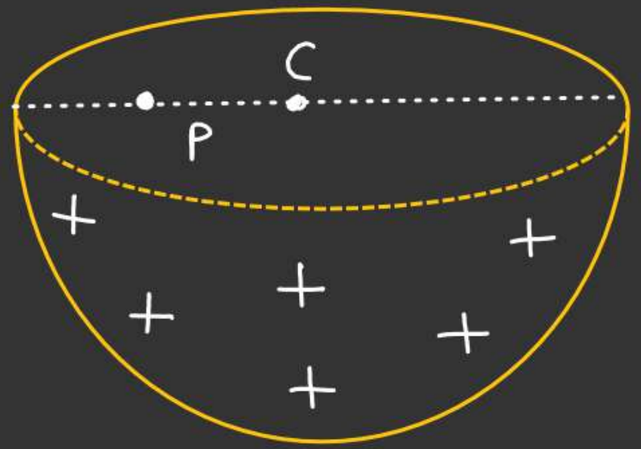
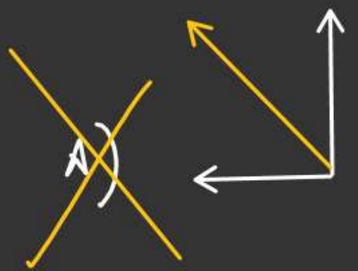
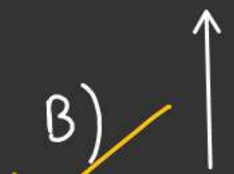




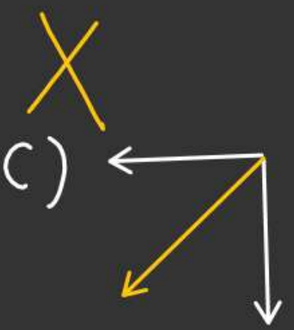
A)



B)



C)



D)

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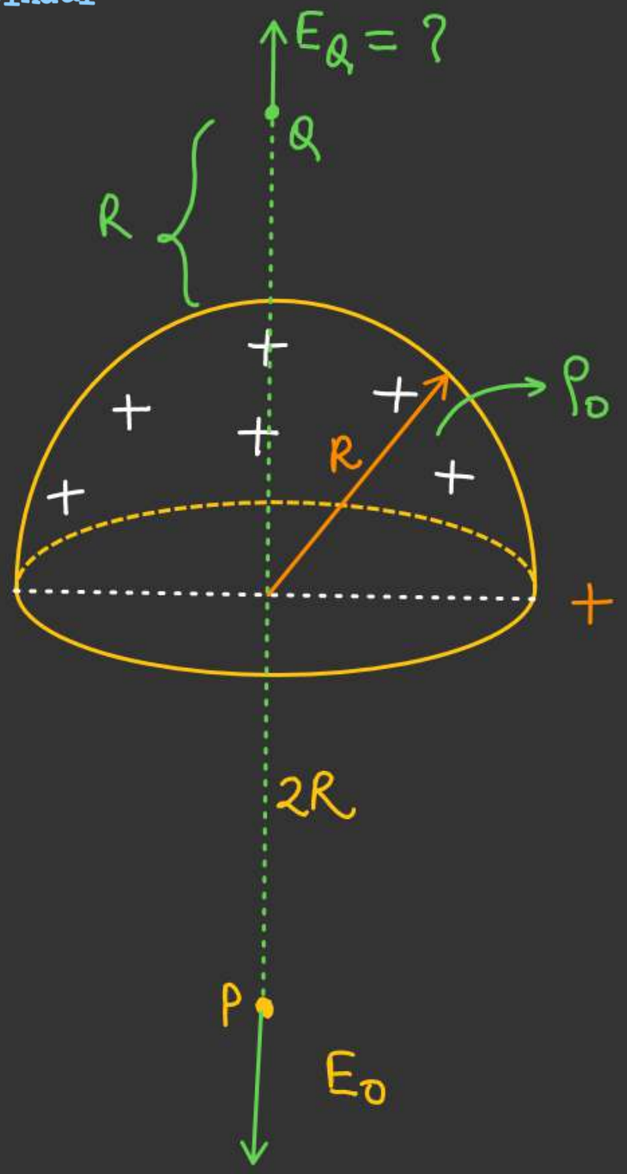
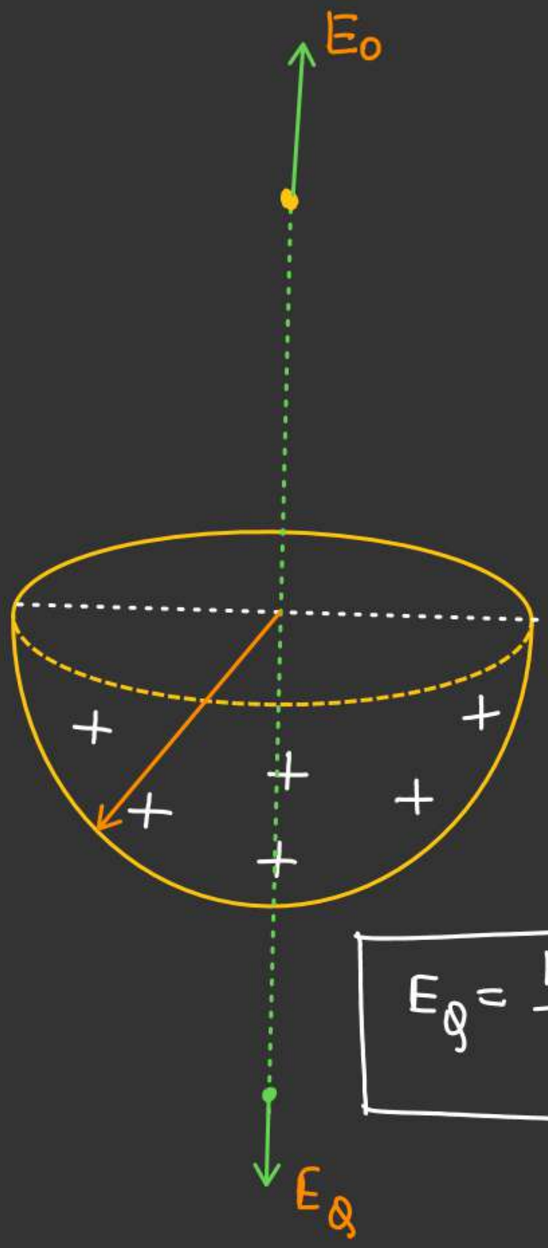
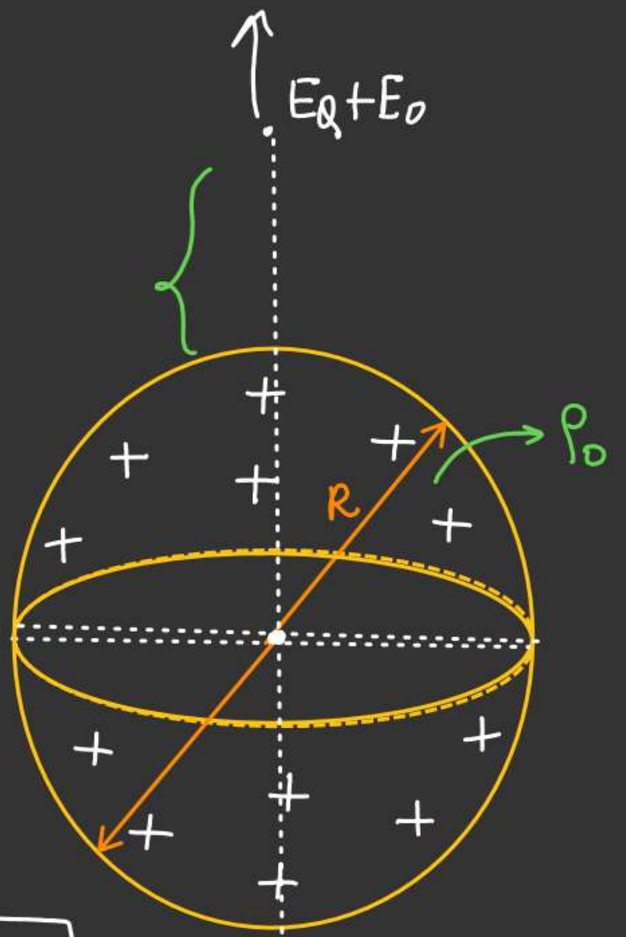




$$E_Q = \frac{KQ}{4R^2} - E_0$$



Ans

$$E_Q + E_0 = \frac{K \cdot Q}{(2R)^2}$$

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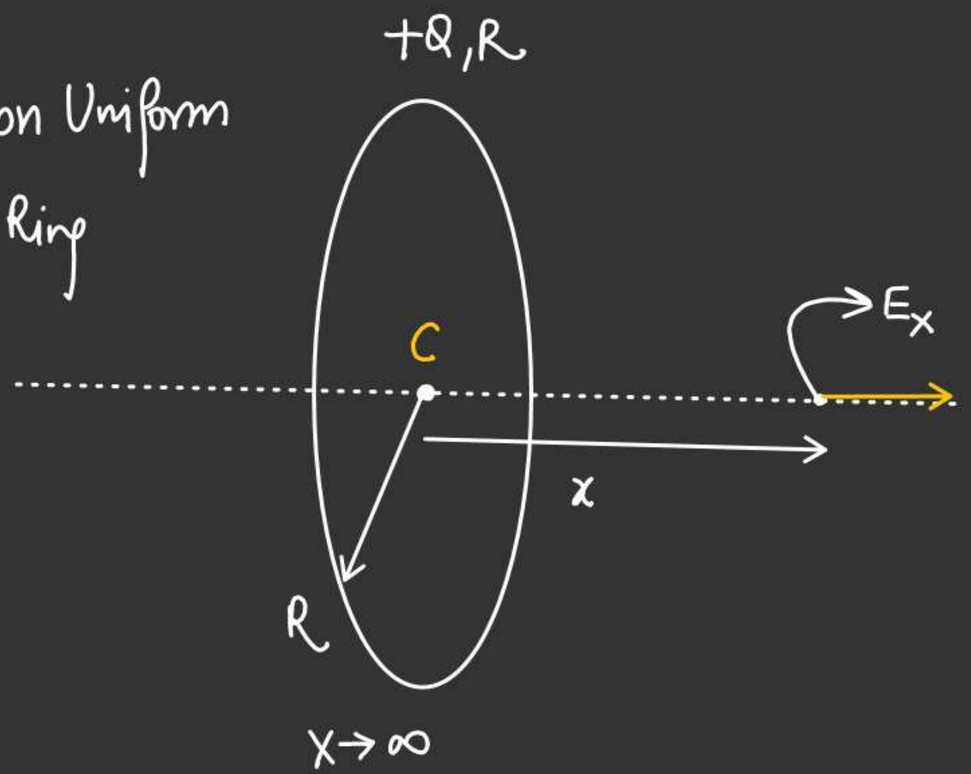
*

Non Uniform Ring



$X \to \infty$

$$\int_{X=0} E_x \, dx = ?$$

$$V_f - V_i = -\int \vec{E} \cdot d\vec{r}$$

$$V_\infty - V_C = -\int_0^\infty E_x \cdot dx$$

$$-\frac{KQ}{R} = -\int_0^\infty E_x \cdot dx$$

$$\boxed{\int_0^\infty E_x \cdot dx = \frac{KQ}{R}}$$

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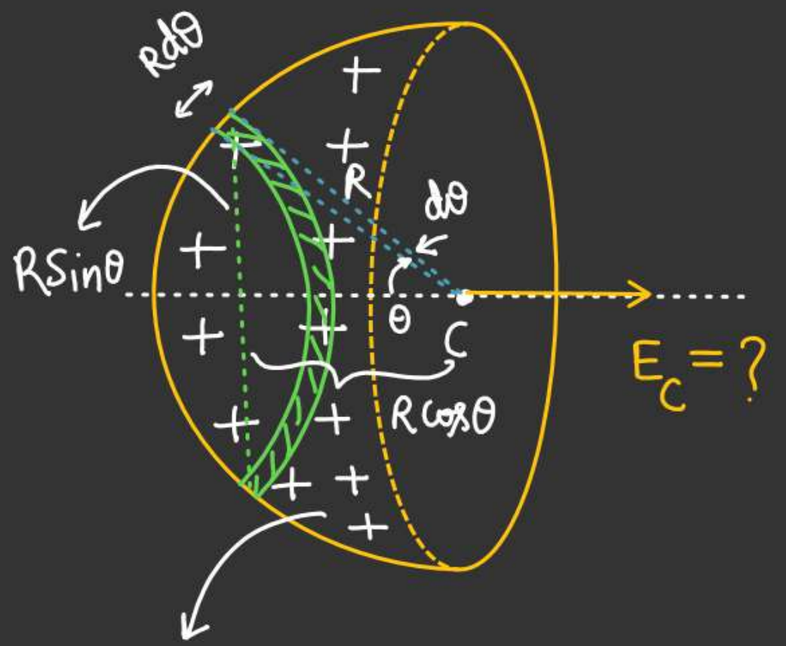
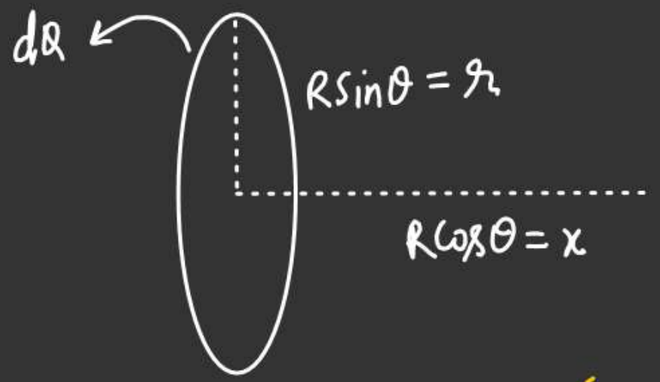
Hemisphere :-





$$\frac{KQx}{(R^2+x^2)^{3/2}}$$

$$dE_C = \frac{K\cdot(\sigma_0\cdot 2\pi R^2 \sin\theta\cdot d\theta)\cdot R\cos\theta}{(R^2)^{3/2}}$$

$$dE_C = \frac{1}{4\pi\epsilon_0}\cdot 2\pi\sigma_0\cdot \sin\theta\cdot\cos\theta\cdot d\theta$$

$$dE_C = \frac{\sigma_0}{2\epsilon_0}\cdot \sin\theta\cdot\cos\theta\cdot d\theta$$

σ_0 Uniform

$dQ = \sigma_0\cdot dA$

$dQ = \sigma_0\cdot 2\pi R^2 \sin\theta\cdot d\theta$

★ $E_C = \frac{\sigma_0}{4\epsilon_0}$

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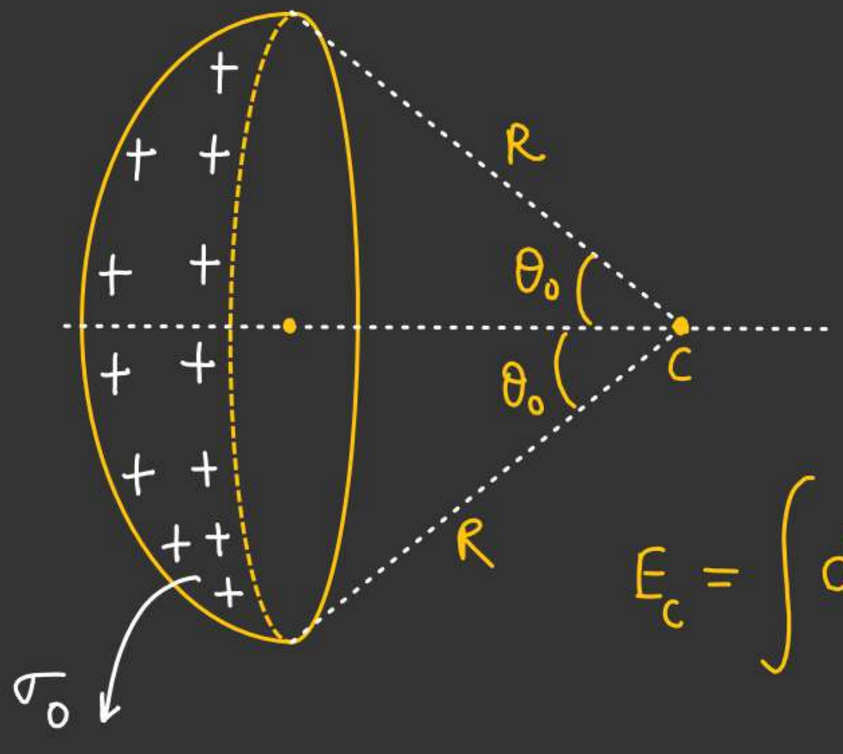
Spherical Cap :-



$$E_C = \int dE_C = \frac{\sigma_0}{2\epsilon_0} \cdot \int_0^{\theta_0} \underbrace{\sin\theta}_{u} \cdot \underbrace{\cos\theta \cdot d\theta}_{du}$$

$$E_C = \frac{\sigma_0}{2\epsilon_0} \cdot \frac{\sin^2\theta_0}{2} \Rightarrow$$

$$\boxed{E_C = \frac{\sigma_0}{4\epsilon_0} \sin^2\theta_0} \quad *$$

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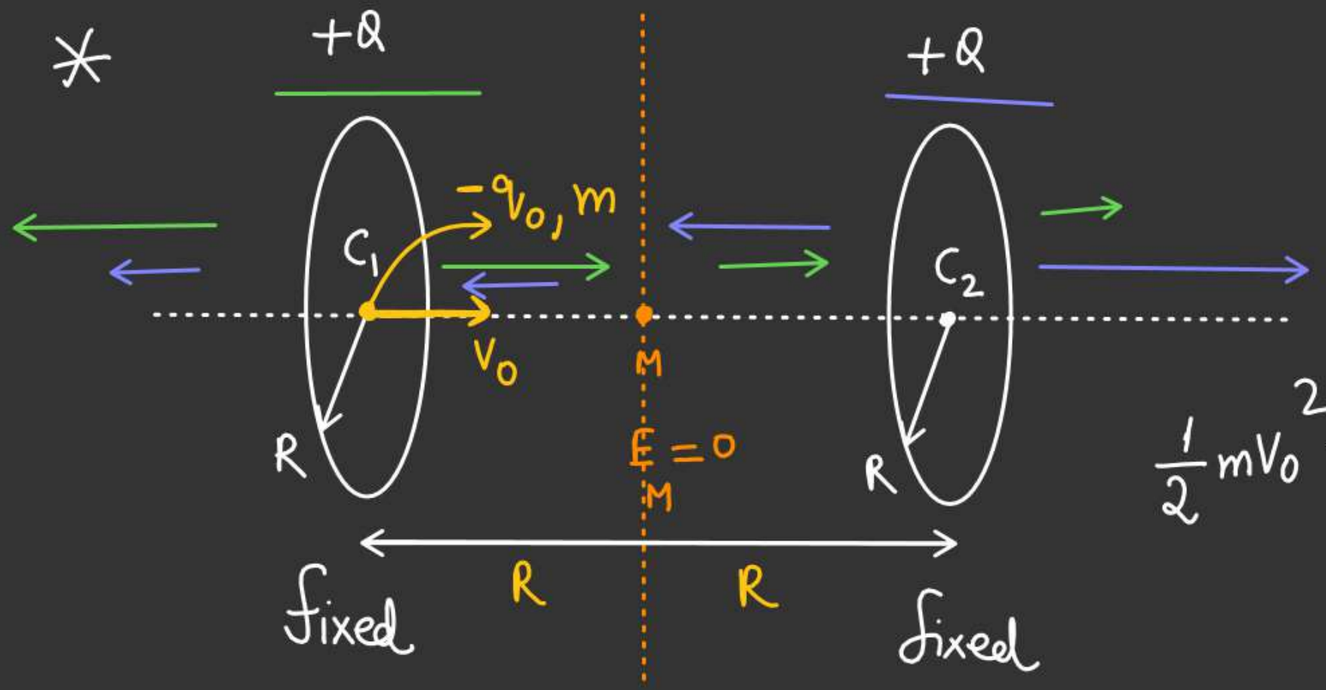

Find min V_0 so that Particle will reach C_2.

For min V_0

$$E_{C_1} = E_M$$

$$\frac{1}{2}mV_0^2 + \frac{KQ}{R}(-q_0) + \frac{KQ}{\sqrt{5}R}(-q_0) = 0 + \left[\frac{KQ}{\sqrt{2}R}(-q_0)\right] \times 2$$

$V_0 = \checkmark$

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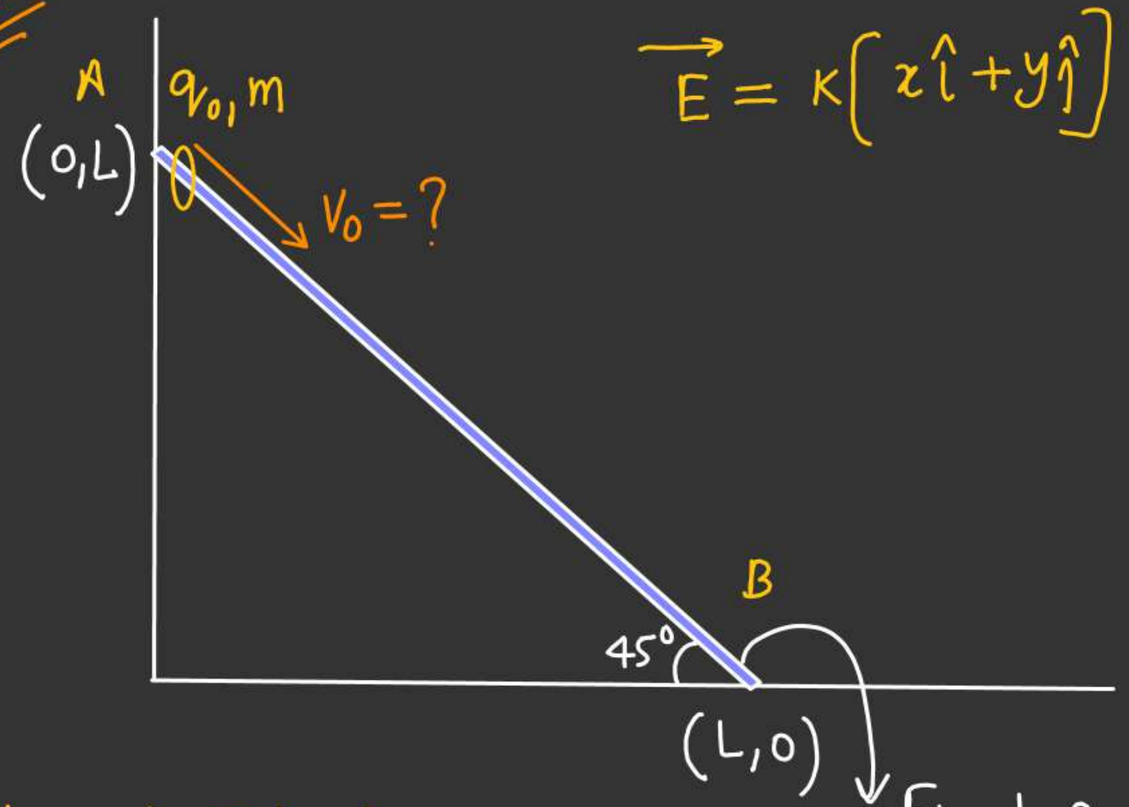
* H.C.



$$\vec{E} = K\left[x\hat{i} + y\hat{j}\right]$$

* find min velocity given to the Ring so that it can reach B

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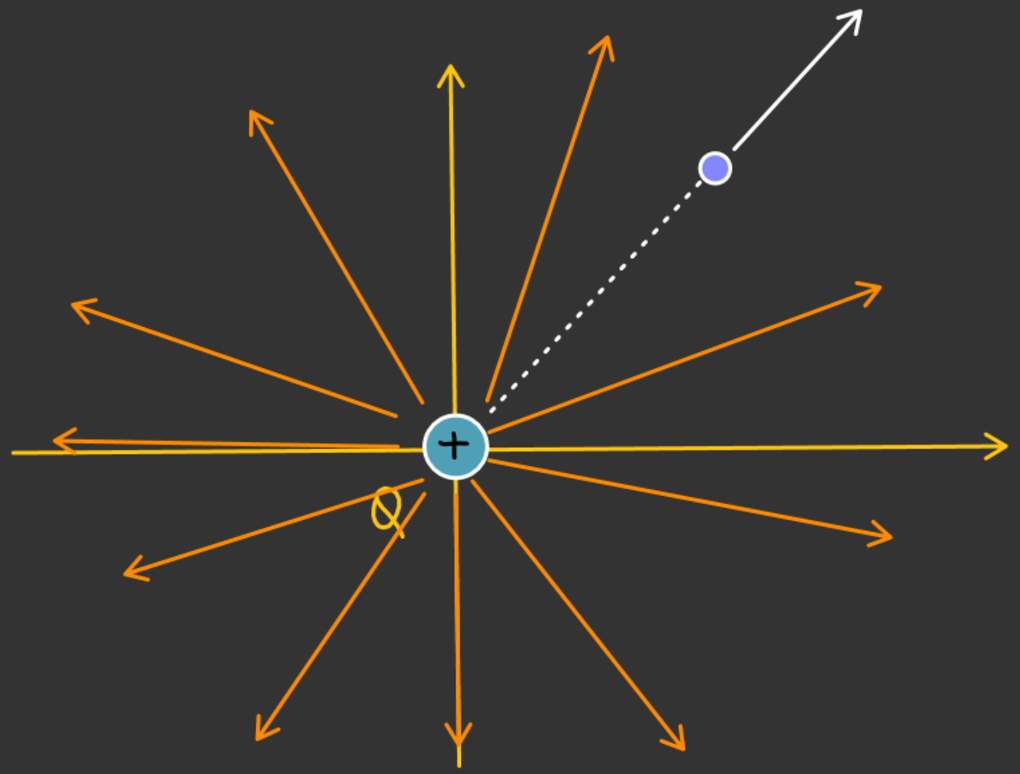
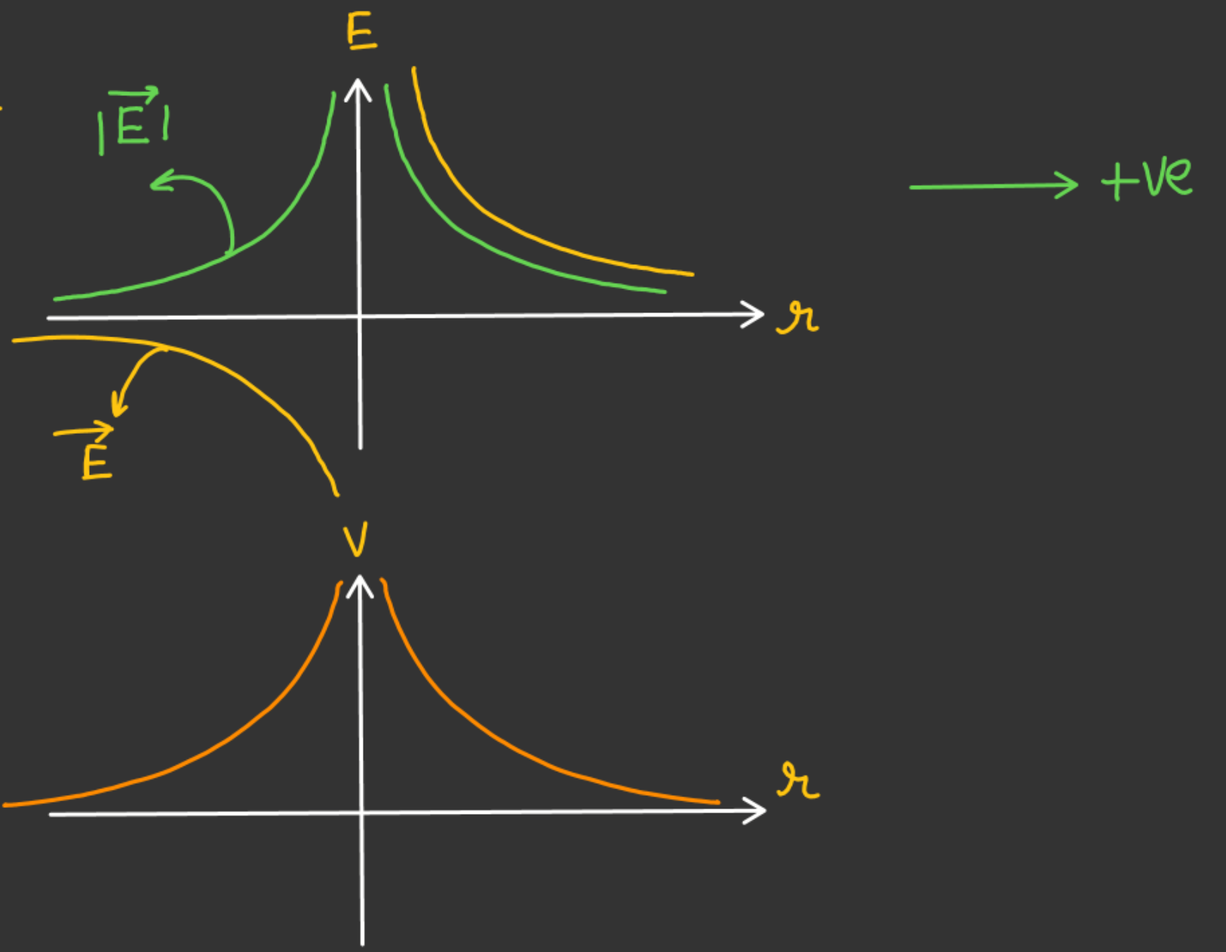
E-r & V-r graphs of point charges :-





$$\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}}{q_0} = \frac{kq}{r^2}(\hat{r})$$

$$V = \frac{kQ_0}{r}$$

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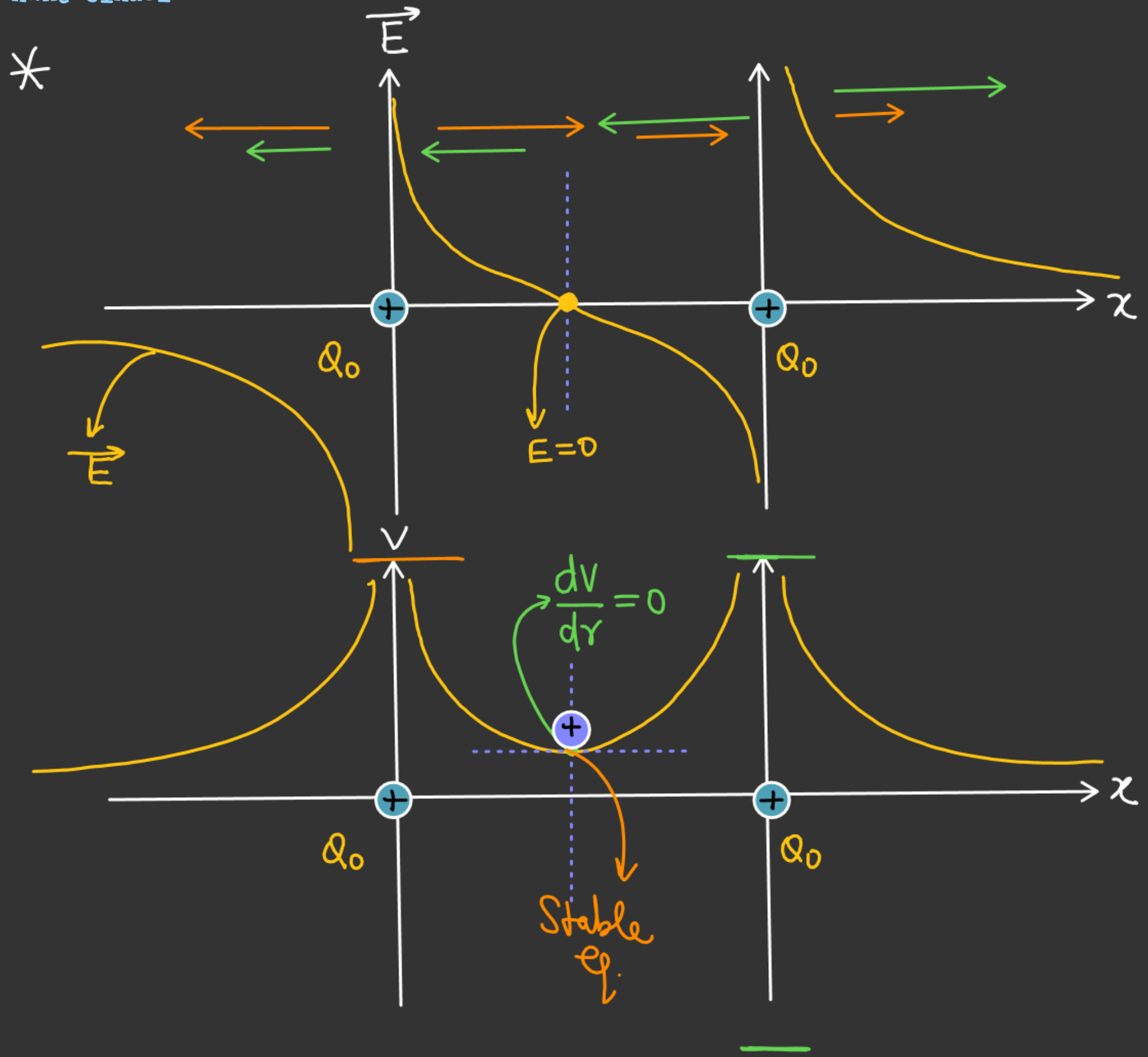

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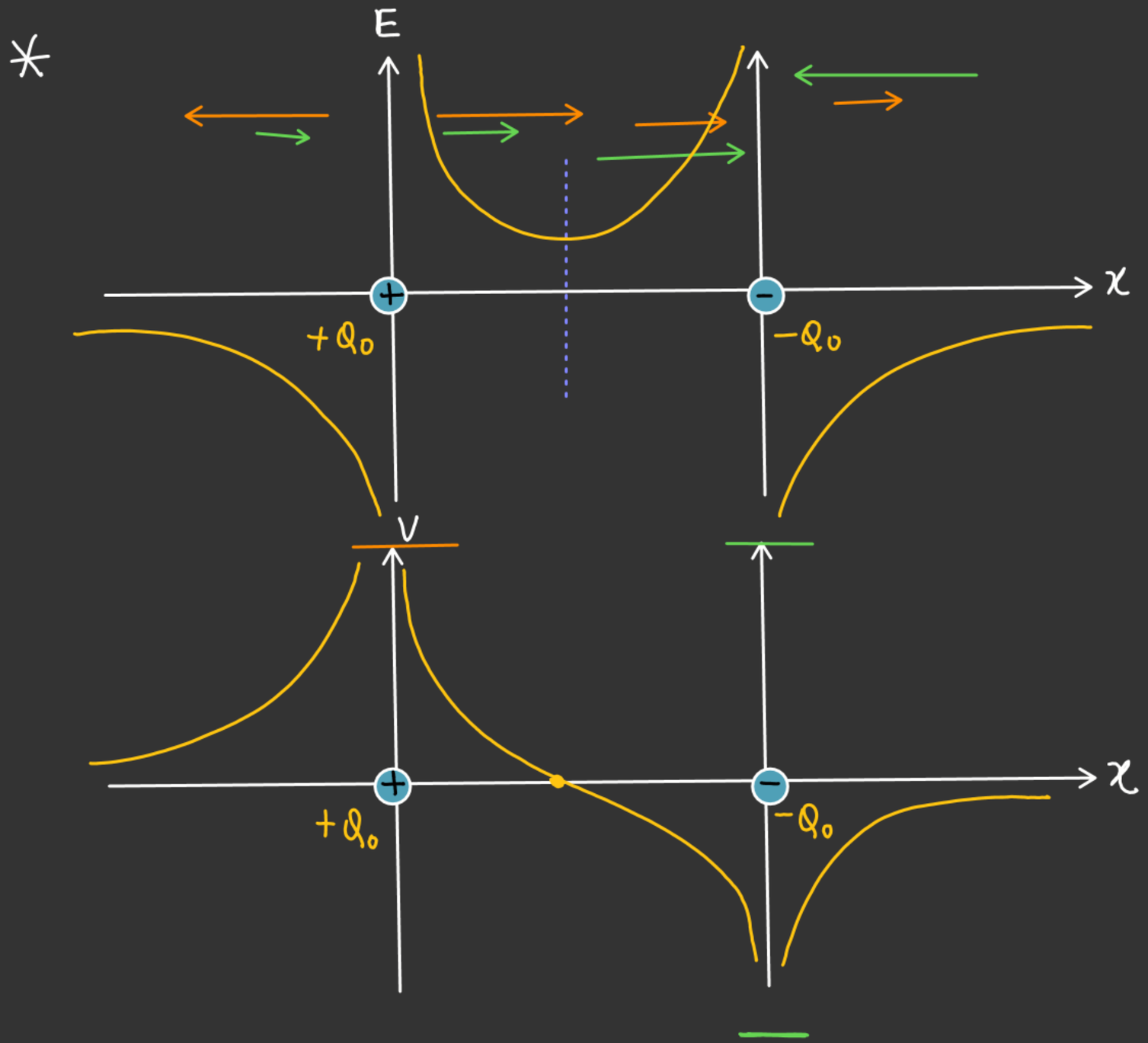
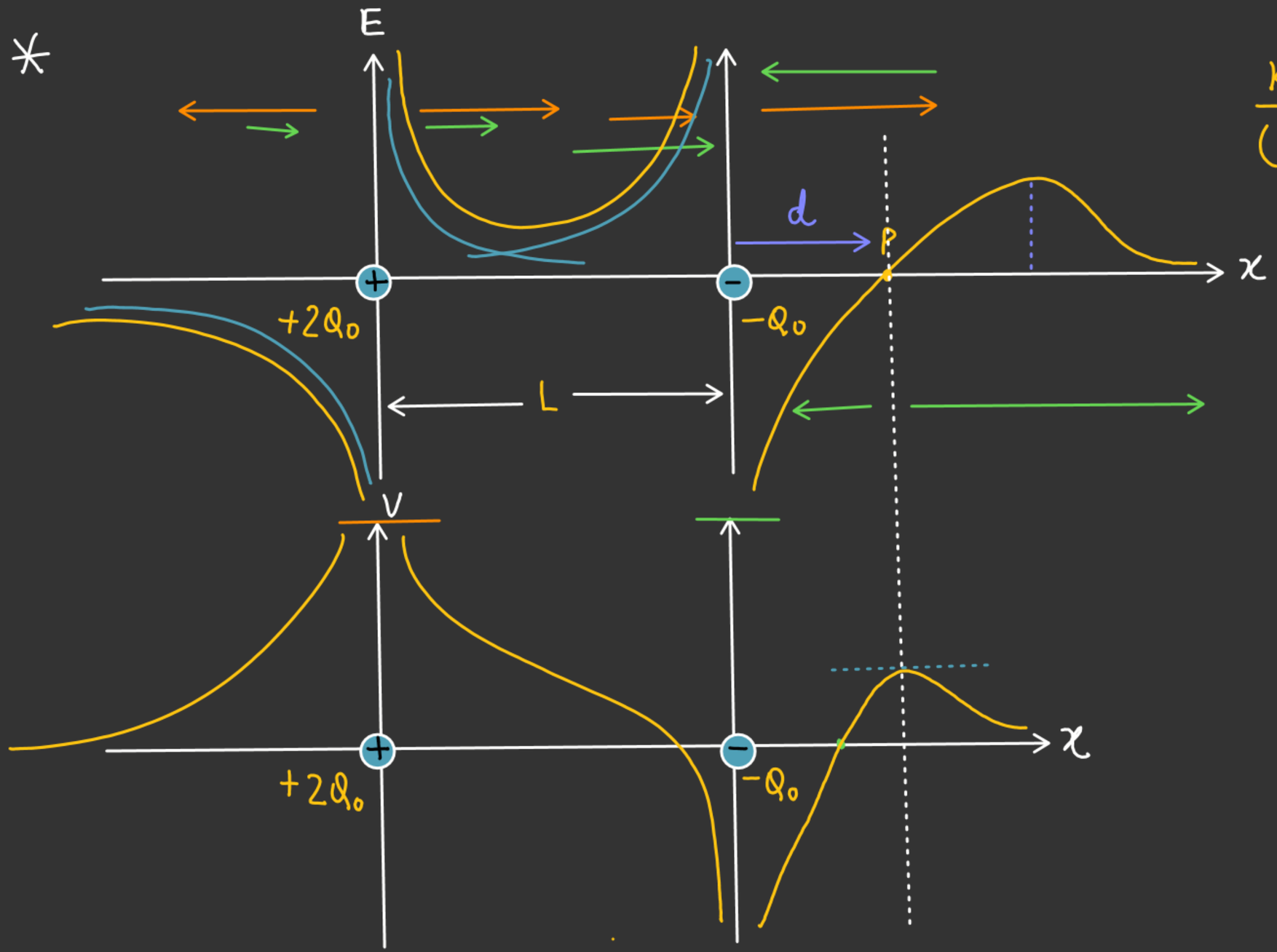


Nishant Jindal



$$\frac{K \cdot 2Q_0}{(L+d)^2} = \frac{K \cdot Q_0}{d^2}$$

$$\frac{\sqrt{2}}{L+d} = \frac{1}{d}$$

$$d = + \frac{L}{\sqrt{2}-1}$$

$$\frac{K \cdot 2Q}{L+d} = \frac{KQ}{d}$$

$$2d = L+d$$

$$\boxed{d = L}$$

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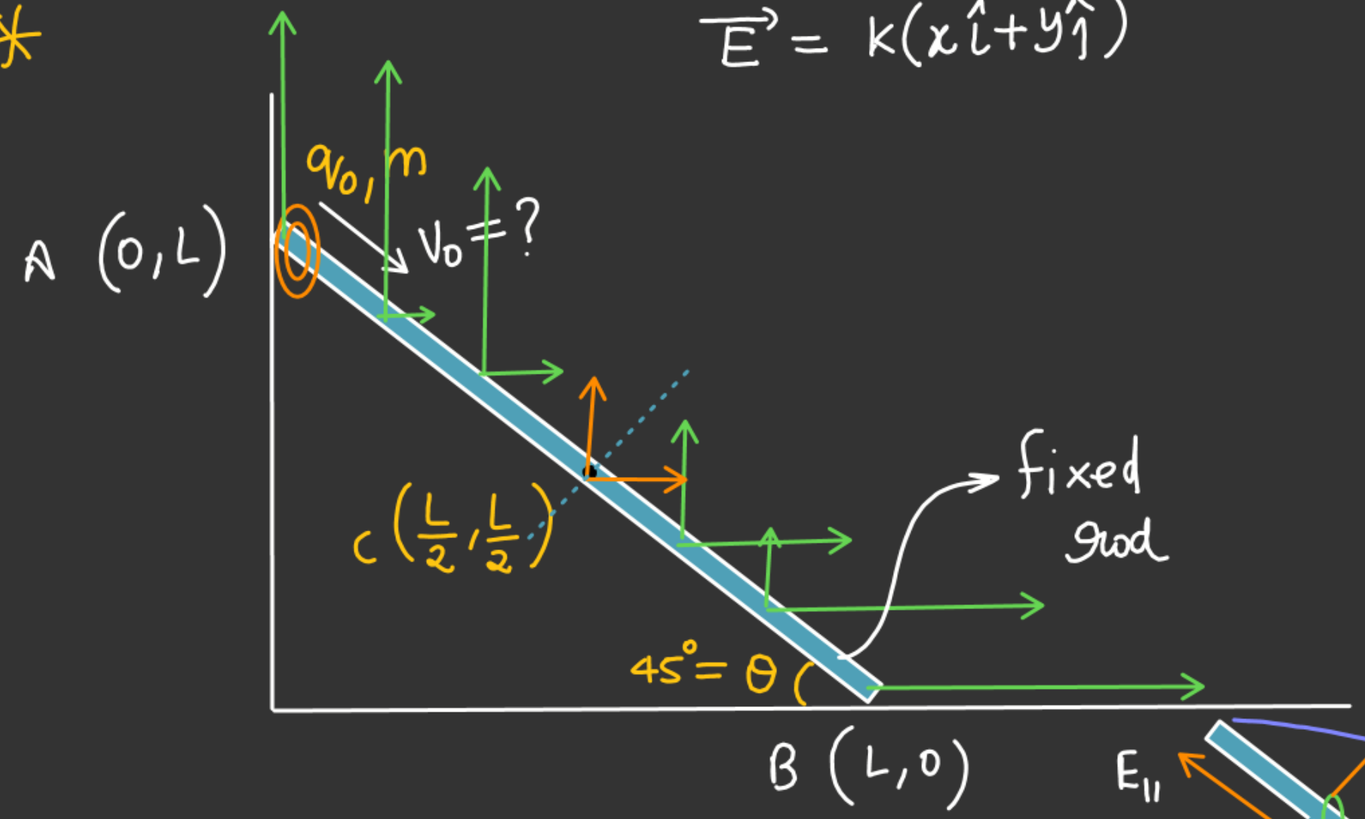
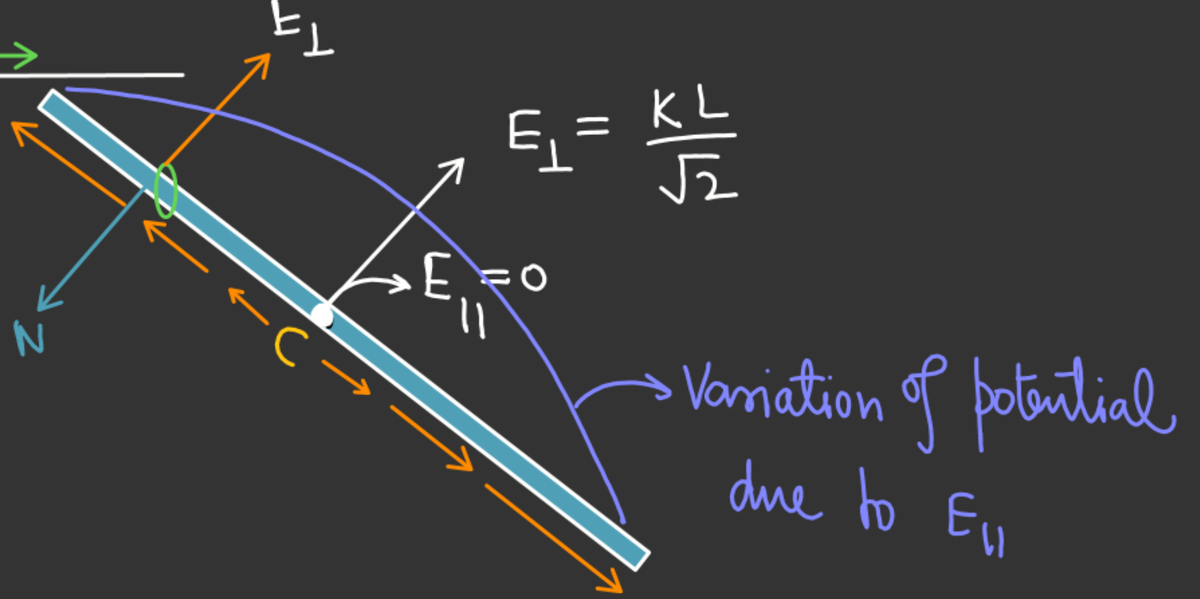
$\vec{E} = k(x\hat{i} + y\hat{j})$





Electric field along the rod :

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$E_A = E_B$

$K_A + U_A = \cancel{K_B}^{0} + U_B$

Wrong

Correct App.

$K_A + U_A = K_C + U_C$ ($K_C > 0$)

$K_A + U_A - U_C > 0$

$K_A > U_C - U_A$

$\frac{1}{2} m v_0^2 > q_0 (V_C - V_A)$

$v_0 > \sqrt{\frac{2q_0}{m}(V_C - V_A)}$

$\vec{E} = kx\hat{i} + ky\hat{j}$

$d\vec{r} = dx\hat{i} + dy\hat{j}$

$V_f - V_i = -\int \vec{E} \cdot d\vec{r}$

$V_C - V_A = -\int_0^{L/2} kx\,dx - \int_L^{L/2} ky\,dy$

$V_C - V_A = -\frac{k}{2} \cdot x^2\Big|_0^{L/2} - \frac{k}{2}\, y^2\Big|_L^{L/2}$

$V_C - V_A = -\frac{kL^2}{8} - \frac{kL^2}{8} + \frac{kL^2}{2} \Rightarrow \frac{kL^2}{4}$

$v_0 > \sqrt{\frac{2q_0 kL^2}{4m}}$

$$\boxed{v_0 = \sqrt{\frac{q_0 kL^2}{2m}}}$$

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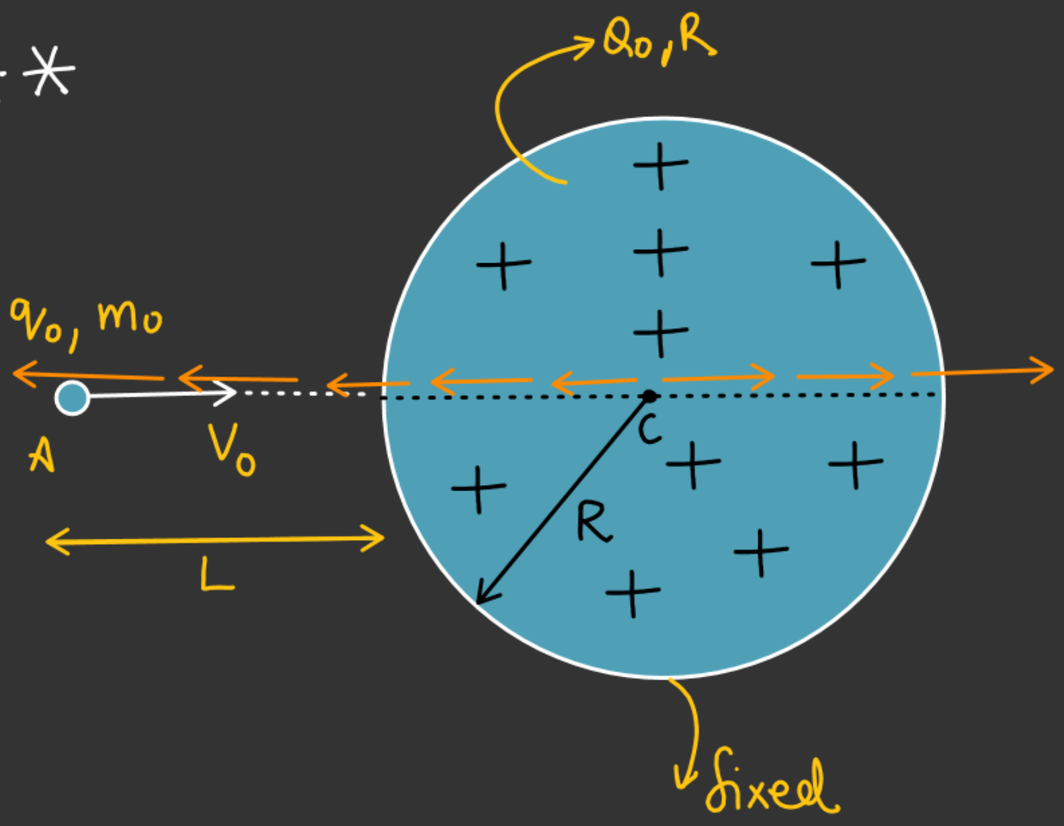
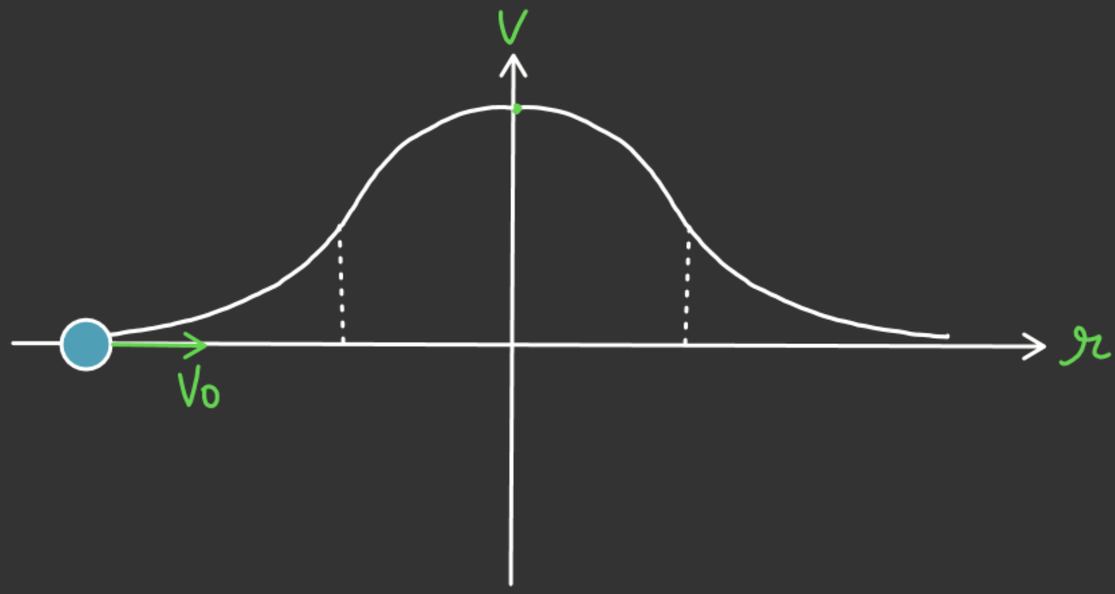
**





$$\frac{1}{2}mV_0^2 + \frac{KQq_0}{L+R} = \cancel{K_C}_0 + q_0\left(\frac{3}{2}\frac{KQ}{R}\right)$$

Find min V_0
So that bullet
Can penetrate the sphere.

* $\boxed{V_0 = \checkmark}$

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H.C. Reading of Elliptical orbits of gravitation

Nishant Jindal

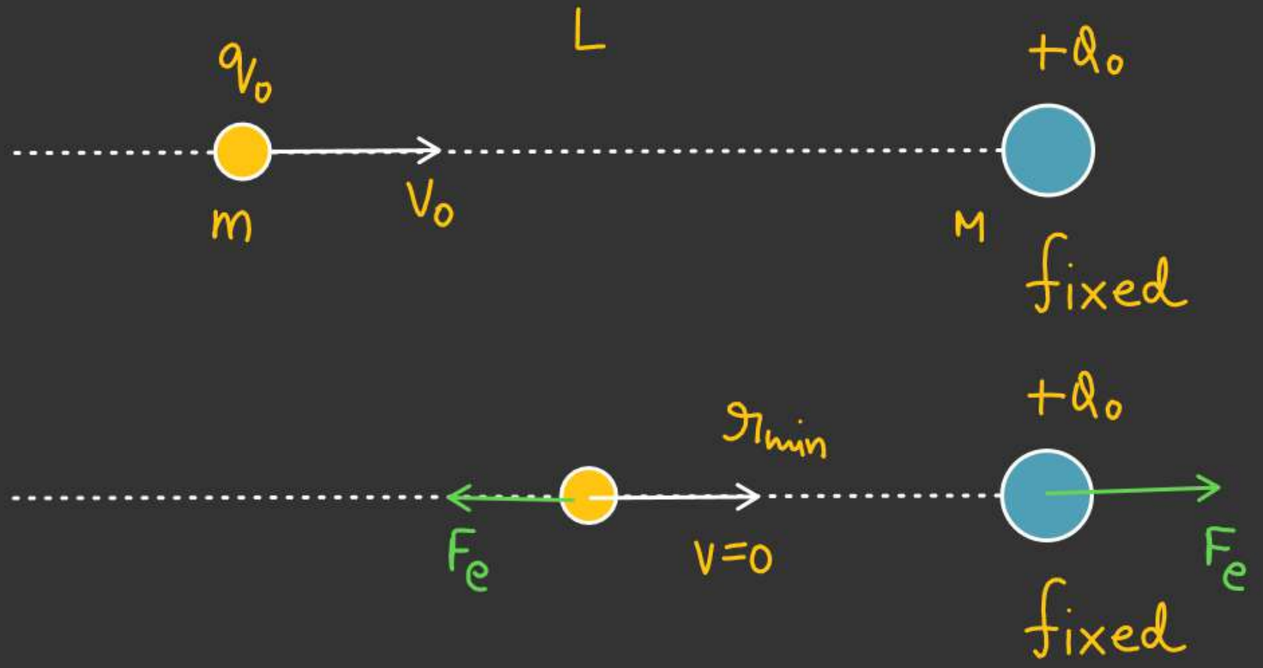
✳ Approach Problems :-

$P_i = P_f$ ✗

$E_i = E_f$ ✓

$L_i = L_f$ ✗

min distance = ?



$$\frac{1}{2}mv_0^2 + \frac{KQq_0}{L} = 0 + \frac{KQ_0q_0}{r_{min}}$$

✳ $r_{min} = L$

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✳ Approach Problems :-

max distance = ?

$$\frac{1}{2}mV_0^2 - \frac{GMm}{L} = 0 - \frac{GMm}{r_{max}}$$

✳

$$r_{max} = L$$

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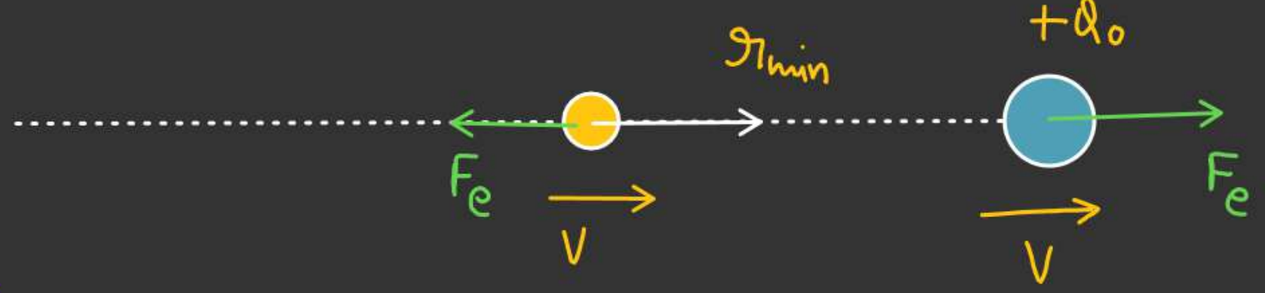
* Approach Problems :-

Case (ii) min distance = ?



$P_i = P_f$ ✓

$E_i = E_f$ ✓

$L_i = L_f$ ✗

$P_i = P_f$

$mV_0 + 0 = mV + MV$

$$V = \left(\frac{m}{m+M}\right) V_0$$

$E_i = E_f$

$$\frac{1}{2}mV_0^2 + \frac{KQq}{L} = \frac{1}{2}mV^2 + \frac{1}{2}MV^2 + \frac{KQq}{r_{min}}$$

$r_{min} = L$

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$$K_i + U_i = K_f + U_f$$

$$K_i - K_f = U_f - U_i$$

$$\frac{1}{2}\left(\frac{mM}{m+M}\right) \cdot v_0^2 = KQq_0\left(\frac{1}{r_{min}} - \frac{1}{L}\right)$$

$r_{min} = L$

★

$$-\Delta K = +\Delta U$$

$$W_{Fe}\big|_{q_0} = \frac{1}{2}mV^2 - \frac{1}{2}mv_0^2$$

$$W_{Fe}\big|_{Q_0} = \frac{1}{2}MV^2$$

$$W_{Fe}\big|_{q_0} + W_{Fe}\big|_{Q} = \underbrace{\frac{1}{2}mV^2 + \frac{1}{2}MV^2} - \frac{1}{2}mv_0^2$$

$$= K_f - K_i = U_i - U_f$$

$$W_{Fe}\big|_{q_0} + W_{Fe}\big|_{Q} = \int \vec{F_e} \cdot \vec{dr} = -\Delta U$$

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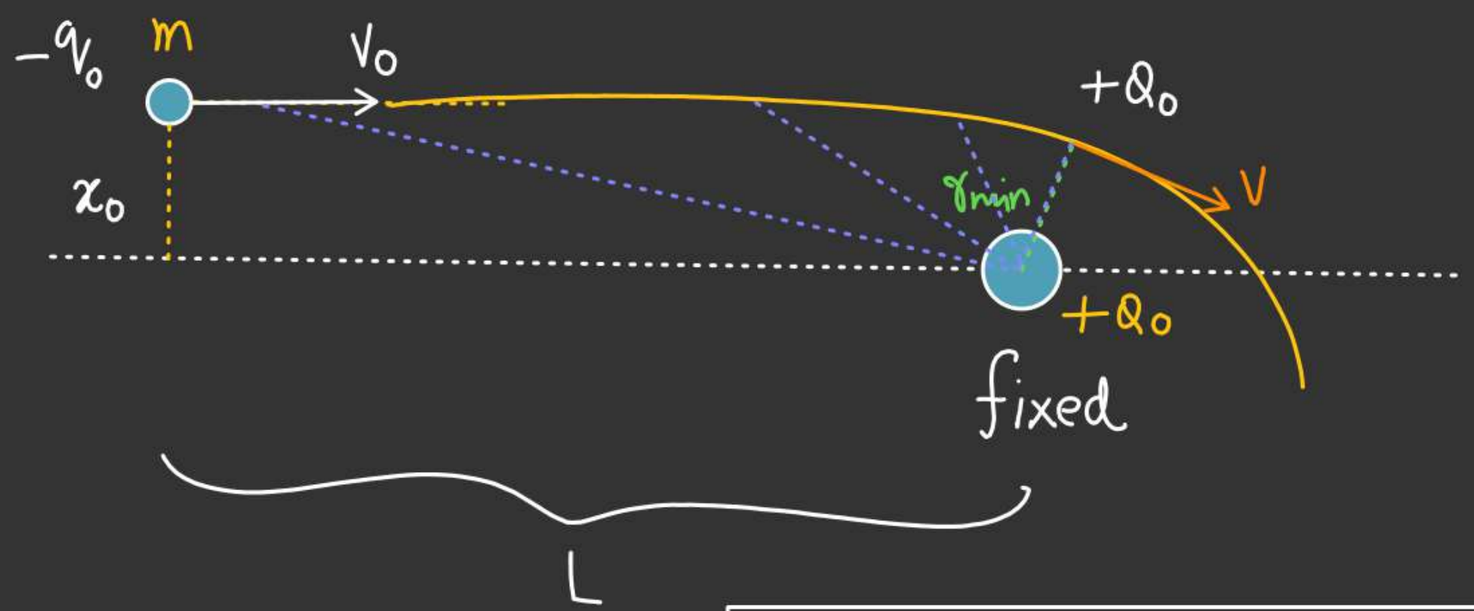
*



$$L_i = L_f$$

$$mv_0x_0 = mv\,r_{min}$$

$$E_i = E_f$$

$$+\frac{1}{2}mv_0^2 - \frac{kQq_0}{\sqrt{L^2+x_0^2}} = \frac{1}{2}mv^2 - \frac{kQq_0}{r_{min}}$$

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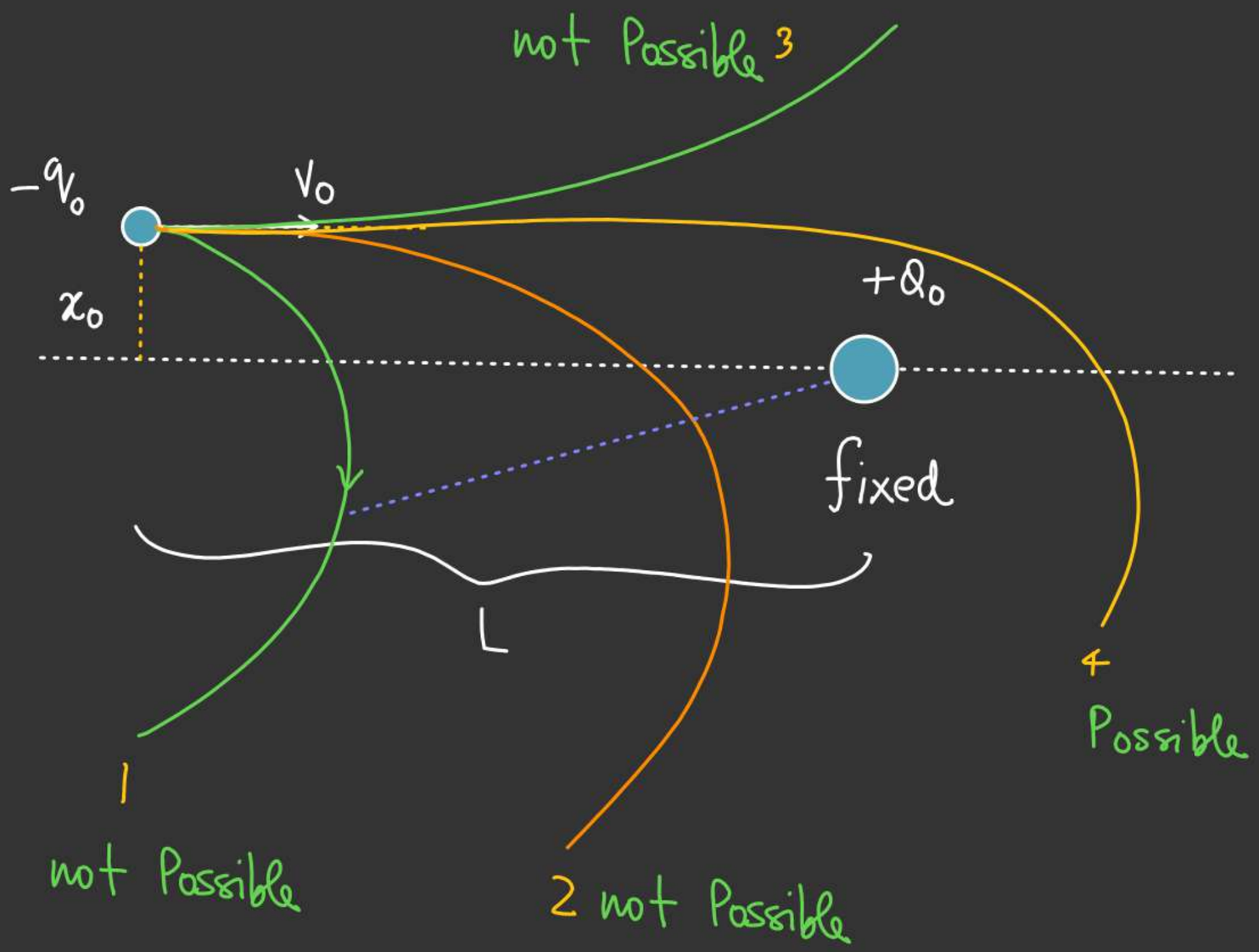

Nishant Jindal

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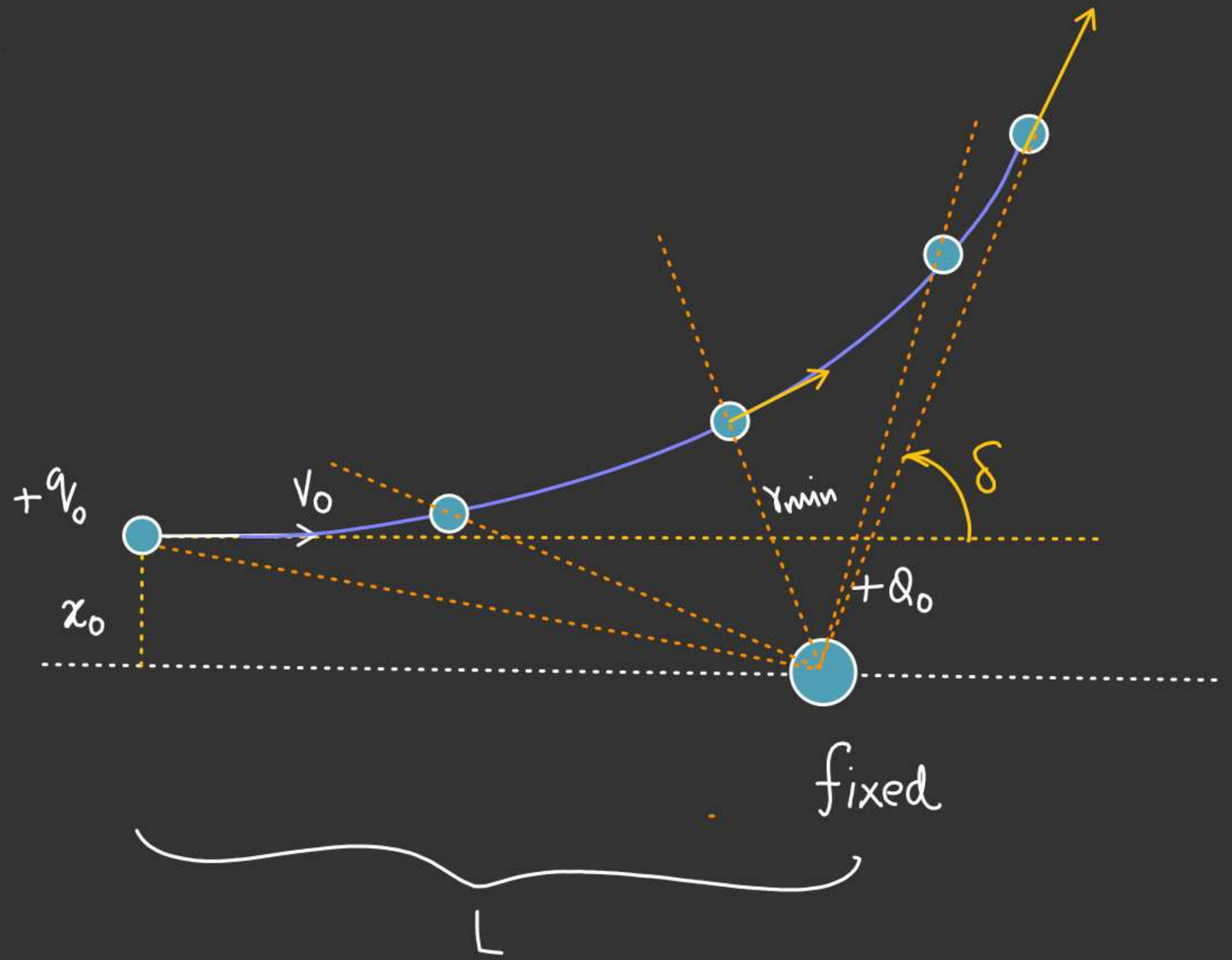

[derivation of δ
↳ Check INSP YT]

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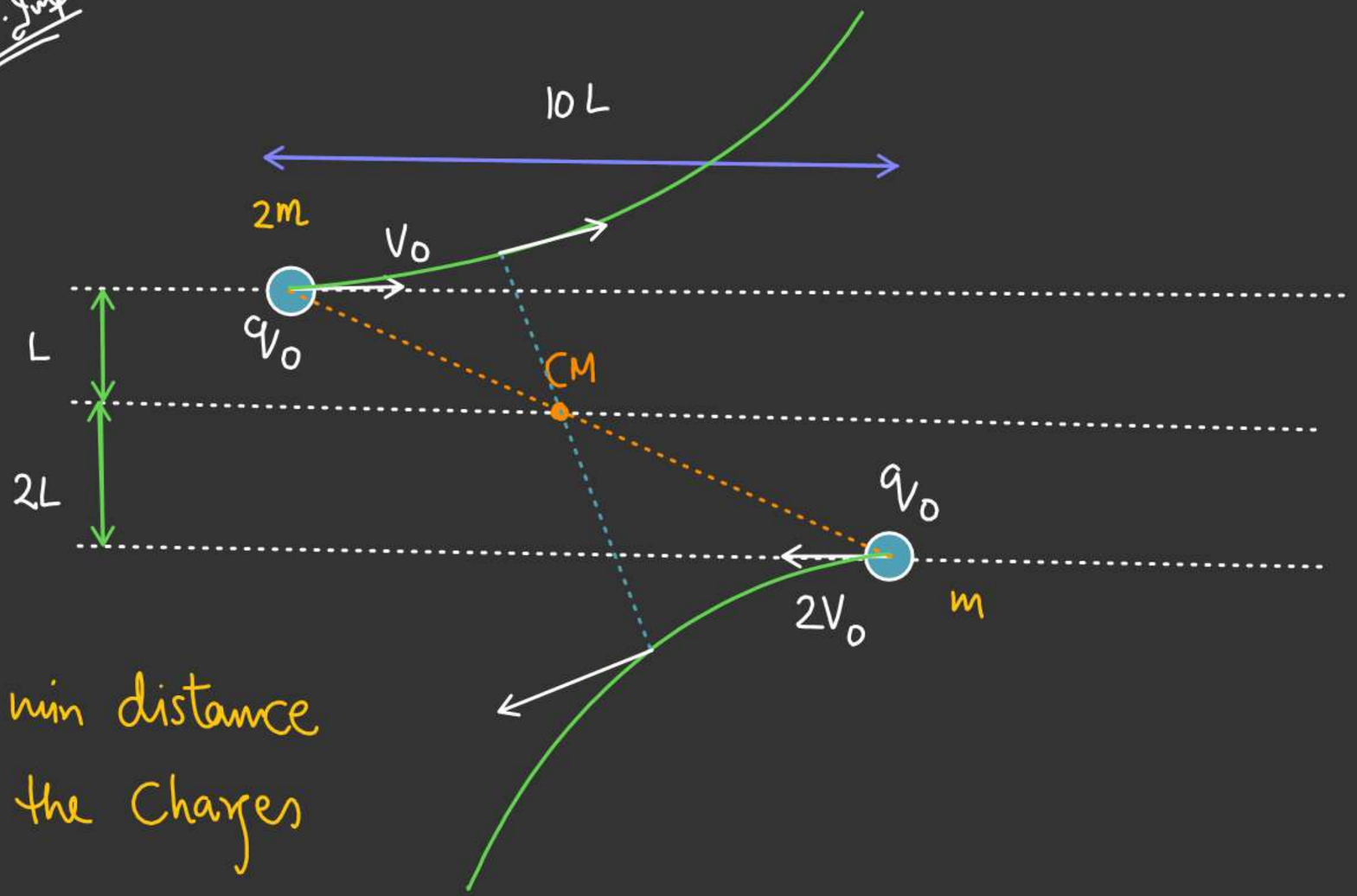
* V. Imp
H.C.



* Find min distance b/w the Charges

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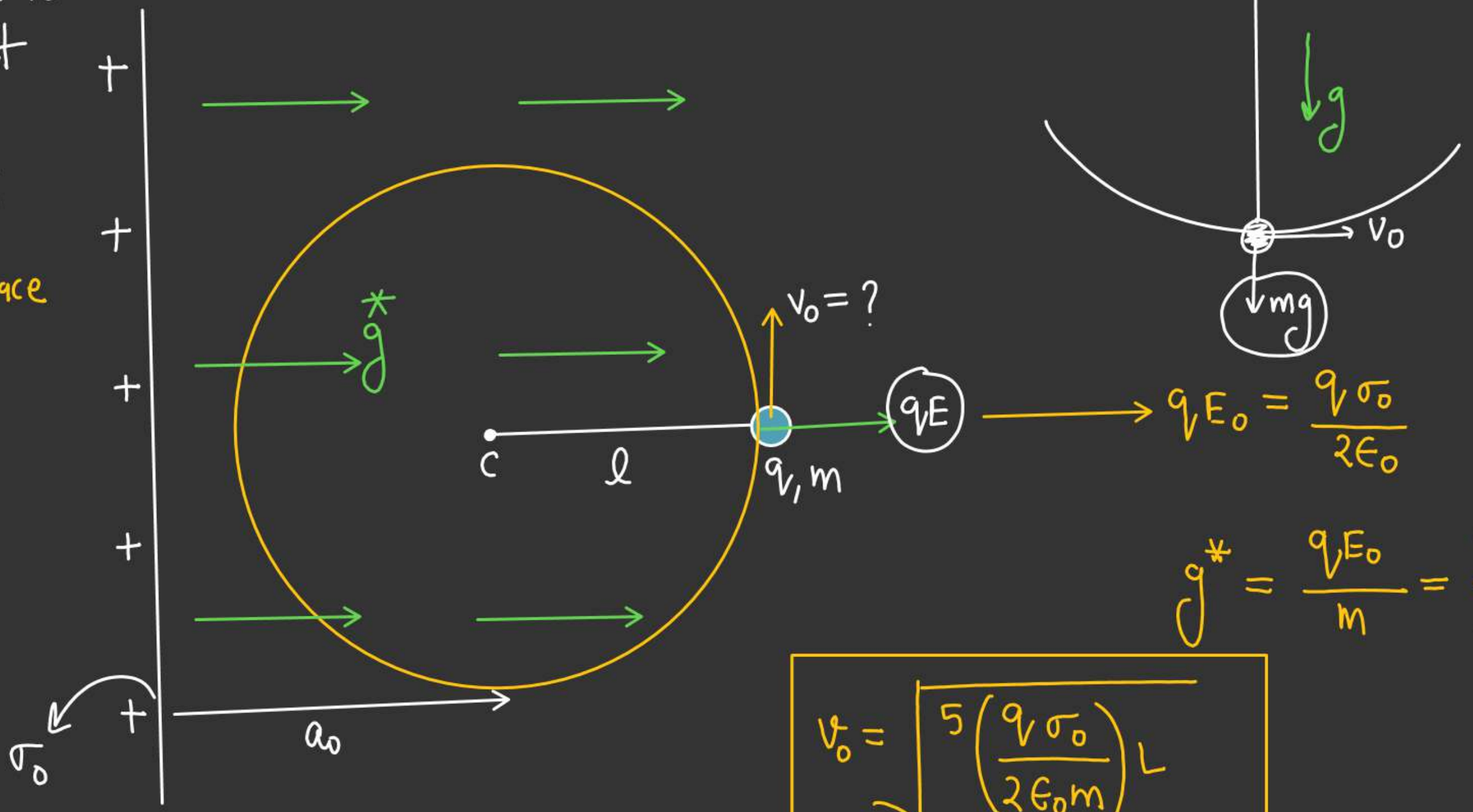
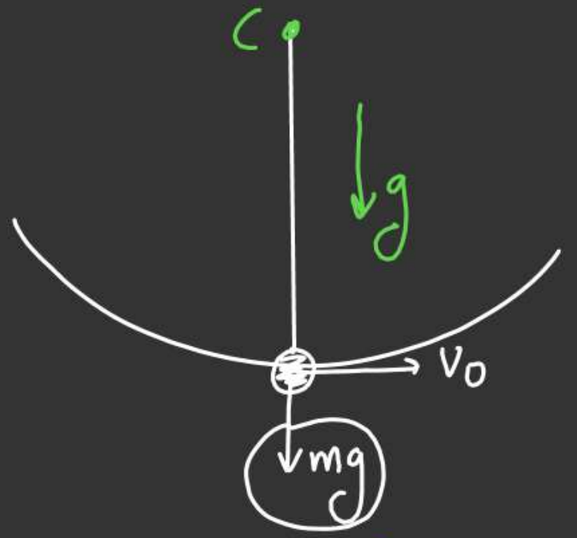
* Infinite sheet

gravity free space





$$qE_0 = \frac{q\sigma_0}{2\epsilon_0}$$

$$g^* = \frac{qE_0}{m} = \frac{q\sigma_0}{2\epsilon_0 m}$$

$$v_0 = \sqrt{5\left(\frac{q\sigma_0}{2\epsilon_0 m}\right)L}$$

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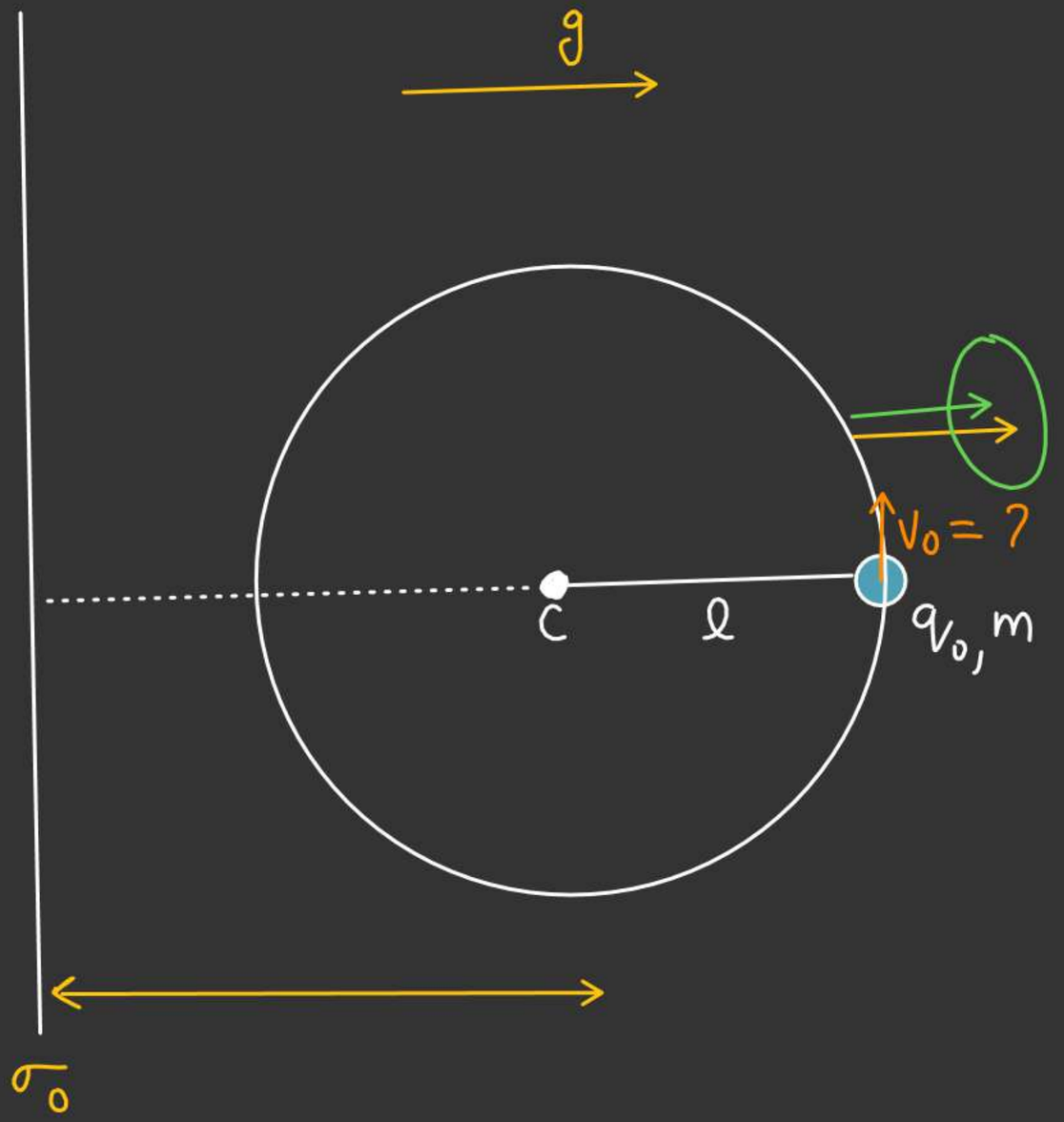

$$(qE + mg) = mg^*$$

$$V_0 = \sqrt{5 g^* L}$$

$$= \sqrt{5\left[g + \frac{q\sigma_0}{2\epsilon_0 m}\right] L}$$

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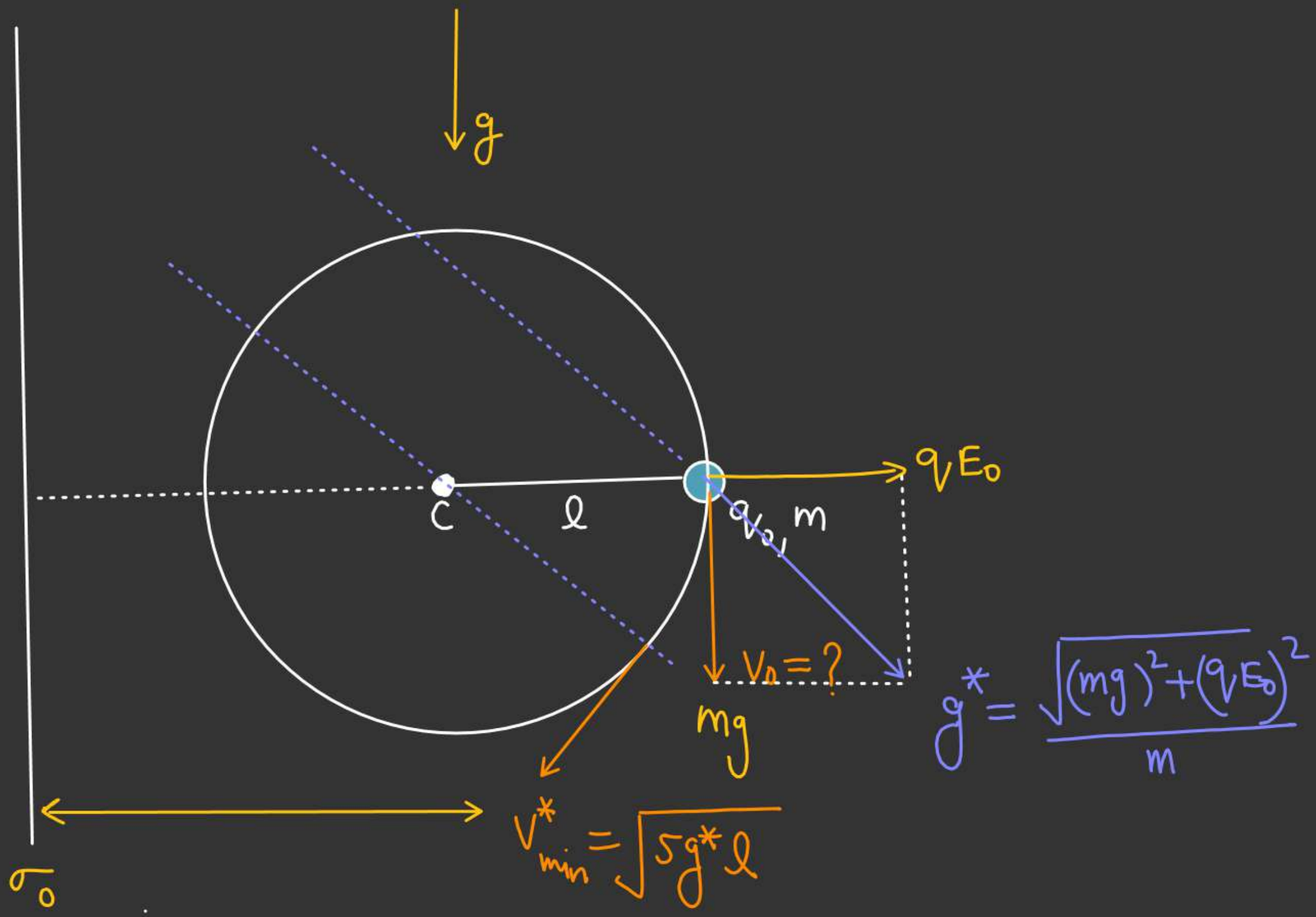

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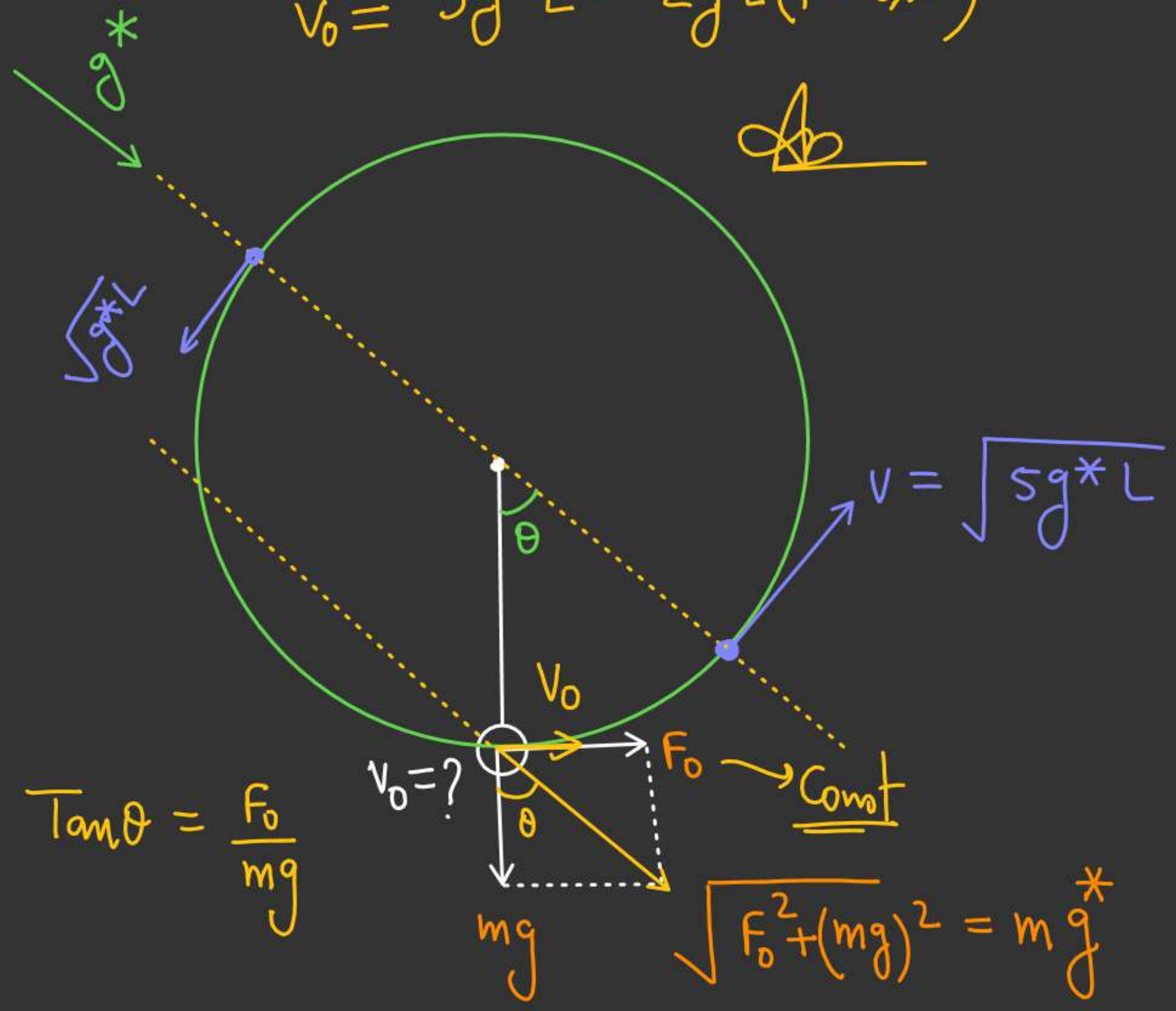
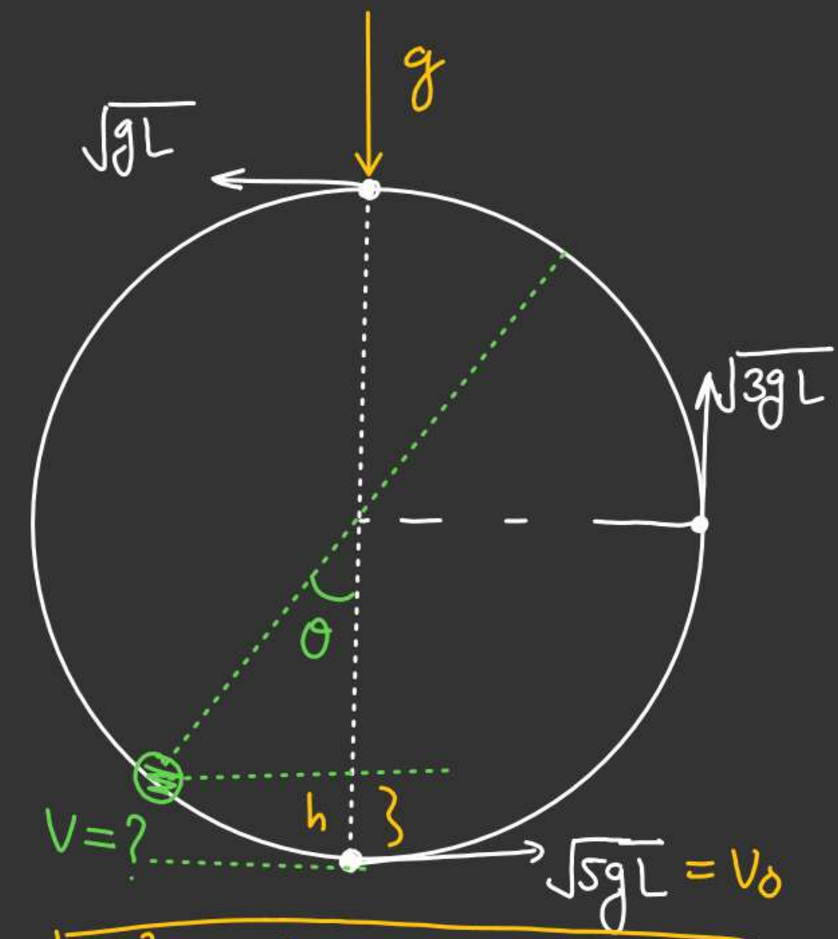
$$V_0^2 = 5g^*L - 2g^*L(1-\cos\theta)$$



$$\tan\theta = \frac{F_0}{mg}$$



$$V^2 = V_0^2 - 2gL(1-\cos\theta)$$

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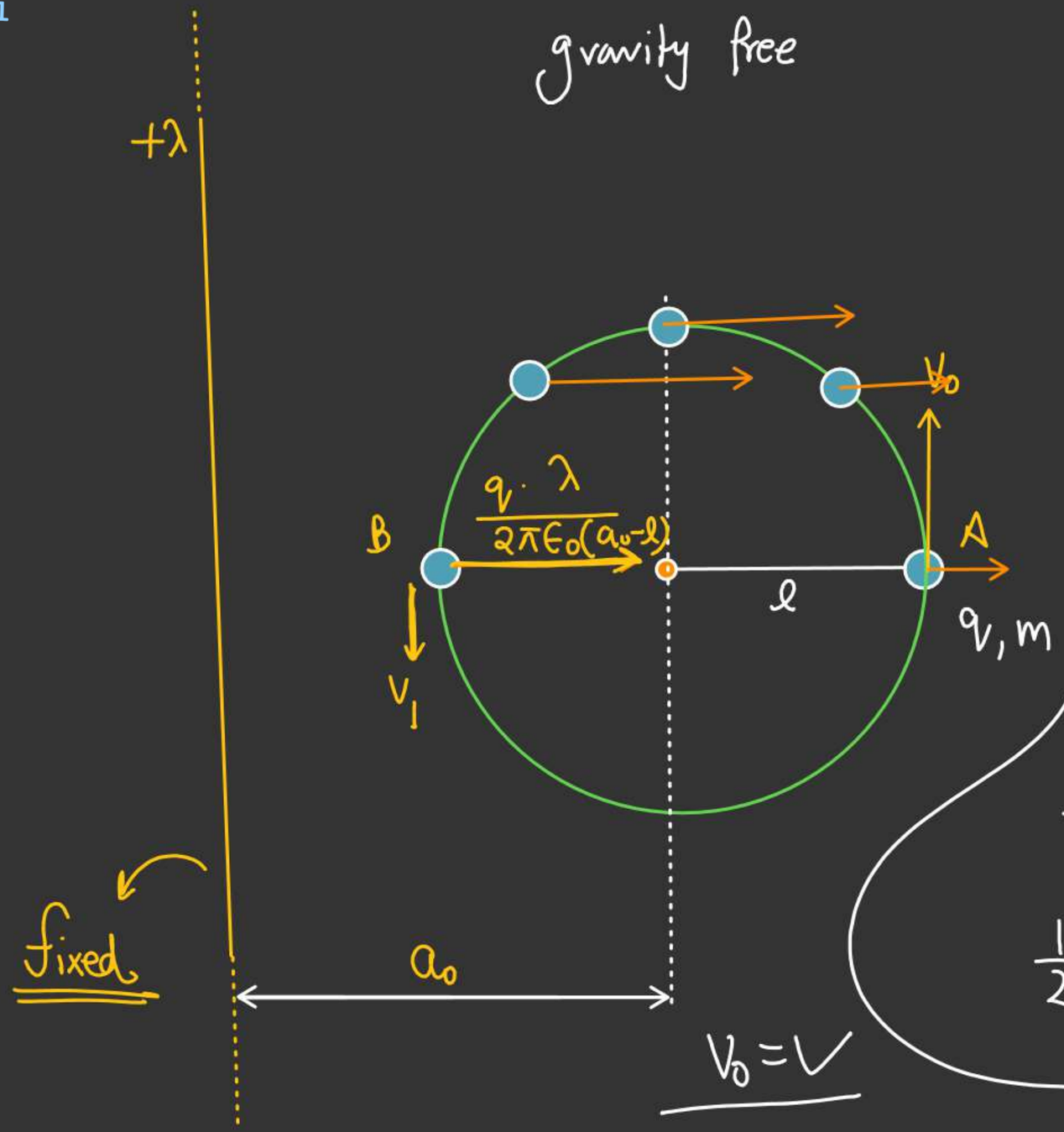
gravity free



$V_0 = V$

$$\frac{q\lambda}{2\pi\epsilon_0(a_0-\ell)} = \frac{mV_1^2}{\ell}$$

$V_1 = V$

$$K_A + U_A = K_B + U_B$$

$$\frac{1}{2}mV_0^2 = \frac{1}{2}mV_1^2 + (U_B - U_A)$$

$$\frac{1}{2}mV_0^2 = \frac{1}{2}mV_1^2 + q_0(V_B - V_A)$$

$$\frac{1}{2}mV_0^2 = \frac{1}{2}mV_1^2 + q_0\cdot\frac{\lambda}{2\pi\epsilon_0}\left[\ln\left(\frac{a_0+L}{a_0-L}\right)\right]$$

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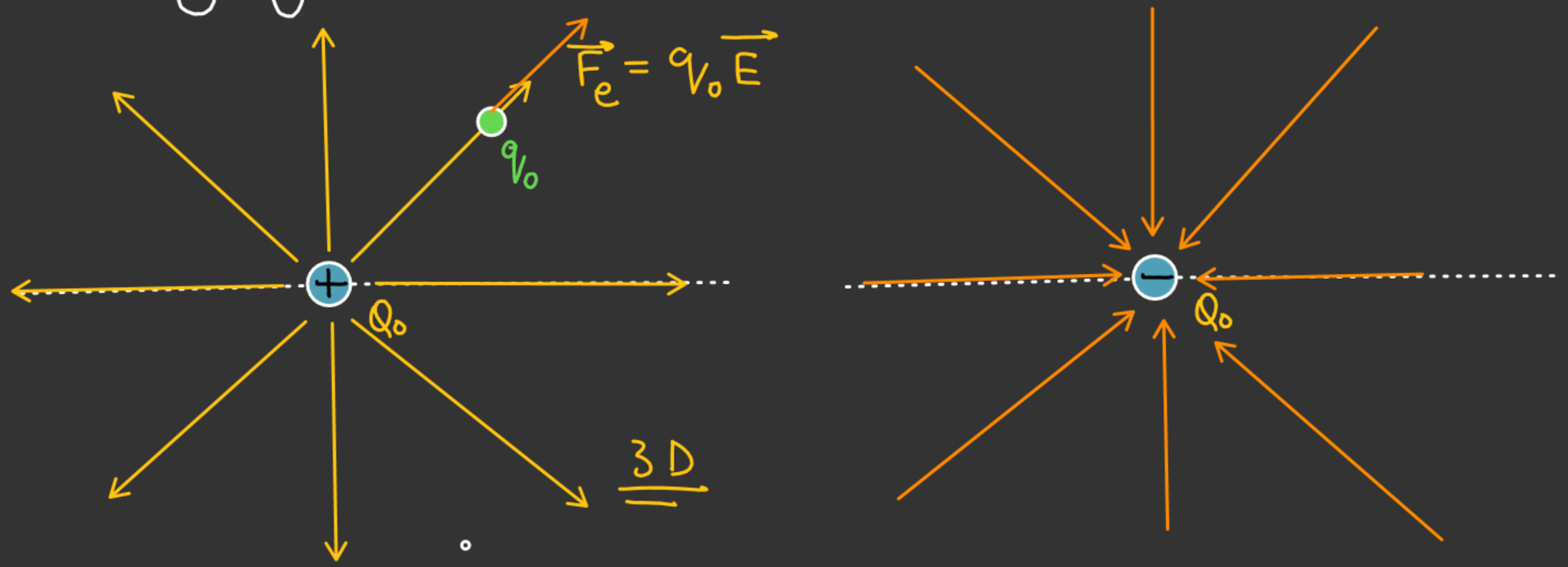
* Electric field lines :-

* Imaginary field lines (represents the direction of Force on a test charge)

* they generate at +ve charge and terminate at -ve charges



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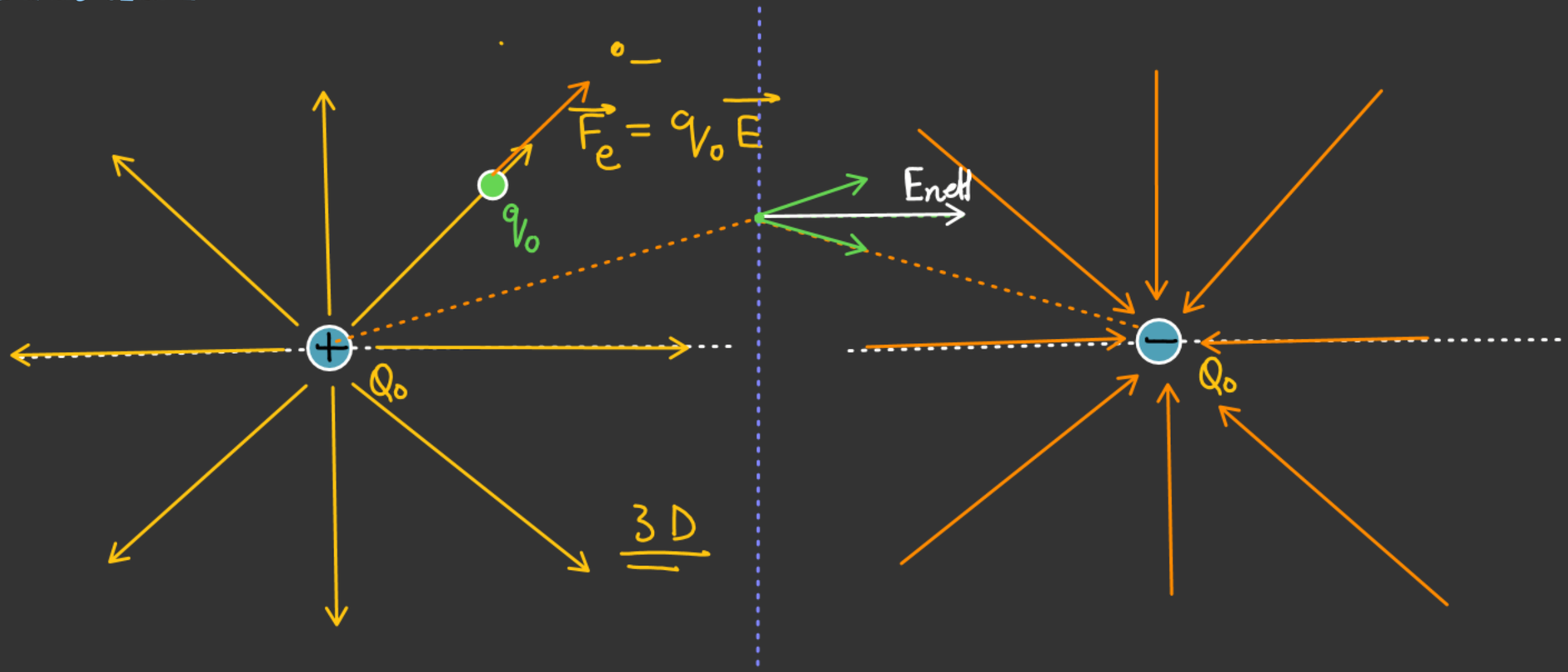

* At any point there will be a unique direction of field lines.

⇒ two field lines can never Intersect each other.

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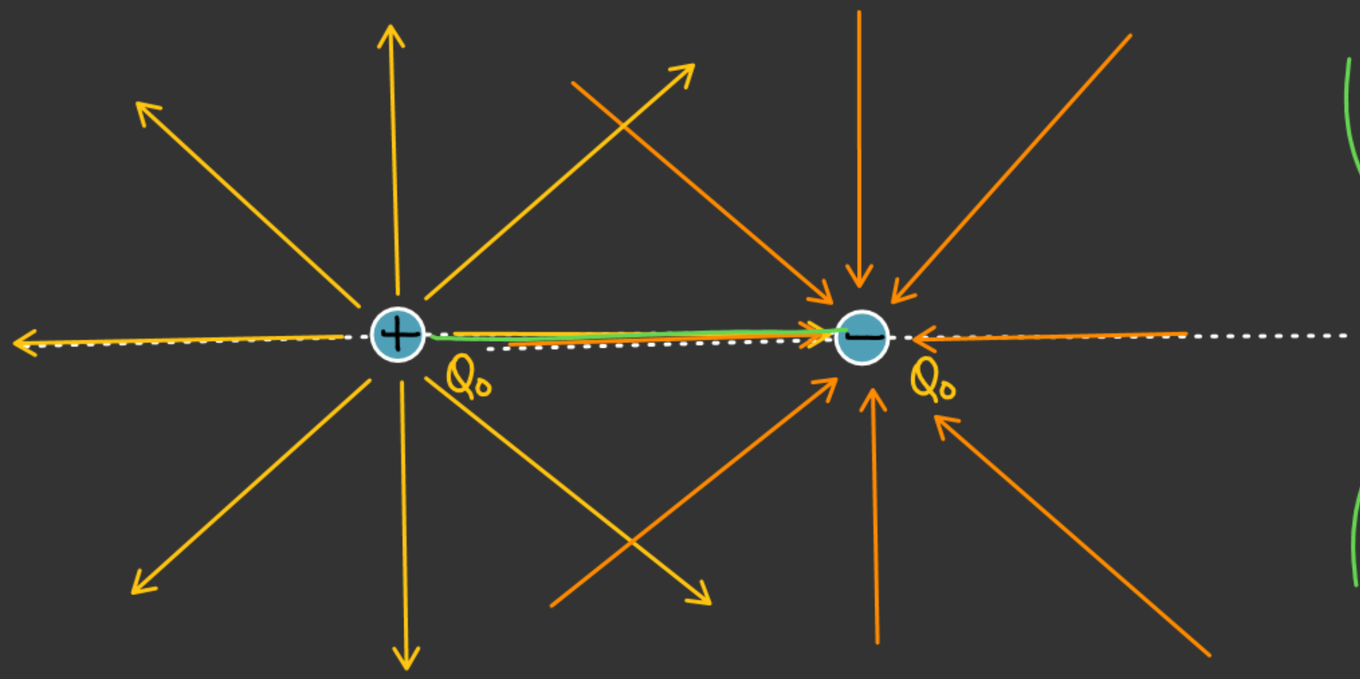

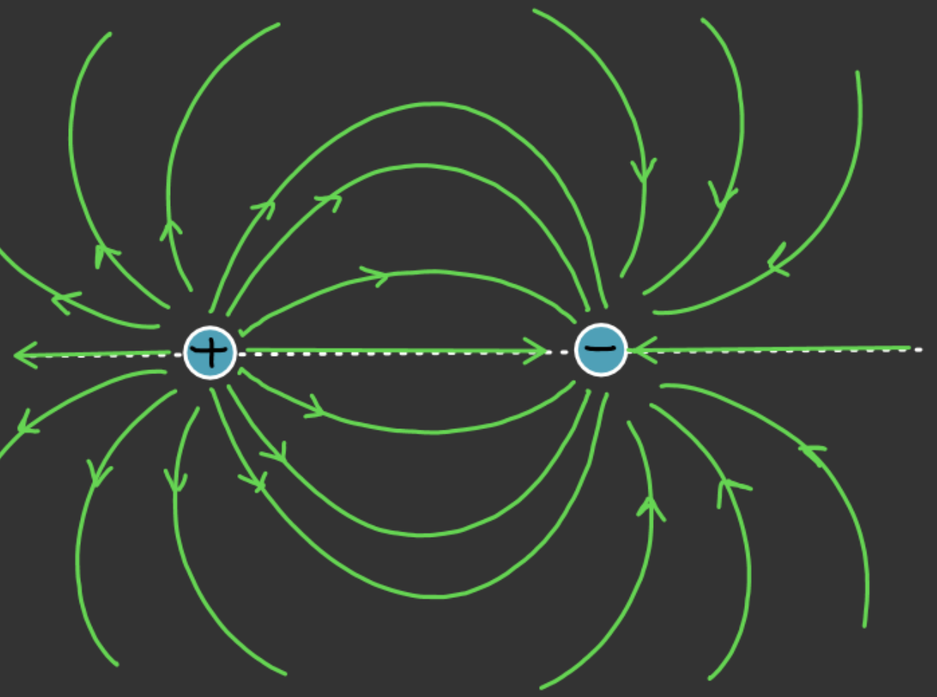

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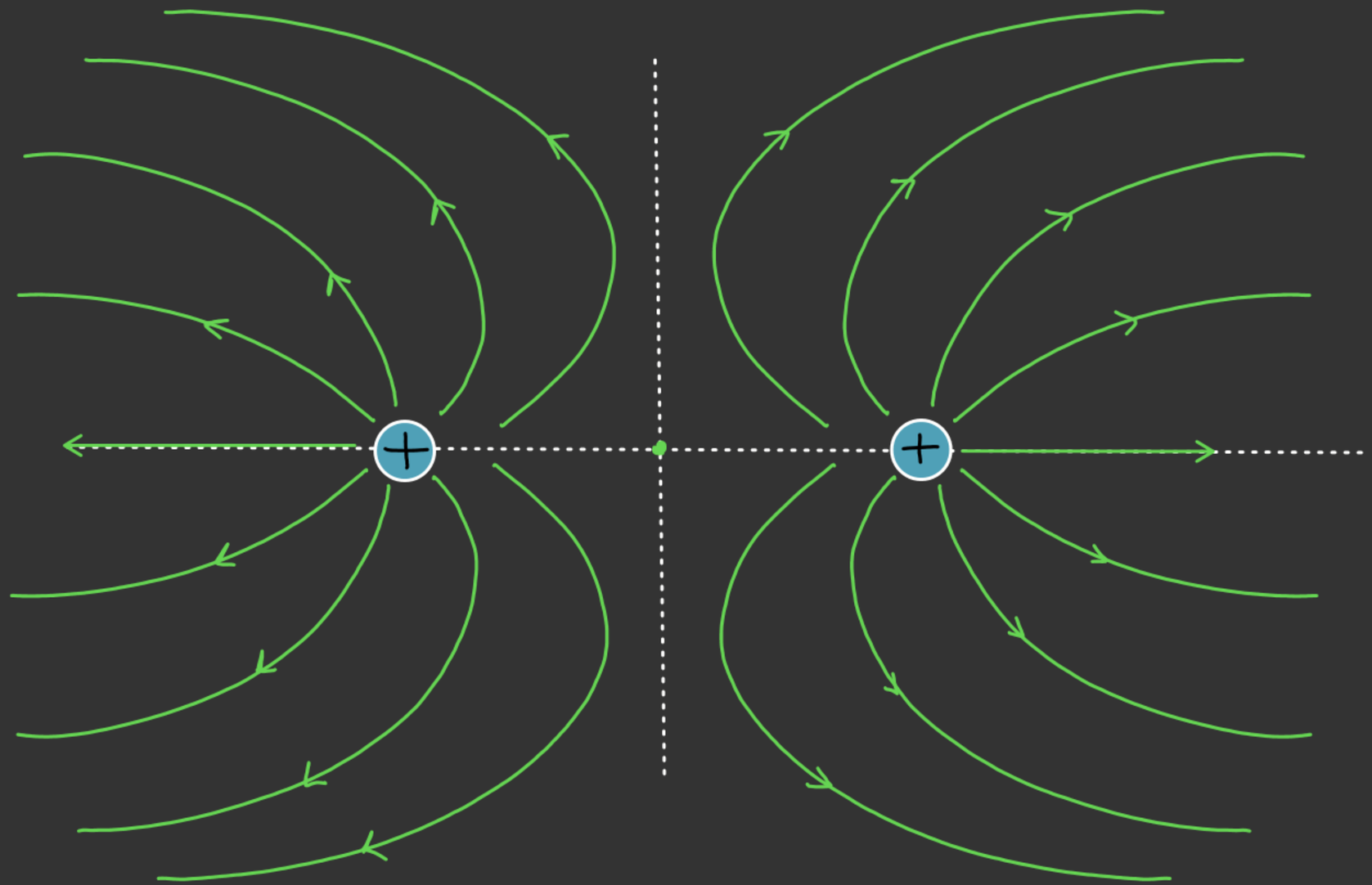

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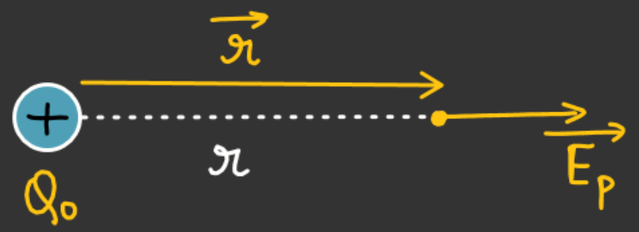
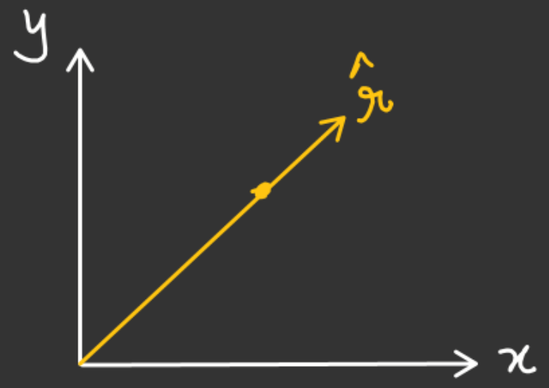
3D



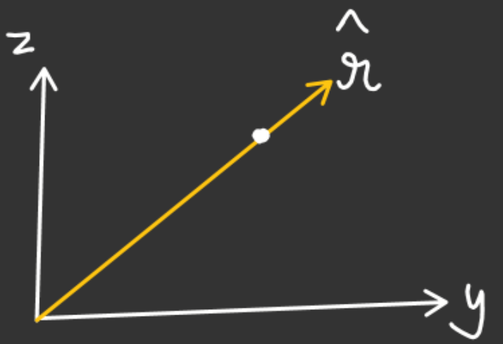
$$\vec{E_p} = \frac{kQ_0}{r^2}\cdot(\hat{r})$$

$$\hat{r} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{\sqrt{x^2+y^2+z^2}}$$

2D:



$$\hat{r} = \frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}$$



$$\hat{r} = \frac{y\hat{j}+z\hat{k}}{\sqrt{y^2+z^2}}$$

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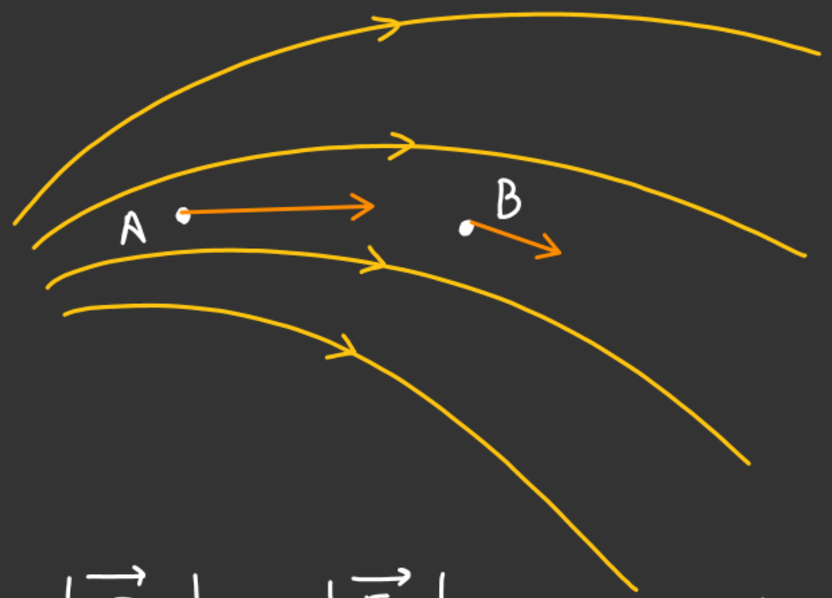

no. of field lines $\propto$ E

* $|\vec{E}_A| > |\vec{E}_B|$

* $V_B < V_A$

$$V_f - V_i = -\int \vec{E} \cdot \vec{dr}$$

$$V_B - V_A = -\underbrace{\int \vec{E} \cdot \vec{dr}}_{>0}$$

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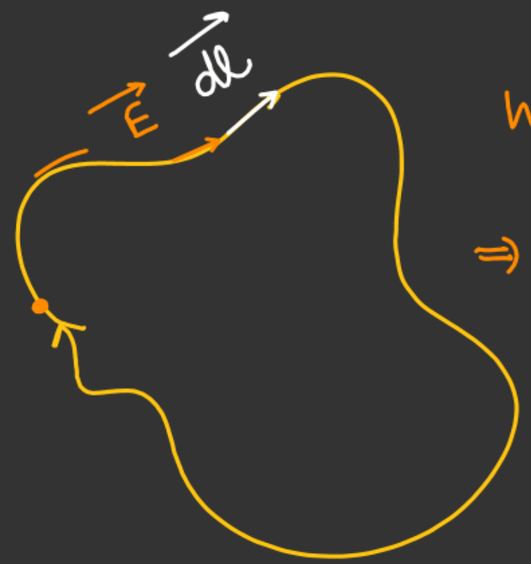
* Electrostatic field lines can never form close loop:



$W_{cyc} > 0$

$\Rightarrow$ Cannot be electrostatic field.

for Conservative force fields

$W_{cyc} = 0$

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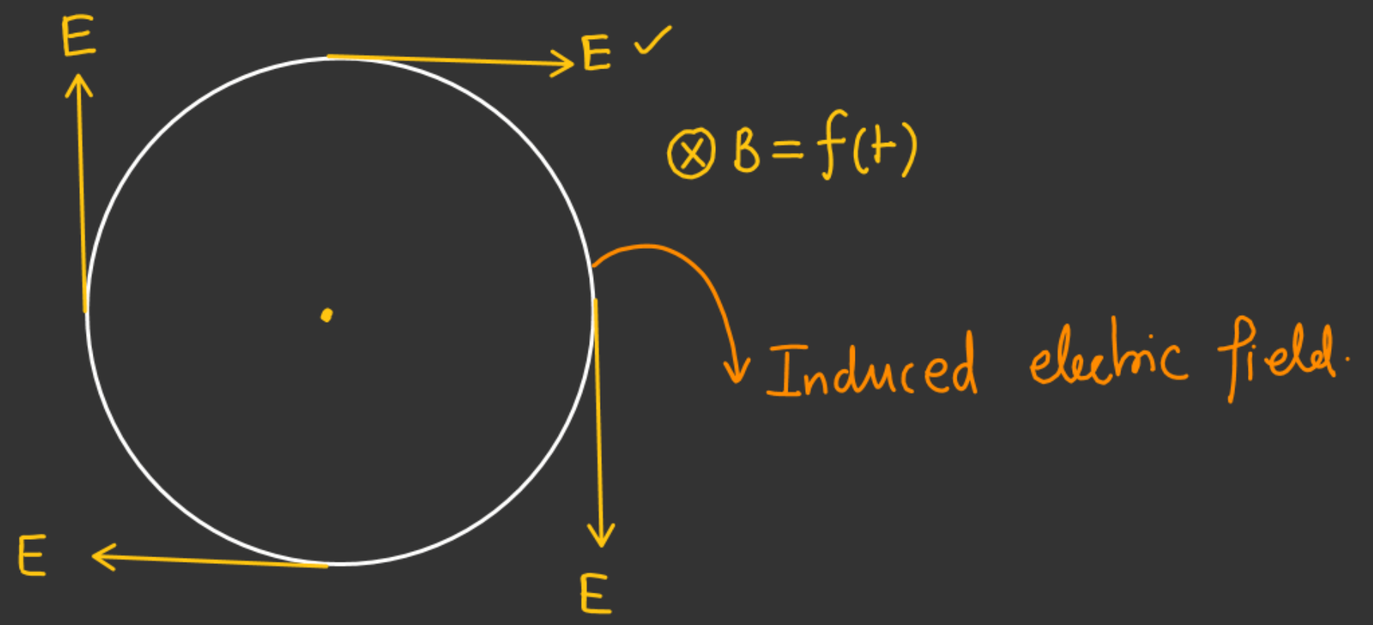

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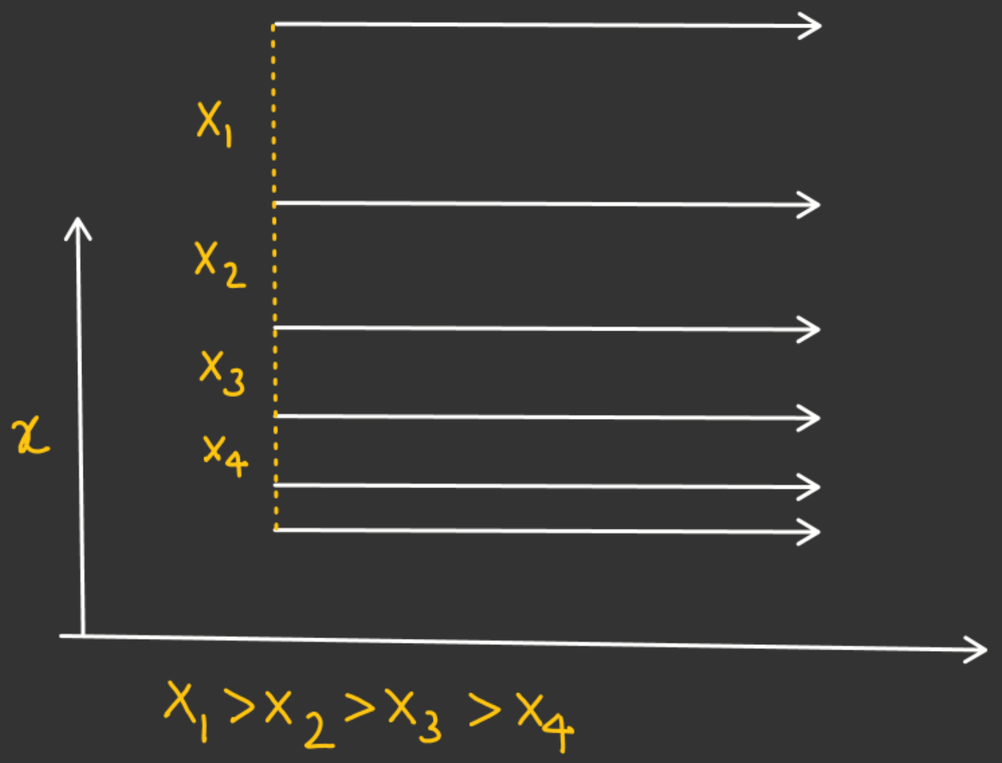
If this field represents electrostatic field then find the variation of E with x.



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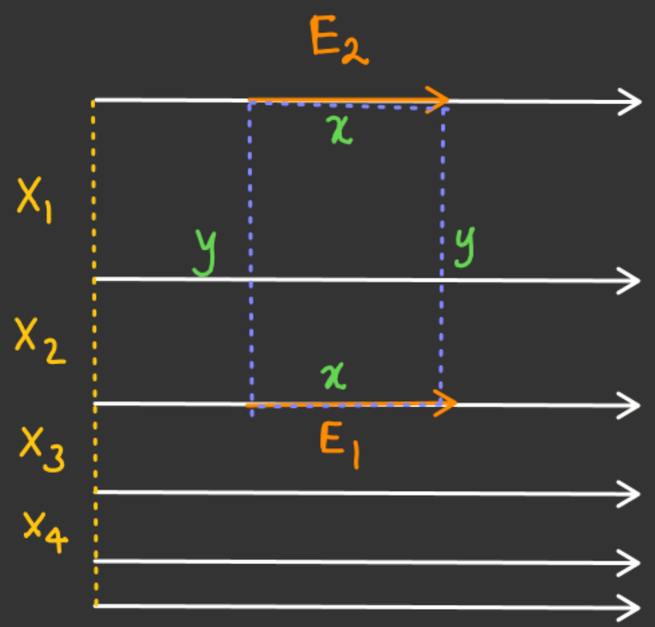

$$W_{cyc} = +E_2 x - E_1 x$$

$$\Rightarrow (E_2 - E_1)\, x \neq 0 \quad \Rightarrow \text{Can Never be an electrostatic field}$$

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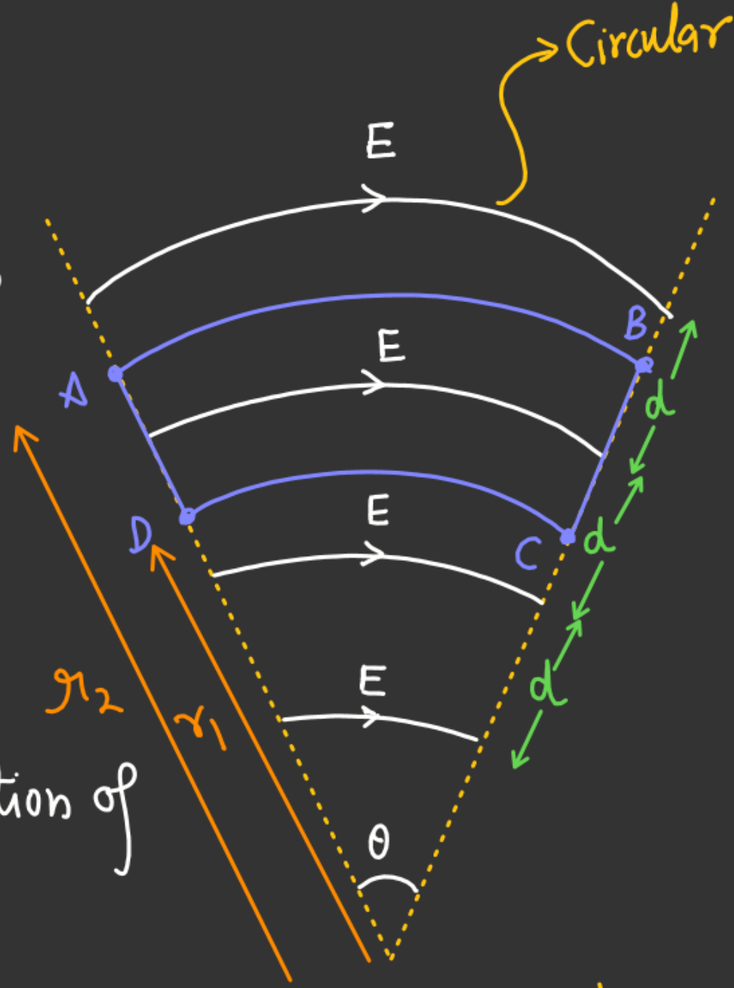
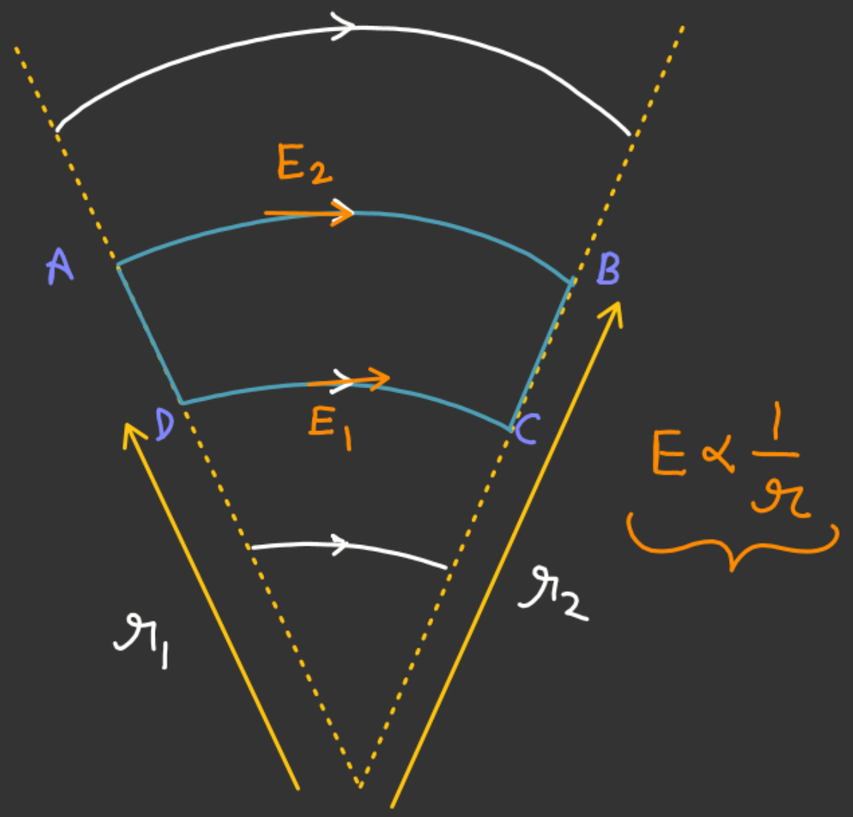
electrostatic



find variation of $\vec{E}$

$$W_{ABCDA} = W_{AB} + \cancel{W_{BC}}^{0} + W_{CD} + \cancel{W_{DA}}$$

$$= E(R_2\theta) - E(R_1\theta) \neq 0 \Rightarrow E \text{ will be Variable}$$



$$W_{ABCDA} = 0 = E_2(r_2\theta) + 0 - E_1(r_1\theta) + 0$$

$$E_2 r_2 = E_1 r_1 \Rightarrow \underline{E \cdot r = \text{Const}}$$

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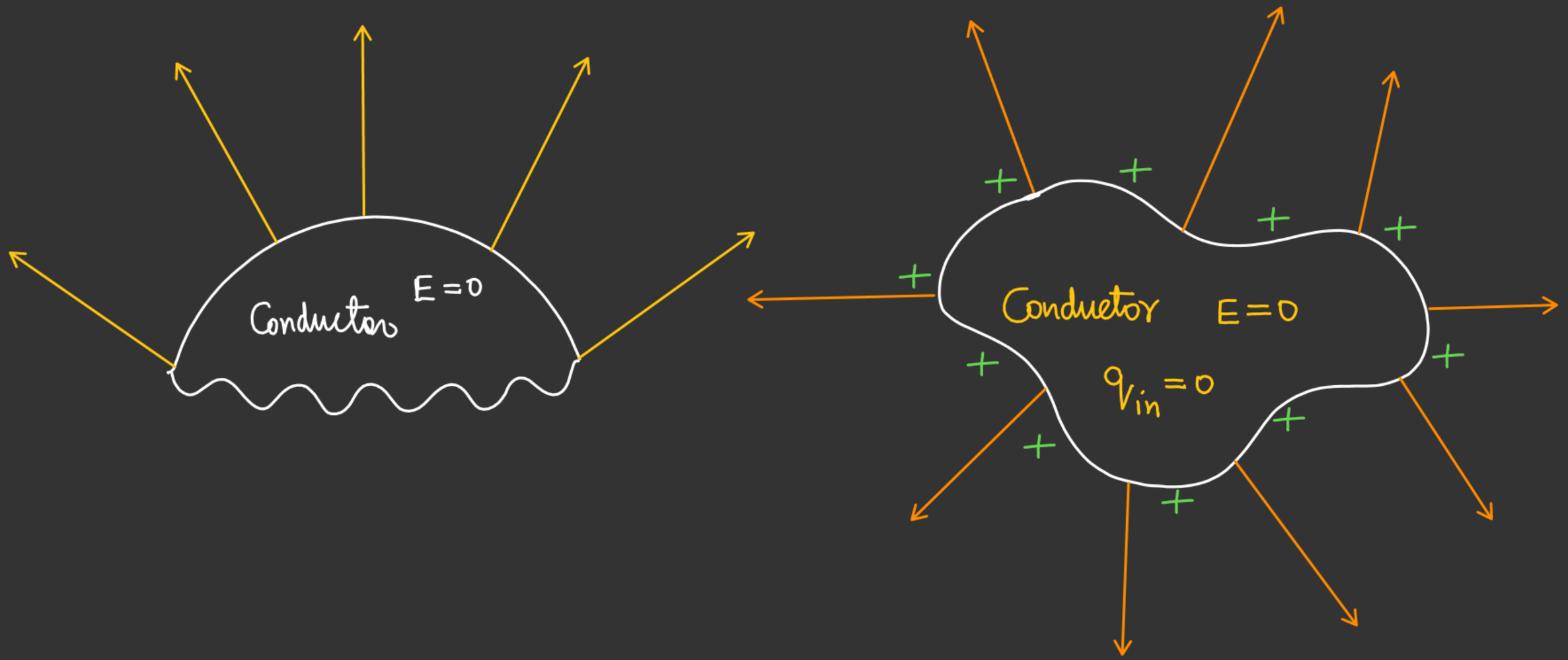
* Electric field lines will be $\perp^r$ to the Conductor Surface :



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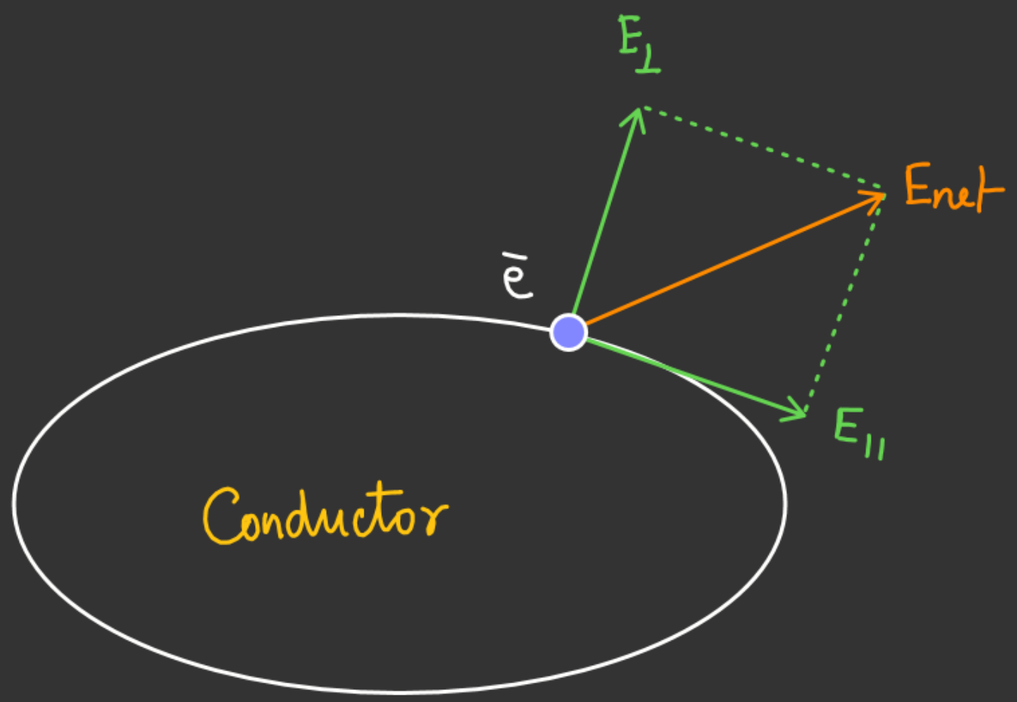

$\vec{F}_{e_\parallel} = -e\vec{E}_\parallel \longrightarrow$ electrons will start moving

$\vec{F}_{e_\perp} \rightsquigarrow$

$\Rightarrow$ In electrostatic Conditions $E_\parallel = 0 \Rightarrow \underbrace{\vec{E}_{net} = \vec{E}_\perp}$

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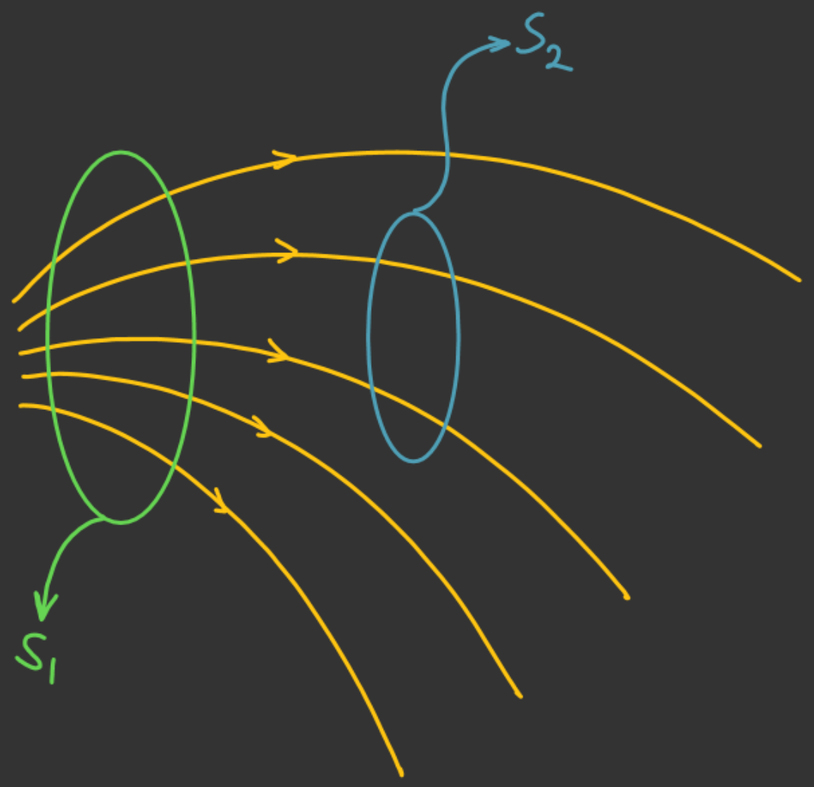
* Electric flux :-



$$\text{Electric flux } \phi = \int \vec{E} \cdot d\vec{A}$$

$$\phi \propto \text{No. of field lines}$$

* $\phi_{S_1} > \phi_{S_2}$

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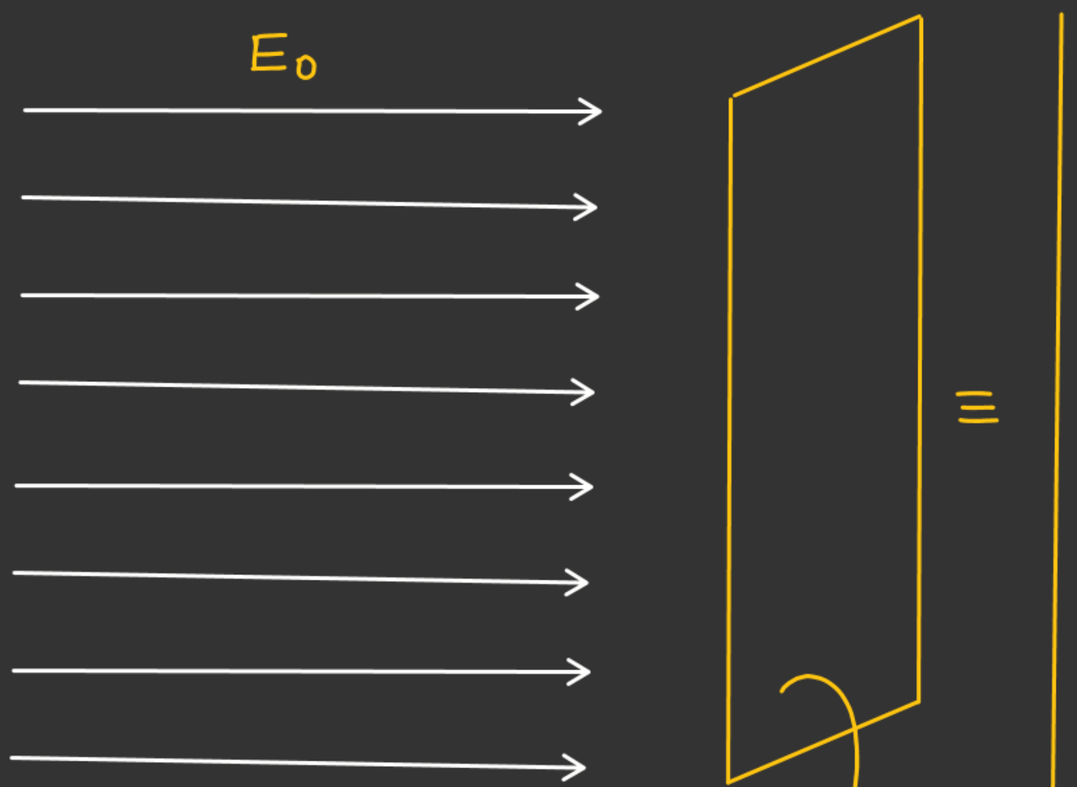

≡

ϕ depends on (i) E
(ii) A
(iii) Orientation

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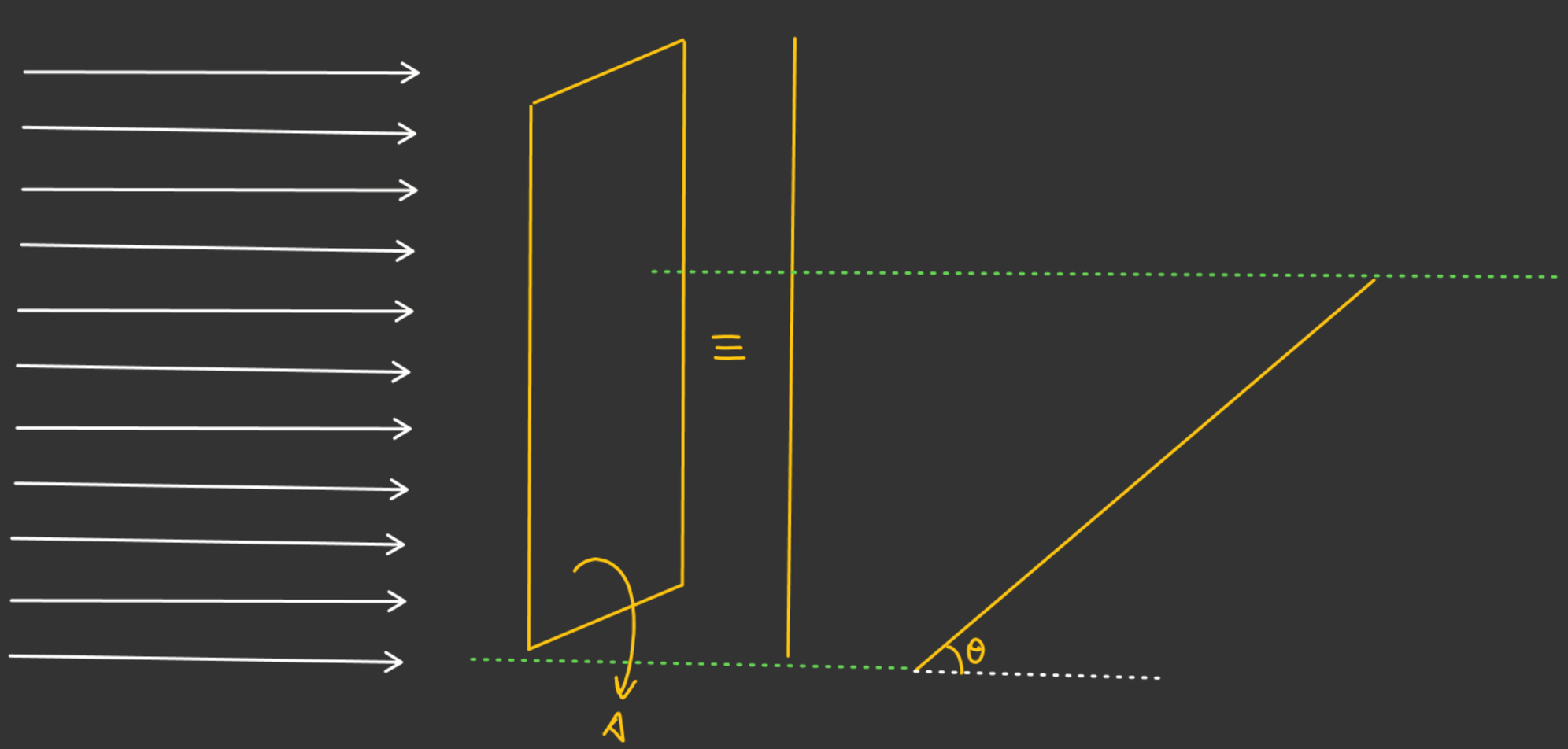

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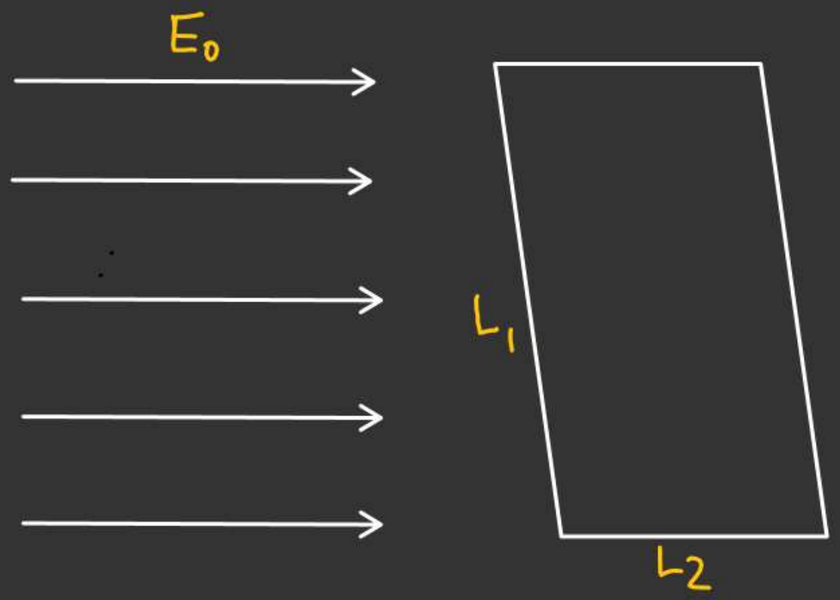

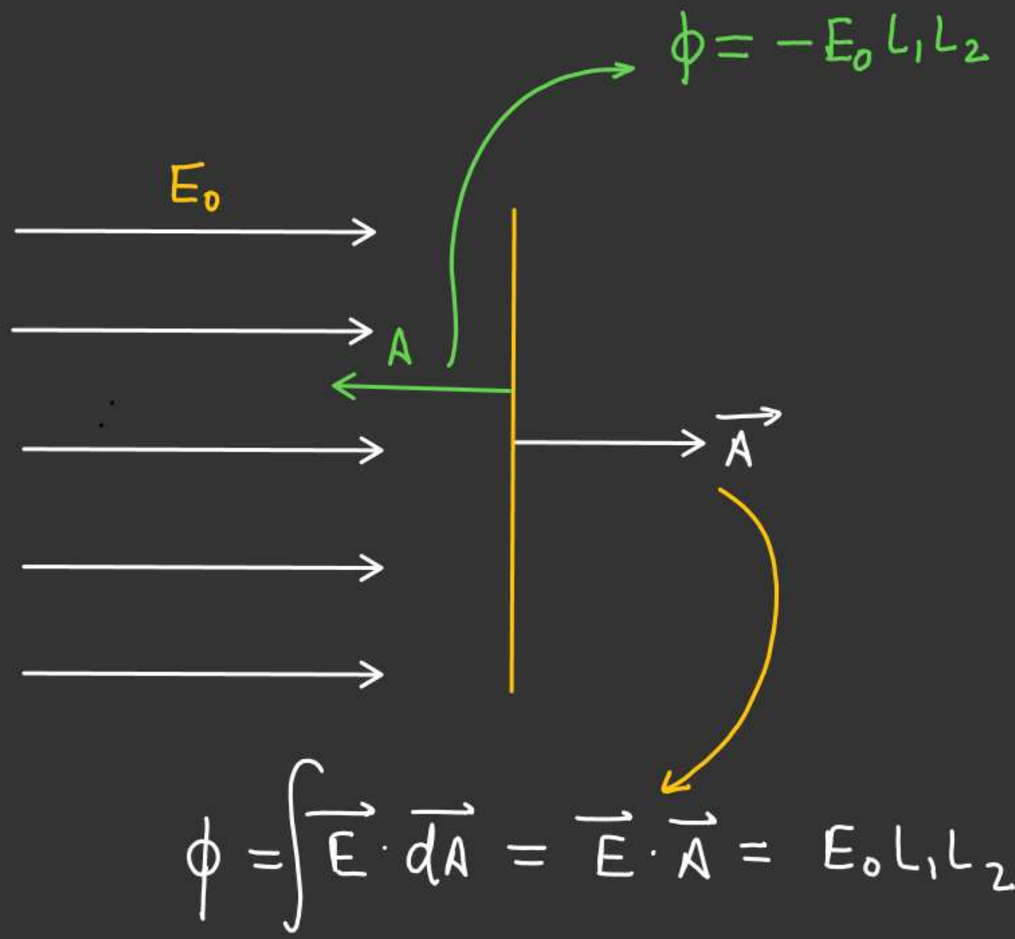

$$\phi = \int \vec{E} \cdot d\vec{A} = \vec{E} \cdot \vec{A} = E_0 L_1 L_2$$

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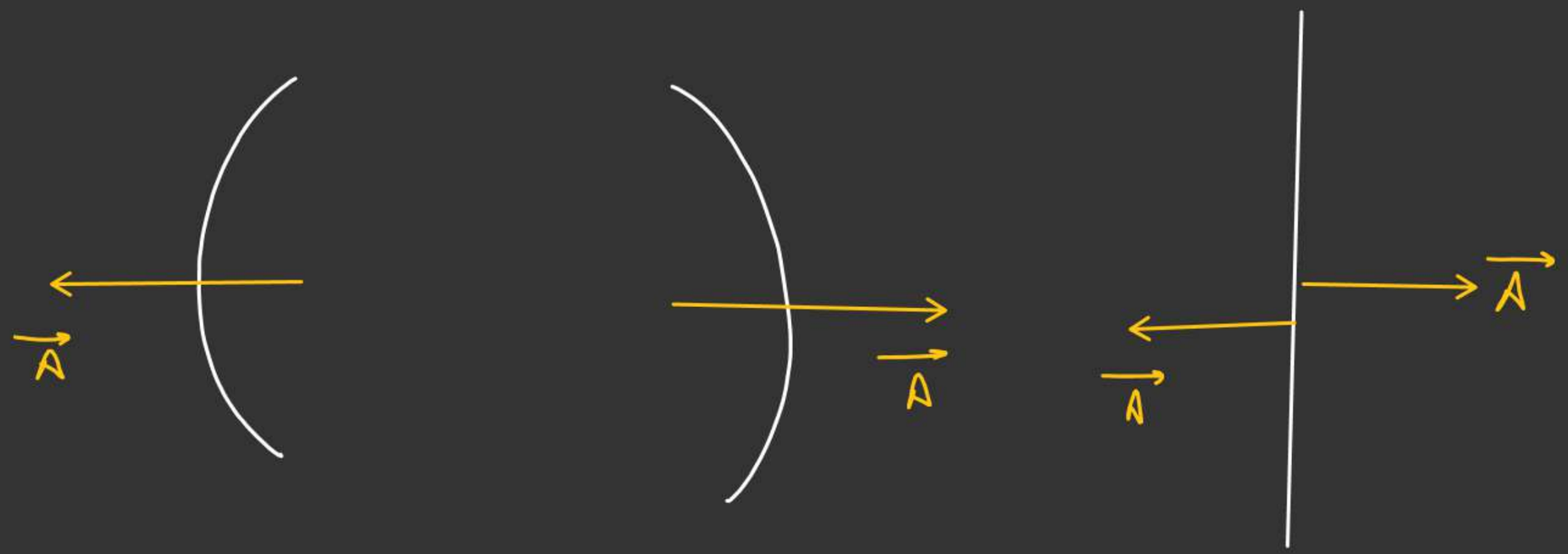
* Sign - Convention :-



open Surfaces

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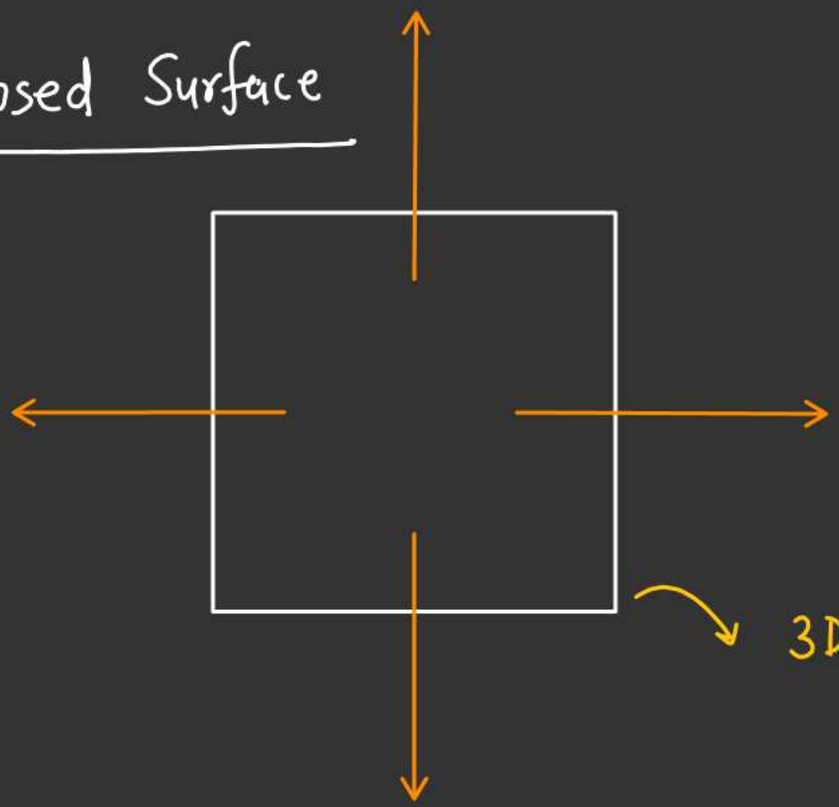
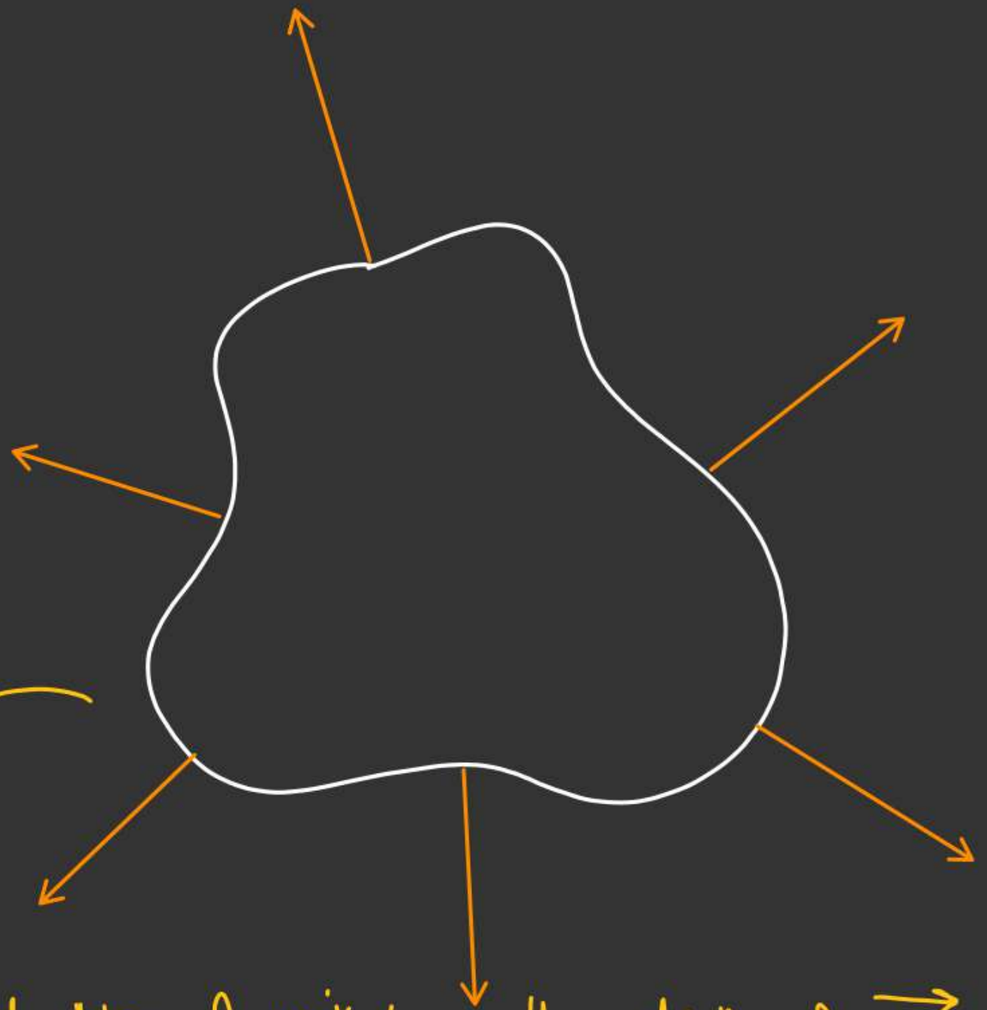
Closed Surface



3D



* for closed Surfaces Always outward Normal will be the dirⁿ. of $\vec{A}$

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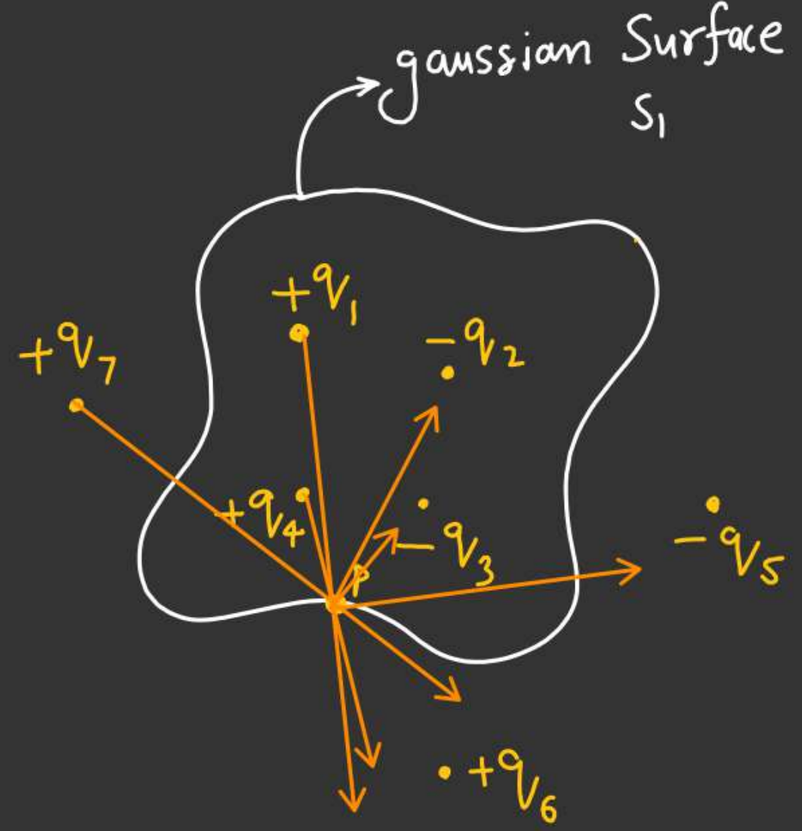
* Gauss's theorem :-



$$\phi_{G.S.} = \oint_{S_1} \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = \frac{q_1 - q_2 - q_3 + q_4}{\epsilon_0}$$

* $\vec{E}$ is due to all the charges present
* q_{in} → strictly inside the gaussian Surface.

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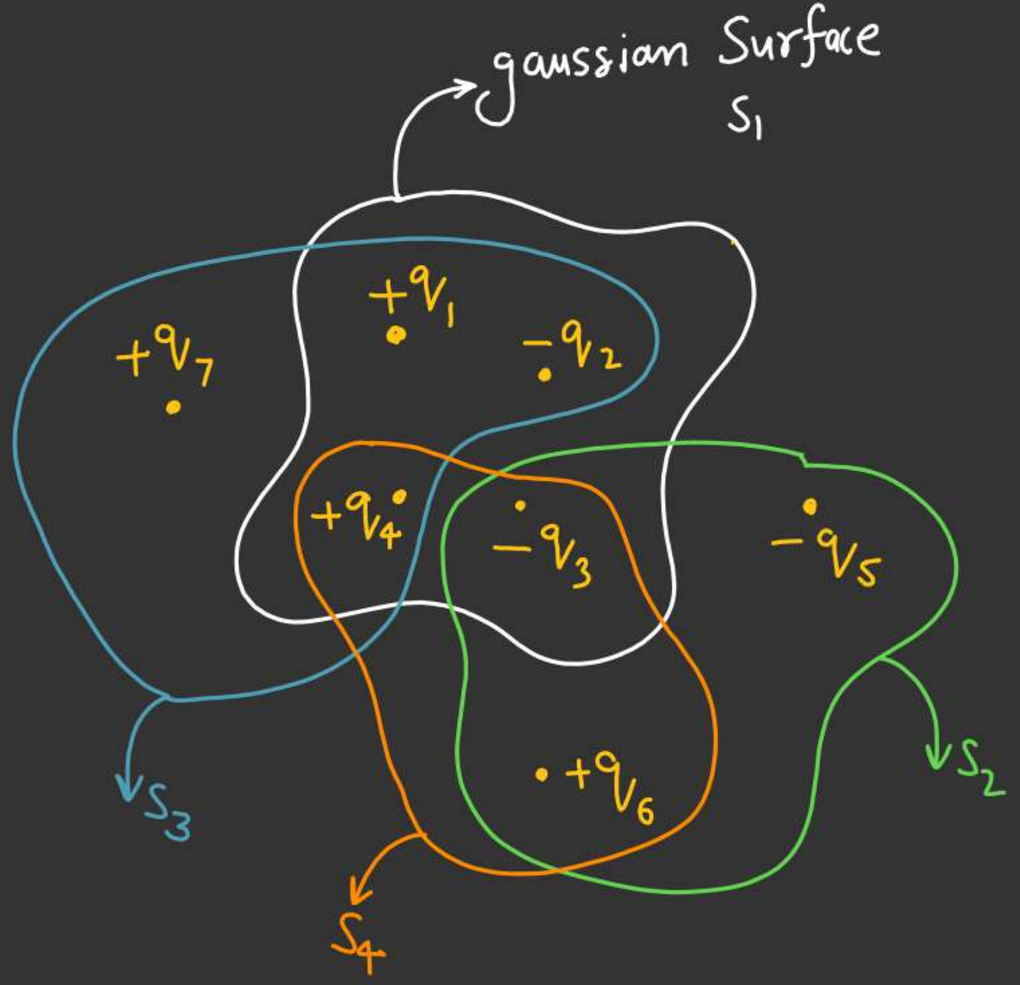
* Gauss's theorem :-



$$\phi_{S_2} = \frac{q_6 - q_3 - q_5}{\epsilon_0}$$

$$\phi_{S_3} = \frac{q_7 + q_1 - q_2 + q_4}{\epsilon_0}$$

$$\phi_{S_4} = \frac{q_4 - q_3 + q_6}{\epsilon_0}$$

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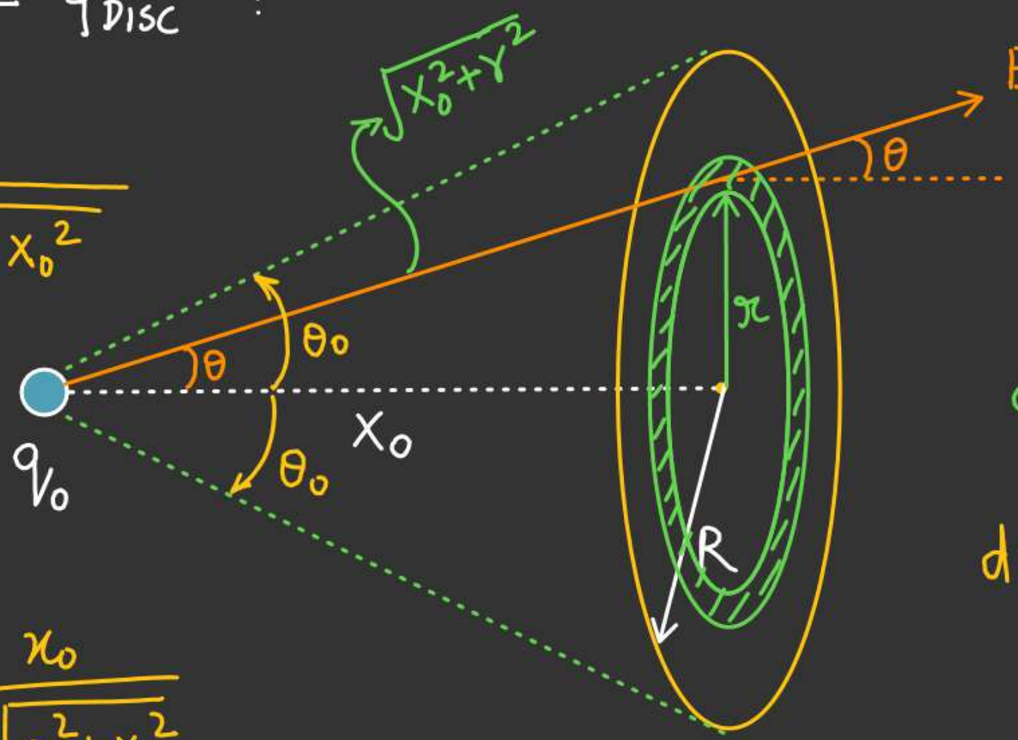
* $\phi_{circle} = \phi_{Disc} = ?$



$$E = \frac{Kq_0}{\left(\sqrt{x_0^2+r^2}\right)^2} = \frac{Kq_0}{x_0^2+r^2}$$

$$\cos\theta_0 = \frac{x_0}{\sqrt{R^2+x_0^2}}$$

$$\cos\theta = \frac{x_0}{\sqrt{r^2+x_0^2}}$$

$$d\phi = E \cdot dA\cos\theta$$

$$d\phi \Rightarrow \frac{Kq_0}{x_0^2+r^2} \cdot 2\pi r\,dr \cdot \frac{x_0}{\sqrt{x_0^2+r^2}}$$

$$\Rightarrow \frac{Kq_0}{x_0^2(1+\tan^2\theta)} \; 2\pi \cdot x_0\tan\theta \cdot \cos\theta \cdot x_0\sec^2\theta \cdot d\theta$$

$$x_0\tan\theta = r$$

$$x_0\sec^2\theta \cdot d\theta = dr$$

$$\phi_{Disc} \Rightarrow 2\pi K q_0 \int_0^{\theta_0} \sin\theta \cdot d\theta = \frac{q}{2\epsilon_0}\left(1-\cos\theta\right)$$

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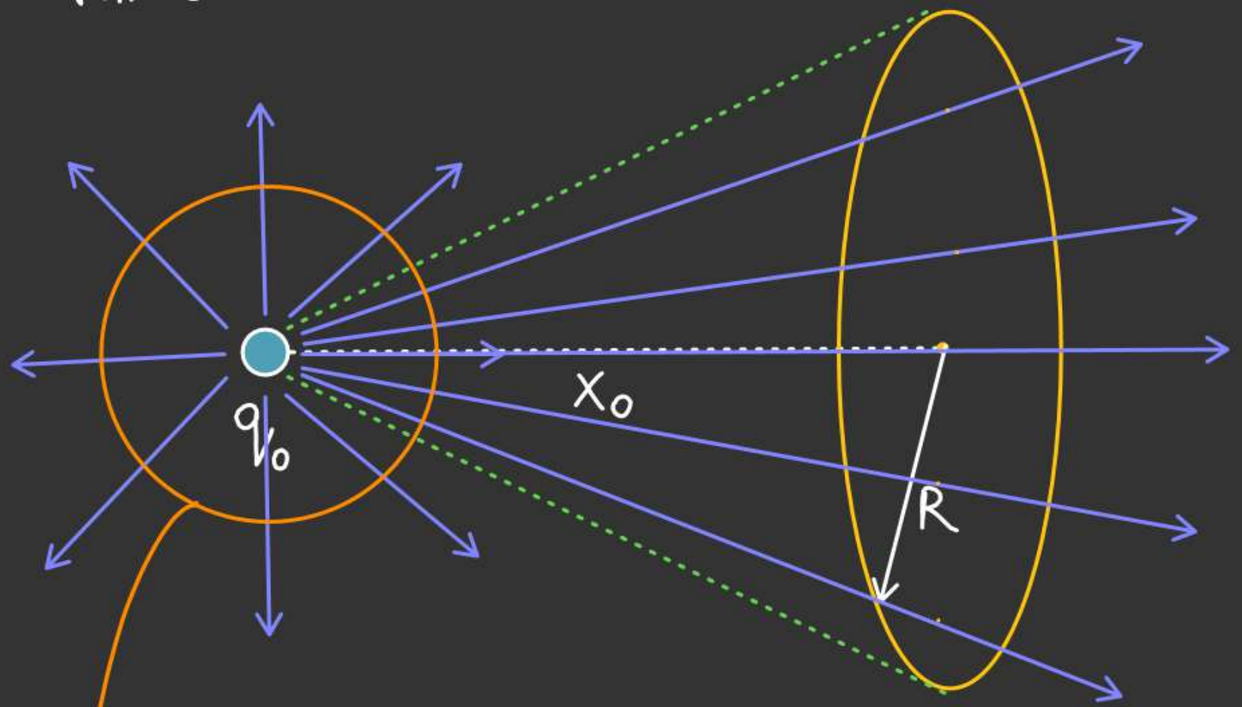
* $\phi_{circle} = \phi_{Disc} = ?$



$$\phi_{sphere} = \frac{q_0}{\epsilon_0}$$

$$\frac{\phi_{Disc}}{\phi_{sphere}} = ?$$

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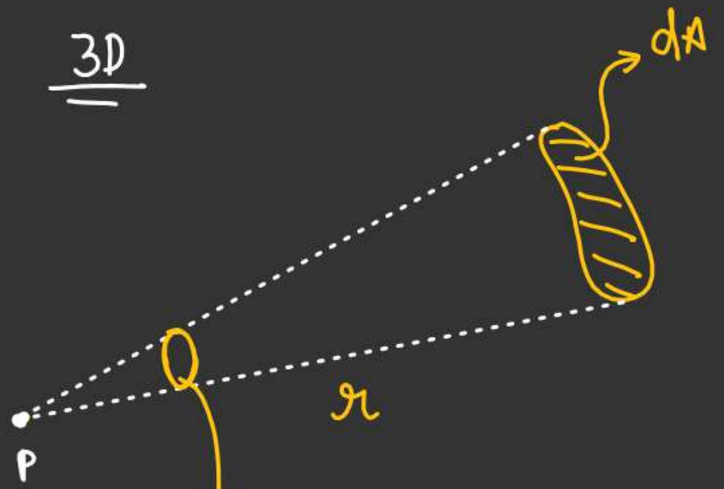
Solid Angle :-

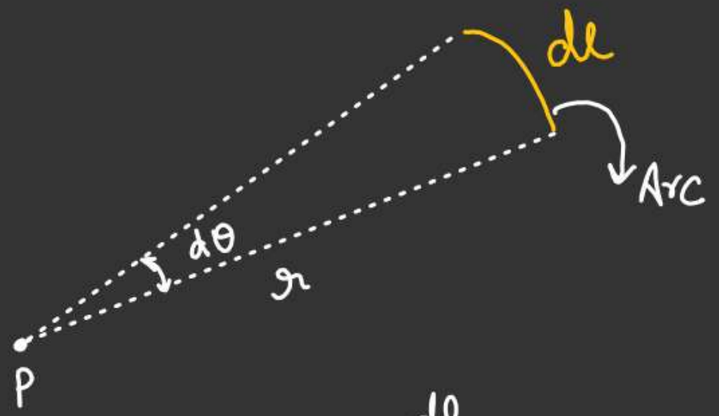
3D



$$d\Omega = \frac{dA}{r^2}$$

all possible directions through the Cone

angle in 2D (Plane)



$$d\theta = \frac{dl}{r}$$

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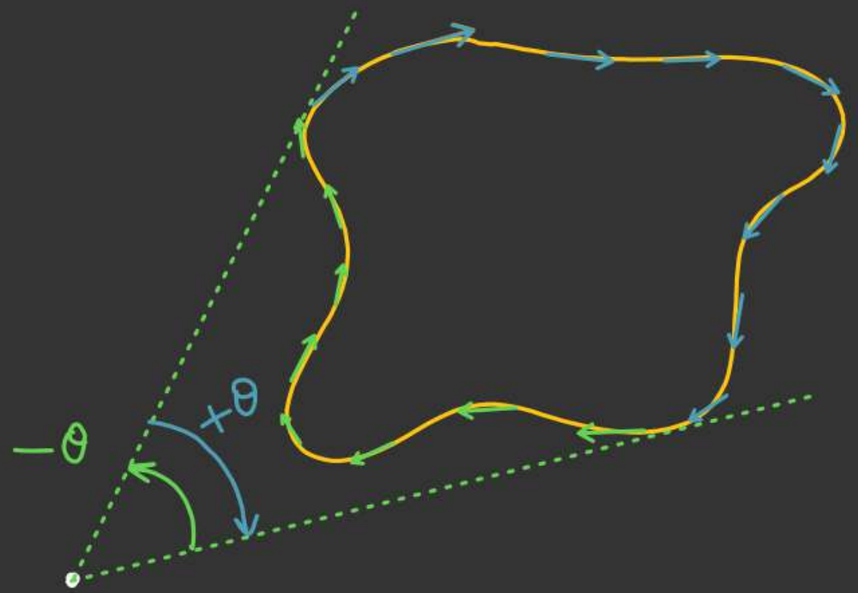
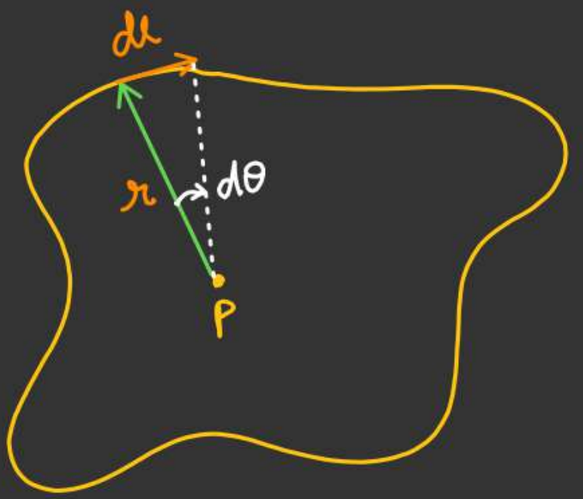


P is outside

$$\int d\theta = 0$$



2D

P is inside

$$\int d\theta = 2\pi$$

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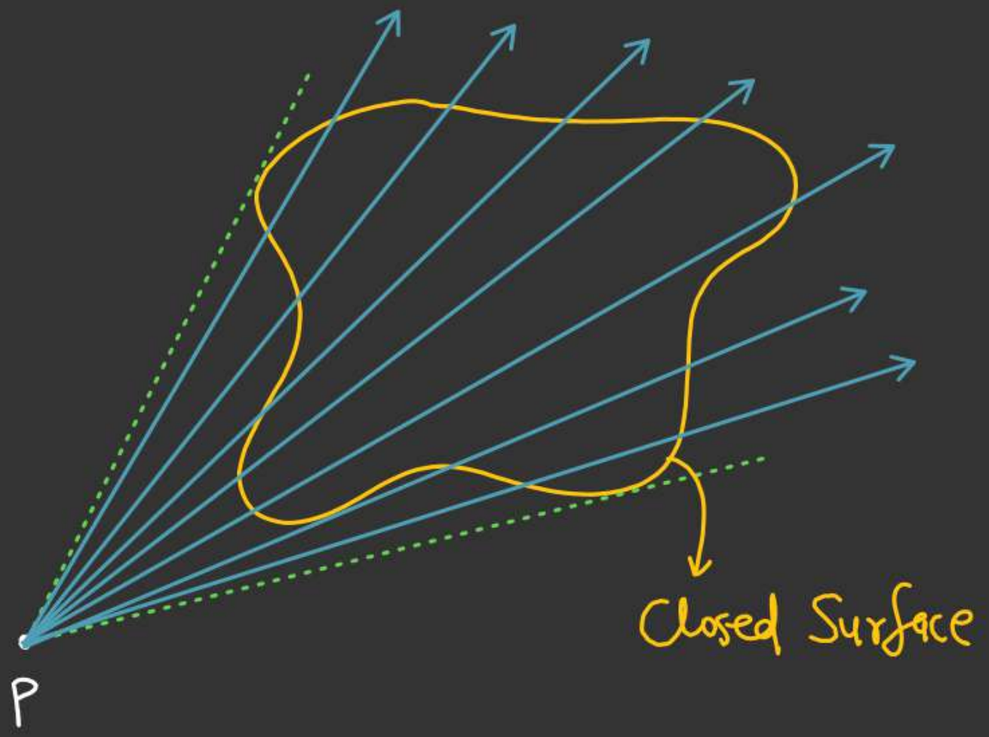
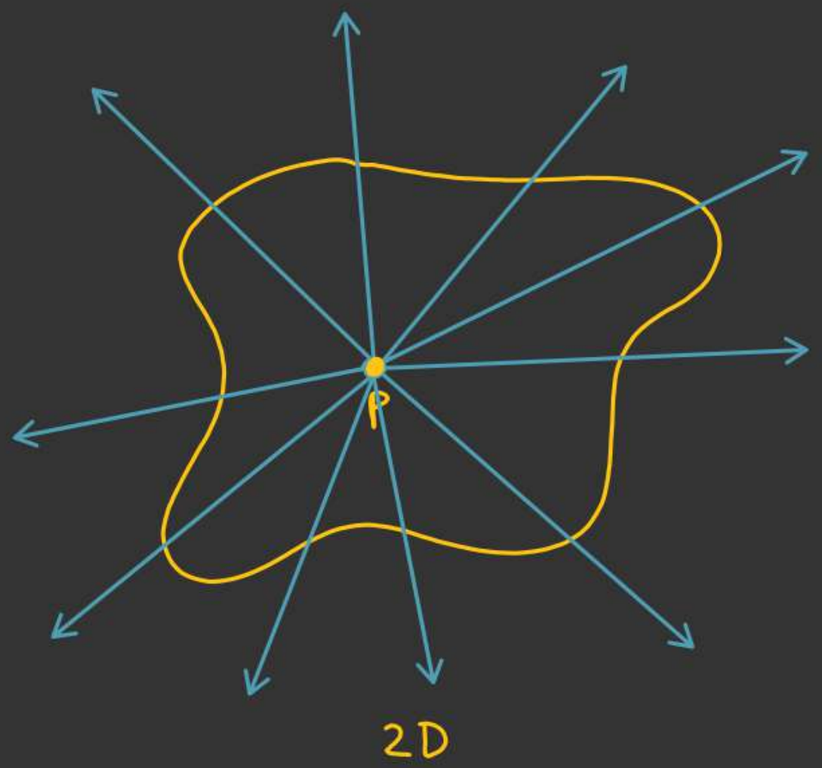


P is outside

$\Omega = 0$



P is inside

$\Omega = 4\pi$

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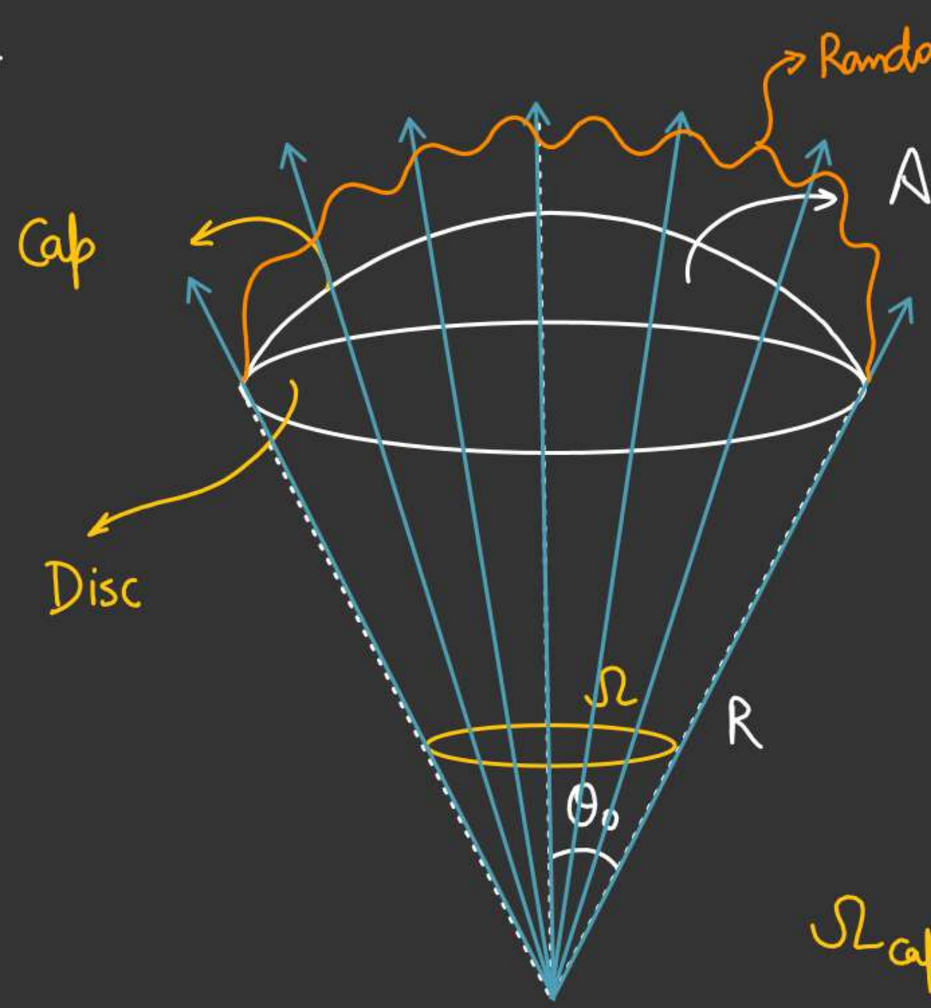

$$A_{cap} = 2\pi R^2 (1 - \cos\theta_0)$$

$$\Omega = \frac{A_{cap}}{R^2} = 2\pi(1 - \cos\theta_0)$$

$$\Omega_{cap} = \Omega_{Disc} = 2\pi(1 - \cos\theta_0) = \Omega_{\text{Random Surface}}$$

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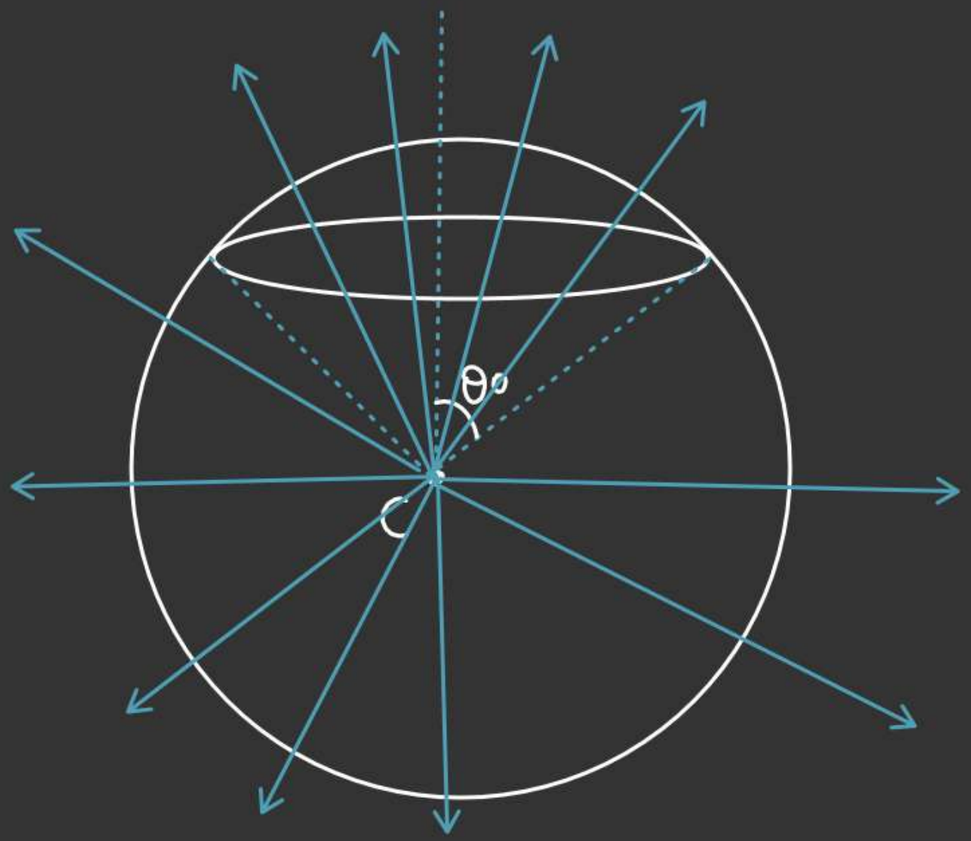
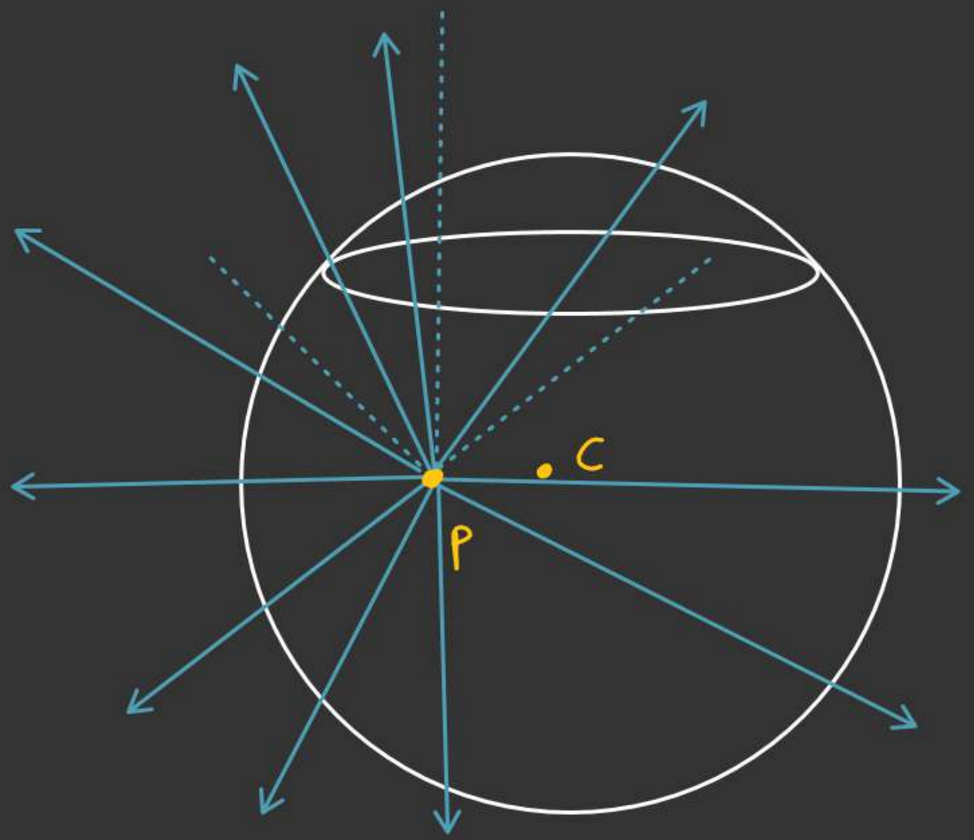


at C $\Omega_{sph} = 2\pi\left(1-\cos\pi\right)$

$= 4\pi$



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$\Omega_P\big|_{sphere} = 4\pi$

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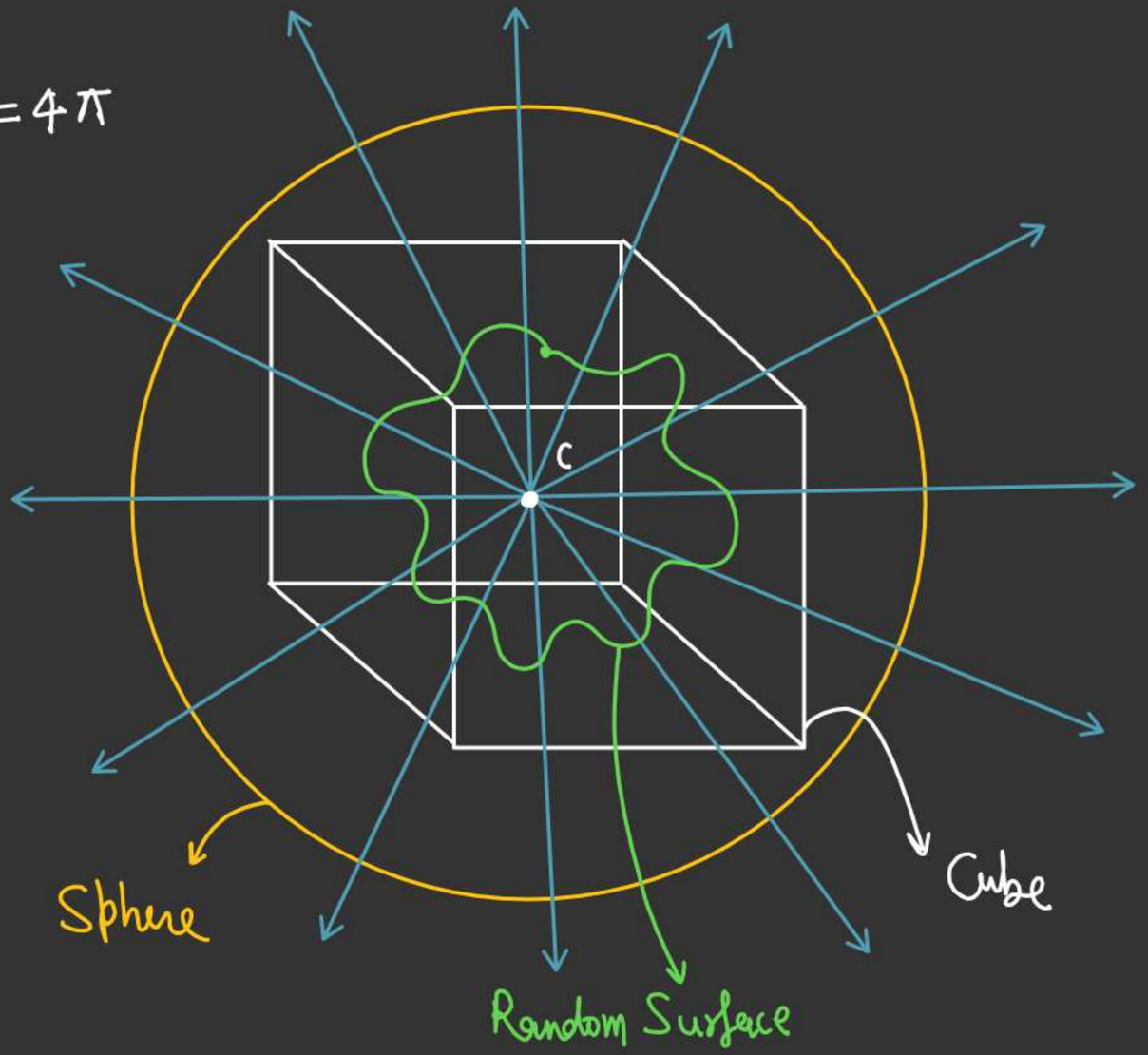

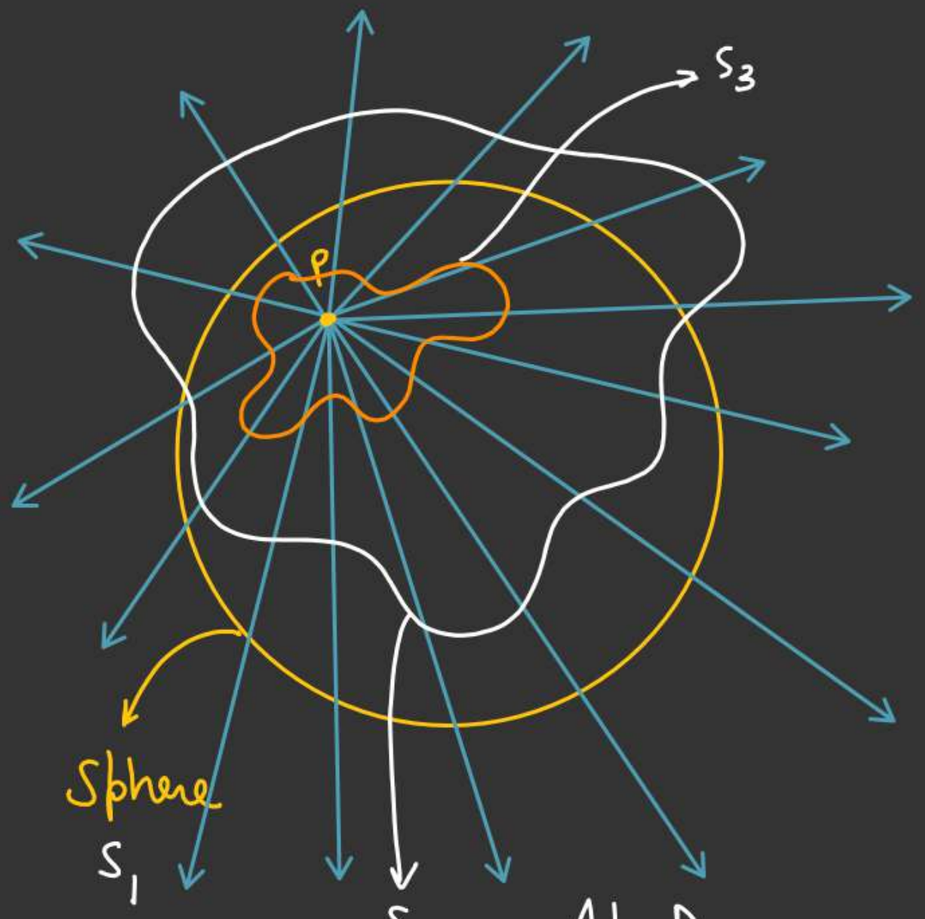

At P

$$\Omega_{S_1} = \Omega_{S_2} = \Omega_{S_3} = 4\pi$$

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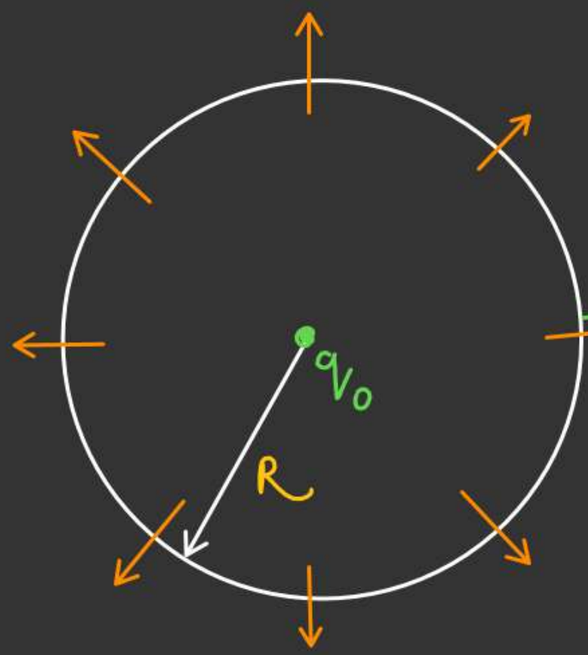

$$E = \frac{kq_0}{R^2}(\hat{r})$$

$$\vec{dA} = dA\,(\hat{r})$$

$$d\phi = \vec{E}\cdot\vec{dA} = \frac{kq_0}{R^2}\cdot dA$$

$$\phi_{sph} = \frac{kq_0}{R^2}\int dA = \frac{1}{4\pi\epsilon_0}\cdot\frac{q_0}{R^2}\cdot 4\pi R^2 = \frac{q_0}{\epsilon_0}$$

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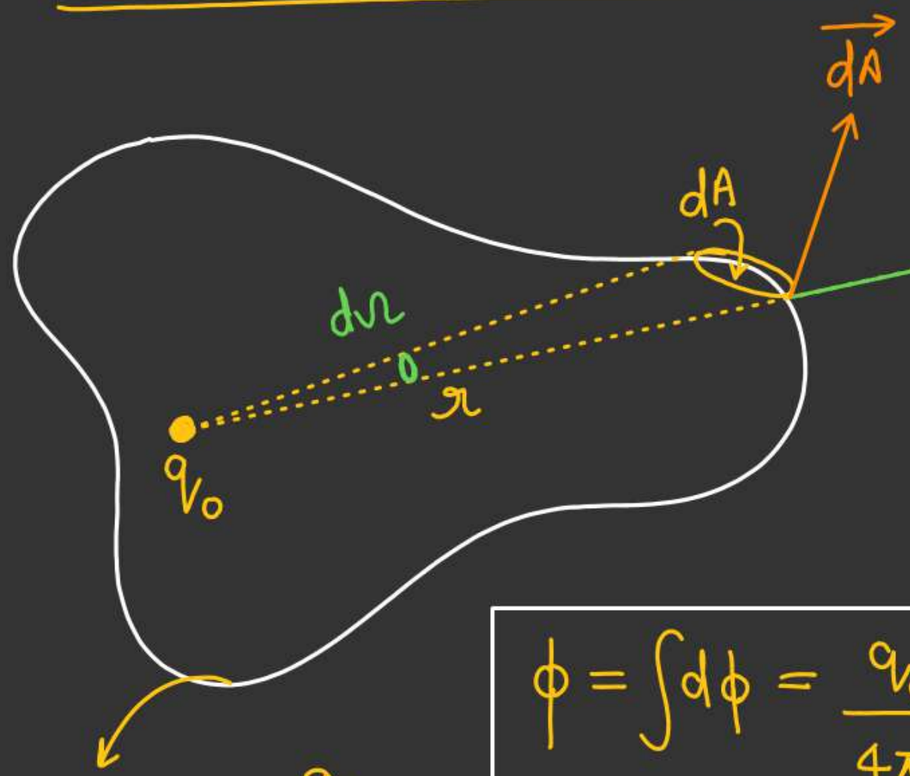
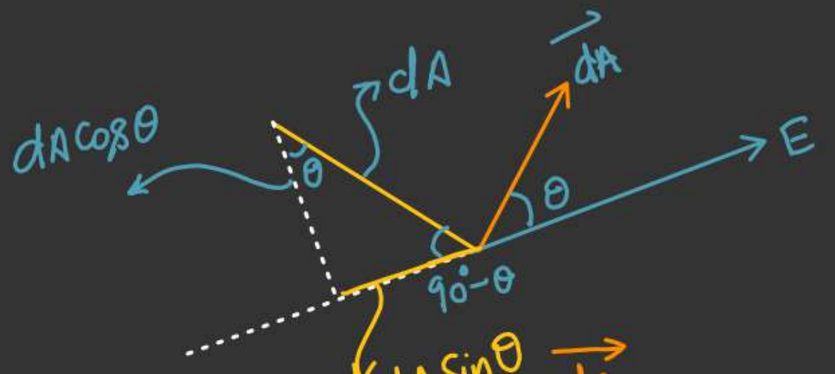
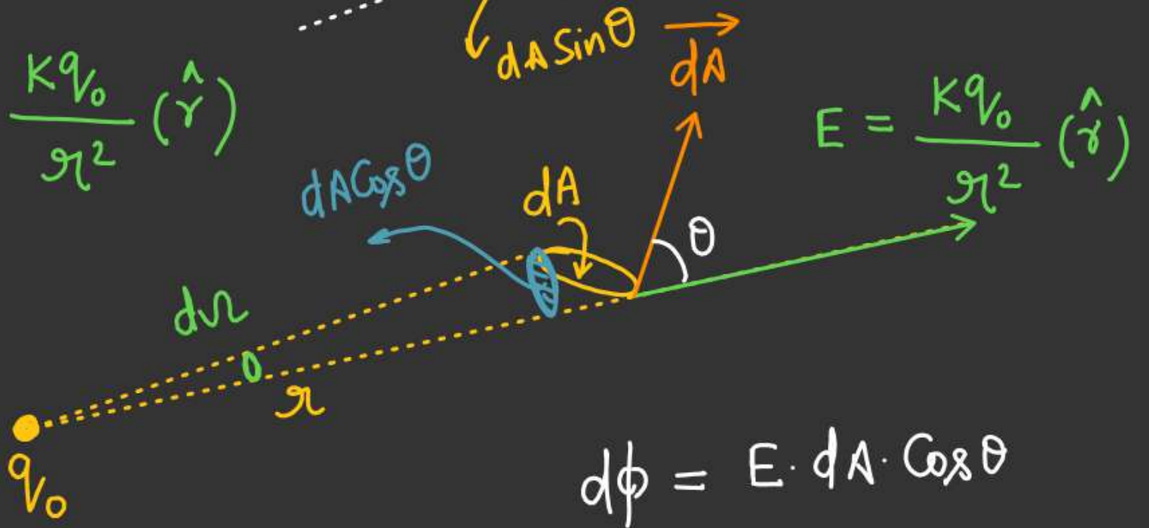
Proof of gauss theorem :-



$$E = \frac{Kq_0}{r^2}(\hat{r})$$





$$E = \frac{Kq_0}{r^2}(\hat{r})$$

$$\phi = \int d\phi = \frac{q_0}{4\pi\epsilon_0}\int d\Omega$$

$$\phi_{surface} = \frac{q_0}{\epsilon_0}$$

$$\phi_{closed} = \frac{q_0}{4\pi\epsilon_0}\cdot 4\pi = \frac{q_0}{\epsilon_0}$$

$$d\phi = E \cdot dA \cdot \cos\theta$$

$$d\phi = Kq_0 \cdot \frac{dA\cos\theta}{r^2}$$

$$\downarrow$$

$$d\Omega$$

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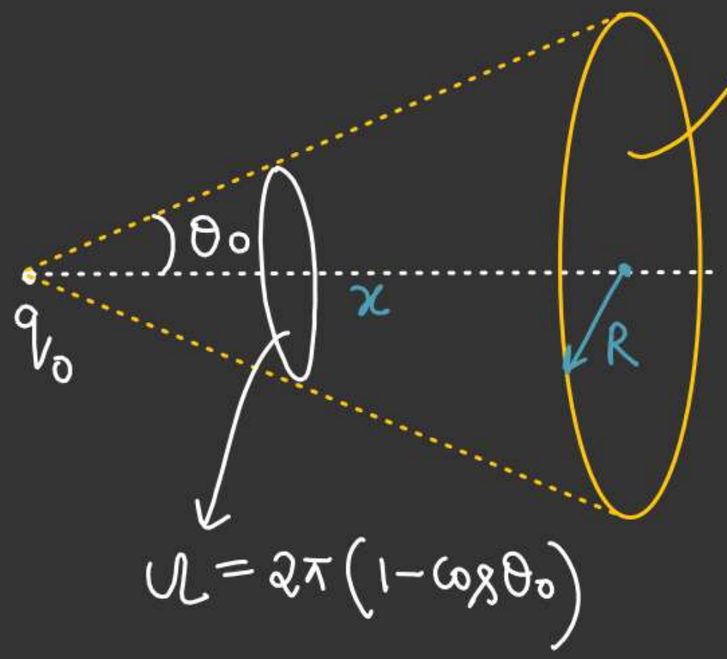
$$\phi = \frac{q_0}{4\pi\epsilon_0} \cdot \Omega$$



$$\phi_{Circle} = \frac{q_0}{4\pi\epsilon_0} \cdot 2\pi(1-\cos\theta_0)$$

$$\phi_{Circle} = \frac{q_0}{2\epsilon_0}\left(1-\cos\theta_0\right)$$

$$\phi_{Circle} = \frac{q_0}{2\epsilon_0}\left(1-\frac{x}{\sqrt{R^2+x^2}}\right)$$

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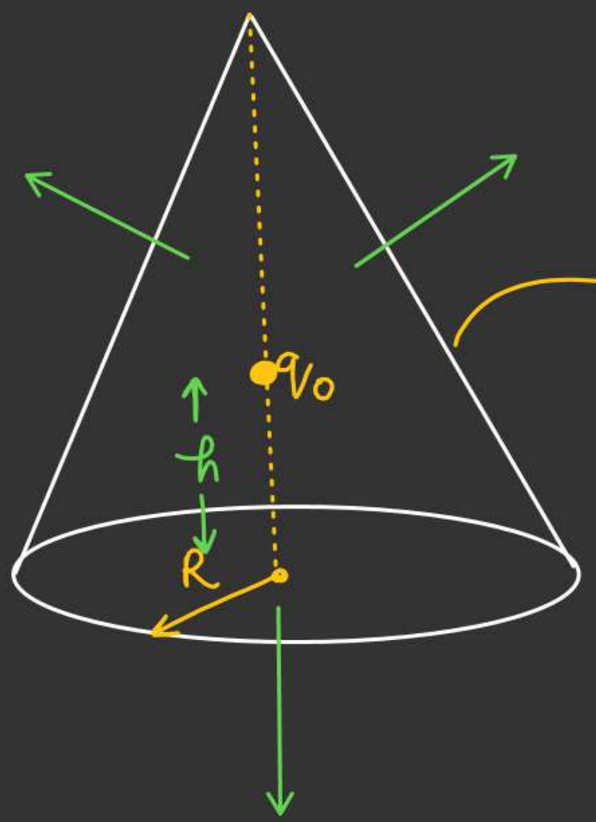

$\phi_{curve} = ?$

$$\phi_{curve} + \phi_{circle} = \frac{q_0}{\epsilon_0}$$

$$\phi_{curve} + \frac{q_0}{2\epsilon_0}\left(1 - \frac{h}{\sqrt{R^2+h^2}}\right) = \frac{q_0}{\epsilon_0}$$

$$\boxed{\phi_{curve} = \frac{q_0}{2\epsilon_0}\left(1 + \frac{h}{\sqrt{R^2+h^2}}\right)}$$

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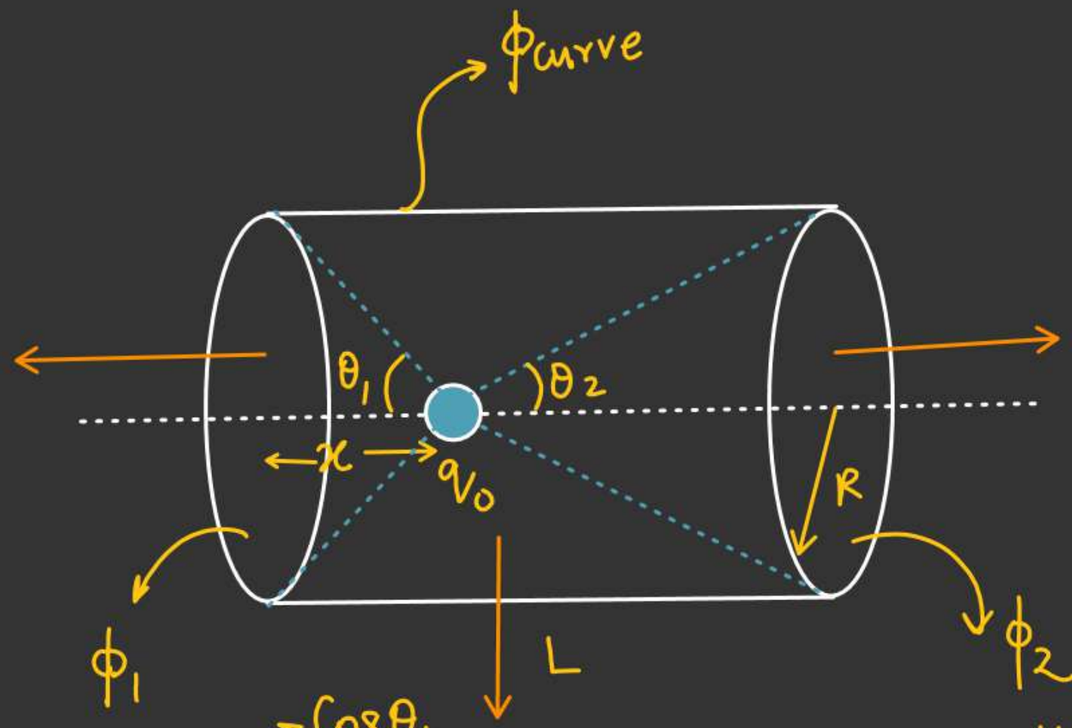

$$\phi_1 + \phi_2 + \phi_C = \frac{q_0}{\epsilon_0}$$

$$\phi_C = \frac{q}{\epsilon_0} - \frac{q_0}{2\epsilon_0}\left(1-\cos\theta_1\right) - \frac{q_0}{2\epsilon_0}\left(1-\cos\theta_2\right)$$

$$\phi_C = \frac{q_0}{2\epsilon_0}\left[\cos\theta_1 + \cos\theta_2\right]$$

$$\boxed{\phi_C = \frac{q_0}{2\epsilon_0}\left(\frac{x}{\sqrt{R^2+x^2}} + \frac{L-x}{\sqrt{R^2+(L-x)^2}}\right)}$$

$$\frac{q_0}{2\epsilon_0}\left(1-\frac{x}{\sqrt{R^2+x^2}}\right)$$

$$\frac{q_0}{2\epsilon_0}\left(1-\frac{L-x}{\sqrt{R^2+(L-x)^2}}\right)$$

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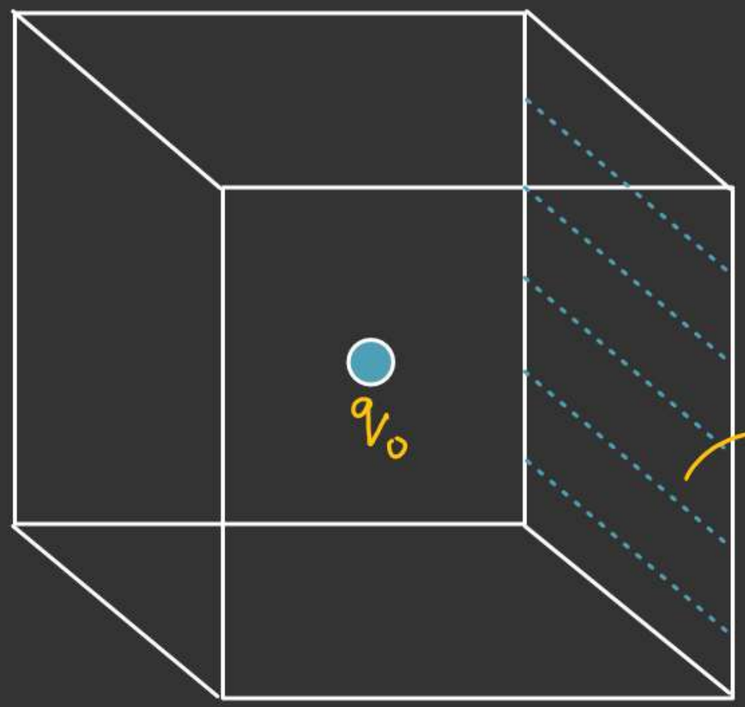

$$6\phi = \frac{q}{\epsilon_0}$$

$$\phi = \frac{q_0}{6\epsilon_0}$$

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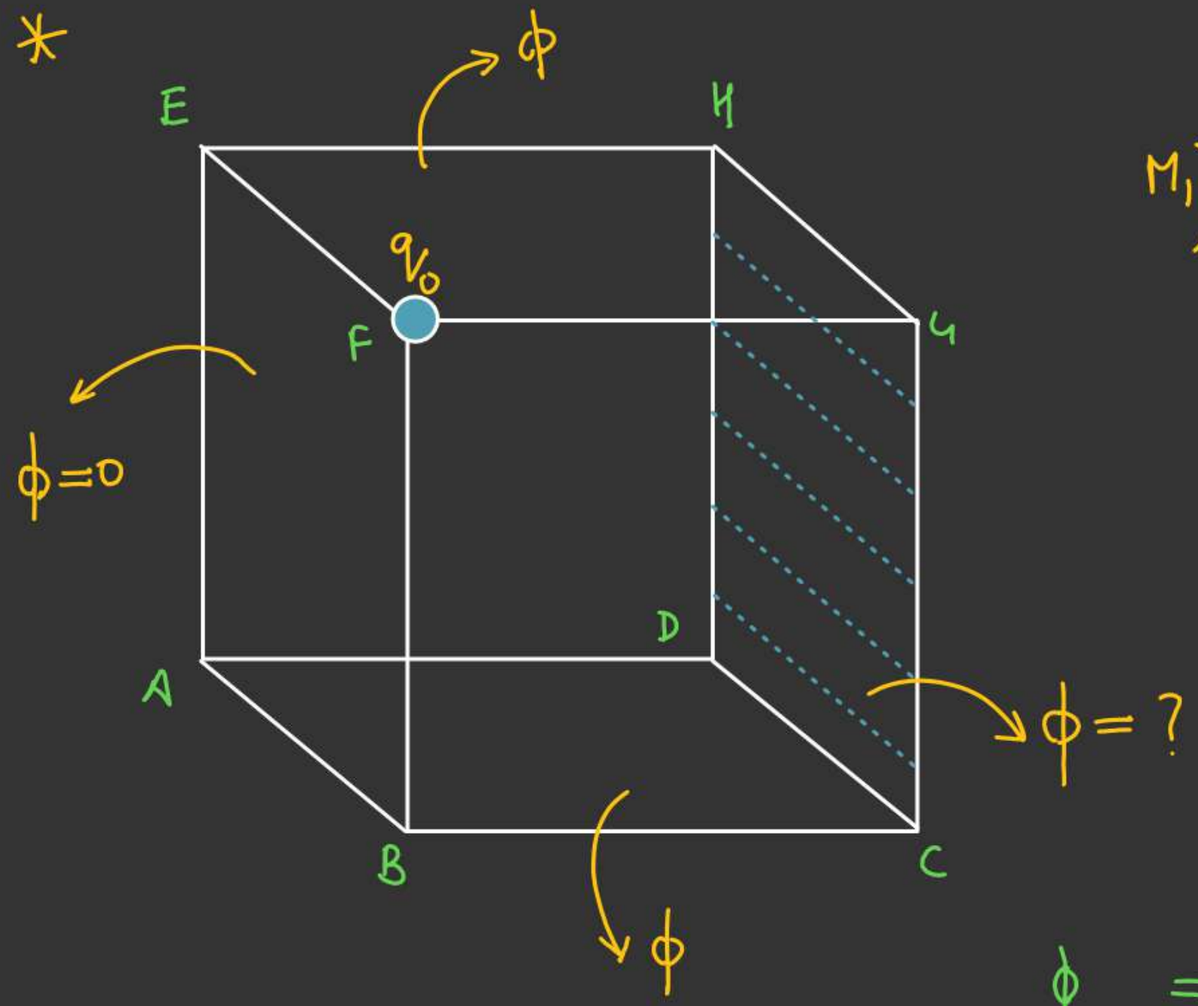


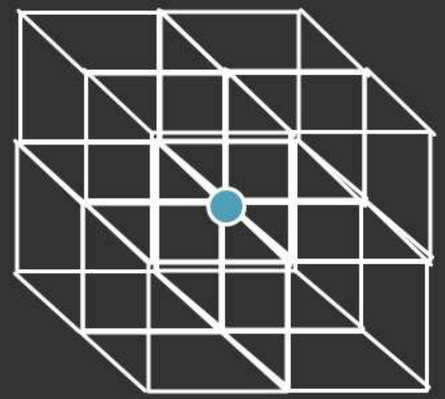
$\phi_{CDGH} = \phi_{EHDA} = \phi_{ABCD} = \phi$

$\phi_{EFGH} = \phi_{ABEF} = \phi_{FGBC} = 0$

$M_1)$ $\quad 3\phi = \dfrac{q_{in}}{\epsilon_0} = \dfrac{q_0/8}{\epsilon_0}$

$$\boxed{\phi = \frac{q_0}{24\epsilon_0}}$$

$M_2)$



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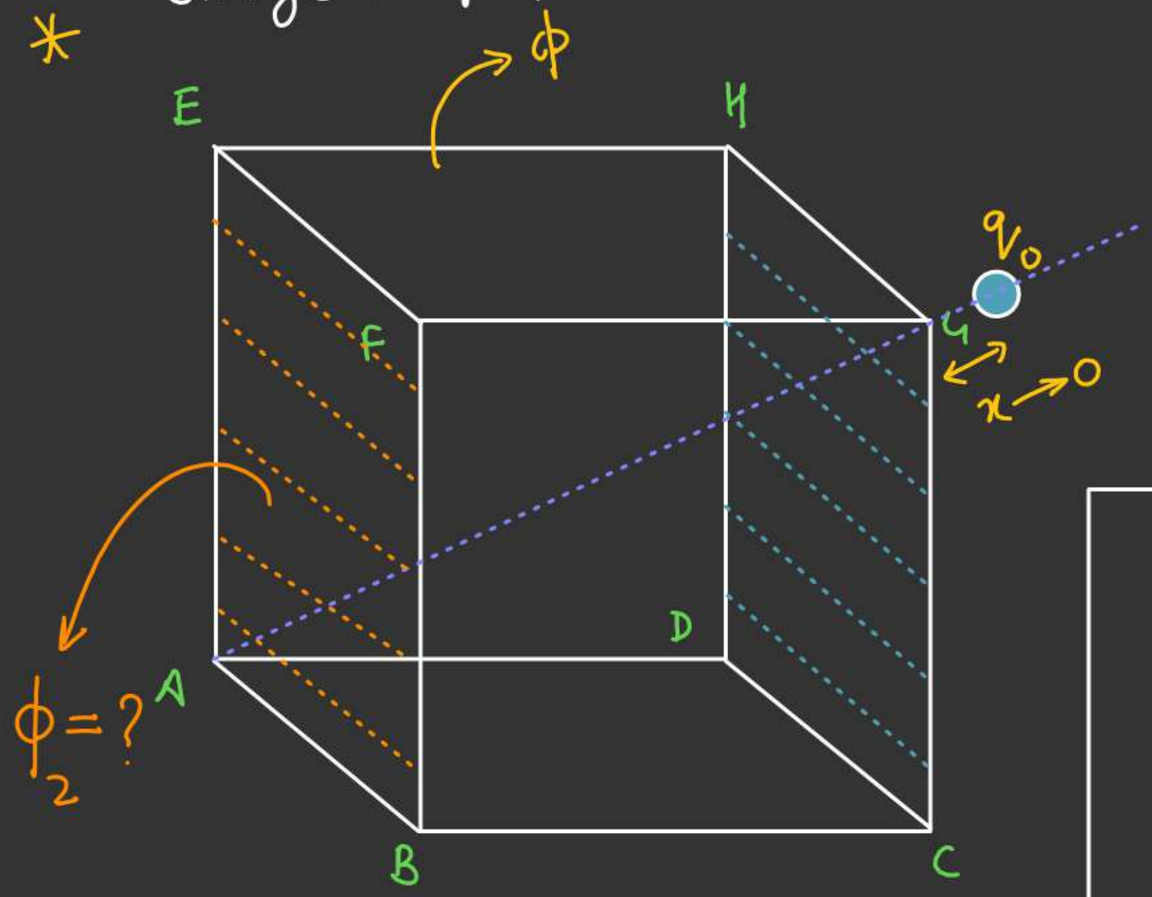
Charge is just outside.

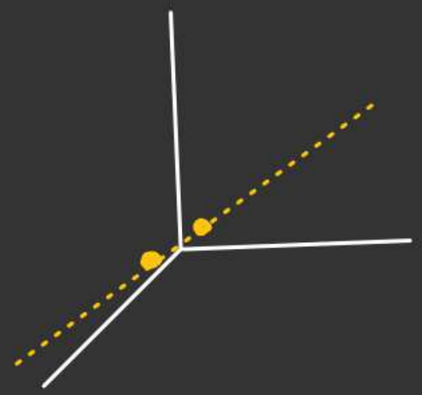
*



$$3\phi_1 + 3\phi_2 = 0$$

$$\phi_2 = \frac{q_0}{24\epsilon_0}$$

$$\phi_1 = -\frac{q_0}{24\epsilon_0}$$



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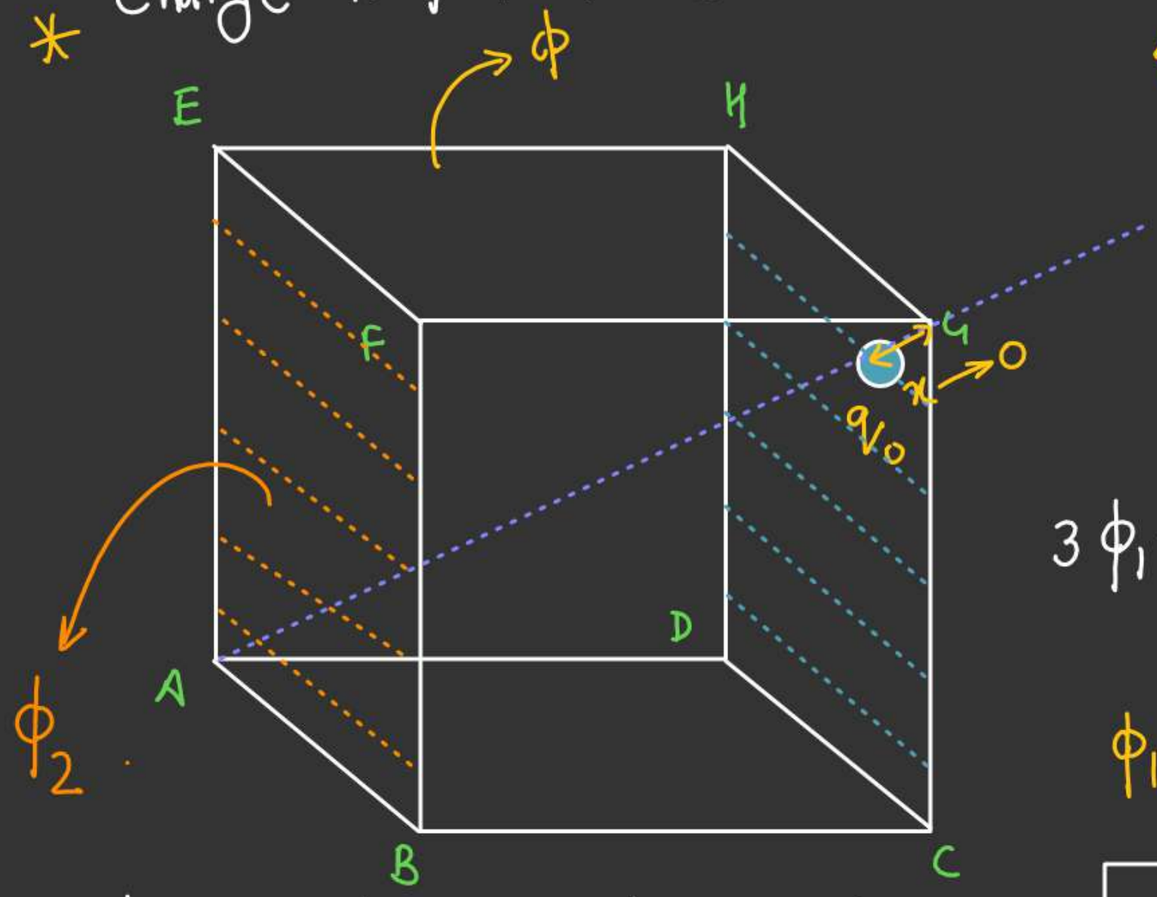
* Charge is just inside :-



$$\phi_{CDGH} = \phi_{EFGH} = \phi_{BCFG} = \phi_1$$

$$\phi_{ADEH} = \phi_{ABEF} = \phi_{ABCD} = \phi_2$$

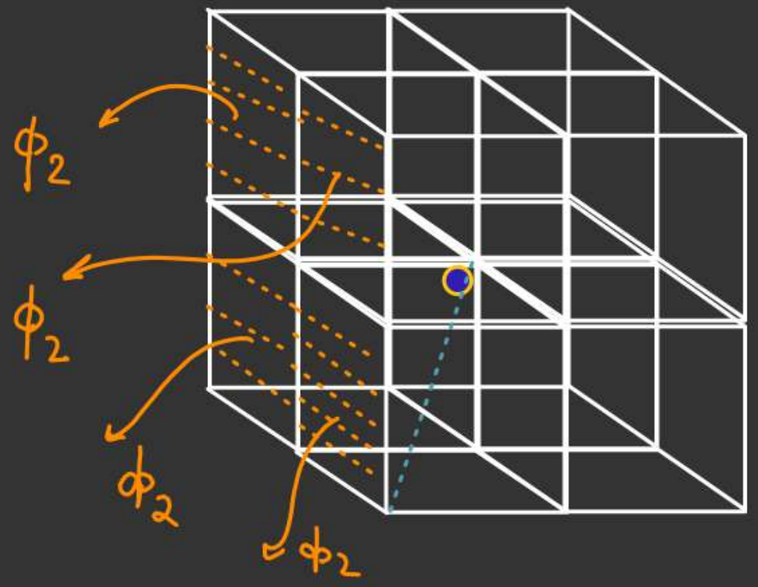
$$4\phi_2 = \frac{q_{in}}{6\epsilon_0}$$

$$\phi_2 = \frac{q_0}{24\epsilon_0}$$

$$3\phi_1 + 3\phi_2 = \frac{q_{in}}{\epsilon_0}$$

$$\phi_1 + \phi_2 = \frac{q_0}{3\epsilon_0}$$

$$\boxed{\phi_1 = \frac{q_0}{3\epsilon_0} - \frac{q_0}{24\epsilon_0} = \frac{7q_0}{24\epsilon_0}}$$



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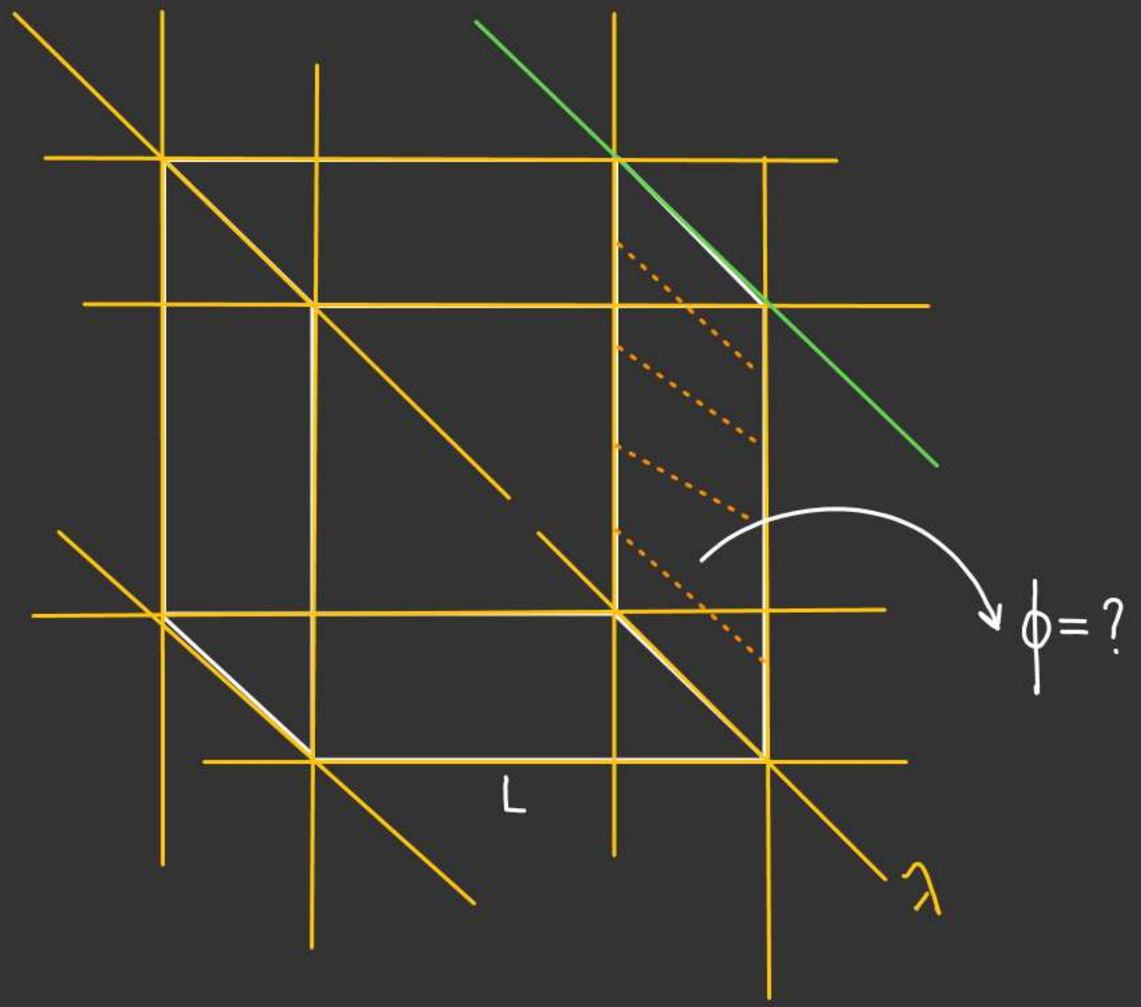
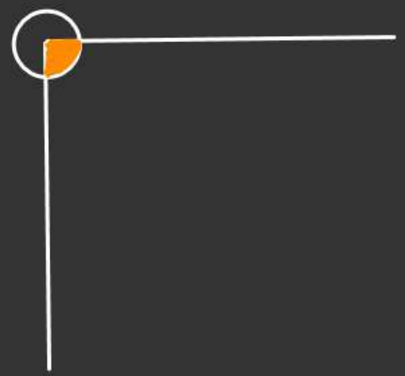


$$6\phi = \frac{q_{in}}{\epsilon_0} = \left[\frac{\lambda L}{\epsilon_0 4}\right] \cdot 12$$

$$\phi = \frac{\lambda L}{2\epsilon_0}$$



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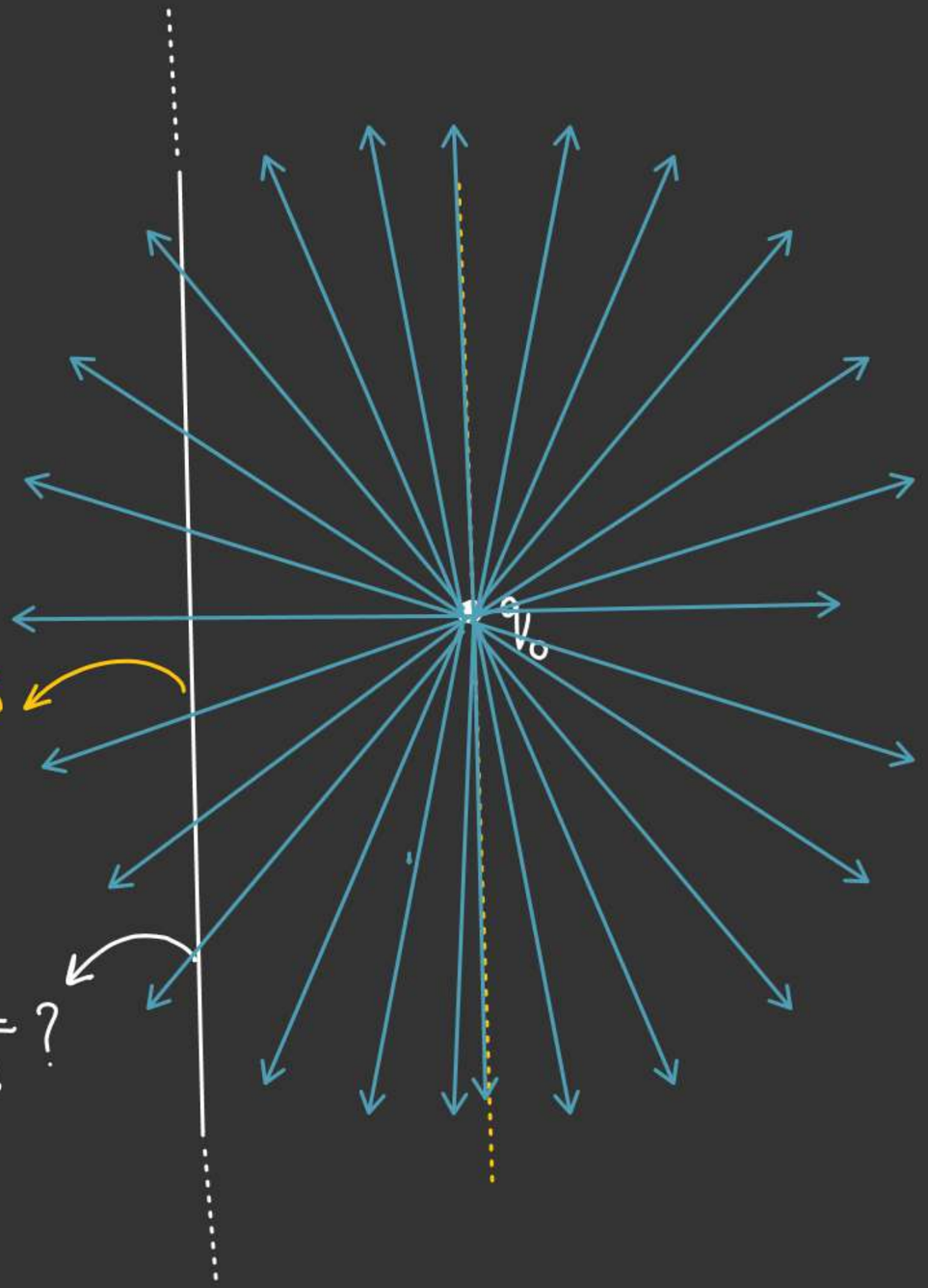
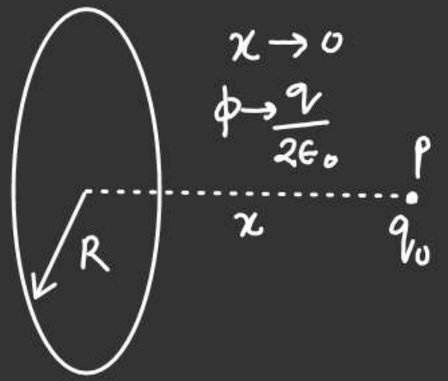


Infinite Plane

$\phi_{Plane} = ?$

$\phi_{Plane} = \frac{q}{2\epsilon_0}$



$x \to 0$

$\phi \to \frac{q}{2\epsilon_0}$

$\phi = \frac{q_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$

$x \to 0$

$\phi \to \frac{q}{2\epsilon_0}$

$x << L$

Finite Plane

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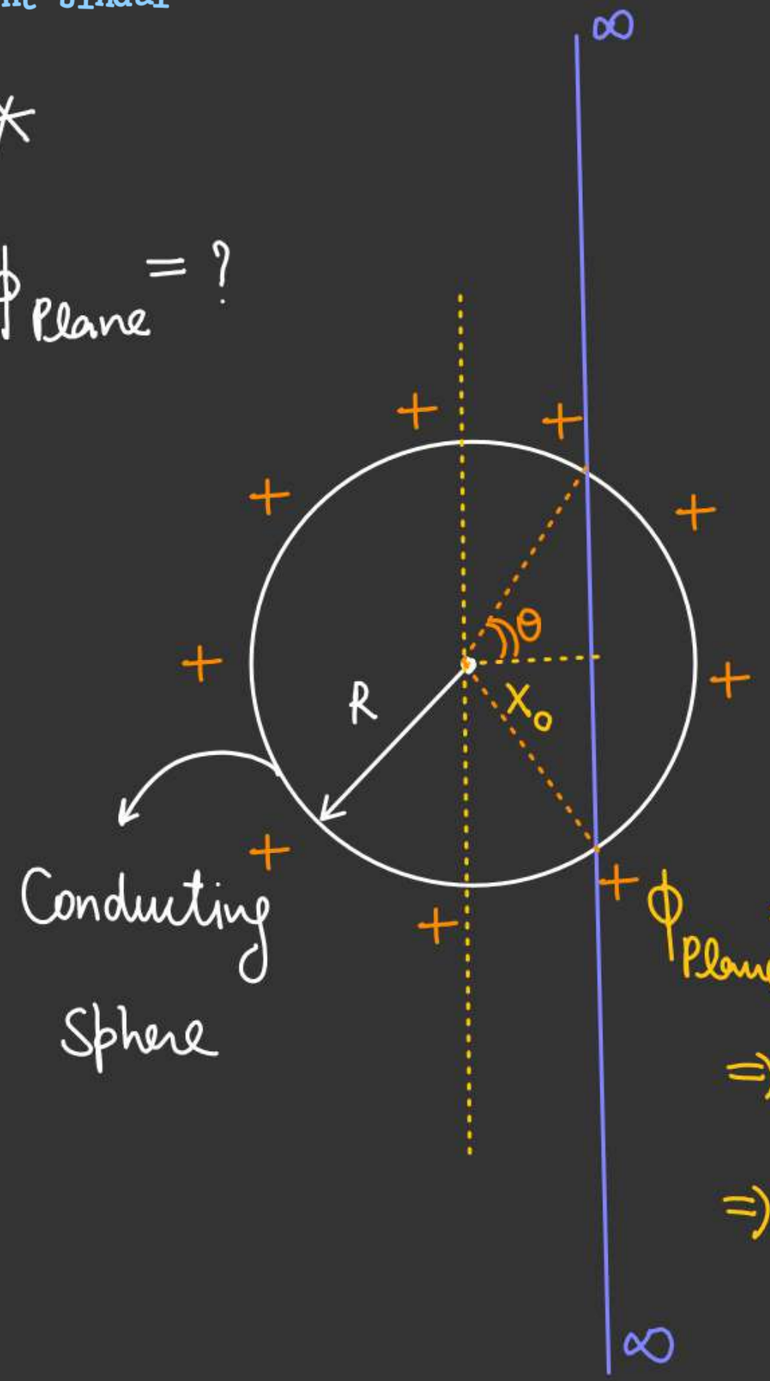
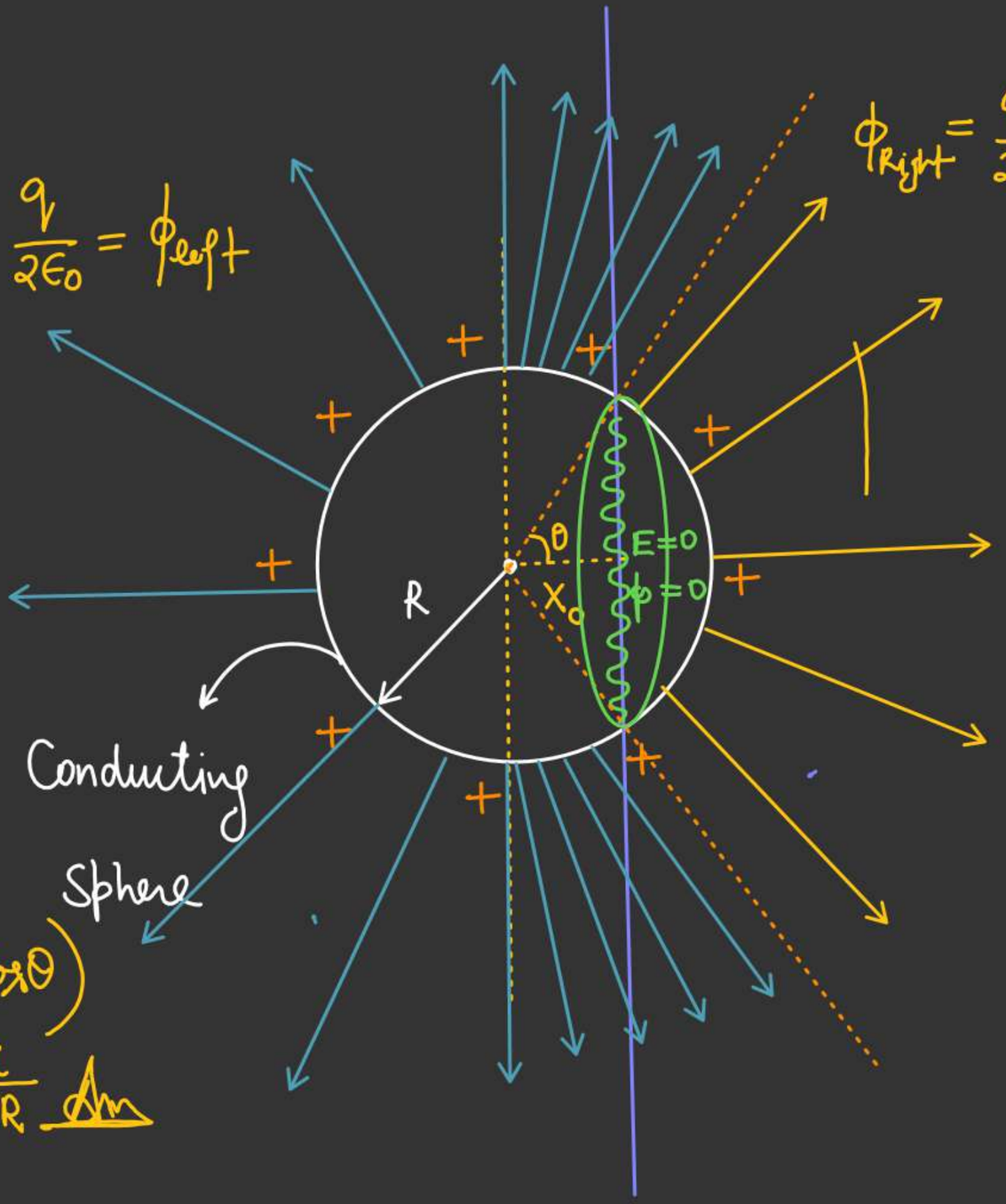
$\phi_{Plane} = ?$



$$\phi_{Plane} = \frac{q}{2\epsilon_0} - \phi_{C.C}$$

$$\Rightarrow \frac{q}{2\epsilon_0} - \frac{q}{2G_0}(1-\cos\theta)$$

$$\Rightarrow \frac{q}{2G_0}\cdot\cos\theta = \frac{qx}{2G_0R}$$ Ans



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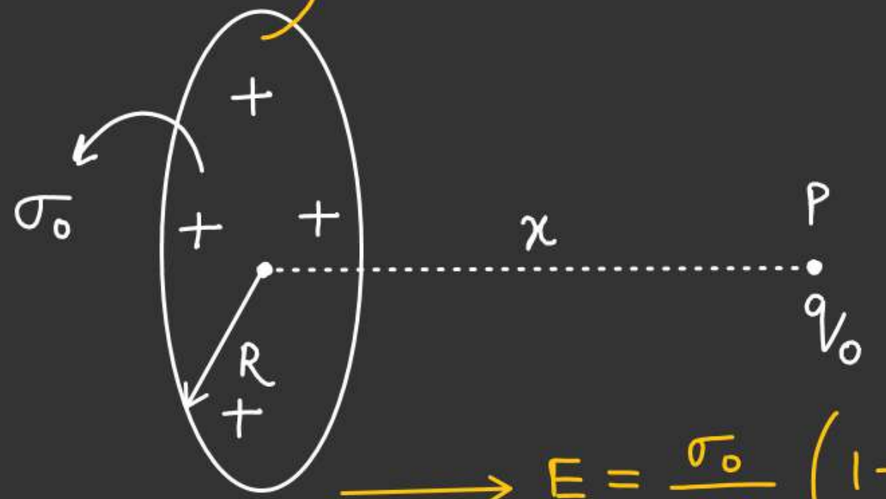
* Force Calculation Using flux :-



$$\phi_{Disc} = \frac{q_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$$

$$E = \frac{\sigma_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$$

$$F_{q_0} = F_{Disc} = q_0 E = \frac{\sigma_0 q_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right) = \sigma\phi$$

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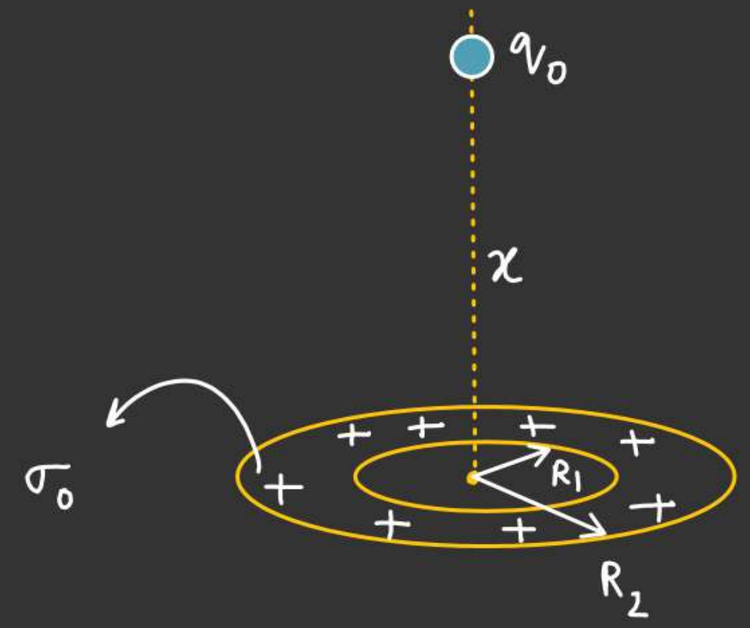

$F = ?$

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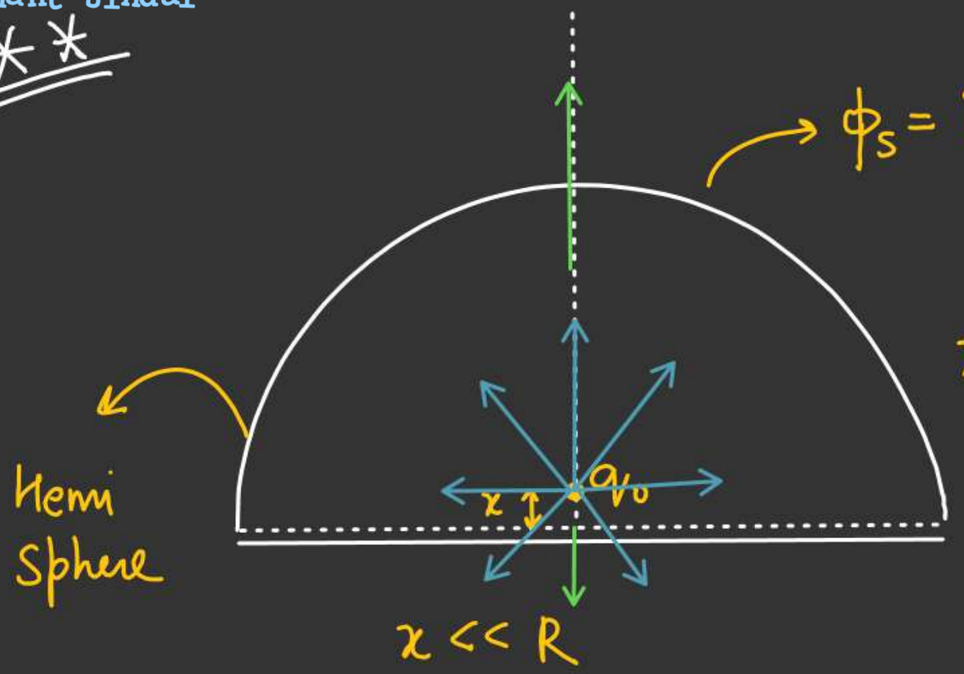


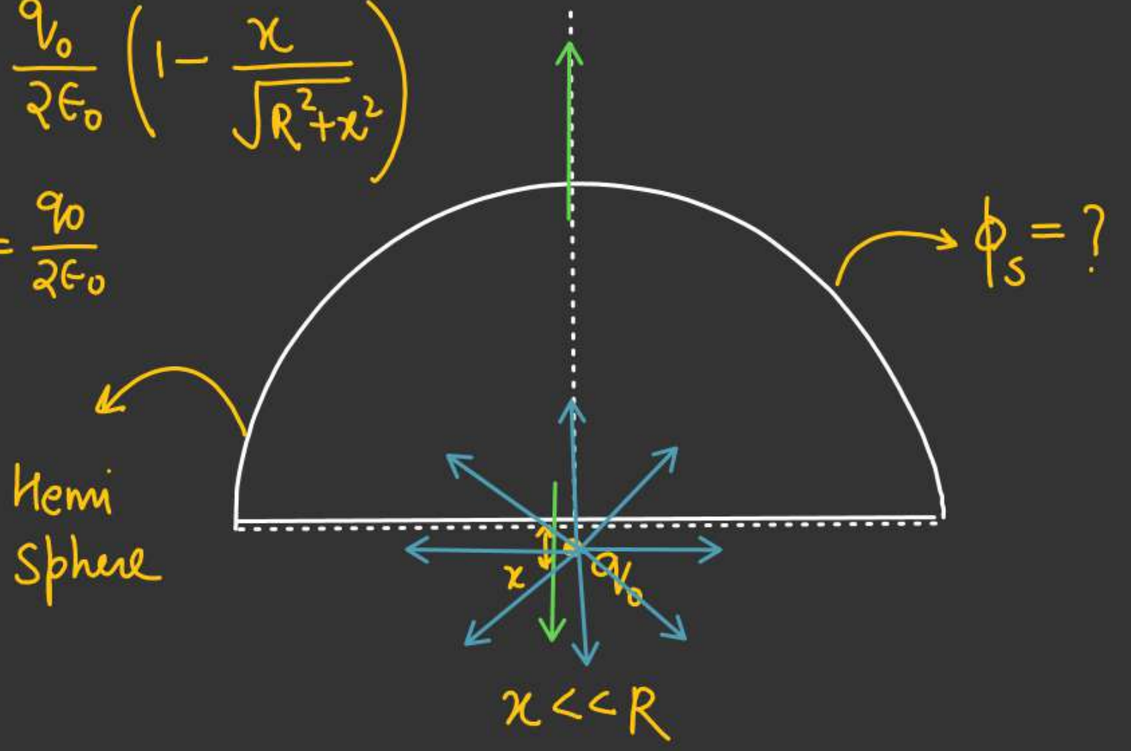
$$\phi_{Disc} = +\frac{q_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$$

$x \to 0 \quad \phi_D = \frac{q_0}{2\epsilon_0}$

$$\phi_{Disc} + \phi_{sph} = \frac{q_0}{\epsilon_0}$$

$$+\frac{q_0}{2\epsilon_0} + \phi_{sph} = \frac{q_0}{\epsilon_0}$$

$$\phi_{sph} = \frac{q_0}{2\epsilon_0}$$



$$\phi_{Disc} + \phi_{sph} = 0$$

$$-\frac{q_0}{2\epsilon_0} + \phi_{sph} = 0$$

$$\boxed{\phi_{sph} = \frac{q_0}{2\epsilon_0}} \quad *$$

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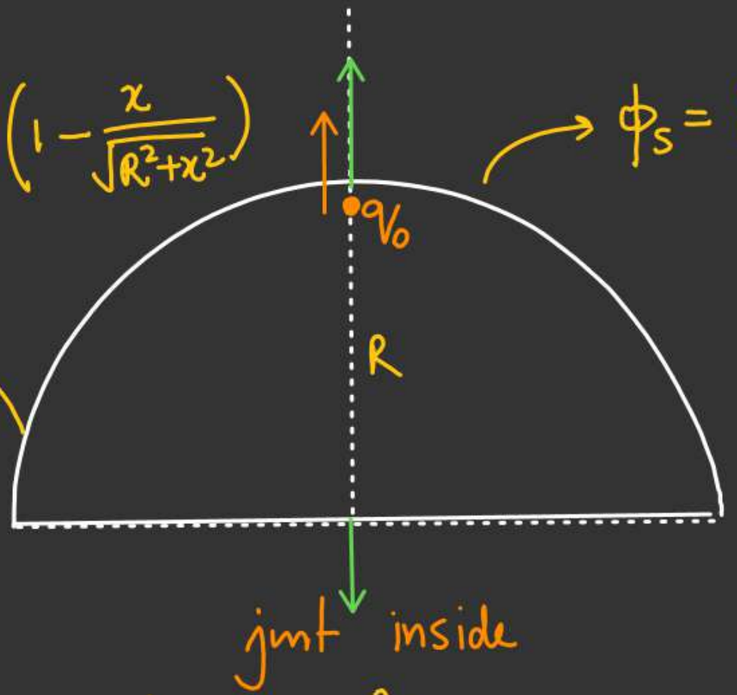
$\phi_{Disc} = \frac{q_0}{2\epsilon_0}\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right)$

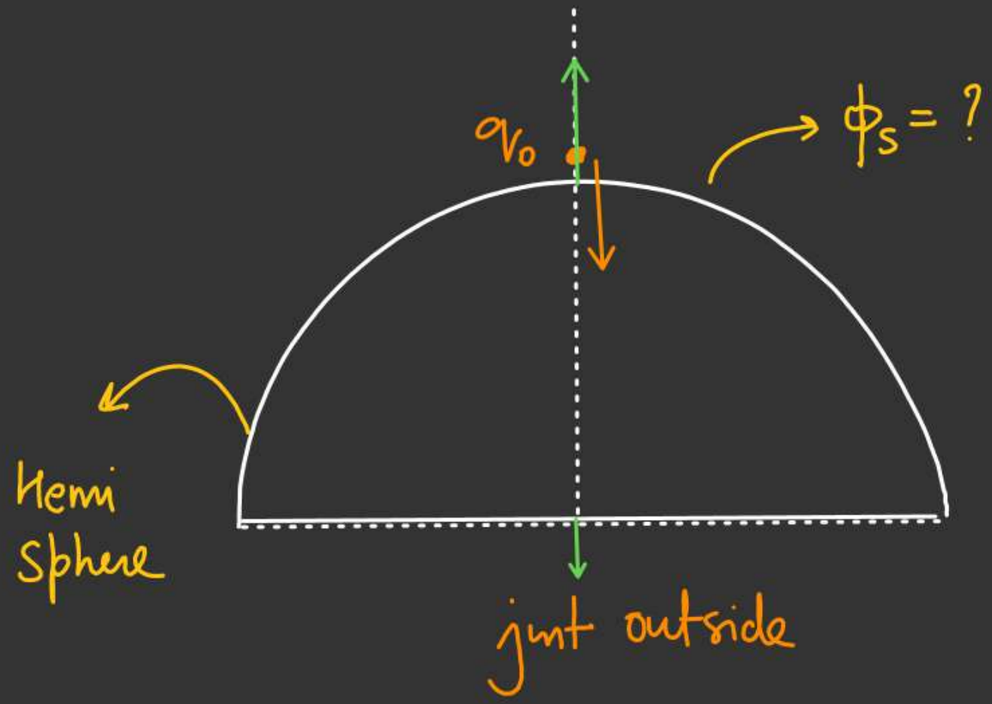
$x = R$



$$\phi_{sph} + \phi_{Disc} = \frac{q_0}{\epsilon_0}$$

$$\phi_{sph} + \frac{q_0}{2\epsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right) = \frac{q_0}{\epsilon_0}$$

$$\boxed{\phi_{sph} = \frac{q_0}{2\epsilon_0}\left(1 + \frac{1}{\sqrt{2}}\right)}$$



$$\phi_{sph} + \phi_{Disc} = 0$$

$$\boxed{\phi_{sph} = -\frac{q_0}{2\epsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)}$$

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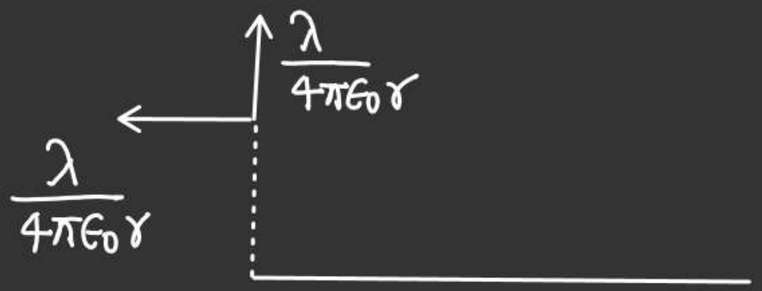
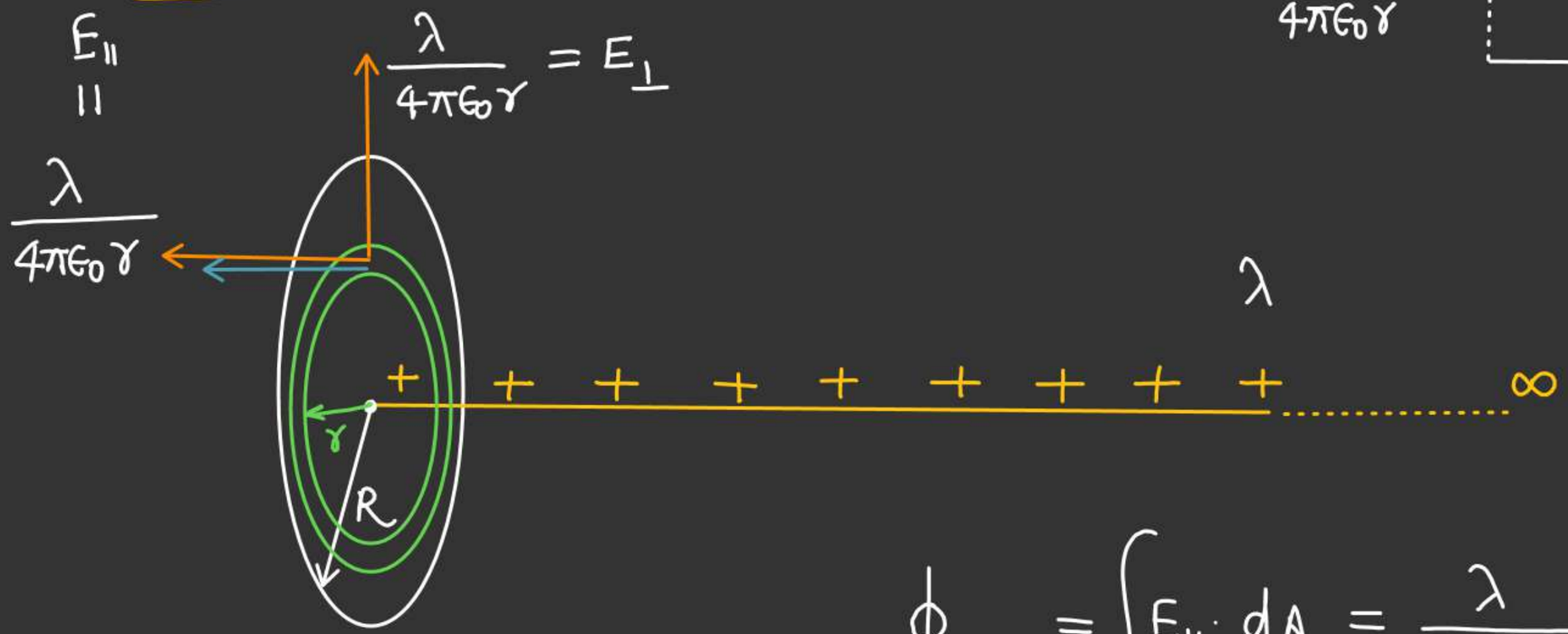
* Disc and Infinite wire :-





$$dA = 2\pi r\, dr$$

$$\phi_{Disc} = \int E_{\parallel} \cdot dA = \frac{\lambda}{4\pi\epsilon_0 r} \cdot 2\pi r\, dr$$

$$\phi_{Disc} = \frac{\lambda}{2\epsilon_0} \int_0^R dr = \frac{\lambda R}{2\epsilon_0}$$

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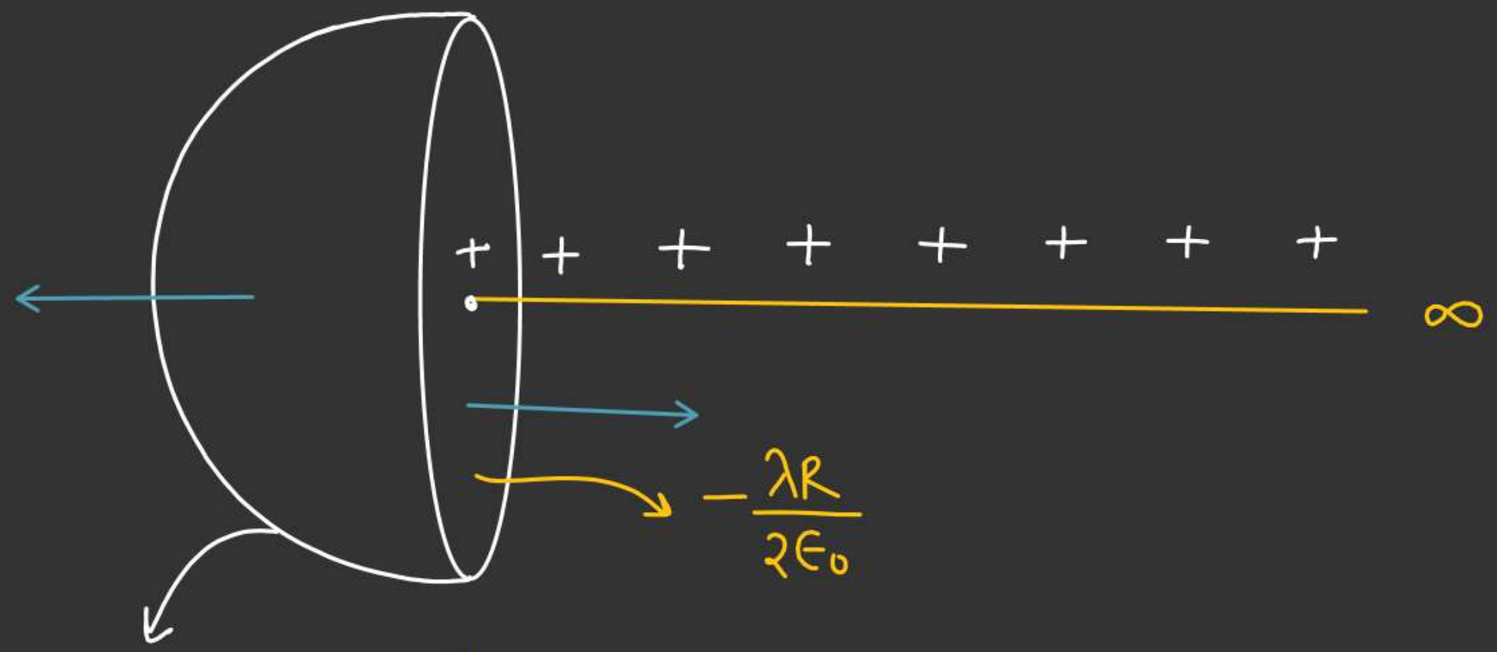

$$\phi_{sphere} = +\frac{\lambda R}{2\epsilon_0}$$

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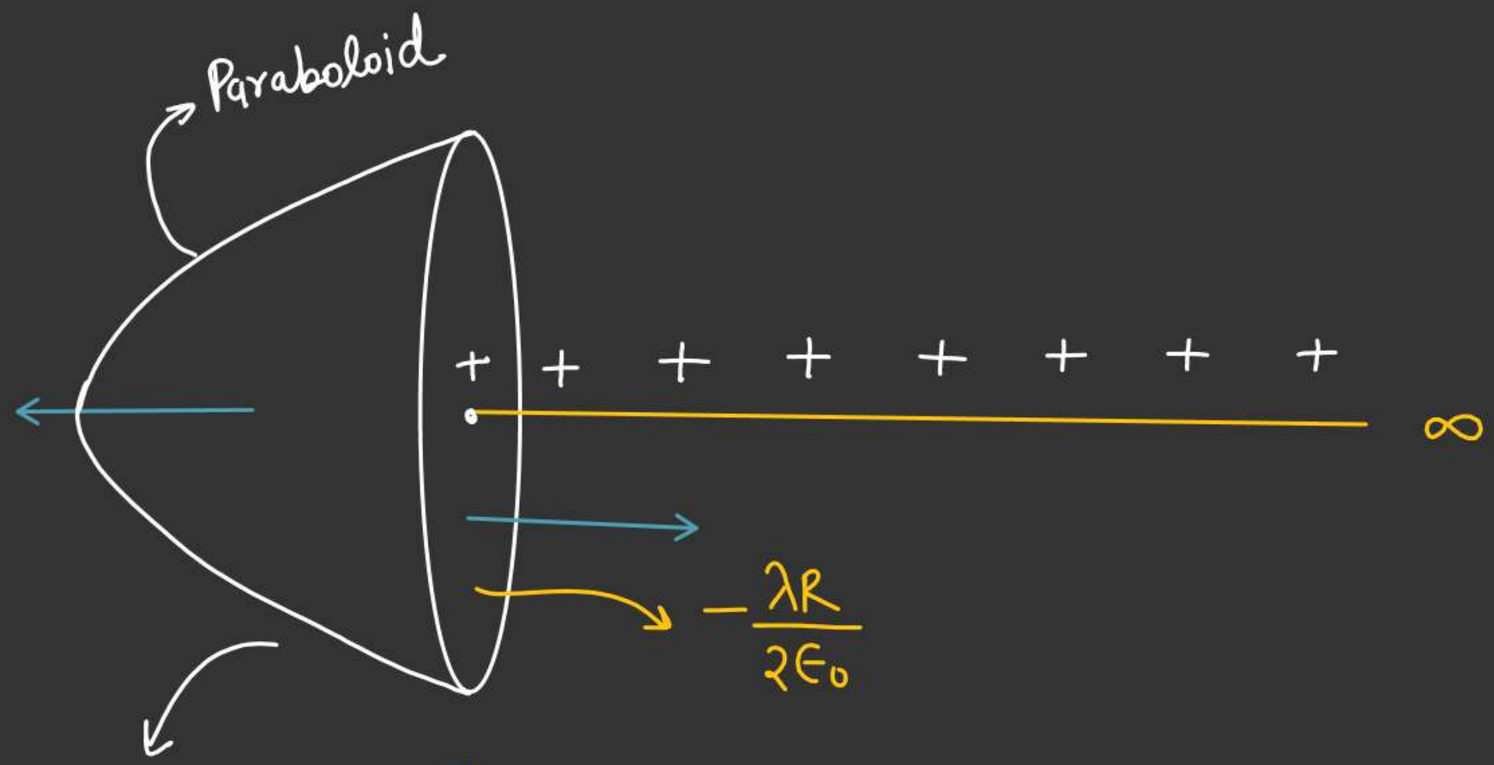

$$\phi_{Para} = +\frac{\lambda R}{2\epsilon_0}$$

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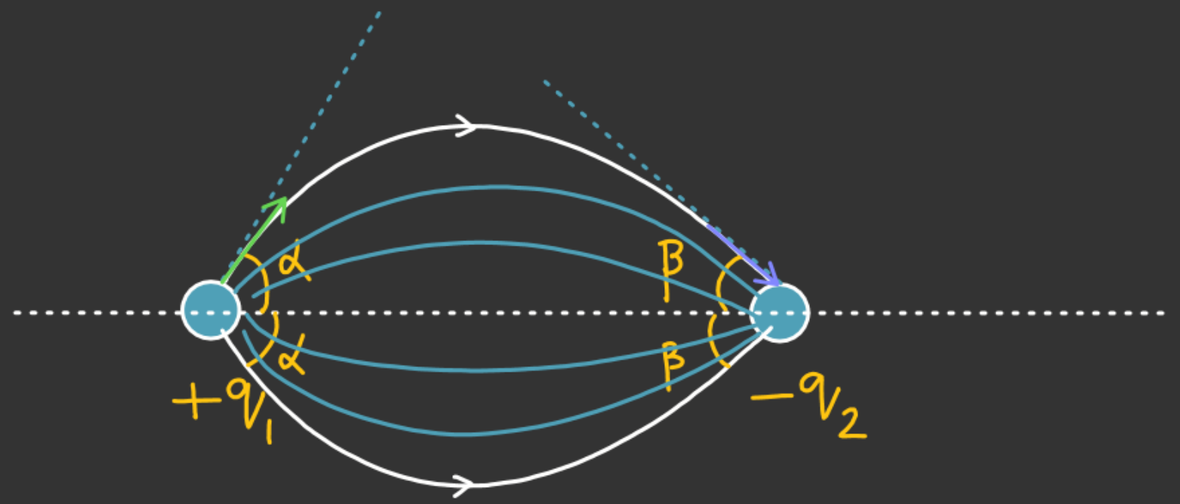
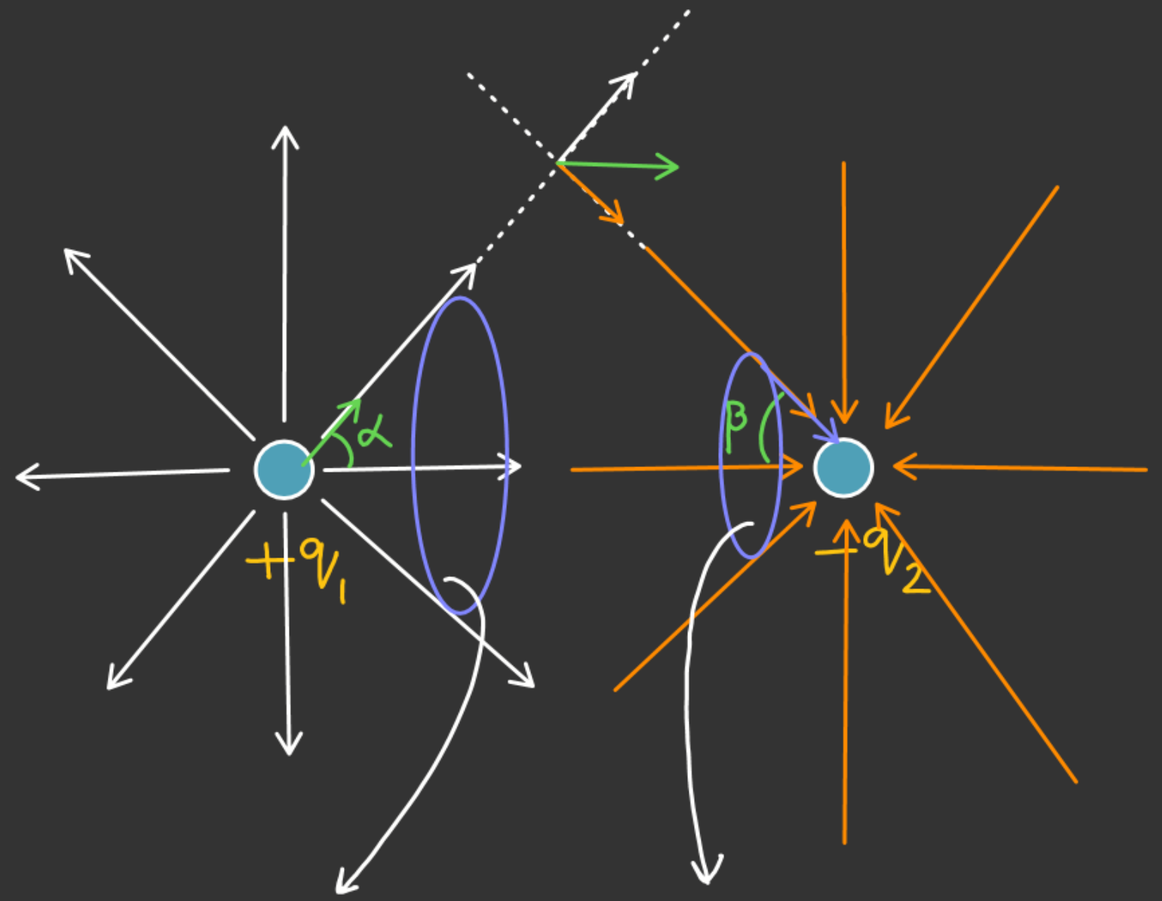
*V. Imp



$$q_1(1-\cos\alpha) = q_2(1-\cos\beta)$$



$$\frac{q_1}{2\epsilon_0}(1-\cos\alpha) = \frac{q_2}{2\epsilon_0}(1-\cos\beta)$$

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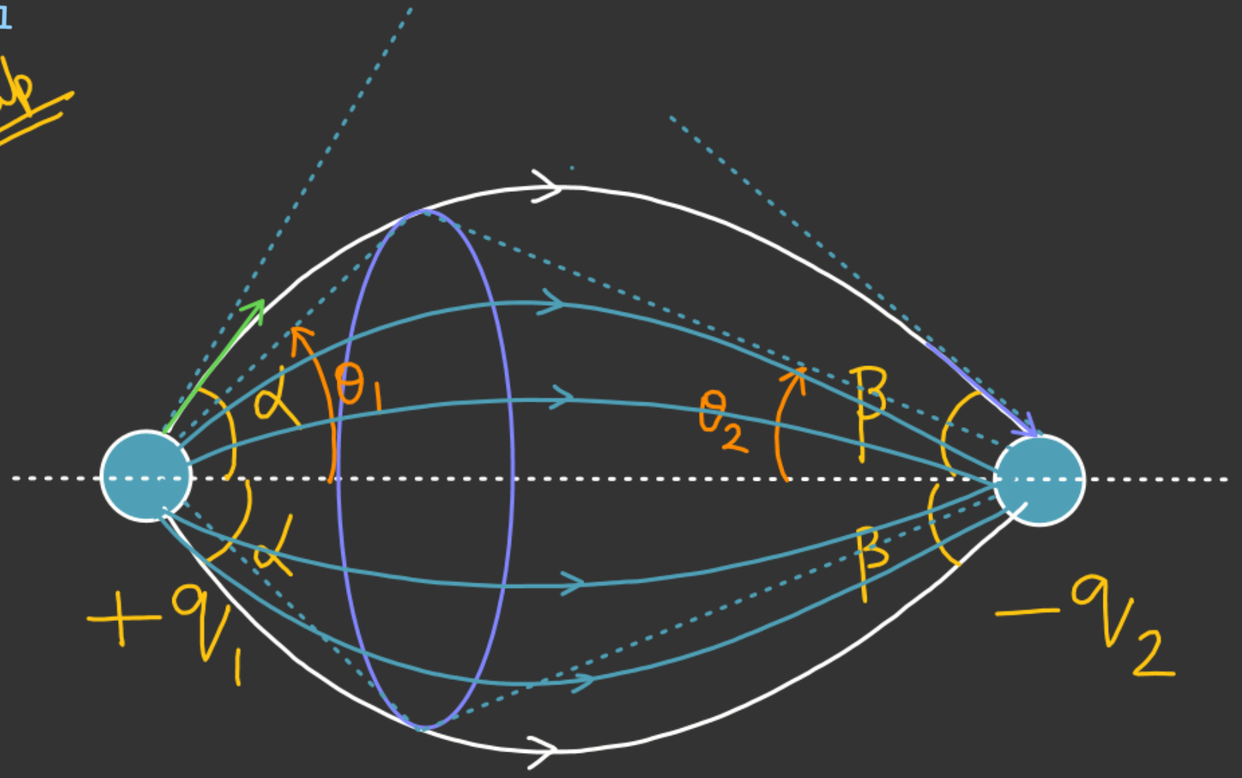
*V. Imp



$$\phi_{q_1}\Big|_C = \phi_{q_2}\Big|_C = \phi_{q_1 \& q_2}\Big|_C$$

$$\frac{q_1}{2\epsilon_0}\left(1-\cos\alpha\right) = \frac{q_2}{2\epsilon_0}\left(1-\cos\beta\right) = \frac{q_1}{2\epsilon_0}\left(1-\cos\theta_1\right) + \frac{q_2}{2\epsilon_0}\left(1-\cos\theta_2\right)$$

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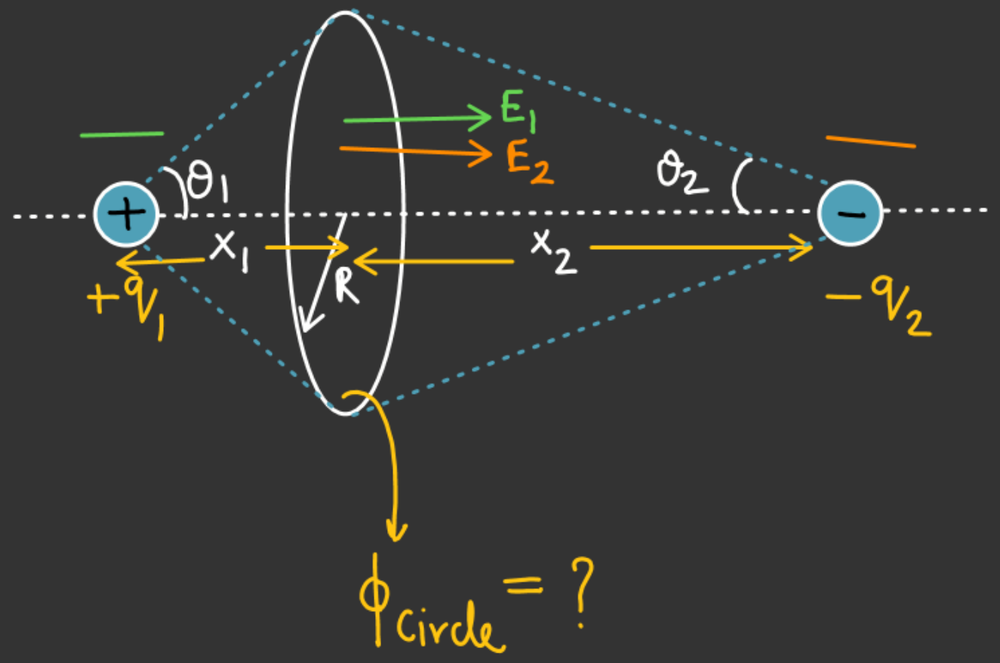

$$\phi_{circle} = \int (\vec{E}_1 + \vec{E}_2) \cdot d\vec{A}_{circle}$$

$$\phi_{circle} = \frac{q_1}{2\epsilon_0}(1-\cos\theta_1) + \frac{q_2}{2\epsilon_0}(1-\cos\theta_2) = \frac{q_1}{2\epsilon_0}\left[1-\frac{x_1}{\sqrt{R^2+x_1^2}}\right] + \frac{q_2}{2\epsilon_0}\left[1-\frac{x_2}{\sqrt{R^2+x_2^2}}\right]$$

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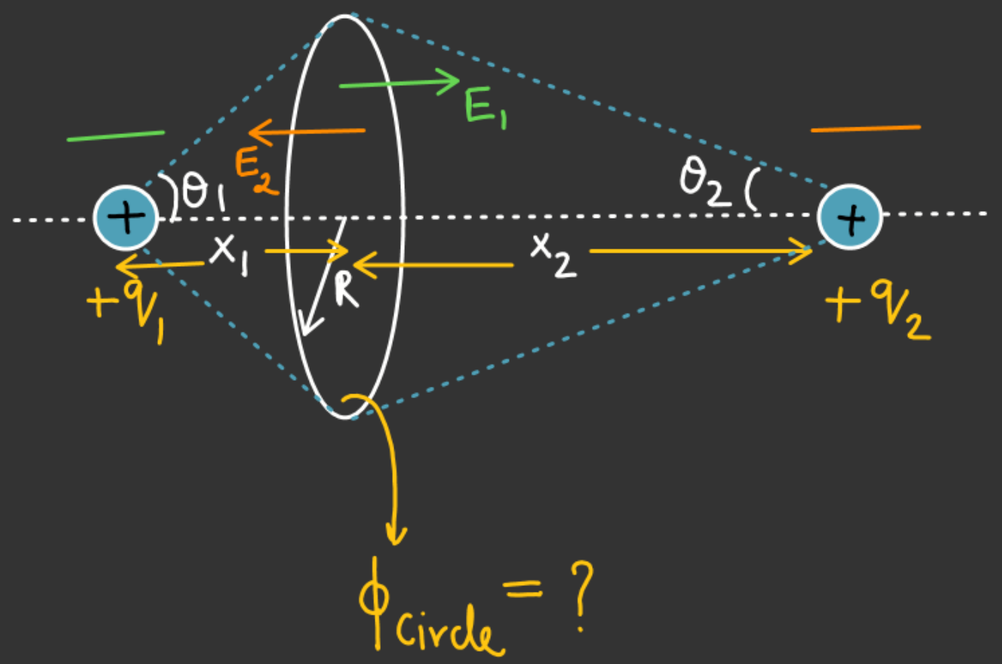

$$\phi_{circle} = \int |E_1 - E_2| \, dA_{circle}$$

$$\phi_{circle} = \left| \frac{q_1}{2\epsilon_0}(1-\cos\theta_1) - \frac{q_2}{2\epsilon_0}(1-\cos\theta_2) \right| = \left| \frac{q_1}{2\epsilon_0}\left(1-\frac{x_1}{\sqrt{R^2+x_1^2}}\right) - \frac{q_2}{2\epsilon_0}\left(1-\frac{x_2}{\sqrt{R^2+x_2^2}}\right) \right|$$

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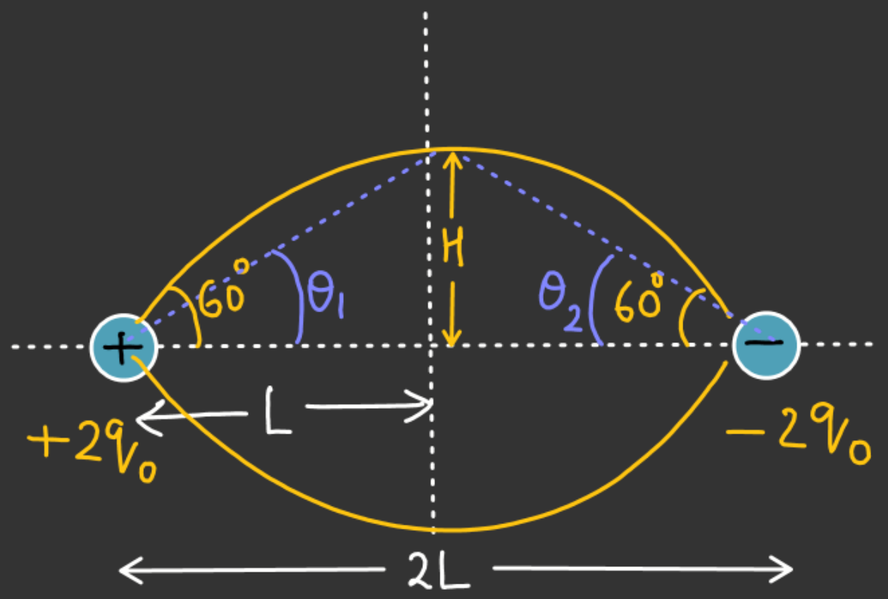

H = ?

$$\frac{2q_0}{2\epsilon_0}\left(1-\cos 60^\circ\right) = \left[\frac{2q_0}{2\epsilon_0}\left(1-\cos\theta_1\right)\right]\cdot 2$$

$$1-\frac{1}{2} = 2-2\cos\theta_1$$

$$2\cdot\frac{L}{\sqrt{L^2+H^2}} = 1+\frac{1}{2}$$

$$\frac{L}{\sqrt{L^2+H^2}} = \frac{3}{4}$$

$$\frac{L^2}{L^2+H^2} = \frac{9}{16}$$

$$7L^2 = 9H^2$$

$$H = \frac{\sqrt{7}L}{3}$$

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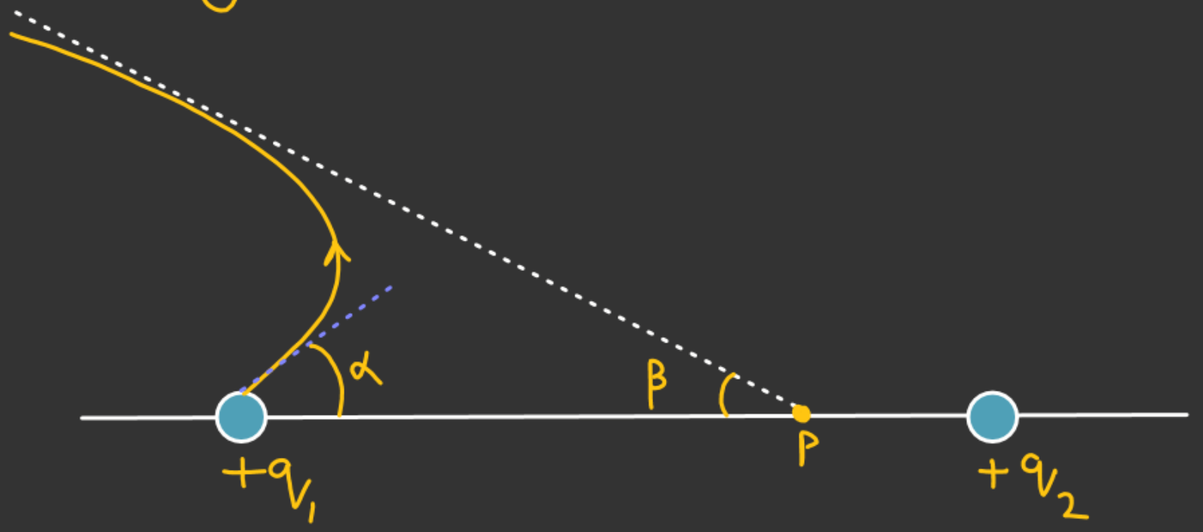
H.C.

Check magic of field lines on INSP YT



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$$\vec{E} = \frac{k\left(x\hat{i}+y\hat{j}+z^2\right)}{\sqrt{x^2+y^2+z^2}}$$

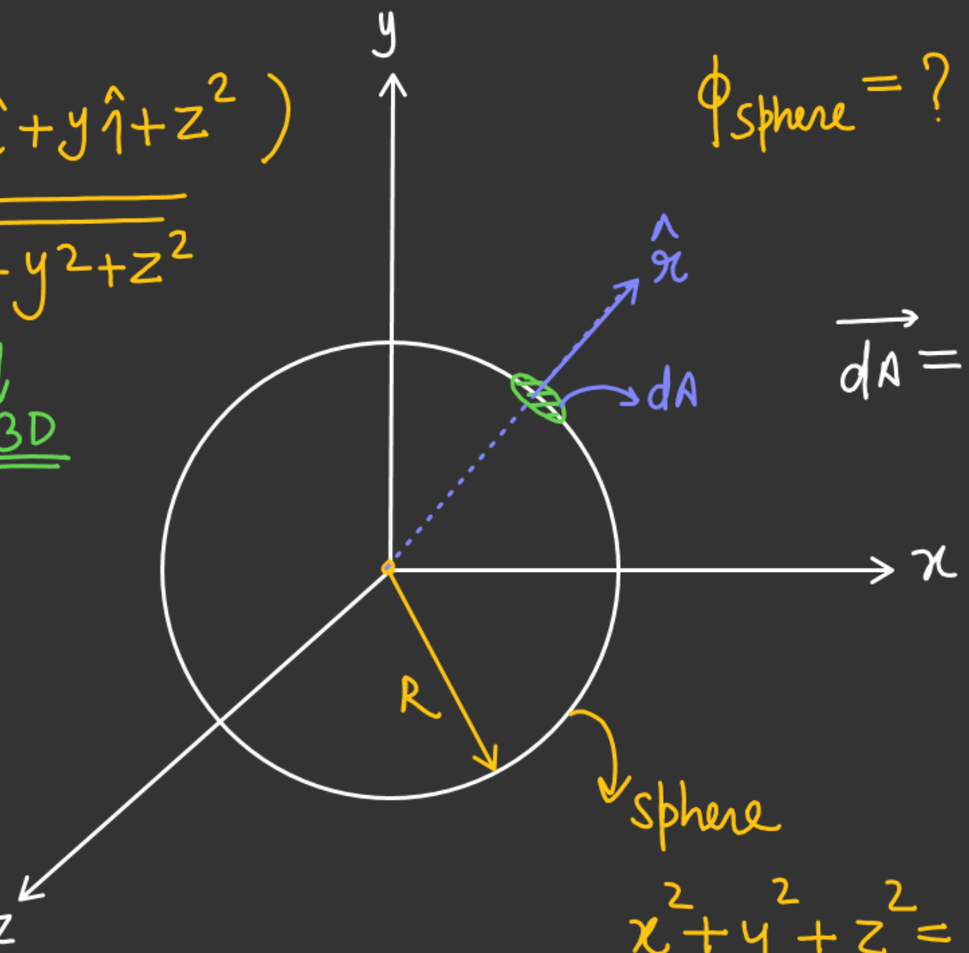
3D

$$\vec{E} = k\left(\hat{r}\right)$$

$$d\phi = \vec{E}\cdot\vec{dA}$$

$$\Rightarrow k\cdot dA$$

$$\phi_{sph} = \int d\phi = k\cdot\int dA = k\cdot 4\pi R^2$$



$\phi_{sphere} = ?$

$$\vec{dA} = dA\cdot\hat{r}$$

$$x^2+y^2+z^2 = R^2$$

$$\frac{q_{in}}{\epsilon_0} = k\cdot 4\pi R^2$$

$$q_{in} = 4\pi\epsilon_0 k R^2$$

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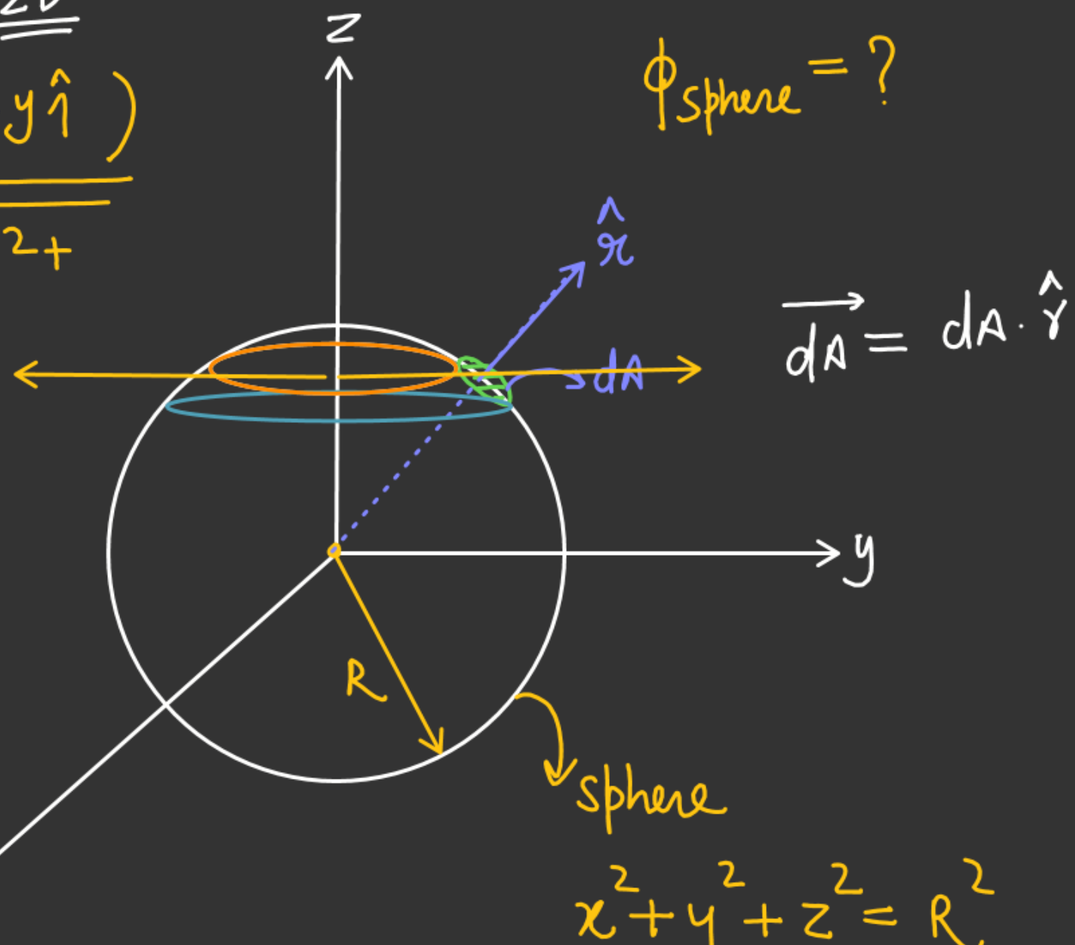
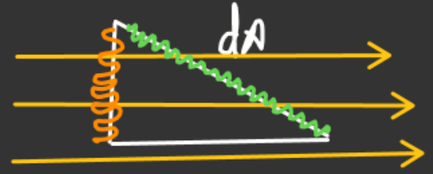
2D

$$\vec{E} = \frac{k(x\hat{i}+y\hat{j})}{\sqrt{x^2+y^2+}}$$

$\phi_{sphere} = ?$



$$\vec{dA} = dA \cdot \hat{r}$$



$$\vec{E} = k(\hat{r})$$

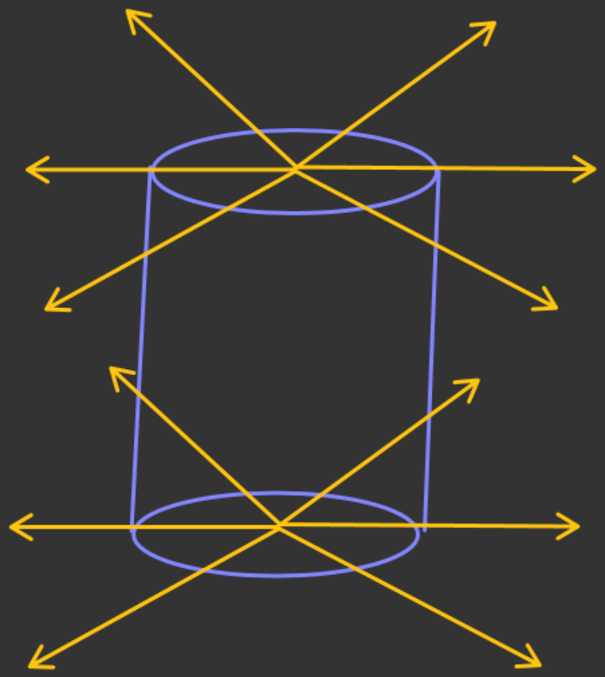
$$\phi_{sph} = \left[k \cdot \pi R^2\right] 2$$

$$= 2\pi k R^2$$

Ans

sphere

$$x^2+y^2+z^2 = R^2$$



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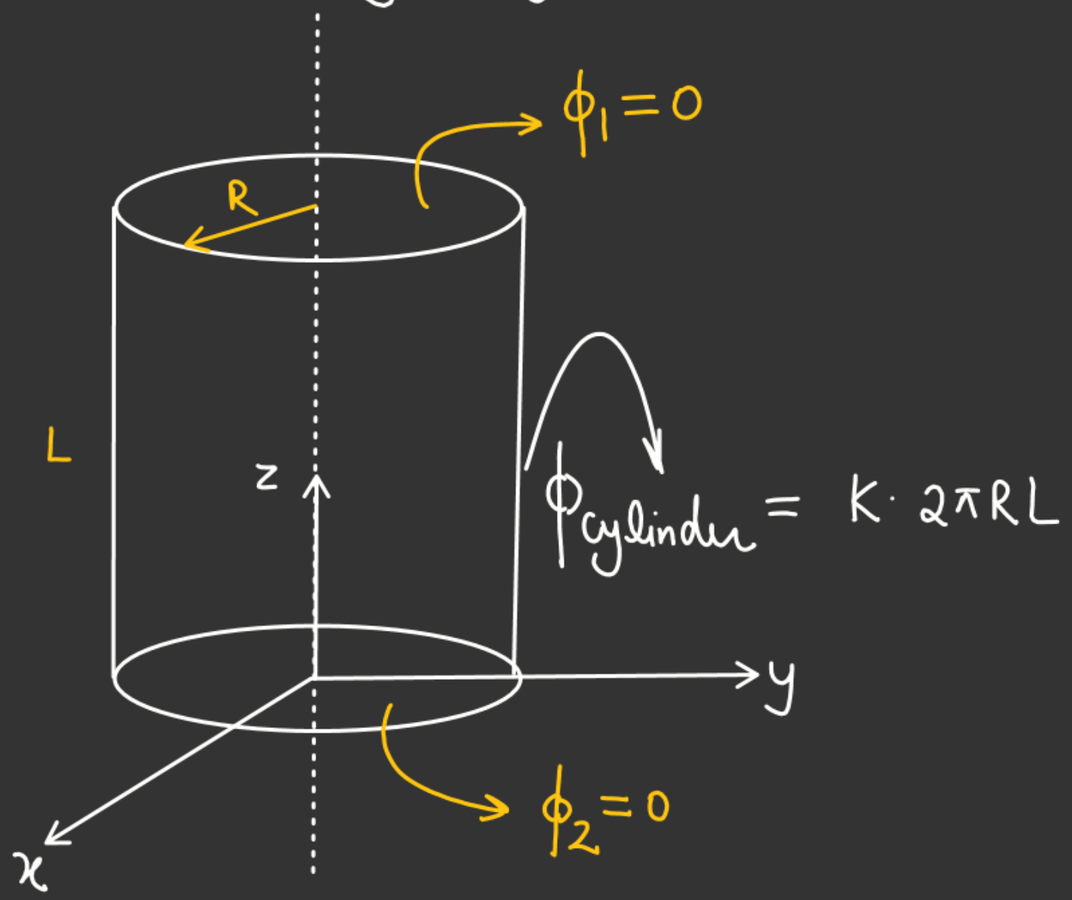
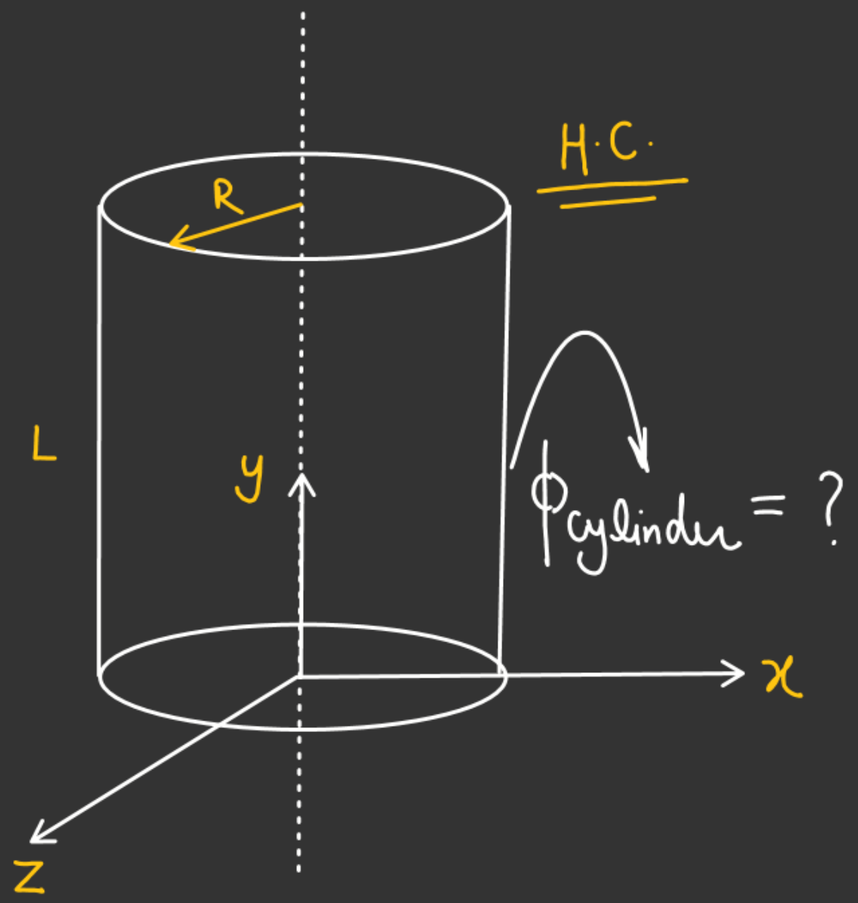
$$* \quad E = \frac{K(x\hat{i}+y\hat{j})}{\sqrt{x^2+y^2}}$$





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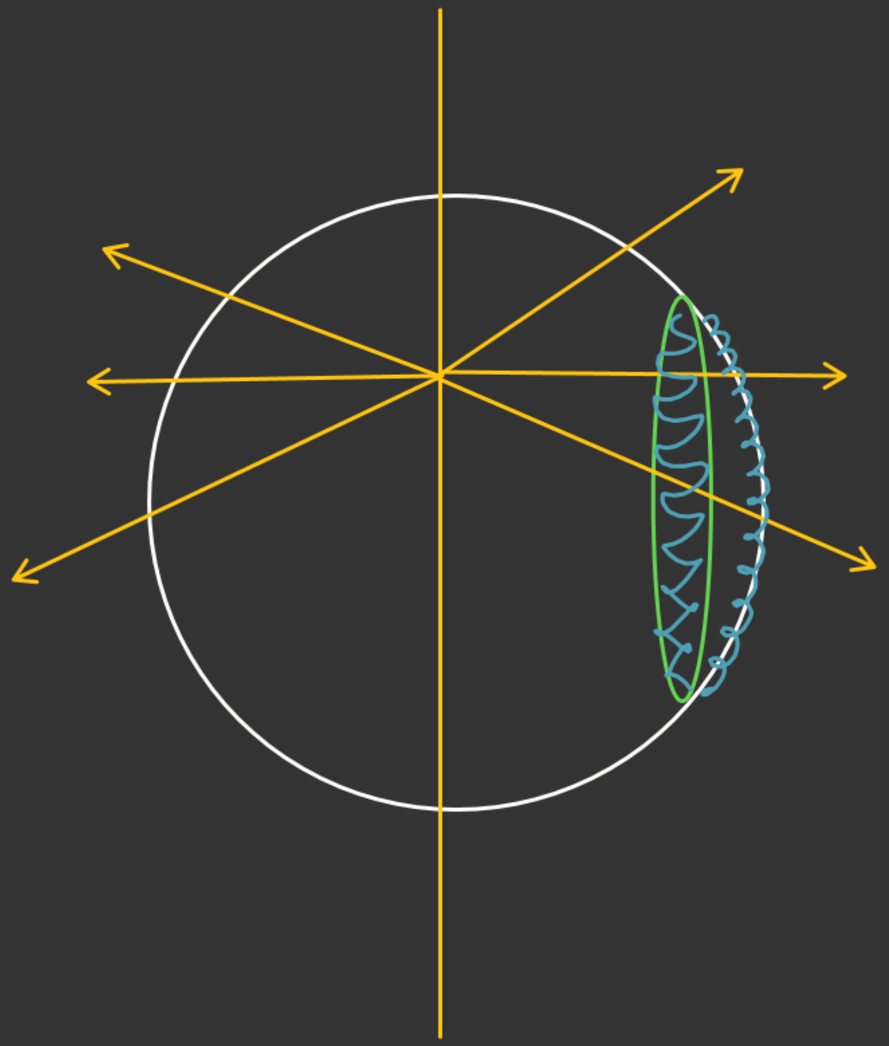

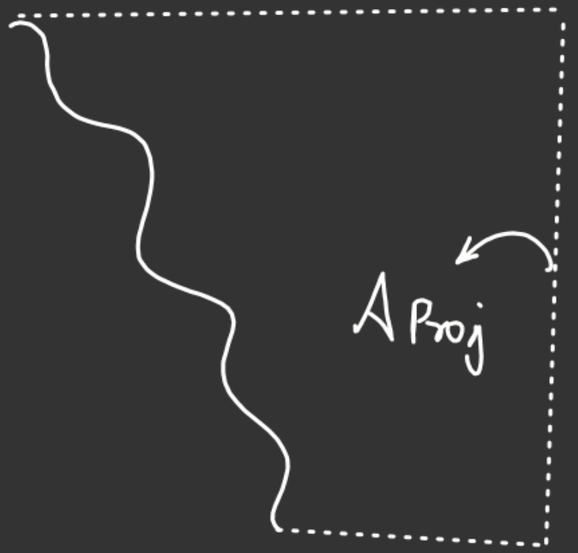

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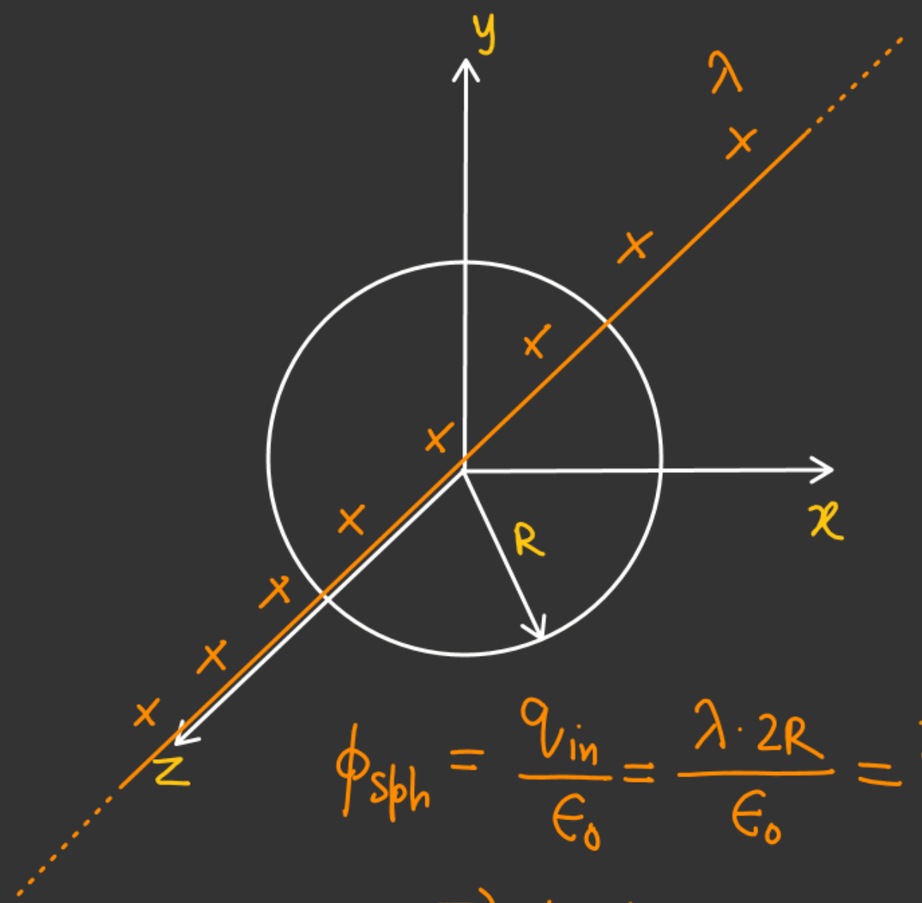
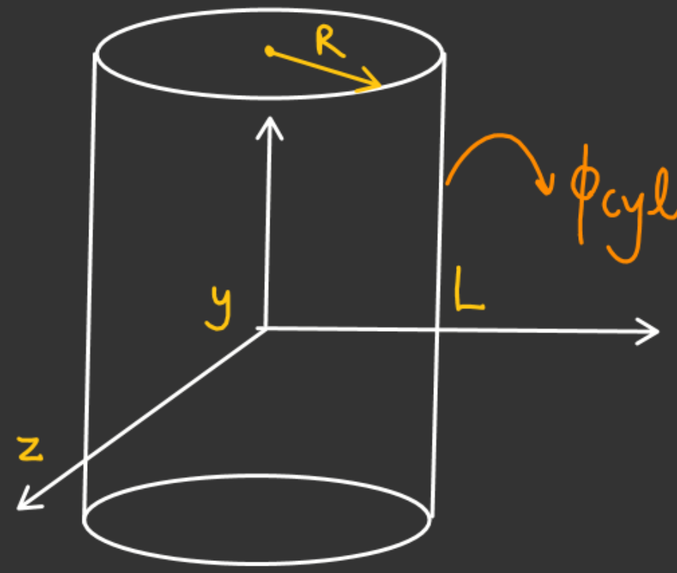
$$\vec{E} = K\left(\frac{x\hat{i}+y\hat{j}}{x^2+y^2}\right) \Rightarrow \frac{K}{\sqrt{x^2+y^2}}\cdot\left(\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}\right) \Rightarrow \frac{k}{r}(\hat{r}) \Rightarrow \frac{\lambda}{2\pi\epsilon_0 r}\cdot\hat{r}$$

$$\lambda = 2\pi\epsilon_0 K$$



$$\phi_{sph} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda \cdot 2R}{\epsilon_0} = \frac{2\pi\epsilon_0 K \cdot 2R}{\epsilon_0}$$

$$\Rightarrow 4\pi kR$$



$$\phi_{cylinder} = 4\pi kR$$

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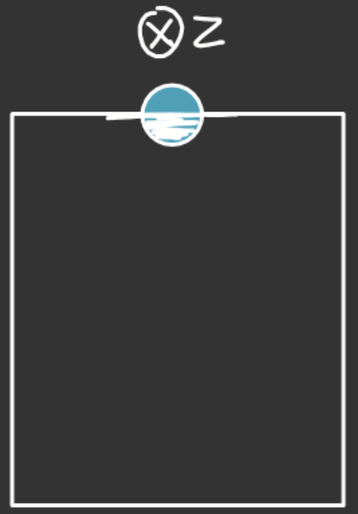
$$\vec{E} = K\left(\frac{x\hat{i}+y\hat{j}}{x^2+y^2}\right) \Rightarrow \frac{K}{\sqrt{x^2+y^2}}\cdot\left(\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}\right) \Rightarrow \frac{K}{r}(\hat{r}) \Rightarrow \frac{\lambda}{2\pi\epsilon_0 r}\cdot\hat{r}$$

$$\lambda = 2\pi\epsilon_0 K$$

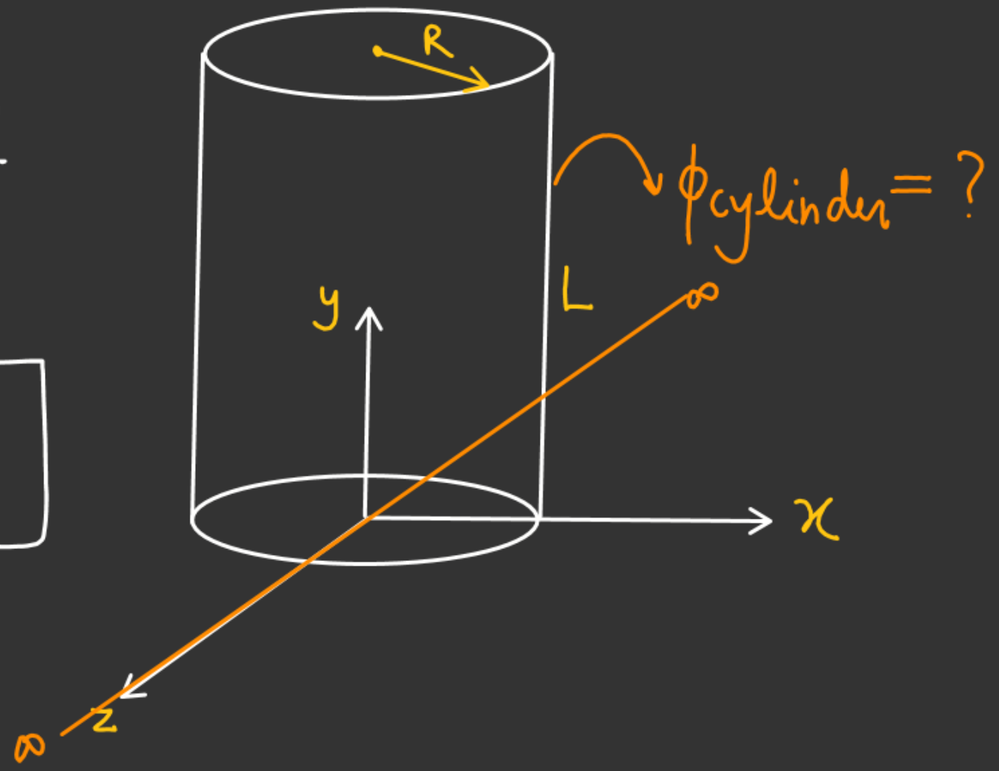
$$\phi_{cyc} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda \cdot 2R}{\epsilon_0} = 4\pi K R$$

Wrong



$$\phi_{cyl} = 2\pi K R$$

Correct



$\phi_{cylinder} = ?$

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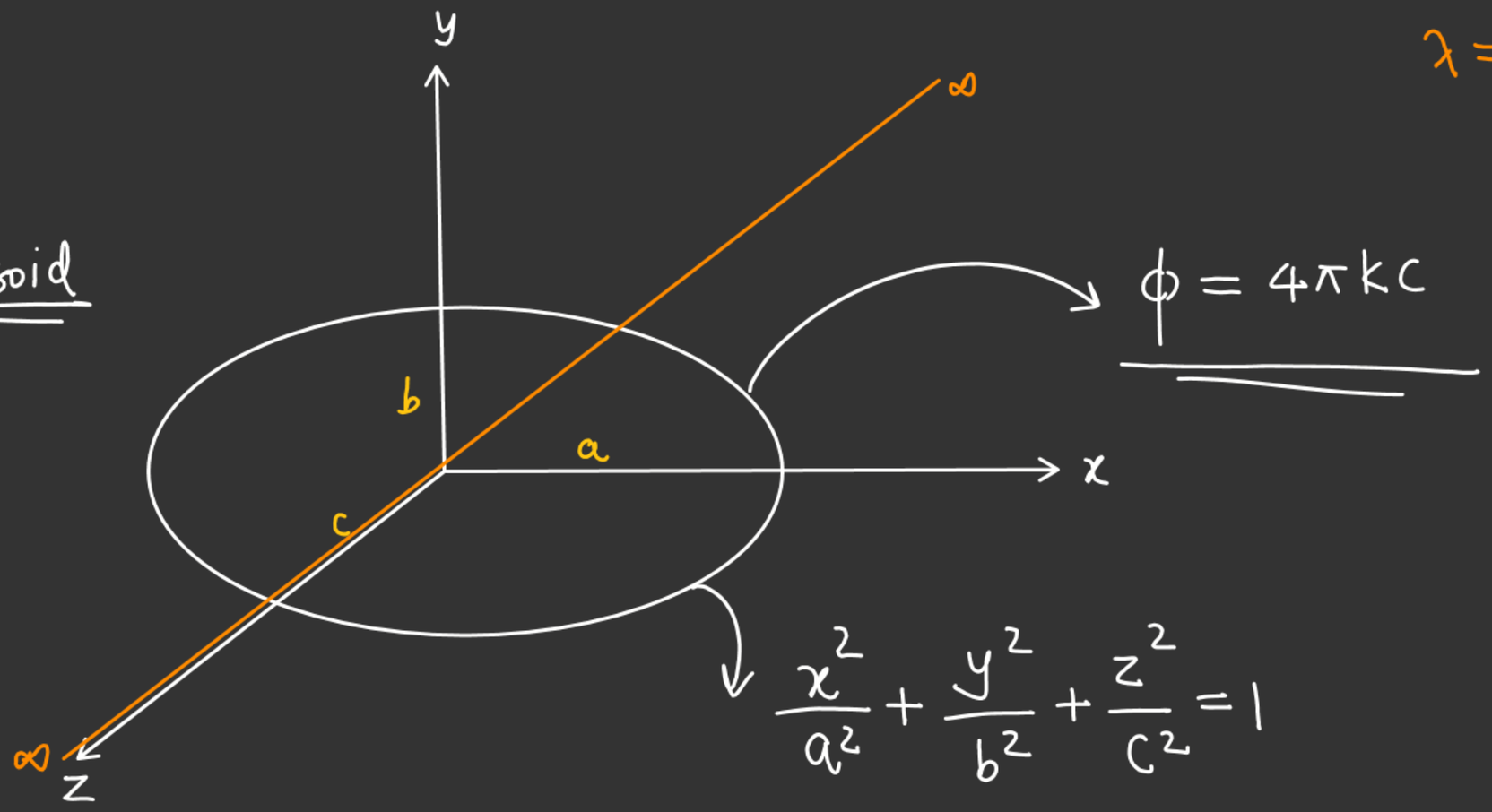
$$\ast \quad \vec{E} = K\left(\frac{x\hat{i}+y\hat{j}}{x^2+y^2}\right) \Rightarrow \frac{K}{\sqrt{x^2+y^2}}\cdot\left(\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}\right) \Rightarrow \frac{k}{r}(\hat{r}) \Rightarrow \frac{\lambda}{2\pi\epsilon_0 r}\cdot\hat{r}$$

$$\lambda = 2\pi\epsilon_0 K$$

ellipsoid



$$\phi = 4\pi kc$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

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* $\vec{E} = K\left(\frac{x\hat{i}+y\hat{j}}{x^2+y^2}\right) \Rightarrow \frac{K}{\sqrt{x^2+y^2}} \cdot \left(\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}\right) \Rightarrow \frac{k}{r}(\hat{r}) \Rightarrow \frac{\lambda}{2\pi\epsilon_0 r} \cdot \hat{r}$

H.C.

$\lambda = 2\pi\epsilon_0 K$

* find the flux through

$$x^4 + y^4 + z^4 = 81$$

Nishant Jindal

T

P C_1 C_2 C_3 M

T/3 T/9 T/9 T/9 T/3

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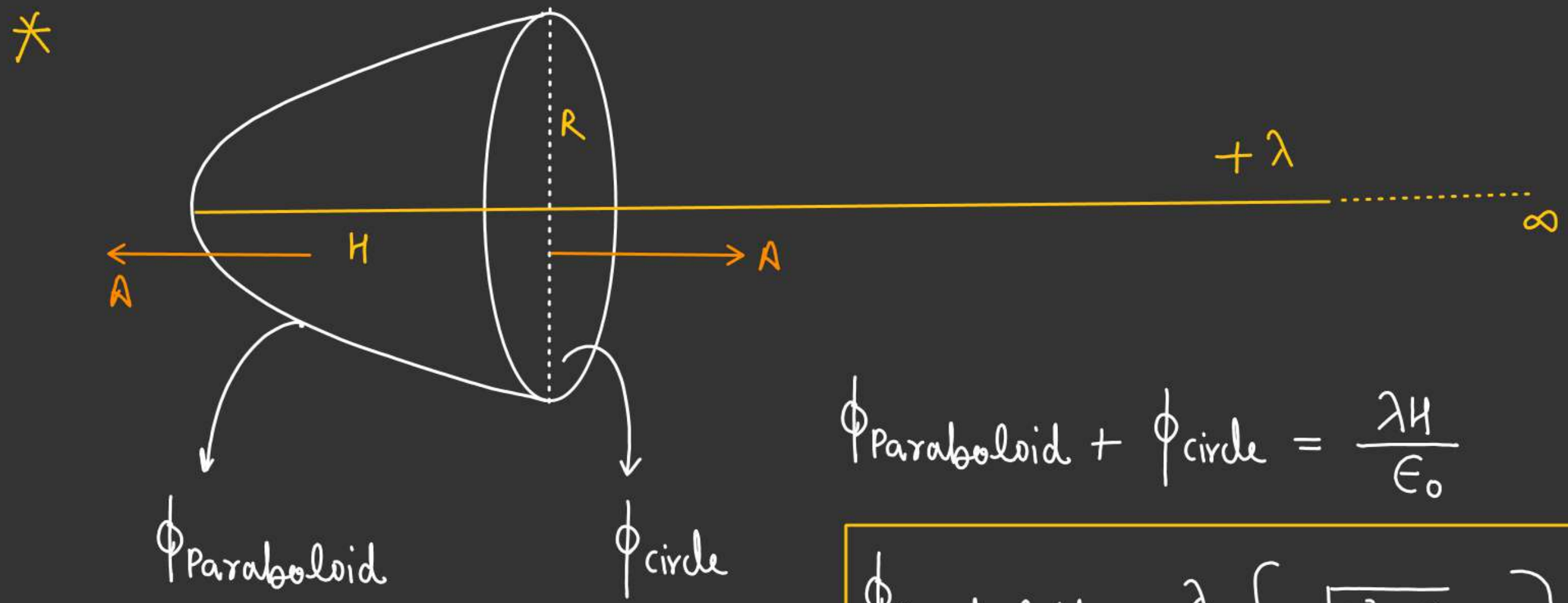

$$\phi_{Paraboloid} + \phi_{circle} = \frac{\lambda H}{\epsilon_0}$$

$$\boxed{\phi_{Paraboloid} - \frac{\lambda}{2\epsilon_0}\left[\sqrt{H^2+R^2} - H\right] = \frac{\lambda H}{\epsilon_0}}$$

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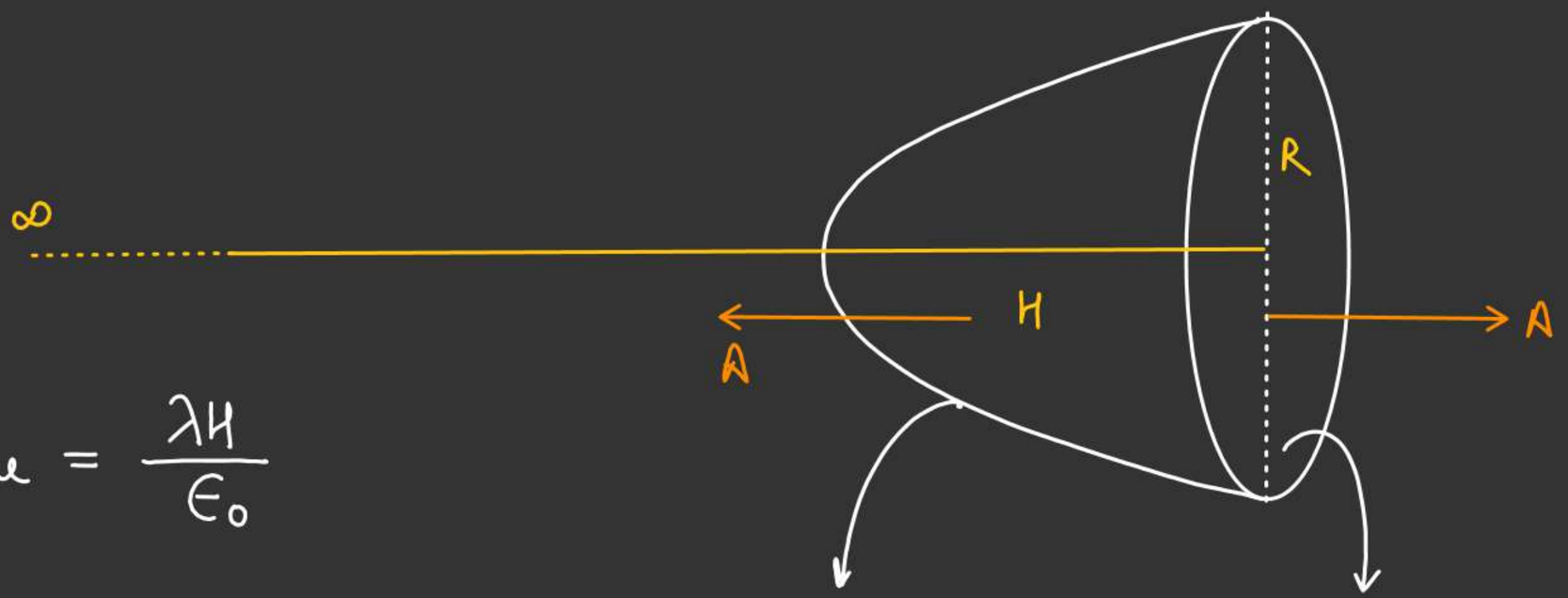

$$\phi_{Paraboloid} + \phi_{circle} = \frac{\lambda H}{\epsilon_0}$$

$$\phi_{Paraboloid} + \frac{\lambda R}{2\epsilon_0} = \frac{\lambda H}{\epsilon_0}$$

$\phi_{Paraboloid} = ?$ ϕ_{circle}

$$\phi_{Paraboloid} = \frac{\lambda}{\epsilon_0}\left[H - \frac{R}{2}\right]$$

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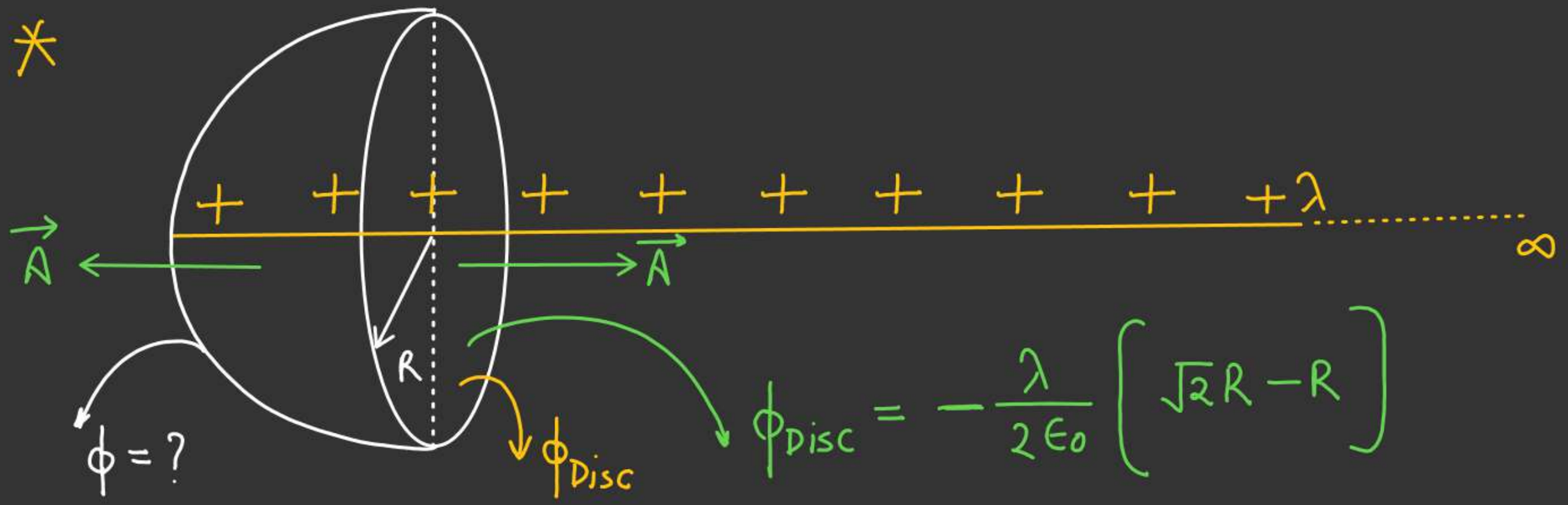

$$\phi_{Disc} = -\frac{\lambda}{2\epsilon_0}\left[\sqrt{2}R - R\right]$$

$$\phi_{sph} + \phi_{Disc} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda R}{\epsilon_0}$$

$$\phi_{sph} - \frac{\lambda R}{2\epsilon_0}\left[\sqrt{2}-1\right] = \frac{\lambda R}{\epsilon_0} \quad \Rightarrow \quad \phi_{sph} = \checkmark$$

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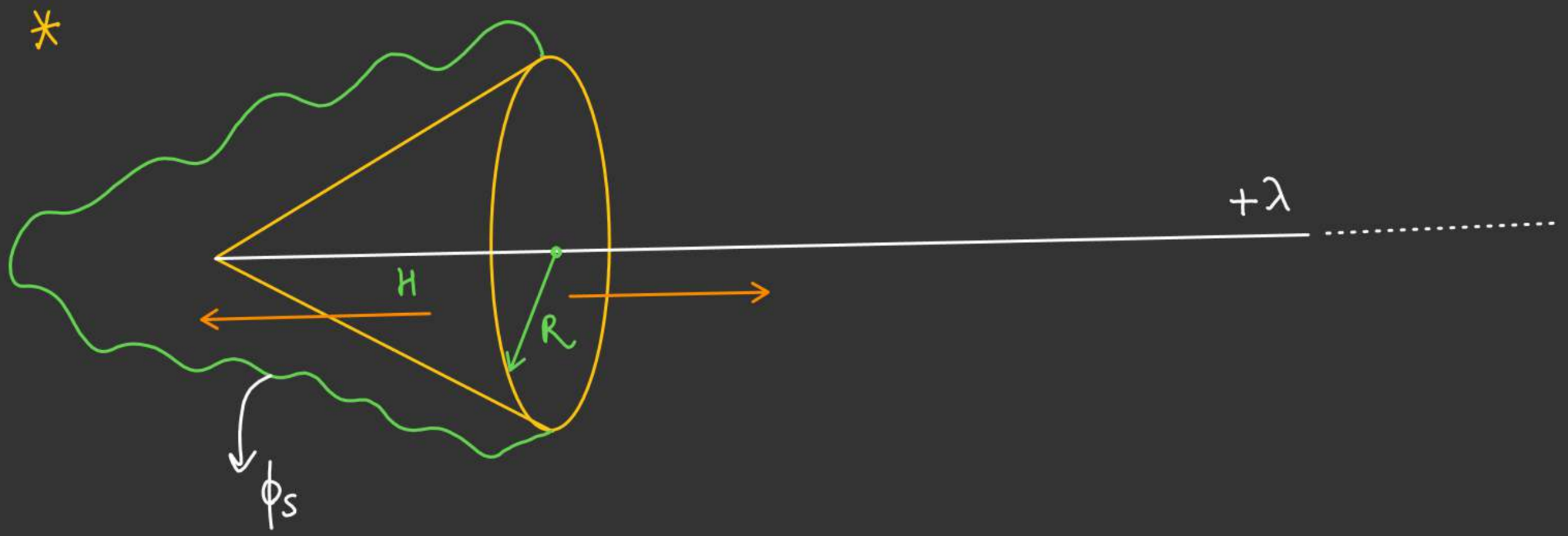

$$\phi_{cone} + \phi_{circle} = \frac{\lambda H}{\epsilon_0}$$

$\phi_{cone} = \phi_s$

$$\phi_{circle} = -\frac{\lambda}{2\epsilon_0}\left[\sqrt{H^2+R^2} - H\right]$$

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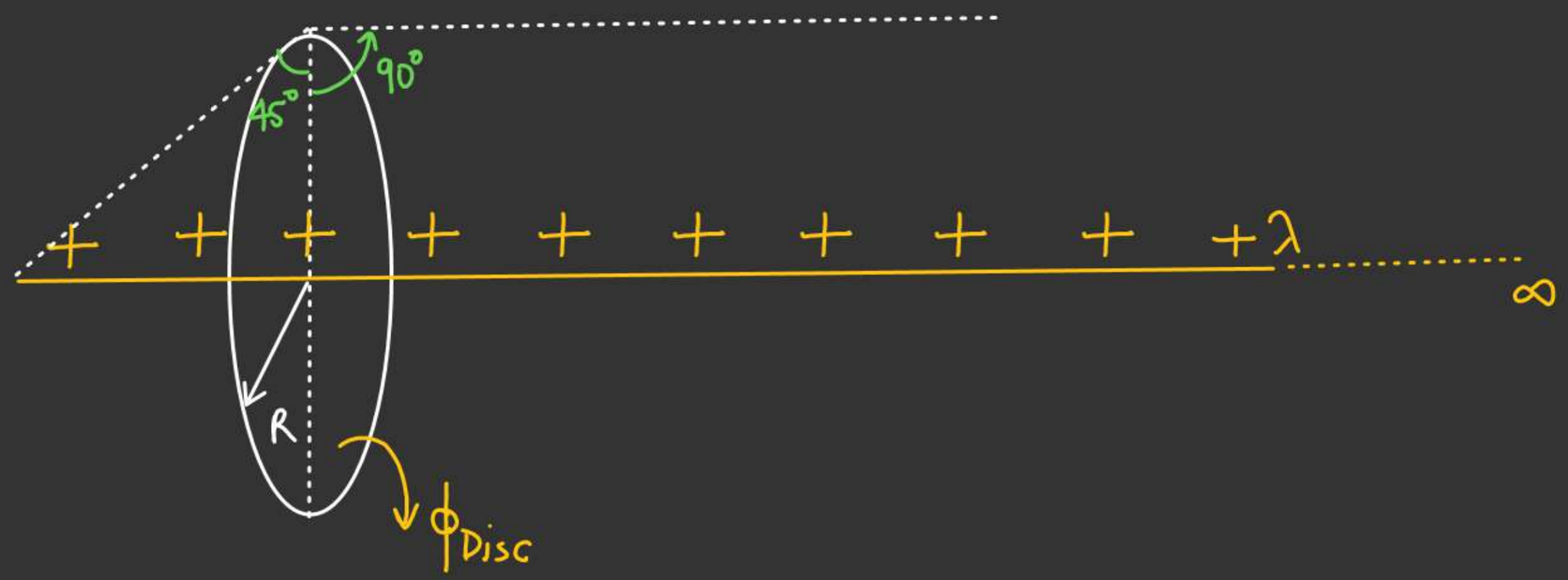

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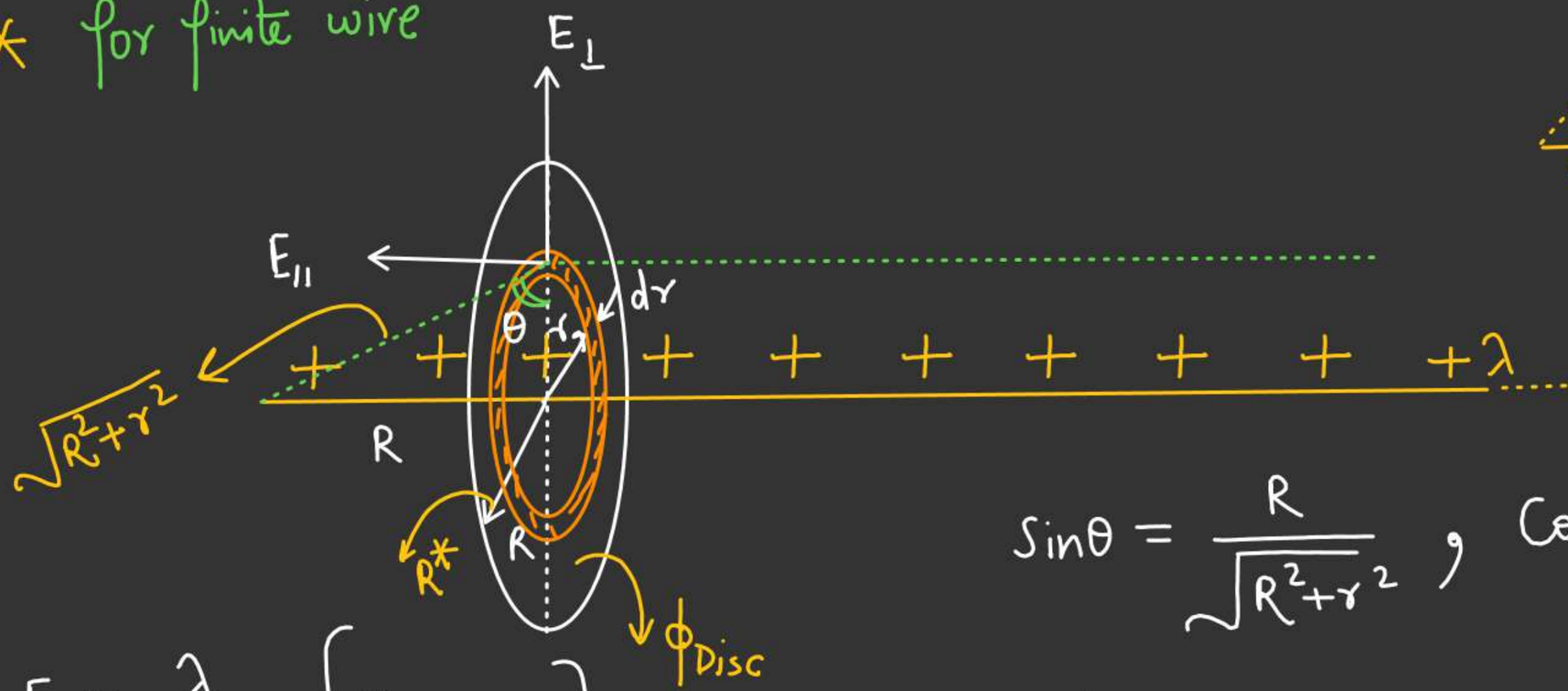
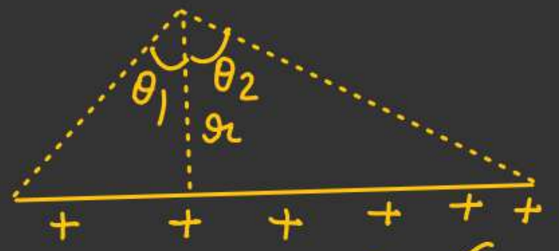
* for finite wire





$$E_{||} = \frac{\lambda}{4\pi\epsilon_0 r}\left[\sin\theta_1 + \sin\theta_2\right]$$

$$\sin\theta = \frac{R}{\sqrt{R^2+r^2}}, \quad \cos\theta = \frac{r}{\sqrt{R^2+r^2}}$$

$$E_{||} = \frac{\lambda}{4\pi\epsilon_0 r}\left[\cos\theta\right]$$

$$\phi_{E_\perp} = 0$$

$$d\phi = E_{||}\cdot 2\pi r\cdot dr = \frac{\lambda}{\cancel{4}\,\cancel{2}\,\pi\epsilon_0 r}\left[\cos\theta\right]\cdot \cancel{2\pi r}\,dr = \frac{\lambda}{2\epsilon_0}\cos\theta\cdot dr = \frac{\lambda}{2\epsilon_0}\cdot\frac{r}{\sqrt{R^2+r^2}}\cdot dr$$

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$$E_{\parallel} = \frac{\lambda}{4\pi\epsilon_0 r}\left[\cos\theta\right]$$

$$\sin\theta = \frac{R}{\sqrt{R^2+r^2}}, \quad \cos\theta = \frac{r}{\sqrt{R^2+r^2}}$$

$$\phi_{E_\perp} = 0$$

$$d\phi = E_{\parallel}\cdot 2\pi r\cdot dr = \frac{\lambda}{4\pi\epsilon_0 r}\left(\cos\theta\cdot\right)\cdot 2\pi r\,dr = \frac{\lambda}{2\epsilon_0}\cos\theta\cdot dr = \frac{\lambda}{2\epsilon_0}\cdot\frac{r}{\sqrt{R^2+r^2}}\cdot dr$$

$$\phi_{circle} = \frac{\lambda}{2\epsilon_0}\int_0^{R^*}\frac{r\,dr}{\sqrt{R^2+r^2}}$$

$$R^2+r^2 = v^2$$

$$r\cdot dr = v\cdot dv$$

$$= \frac{\lambda}{2\epsilon_0}\int\frac{v\cdot dv}{v} = \frac{\lambda}{2\epsilon_0}\int_R^{\sqrt{R^2+R^{*2}}}\cdot dv \Rightarrow \frac{\lambda}{2\epsilon_0}\left[\sqrt{R^2+R^{*2}} - R\right]$$

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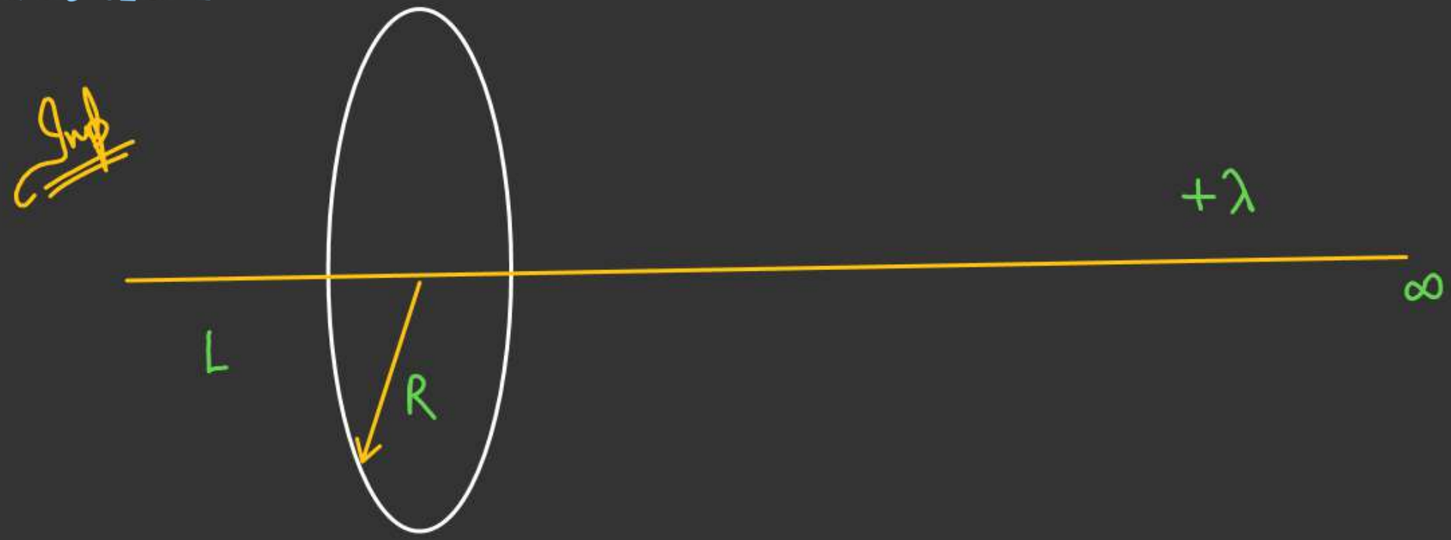
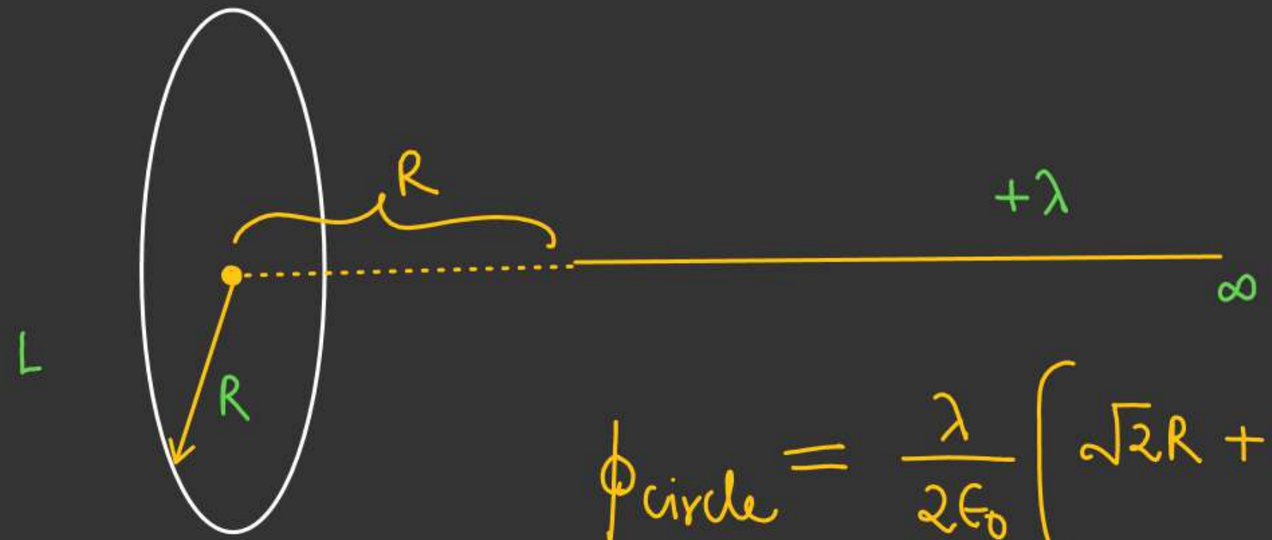


$$\phi_{circle} = \frac{\lambda}{2\epsilon_0}\left[\sqrt{L^2+R^2} - L\right]$$



$$\phi_{circle} = \frac{\lambda}{2\epsilon_0}\left[\sqrt{2}R + R\right]$$

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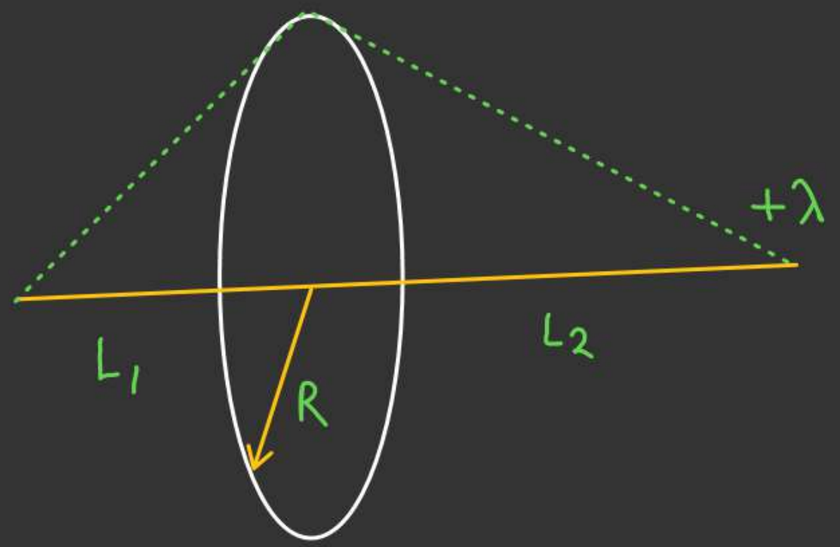

$$\frac{\lambda}{4\pi\epsilon_0 r}\cdot\left[\cos\theta_1-\cos\theta_2\right]\cdot 2\pi r\,dr$$

$$d\phi=\frac{\lambda}{2\epsilon_0}\cos\theta_1\cdot dr-\frac{\lambda}{2\epsilon_0}\cdot\cos\theta_2\cdot dr$$

$$\phi_{circle}=\frac{\lambda}{2\epsilon_0}\left[\left(\sqrt{L_1^2+R^2}-L_1\right)-\left(\sqrt{L_2^2+R^2}-L_2\right)\right]$$

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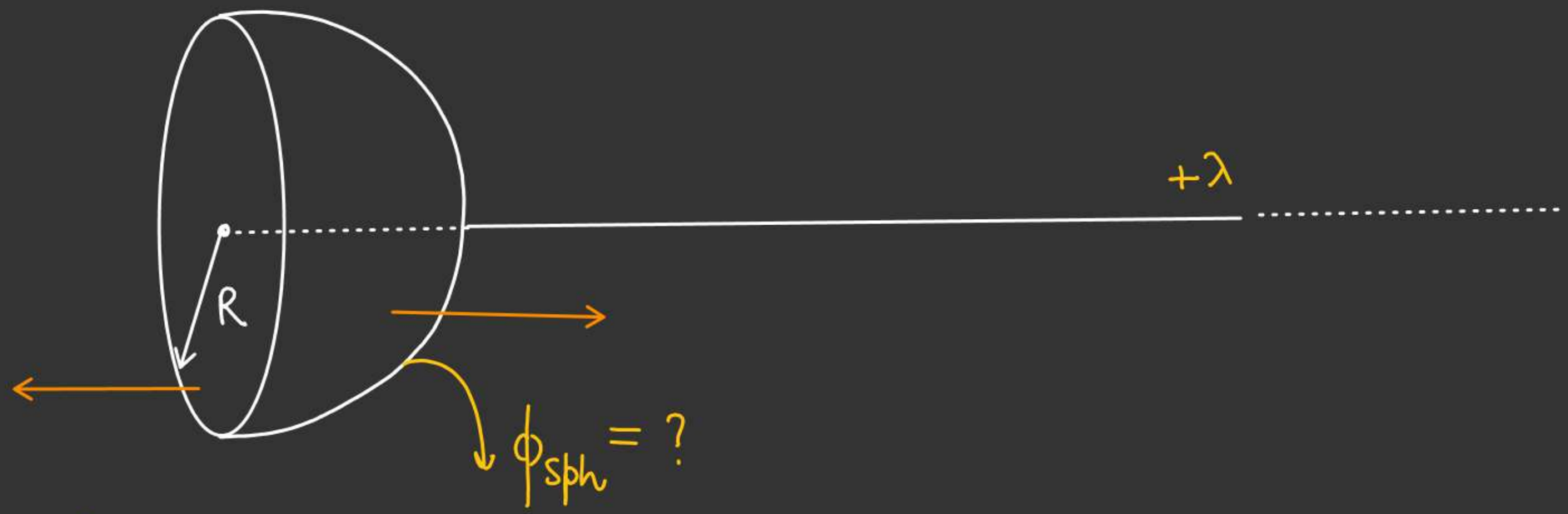

$$\phi_{circle} + \phi_{sph} = 0$$

$$\downarrow$$

$$\frac{\lambda}{2\epsilon_0}\left[\sqrt{R^2+R^2} + R\right] + \phi_{sph} = 0$$

$$\Rightarrow \boxed{\phi_{sph} = -\frac{\lambda R}{2\epsilon_0}\left(\sqrt{2}+1\right)}$$

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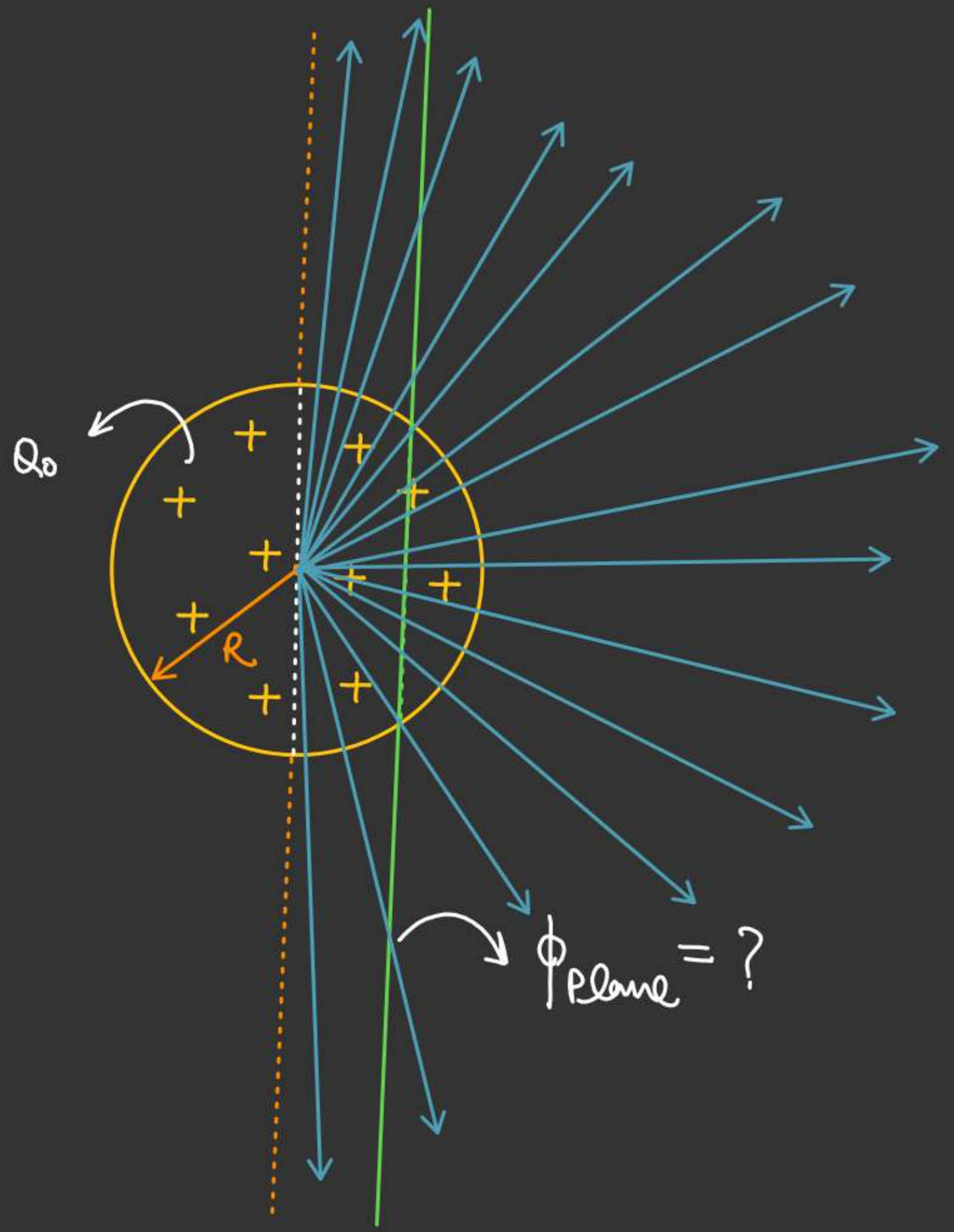
* H.C.

Solid Non Conducting Sphere :-

$$\phi_{Plane} = \frac{Q}{2\epsilon_0}$$



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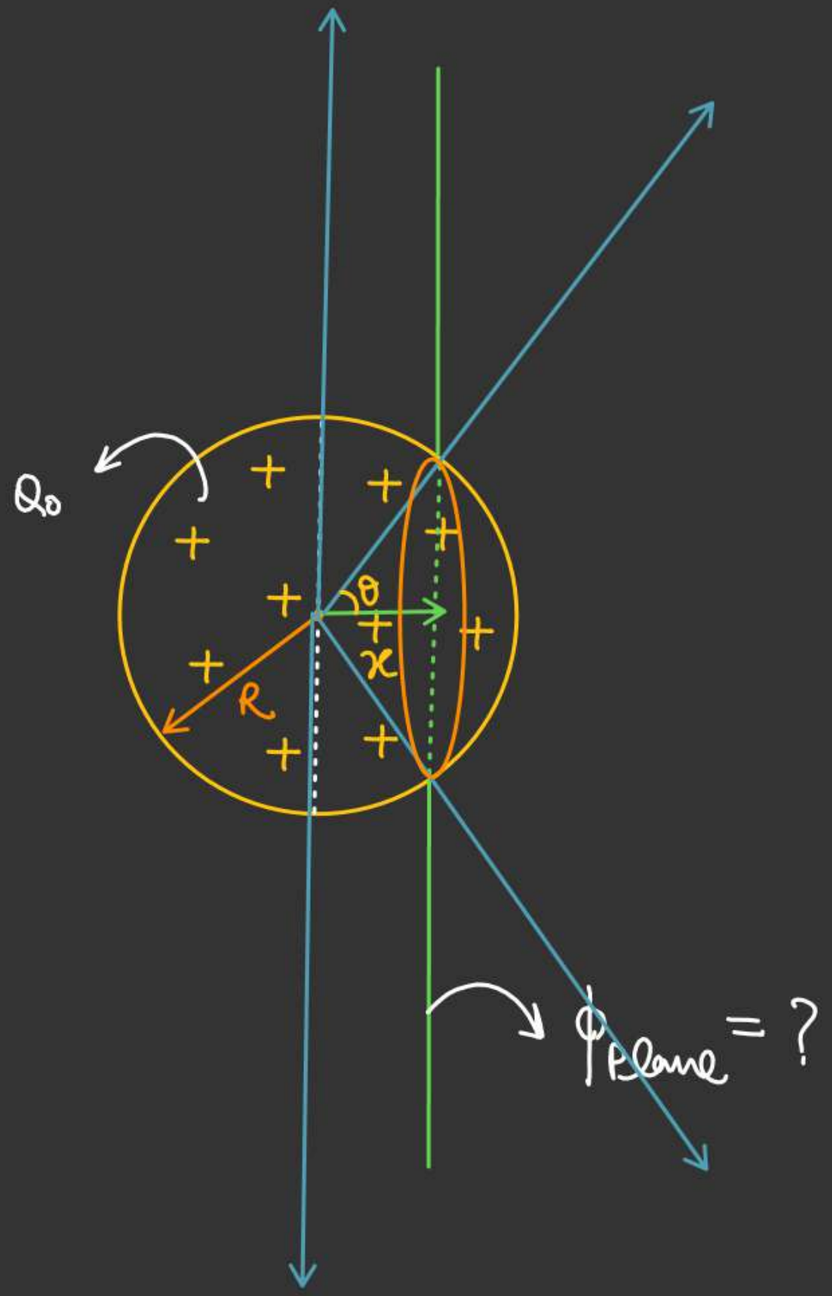
Solid Non Conducting Sphere :-



$$\phi_{Plane} = \frac{Q}{2\epsilon_0} - \phi_{circle}$$

$$\Rightarrow \frac{Q}{2\epsilon_0} - \frac{Q}{2\epsilon_0}(1 - \cos\theta)$$

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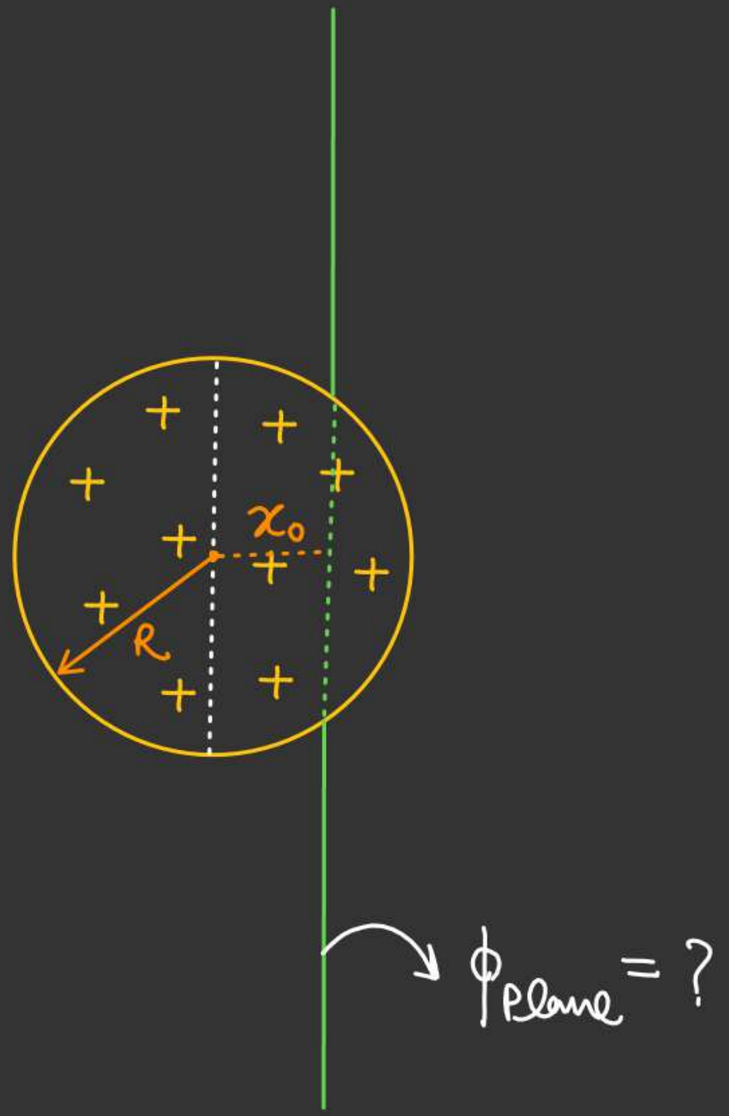
* H.C. Originals

Solid Non Conducting Sphere :-

$$\rho_r = \frac{\rho_0 r}{R}$$



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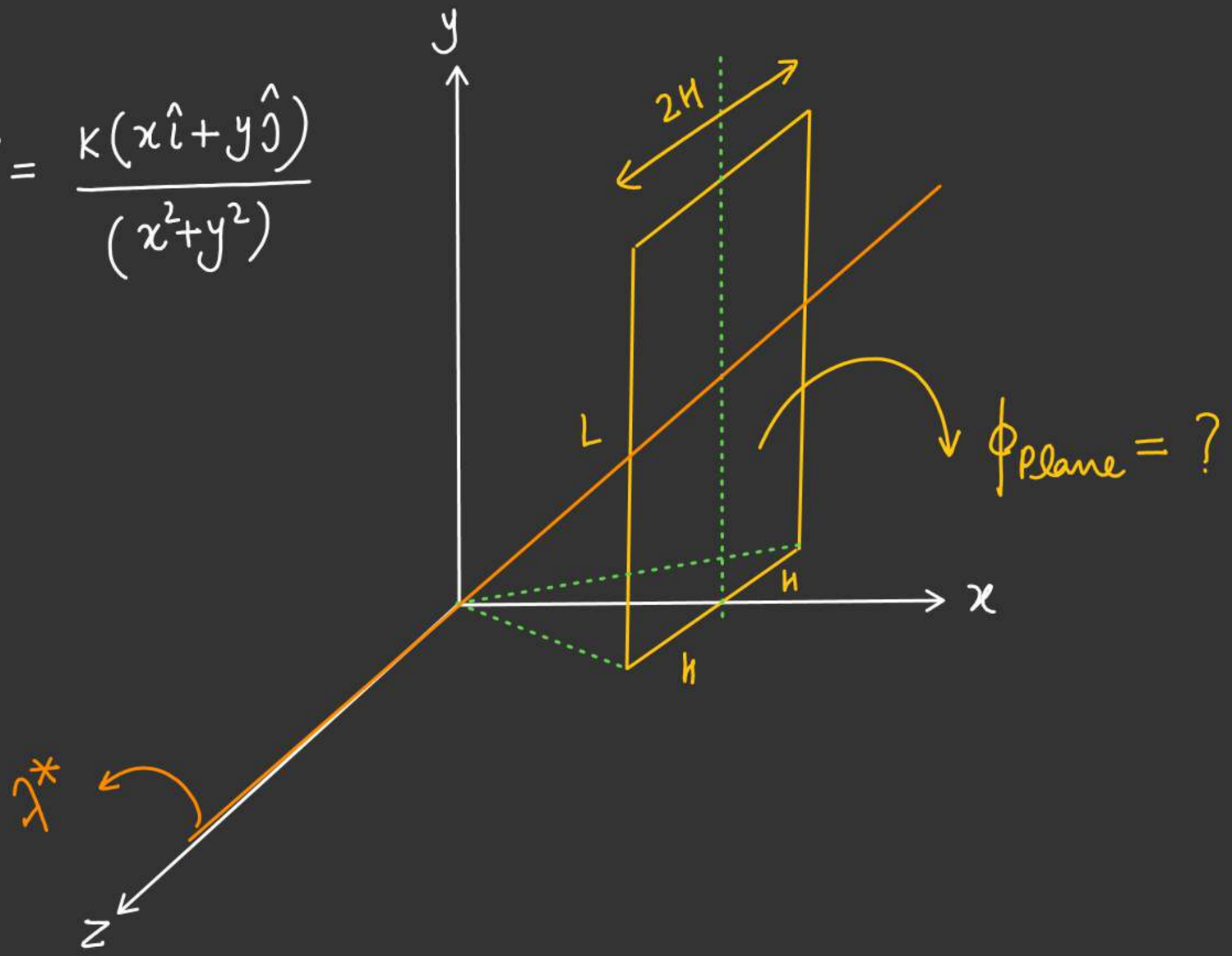
$$\vec{E} = \frac{k(x\hat{i}+y\hat{j})}{(x^2+y^2)}$$



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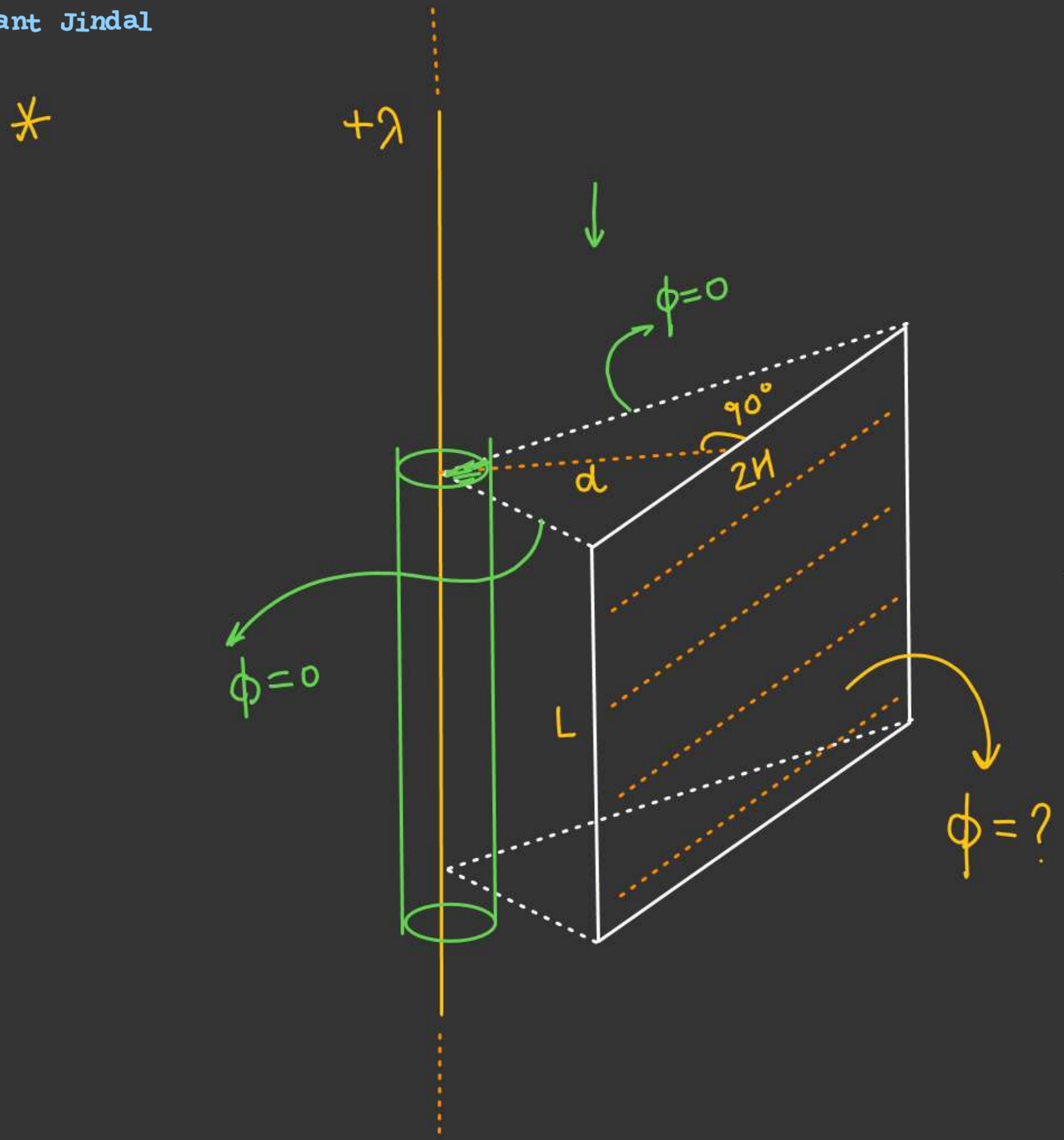
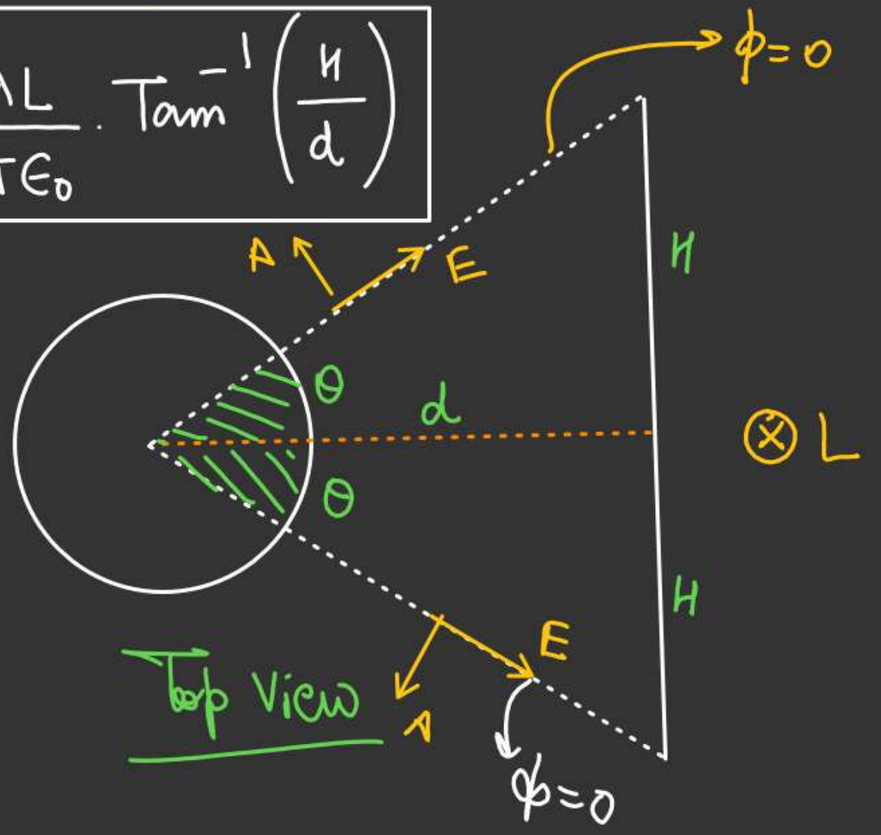


$$\phi = \frac{q_{in}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}\left[\frac{2\theta}{2\pi}\right] = \frac{\lambda L}{\pi\epsilon_0}\cdot\theta$$

$$\boxed{\phi = \frac{\lambda L}{\pi\epsilon_0}\cdot \tan^{-1}\left(\frac{H}{d}\right)}$$



Nishant Jindal

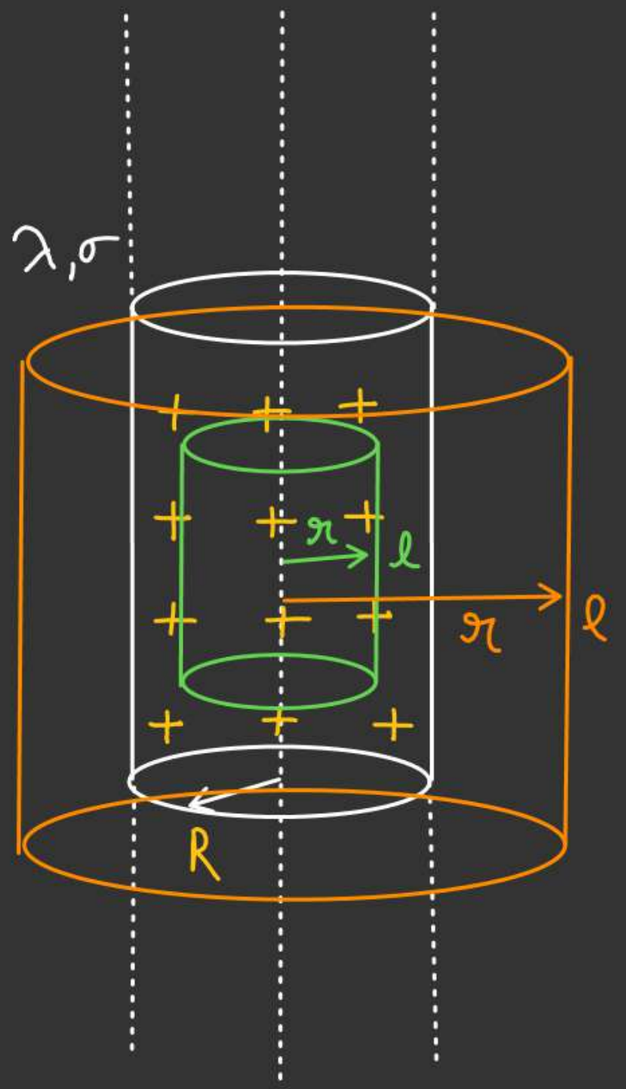
✳ Field and potential of Cylinder :-

(i) Shell

$\lambda \cdot L = \sigma \cdot 2\pi R L$

$\lambda = \sigma \cdot 2\pi R$

Conducting & Non Conducting



$r < R$

$E \cdot 2\pi r l = \frac{q_{in}}{\epsilon_0} = 0$

$E = 0$

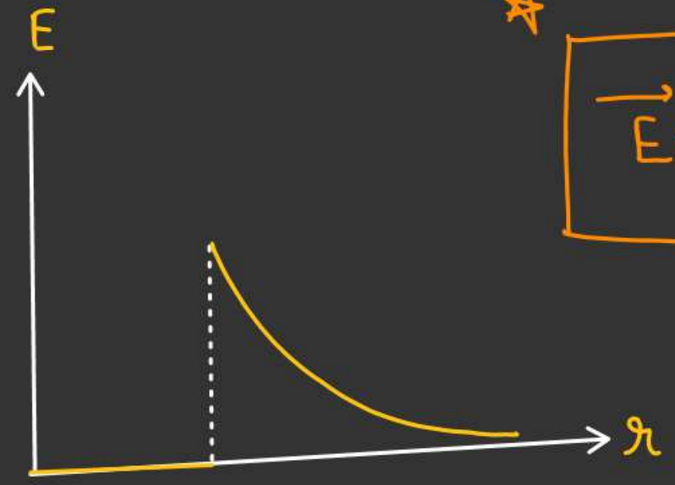
$r > R$

$E \cdot 2\pi r l = \frac{\lambda l}{\epsilon_0}$

$E = \frac{\lambda}{2\pi\epsilon_0 r}$

✳

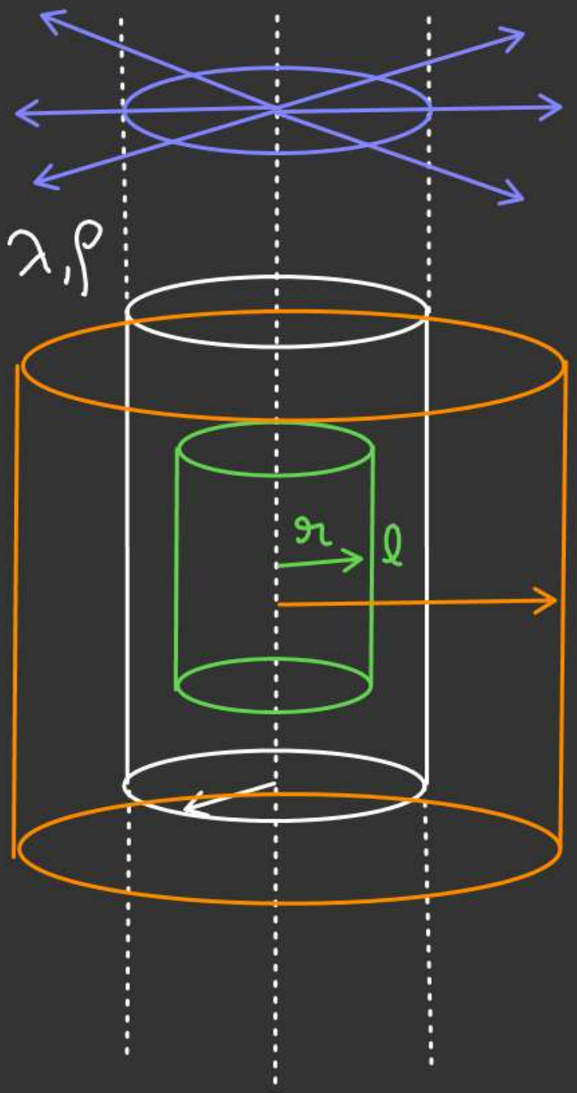
$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}(\hat{r})$$



Nishant Jindal

✳ Field and potential of Cylinder :-

(ii) Solid Non Conducting



$\lambda \cdot L = \rho \cdot \pi R^2 L$

$\lambda = \rho \cdot \pi R^2$

$r < R$

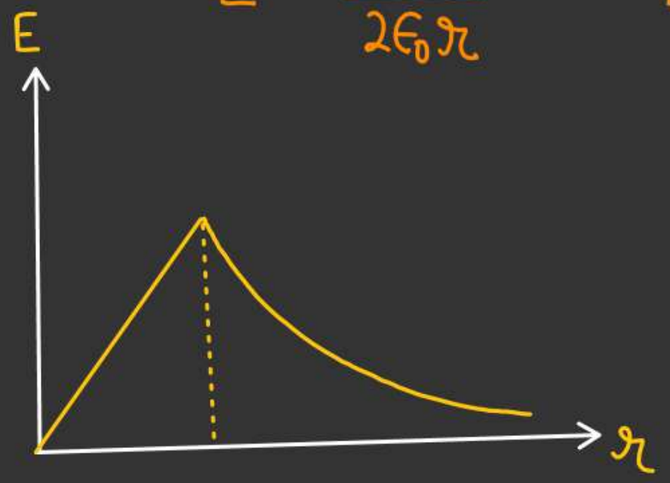
$$E \cdot 2\pi r l = \frac{\rho \cdot \pi r^2 l}{\epsilon_0}$$

$$E = \frac{\rho r}{2\epsilon_0} \Rightarrow \boxed{\vec{E} = \frac{\rho \vec{r}}{2\epsilon_0}} \star$$

$r > R$

$$E \cdot 2\pi r l = \frac{\rho \cdot \pi R^2 l}{\epsilon_0}$$

$$E = \frac{\rho R^2}{2\epsilon_0 r} \qquad \vec{E} = \frac{\rho R^2}{2\epsilon_0 r} \cdot (\hat{r})$$

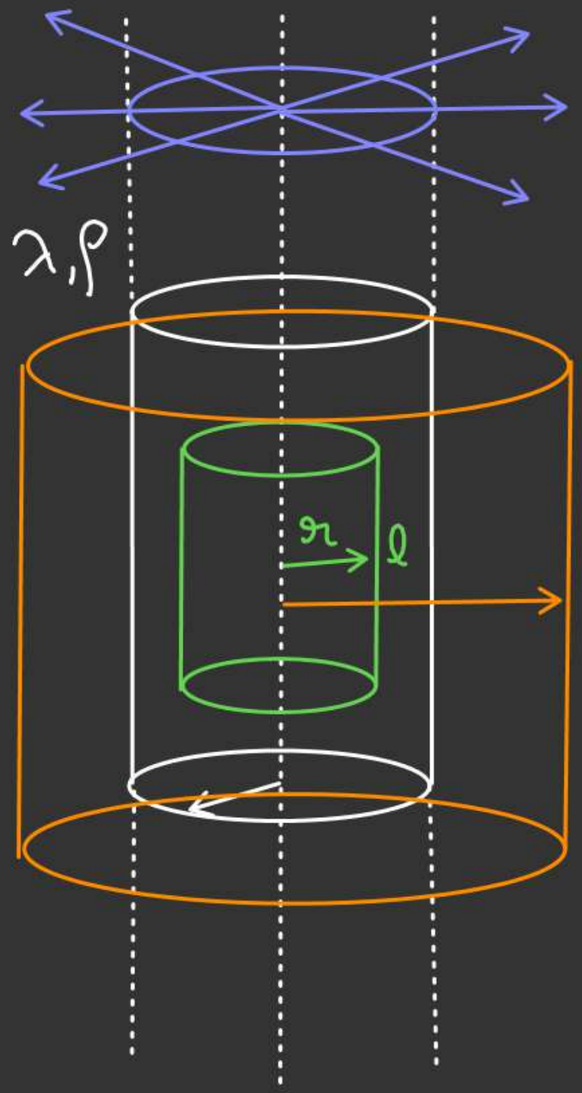


$$E = \frac{\lambda}{2\pi\epsilon_0 r}(\hat{r})$$

Nishant Jindal

✳ Field and potential of Cylinder :-

(ii) Solid Non Conducting



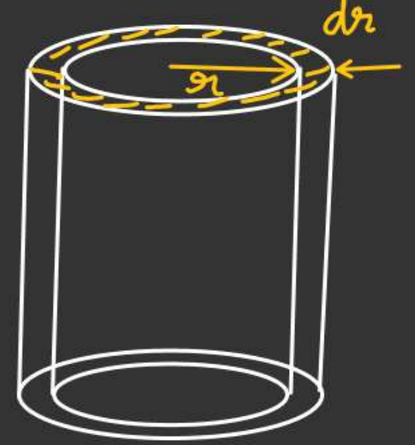
$\rho_r = f(r)$

$\rho_r = \frac{\rho_0 r}{R}$

$r < R$

$$E \cdot 2\pi r l = \frac{q_{in}}{\epsilon_0} = \frac{\int \rho_r \cdot dV}{\epsilon_0}$$

$$E \cdot 2\pi r l = \frac{\int \frac{\rho_0 r}{R} \cdot 2\pi r\, dr \cdot l}{\epsilon_0}$$



$$E \cdot r = \frac{\rho_0}{\epsilon_0 R} \int_0^r r^2 \cdot dr$$

$$E \cdot r = \frac{\rho_0 r^3}{3\epsilon_0 R}$$

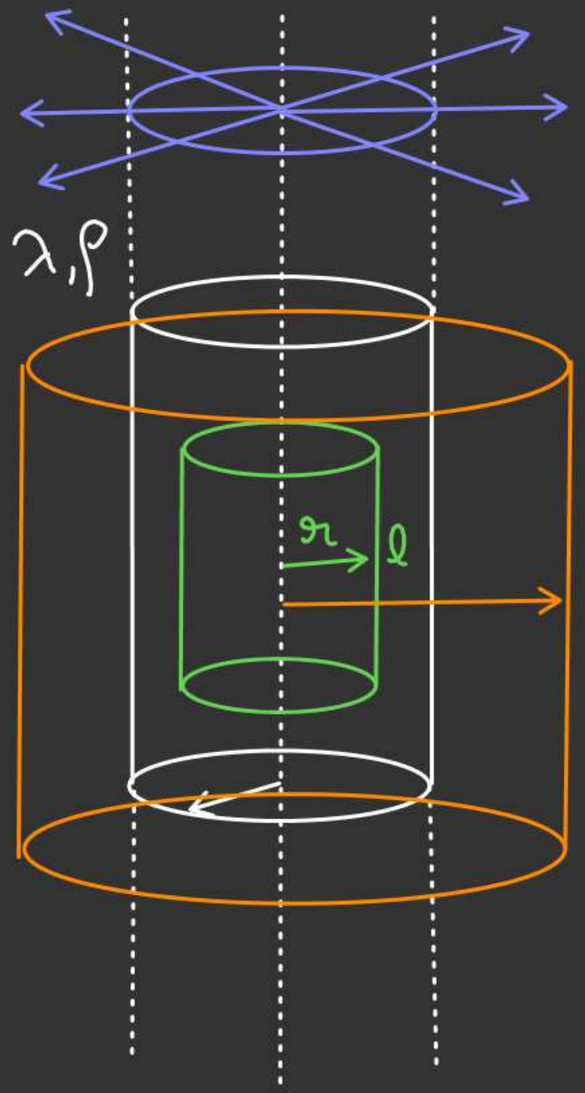
★

$$\boxed{E = \frac{\rho_0 r^2}{3\epsilon_0 R}}$$

Nishant Jindal

✳ Field and potential of Cylinder :-

(ii) Solid Non Conducting



$r > R$

$$E \cdot 2\pi r l = \frac{q_{in}}{\epsilon_0} = \frac{\int \rho_r \cdot dV}{\epsilon_0}$$

$$E \cdot 2\pi r l = \frac{\int_0^R \frac{\rho_0 r}{R} \cdot 2\pi r\, dr \cdot l}{\epsilon_0}$$

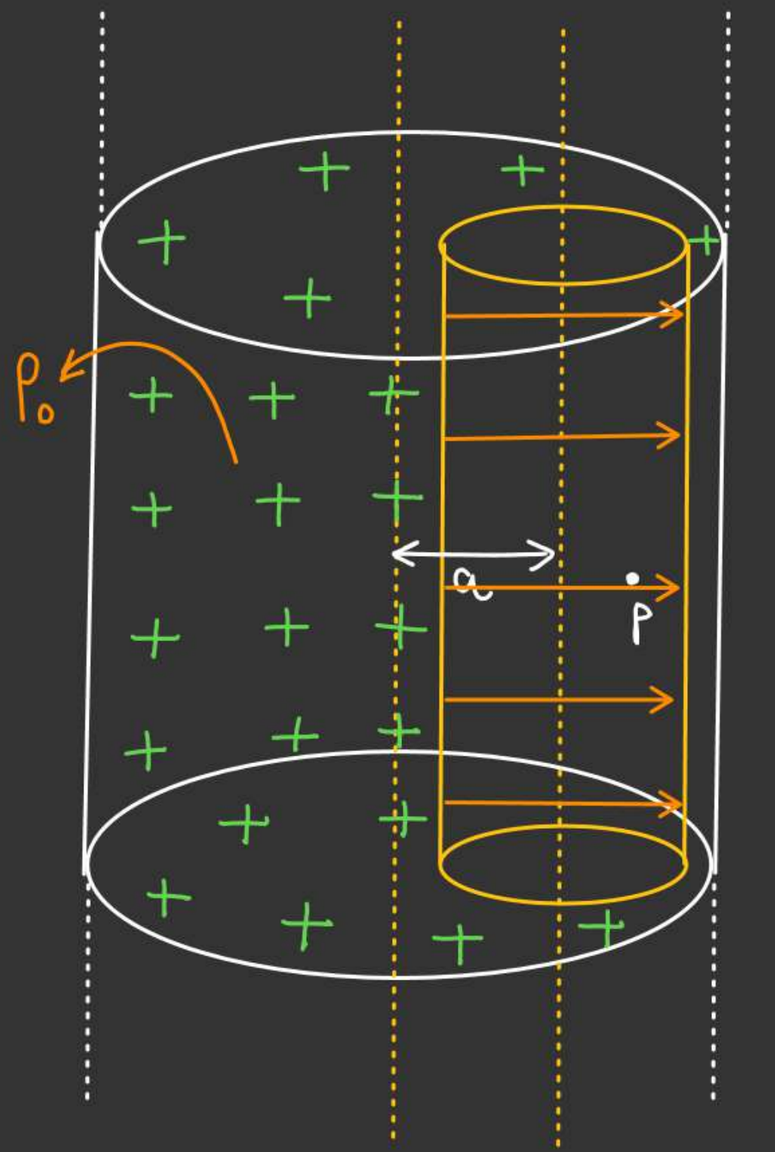
$$E \cdot r = \frac{\rho_0}{\epsilon_0 R} \int_0^R r^2 \cdot dr$$

$$E \cdot r = \frac{\rho_0 R^2}{3\epsilon_0}$$

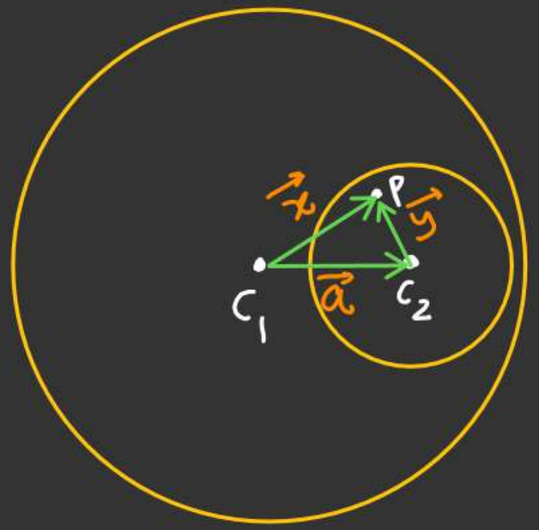
$$E = \frac{\rho_0 R^2}{3\epsilon_0 r}$$

Nishant Jindal

Cylinder with Cavity :-

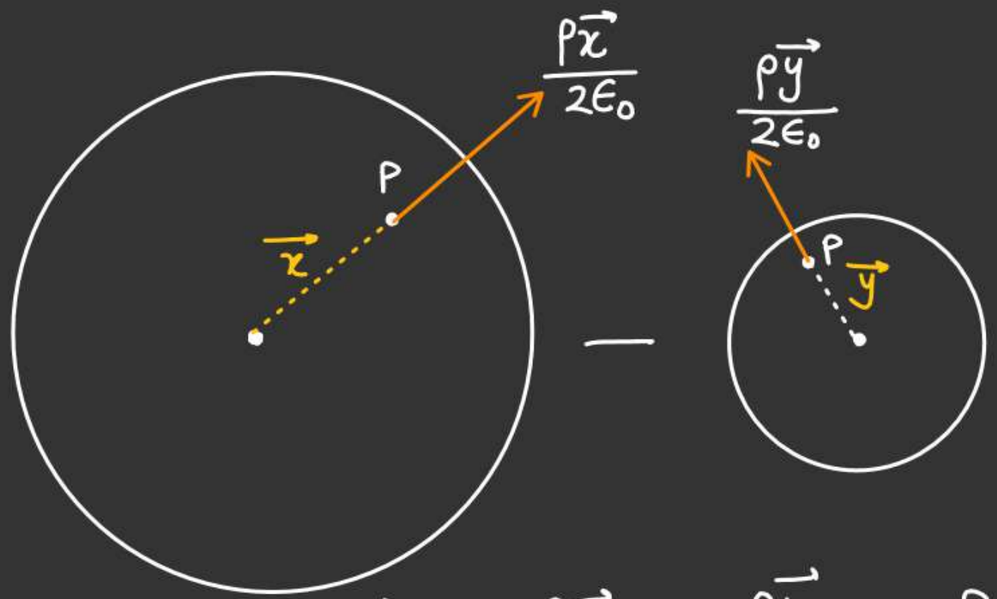


$\equiv$



$$\vec{a}+\vec{y}=\vec{x}$$

$$\vec{x}-\vec{y}=\vec{a}$$



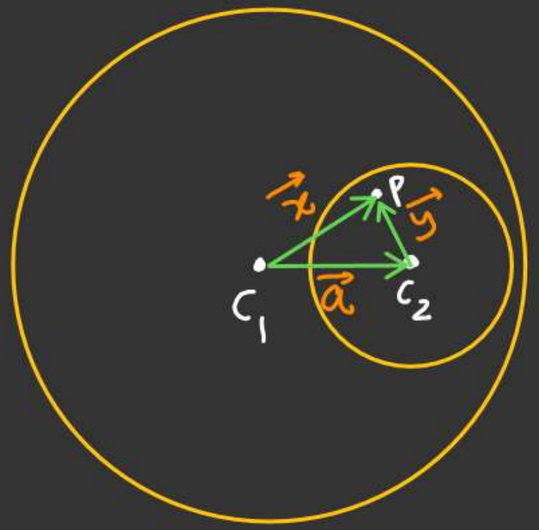
$-$

Const

$$\vec{E_P}=\frac{\rho\vec{x}}{2\epsilon_0}-\frac{\rho\vec{y}}{2\epsilon_0}=\frac{\rho}{2\epsilon_0}(\vec{x}-\vec{y})=\frac{\rho\vec{a}}{2\epsilon_0}$$

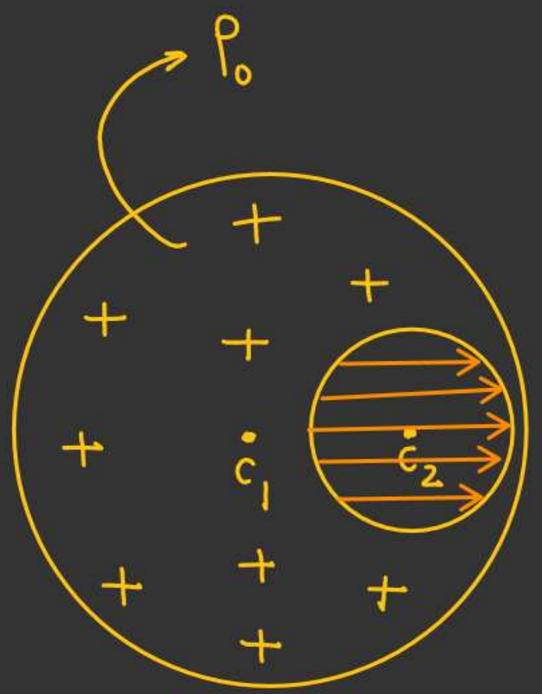
Nishant Jindal

Solid sphere with Cavity :-

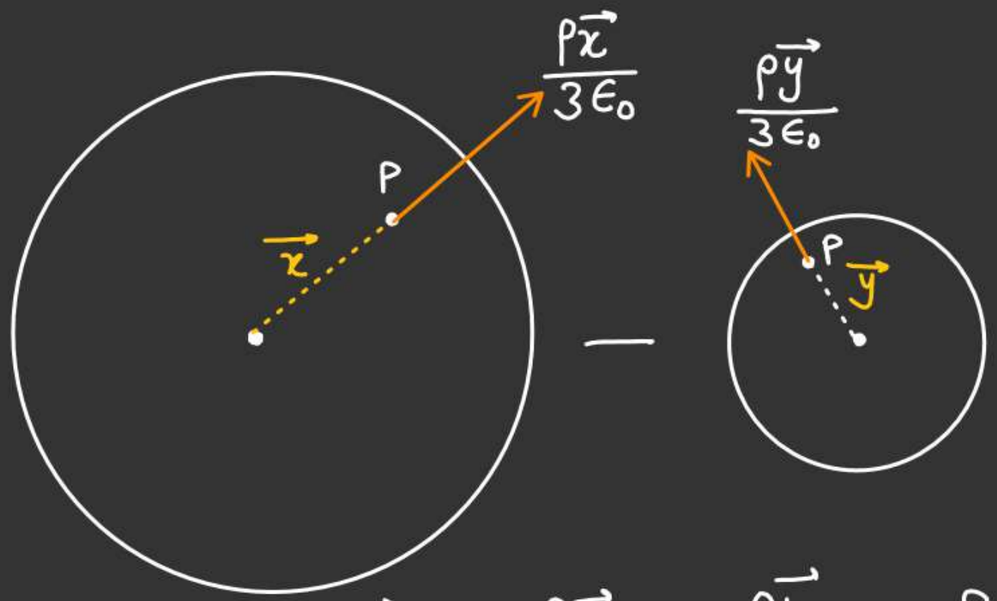


$$\vec{a}+\vec{y}=\vec{x}$$

$$\vec{x}-\vec{y}=\vec{a}$$



$\equiv$



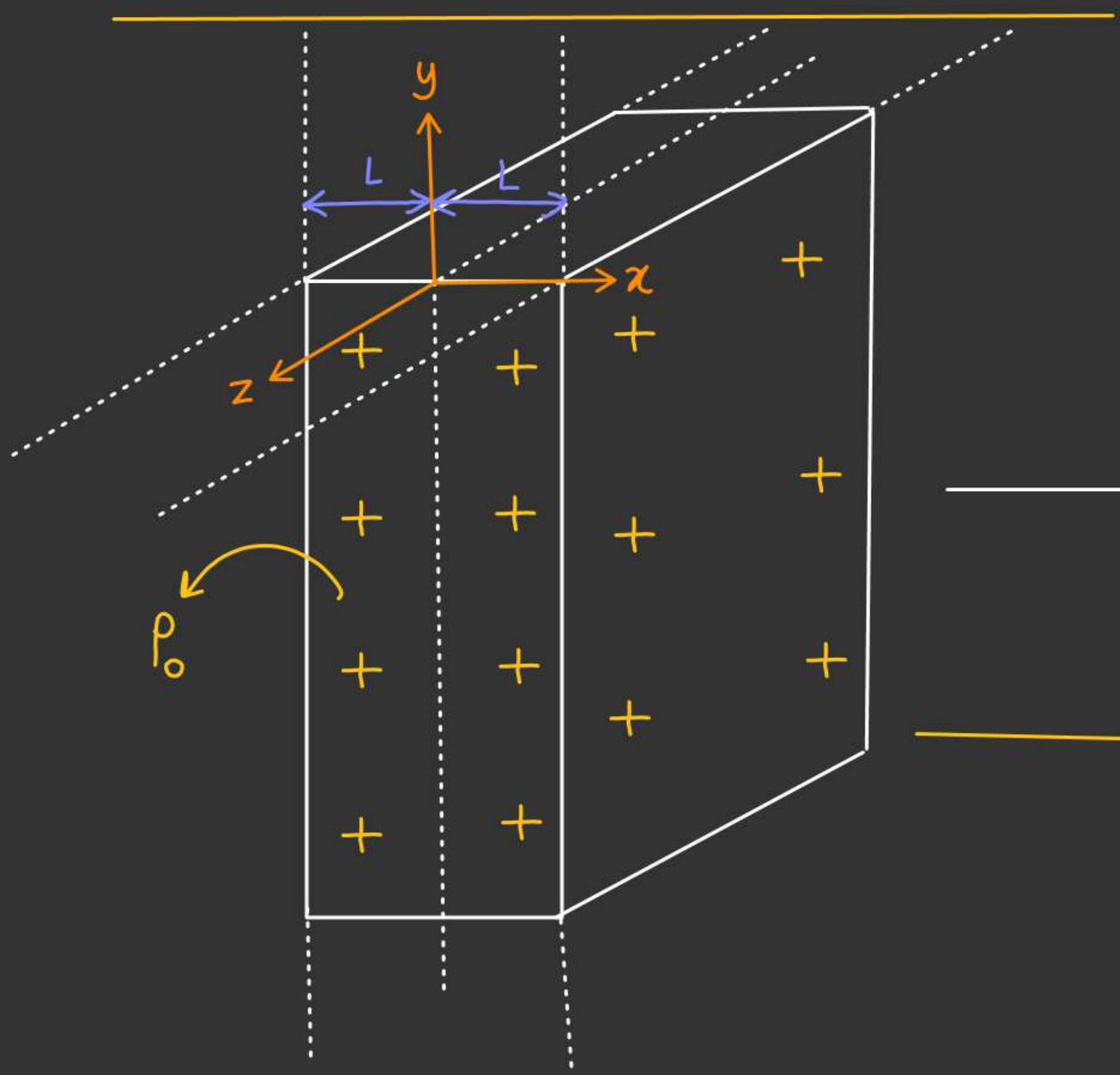
$-$

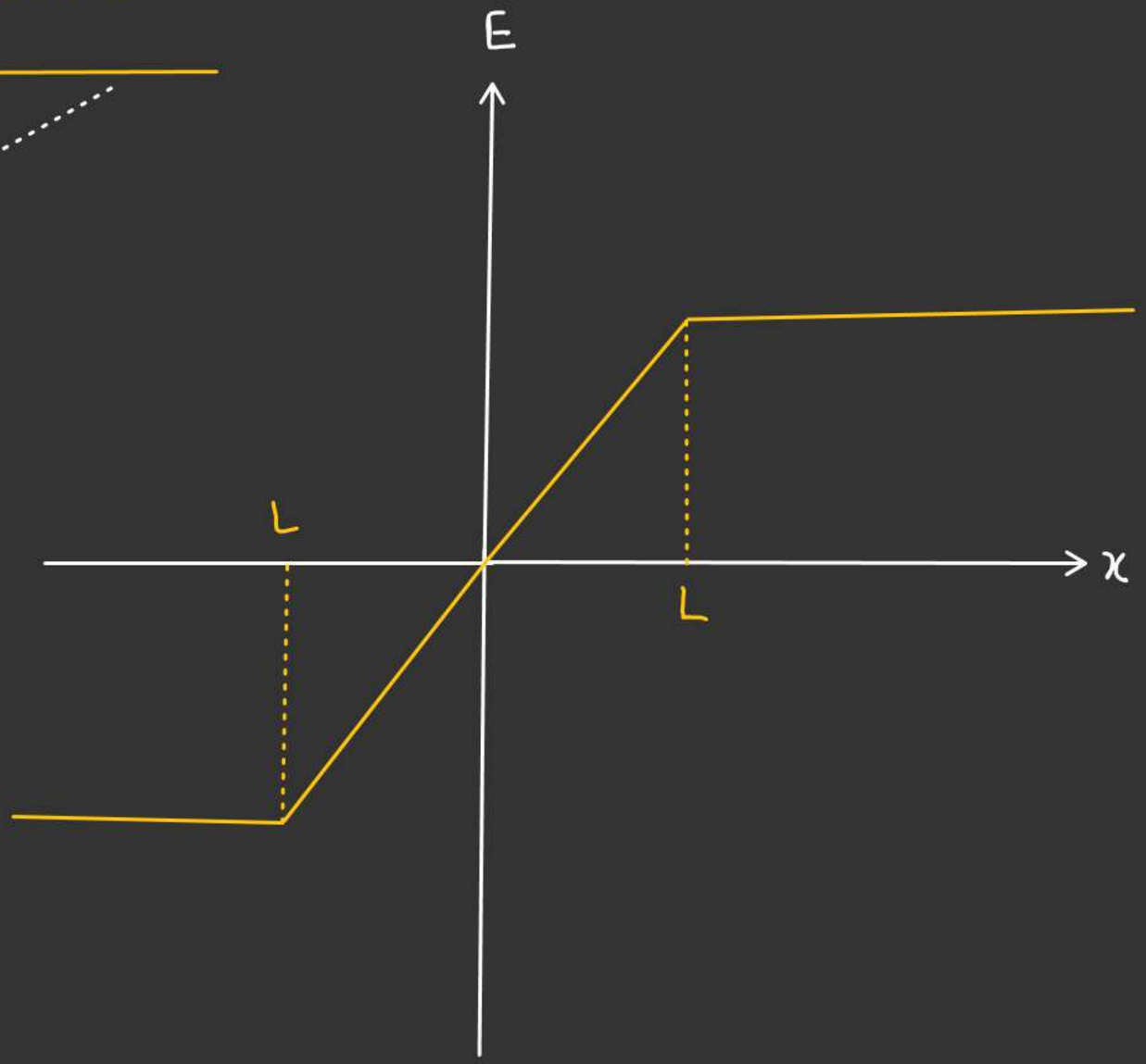
Const

$$\vec{E_P}=\frac{\rho\vec{x}}{3\epsilon_0}-\frac{\rho\vec{y}}{3\epsilon_0}=\frac{\rho}{3\epsilon_0}(\vec{x}-\vec{y})=\frac{\rho\vec{a}}{3\epsilon_0}$$

Nishant Jindal

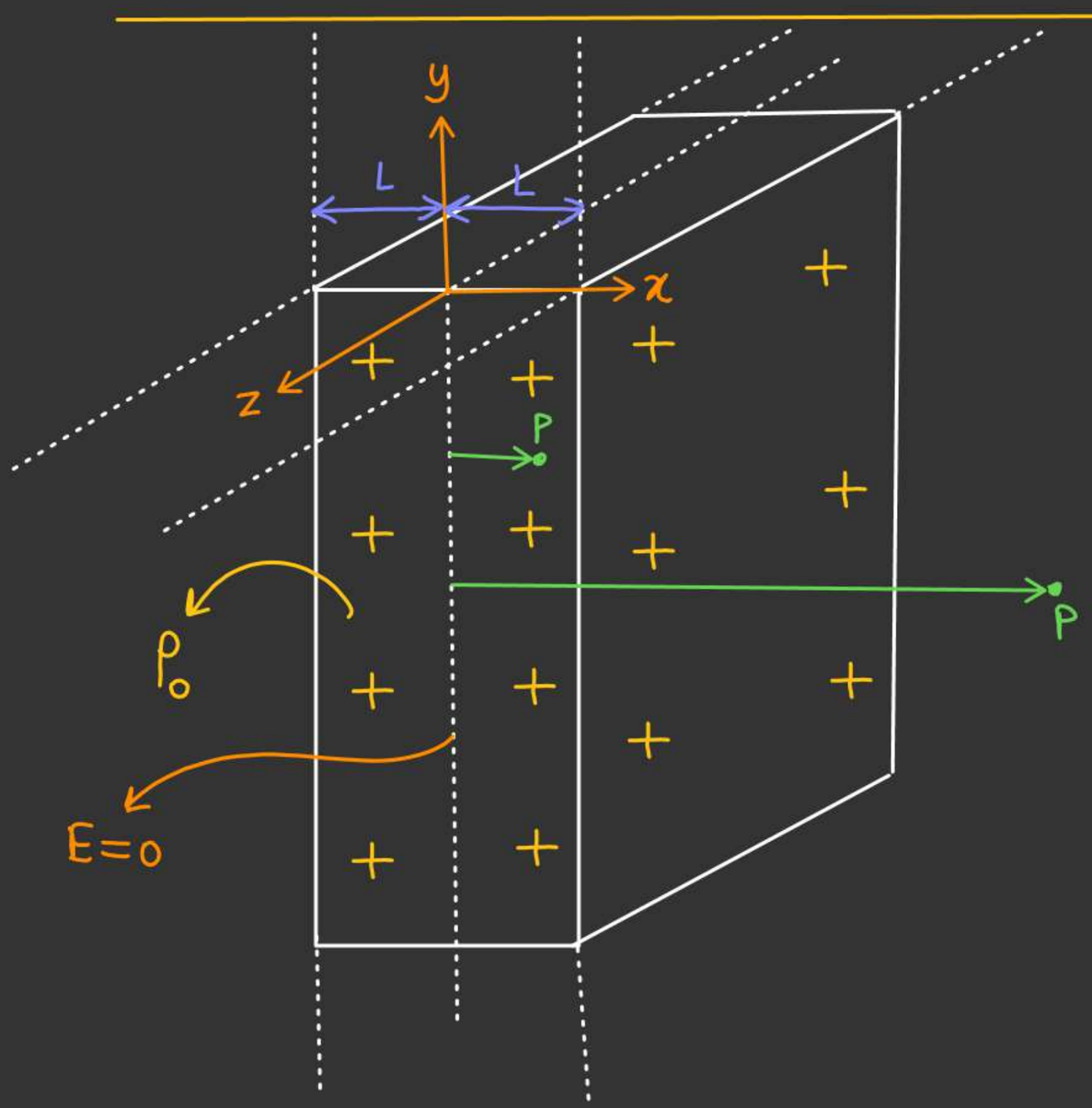
* E and V due to Infinite slab :-



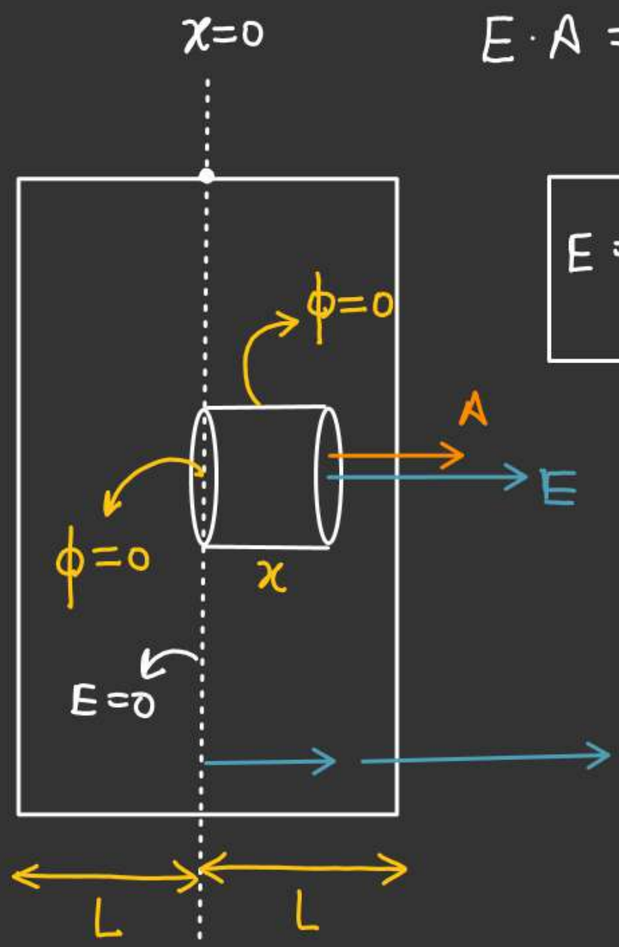


Nishant Jindal

* E and V due to Infinite slab :-



$x < L$

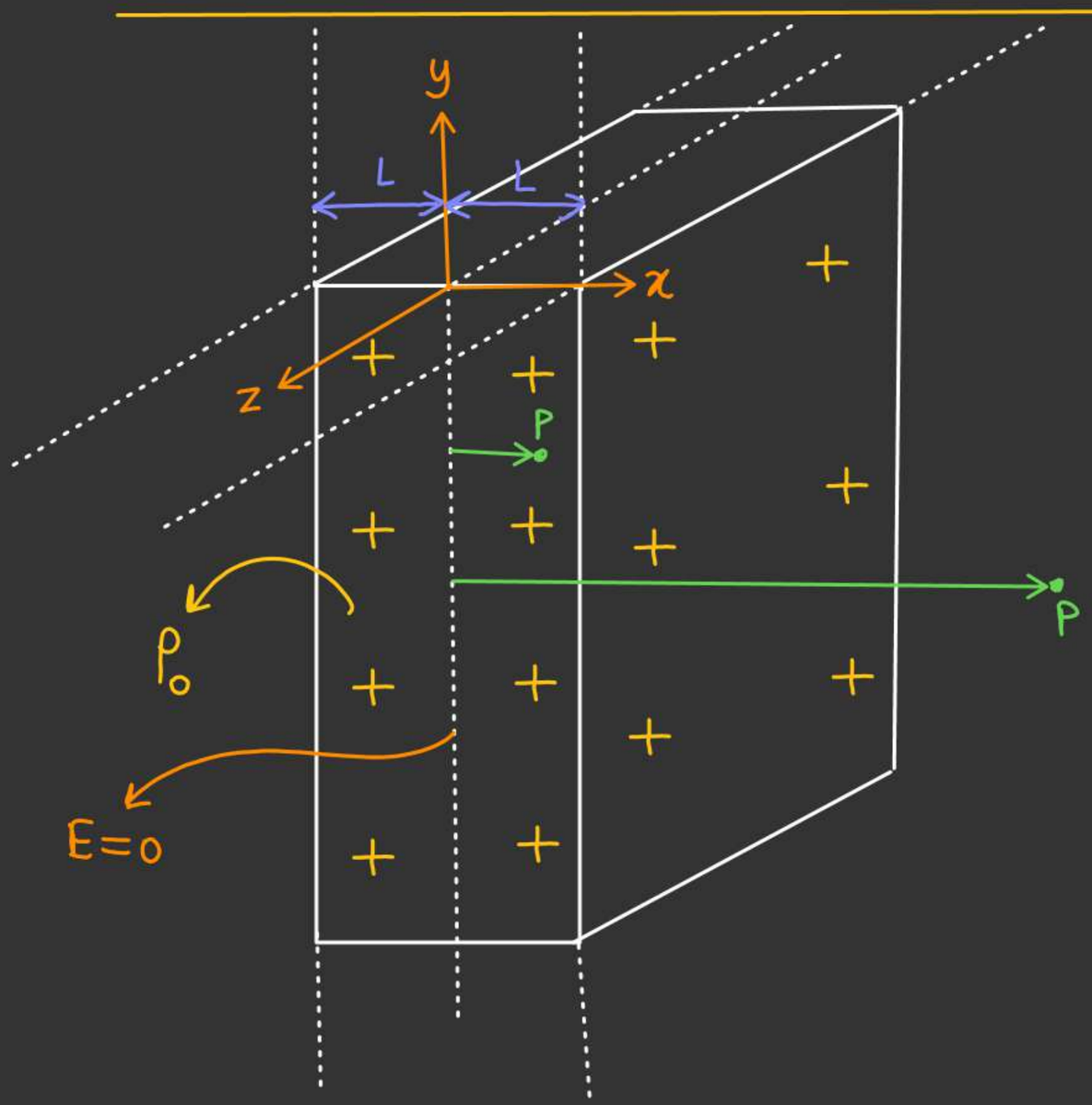


$$E \cdot A = \frac{\rho \cdot A x}{\epsilon_0}$$

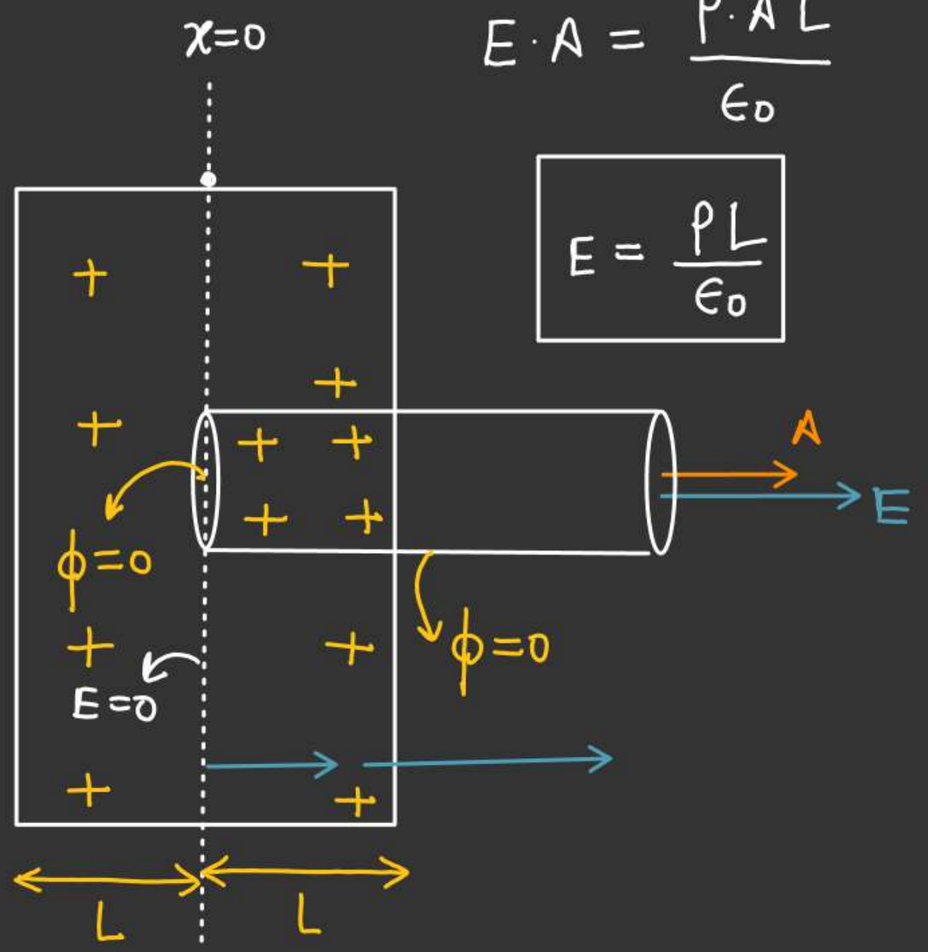
$$E = \frac{\rho x}{\epsilon_0}$$

Nishant Jindal

* E and V due to Infinite slab :-



$x > L$



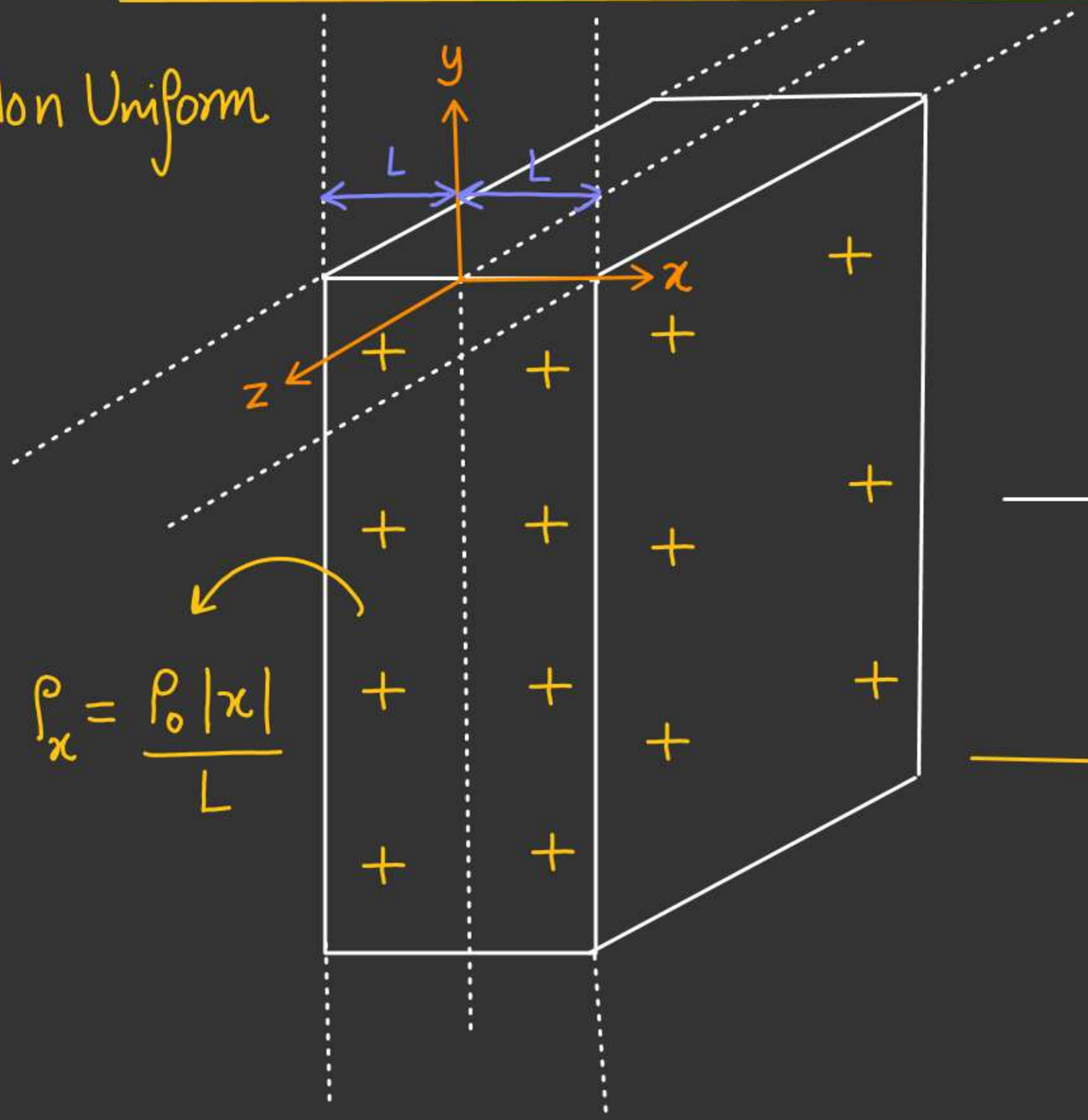
$$E \cdot A = \frac{\rho \cdot A L}{\epsilon_0}$$

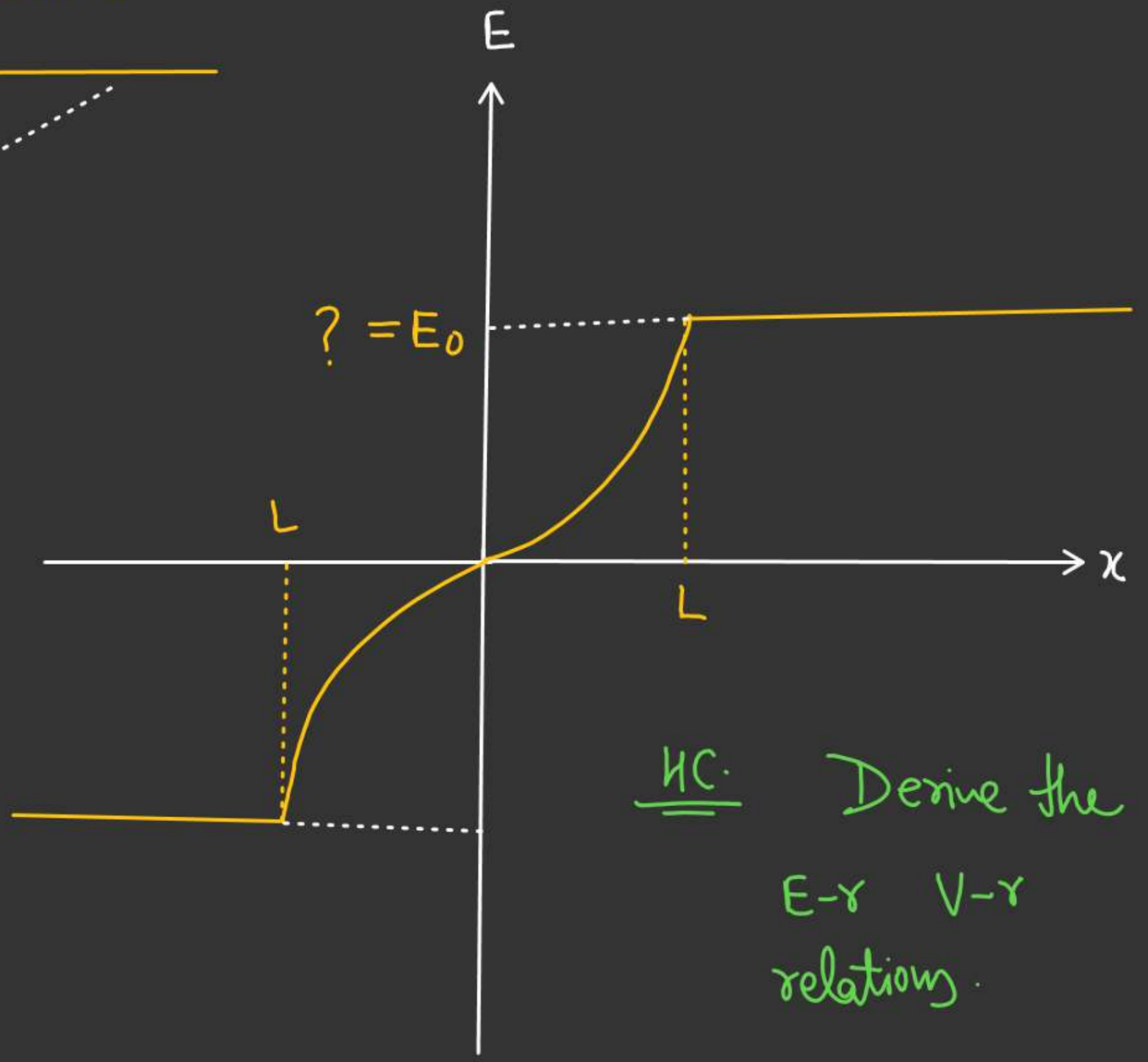
$$E = \frac{\rho L}{\epsilon_0}$$

Nishant Jindal

* E and V due to Infinite slab :-

Non Uniform





HC. Derive the E-r V-r relations.

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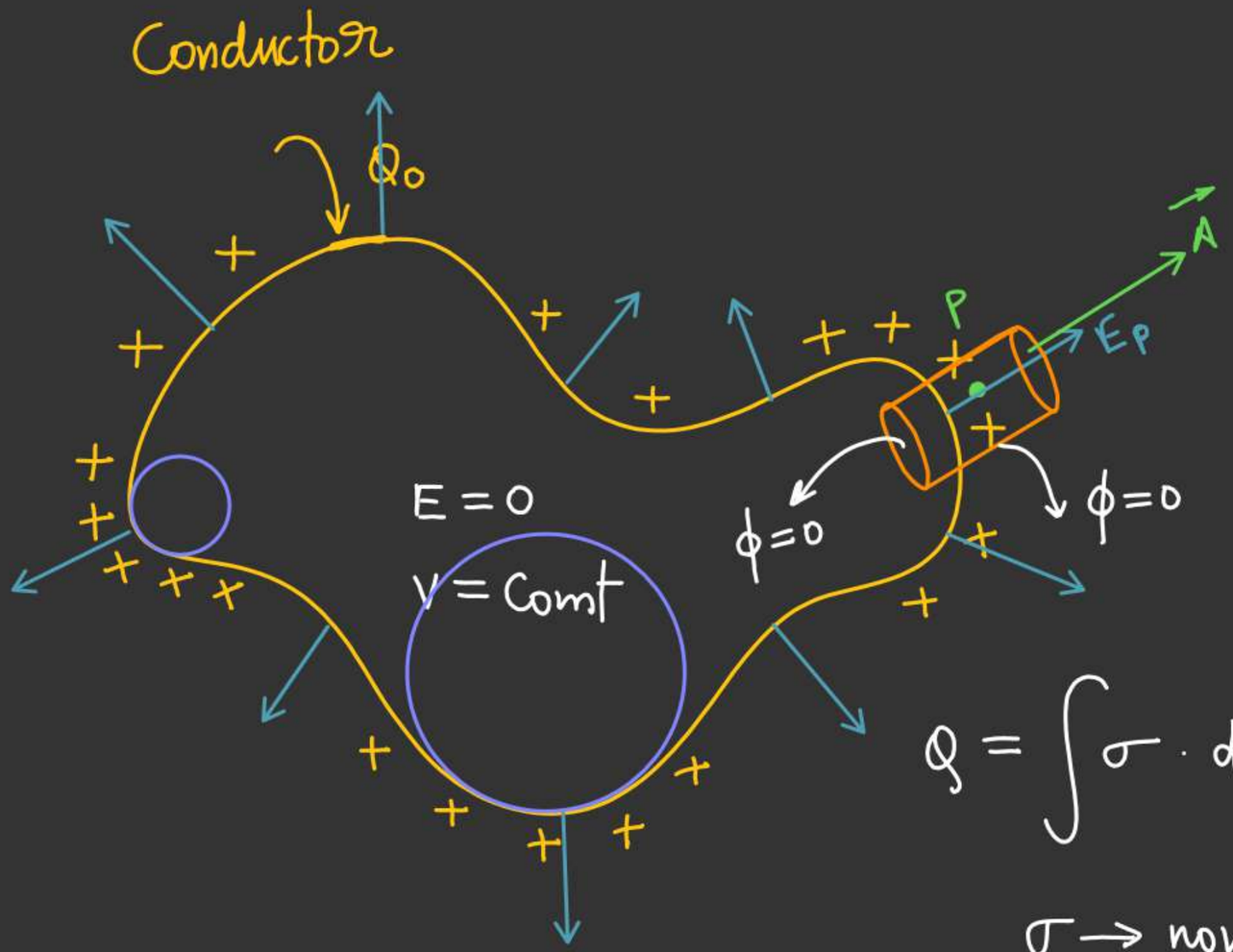
* Electric field in the Vicinity of Conductor :-



Approx.

$\sigma \propto \frac{1}{r}$

$E_P \cdot dA = \frac{\sigma \cdot dA}{\epsilon_0}$

$q = \int \sigma \cdot dA$

$\sigma \rightarrow$ non Uniform

★ $\boxed{E_P = \frac{\sigma}{\epsilon_0}}$

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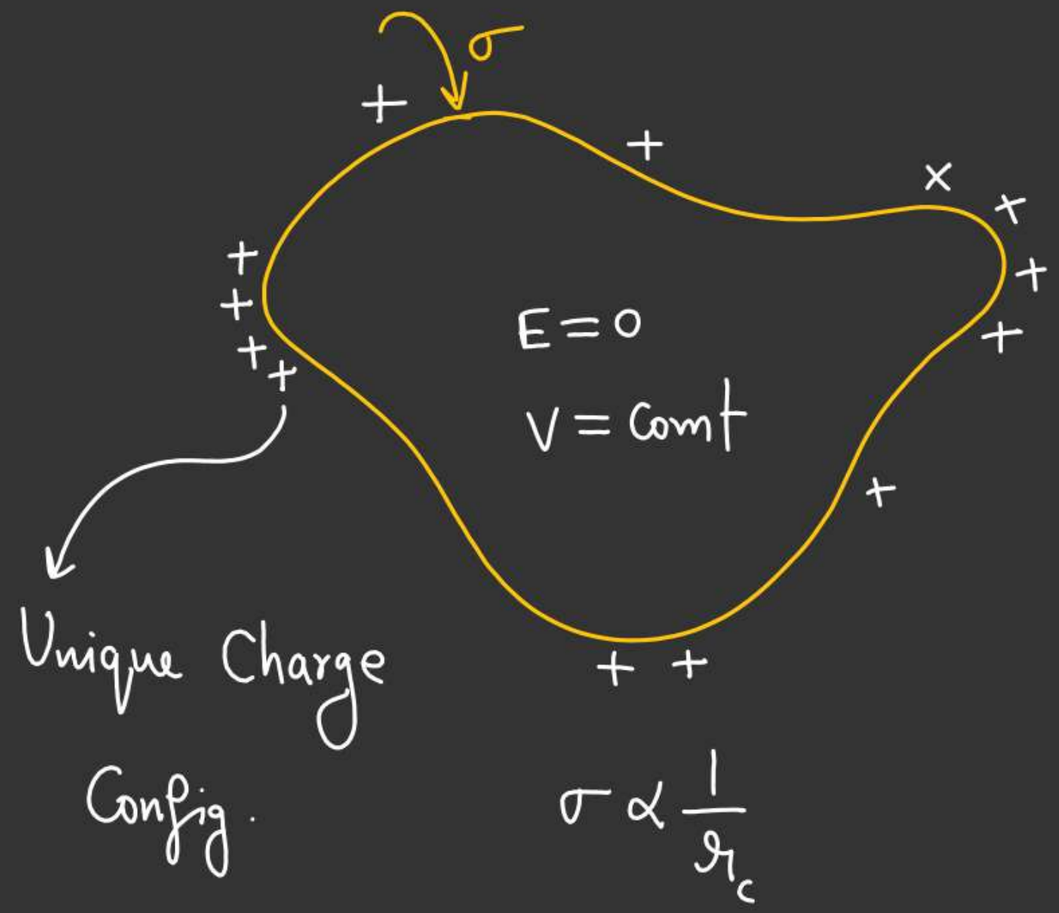
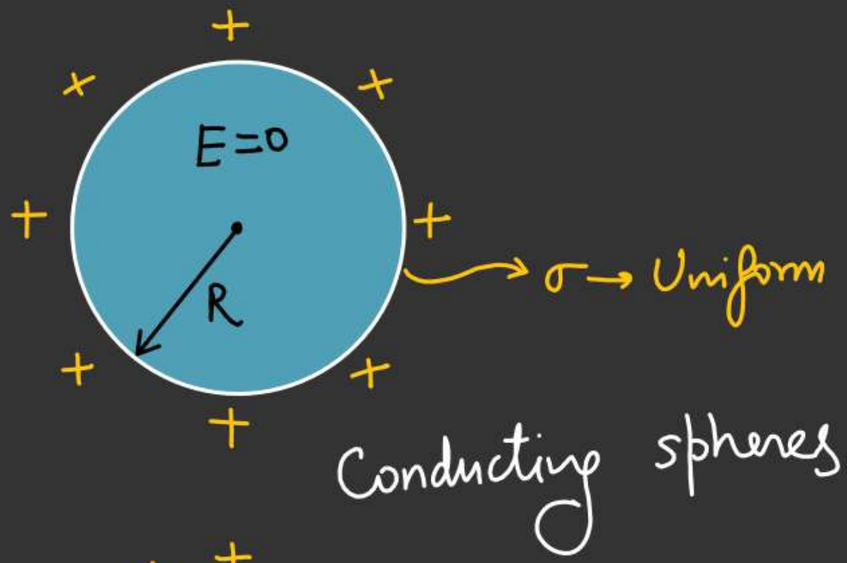
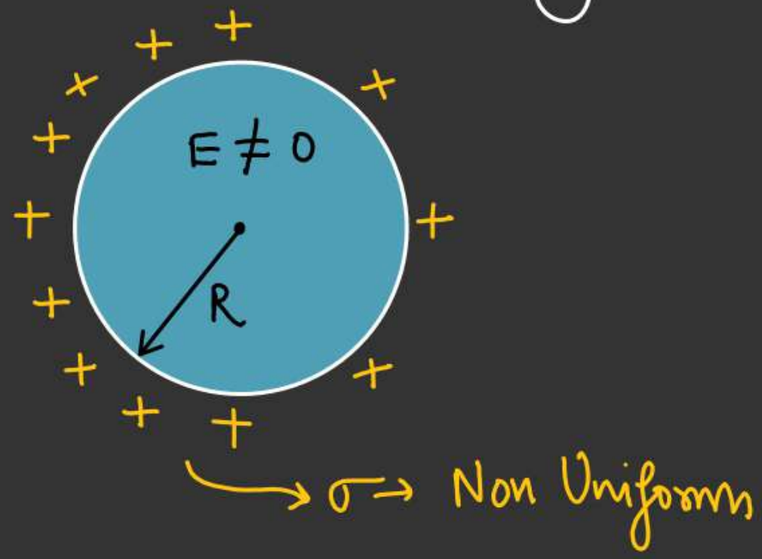
Uniqueness Theorem :-







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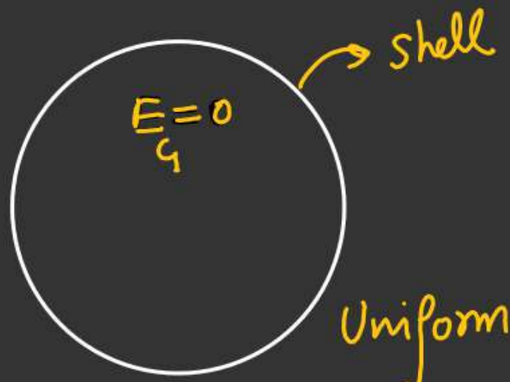
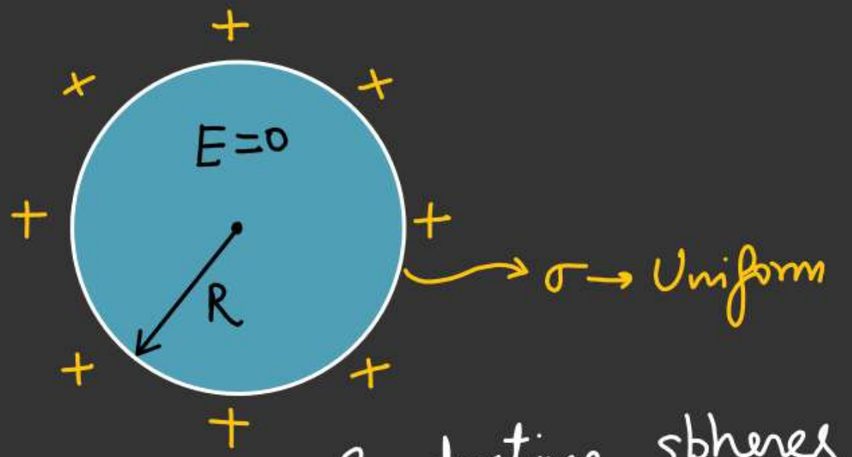
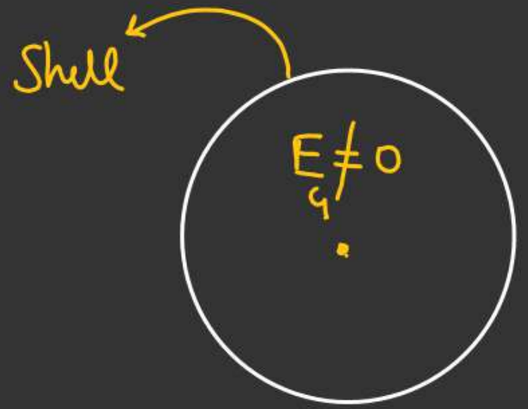


Uniform Mass distribution

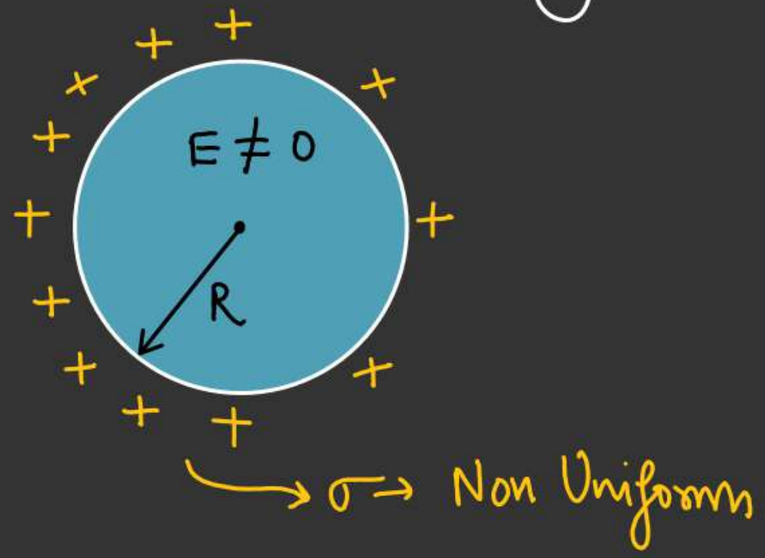


Conducting spheres



Non Uniform

Mass distribution



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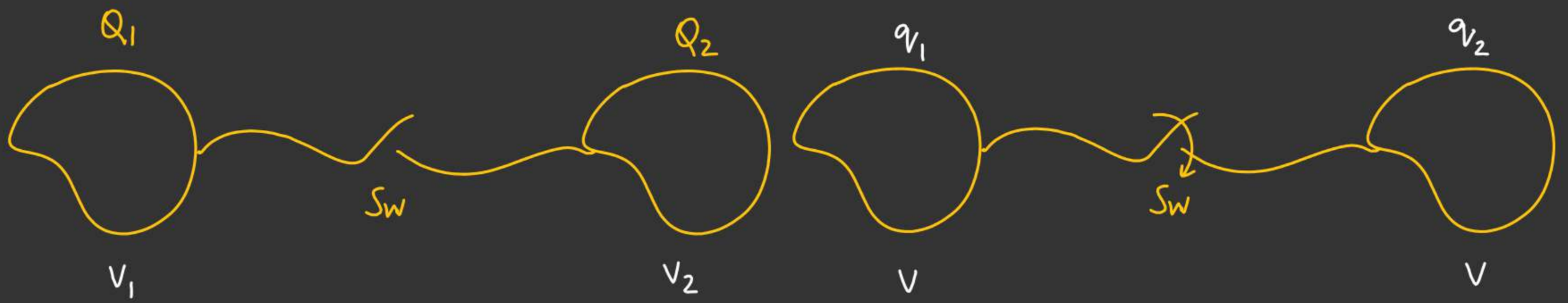
* Conductors in Contact :-

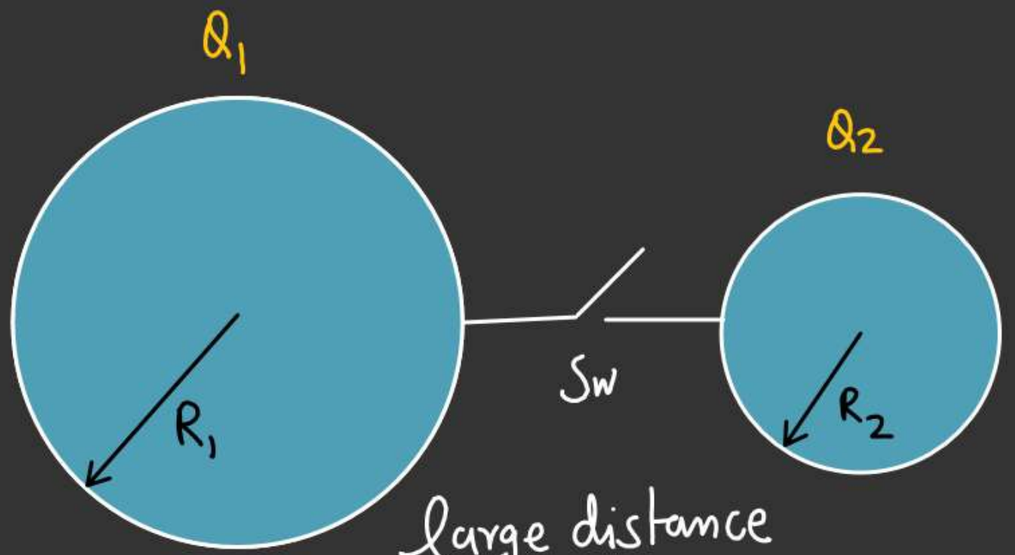


Heat $= V_i - V_f$

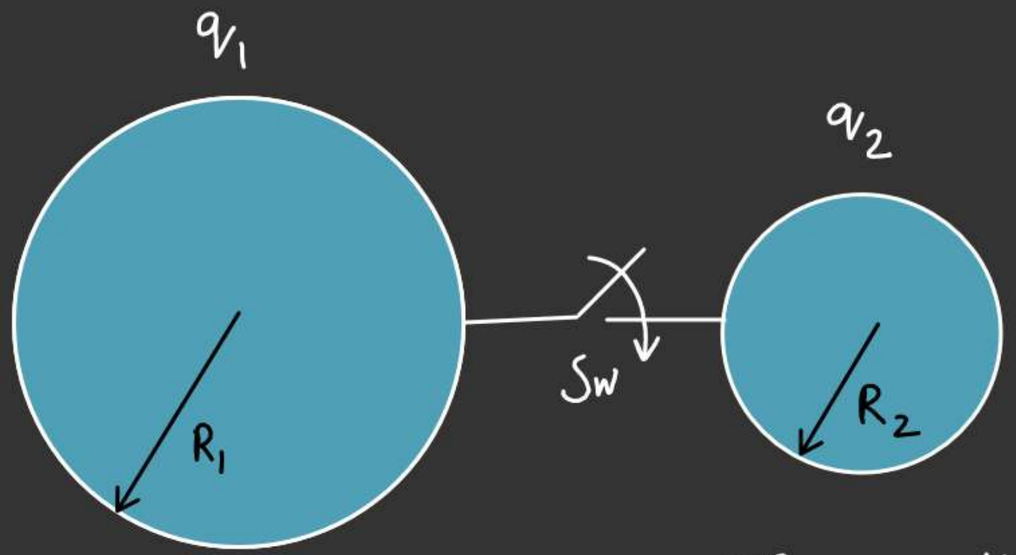
(i) $q_1 + q_2 = Q_1 + Q_2$

(ii) Potential of two Conductors will be Same.

Nishant Jindal



large distance



$$\begin{bmatrix} Q_1 + Q_2 = q_1 + q_2 \\ \frac{q_1}{R_1} = \frac{q_2}{R_2} \end{bmatrix}$$

q_1 ✓

q_2 ✓

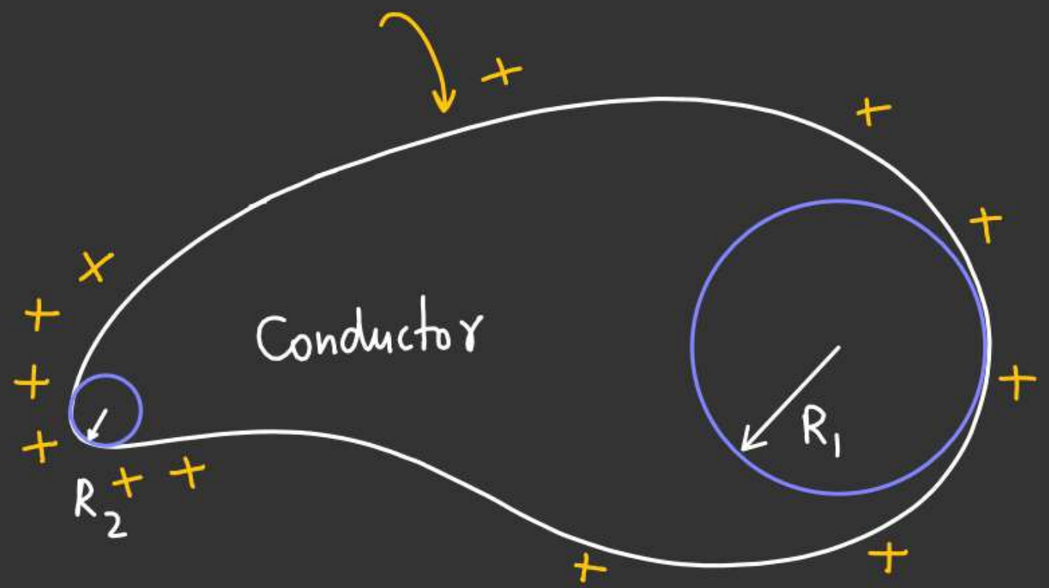
$$\sigma_1 = \frac{q_1}{4\pi R_1^2} \; ; \; \sigma_2 = \frac{q_2}{4\pi R_2^2}$$

$$\frac{Kq_1}{R_1} = \frac{Kq_2}{R_2}$$

★ $\boxed{\sigma_1 R_1 = \sigma_2 R_2}$

★ $\boxed{\sigma \propto \frac{1}{R}}$

Nishant Jindal



$$\sigma \propto \frac{1}{R}$$

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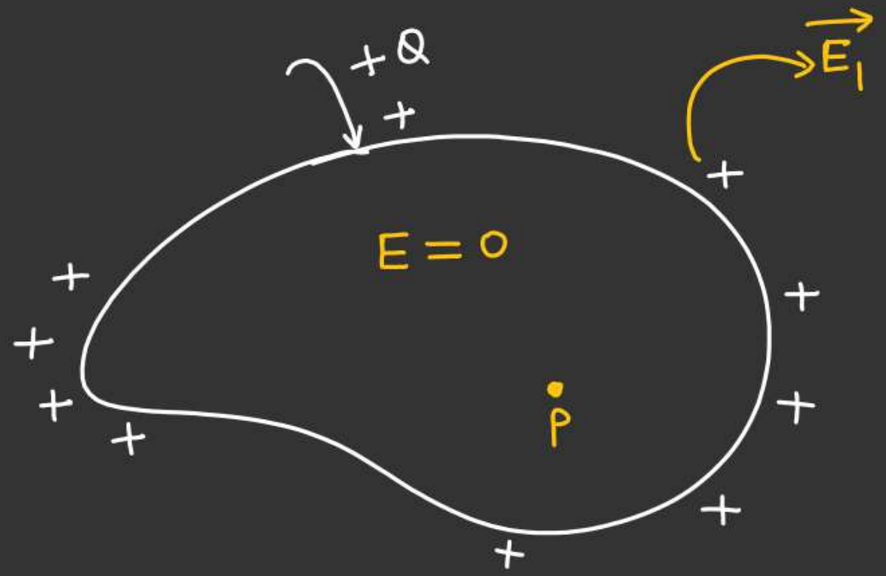
Conductors and Cavity :-

* Case (i)



At P

$\vec{E_1} = 0$

At Q

$\vec{E_1} \neq 0$

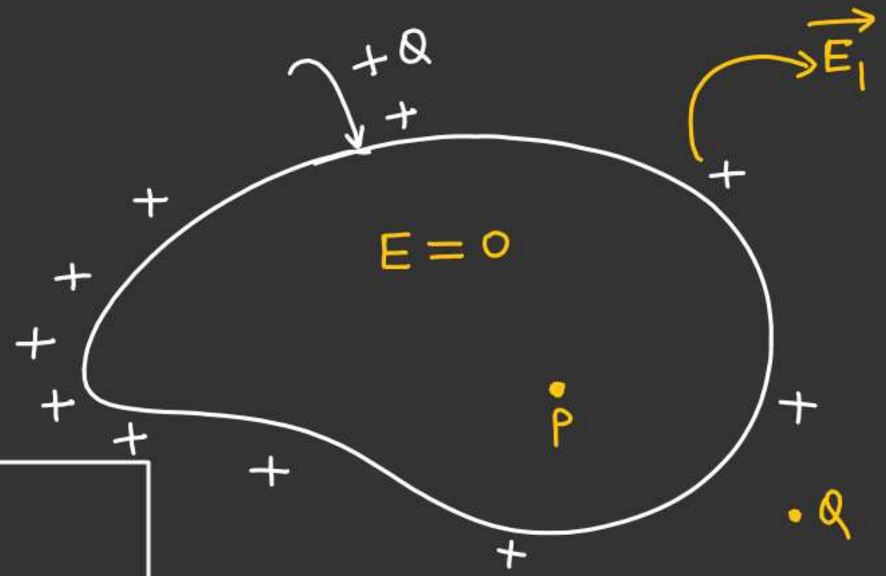
Q

Nishant Jindal

Conductors and Cavity :-

Vidyuth

* Case(ii)



At P

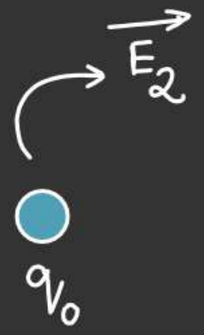
$\vec{E_1} + \vec{E_2} = 0$

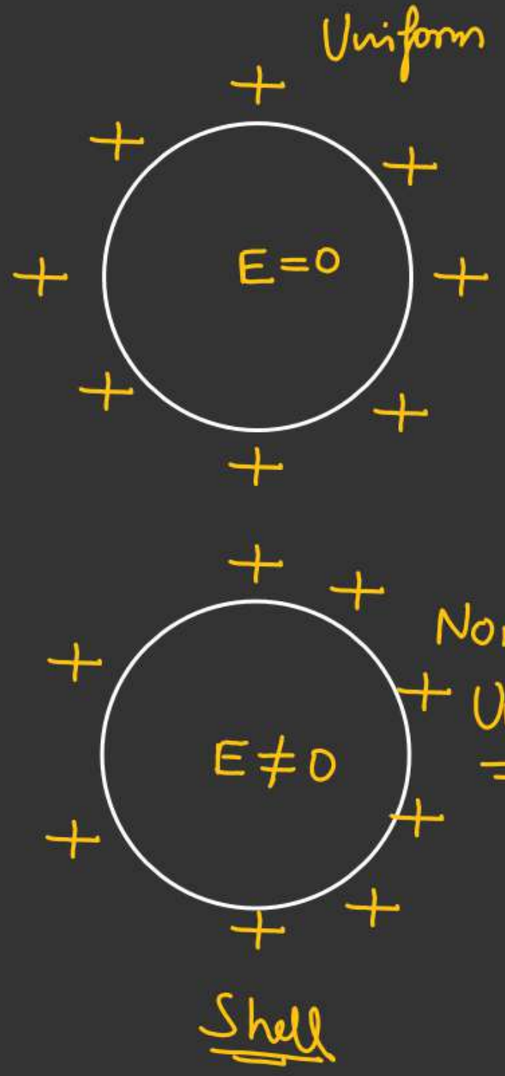
$\vec{E_1} \neq 0 \quad \vec{E_2} \neq 0$

At Q

$\vec{E_1} + \vec{E_2} \neq 0$

$\vec{E_1} \neq 0 ; \vec{E_2} \neq 0$





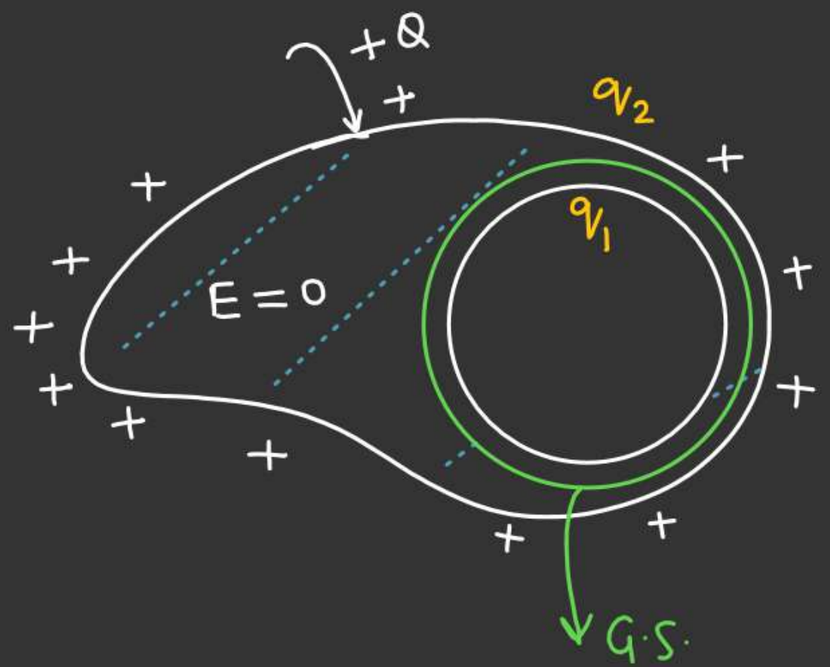
Nishant Jindal

Conductors and Cavity :-

$q_1 + q_2 = Q$

* Case (iii)

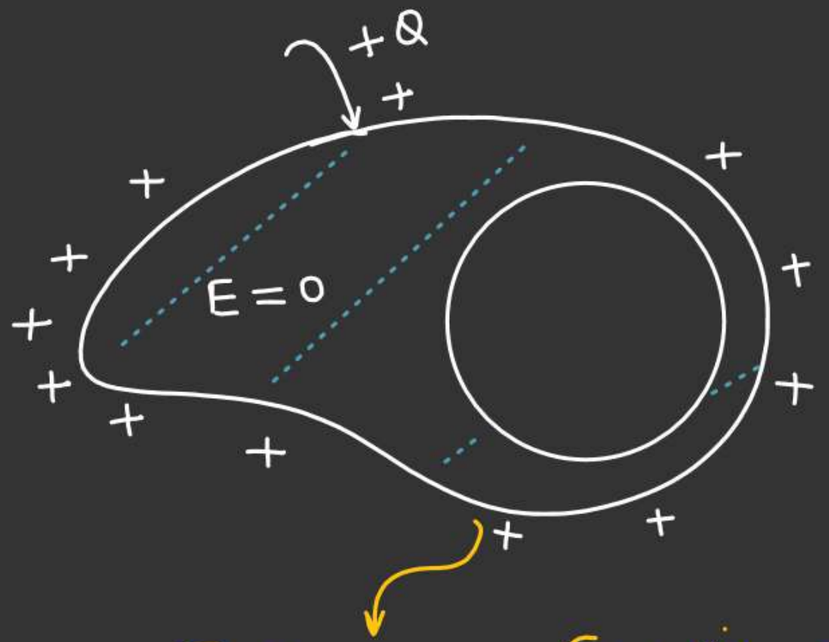
Conductor with Cavity



$q_1 + q_2 = Q$

$E = 0 \Rightarrow \phi = 0 \Rightarrow \frac{q_{in}}{\epsilon_0} = 0$

$\Rightarrow q_1 = 0$



[Same as Case (i)

$E_1 = 0$ Inside the Conductor including Cavity]

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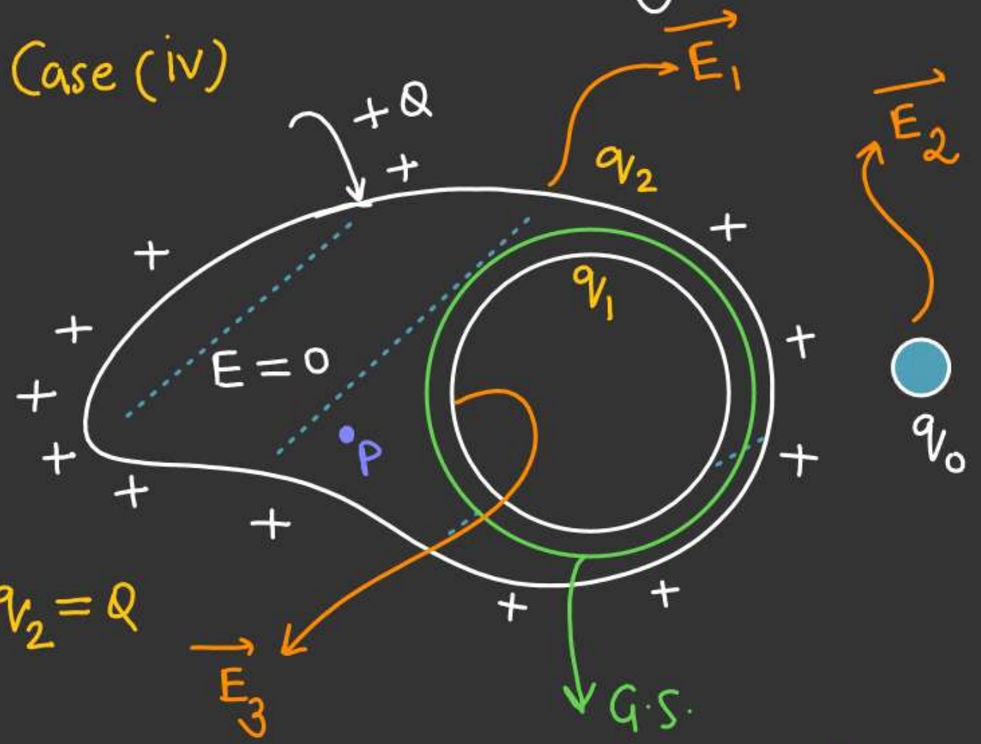
Conductors and Cavity :-

V. Imp

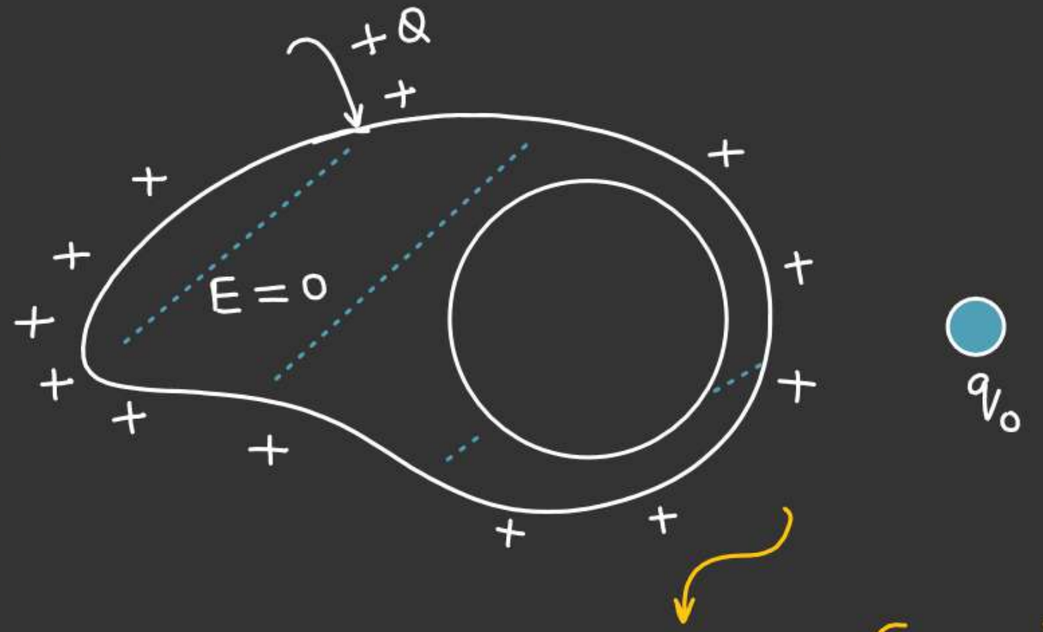
Conductor with a Cavity

* Case (iv)



$E = 0 \Rightarrow \phi = 0 \Rightarrow \frac{q_{in}}{\epsilon_0} = 0$

$\Rightarrow q_1 = 0$



Same as Case (ii)

At P: $\vec{E}_1 + \vec{E}_2 + \vec{E}_3 = 0$

$\vec{E}_1 + \vec{E}_2 = 0$

$\vec{E}_3 = 0$

$\vec{E}_1 \neq 0$

$\vec{E}_2 \neq 0$

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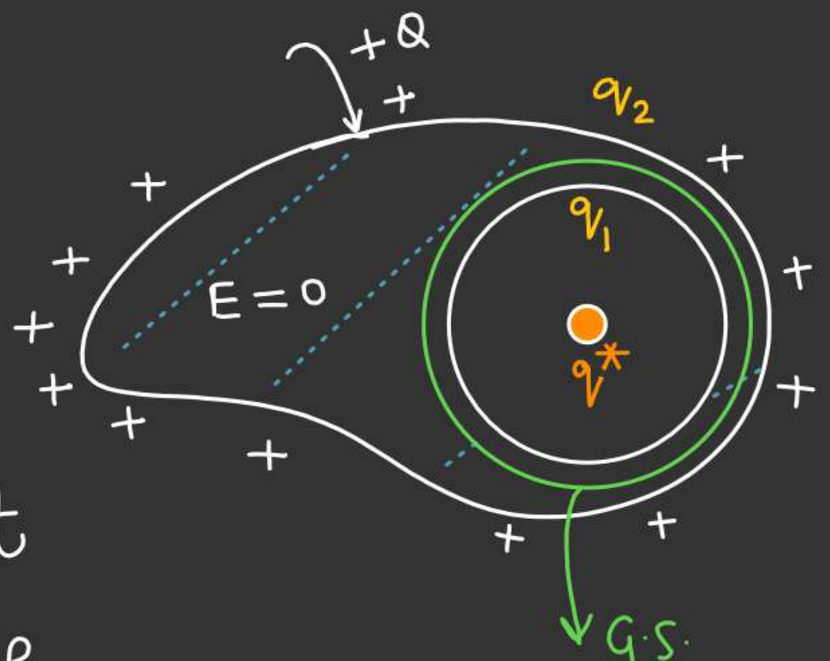
Conductors and Cavity :-

$q_1 + q_2 = Q$

* Case (V)

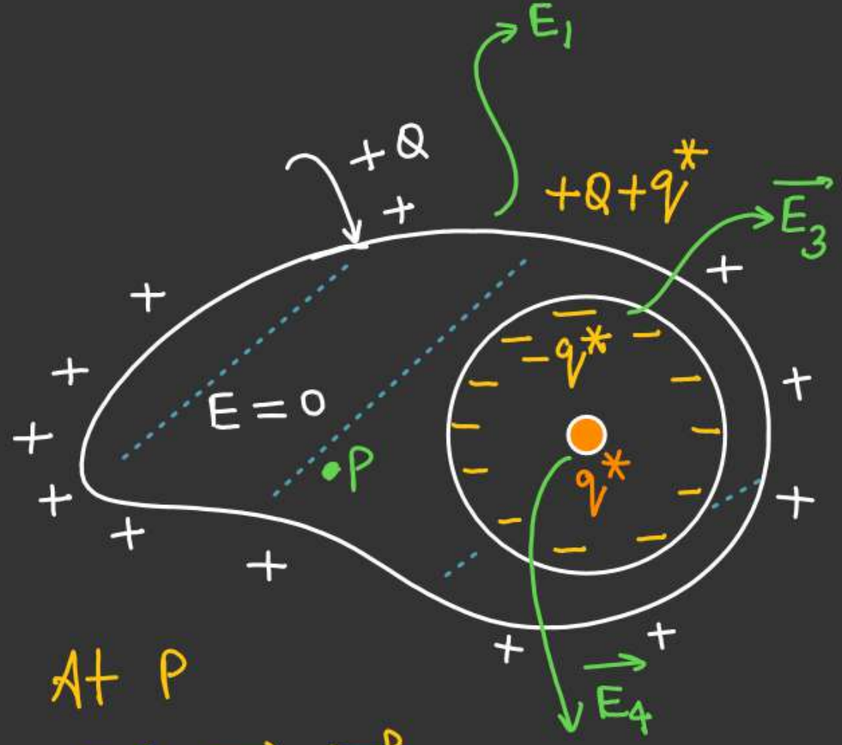
Conductor with Cavity

Charge is present at the Centre of Cavity



$E = 0 \Rightarrow \phi = 0 \Rightarrow \frac{q_{in}}{\epsilon_0} = 0$

$q^* + q_1 = 0 \Rightarrow q_1 = -q^*$



At P

$\vec{E_1} + \vec{E_3} + \vec{E_4} = 0$

$\vec{E_4} \neq 0 \quad \vec{E_3} \neq 0$

$\vec{E_3} + \vec{E_4} = 0$

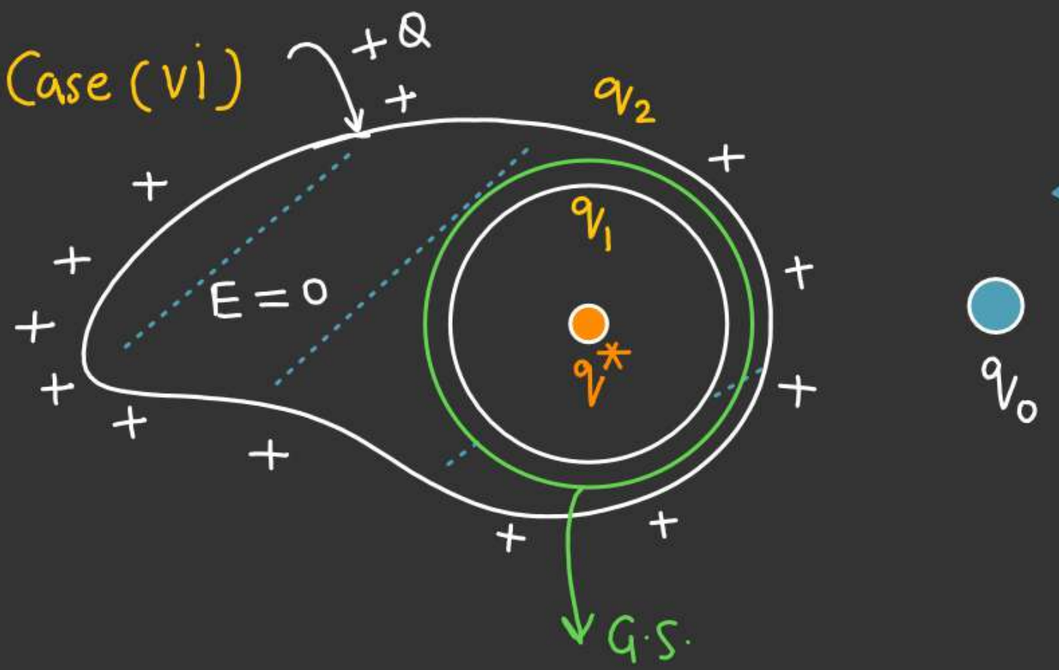
$\vec{E_1} = 0 \Rightarrow$ Same as Case (I)

Nishant Jindal

Conductors and Cavity :-

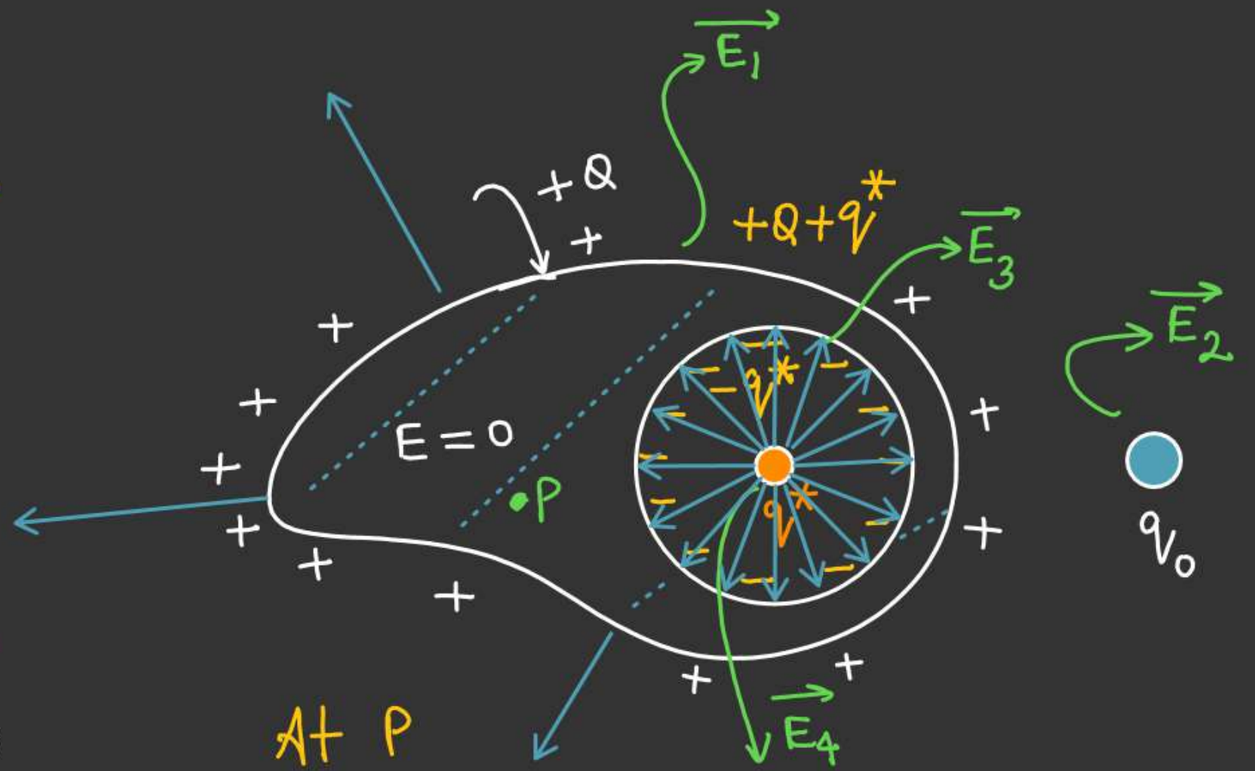
Vi gub

* Case (vi)



$E=0 \Rightarrow \phi = 0 \Rightarrow \frac{q_{in}}{\epsilon_0} = 0$

$q^* + q_1 = 0 \Rightarrow q_1 = -q^*$



At P

$\vec{E_1} + \vec{E_2} + \vec{E_3} + \vec{E_4} = 0$

$\vec{E_3} + \vec{E_4} = 0$

$\vec{E_3} \neq 0, \vec{E_4} \neq 0$

$\vec{E_1} + \vec{E_2} = 0$

$\vec{E_1} \neq 0 \quad \vec{E_2} \neq 0$

Same as Case (ii)

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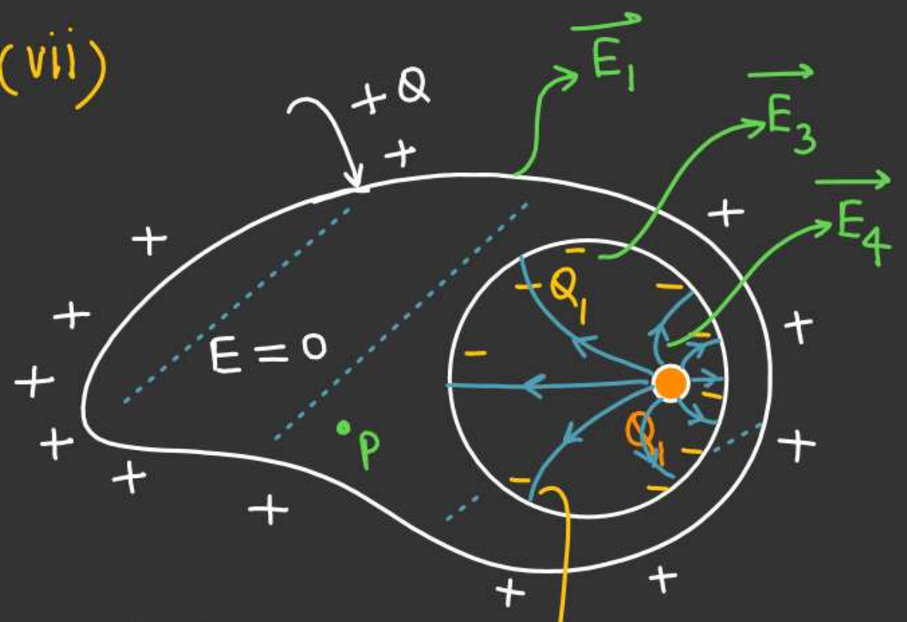
Conductors and Cavity :-

V. Imp

* Case (vii)



V. Imp

* Centre of charge of Induce charges on Cavity Surface will Coincide with the charges present inside the cavity.

non Uniform

At P

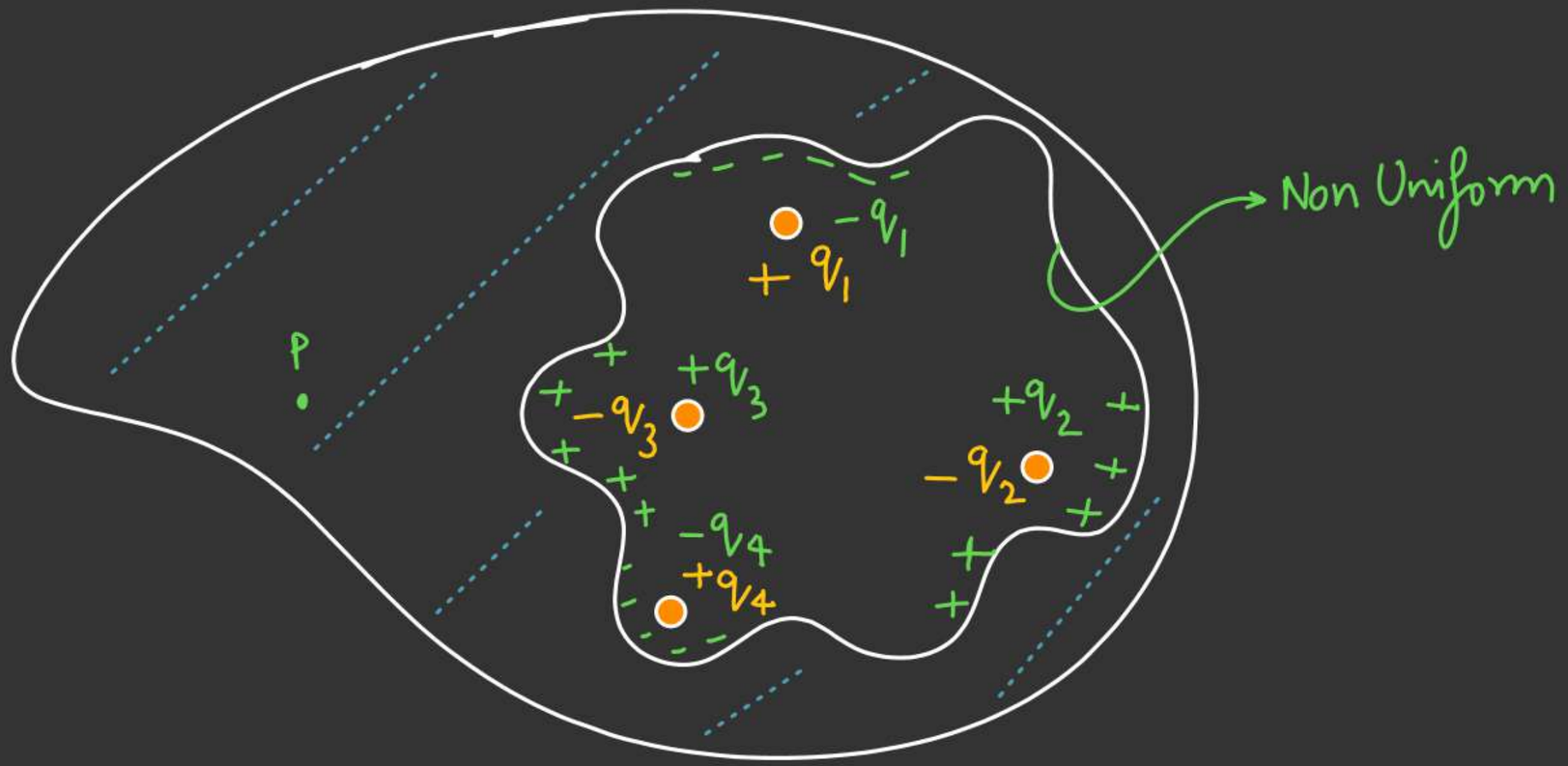
$$\vec{E_1} + \vec{E_3} + \vec{E_4} = 0$$

$$\vec{E_3} + \vec{E_4} = 0 \quad \text{(always true)}$$

V. Imp

(S1) induced charge on the Surface of Cavity and charges present inside the Cavity will always Produce zero field outside the Cavity.

Nishant Jindal



Nishant Jindal

S_2) Charge on the outer surface of Conductor and charges present outside the boundry of Conductor will not Produce any field Inside the Conductor boundry including Cavity.

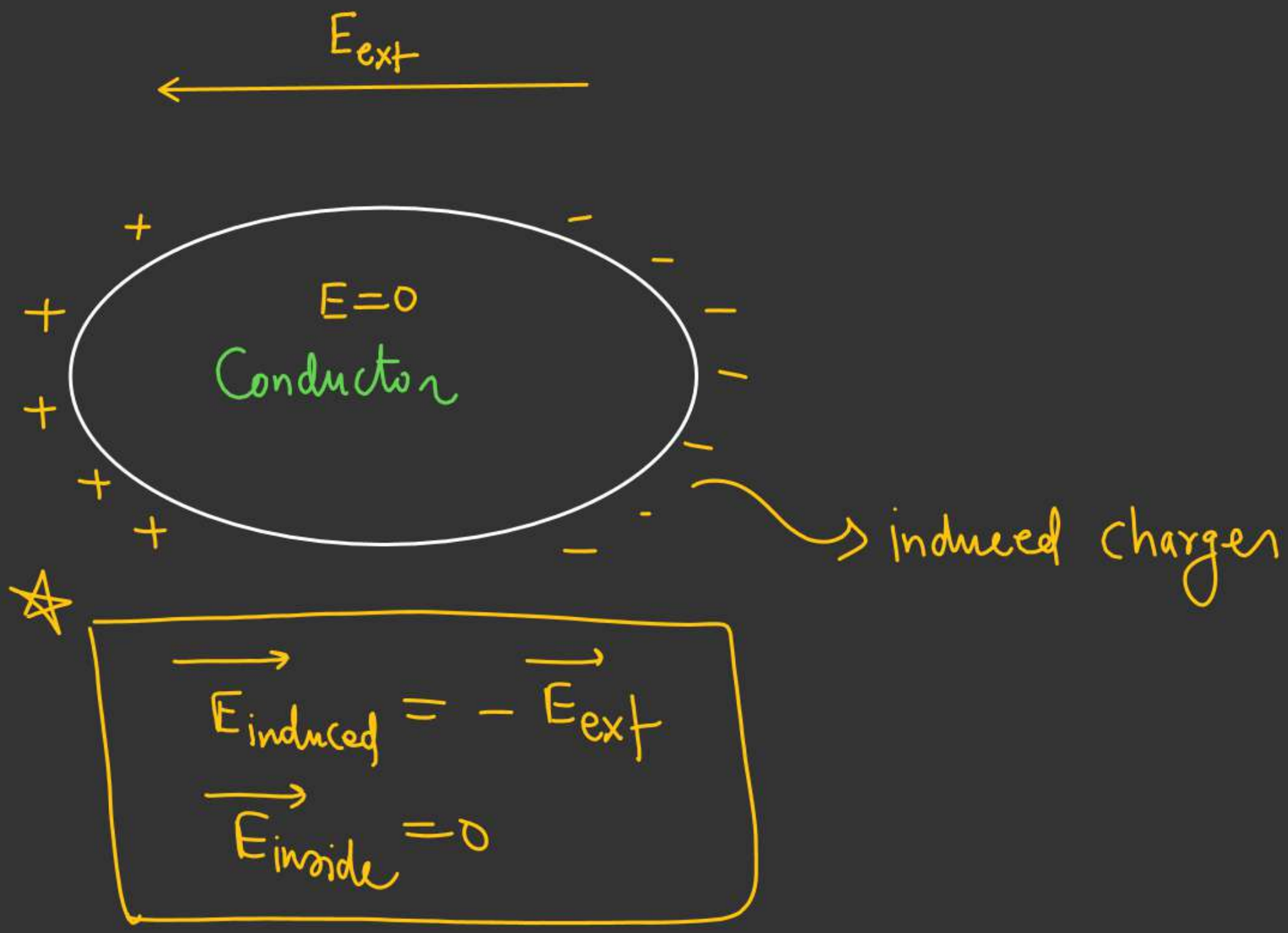
$\vec{E_1} + \vec{E_2} = 0$

$\vec{E_1} \neq 0$, $\vec{E_2} \neq 0$

} → electrostatic Shielding.

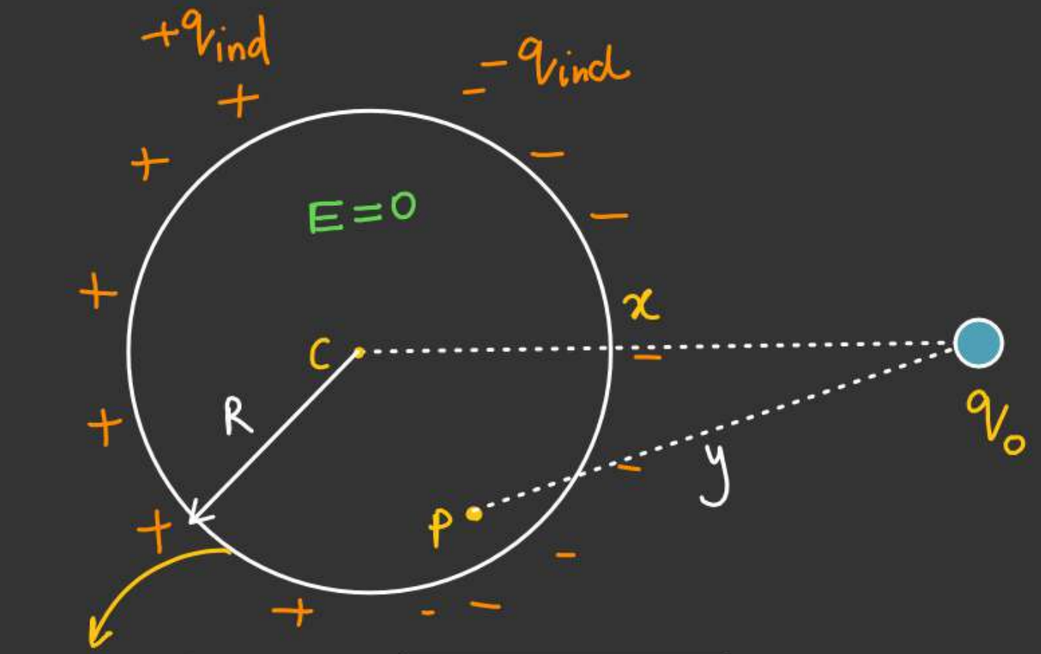
Inside the Conductor boundry including Cavity.

Nishant Jindal



Nishant Jindal

* Potential and field due to induced charges.



$$V_P = V_C$$

$$V_P|_{q_o} + V_P|_{ind} = V_C|_{q_o} + V_C|_{ind}$$

$$\frac{Kq_o}{y} + V_P|_{ind} = \frac{Kq_o}{x} + 0$$

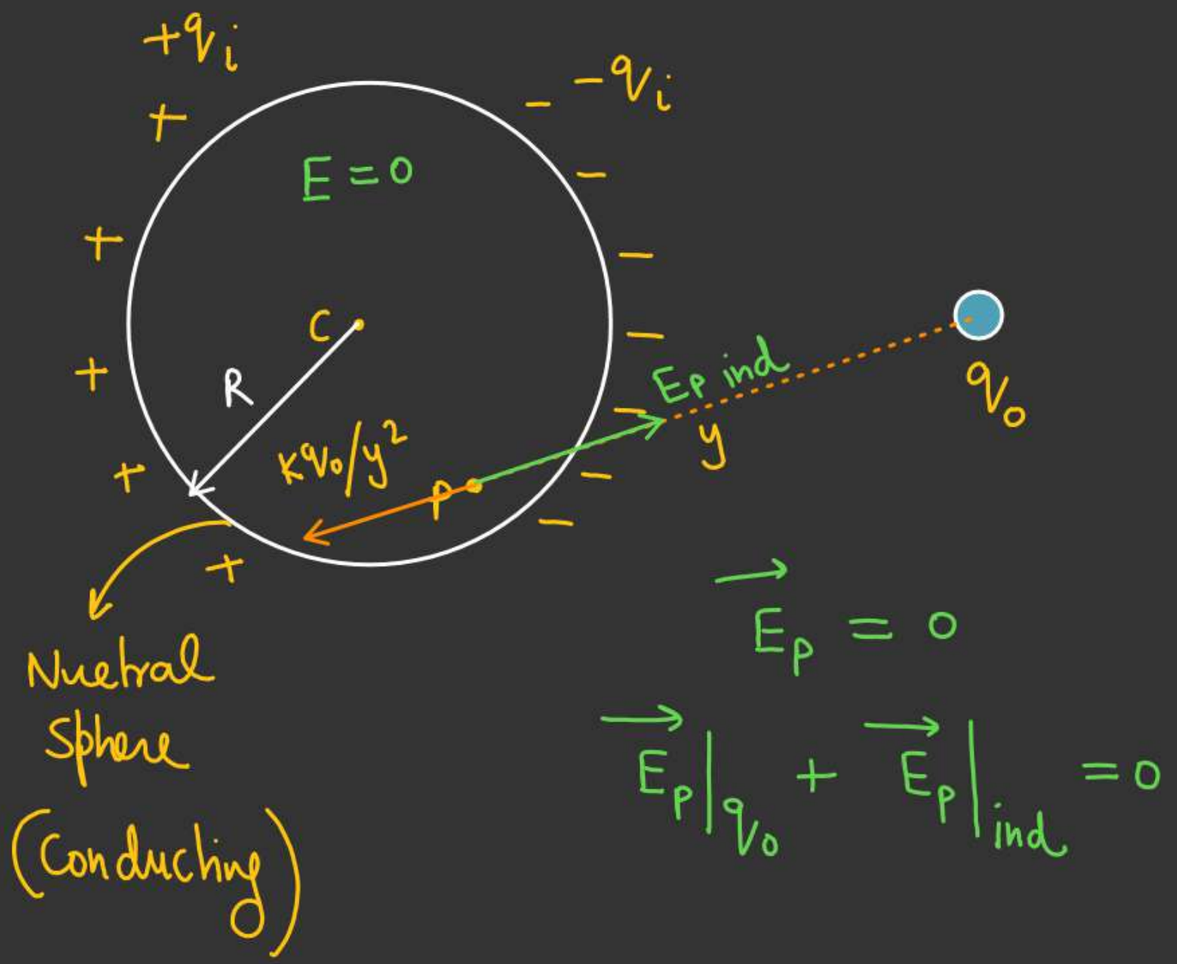
$$\boxed{V_P|_{ind} = ?}$$

$$\boxed{V_P|_{ind} = \frac{Kq_o}{x} - \frac{Kq_o}{y}}$$

$$V_C|_{ind} = \frac{K}{R}\int dq_{ind} = 0$$

Nishant Jindal

* Potential and field due to induced charges.

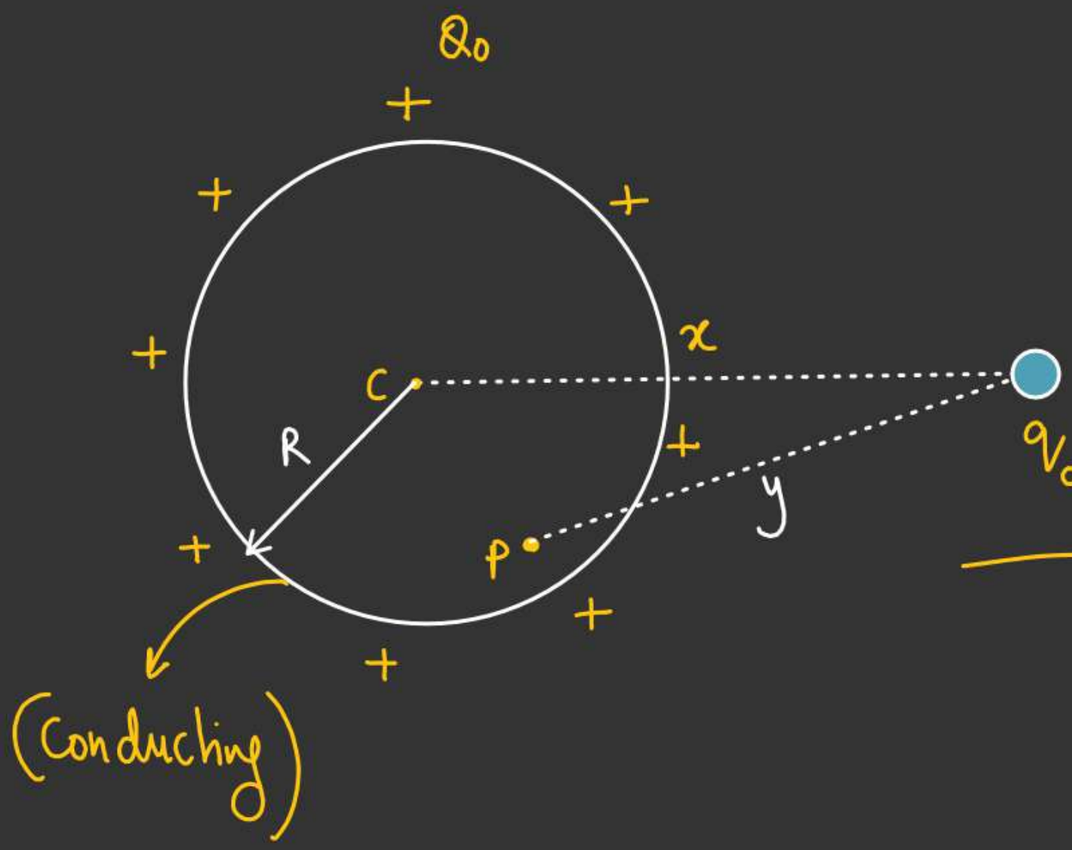


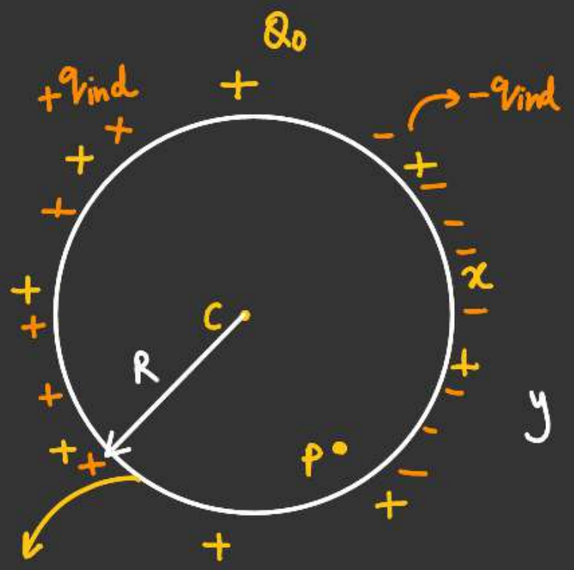
$$\vec{E}_p = 0$$

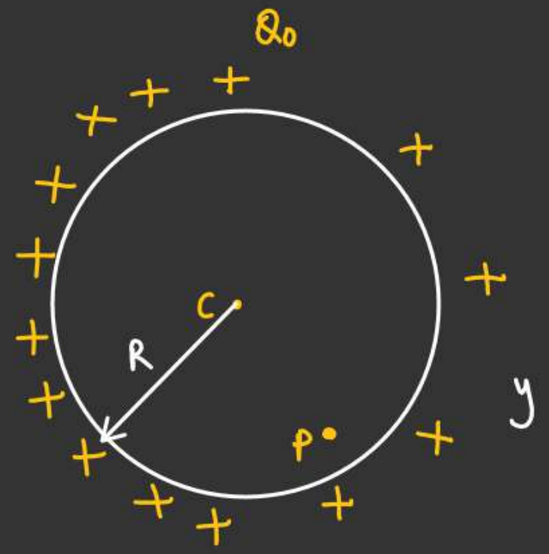
$$\vec{E}_p\big|_{q_0} + \vec{E}_p\big|_{ind} = 0$$

Nishant Jindal

Potential and field due to induced charges.







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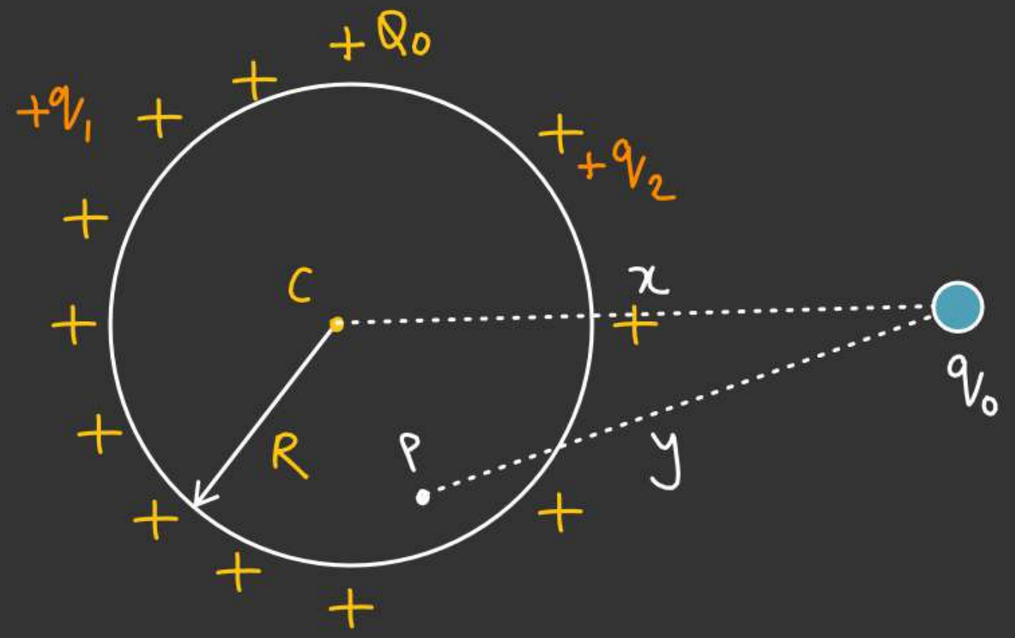
* Potential due to Induced charges :-



$q_1 + q_2 = Q_0$

$$V_P = V_C$$

$$V_P|_{ind} + V_P|_{q_0} = V_C|_{ind} + V_C|_{q_0}$$

$$V_P|_{ind} + \frac{Kq_0}{y} = \frac{KQ_0}{R} + \frac{Kq_0}{x}$$

$$V_P|_{ind} = \frac{KQ_0}{R} + \left(\frac{Kq_0}{x} - \frac{Kq_0}{y}\right)$$

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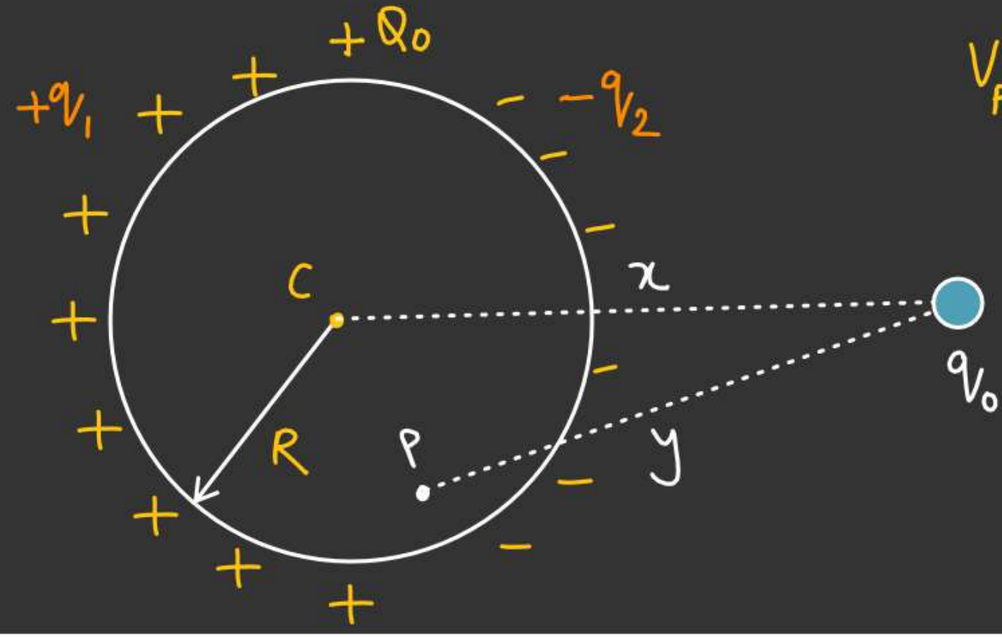
* Potential due to Induced charges :-

$q_1 - q_2 = Q_0$



$$V_P = V_C$$

$$V_P|_{Q_0} + V_P|_{ind} + V_P|_{q_0} = V_C|_{Q_0} + V_C|_{ind} + V_C|_{q_0}$$

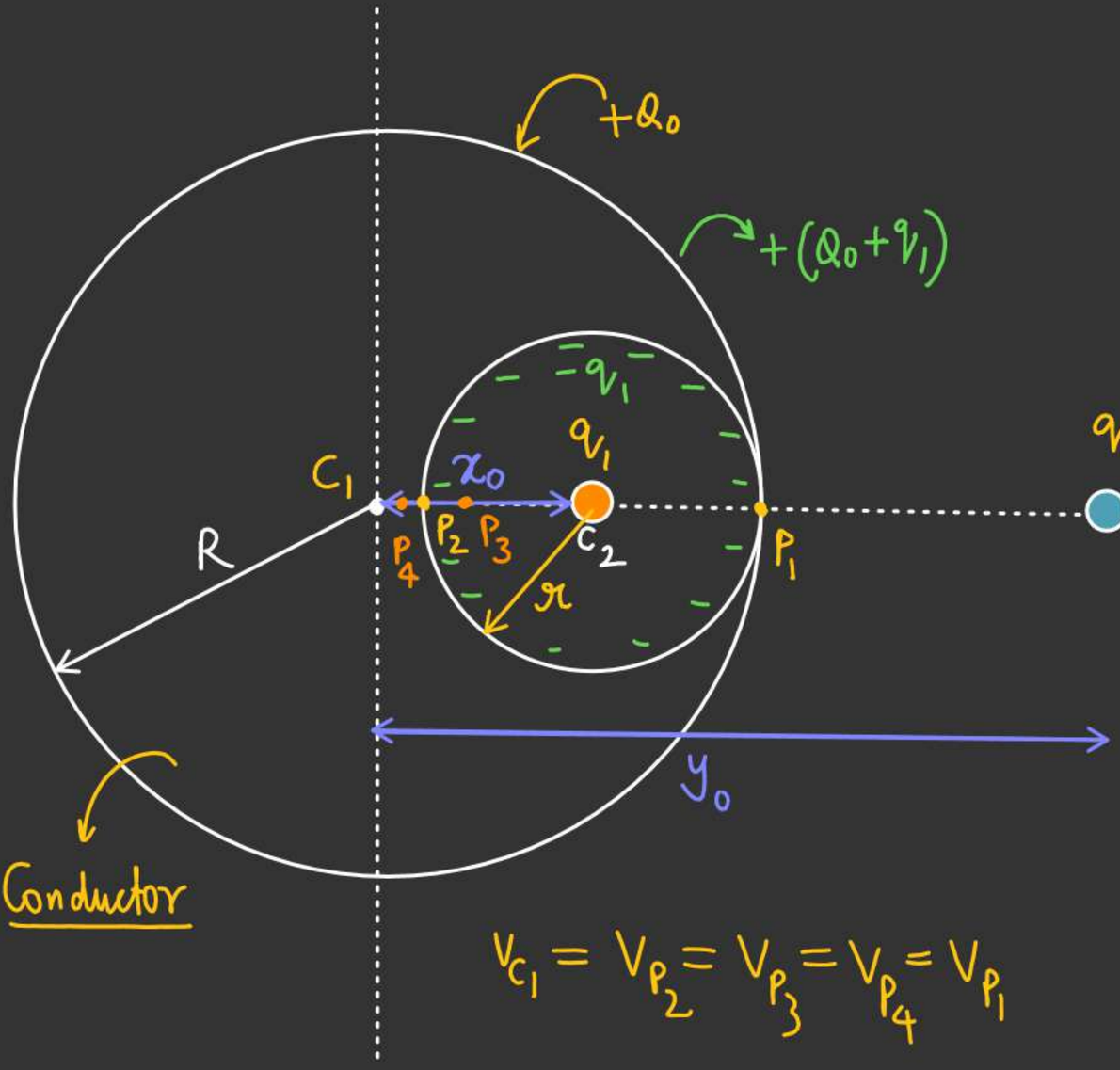
$$\frac{KQ_0}{R} + V_P|_{ind} + \frac{Kq_0}{y} = \frac{KQ_0}{R} + 0 + \frac{Kq_0}{x}$$

$$V_P|_{ind} = Kq_0\left(\frac{1}{x} - \frac{1}{y}\right)$$

$$V_P\Big|_{\text{Conductor Surface}} = \frac{KQ_0}{R} + \left(\frac{Kq_0}{x} - \frac{Kq_0}{y}\right)$$

Nishant Jindal

*



(I) $\vec{E}_{P_2} = 0$

(II) $E_{P_4} = 0$

(III) $E_{P_3} = \dfrac{Kq_1}{r^2}$

P_3 is just inside the cavity

$P_4 \rightarrow$ just outside

(IV) $V_{C_1} = \dfrac{Kq_0}{y_0} + \dfrac{K(Q_0+q_1)}{R}$

(V) $\boxed{V_{P_3} = \dfrac{Kq_0}{y_0} + \dfrac{K(Q_0+q_1)}{R}}$ Jump

(VI) $V_{P_1} = V_{P_4} = \dfrac{Kq_0}{y_0} + \dfrac{K(Q_0+q_1)}{R}$

$V_{C_1} = V_{P_2} = V_{P_3} = V_{P_4} = V_{P_1}$

Nishant Jindal

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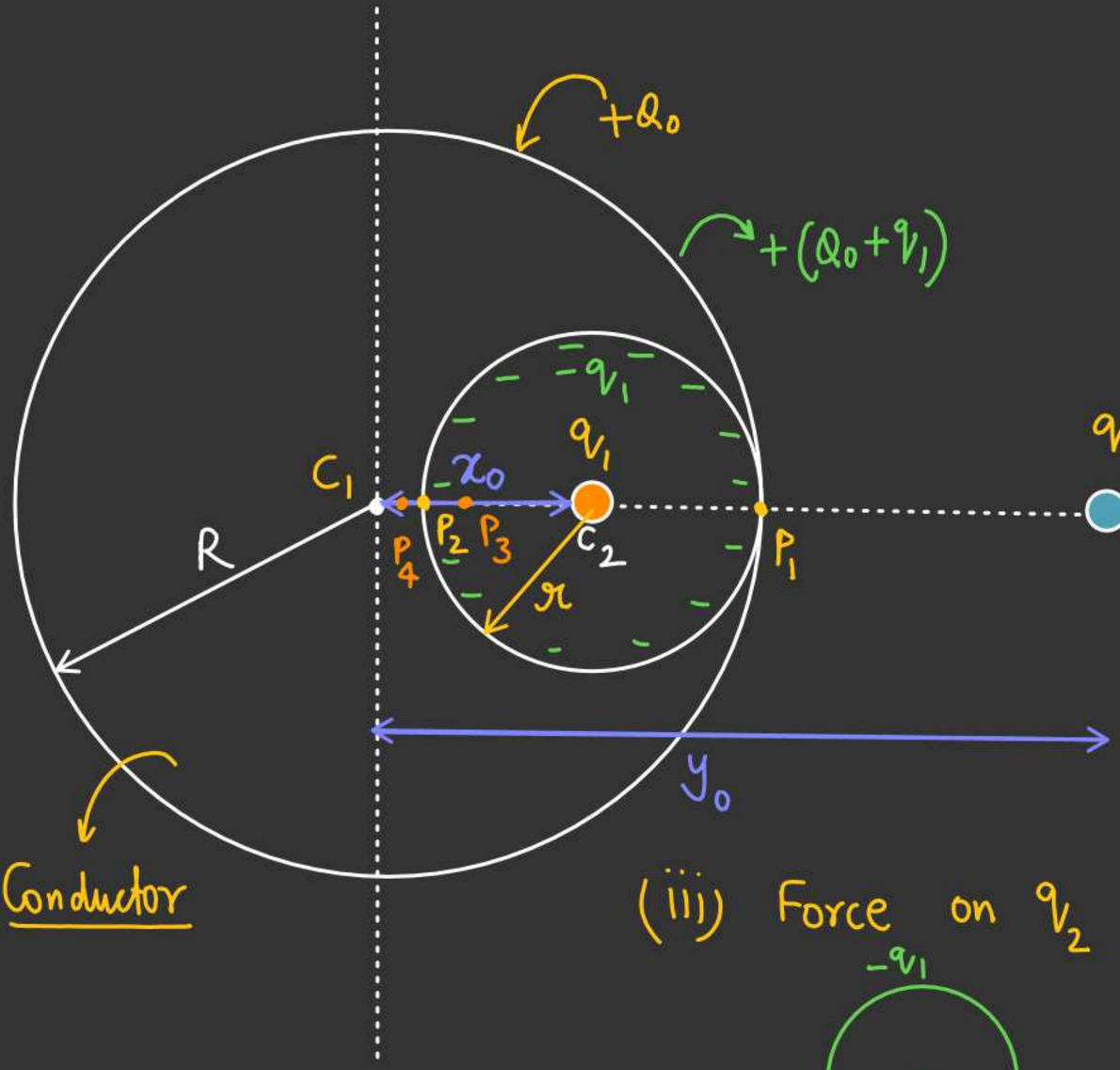

(i) $\vec{F}_{q_1} = 0$

(ii) Force on q_1 by sphere with cavity

$$\vec{F}_{q_1}\Big|_{sph} + \vec{F}_{q_1}\Big|_{q_2} = 0$$

$$\vec{F}_{q_1}\Big|_{sph} = \frac{K q_1 q_2}{(y_0 - x_0)^2} \longrightarrow$$

(iii) Force on q_2 by Charge on Surface of Cavity

$$\frac{K q_1 q_2}{(y_0 - x_0)^2} \longleftarrow$$

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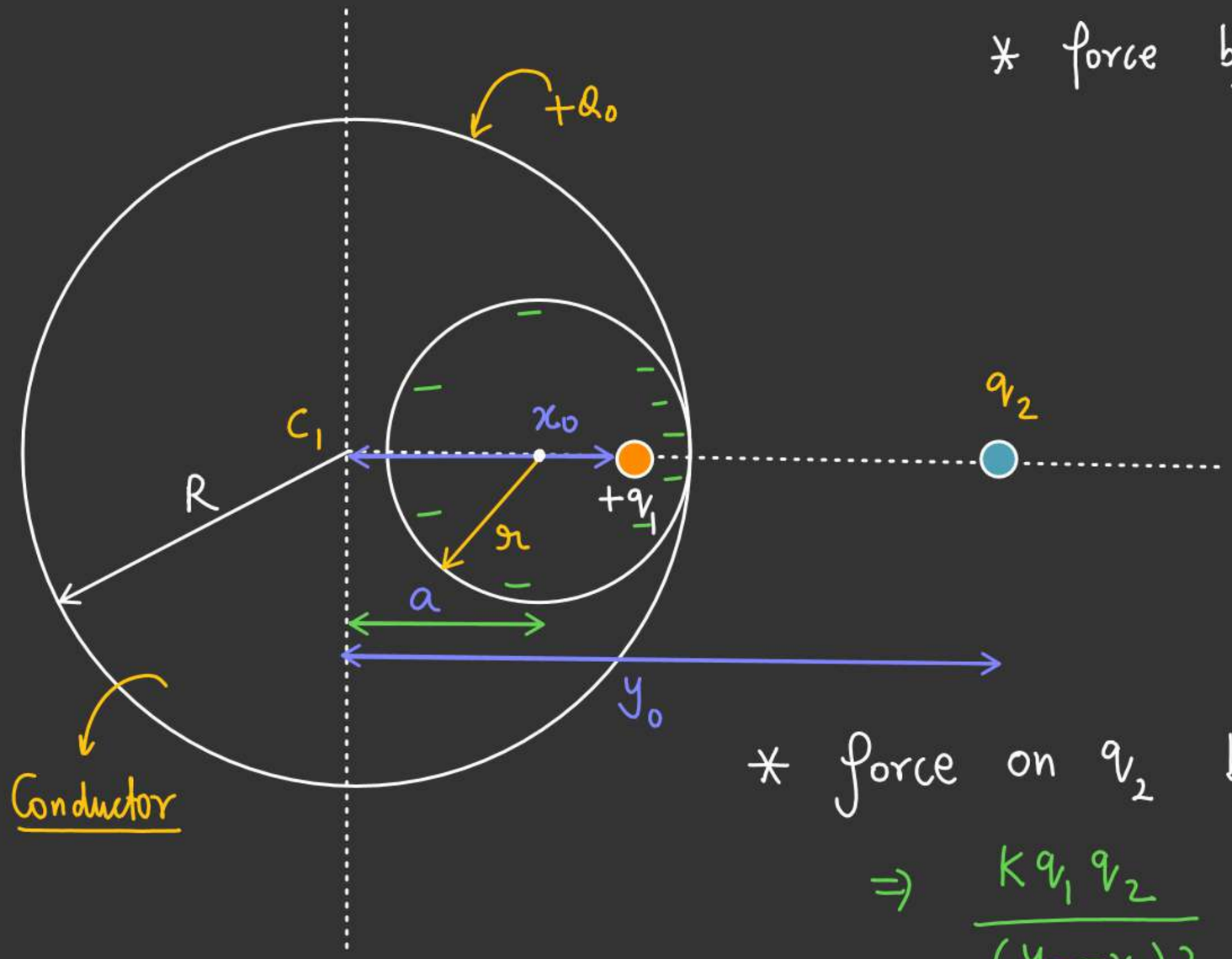
*



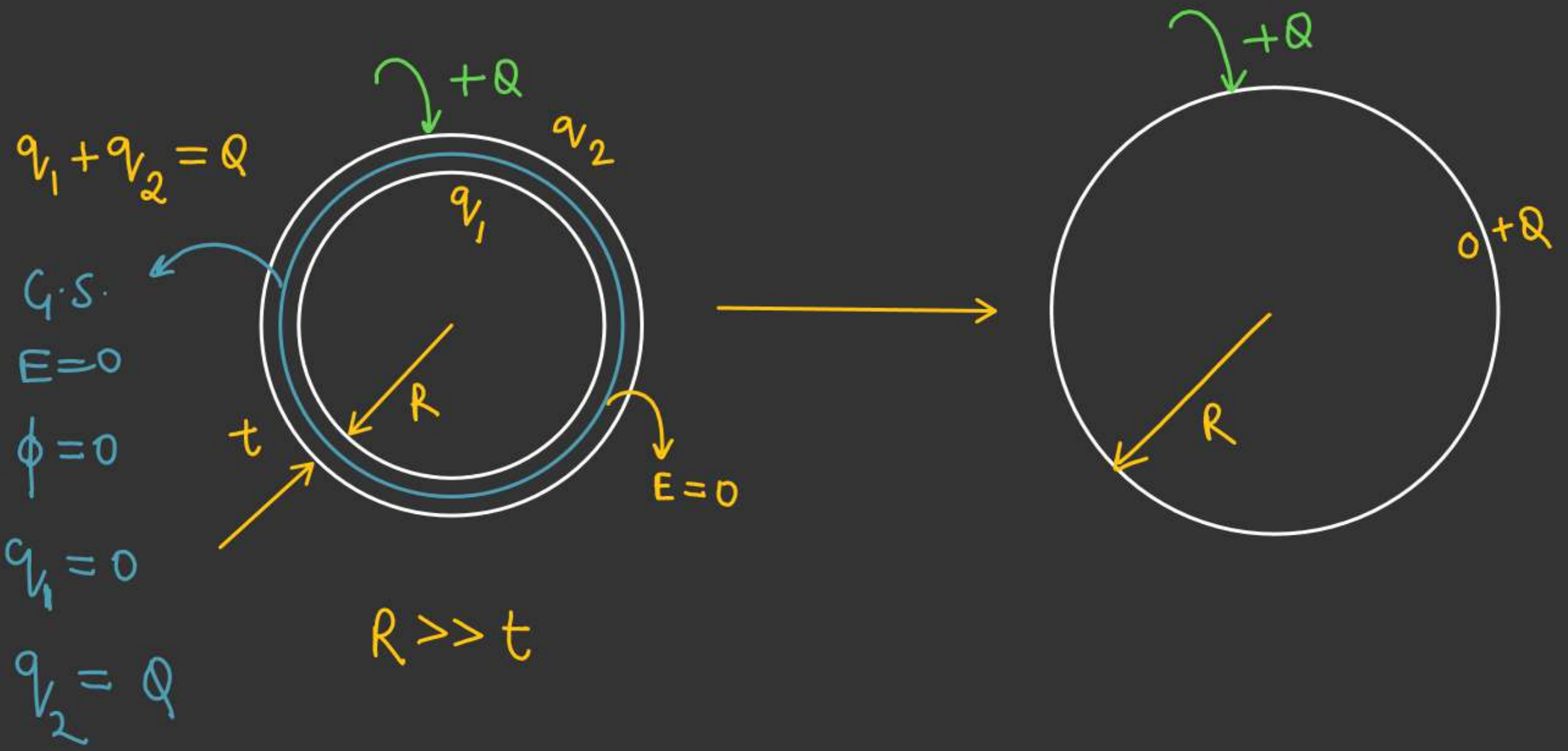
* force by sphere on q_1

* force on q_2 by charge on the Surface of Cavity

$$\Rightarrow \frac{K q_1 q_2}{(y_0 - x_0)^2} \leftarrow$$

Nishant Jindal

* System of shells :-



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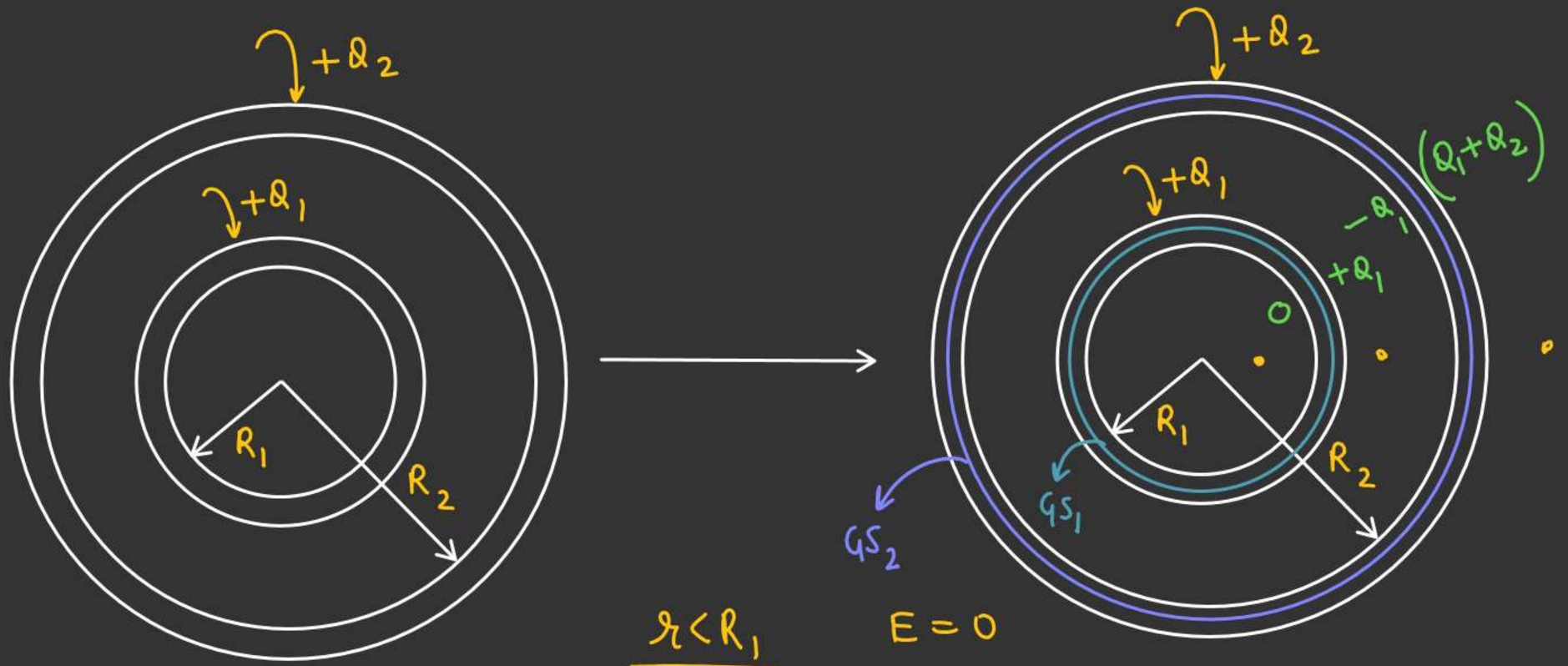
* Concentric shell



$r < R_1$ $\quad E = 0$

$R_1 < r < R_2$ $\quad E = \frac{kQ_1}{r^2}$

$r > R_2$ $\quad E = \frac{k(Q_1+Q_2)}{r^2}$

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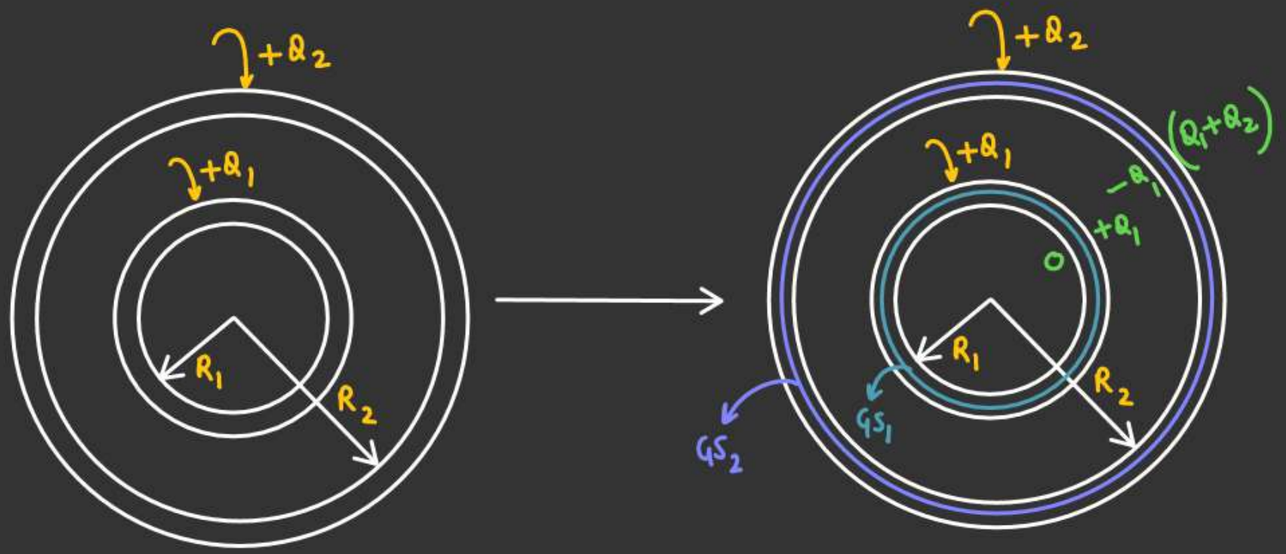
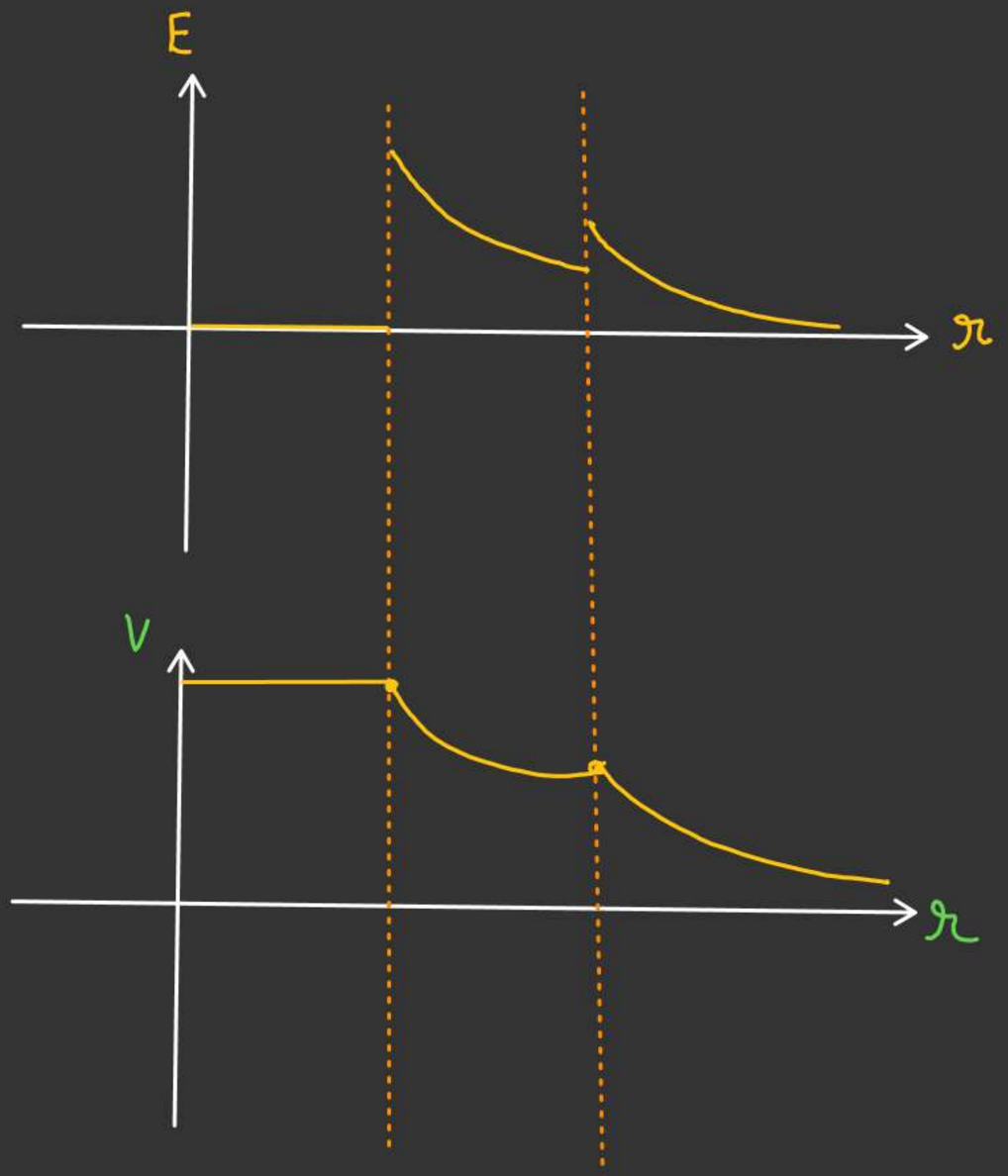
* Concentric shell





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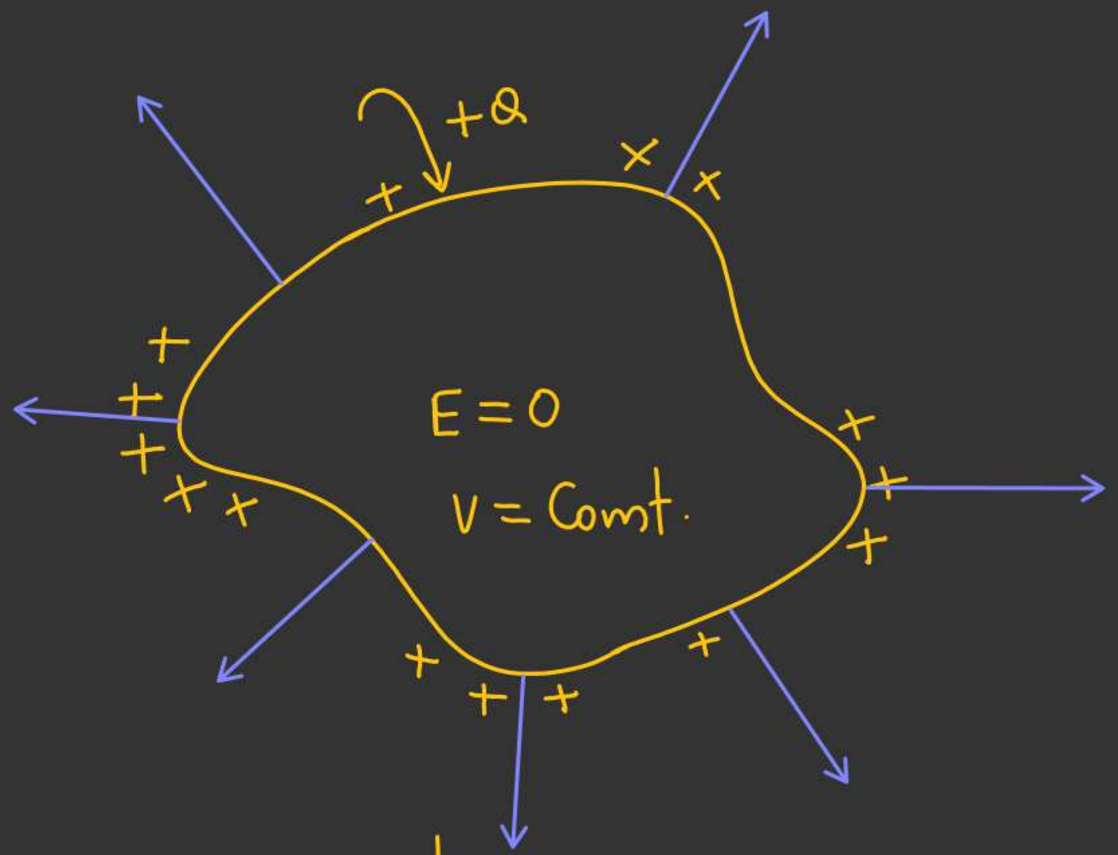
* Electrostatic Pressure :-



$\sigma \propto \frac{1}{R_c}$ → approx.

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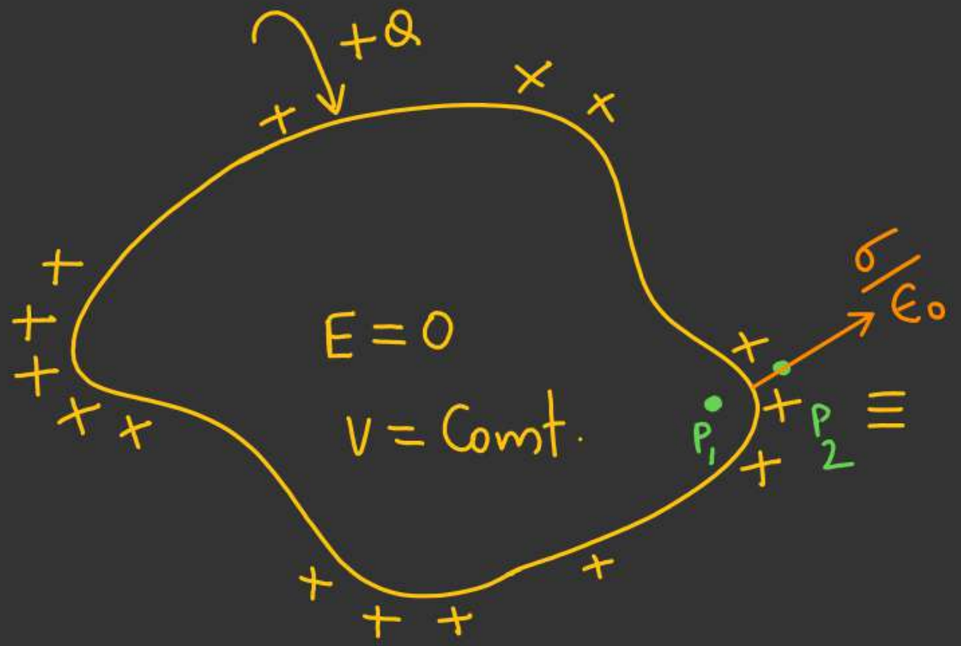
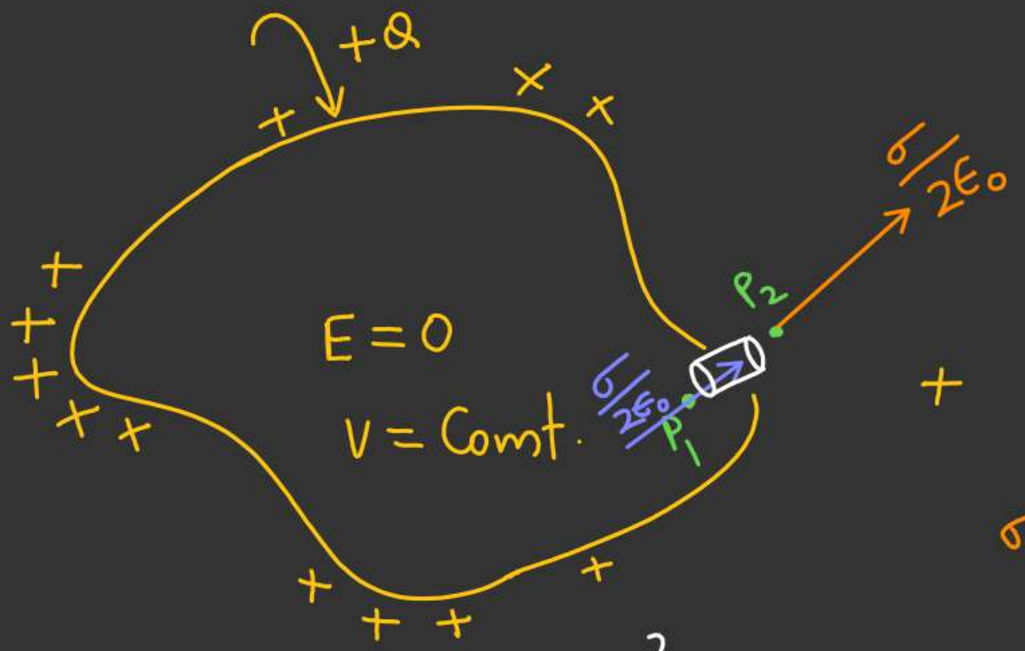
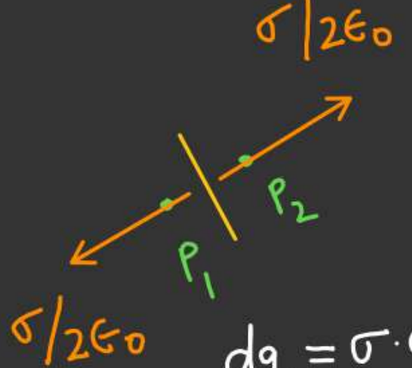
* Electrostatic Pressure :-







$dq = \sigma \cdot dA$

$$dF = dq \cdot \frac{\sigma}{2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0} \cdot dA$$

$$\sigma \propto \frac{1}{R_c} \rightarrow \text{approx.}$$

$$\boxed{P_e = \frac{dF}{dA} = \frac{\sigma^2}{2\epsilon_0}}$$

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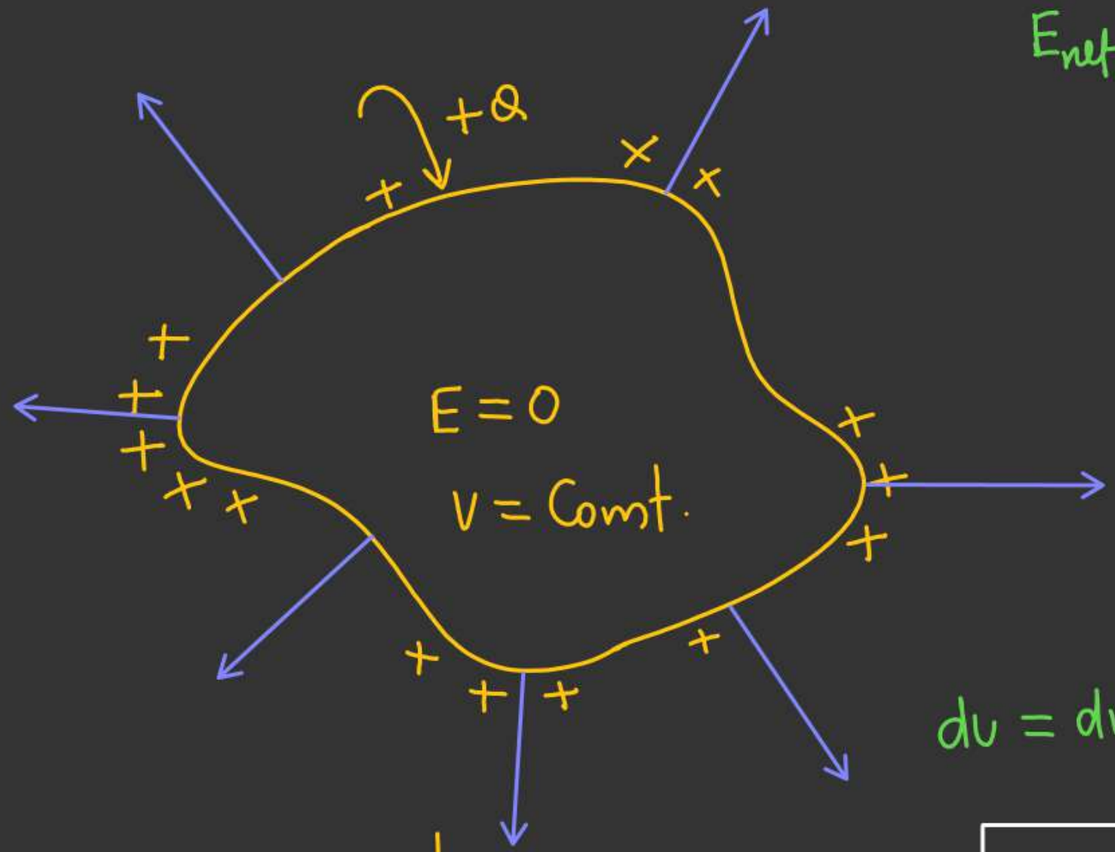
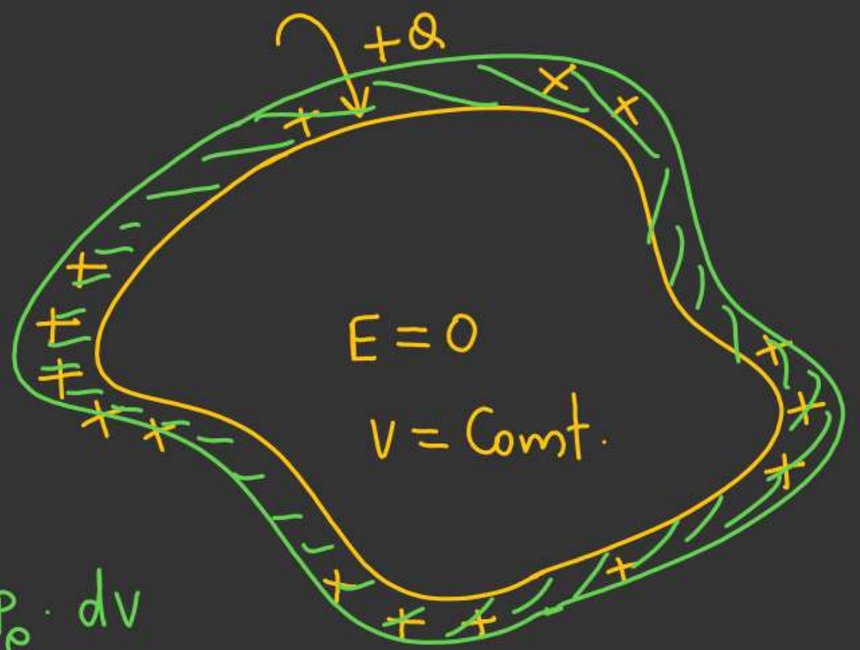
* Electrostatic Pressure :-



$$E_{net} = \frac{\sigma}{\epsilon_0}$$



$$dU = dW_{ext} = P_e \cdot dV$$

$$\sigma \propto \frac{1}{R_c} \rightarrow \underline{\text{approx.}}$$

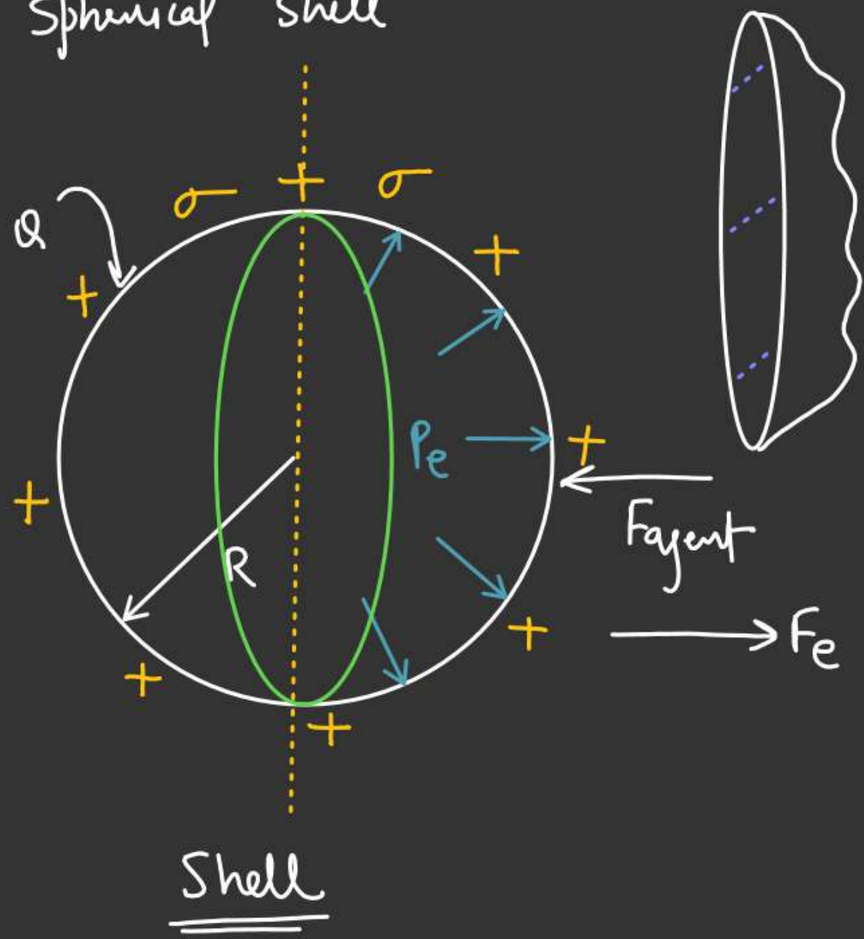
$$u_e = P_e = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2}\epsilon_0 E_{net}^2$$

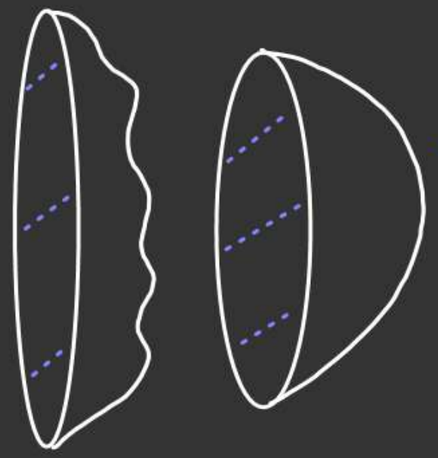
Nishant Jindal

**

V. Imp

Spherical Shell





* force req. to hold the hemispheres together

* force b/w the two halves

$$F_{agent} = F_e = P_e \cdot \pi R^2 = \frac{\sigma^2}{2\epsilon_0} \cdot \pi R^2$$

$$\sigma = \frac{Q}{4\pi R^2}$$

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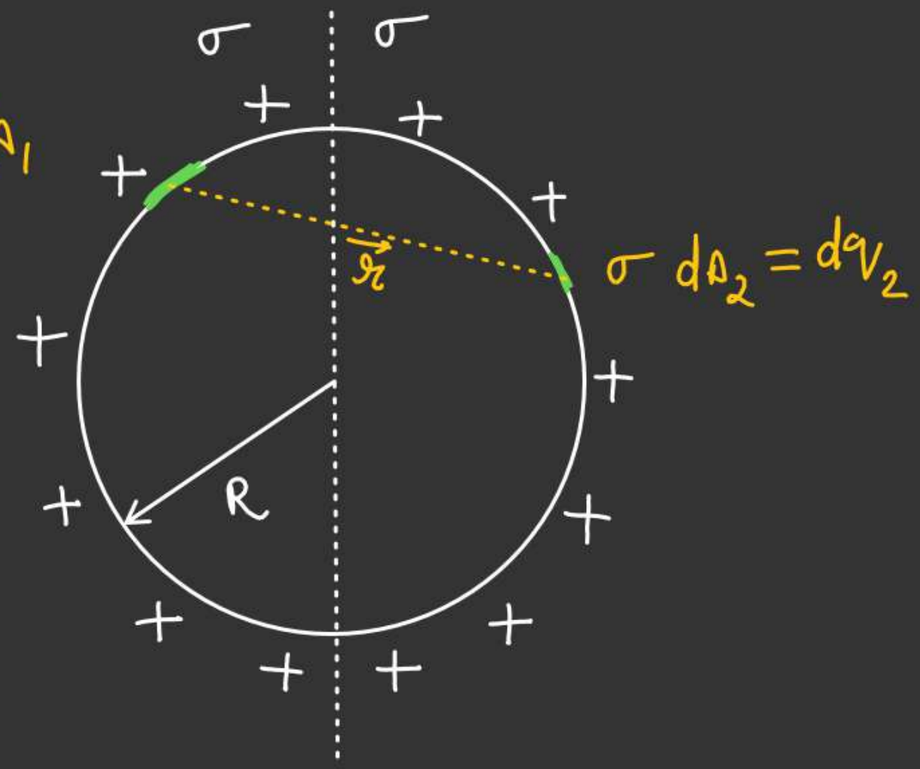
*

Force b/w the two Halves



$dq_1 = \sigma\, dA_1$

$\sigma\, dA_2 = dq_2$

$$\vec{F_e} = K\sigma\cdot\sigma \int \frac{dA_1\, dA_2\, \vec{r}}{r^3} = \frac{\sigma^2}{2\epsilon_0}\cdot \pi R^2$$

↓ depends on geometry and not on charge distribution.

$$\vec{dF_e} = \frac{K\cdot dq_1\, dq_2\, \vec{r}}{r^3} \Rightarrow K\sigma\sigma\, \frac{dA_1\, dA_2\, \vec{r}}{r^3}$$

$$\int \frac{dA_1\, dA_2}{r^3}\vec{r} = \frac{\pi R_1^2}{2\epsilon_0 K}$$

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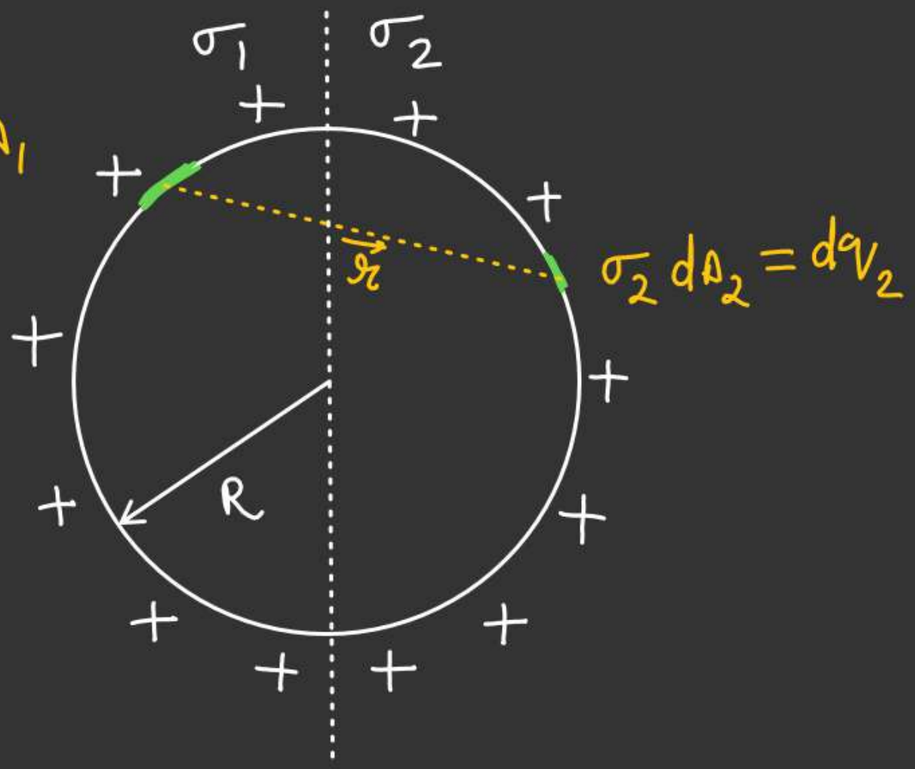
*

Force b/w the two Halves



$$\vec{F_e} = K\sigma_1\sigma_2 \int \frac{dA_1\, dA_2\, \vec{r}}{r^3}$$

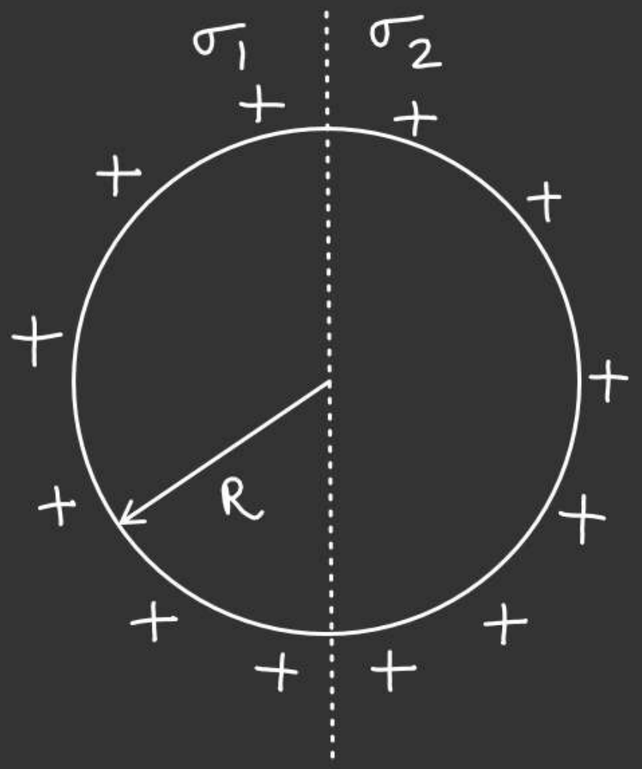
$$F \Rightarrow \frac{\sigma_1\sigma_2}{2\epsilon_0} \cdot \pi R^2$$

$$\vec{dF_e} = \frac{K \cdot dq_1\, dq_2\, \vec{r}}{r^3} \Rightarrow K\sigma_1\sigma_2 \frac{dA_1\, dA_2\, \vec{r}}{r^3}$$

Nishant Jindal

*

Force b/w the two Halves



A) $\frac{\sigma_1^2+\sigma_2^2}{4\epsilon_0} \cdot \pi R^2$

B) $\frac{\sigma_1\sigma_2}{2\epsilon_0} \cdot \pi R^2$

C) $\frac{(\sigma_1+\sigma_2)^2}{8\epsilon_0} \cdot \pi R^2$

D) None

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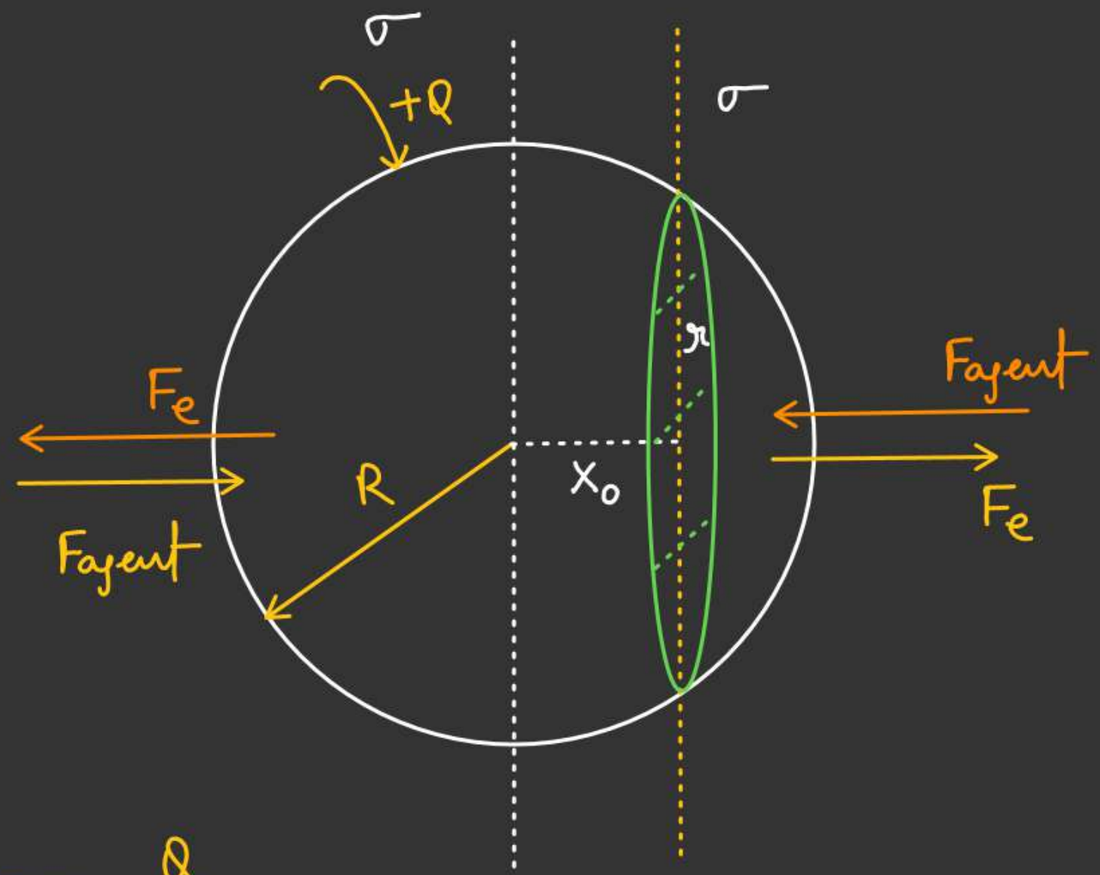


$$\sigma = \frac{Q}{4\pi R^2}$$

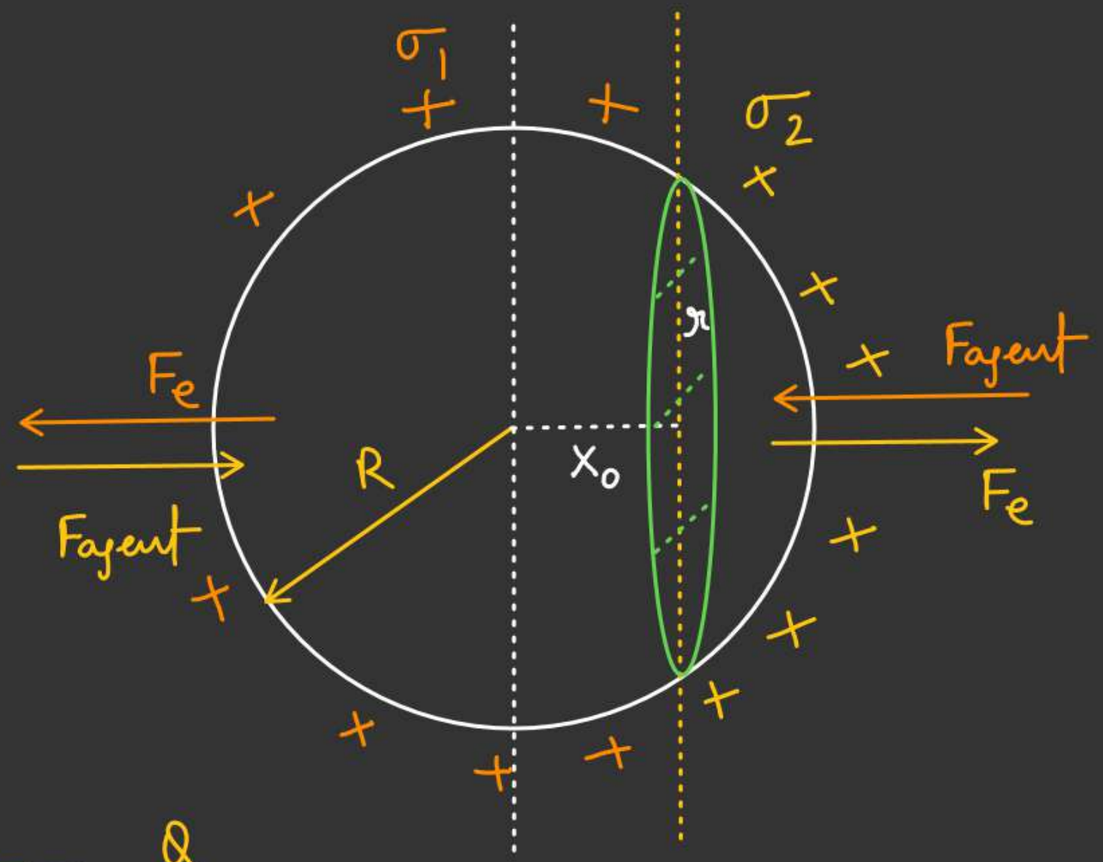
$$r = \sqrt{R^2 - x_0^2}$$

$$F_{agent} = F_e = \frac{\sigma^2}{2\epsilon_0} \cdot \pi r^2$$

$$= \frac{\sigma^2}{2\epsilon_0} \cdot \pi \left(R^2 - x_0^2\right)$$

Nishant Jindal

*



$$\sigma = \frac{Q}{4\pi R^2}$$

$$r = \sqrt{R^2 - x_0^2}$$

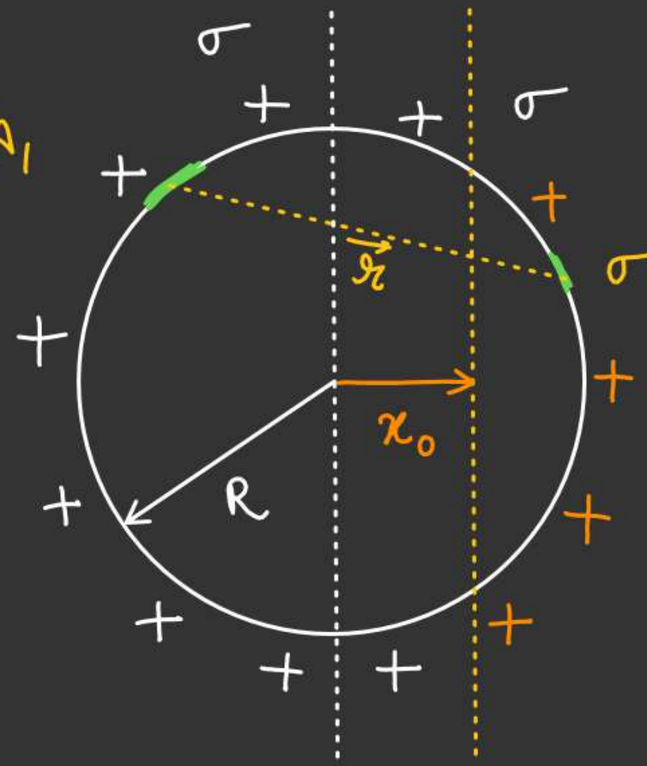
$$F_{agent} = F_e = \frac{\sigma_1 \sigma_2}{2\epsilon_0} \cdot \pi r^2$$

$$= \frac{\sigma_1 \sigma_2}{2\epsilon_0} \cdot \pi \left(R^2 - x_0^2 \right)$$

Nishant Jindal

*

Force b/w the two Halves



$$\vec{F_e} = K\sigma\sigma \int \frac{dA_1\, dA_2\, \vec{r}}{r^3}$$

$$\Rightarrow \frac{\sigma^2}{2\epsilon_0} \cdot \pi\left(R^2 - x_0^2\right)$$

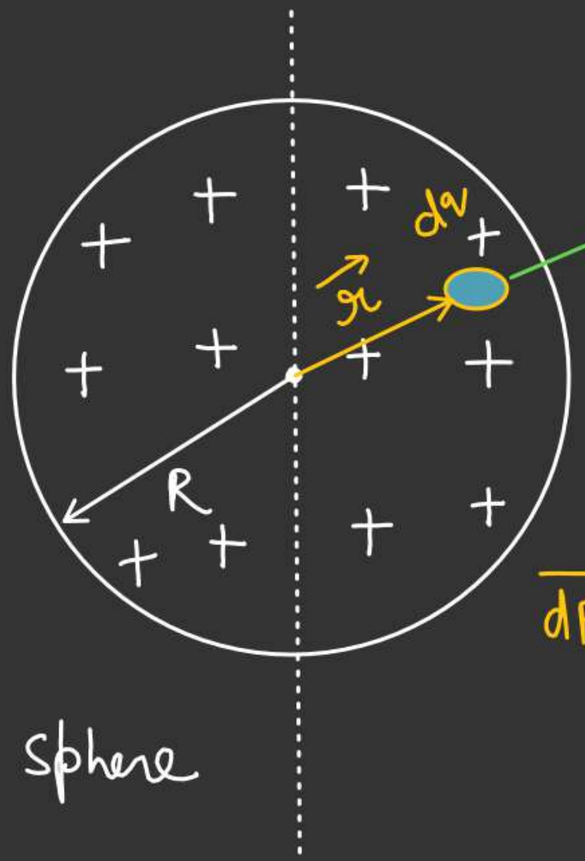
$$\vec{dF_e} = \frac{K \cdot dq_1\, dq_2\, \vec{r}}{r^3} \Rightarrow K\sigma\sigma \frac{dA_1\, dA_2\, \vec{r}}{r^3}$$

$$\int \frac{dA_1\, dA_2\, \vec{r}}{r^3} = \frac{\pi\left(R^2 - x_0^2\right)}{2\epsilon_0 K}$$

Nishant Jindal

Force b/w Solid Hemispheres.

Q_0, ρ_0



Solid sphere

$$Q_0 = \rho_0 \cdot \frac{4}{3}\pi R^3$$

$$\vec{E} = \frac{KQ_0\vec{r}}{R^3} = \frac{\rho\vec{r}}{3\epsilon_0}$$

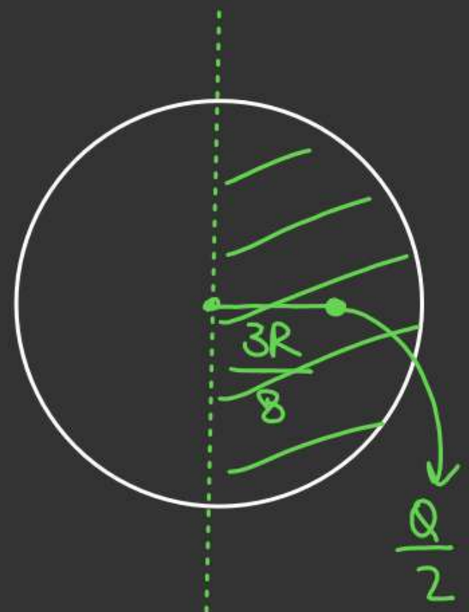
$$\vec{dF} = dq \cdot \vec{E}$$

$$\vec{dF} = \frac{\rho}{3\epsilon_0} \cdot \vec{r} \cdot dq = \frac{KQ}{R^3} \cdot \vec{r} \cdot dq$$

$$\vec{F} = \int \vec{dF} = \frac{\rho}{3\epsilon_0} \underbrace{\int \vec{r} \cdot dq}_{\text{Hemisphere}}$$

$$\Rightarrow \frac{\rho}{3\epsilon_0} \cdot \frac{3R}{8} \cdot \rho_0 \cdot \frac{2}{3}\pi R^3 \Rightarrow \frac{\pi \rho_0^2 R^4}{12\epsilon_0}$$

$$\int \vec{r} \cdot dm = \vec{r}_{cm} \cdot M$$



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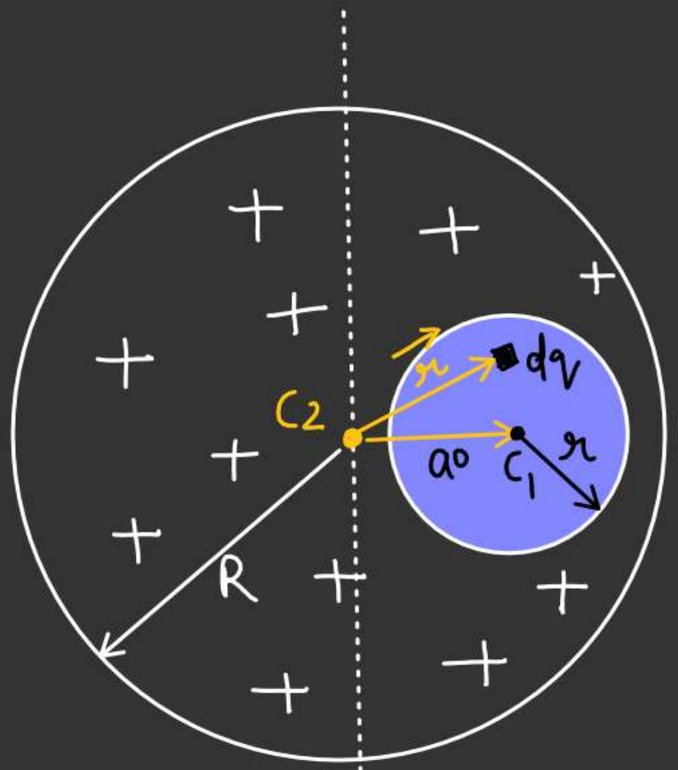

$$\vec{dF} = \frac{KQ}{R^3}\int \vec{r}\cdot dq = \frac{\rho_0}{3\epsilon_0}\int \vec{r}\cdot dq$$

$$\vec{F} \Rightarrow \frac{KQ}{R^3}\cdot a_0 \cdot \frac{Q_0 r^3}{R^3} \longrightarrow$$

$$\boxed{\vec{F} = \frac{\rho_0}{3\epsilon_0}\cdot a_0 \cdot \rho_0 \cdot \frac{4}{3}\pi r^3 \longrightarrow}$$

* force on shaded (blue) due to remaing Sphere.

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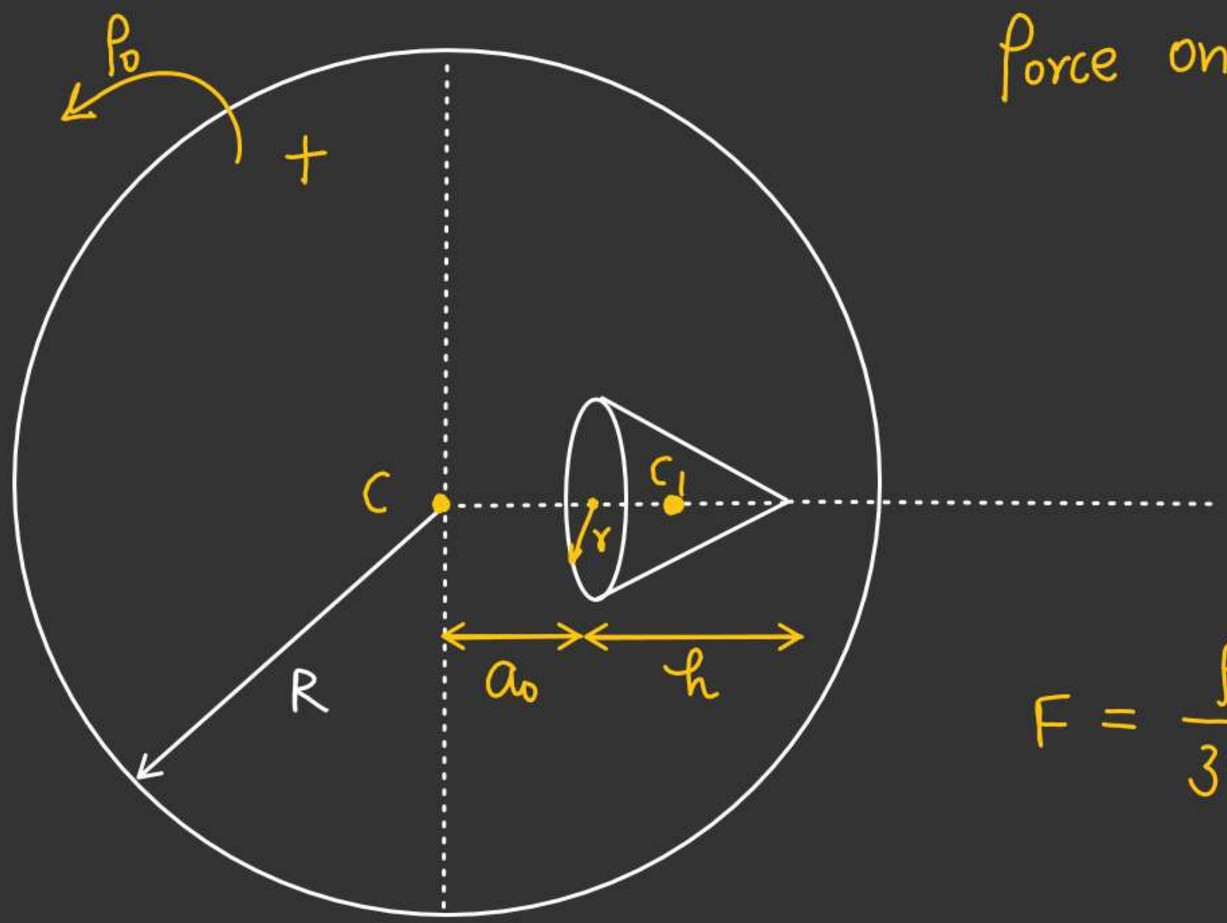

Force on Cone due to remaining sphere.

$$F = \frac{\rho_0}{3\epsilon_0} \cdot \left(a_0 + \frac{h}{4}\right) \cdot \rho_0 \cdot \frac{1}{3}\pi r^2 h \longrightarrow$$

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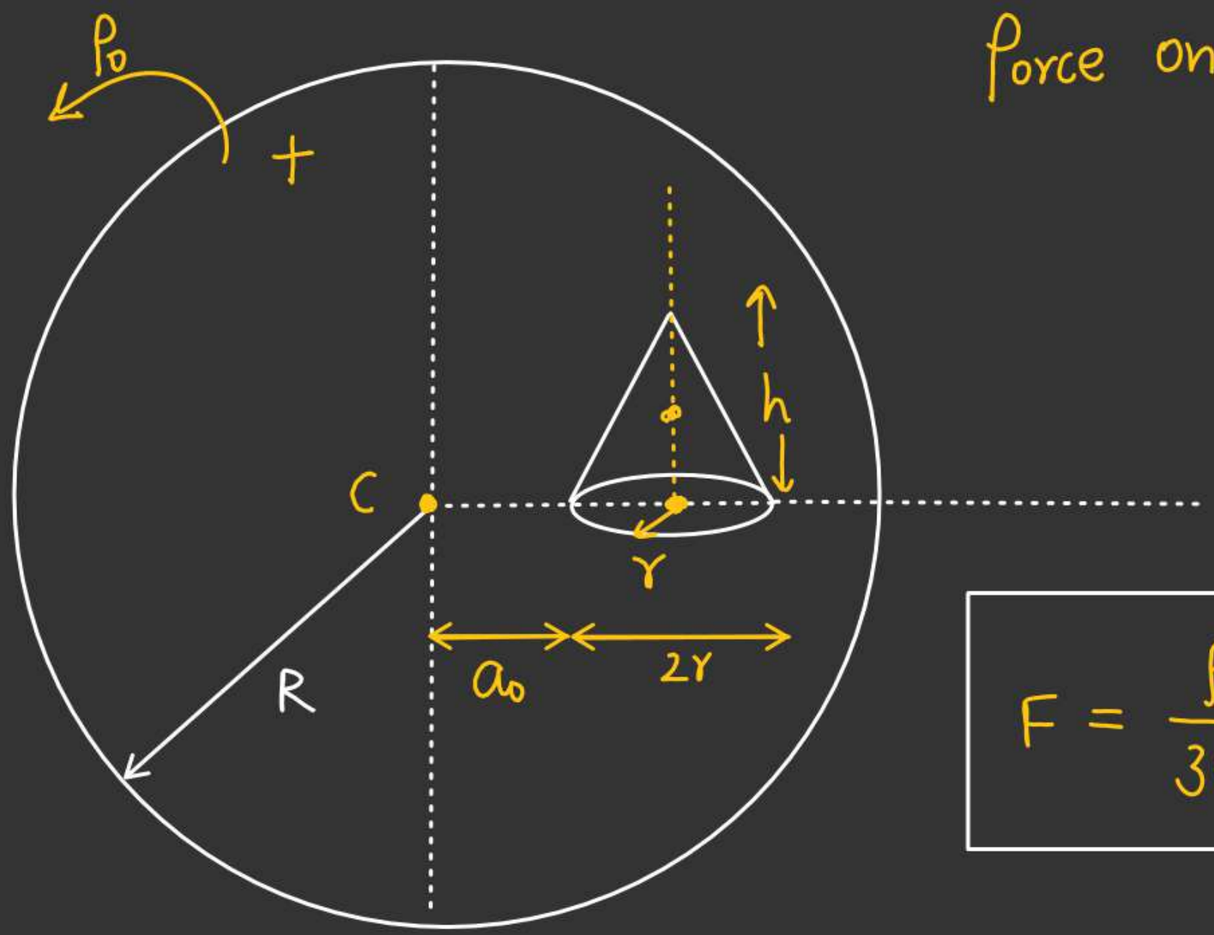

Force on Cone due to remaining sphere.

$$F = \frac{\rho_0}{3\epsilon_0} \cdot \left((a_0 + r)\hat{i} + \frac{h}{4}\hat{j} \right) \cdot \rho_0 \cdot \frac{1}{3}\pi r^2 h$$

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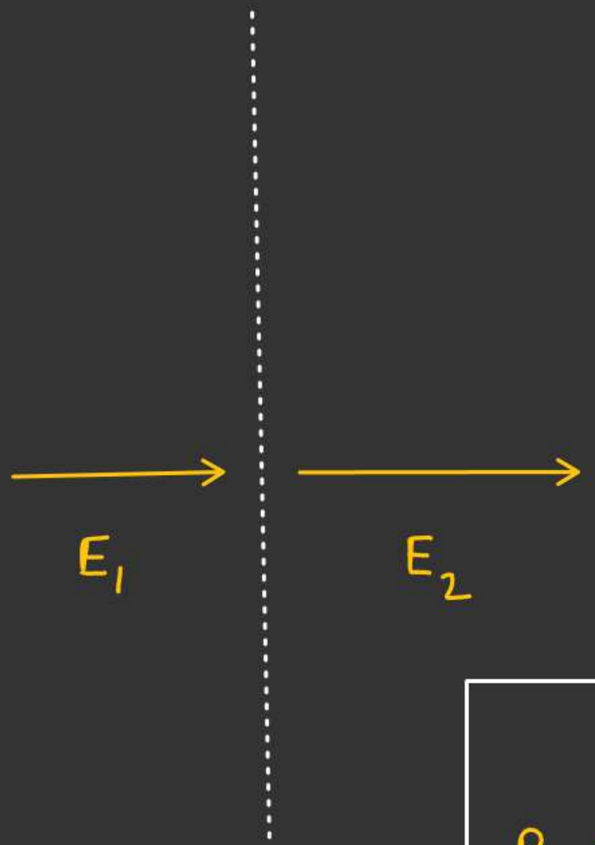

$$\rho_s = \left| \frac{1}{2} \epsilon_0 \left(E_1^2 - E_2^2 \right) \right|$$

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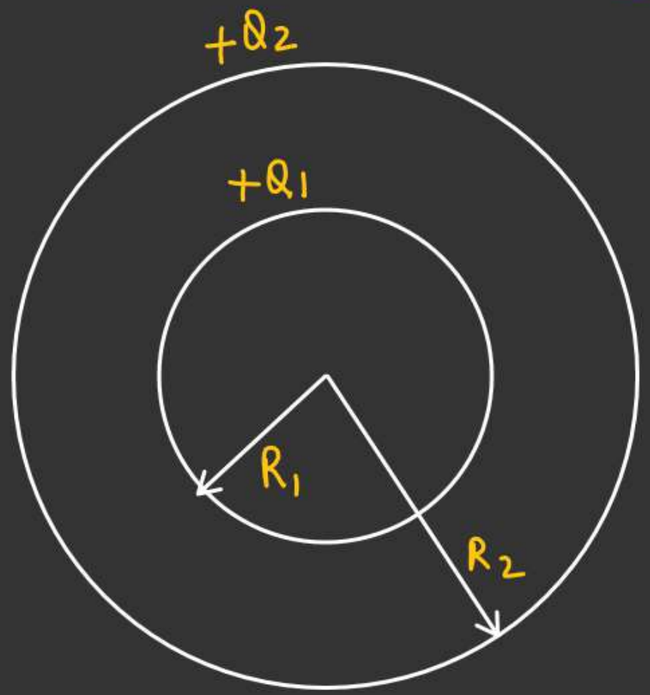
* H.C.

Concentric shells



$P_e|_{S_1} = ?$

$P_e|_{S_2} = ?$

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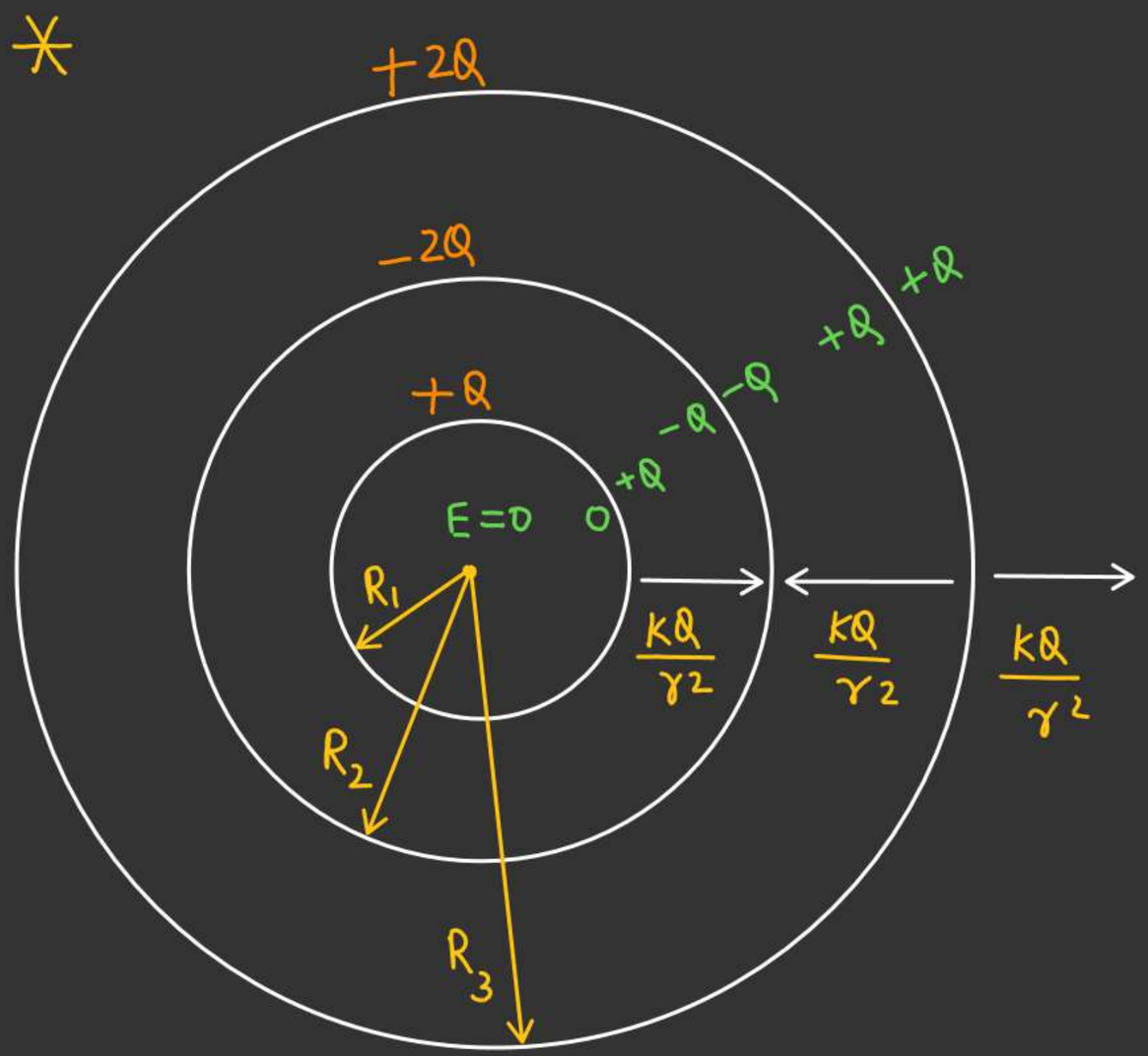

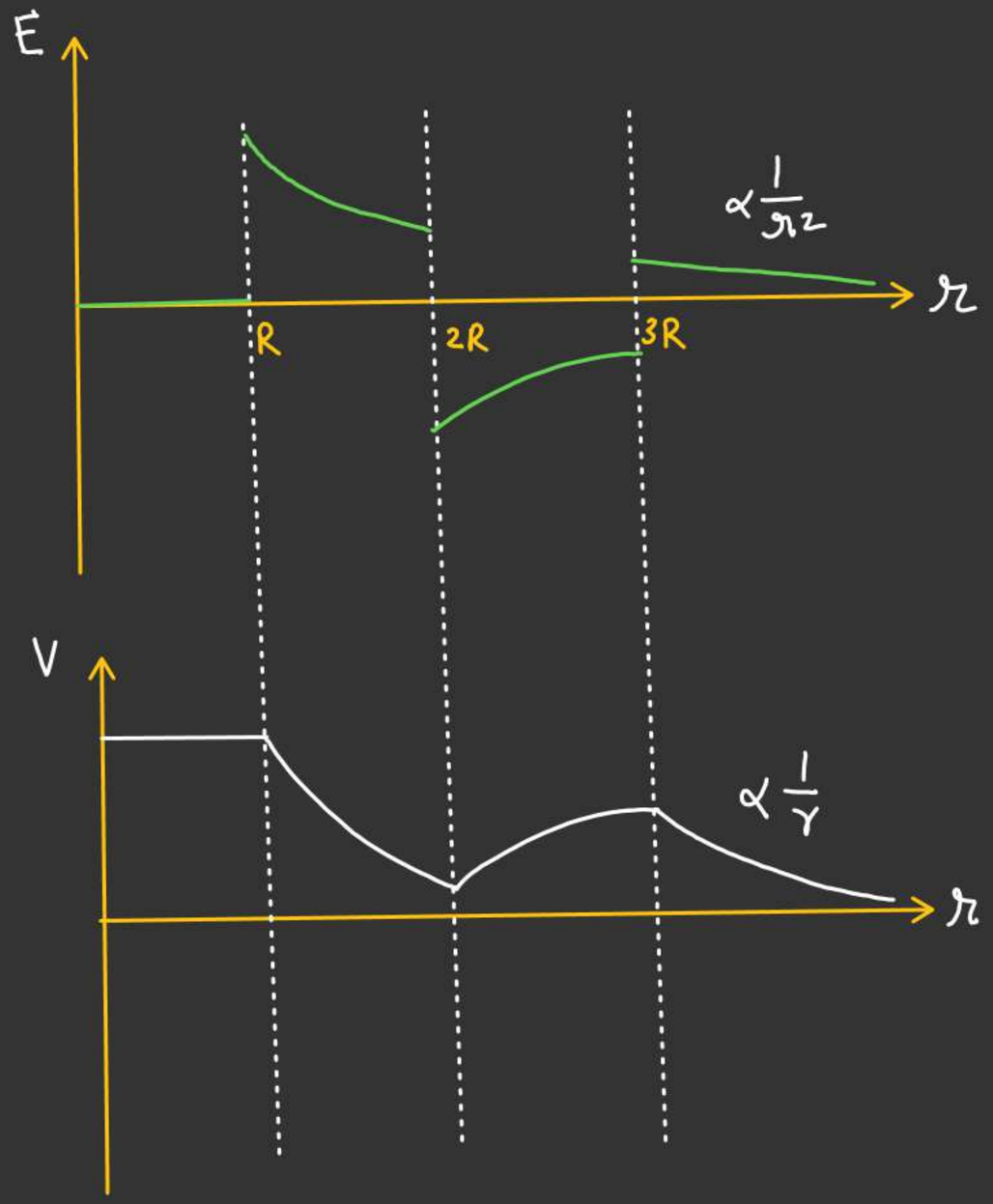

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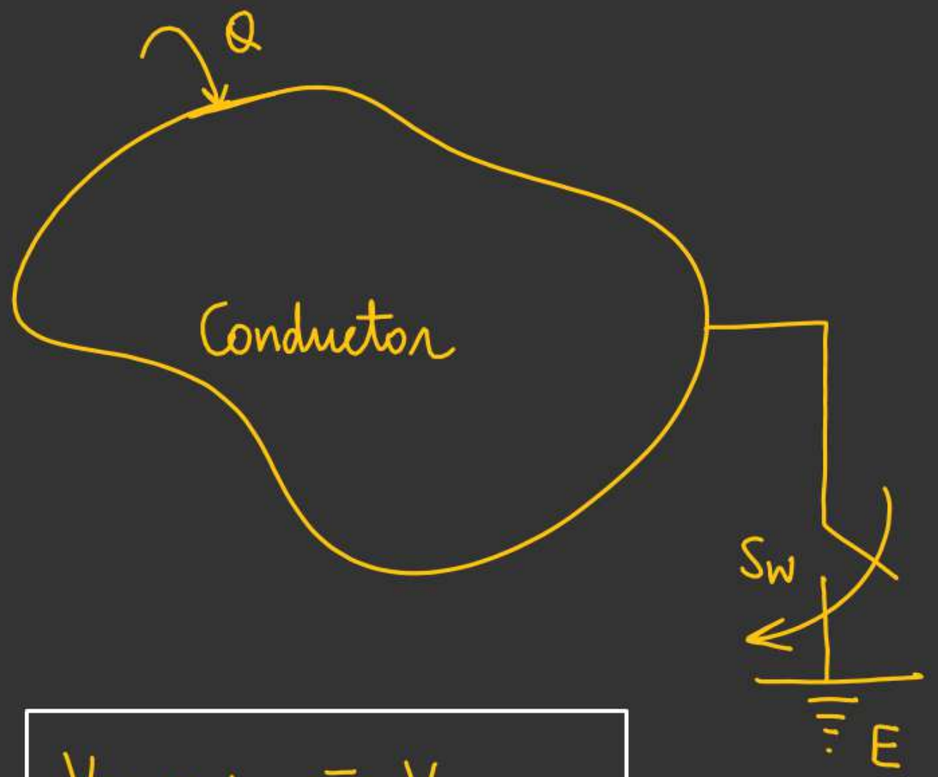
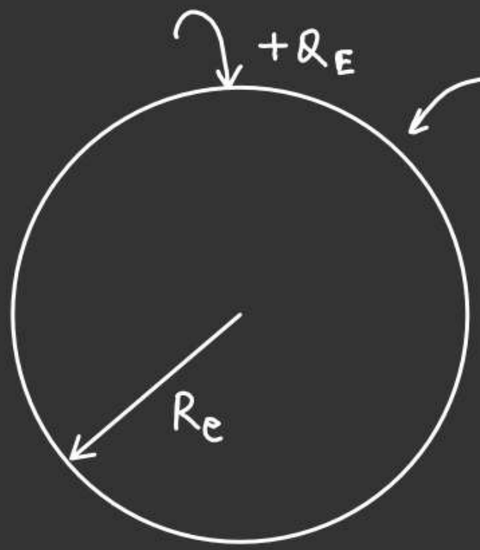
Earthing of Conductors :-



$$V_{conductor} = V_{Earth}$$



$$V_E|_0 = \frac{kQ_E}{R_e}$$

$$V_E|' = \frac{k(Q_E + q_{ext})}{R_e}$$

$$= \frac{kQ_E}{R_e} + \frac{k \cdot q_{ext}}{R_e} \approx \frac{kQ_E}{R_e}$$

$$V_{E'} \approx V_{E^0}$$

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* Since potential of earth doesn't change on giving/taking a finite charge

$\Rightarrow$ Ref. of potential at earth.

$\Rightarrow$ $\boxed{V_{Earth} = 0}$

* If Ref. is taken at earth Surface then potential at infinity may or may not be zero.

* Charge on the Conductor may or may not be zero.

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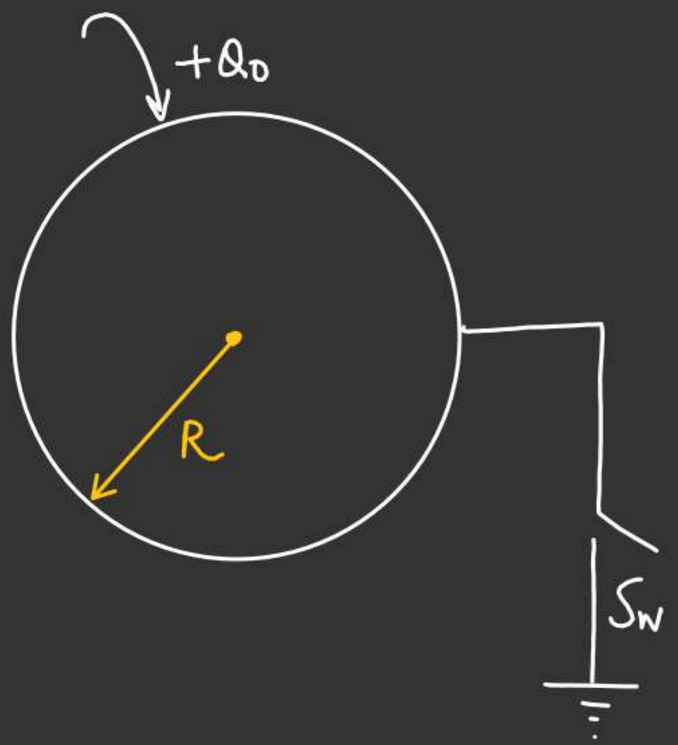

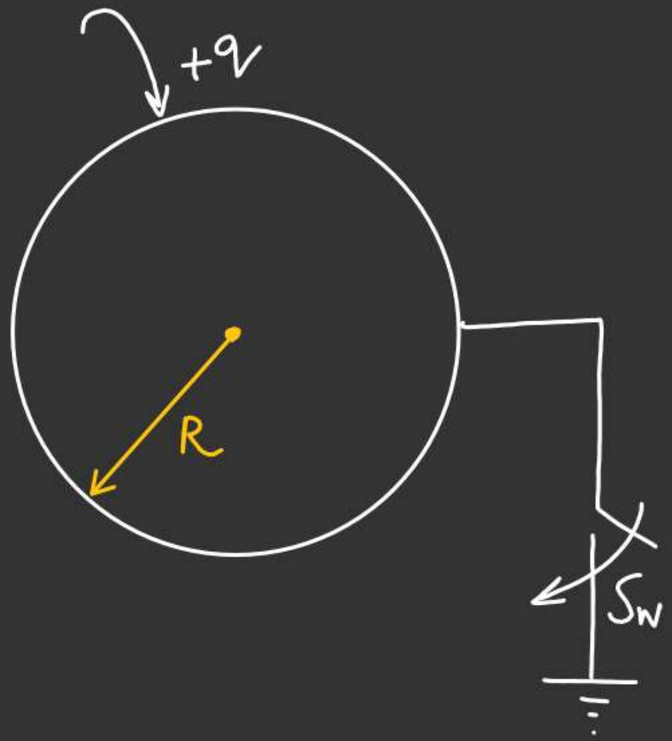

$$V_{Sph} = V_{Earth} = \frac{kq}{R} = 0$$

$$\Rightarrow \boxed{q = 0}$$

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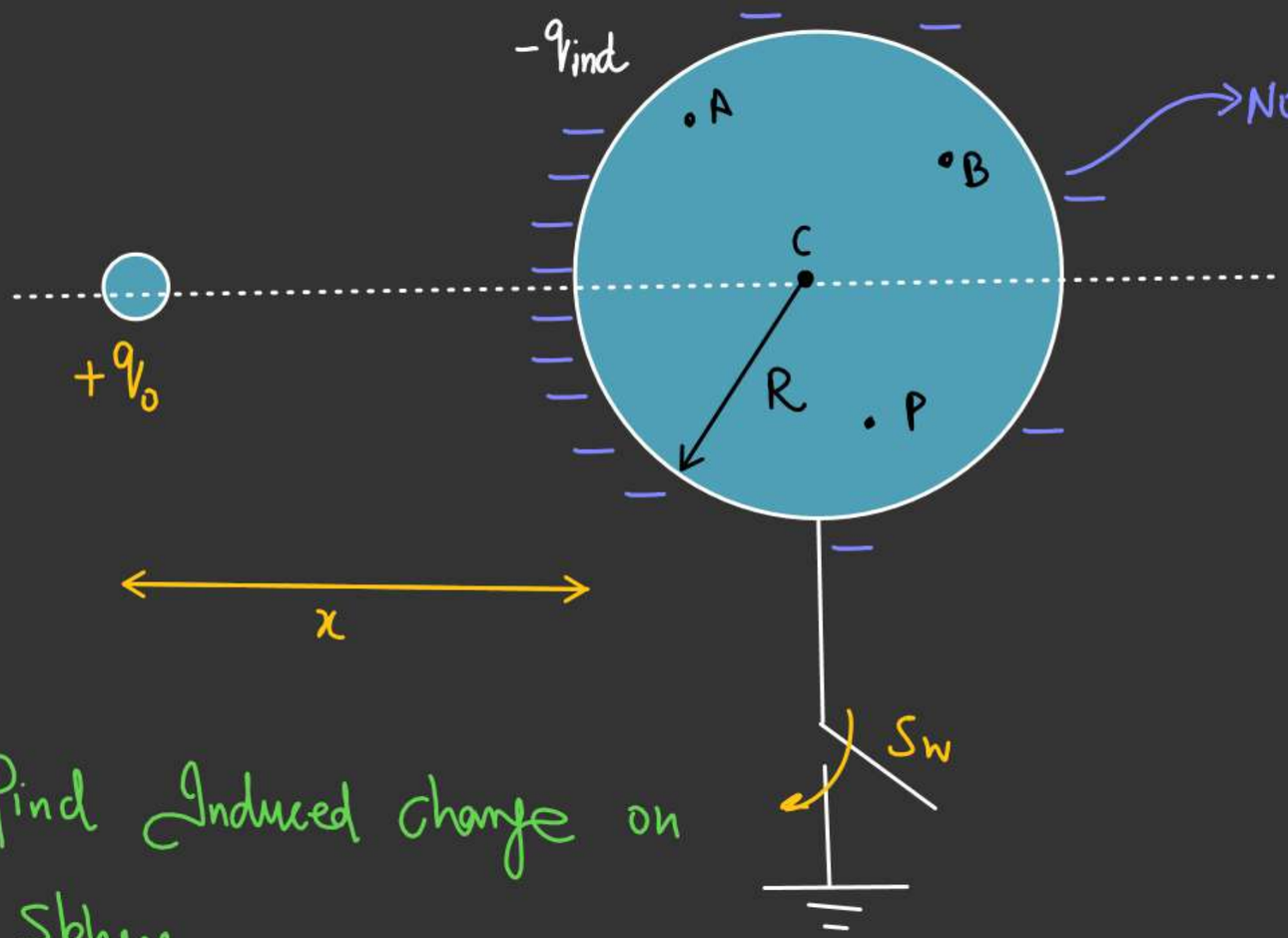
*



→ Non Uniform

$V_{sphere} = 0$

$V_A = V_B = V_C = V_P = 0$

$$\left(\frac{kq_0}{R+x}\right) - \frac{kq_{ind}}{R} = 0$$

$$\boxed{q_{ind} = q_0\left(\frac{R}{R+x}\right)}$$

* find Induced charge on Sphere.

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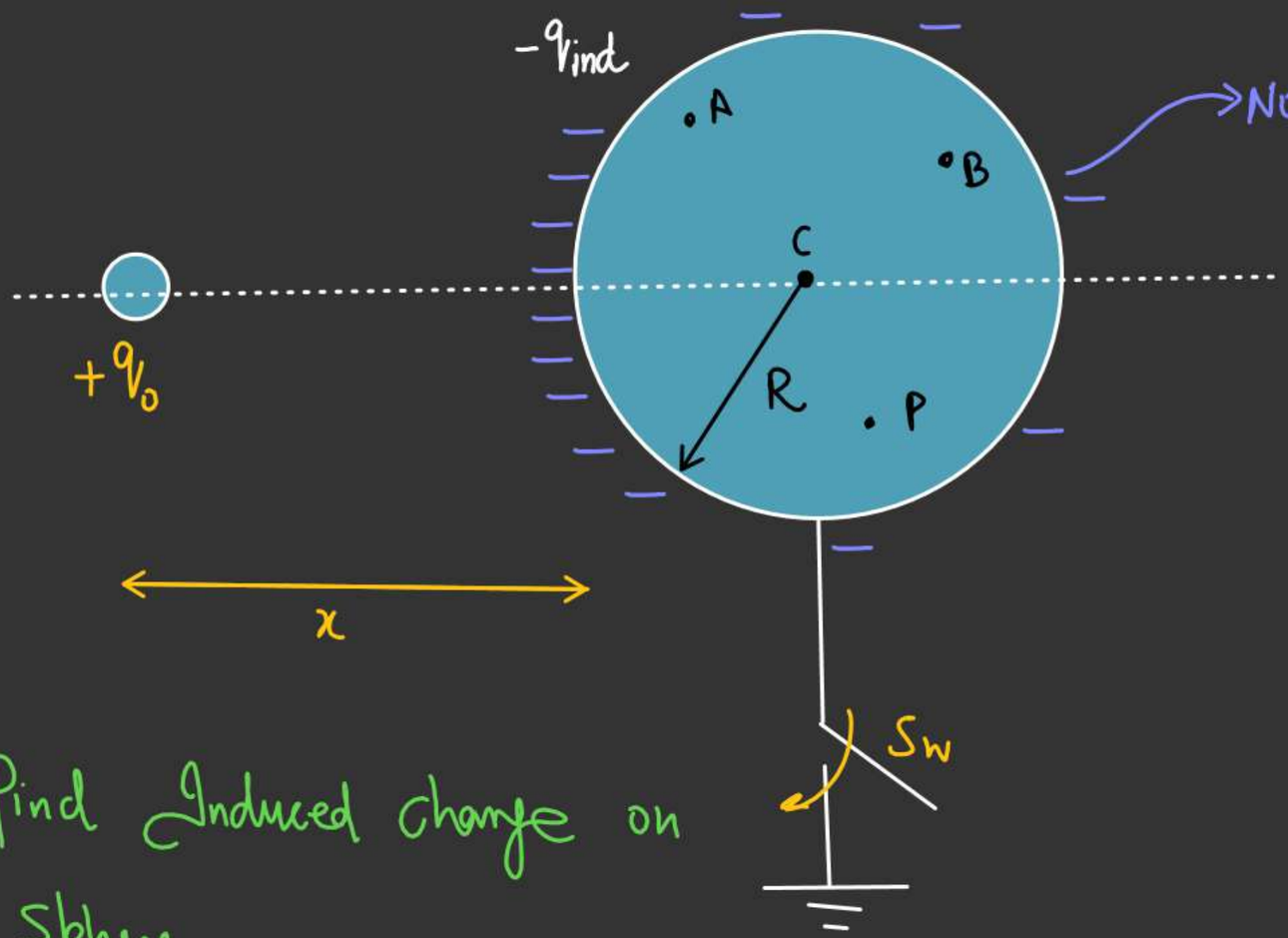
*



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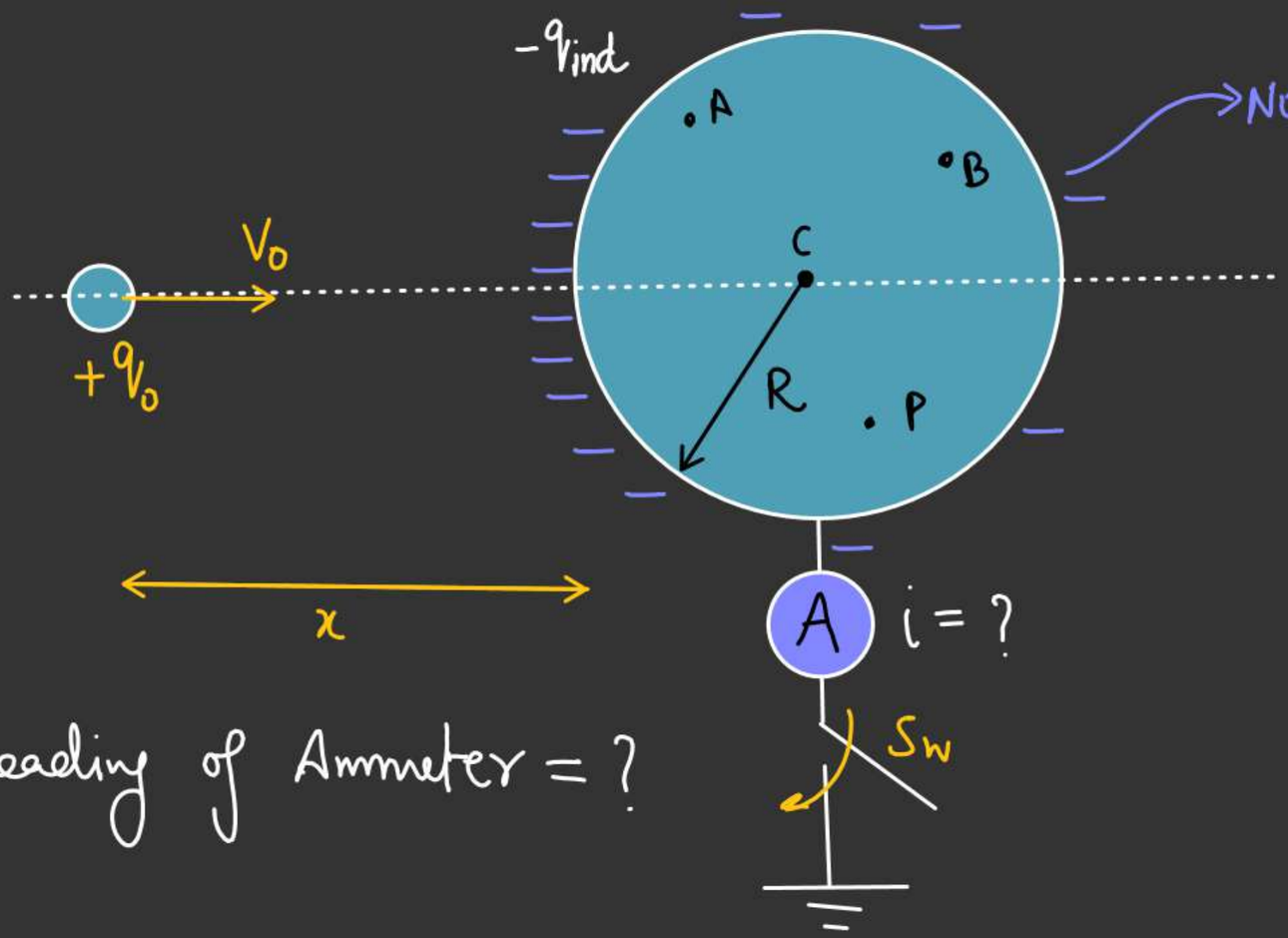
* find Induced charge on Sphere.

Nishant Jindal

*



→ Non Uniform

$V_{sphere} = 0$

$V_A = V_B = V_C = V_P = 0$

$$\left(\frac{Kq_0}{R+x}\right) - \frac{Kq_{ind}}{R} = 0$$

$$\boxed{q_{ind} = q_0\left(\frac{R}{R+x}\right)}$$

* Reading of Ammeter = ?

$x \downarrow \quad -\frac{dx}{dt} = V_0$

$$i = \frac{dq_{ind}}{dt} = \frac{q_0 R}{(R+x)^2}\left(-\frac{dx}{dt}\right) = \frac{qRV_0}{(R+x)^2}$$ Ans

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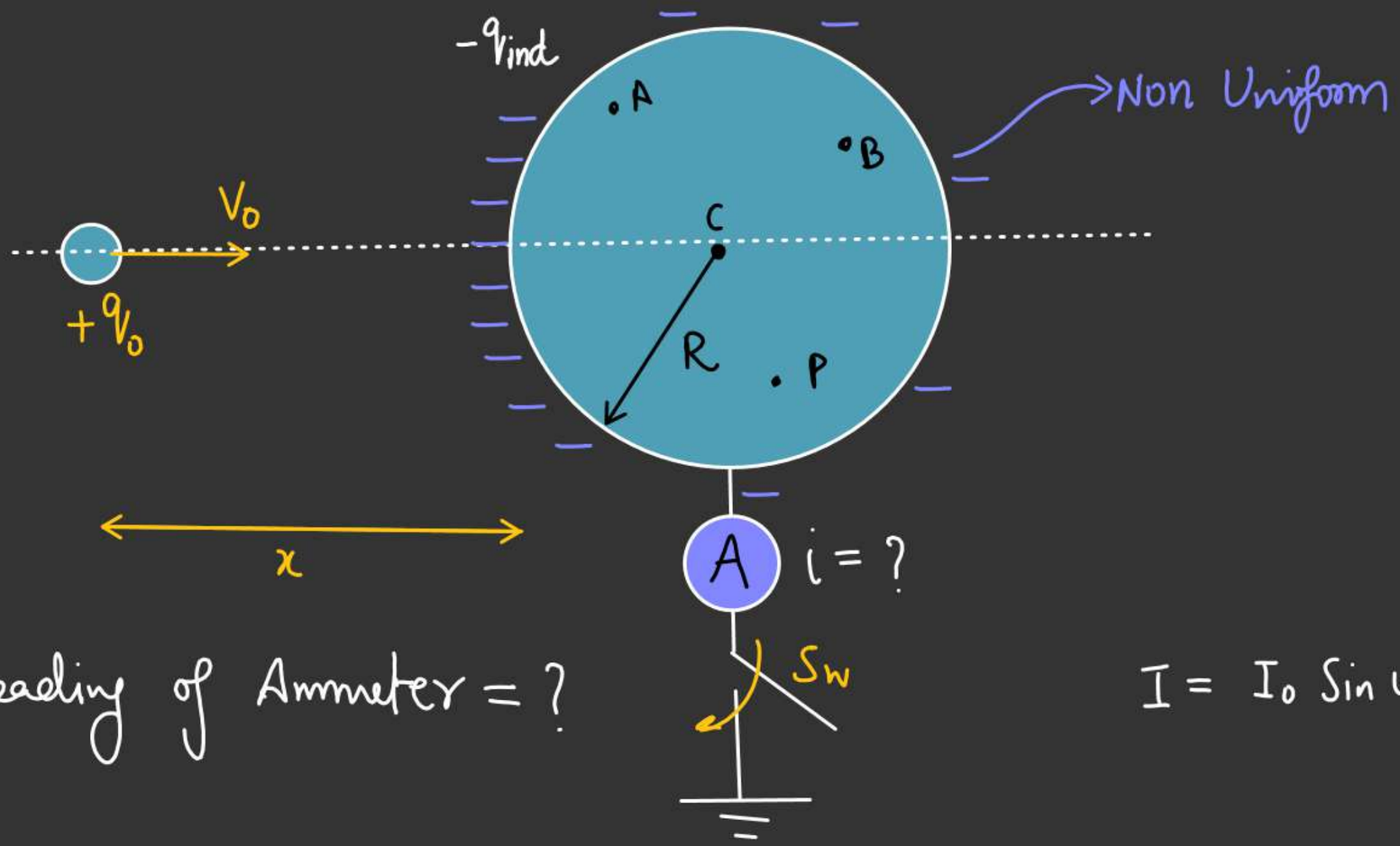
*



* Reading of Ammeter = ?

$I = I_0 \sin \omega t = f(t)$

$x \downarrow \quad -\frac{dx}{dt} = V_0$

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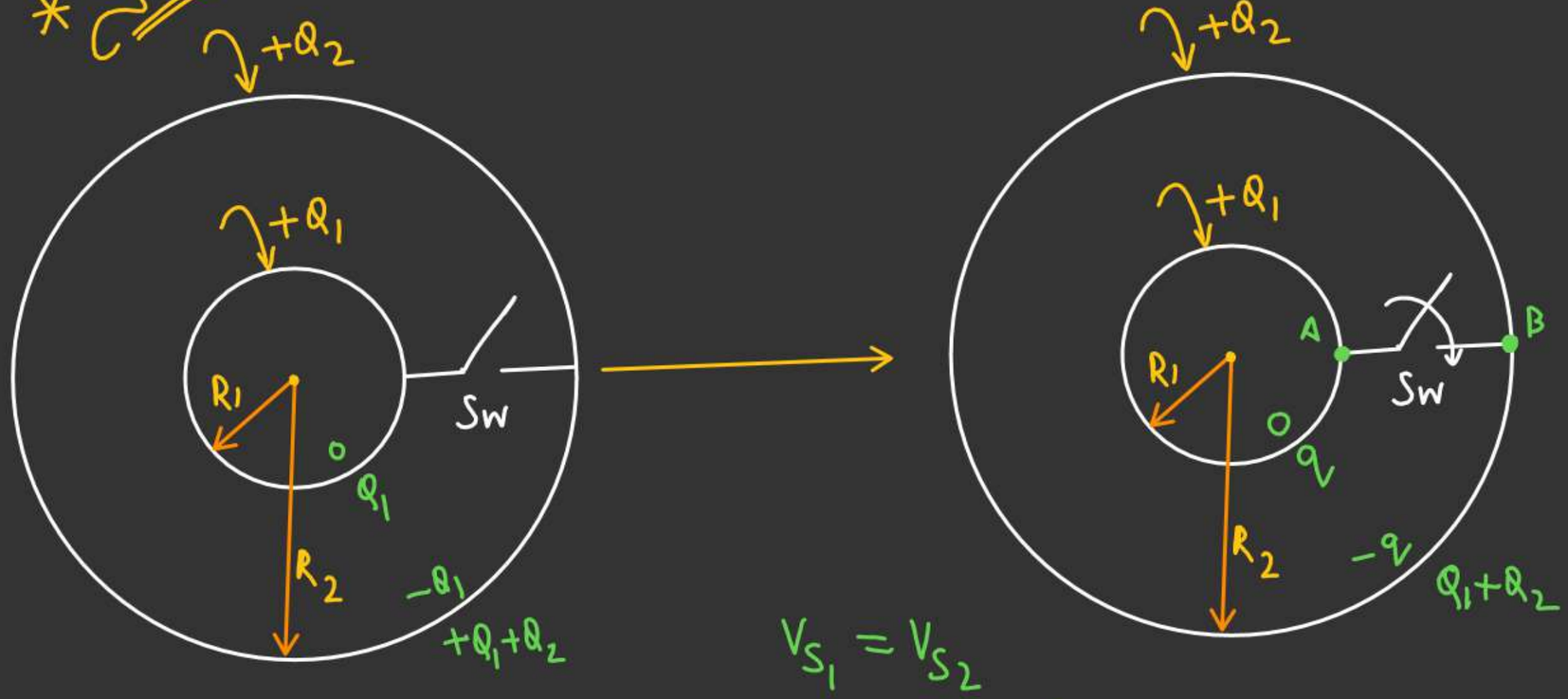
* Imp

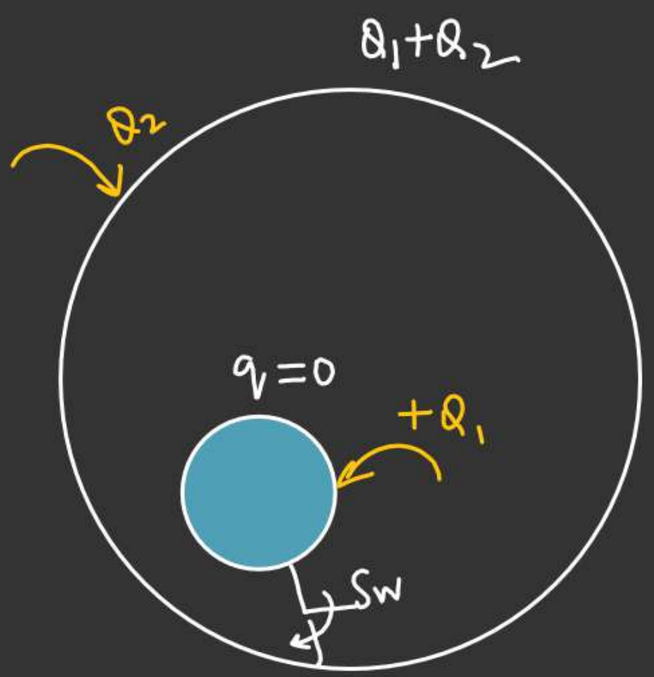


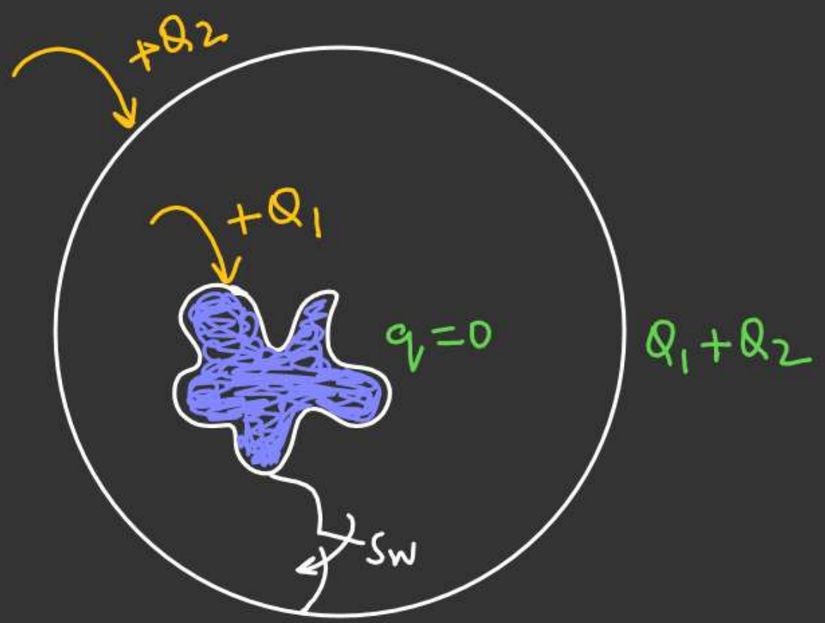
$$V_{S_1} = V_{S_2}$$

$$\frac{kq}{R_1} + \frac{k(Q_1+Q_2-q)}{R_2} = \frac{kq}{R_2} + \frac{k(Q_1+Q_2-q)}{R_2}$$

$$kq\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0 \quad \Rightarrow \boxed{q = 0}$$

Nishant Jindal





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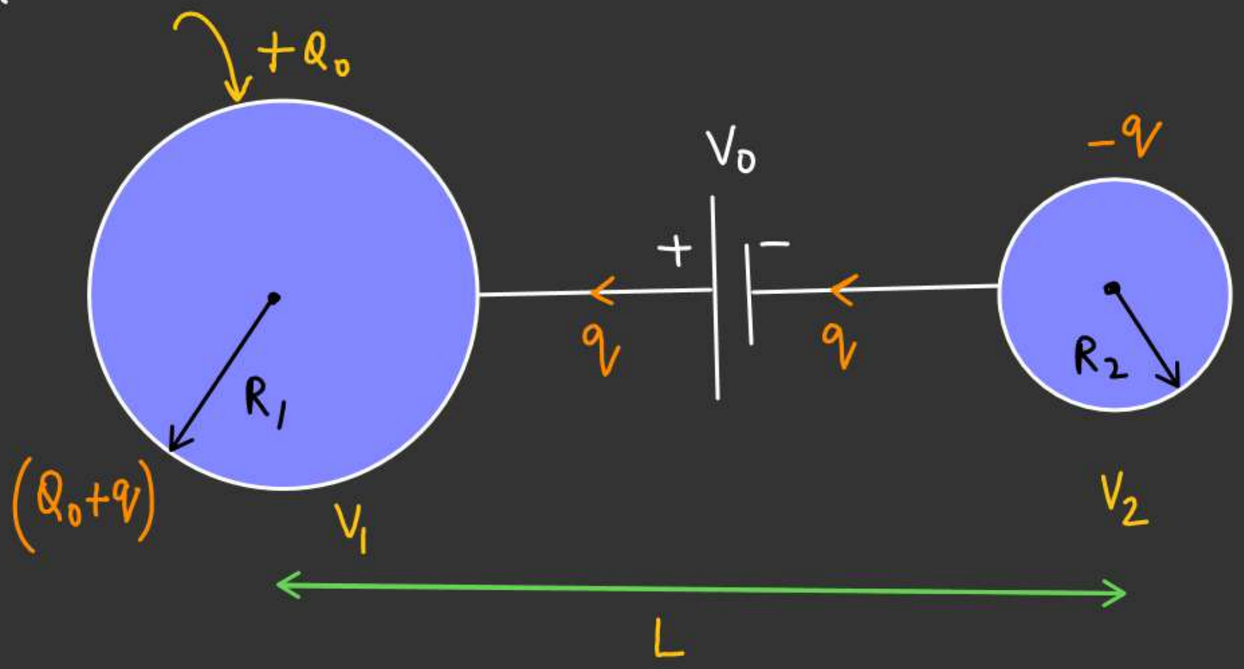


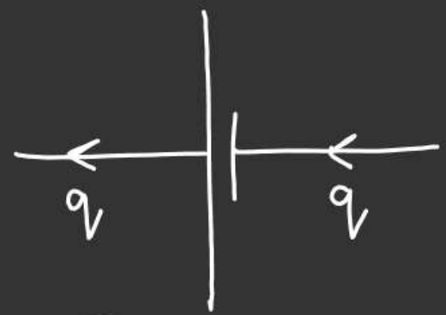
$$V_1 - V_2 = V_0$$

$$\left[\begin{array}{l} L >> R_1 \\ L >> R_2 \end{array}\right]$$

$$\frac{K(Q_0+q)}{R_1} - \left(\frac{-Kq}{R_2}\right) = V_0$$

$$\boxed{q = V}$$

* battery will function only if charge given by it comes back to it again.



$$W_b = +q \cdot V_0$$

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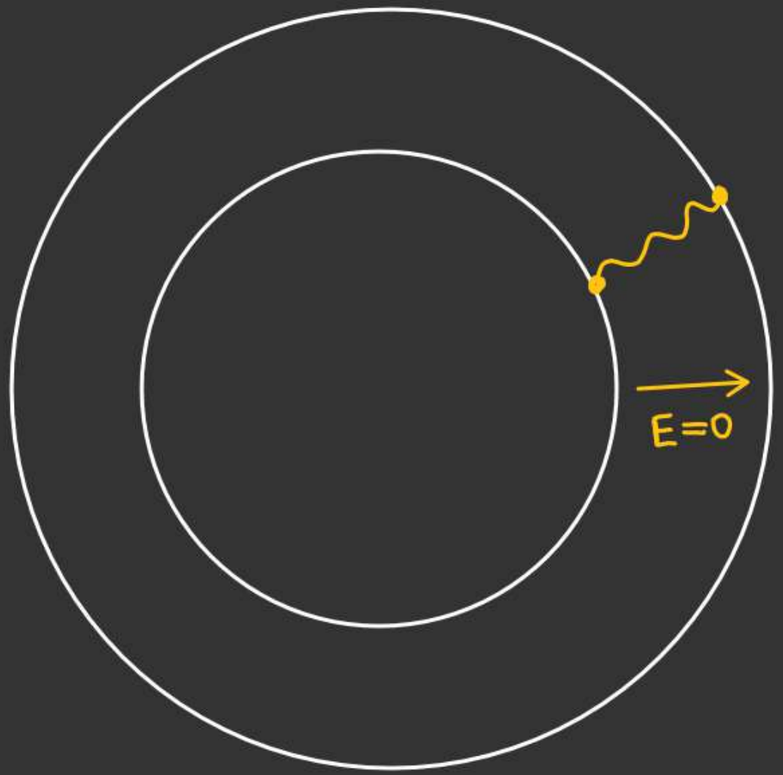

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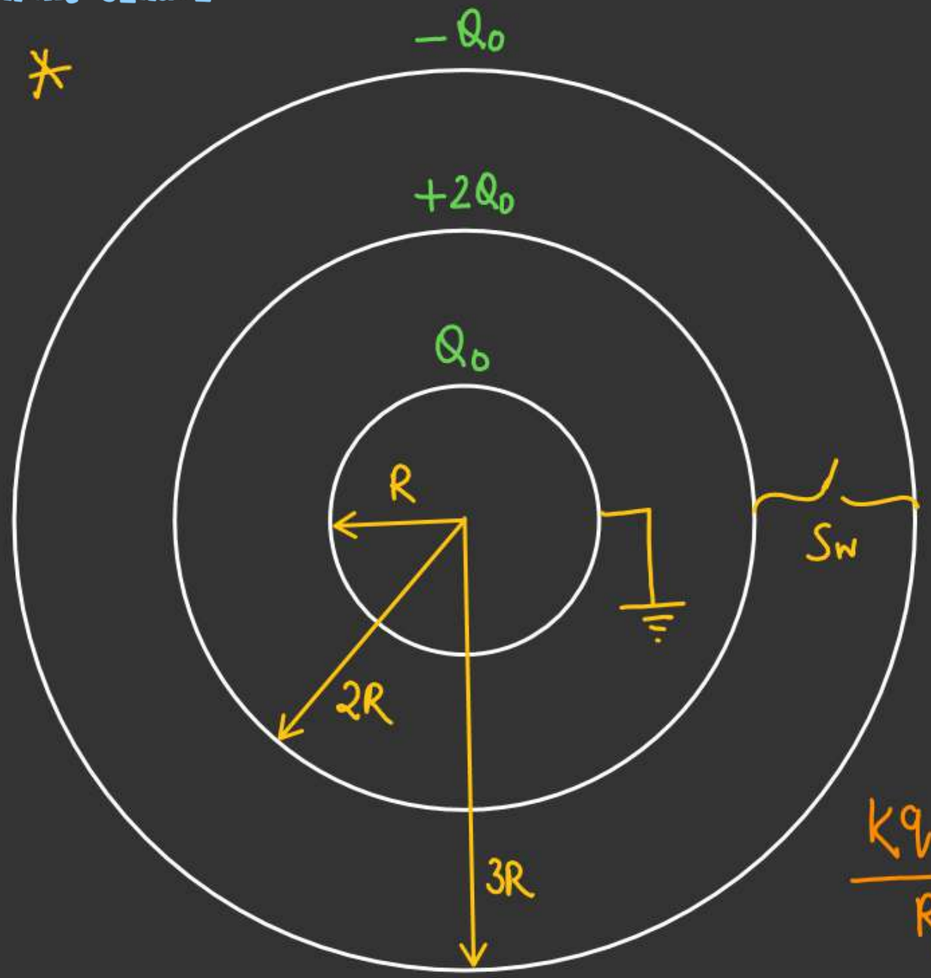

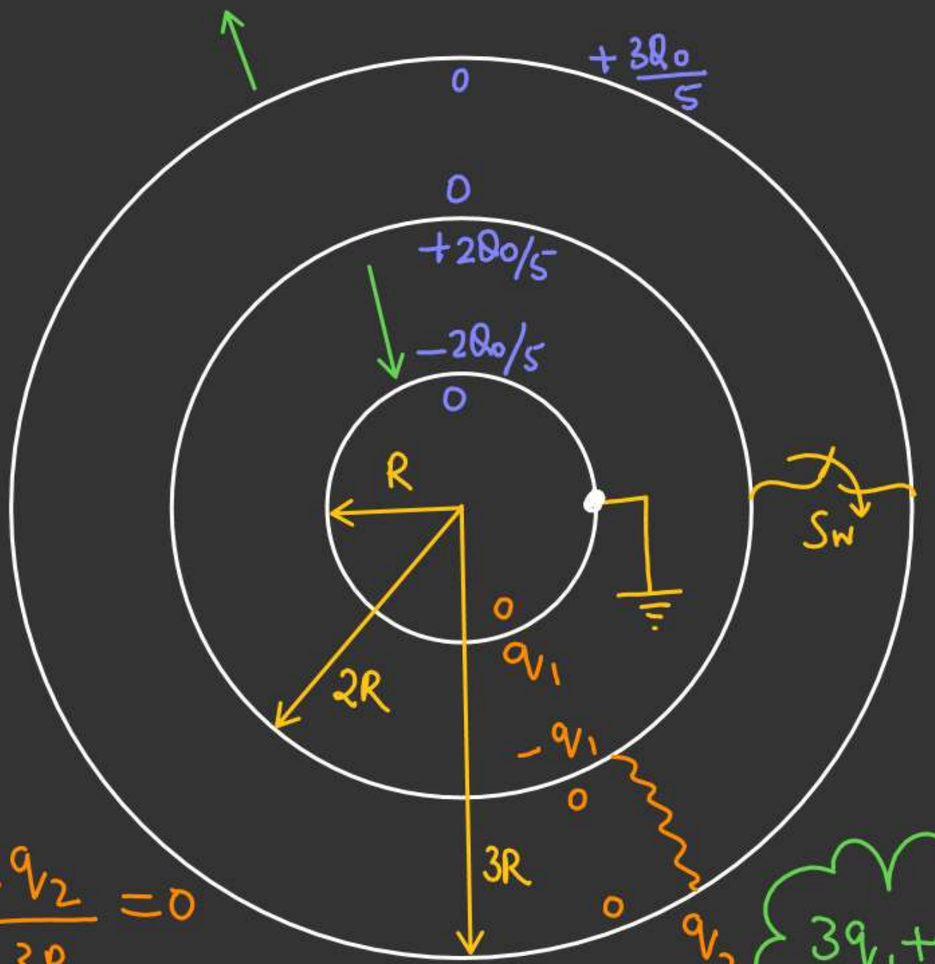

$V_1 = 0$

$$\frac{kq_1}{R} - \frac{kq_1}{2R} + \frac{kq_2}{3R} = 0$$

$3q_1 + 2q_2 = 0$

2 & 3 :-

$+2Q_0 - Q_0 = -q_1 + q_2$

$q_2 - q_1 = Q_0$

$q_2 - \left(\frac{-2q_2}{3}\right) = Q_0$

$\frac{5q_2}{3} = Q_0 \Rightarrow q_2 = \frac{3Q_0}{5}$

$q_1 = -\frac{2Q_0}{5}$

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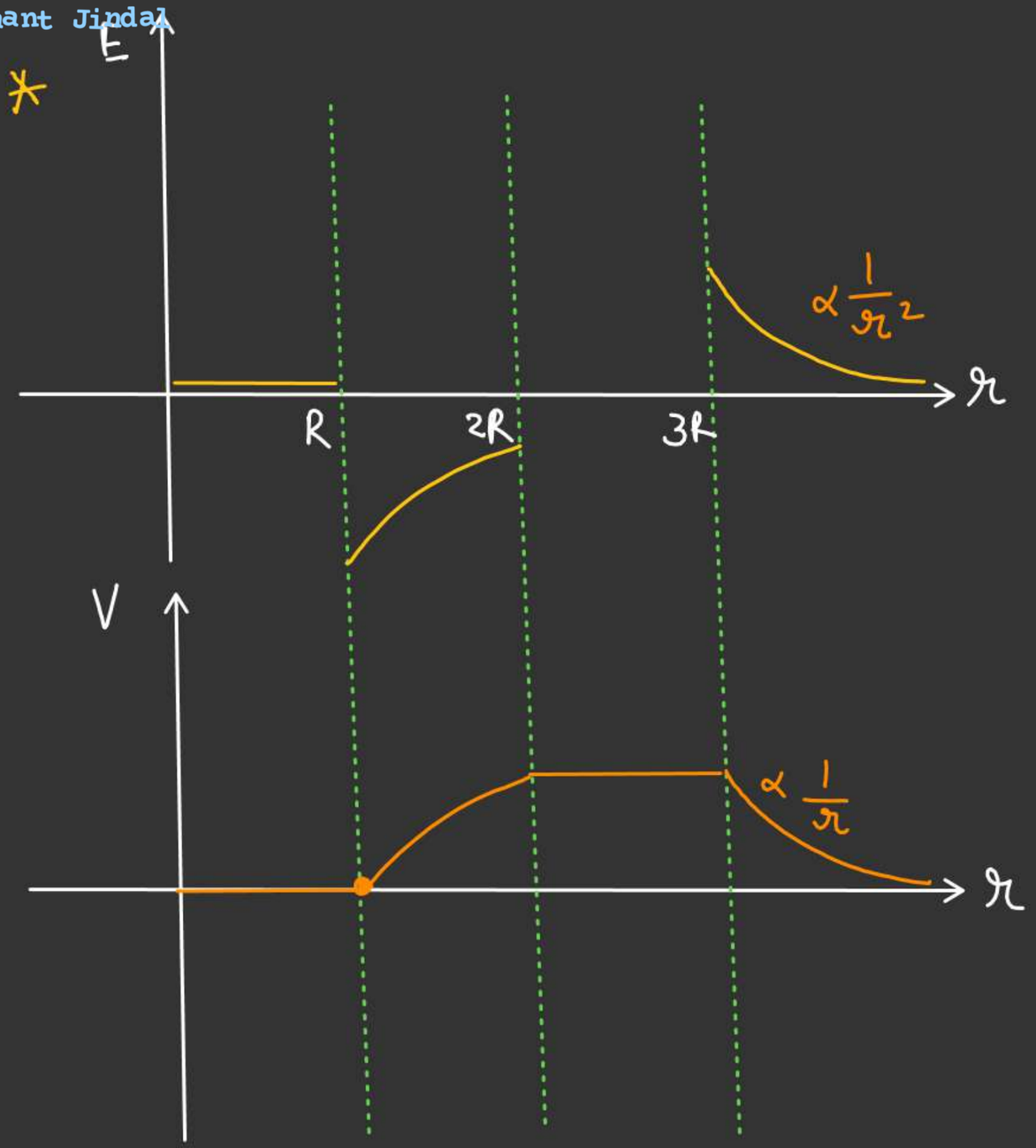

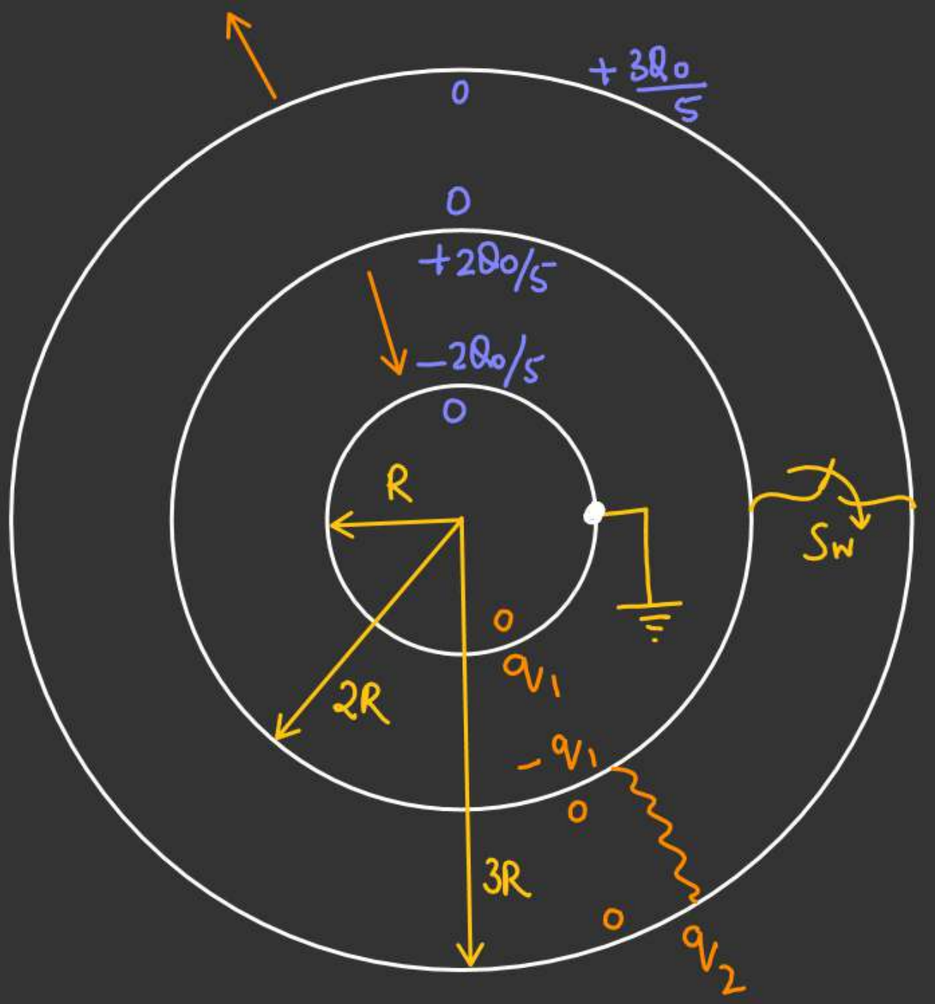

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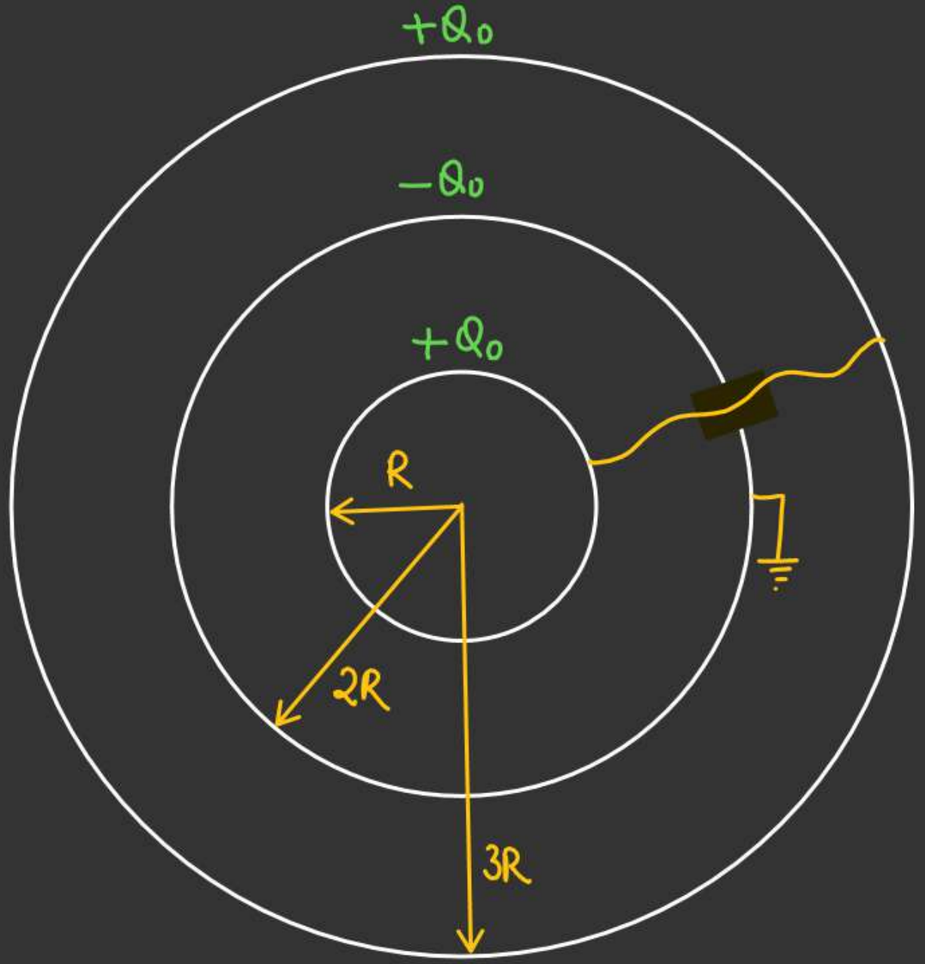

* Draw E-r V-r graphs

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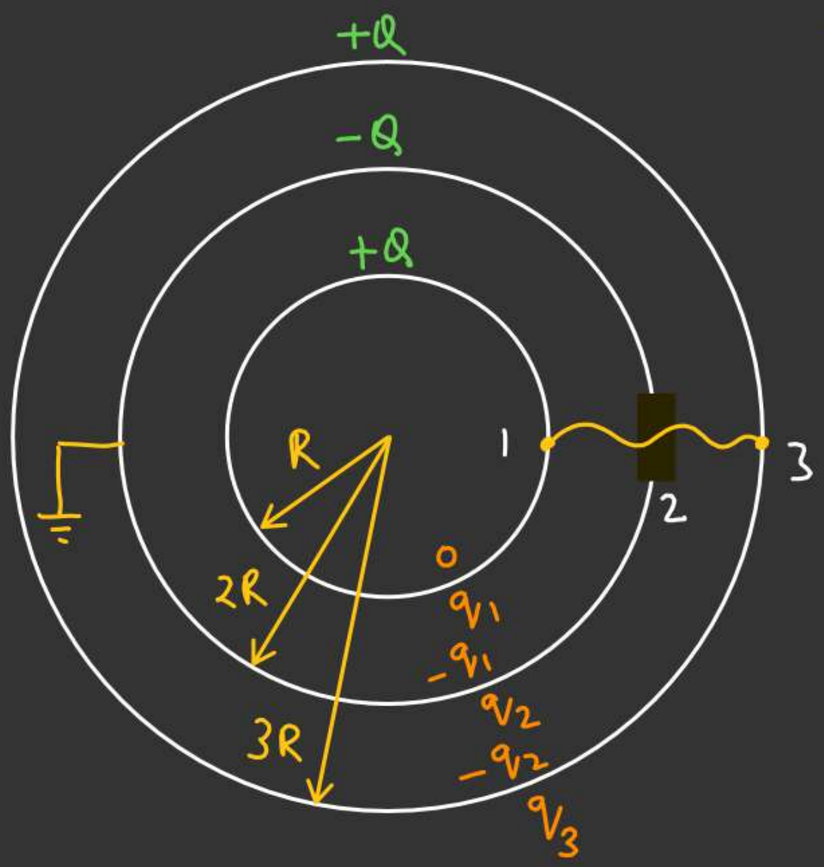

1 & 3 Charge cons.

$$2Q = q_1 - q_2 + q_3 \longrightarrow (i)$$

$$V_1 = V_3$$

$$\frac{Kq_1}{R} + \frac{K(q_2-q_1)}{2R} + \frac{K(q_3-q_2)}{3R} = \frac{Kq_3}{3R}$$

$$q_1 + \frac{q_2-q_1}{2} - \frac{q_2}{3} = 0$$

$$6q_1 + 3q_2 - 3q_1 - 2q_2 = 0$$

$$\boxed{3q_1 + q_2 = 0} \longrightarrow (ii)$$

$V_2 = 0$

$$\frac{Kq_1}{2R} + \frac{K(q_2-q_1)}{2R} + \frac{K(q_3-q_2)}{3R} = 0 \Rightarrow q_2 + 2q_3 = 0 \longrightarrow (iii)$$

$$3q_1 + 3q_2 - 3q_1 + 2q_3 - 2q_2 = 0$$

$$q_3 = -\frac{q_2}{2} = +\frac{3q_1}{2}$$

* $2Q = q_1 + 3q_1 + \frac{3q_1}{2}$

$$2Q = 4q_1 + \frac{3q_1}{2}$$

$$2Q = \frac{11q_1}{2}$$

* $q_1 = \frac{4Q}{11}$

$q_2 = \frac{-12Q}{11}$, $q_3 = \frac{6Q}{11}$

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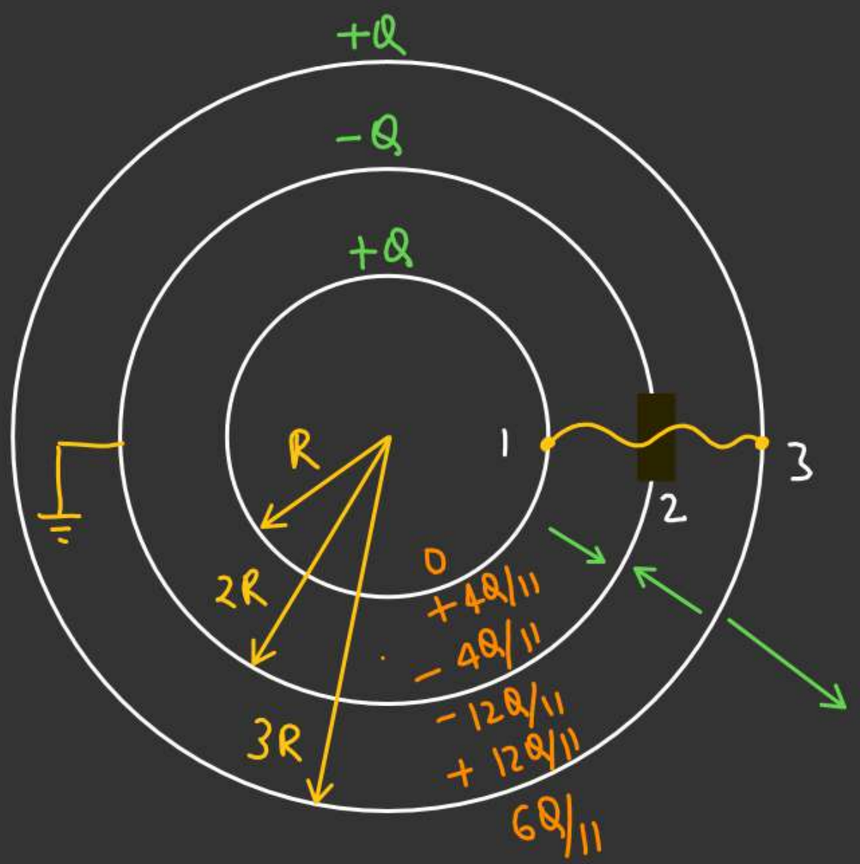

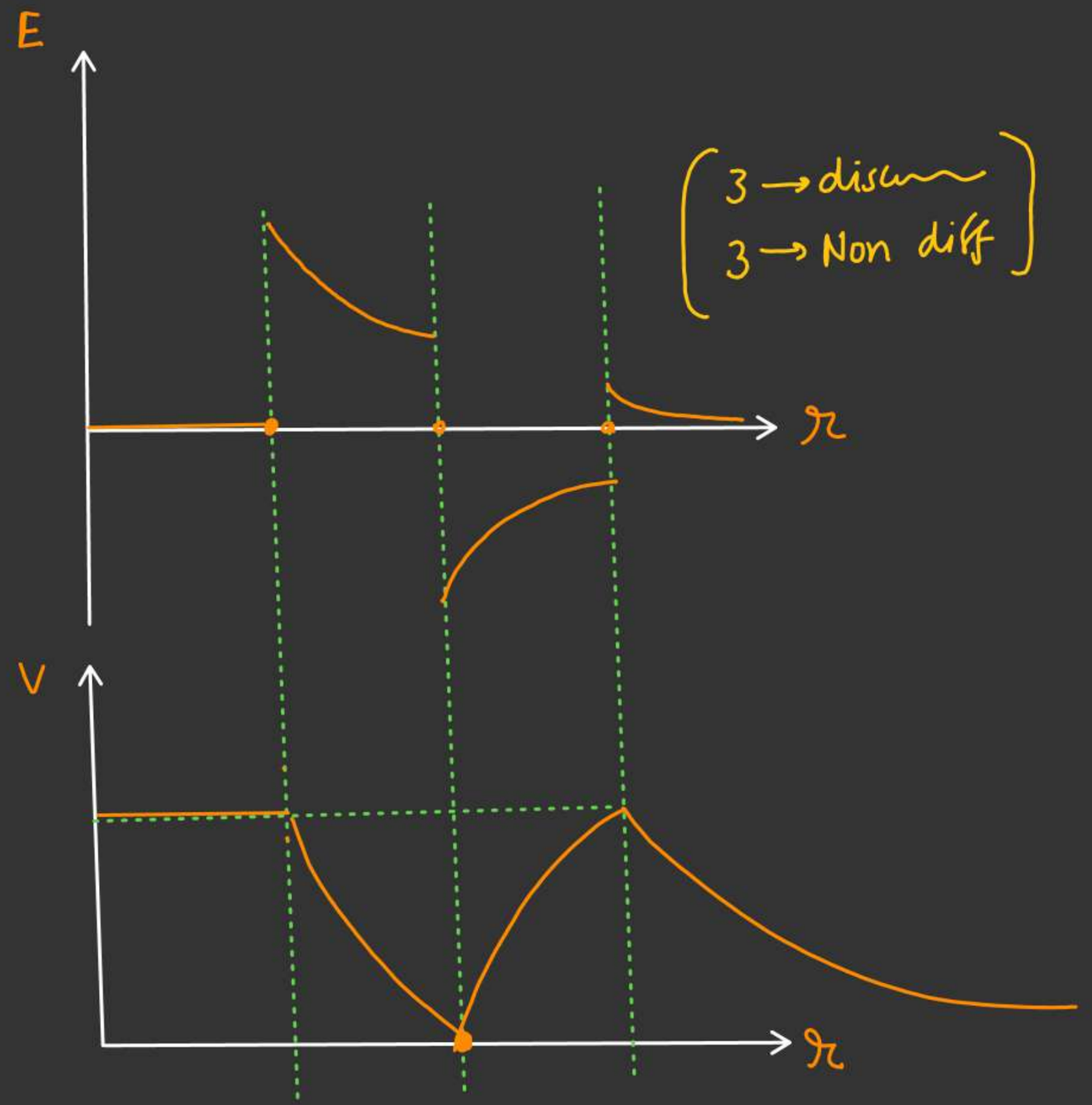

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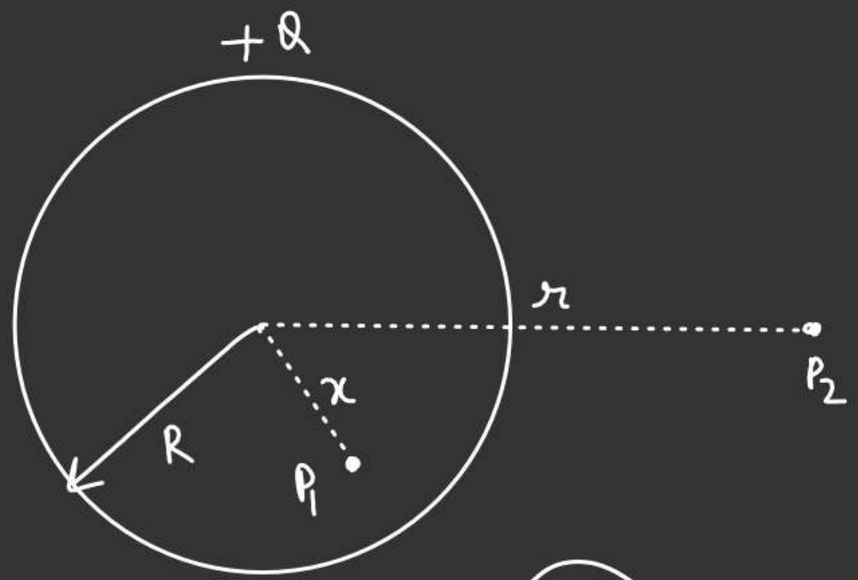

$E_{P_1} = 0$ $\quad V_{P_1} = \frac{KQ}{R}$

$E_{P_2} = \frac{KQ}{r^2}$ $\quad V_{P_2} = \frac{KQ}{r}$

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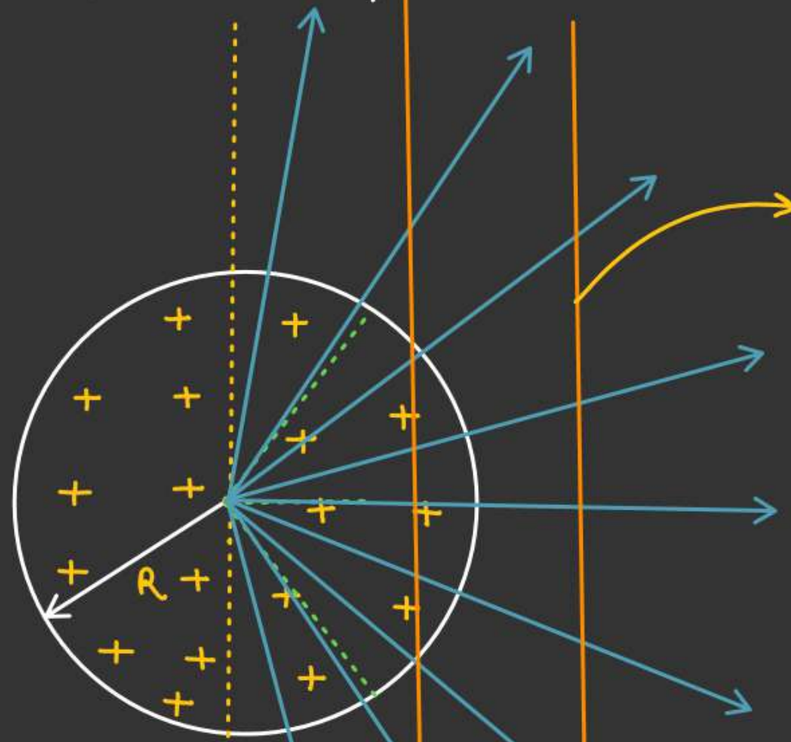
* $\phi_{Plane} = ?$

$\phi_{total} = \frac{Q}{2\epsilon_0}$

$\frac{Q}{2\epsilon_0}$ ✓✓ Correct



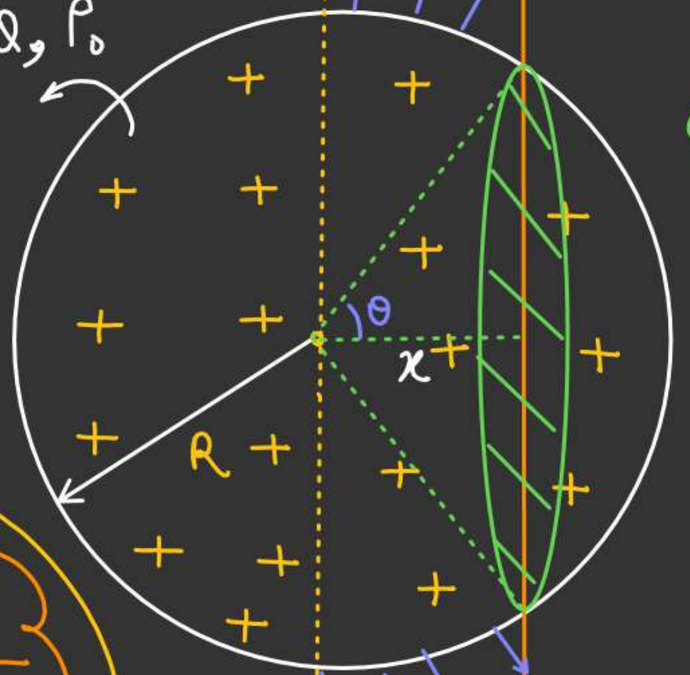
*

expected $\phi_{Plane} = \frac{Q}{2\epsilon_0}$

Wrong

Non Conducting

$\phi_1 = \frac{Q_0}{2\epsilon_0}\frac{x}{R}$ ✓✓



$\phi_{Plane} = \underbrace{\phi_1 + \phi_{circle}}$

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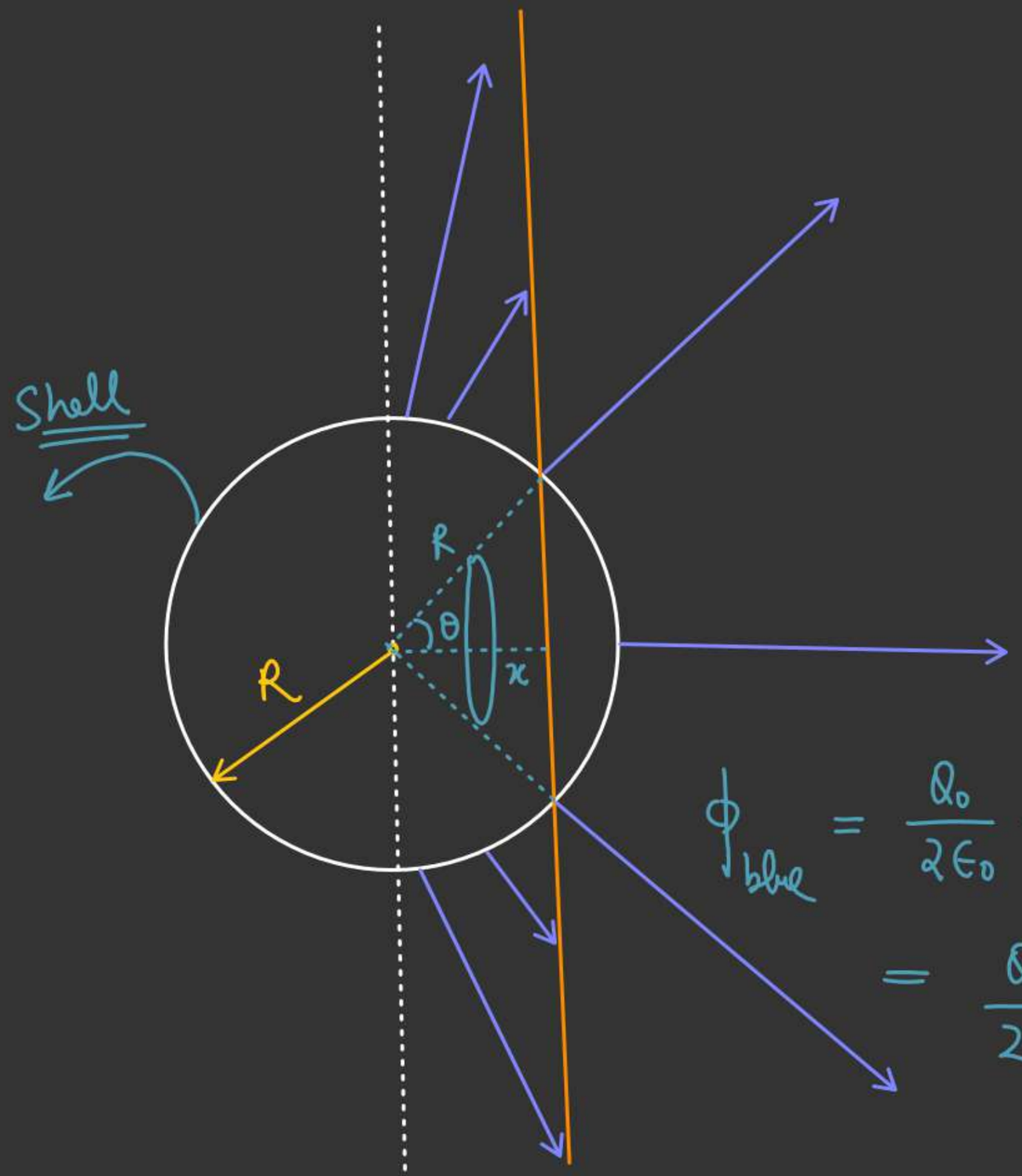

$$\phi_{blue} = \frac{Q_0}{2\epsilon_0} - \frac{Q_0}{2\epsilon_0}(1-\cos\theta)$$

$$= \frac{Q}{2\epsilon_0}\cdot\cos\theta = \frac{Qx}{2\epsilon_0 R}$$

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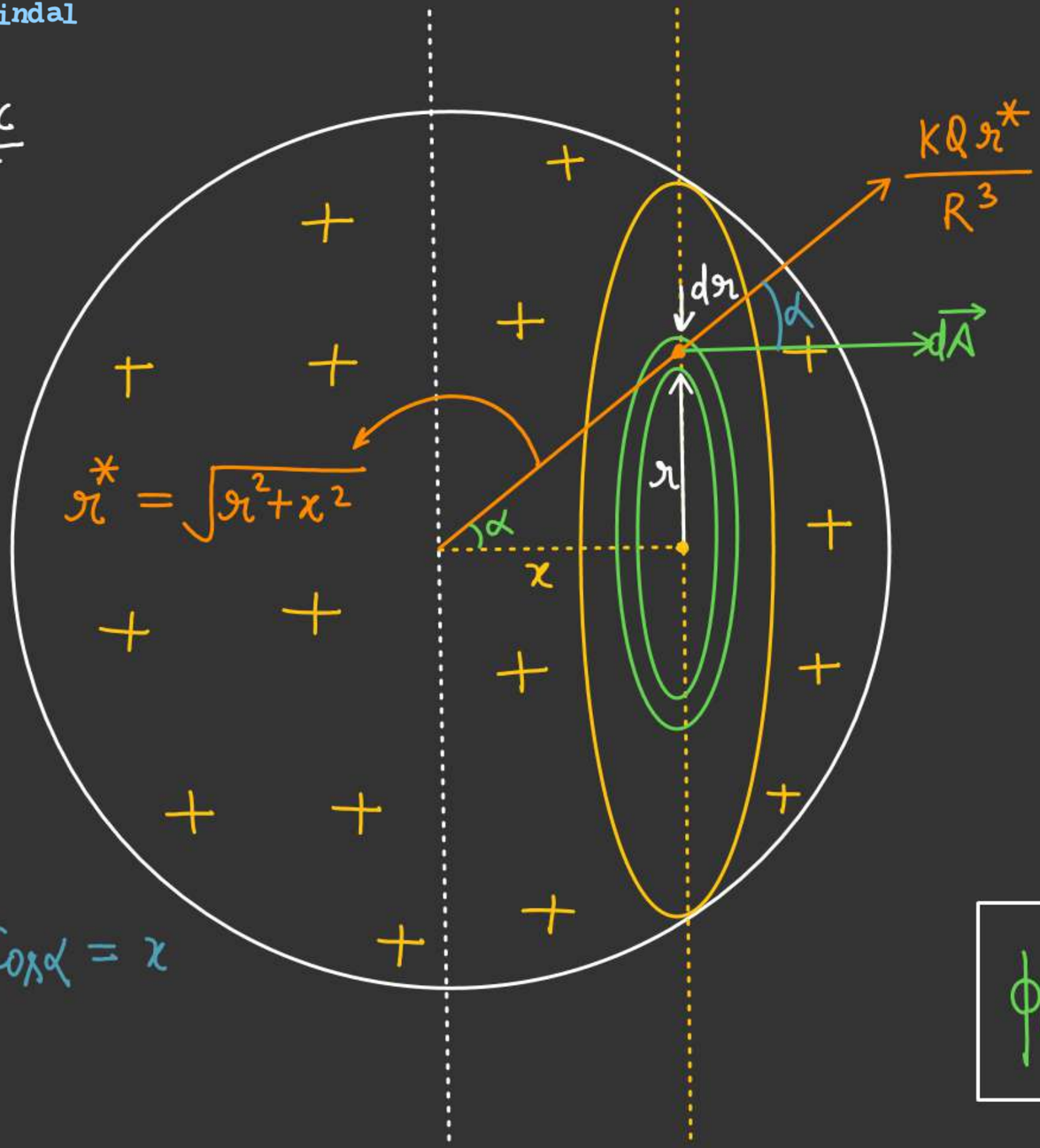
ϕ_{Disc}



$$d\phi = \frac{KQr^*}{R^3} \cdot 2\pi r\,dr \cdot \cos\alpha$$

$$d\phi = \frac{KQx}{R^3} \cdot 2\pi r\,dr$$

$$\phi_{circle} = \frac{KQ \cdot x \cdot 2\pi}{R^3} \int_0^{\sqrt{R^2-x^2}} r \cdot dr$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \cdot \frac{Q \cdot x \cdot 2\pi}{R^3} \left(\frac{R^2-x^2}{2}\right)$$

$$\boxed{\phi_{circle} = \frac{Qx}{4\epsilon_0 R^3}\left(R^2-x^2\right)}$$

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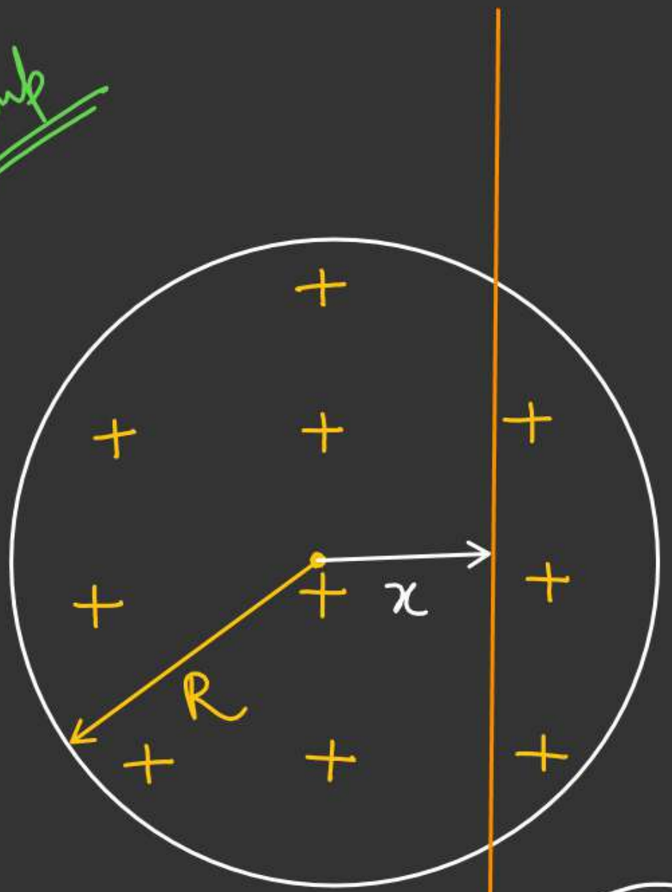
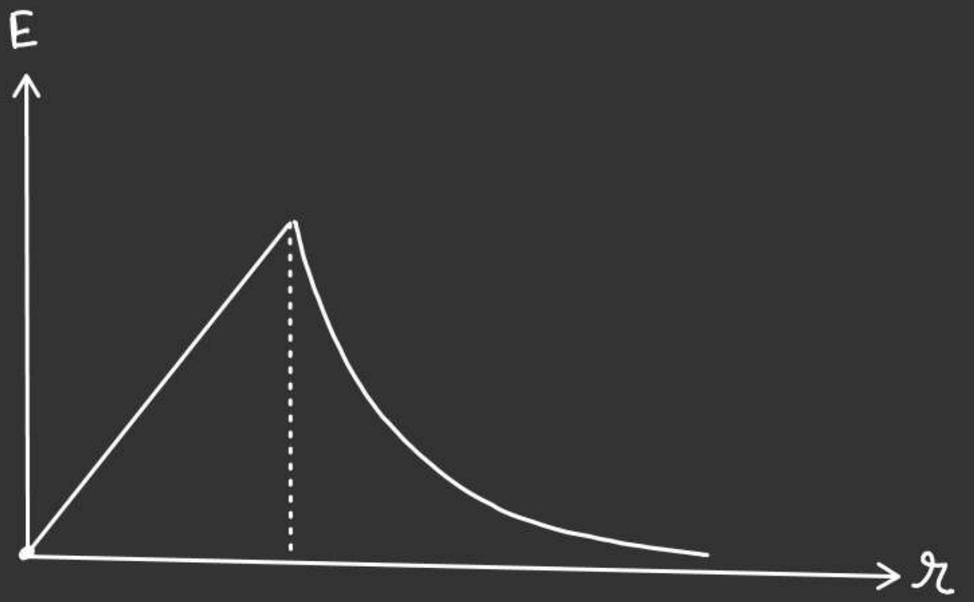
Imp





$$\phi_{Plane} = \frac{Q}{2\epsilon_0}\cdot\frac{x}{R} + \frac{Qx}{4\epsilon_0 R^3}\left(R^2 - x^2\right) \quad \text{if } x < R$$

$$\phi_{plane} = \frac{Q}{2\epsilon_0} \quad \text{if } x > R$$

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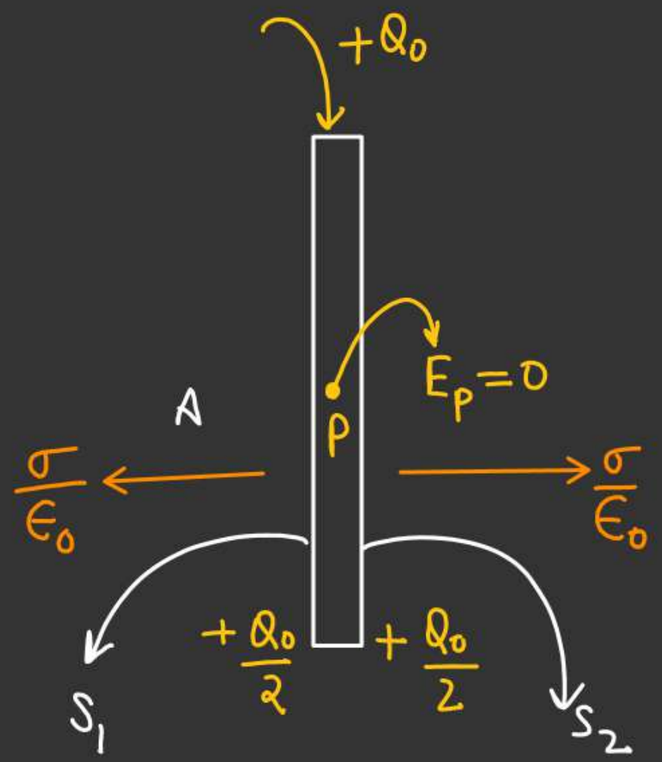
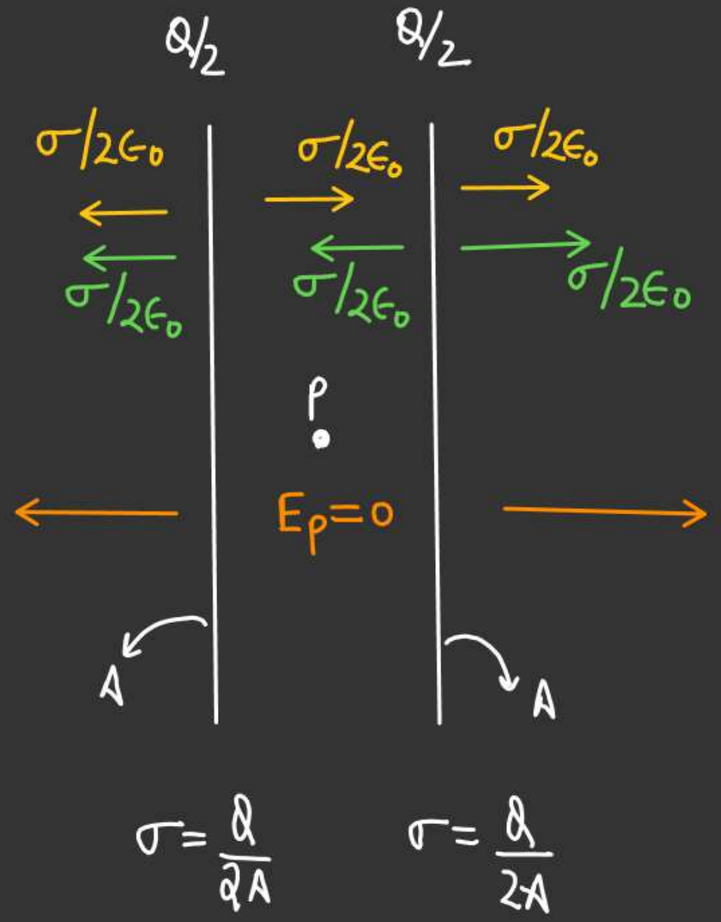
* Conducting Plates :-



$$\sigma = \frac{Q_0}{2A}$$



$$\sigma = \frac{Q}{2A} \qquad \sigma = \frac{Q}{2A}$$

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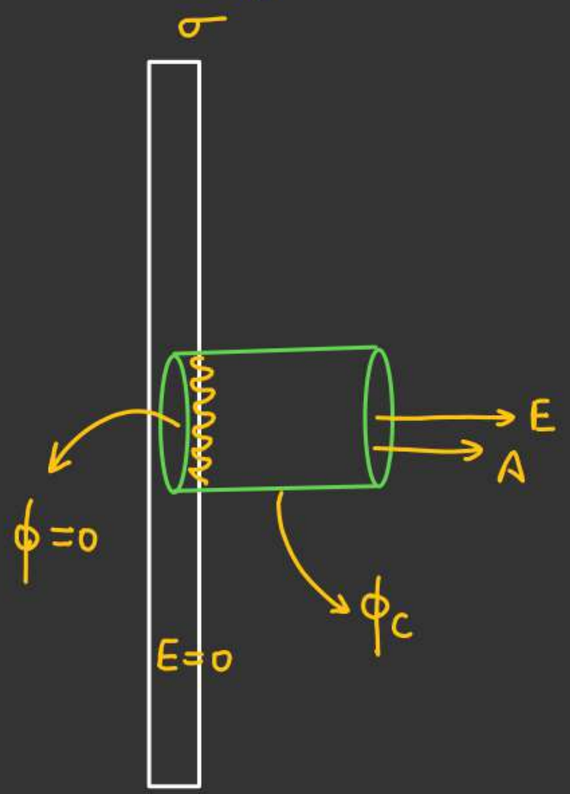
Conducting Plate



* Finite Plate & Nearby points
* Very large Plate & Far away.

$$E \cdot A = \frac{\sigma \cdot A}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

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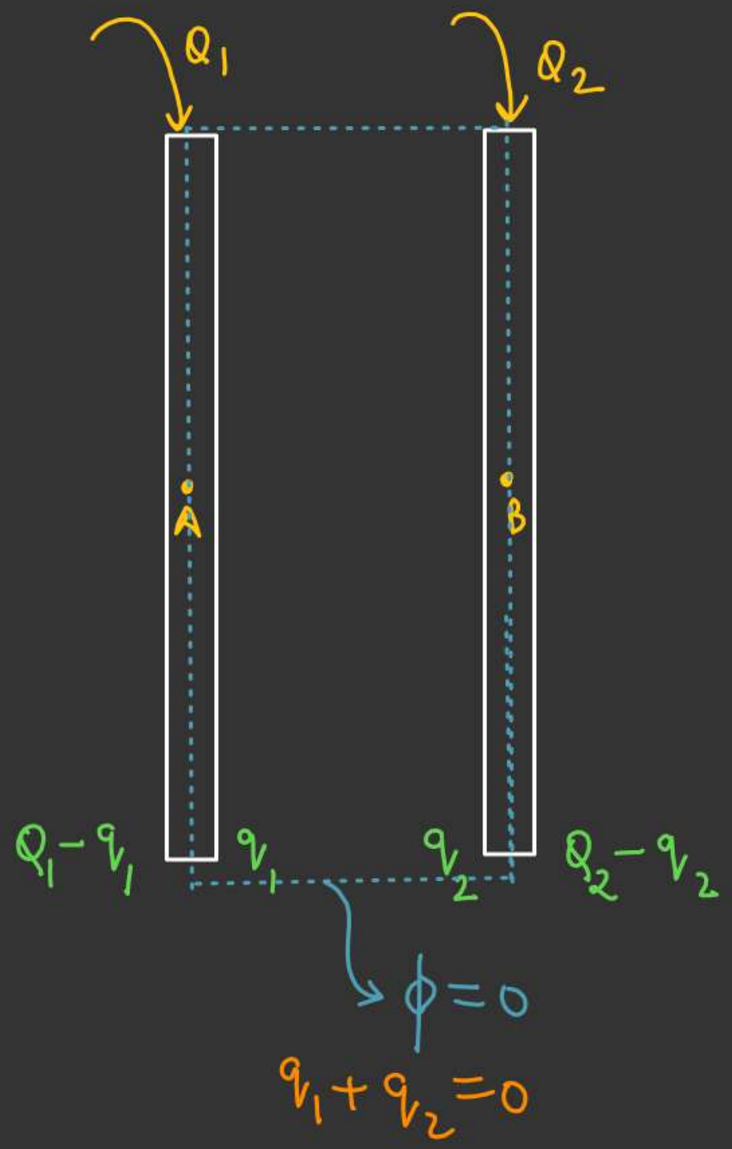

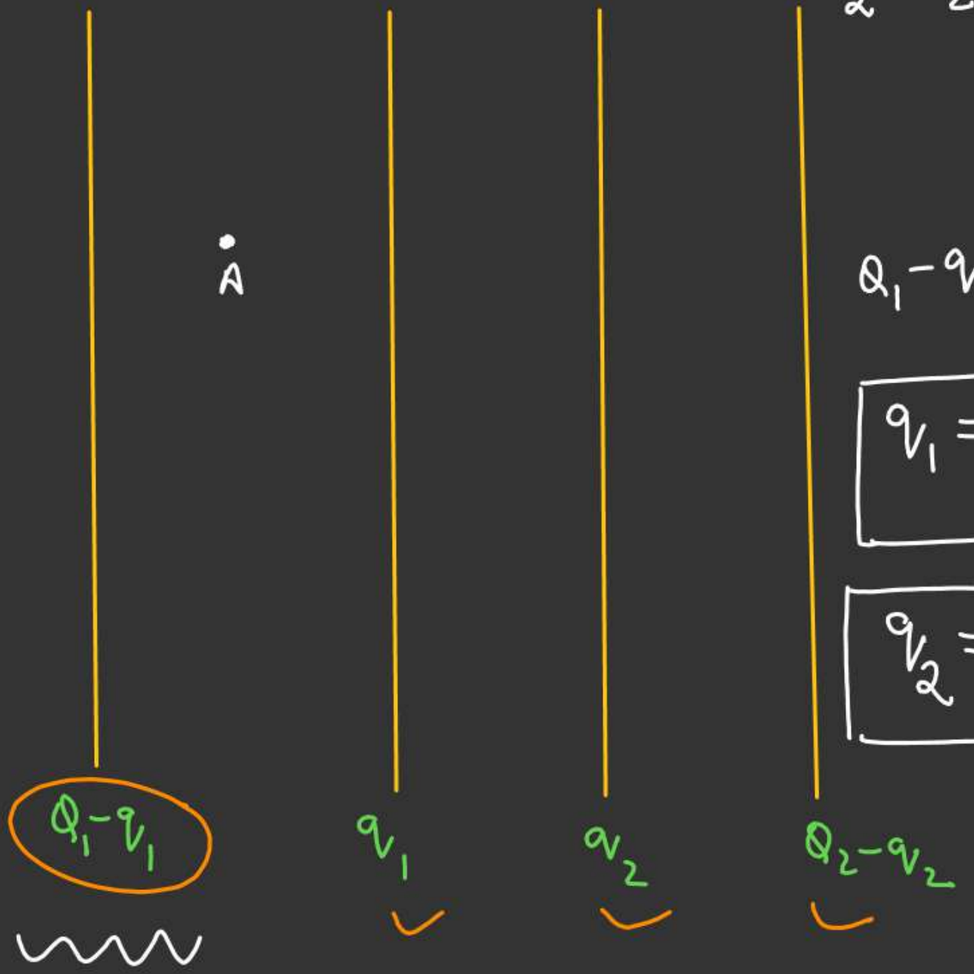

$$Q_1-q_1=\frac{Q_1+Q_2}{2}$$

$$Q_2-q_2=\frac{Q_1+Q_2}{2}$$

$$Q_1-q_1=q_1+Q_2$$

$$\boxed{q_1=\frac{Q_1-Q_2}{2}}\ \star$$

$$\boxed{q_2=-q_1}$$

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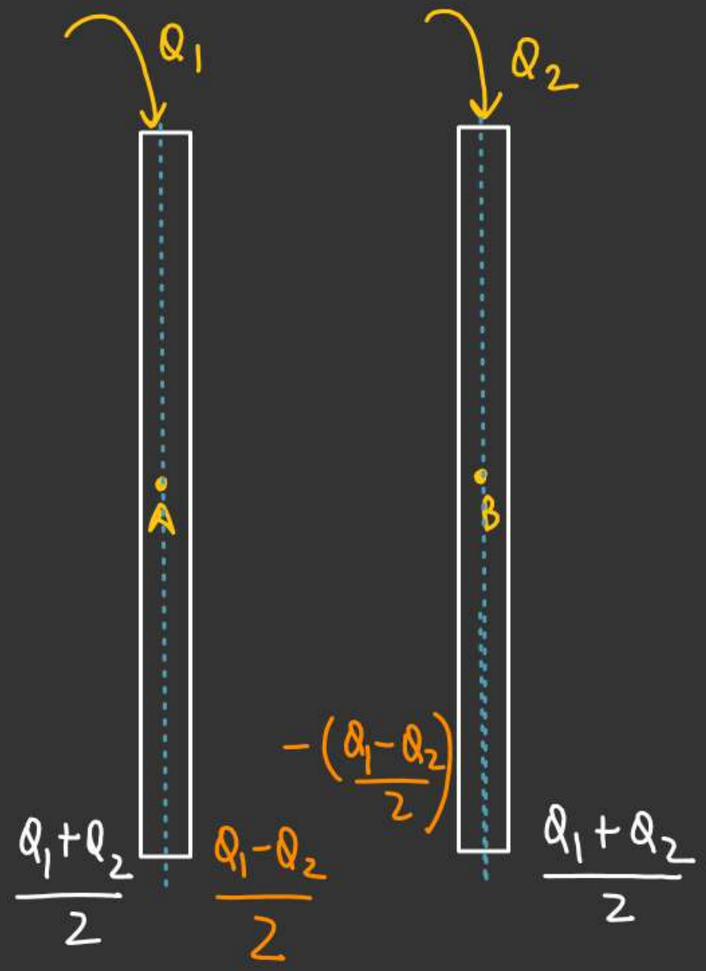

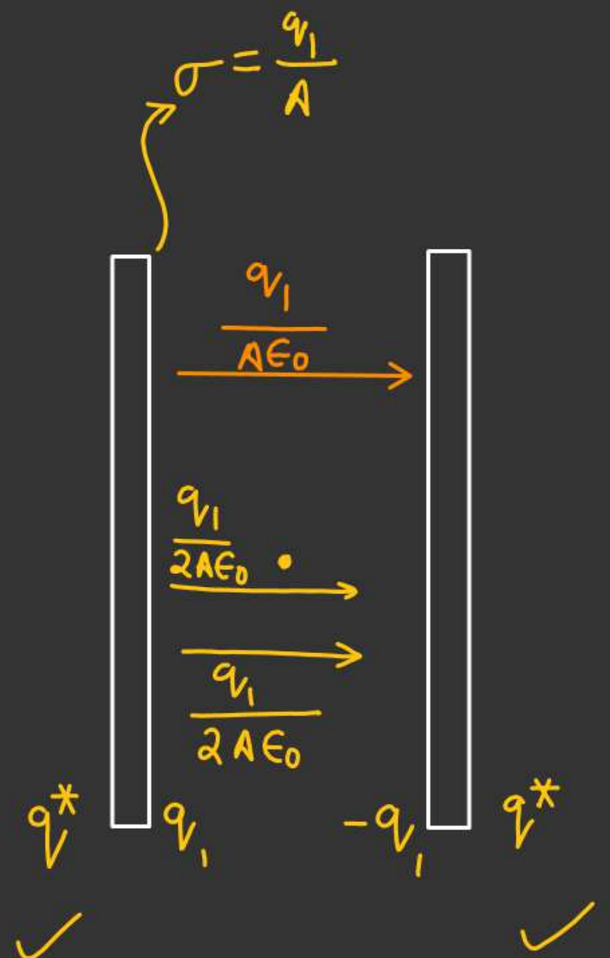

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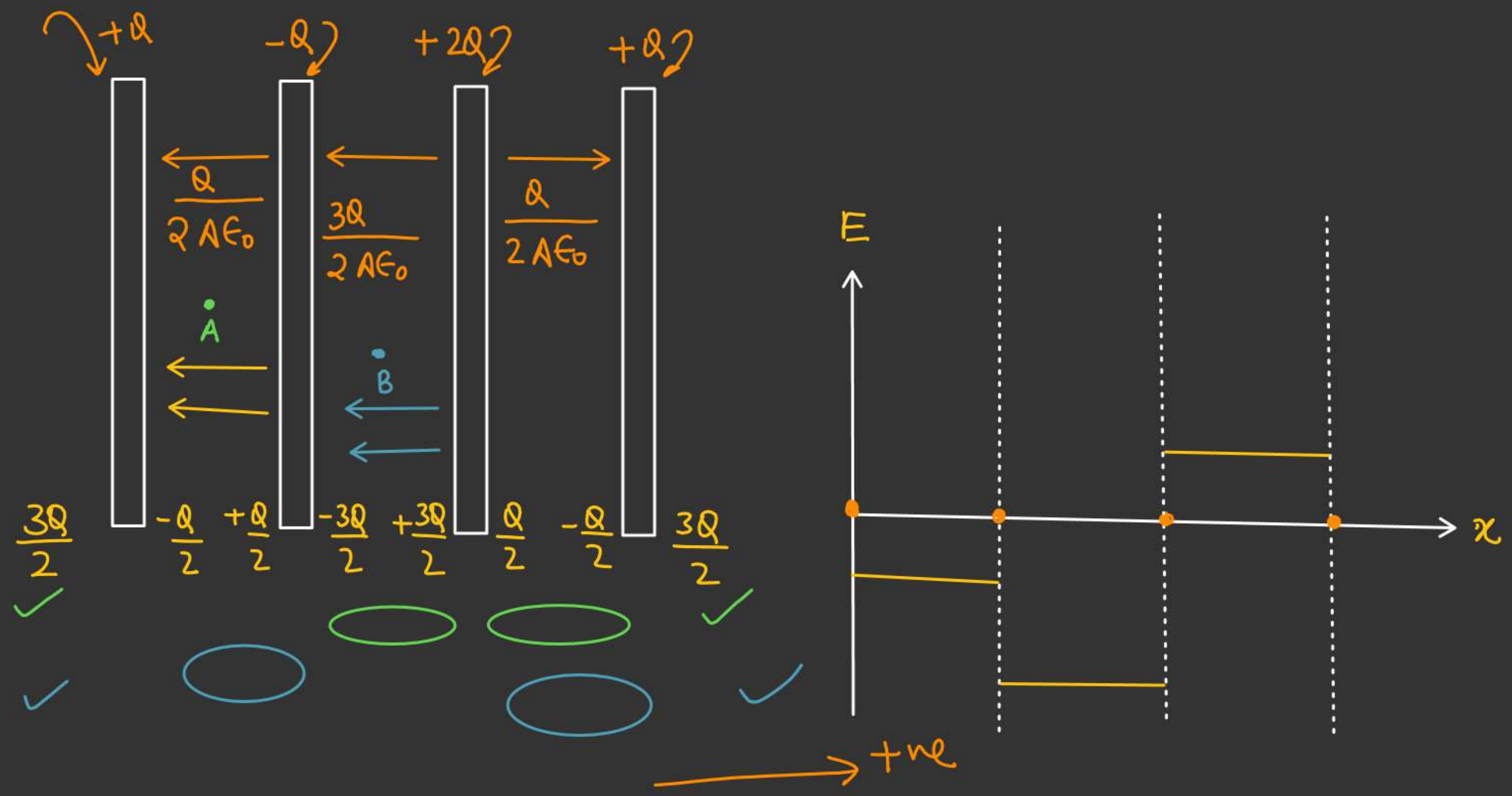

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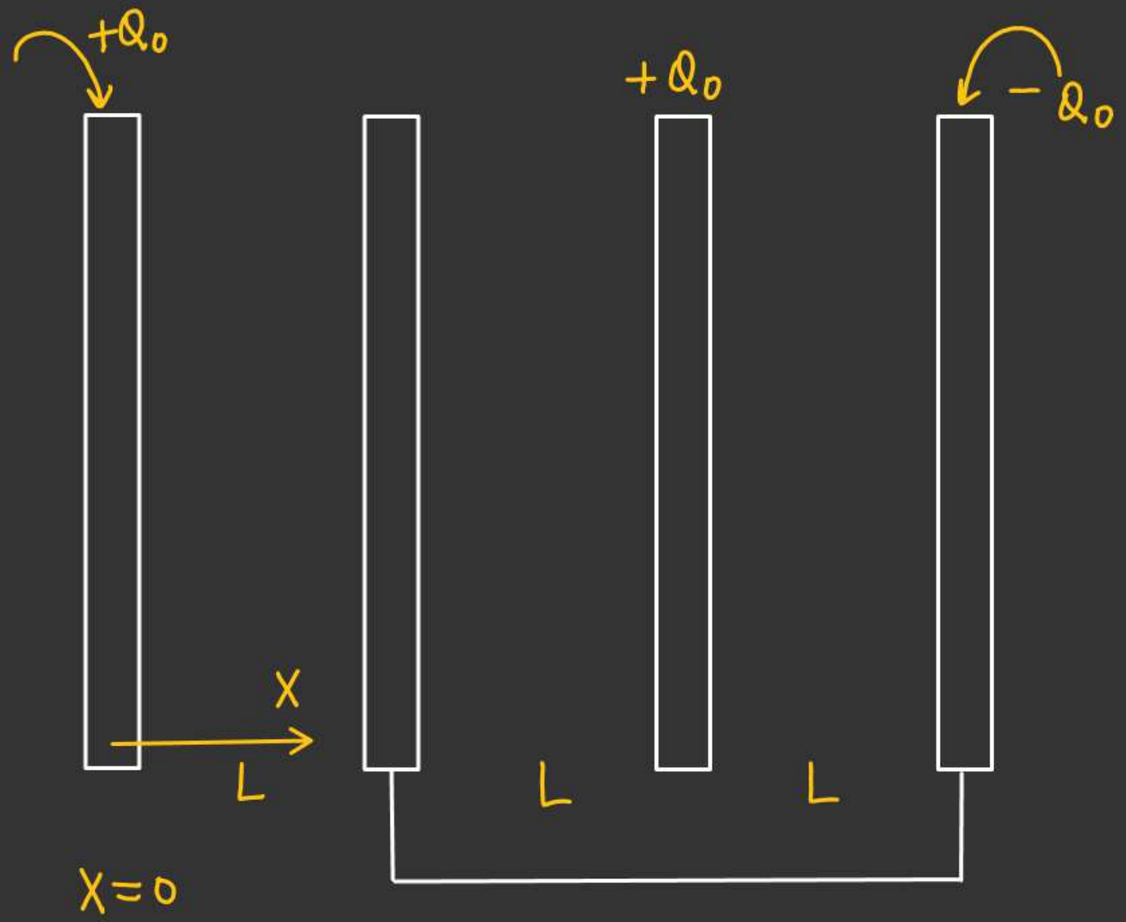

Draw E-x
V-x graph.

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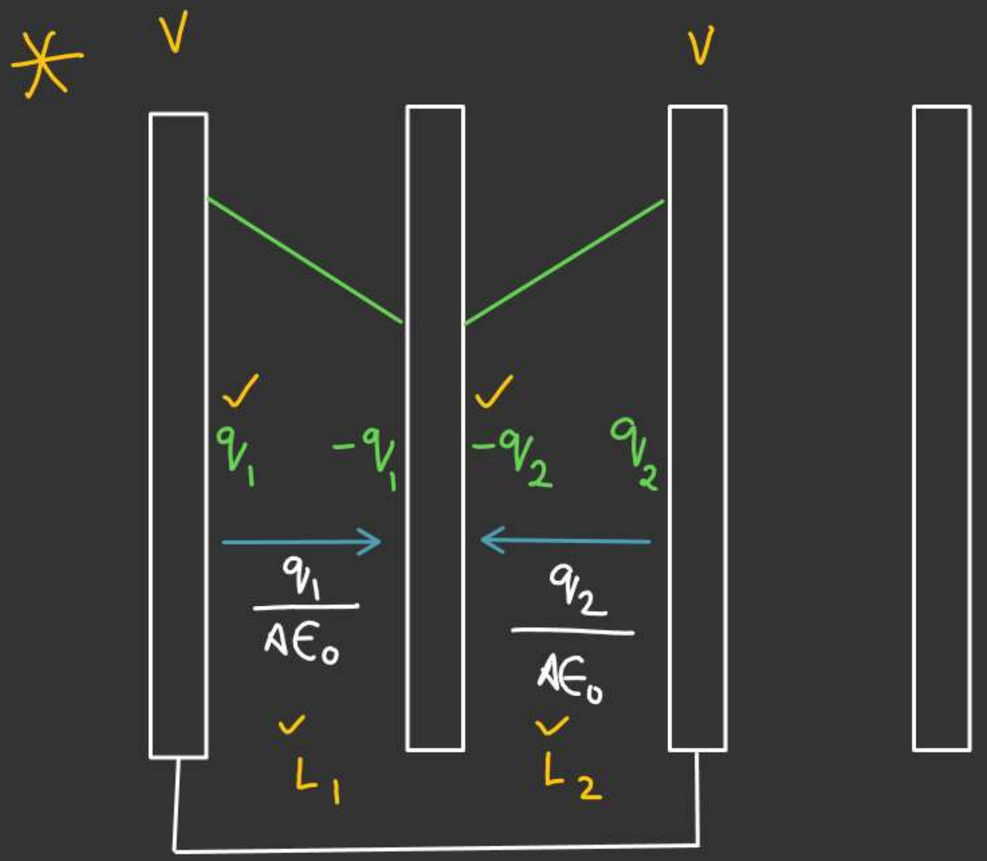

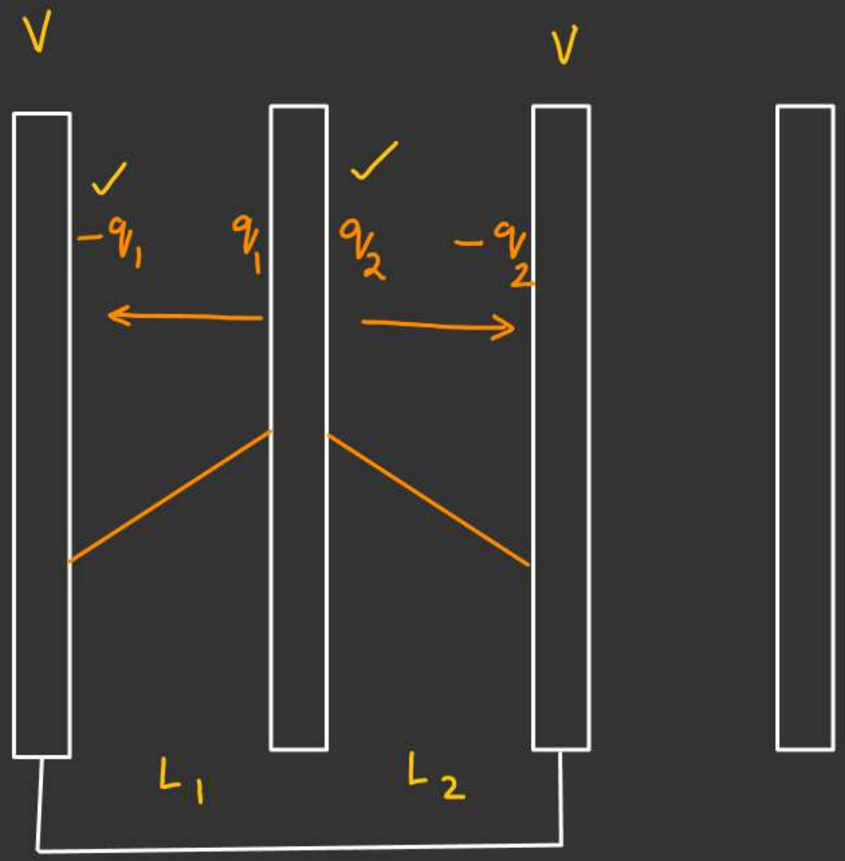

$$0 = -\left[\frac{q_1}{A\epsilon_0}\cdot L_1 - \frac{q_2}{A\epsilon_0}\cdot L_2\right]$$

$$q_1 L_1 - q_2 L_2 = 0$$

$$V_f - V_i = -\int \vec{E}\cdot d\vec{x}$$

$$-q_1 L_1 + q_2 L_2 = 0$$

$$\boxed{\sum q_i x_i = 0}$$

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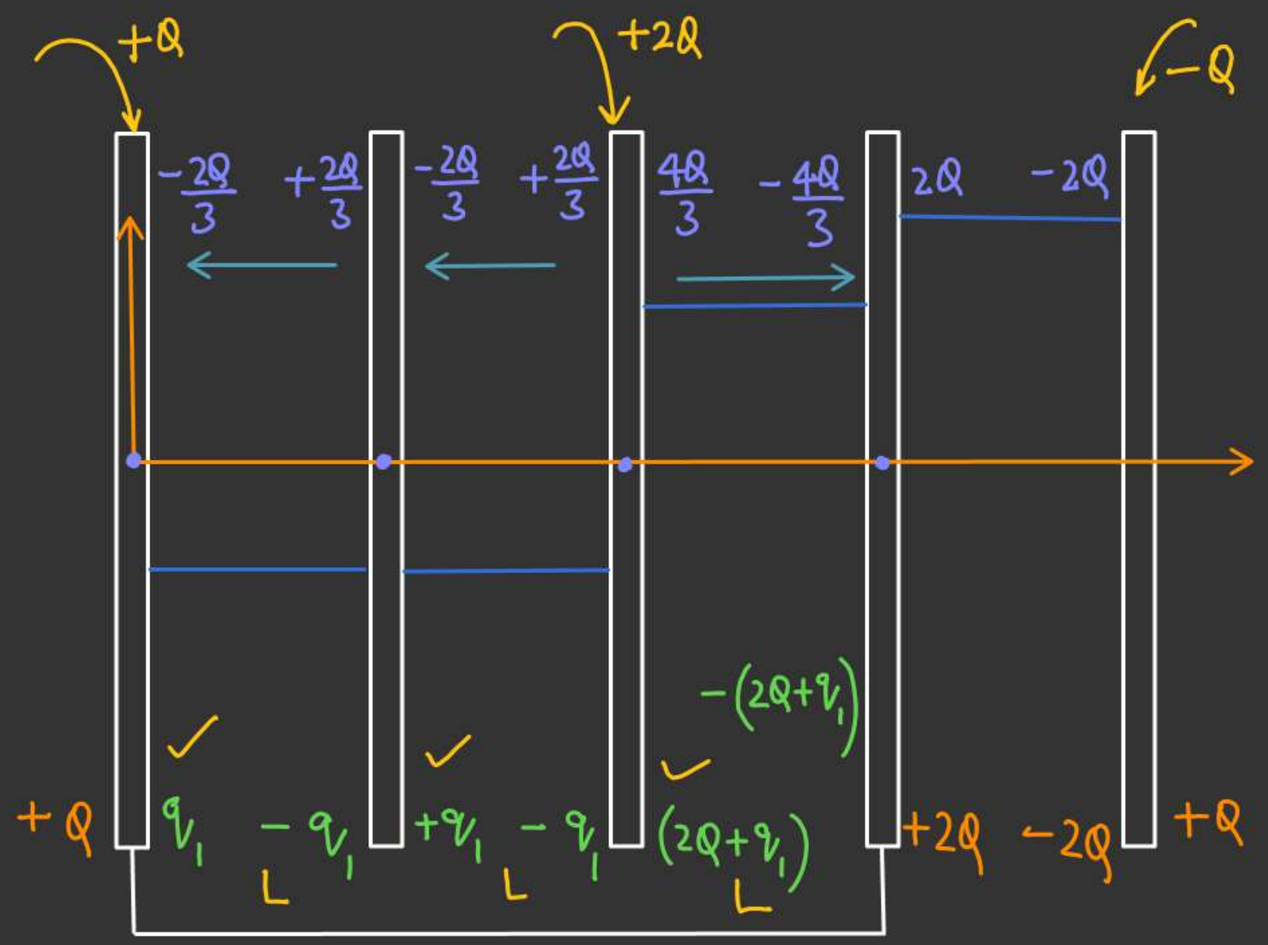

1 & 4 ;

$+Q = +Q + q_1 + 2Q - 2Q - q_1$

$Q = Q$

Jump *

In case of plates

$$\frac{\text{total Charge}}{2}$$

on outer Surface will be valid till plates are earthed.

$\sum q_i x_i = 0 \Rightarrow \sum q_i = 0$

$q_1 + q_1 + 2Q + q_1 = 0$

$q_1 = -2Q/3$

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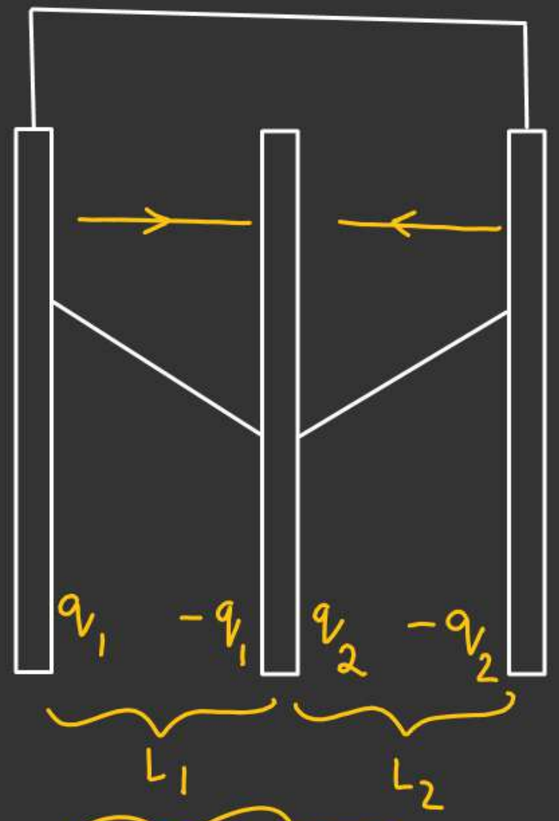
* Adjacent plates

Alternate plates



$$q_1 L_1 + q_2 L_2 = 0$$

$$q_2 = -q_1\left(\frac{L_1}{L_2}\right)$$

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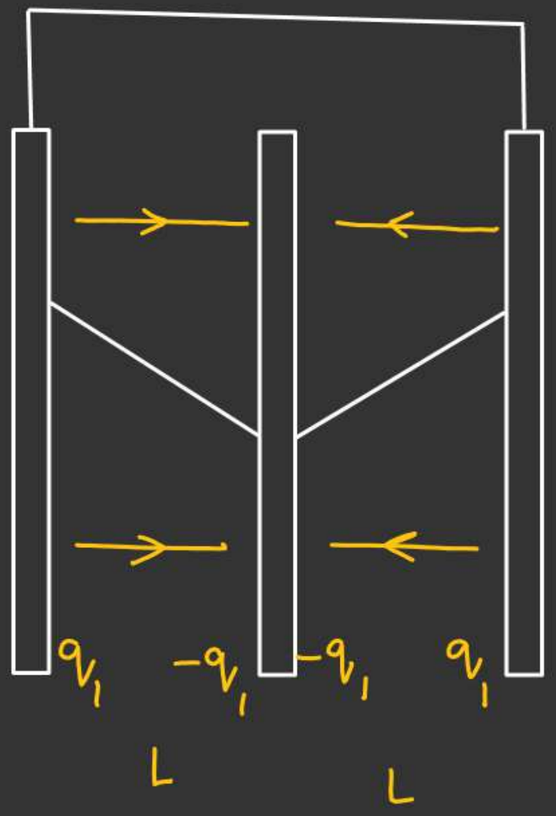
Alternate plates



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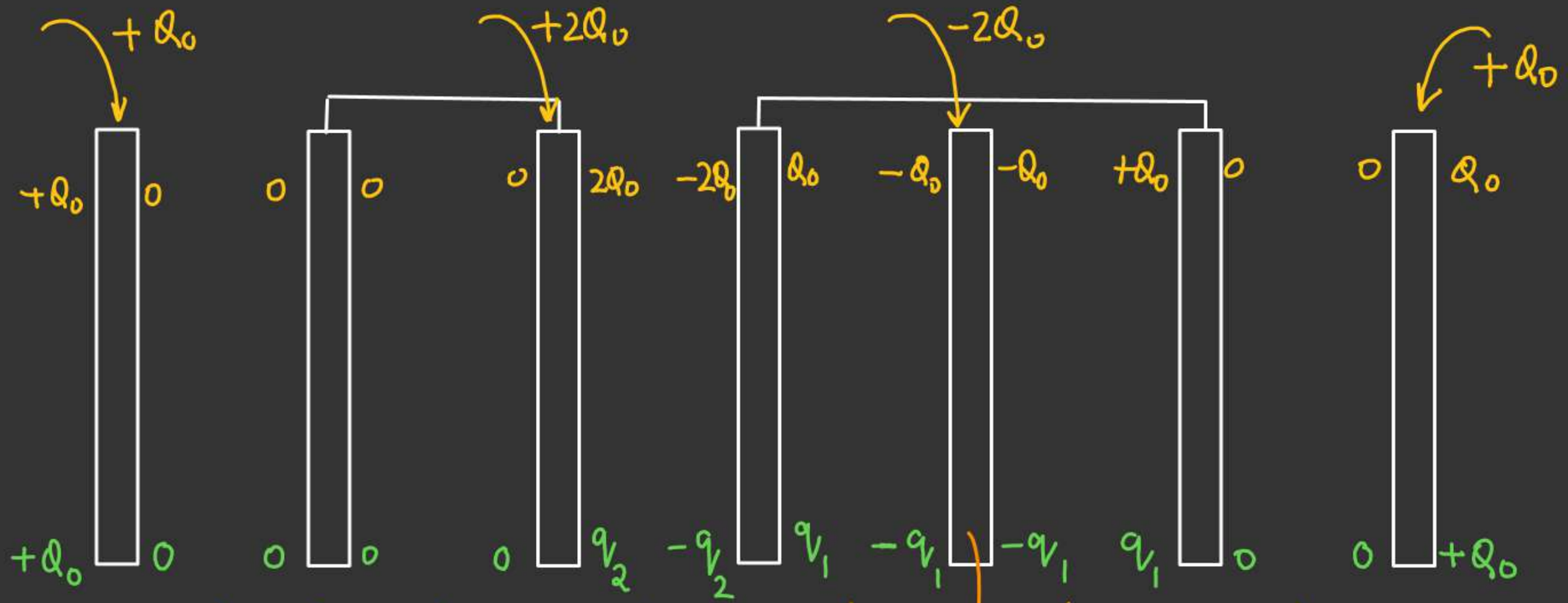

Plate 2 & 3

$2Q_0 = q_2$

$q_1 = Q_0$

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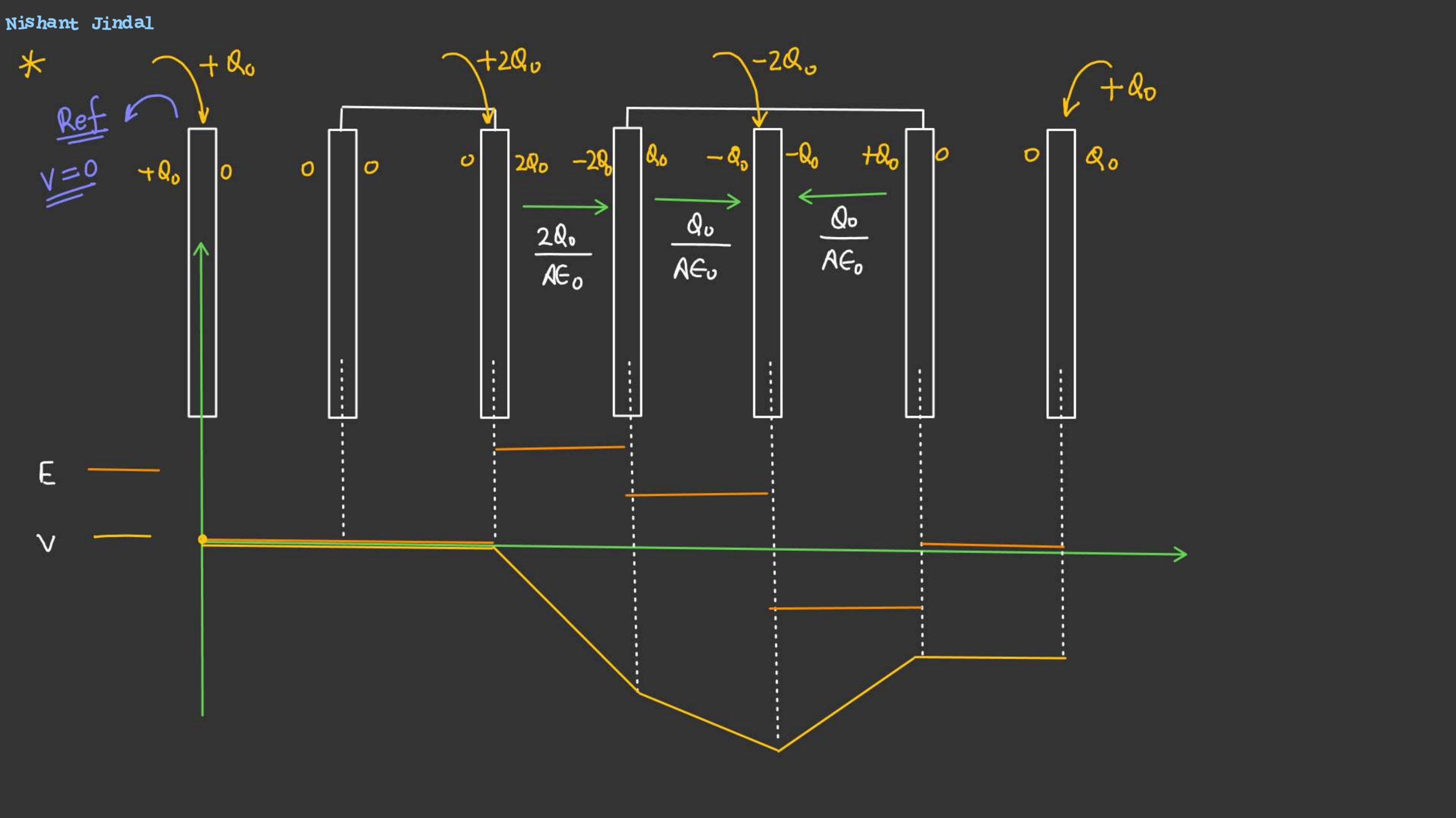

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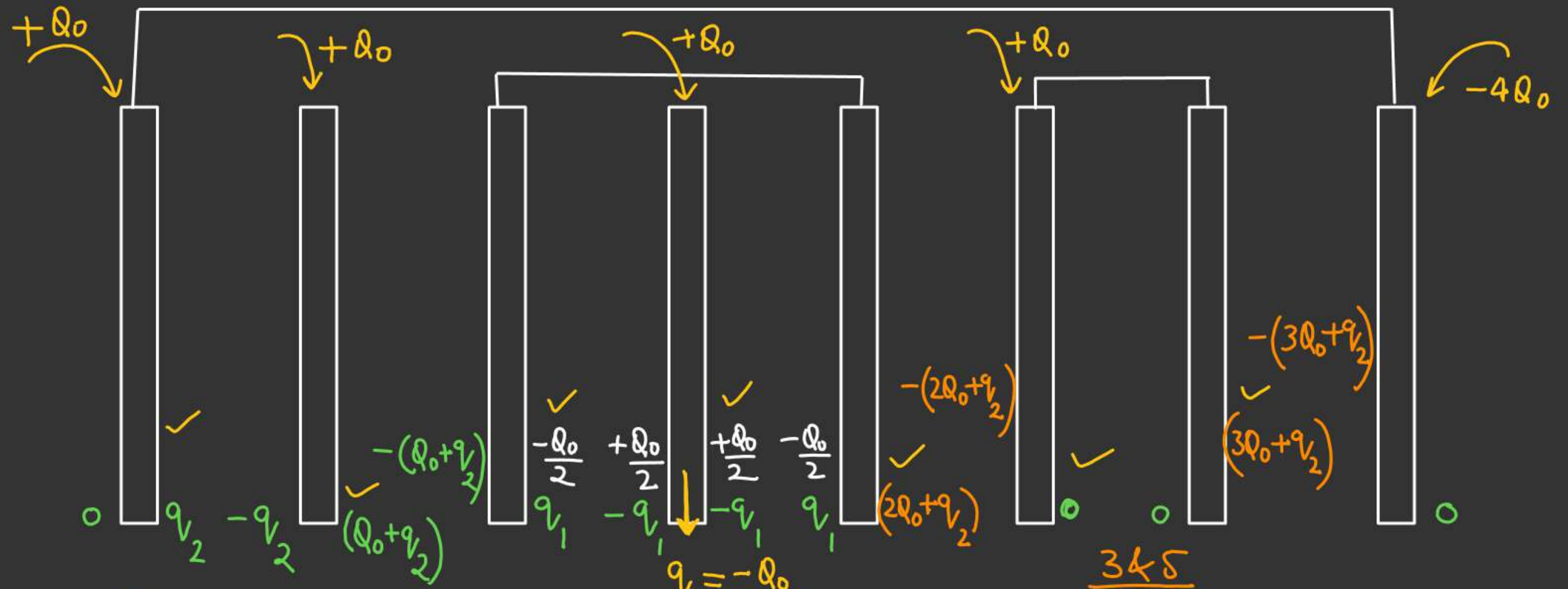
* equidistant Plates



$q_1 = \frac{-Q_0}{2}$

6 & 7 $+Q_0 = -2Q_0 - q_2 + 0 + 0 + (\ )$

Plate 1 & 8 $\sum q_i = 0$

$$q_2 + Q_0 + q_2 - \frac{Q_0}{2} + \frac{Q_0}{2} + 2Q_0 + q_2 + 3Q_0 + q_2 = 0$$

$$4q_2 + 6Q_0 = 0$$

$$\boxed{q_2 = \frac{-3Q_0}{2}}$$

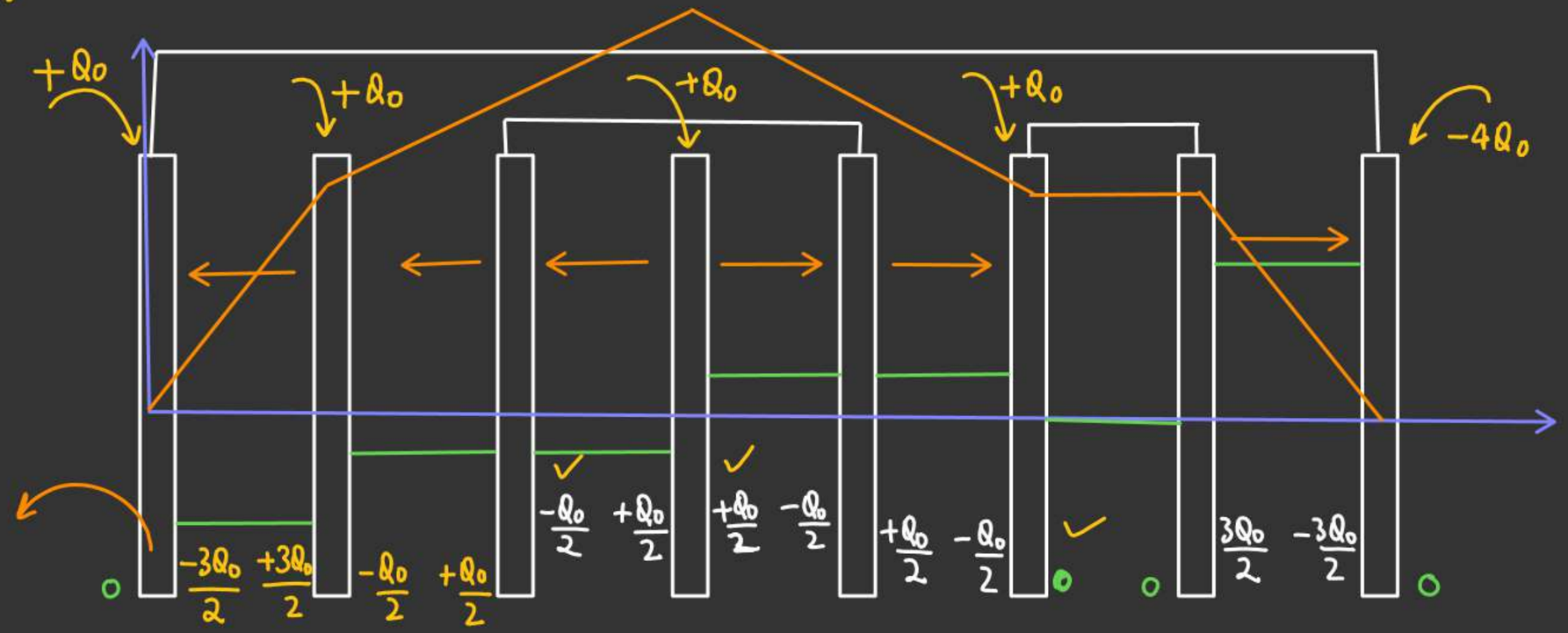
3 & 5

$$-Q_0 - q_2 + q_1 + q_1 + (\ ) = 0$$

$$-Q_0 - q_2 - Q_0 + (\ ) = 0$$

Nishant Jindal

* equidistant Plates



E ——

V ——

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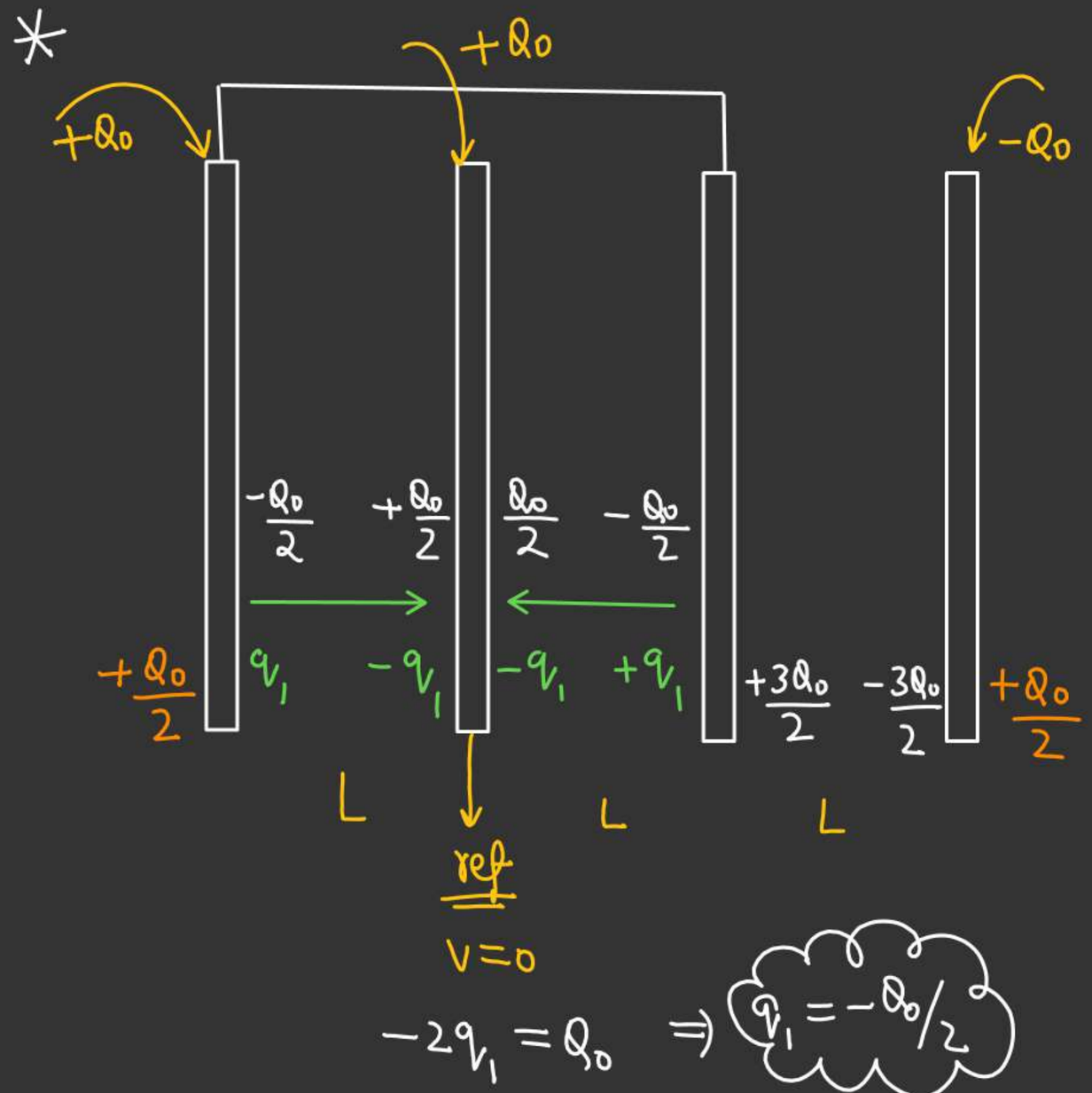

$$-2q_1 = Q_0 \Rightarrow q_1 = -Q_0/2$$

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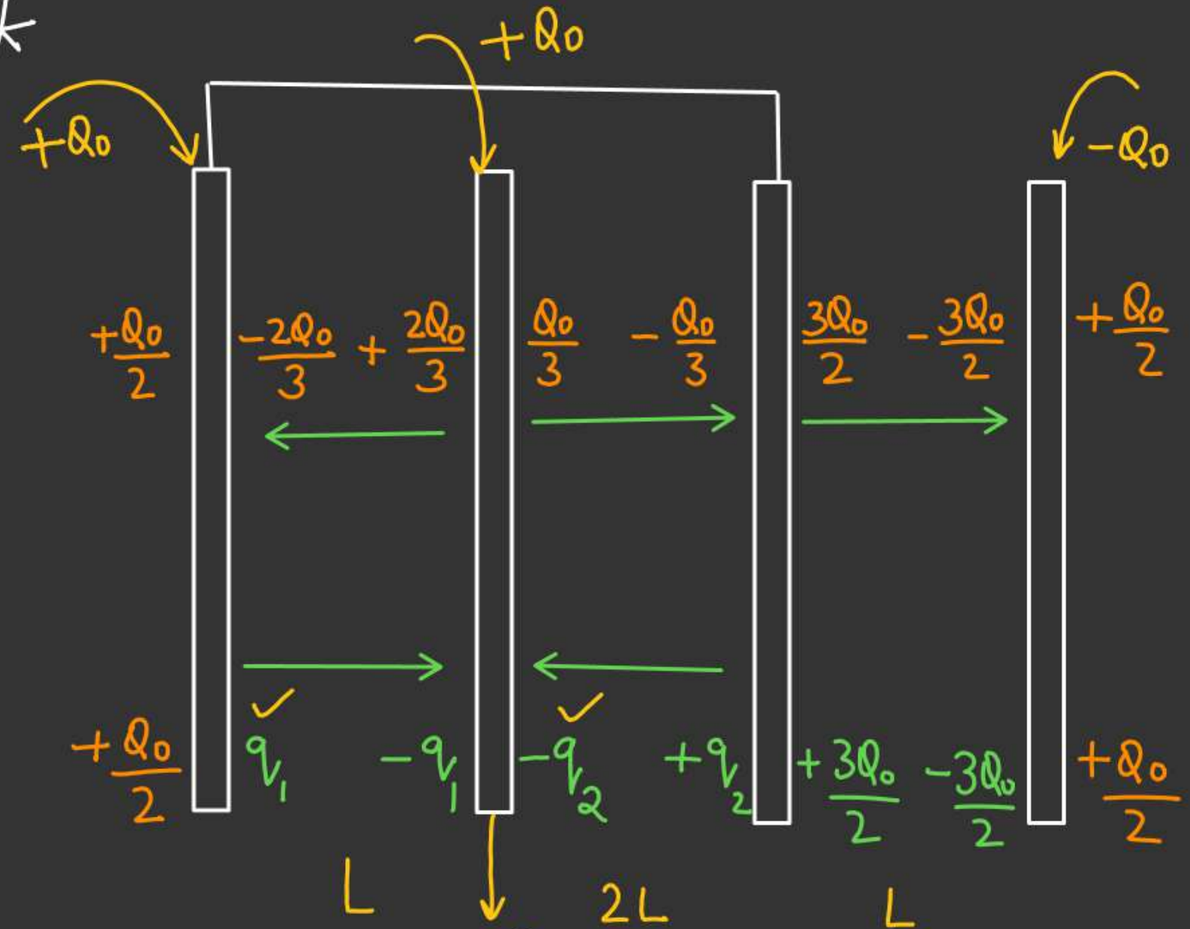
*



$$-q_1 - q_2 = +Q_0$$

$$-2q_2 - q_2 = +Q_0$$

$$\left[\begin{array}{c} q_2 = -Q_0/3 \\ q_1 = -2Q_0/3 \end{array}\right]$$

$$q_1 \cdot L - q_2 \cdot 2L = 0$$

$$q_1 = 2q_2$$

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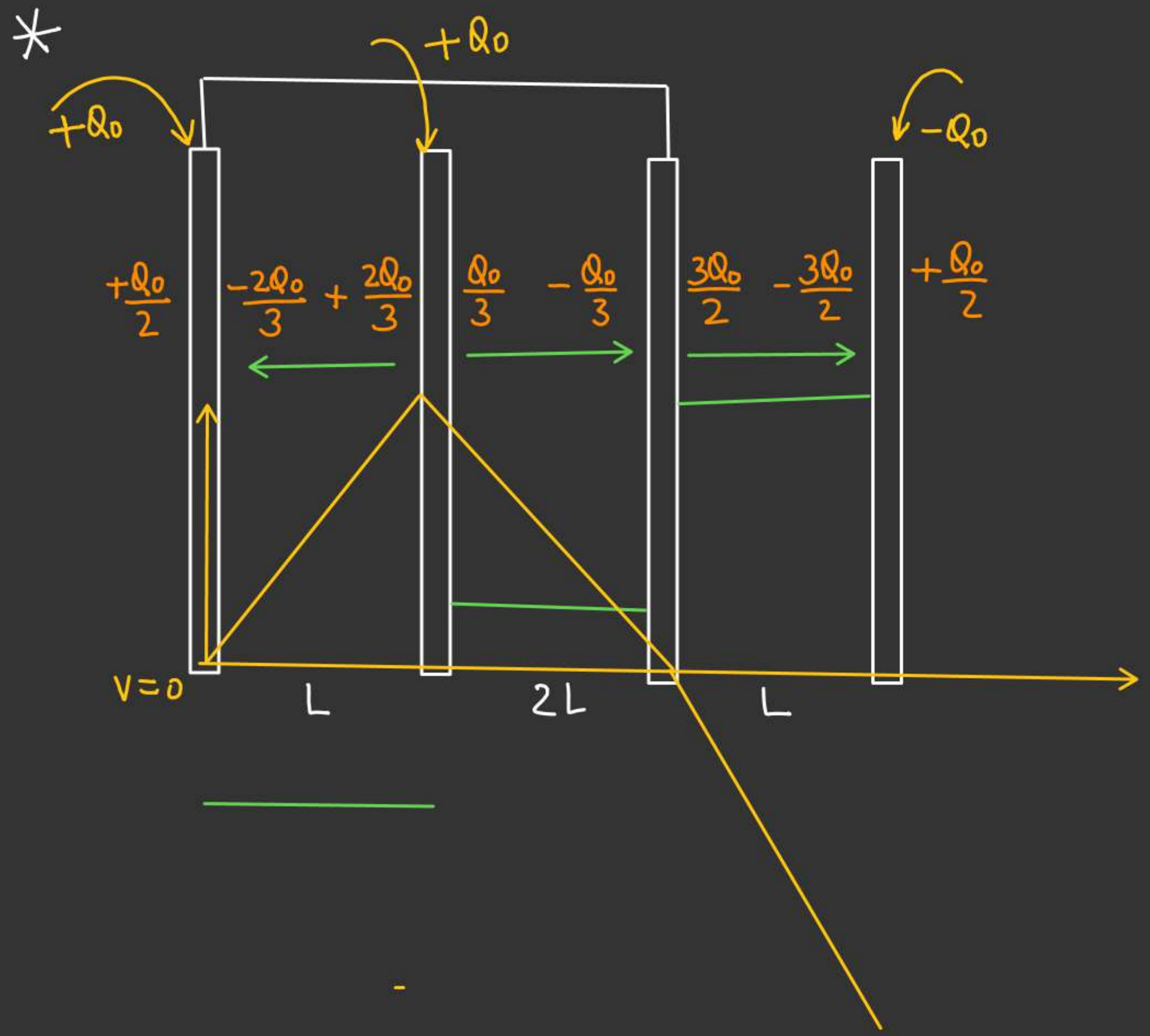

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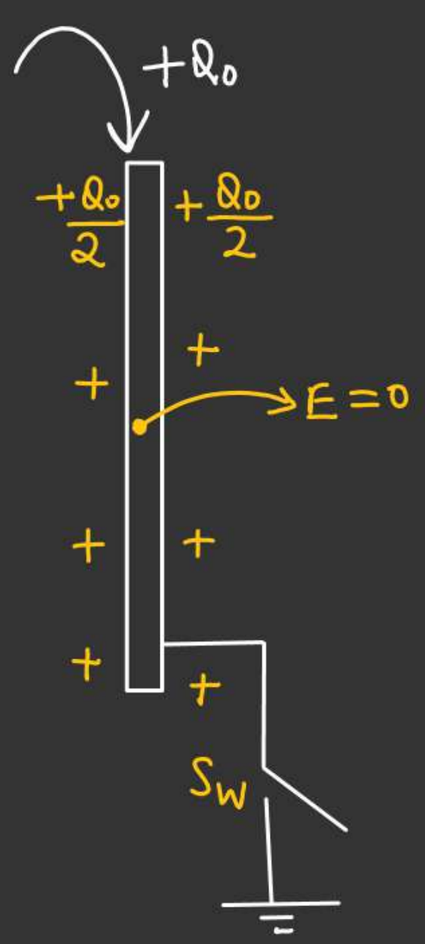
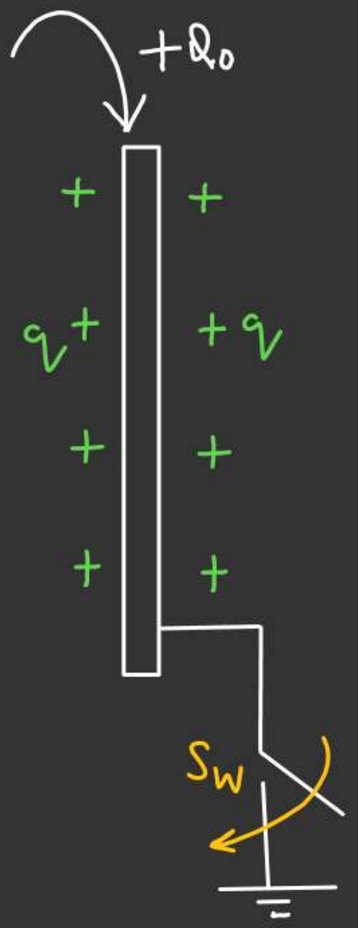
* Earthing of Plates :-

Conducting Plates





$$\alpha \frac{Kq}{L} = 0$$

$$q = 0$$

$$V_{Plate} = 0$$

$$Q_{Plate} = 0$$

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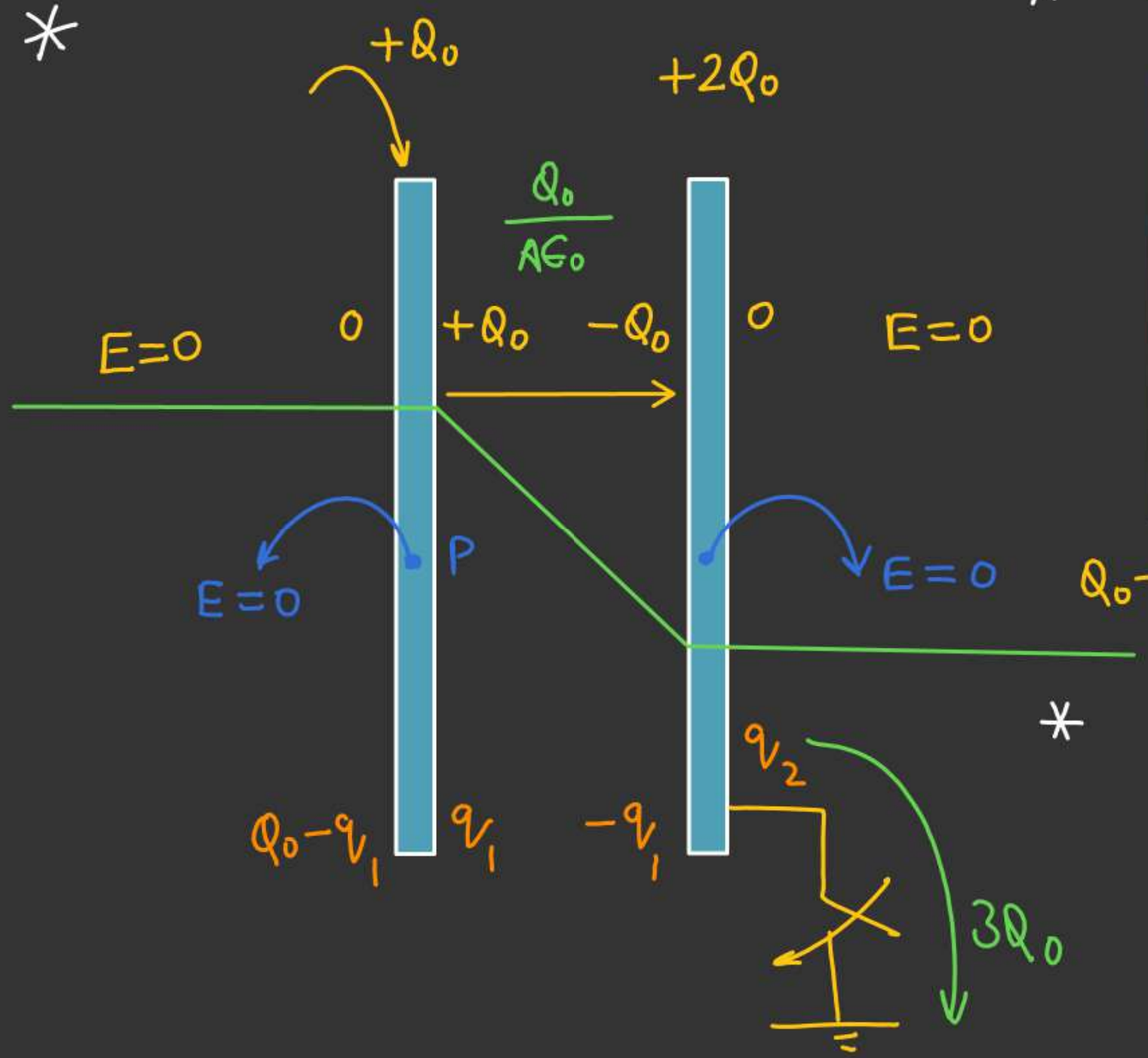
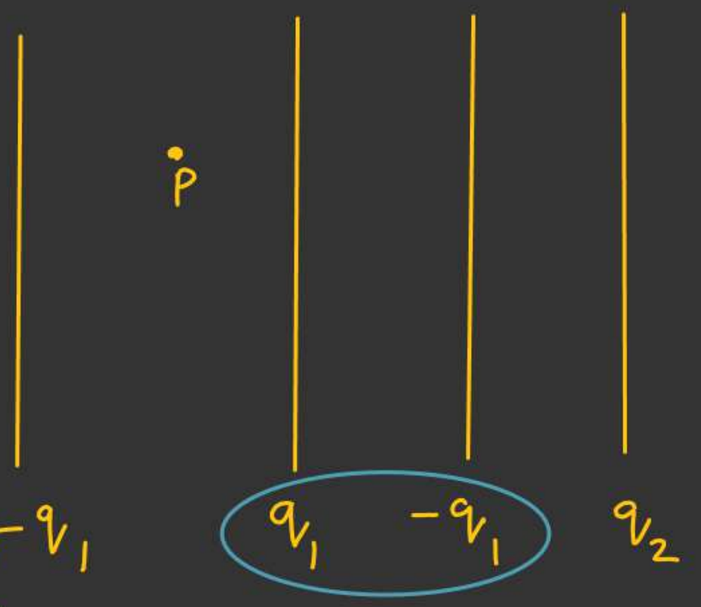


* $E_p = 0$



$Q_0 - q_1 = q_2$

$q_1 + q_2 = Q_0 \longrightarrow (i)$

* $\sum q_{plates} = 0 \qquad \sum q_i = 0$

$Q_0 - q_1 + q_2 = 0 \Rightarrow q_2 - q_1 = -Q_0 \longrightarrow (ii)$

* $\boxed{q_1 = Q_0 \ , \ q_2 = 0}$

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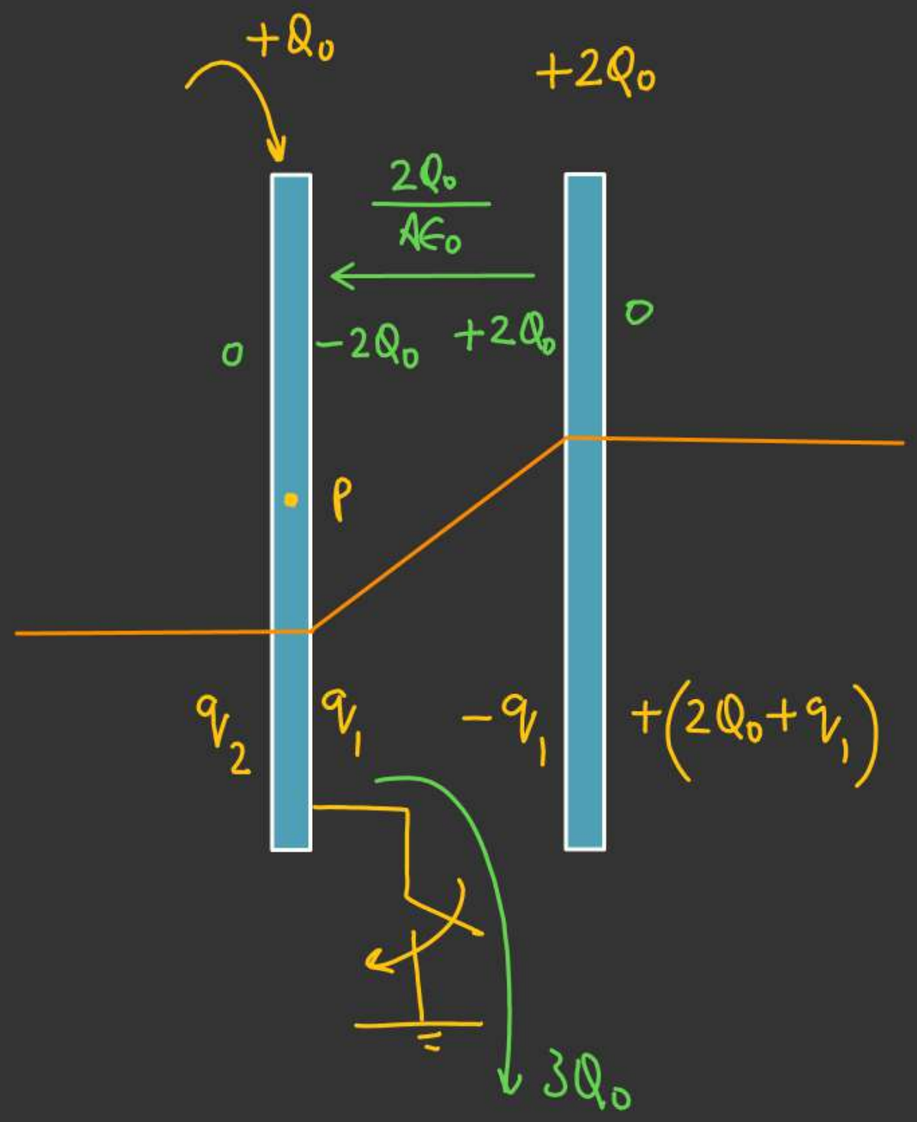

$E_p = 0$

$q_2 = 2Q_0 + q_1 \longrightarrow q_2 - q_1 = 2Q_0$

$\sum q_i = 0$

$q_2 + 2Q_0 + q_1 = 0 \longrightarrow q_2 + q_1 = -2Q_0$

$$\boxed{q_2 = 0, \quad q_1 = -2Q_0}$$

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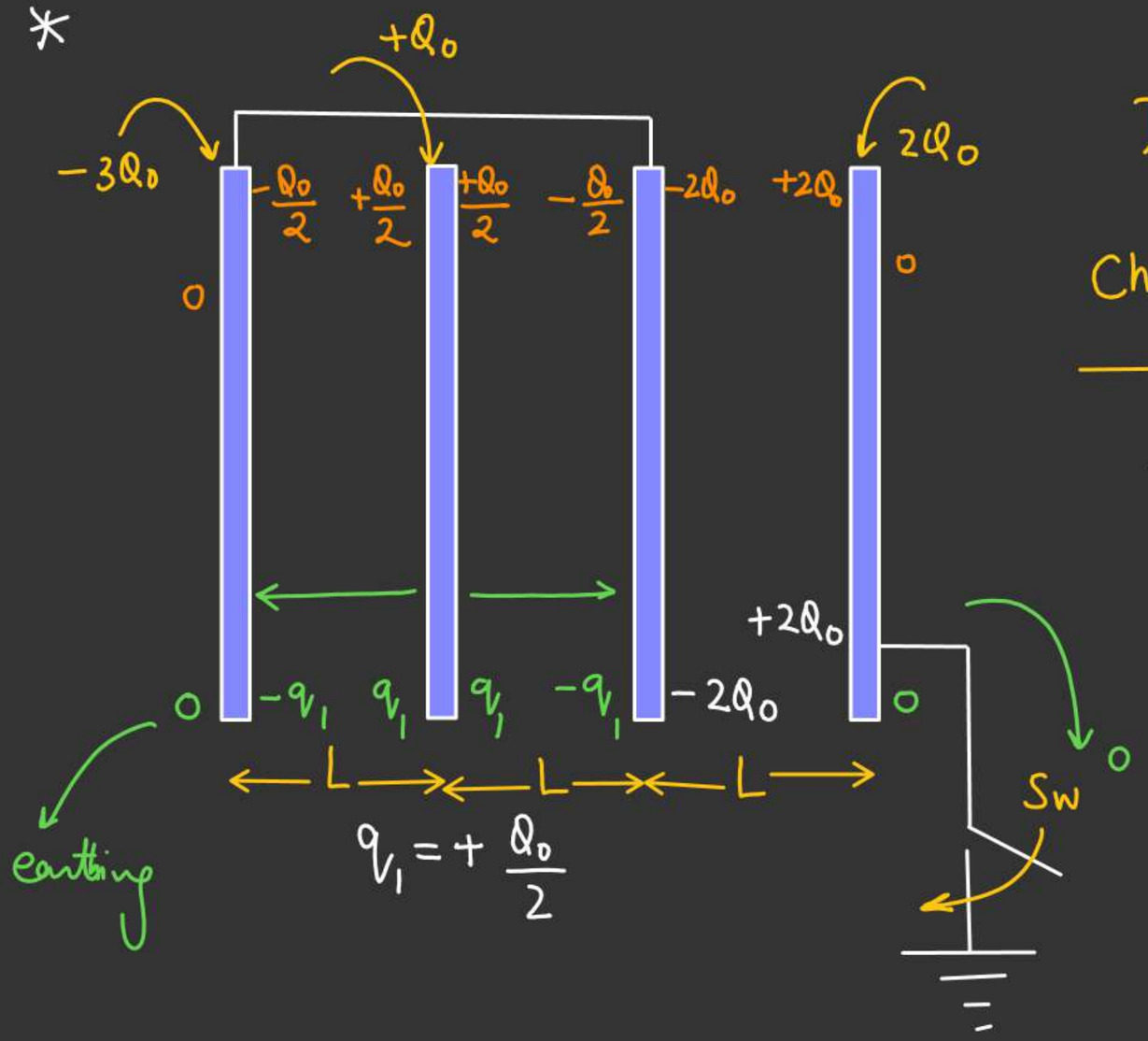
*



Draw E-r ; V-r graphs

Charge Conservation on 1 & 3

$$-3Q_0 = 0 - q_1 - q_1 + (\quad)$$

$$(\quad) = 2q_1 - 3Q_0 = -2Q_0$$

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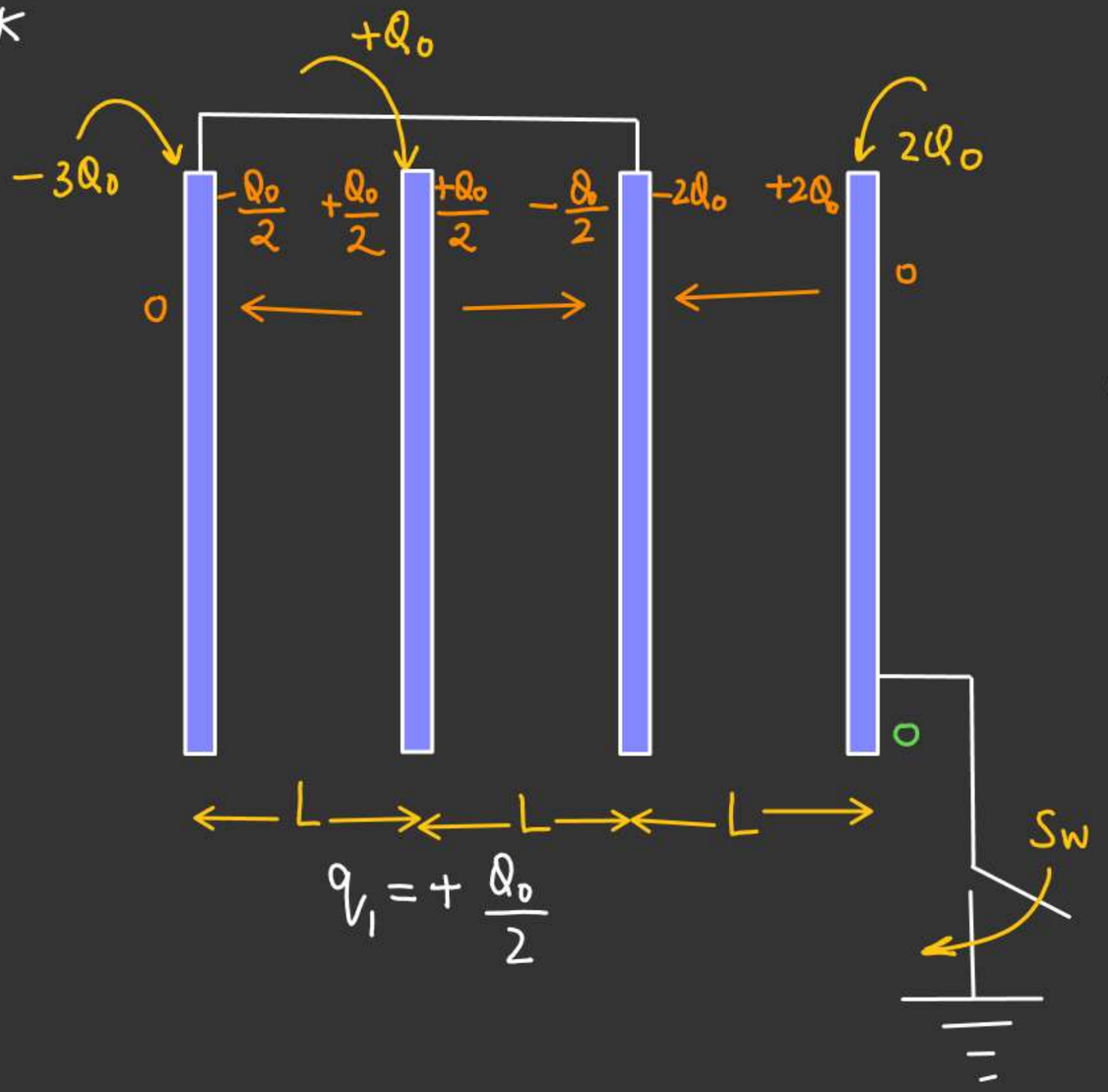

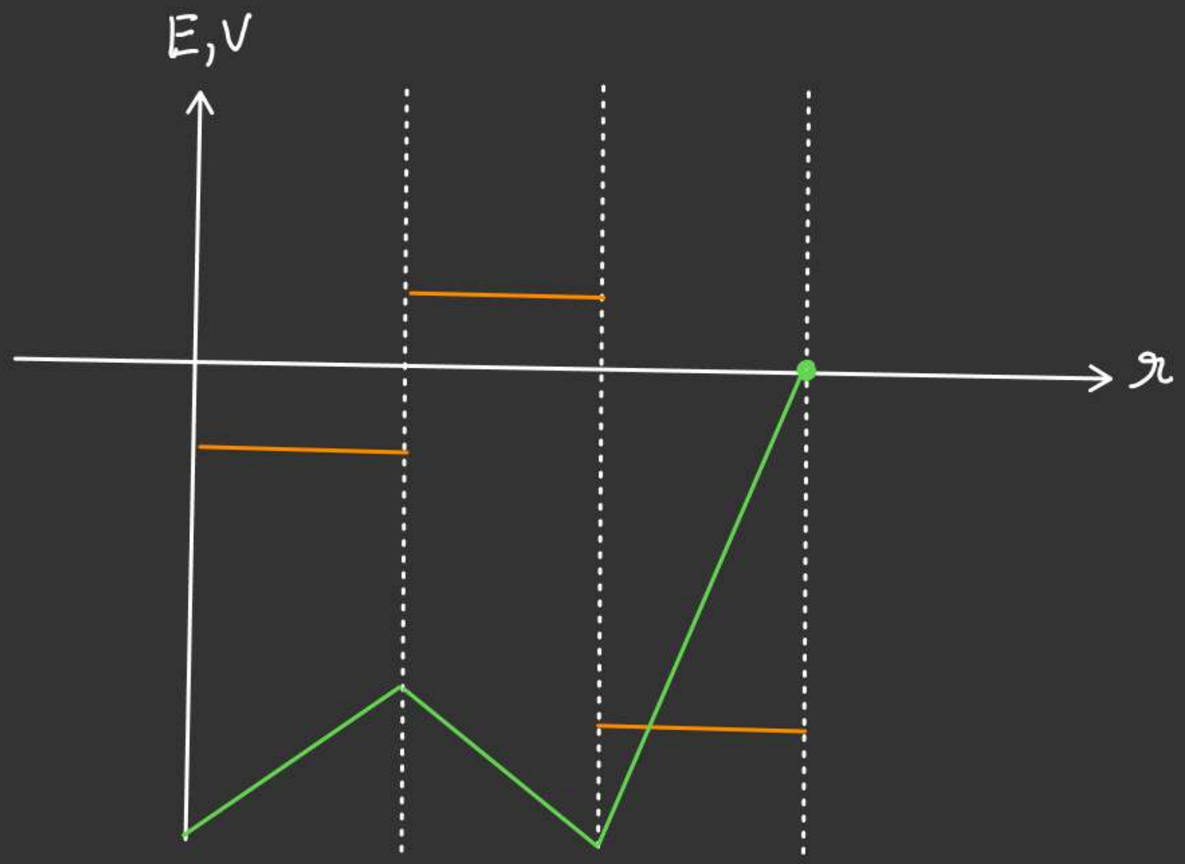

V ——

E ——

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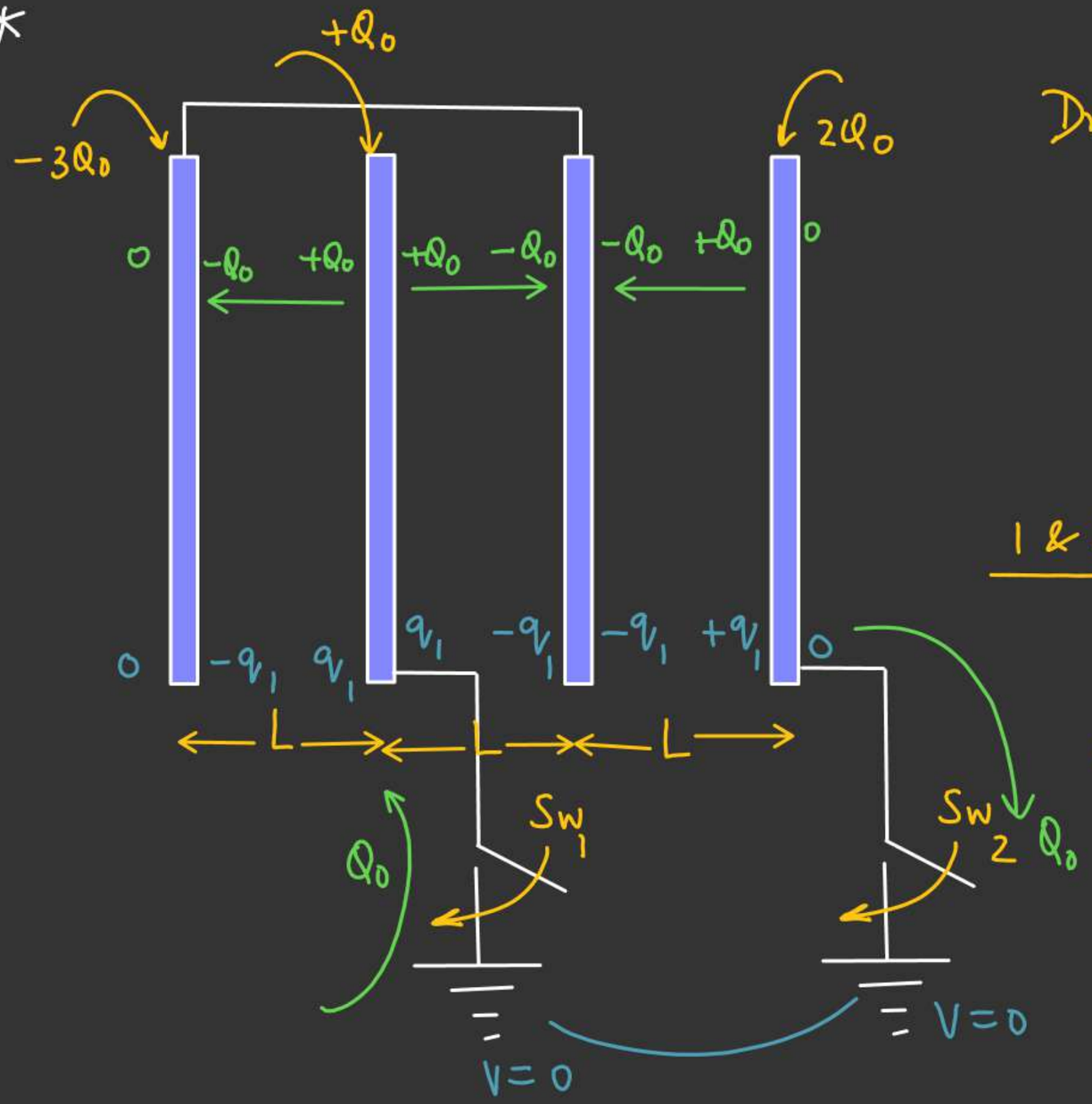

Draw E-r ; V-r graphs

$$\frac{|Q_{s_1}|}{|Q_{s_2}|} = 1$$

$$|Q_{s_1} + Q_{s_2}| = 0$$

1 & 3 :

$$-3Q_0 = 0 - q_1 - q_1 - q_1$$

$$q_1 = Q_0$$

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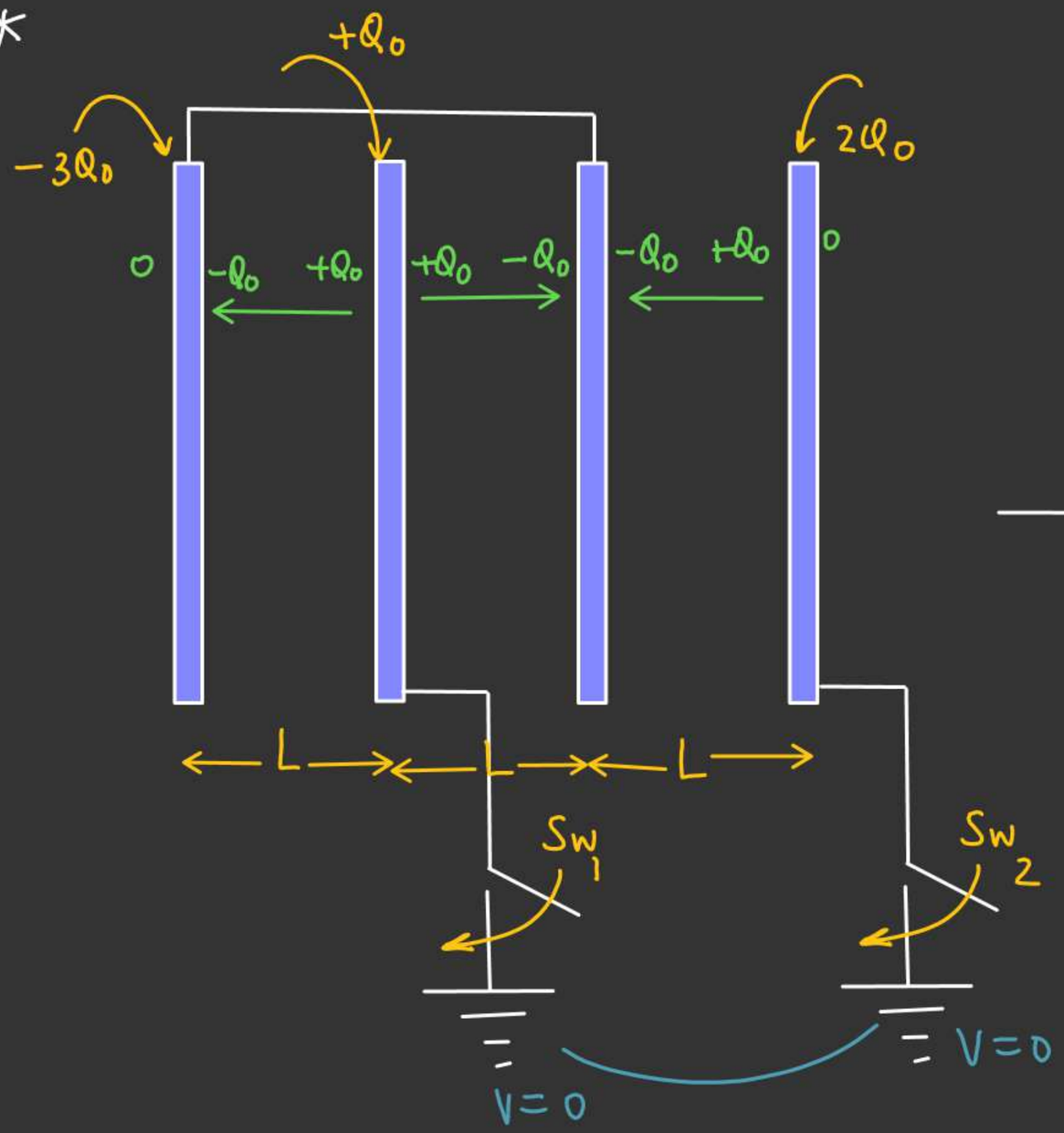

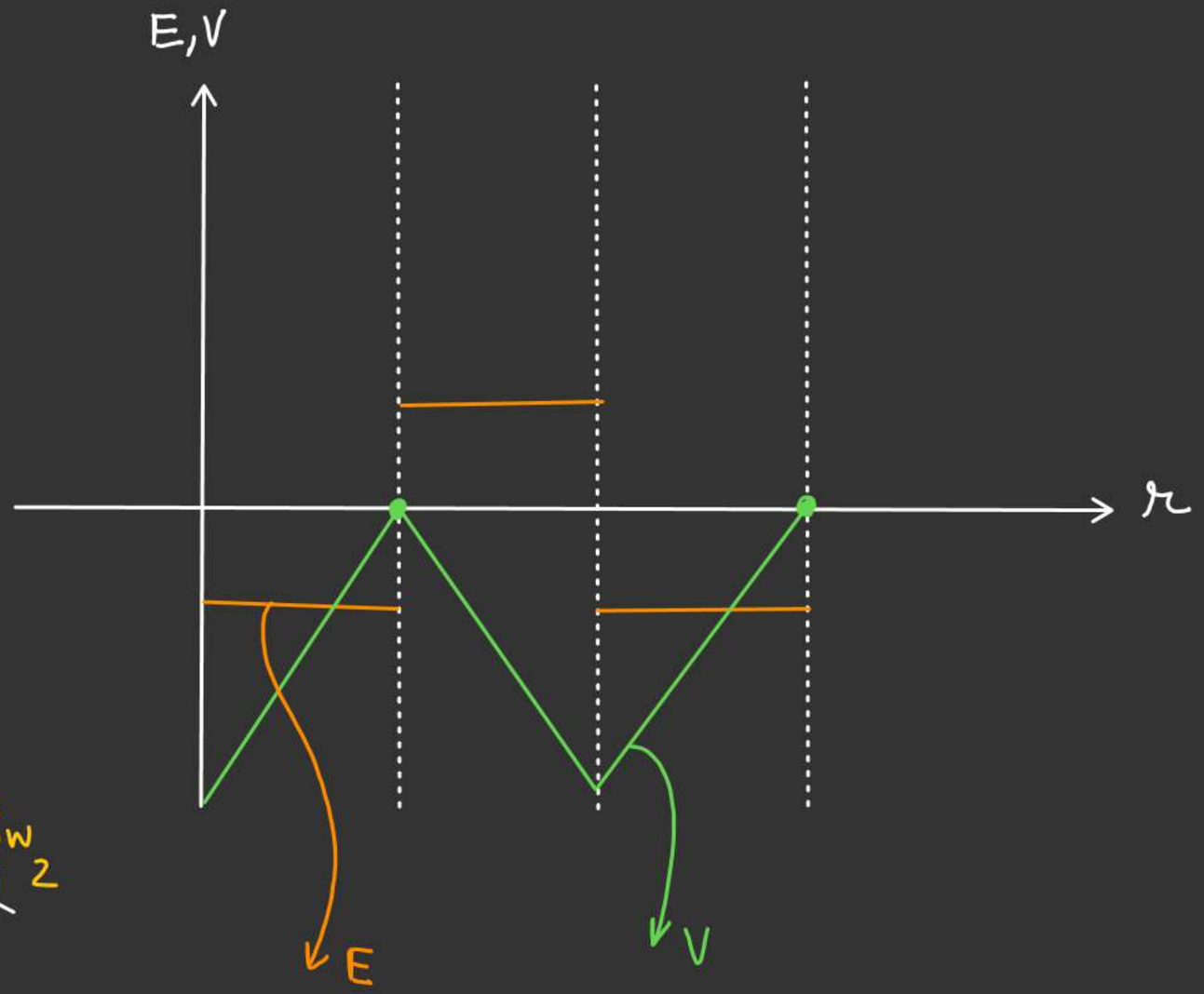

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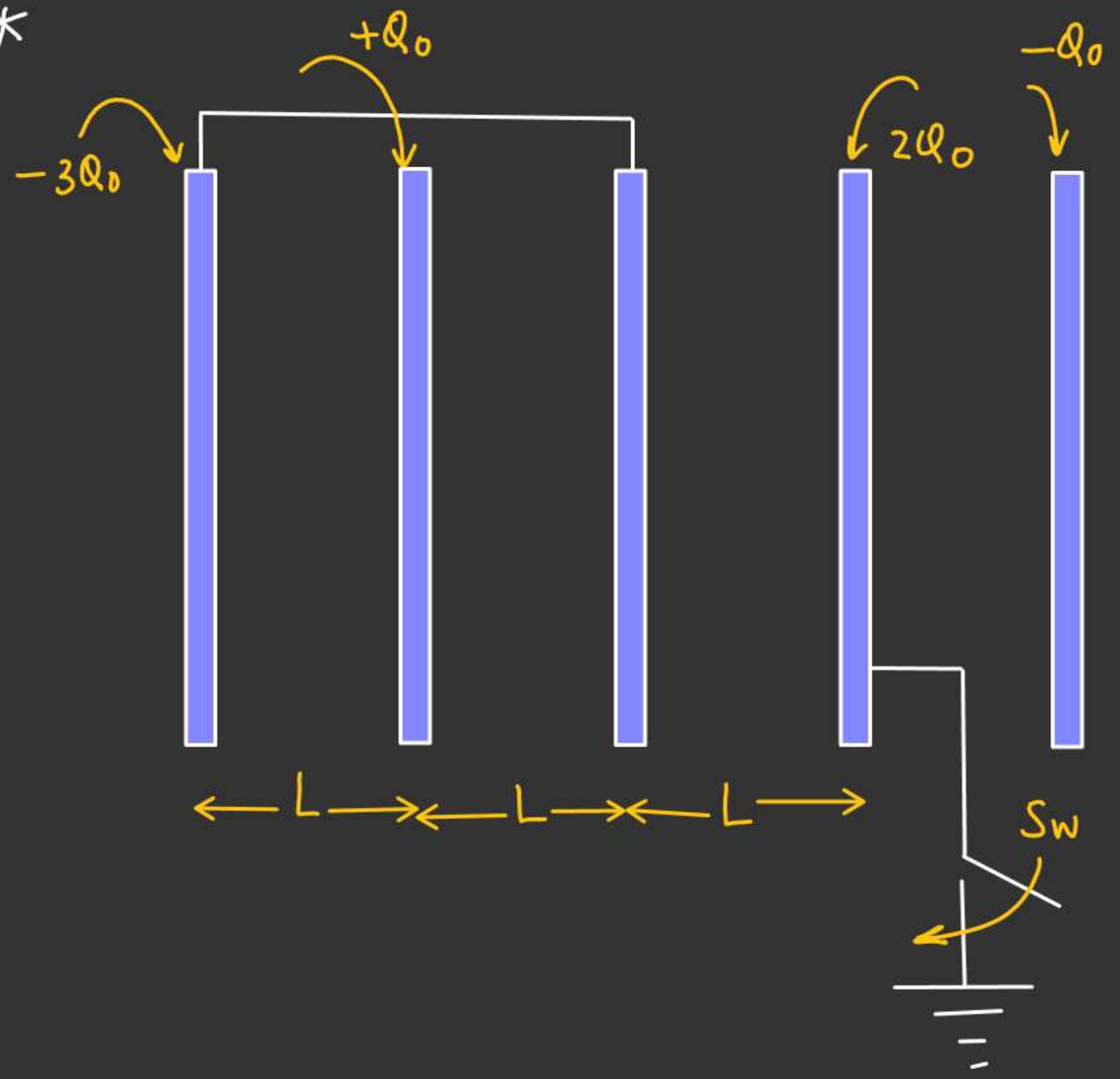

Draw E-r ; V-r graphs

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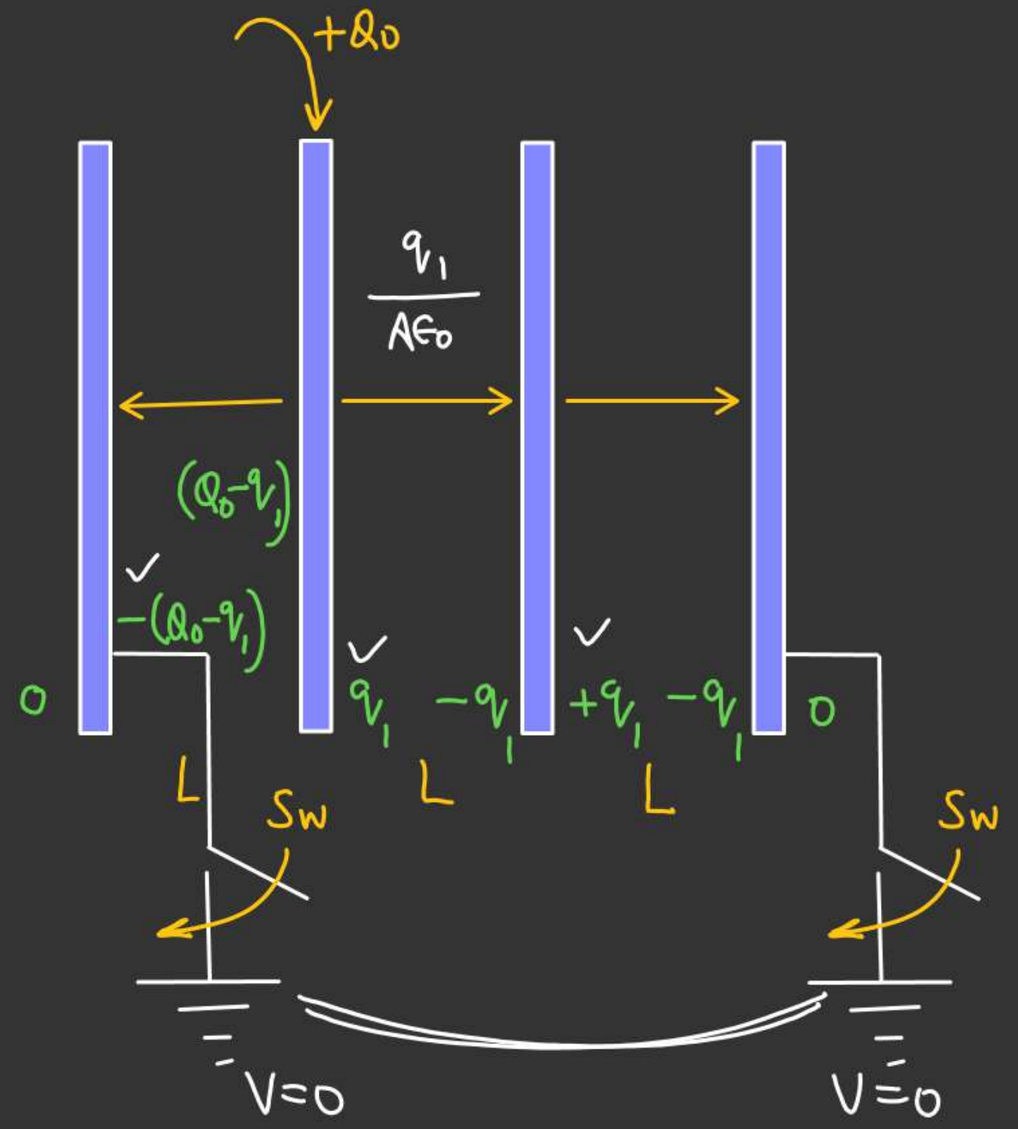

$V_2 - V_3 = ?$

$$\sum q_i L_i = 0 \quad \Rightarrow \sum q_i = 0$$

$$-(Q_0 - q_1) + q_1 + q_1 = 0$$

$$q_1 = \frac{Q_0}{3}$$

$$V_2 - V_3 = \frac{q_1}{A\epsilon_0} \cdot L = \frac{Q_0 L}{3A\epsilon_0}$$

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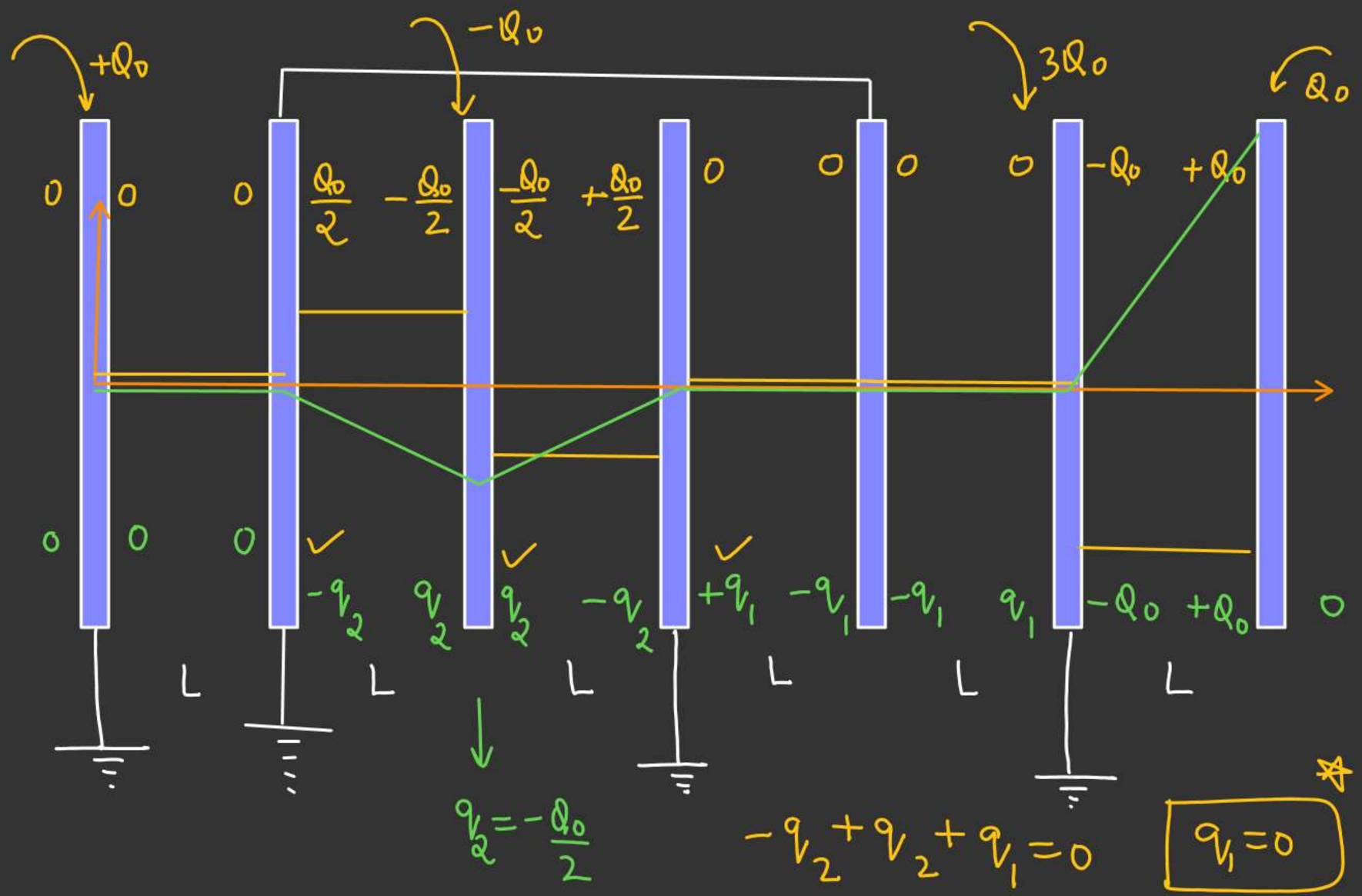

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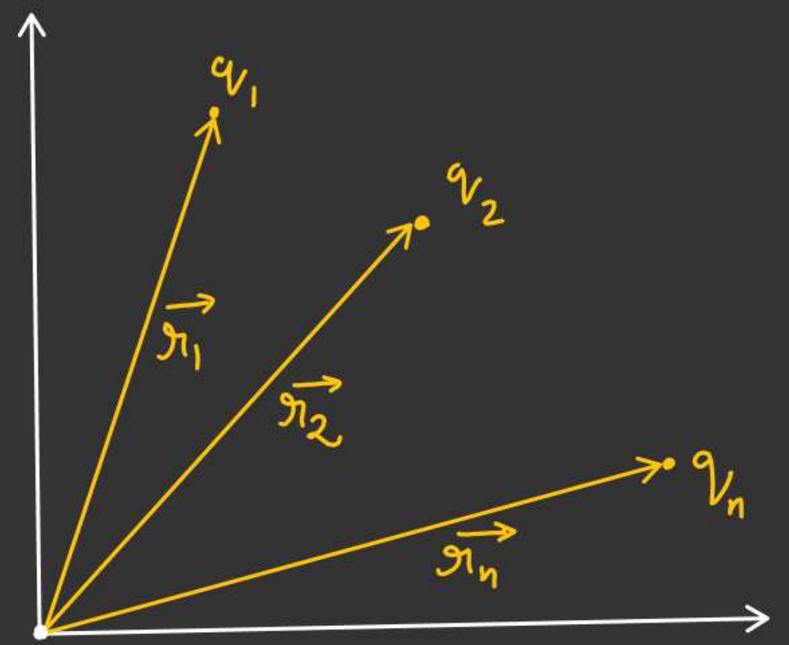
Electric Dipole :-

$q_{total} \cdot \vec{r}_{cm}$

$$\vec{p} = q_1\vec{r}_1 + q_2\vec{r}_2 + \quad + q_n\vec{r}_n$$

$$\vec{p} = \sum q_i \vec{r}_i$$

→ electric dipole moment



* This defⁿ is dependent on choice of origin.
* If net charge on the system is zero then $\vec{p}$ will be independent of Choice of origin.

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* $q_1 + q_2 + \quad + q_n = 0$

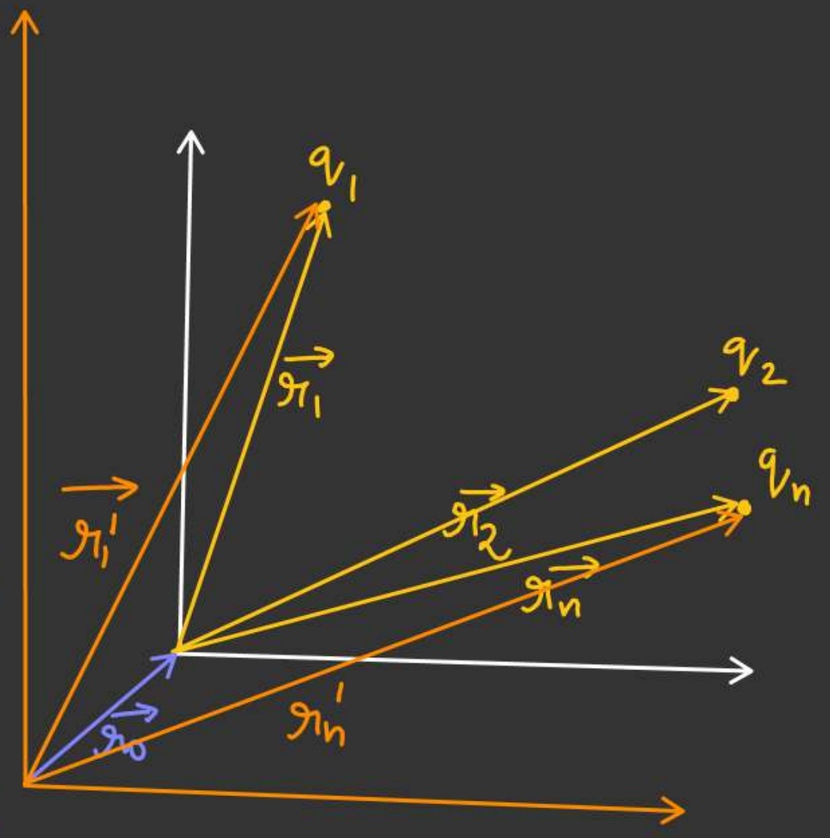
$$\vec{p_0} = q_1\vec{r_1} + q_2\vec{r_2} + \quad + q_n\vec{r_n}$$

$$\vec{p'} = q_1\vec{r_1'} + q_2\vec{r_2'} + \quad + q_n\vec{r_n'}$$

$$\vec{p'} = q_1(\vec{r_0}+\vec{r_1}) + q_2(\vec{r_0}+\vec{r_2'}) + \quad + q_n(\vec{r_0}+\vec{r_n'})$$

$$\vec{p'} = (q_1+q_2+\quad+q_n)\vec{r_0} + (\underbrace{q_1\vec{r_1}+q_2\vec{r_2}+\quad+q_n\vec{r_n}}_{\vec{p_0}})$$

$$\boxed{\vec{p'} = (\Sigma q_i)\,\vec{r_0} + \vec{p_0}}$$



$$\vec{r_1'} = \vec{r_0} + \vec{r_1}$$
$$\vec{r_2'} = \vec{r_0} + \vec{r_2}\,,$$

$$\vec{r_n'} = \vec{r_0} + \vec{r_n}$$

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$$\sum q_i = 0$$

$$\vec{p'} = \vec{p_0}$$

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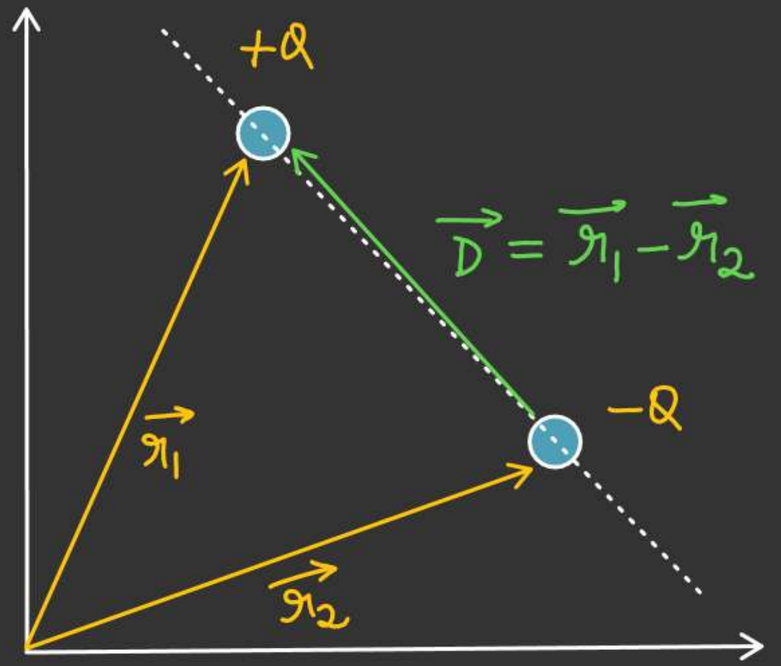
$$\vec{p} = +q \cdot \vec{r}_1 - q \cdot \vec{r}_2$$

$$\Rightarrow q(\vec{r}_1 - \vec{r}_2)$$

$$\boxed{\vec{p} = q\vec{D}}$$

$(-ve \text{ to } +ve)$



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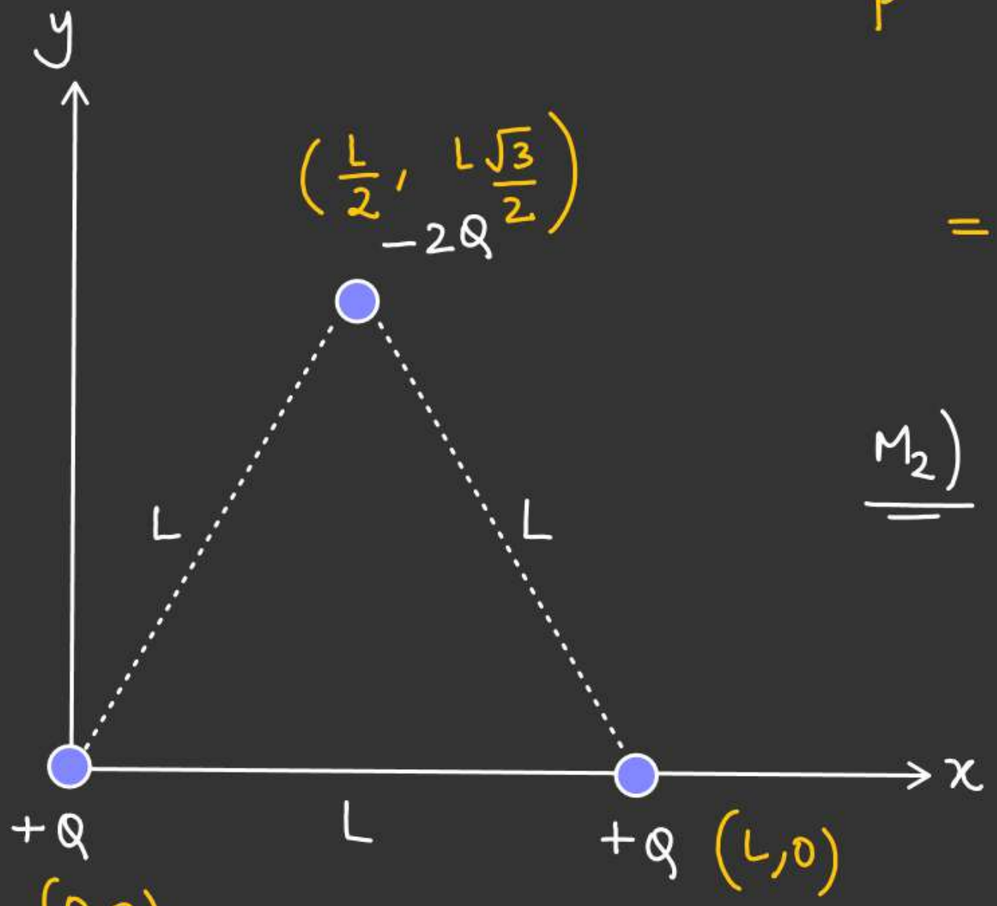
*

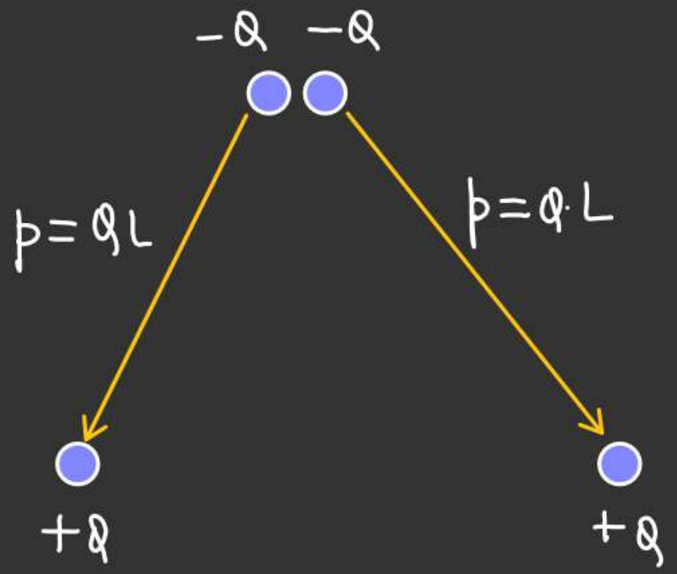
$M_1)$

$$\vec{p} = +Q(0,0) + Q(L,0) - 2Q\left(\frac{L}{2}, \frac{L\sqrt{3}}{2}\right)$$

$$= -QL\sqrt{3} \downarrow$$



$M_2)$



$$p_{net} = 2p\cos 30°$$

$$\Rightarrow \sqrt{3}p = \sqrt{3}QL \downarrow$$

$$\vec{r} = x\hat{i} + y\hat{j} = (x, y)$$

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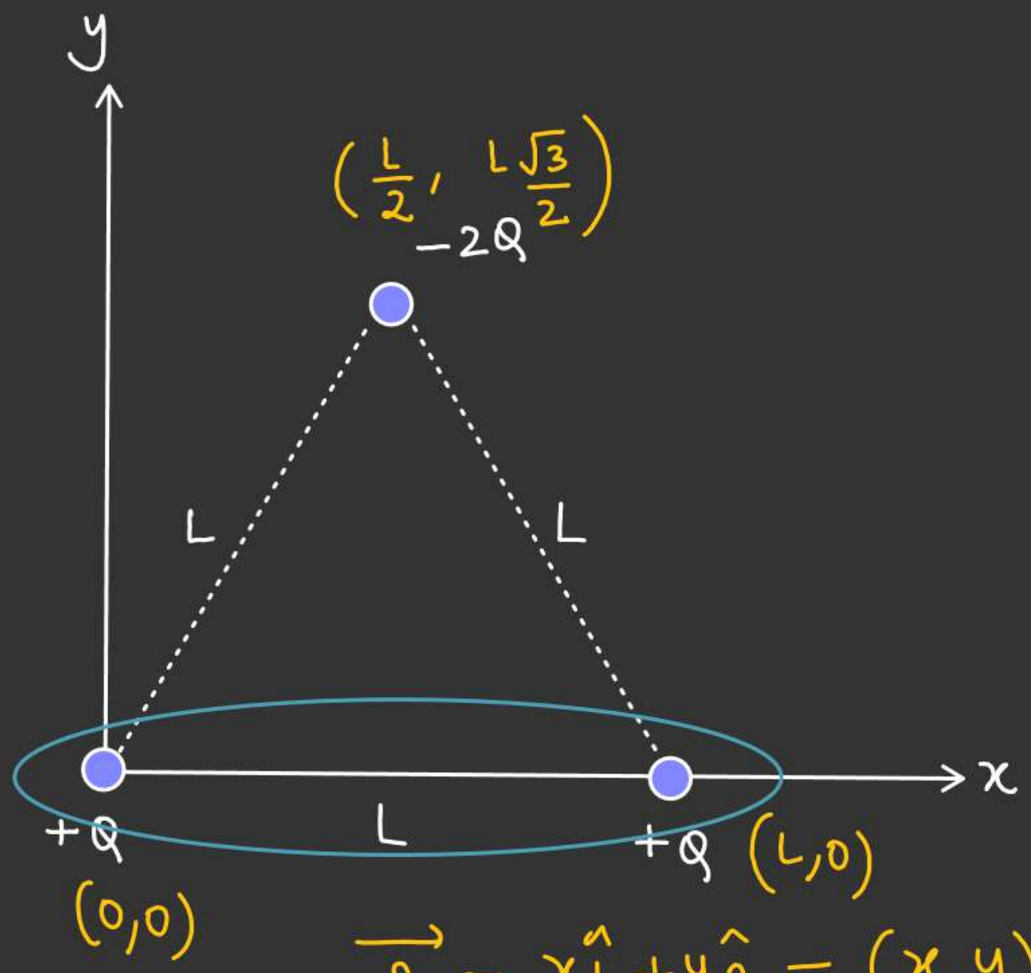
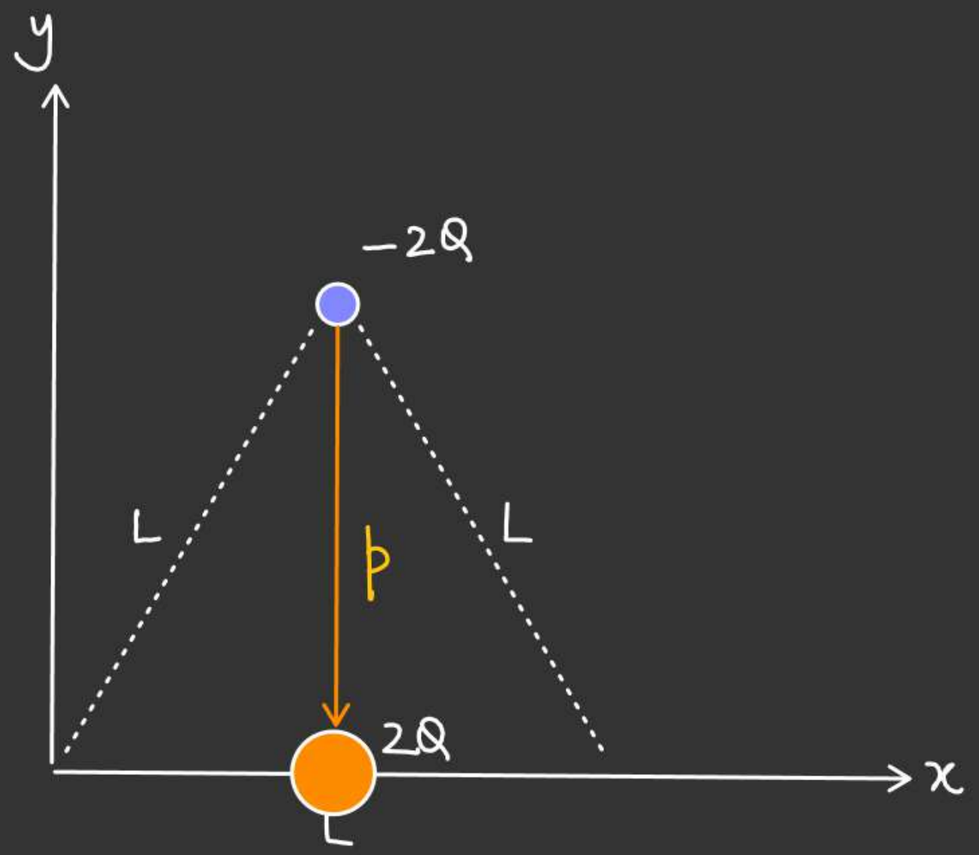
* M_3)



$$\vec{r} = x\hat{i} + y\hat{j} = (x, y)$$



$$p = 2Q \cdot L \sin 60 = \sqrt{3}QL$$

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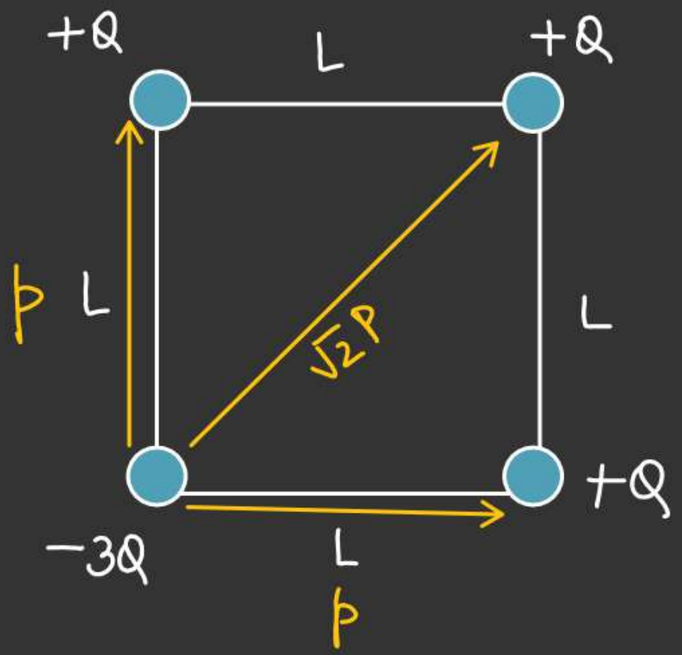
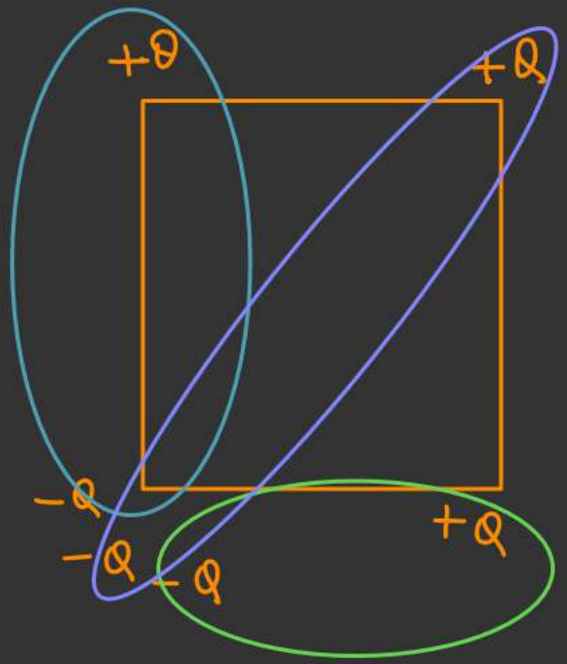
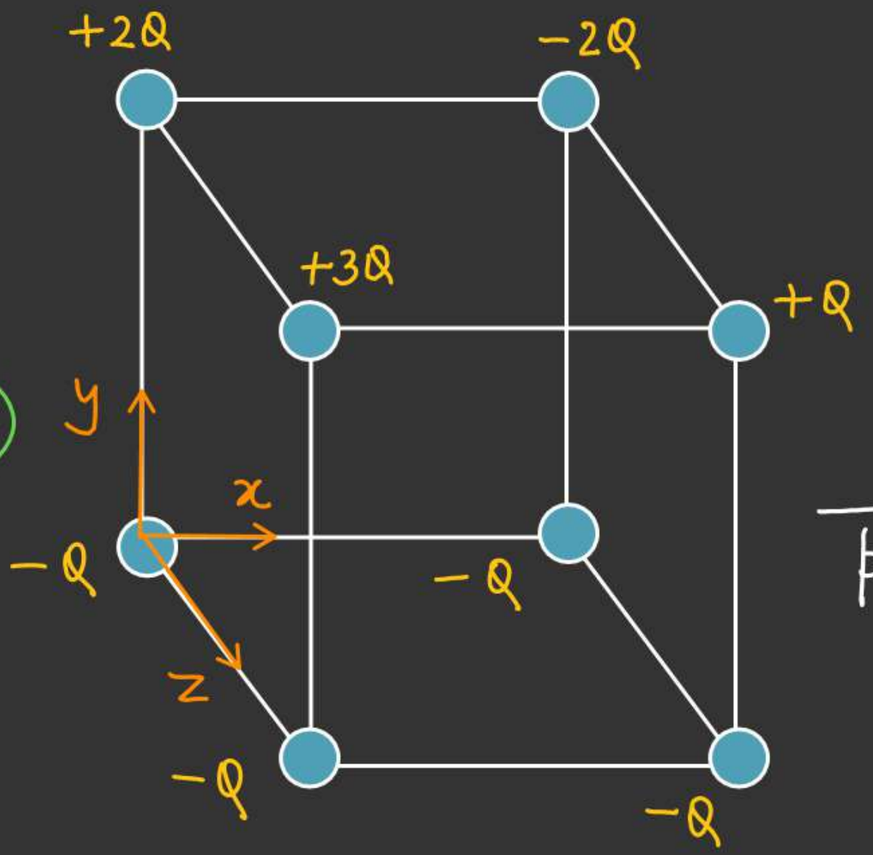


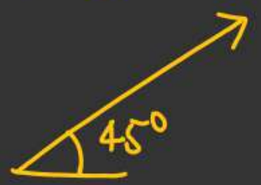


H.C.



$\vec{p} = ?$

$$p_{net} = (\sqrt{2}\,p)2 = 2\sqrt{2}\,QL$$



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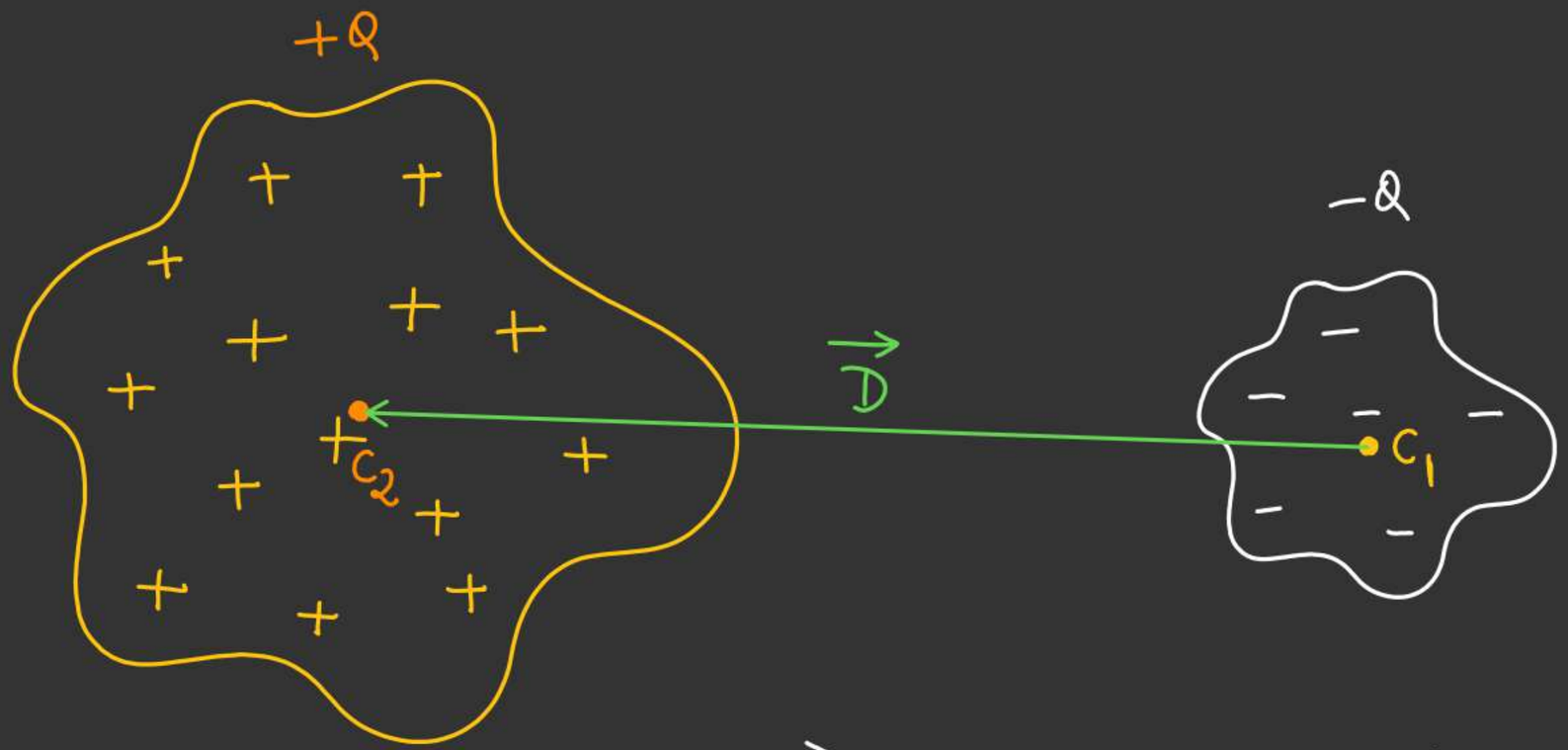
Distributed Dipoles :-



$$\vec{p} = Q \cdot \vec{C_1C_2} = Q \cdot \vec{D}$$

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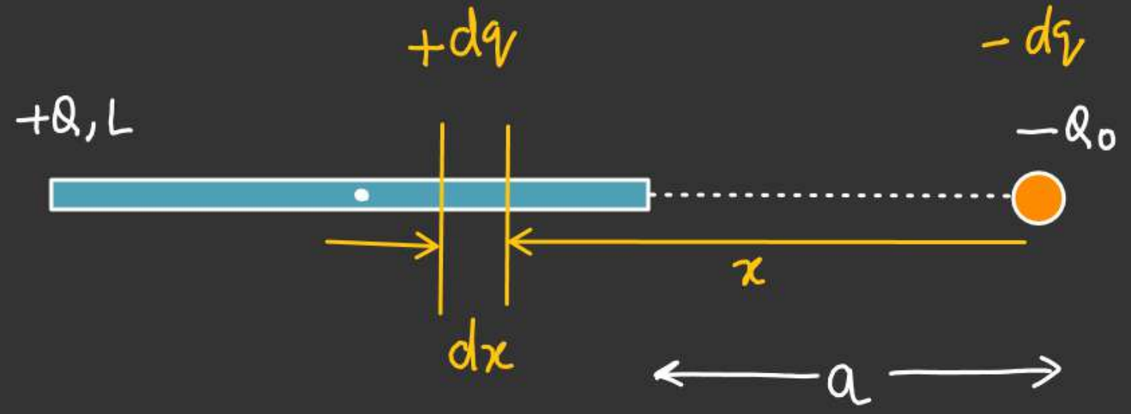
*



$M_1)$ $\quad P = Q\left(a + \frac{L}{2}\right)$ ⟵

$M_2)$ $\quad dP = dq \cdot x$

$\Rightarrow \frac{Q}{L} \cdot dx \cdot x$

$$P = \int dP = \frac{Q}{L} \int_{a}^{a+L} x\,dx = Q\left(a + \frac{L}{2}\right)$$

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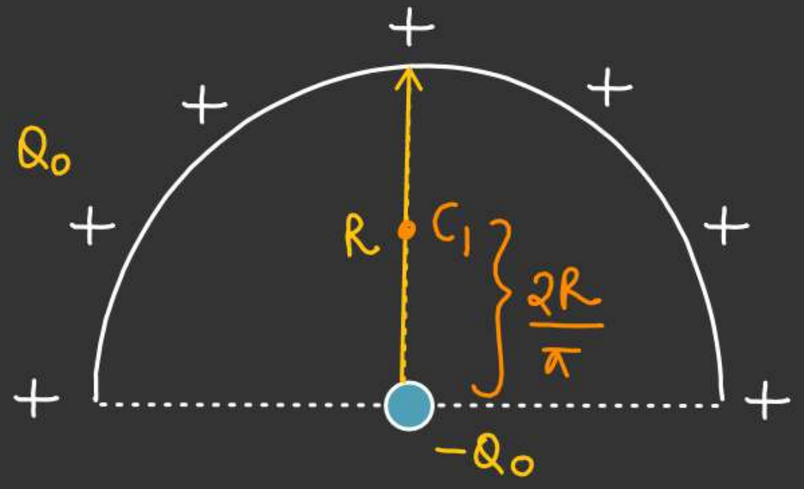
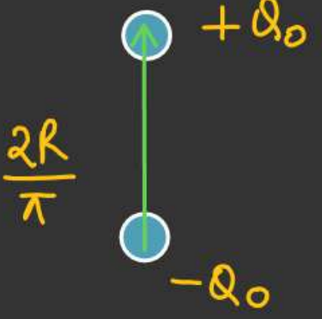
*

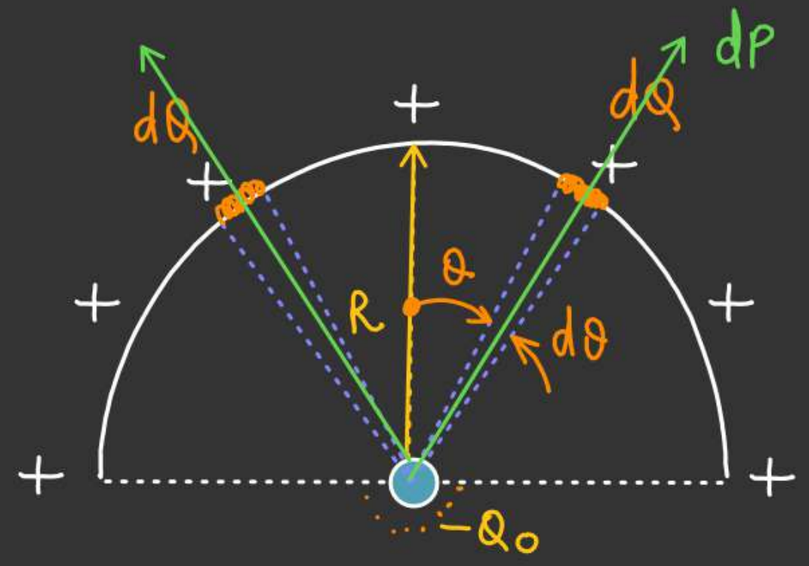
M_1)



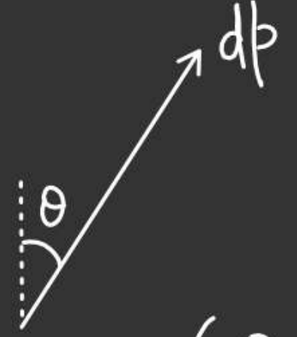


$$\vec{p} = Q_0 \cdot \frac{2R}{\pi} \uparrow$$

M_2)



$$dp = dQ \cdot R$$



$$dQ = \frac{Q_0}{\pi} \cdot d\theta$$

$$p_{net} = 2\,dp\cos\theta = 2 \cdot \left(\frac{Q_0}{\pi} \cdot d\theta\right) \cdot R \cdot \cos\theta$$

$$p_{net} = \frac{2Q_0R}{\pi} \int_0^{\pi/2} \cos\theta \cdot d\theta = Q_0\left(\frac{2R}{\pi}\right) \uparrow$$

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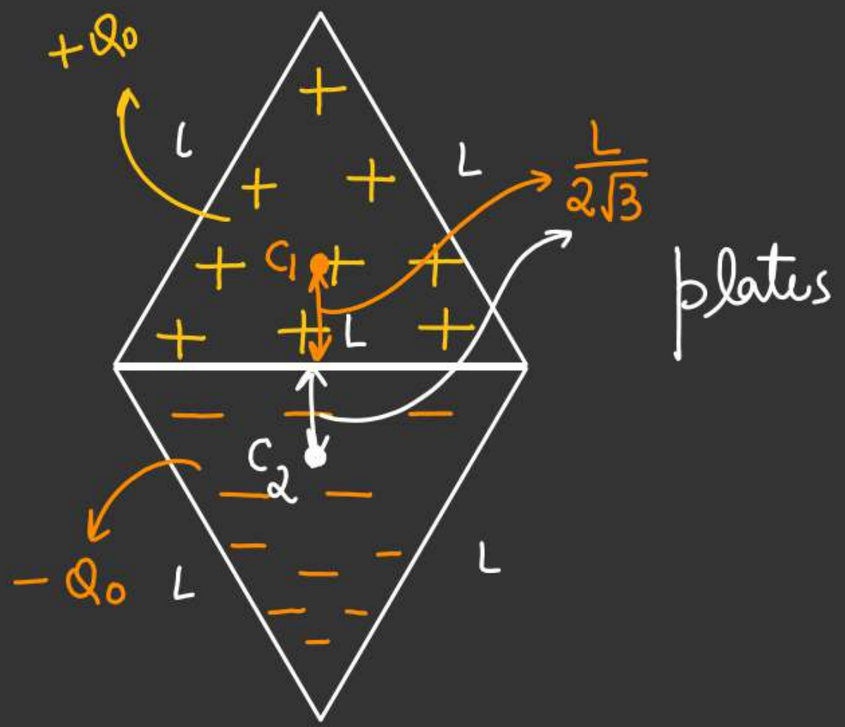
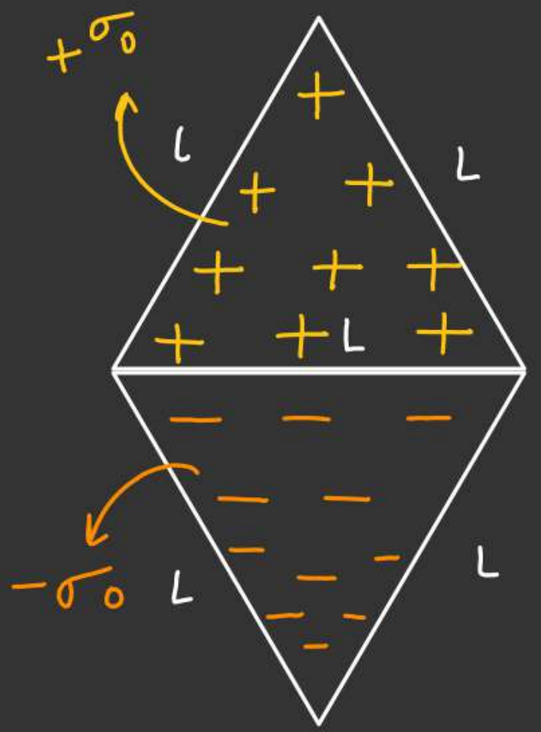


plates



$$\vec{p} = Q_0 \cdot \frac{L}{\sqrt{3}} \uparrow$$

$$\vec{p} = \sigma_0 \cdot \frac{\sqrt{3}}{4} L^2 \cdot \frac{L}{\sqrt{3}} \Rightarrow \frac{\sigma_0 L^3}{4} \uparrow$$

Nishant Jindal

* Field & Potential due to dipole :-

(Tiny dipoles)

$$\vec{E_p} = \frac{K \cdot 2\vec{P}}{r^3}$$

$E_p = ?$

$V_p = ?$

Case (i) Along the axis $\quad r >> L$

$$P = q \cdot 2L$$

$$E_p = E^+ - E^- = \frac{Kq}{(r-L)^2} - \frac{Kq}{(r+L)^2} = \frac{Kq\left[r^2 + L^2 + 2Lr - r^2 - L^2 + 2Lr\right]}{(r^2 - L^2)^2}$$

$$E_p \Rightarrow \frac{2Kq(2Lr)}{(r^2 - L^2)^2}, \quad r >> L \Rightarrow E_p = \frac{2K \cdot q \cdot 2Lr}{r^4} = \frac{2K(q \cdot 2L)}{r^3} = \frac{2KP}{r^3}$$

Nishant Jindal

* Field & Potential due to dipole :-

(Tiny dipoles)

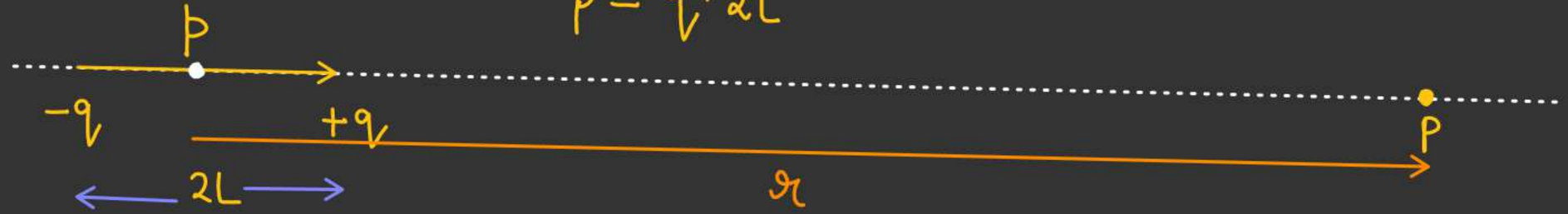
$E_p = ?$

$V_p = ?$

Case (i) Along the axis

$r >> L$

$p = q \cdot 2L$



$$V_p = V^+ + V^- = \frac{Kq}{r-L} - \frac{Kq}{r+L} = \frac{Kq}{r^2-L^2}(r+L-r+L) = \frac{K \cdot q \cdot 2L}{r^2-L^2}$$

$r >> L$

$$\boxed{V_p = \frac{Kp}{r^2}}$$

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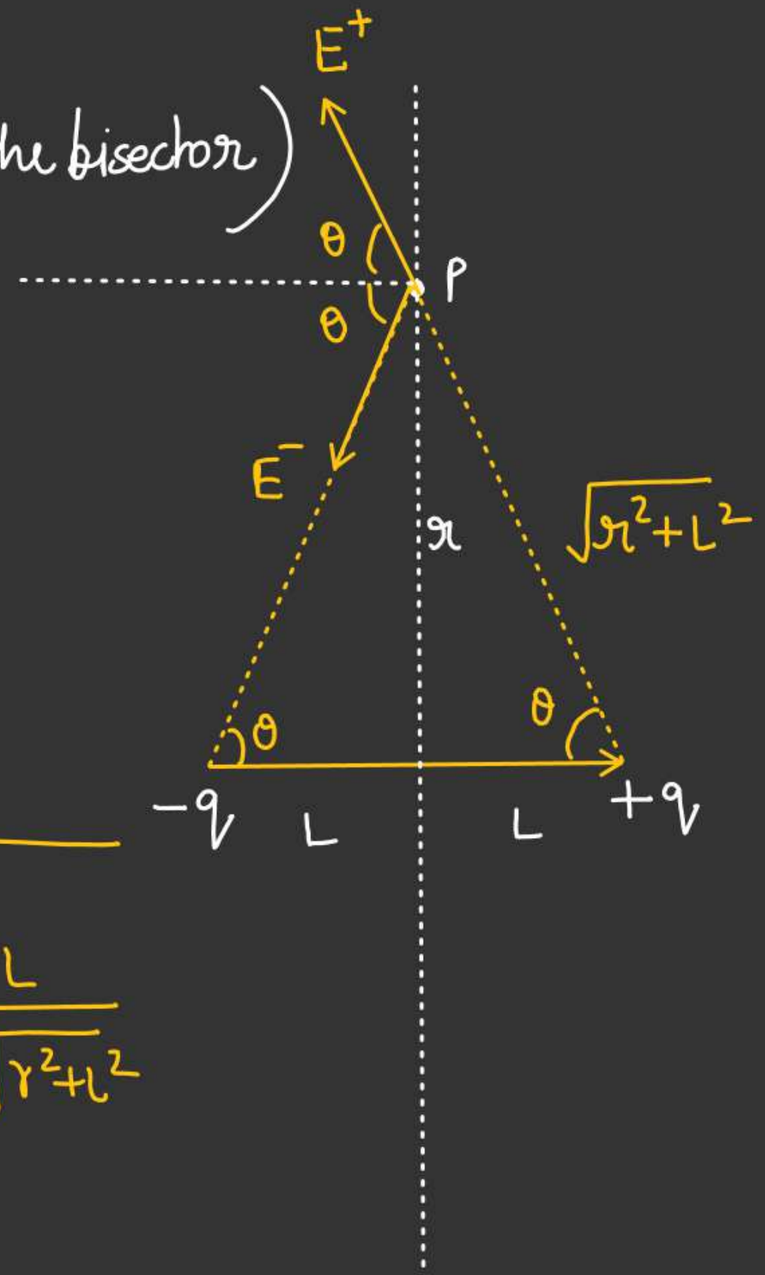
**

Case (ii) (along the bisector)



$p = q \cdot 2L$

$E_p = 2E\cos\theta$ ←

$= 2 \cdot \frac{KQ}{(r^2+L^2)} \cdot \frac{L}{\sqrt{r^2+L^2}}$

$E_p = \frac{2KqL}{(r^2+L^2)^{3/2}}$

$r >> L$

$E_p = \frac{K \cdot (2qL)}{r^3} = \frac{Kp}{r^3}$

$$\boxed{\vec{E_p} = -\frac{K\vec{p}}{r^3}}$$

$$\boxed{V_p = 0}$$

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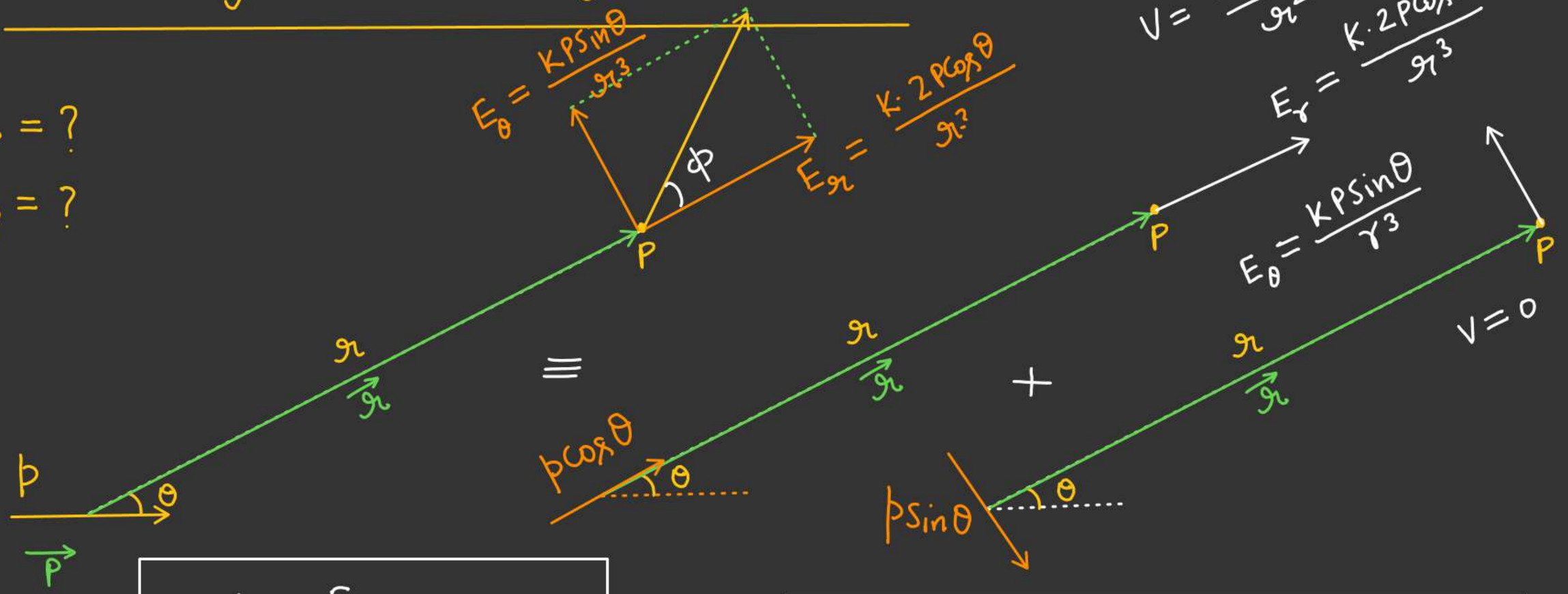
* E & V of Dipole at general point :-

$E_p = ?$

$V_p = ?$



$$\tan\phi = \frac{E_\theta}{E_r} = \frac{\tan\theta}{2}$$

$$V_p = \frac{KP\cos\theta}{r^2} + 0 = \frac{KP\cos\theta}{r^2} = \frac{K(\vec{P}\cdot\hat{r})}{r^2}$$

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Vector form of $\vec{E_p}$:-

$$E_p = \sqrt{E_r^2 + E_\theta^2} = \frac{KP}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta} = \frac{KP}{r^3}\sqrt{1+3\cos^2\theta}$$

$$E_r = \frac{K \cdot 2P\cos\theta}{r^3}$$

$$E_\theta = \frac{K \cdot P\sin\theta}{r^3}$$

$\vec{p} = \hat{i}+\hat{j}+\hat{k}$

$\vec{r} = -2\hat{i}-7\hat{j}-\hat{k}$

P (−1, −5, +2)

$\vec{p}$

(1, 2, 3)

$\vec{E_p} =$

$$V_p = \frac{K(-2-7-1)}{(4+49+1)^{3/2}}$$

$\vec{p}\cdot\hat{r} = P\cos\theta$

$\vec{p}\cdot\vec{r} = Pr\cos\theta$

$$V_p = \frac{KP\cos\theta}{r^2} = \frac{K(\vec{p}\cdot\vec{r})}{r^3}$$

$$\Rightarrow \frac{K(\vec{p}\cdot\hat{r})}{r^2}$$

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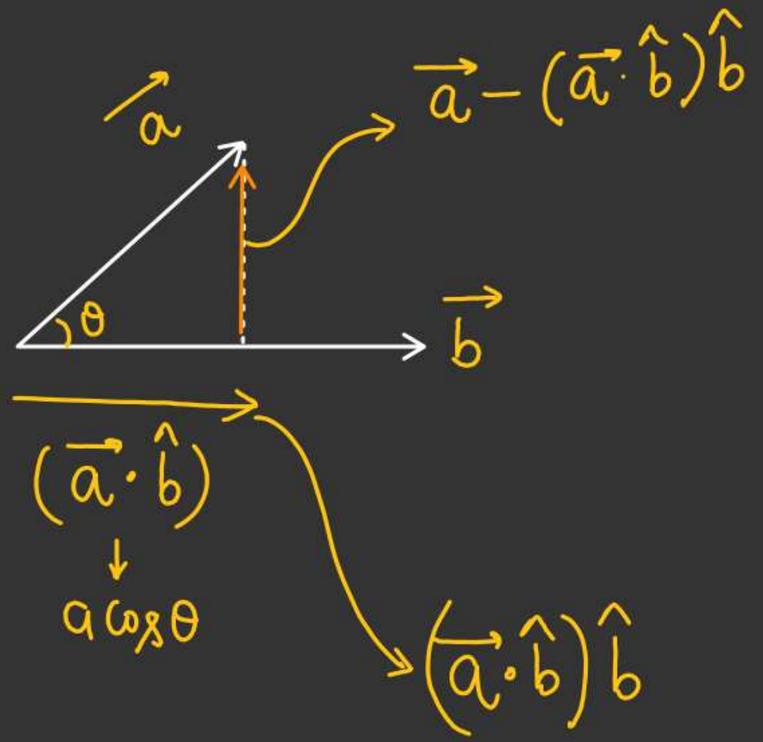
$$\vec{E}_P = \vec{E}_r + \vec{E}_\theta$$

$$\Rightarrow \frac{K2P\cos\theta}{r^3}\cdot\hat{r} + \frac{K\cdot P\sin\theta}{r^3}\cdot\hat{\theta}$$

$$\vec{P}\cdot\hat{r} = P\cos\theta$$



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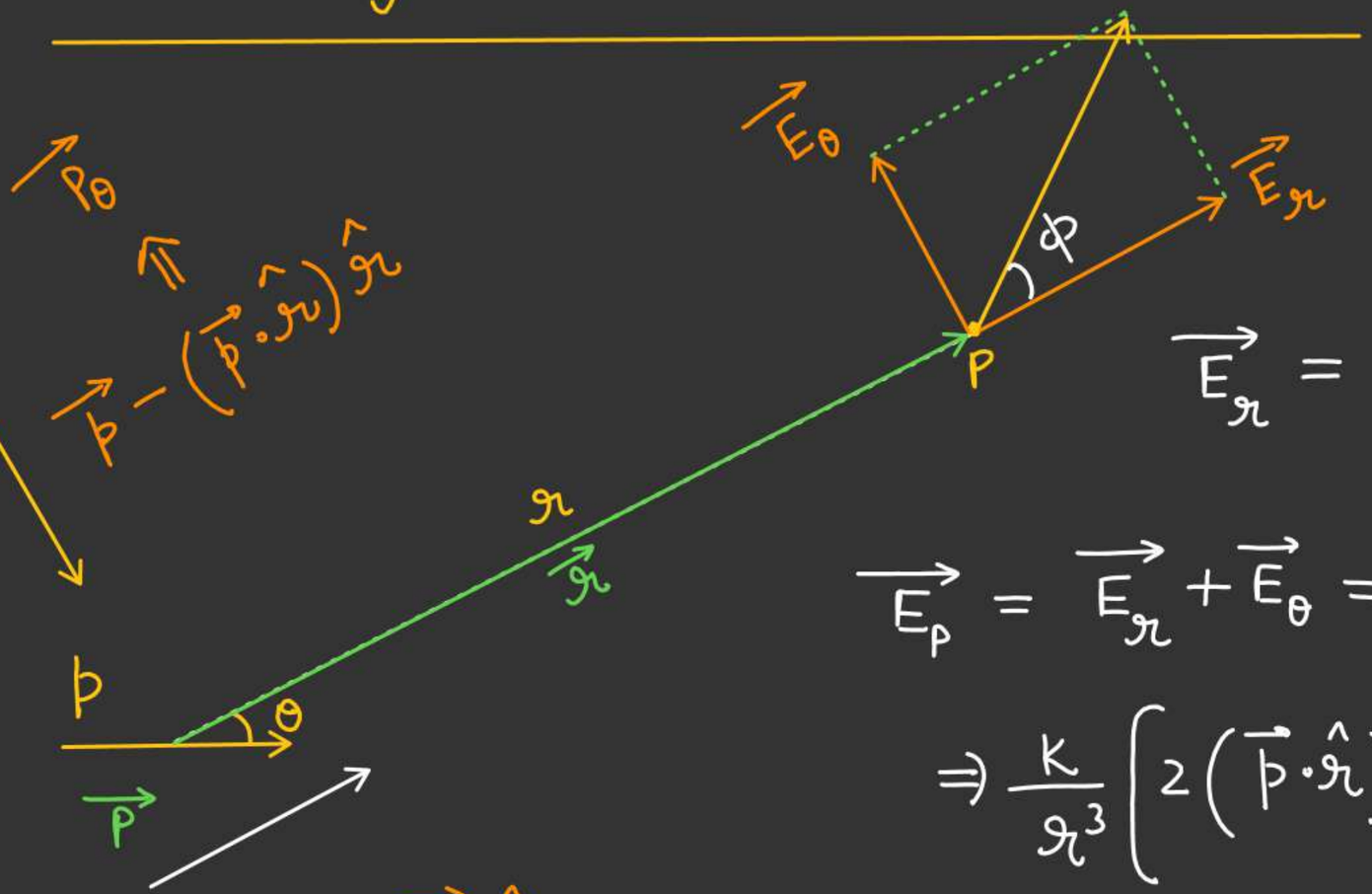
* E & V of Dipole at general point :-



$$\vec{E}_r = \frac{K \cdot 2\vec{p}_r}{r^3} \quad , \quad \vec{E}_\theta = -\frac{K \cdot \vec{p}_\theta}{r^3}$$

$$\vec{E}_p = \vec{E}_r + \vec{E}_\theta = \frac{K}{r^3}\left[2\vec{p}_r - \vec{p}_\theta \right]$$

$$\Rightarrow \frac{K}{r^3}\left[2(\vec{p}\cdot\hat{r})\hat{r} - \left[\vec{p} - (\vec{p}\cdot\hat{r})\hat{r}\right] \right] \quad \text{☆}$$

$$\boxed{\vec{E}_p = \frac{K}{r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p} \right]}$$

$$p\cos\theta = (\vec{p}\cdot\hat{r})$$

$$\vec{p}_r = (\vec{p}\cdot\hat{r})\hat{r}$$

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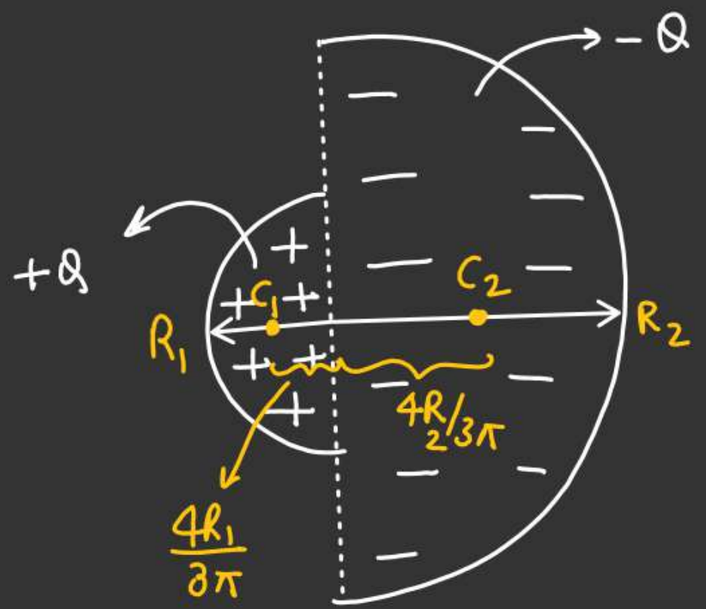
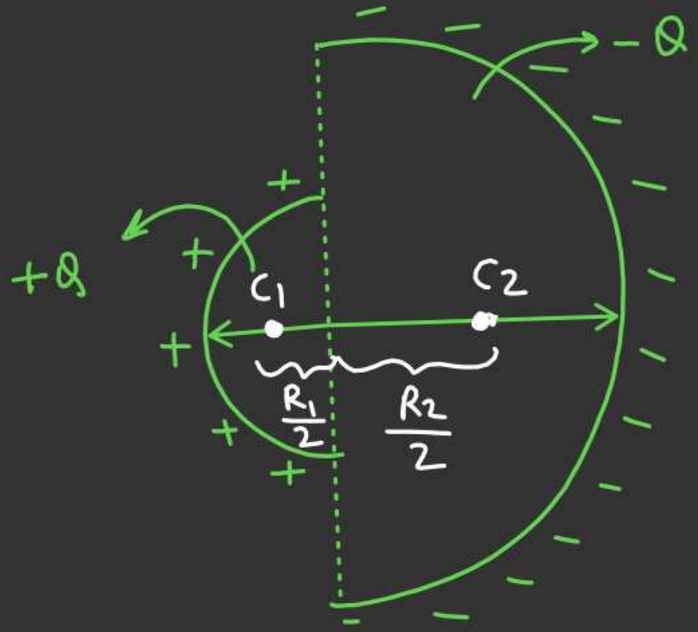
*

Disc



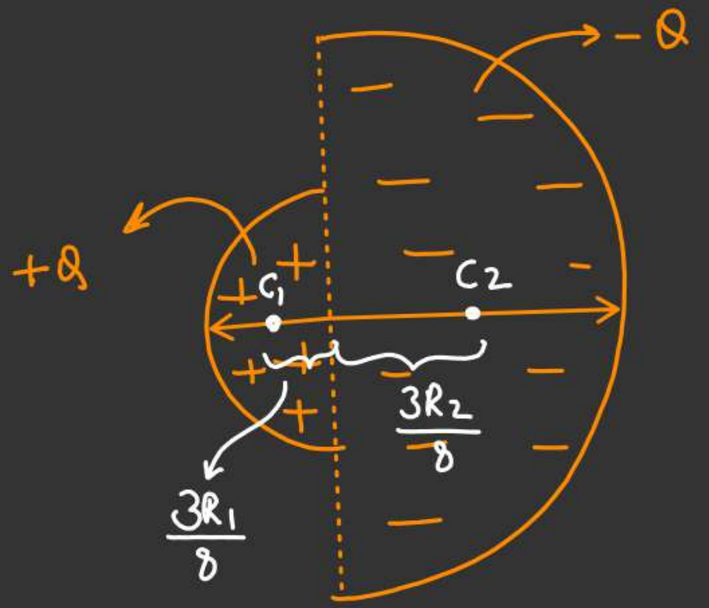
$$Q \cdot \frac{4}{3\pi}(R_1+R_2)$$

Shell



$$p = \frac{Q \cdot (R_1+R_2)}{2}$$

Solid sphere



$$p = Q \cdot \frac{3}{8}(R_1+R_2)$$

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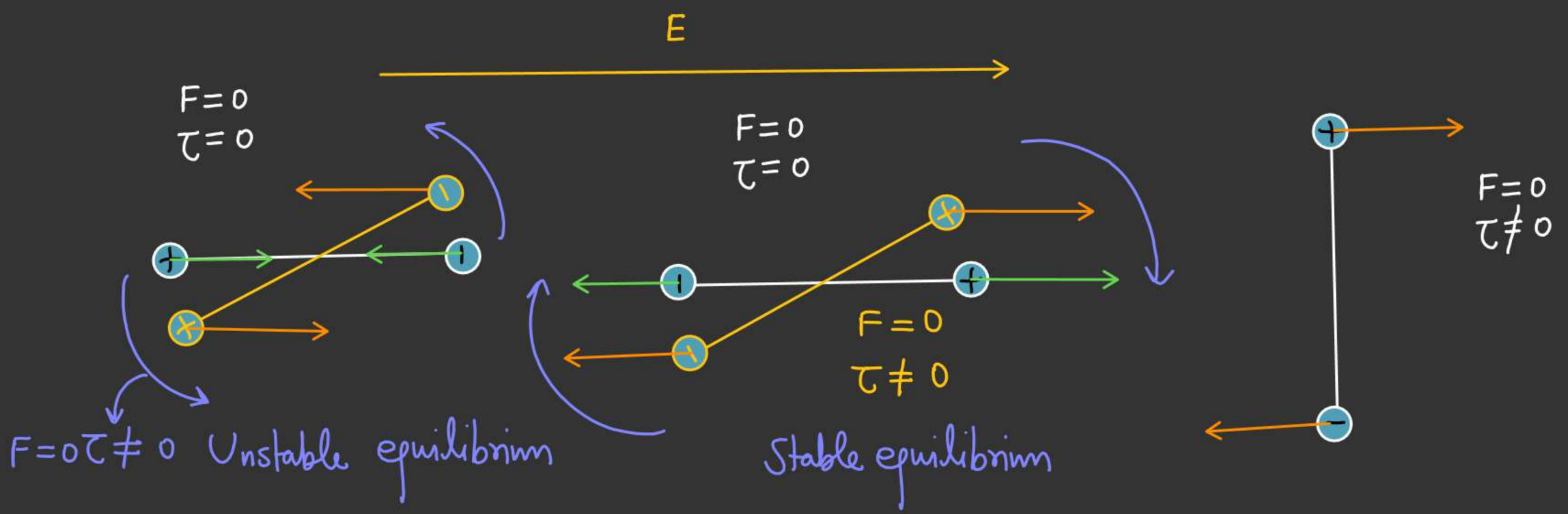
* Dipole in electric field :-



* for a dipole in Uniform E net force will always be zero.

⇒ Dipole will be in translational equilibrium.

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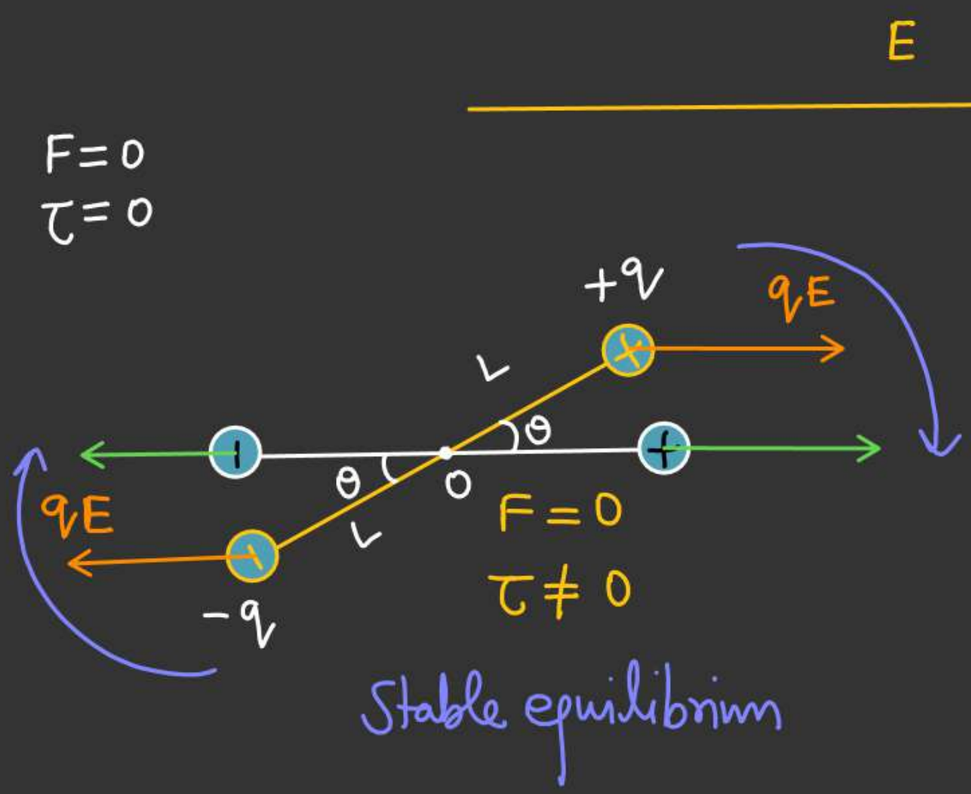
* Dipole in electric field :-



$$\vec{\tau}_0 = qE(L\sin\theta)\cdot 2 \otimes = pE\sin\otimes = \vec{p}\times\vec{E}$$

$$\vec{\tau}_q = qE(2L\sin\theta)\otimes = \vec{p}\times\vec{E}$$

$$\vec{\tau}_{-q} = \vec{p}\times\vec{E}$$

$$p = q\cdot 2L$$

★ $$\vec{\tau}_E = \vec{p}\times\vec{E}$$

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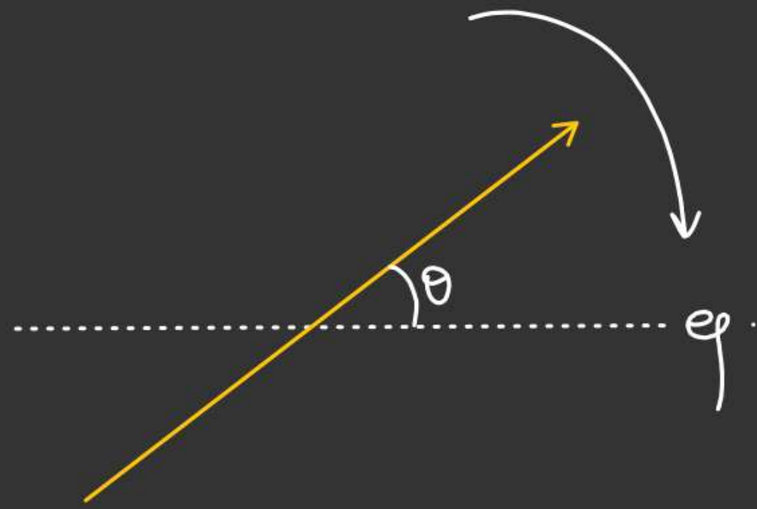
* SHM for Small Osc :-



for small osc.

$\sin\theta \approx \theta$

SHM

$$\tau_{res} = -\left[pE\sin\theta \right]$$

$$I_{axis} \cdot \frac{d^2\theta}{dt^2} = -pE\cdot\theta$$

$$\frac{d^2\theta}{dt^2} + \left(\frac{pE}{I_{axis}}\right)\cdot\theta = 0$$

$$\downarrow \omega^2$$

$$T_{osc} = 2\pi\sqrt{\frac{I_{axis}}{p\cdot E}}$$

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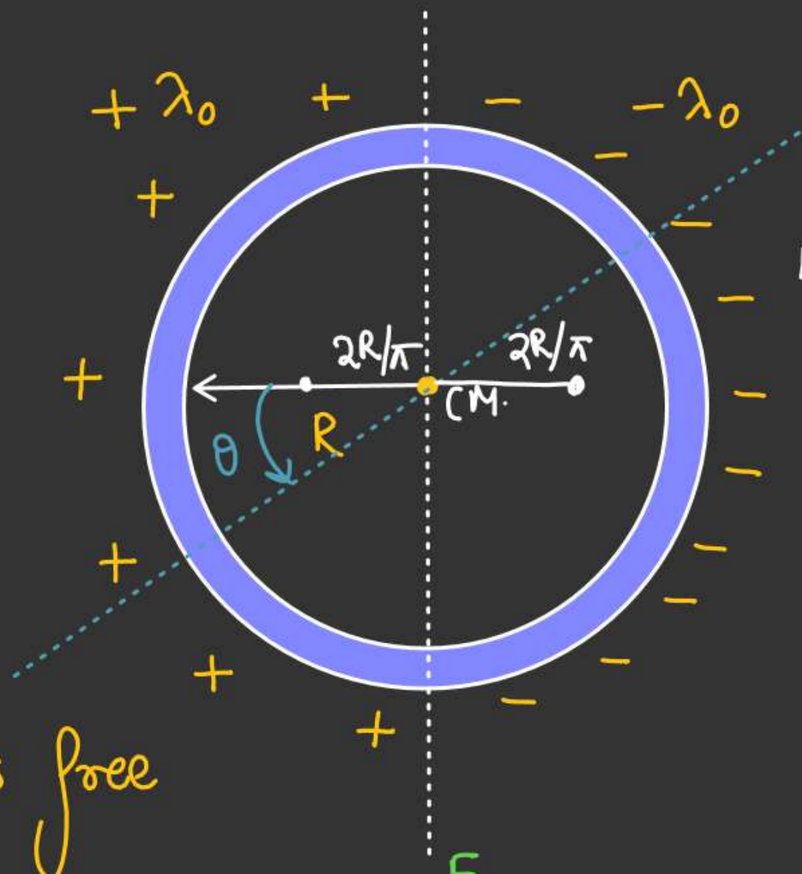

$T_{osc} = ?$

$$p = Q \cdot \frac{4R}{\pi} = \lambda_0 \cdot \pi R \cdot \frac{4R}{\pi} = 4\lambda_0 R^2$$

$$T_{osc} = 2\pi \sqrt{\frac{I_{axis}}{PE}}$$

$$= 2\pi \sqrt{\frac{MR^2}{4\lambda_0 R^2 \cdot E_0}} = 2\pi \sqrt{\frac{M}{4\lambda_0 E_0}}$$

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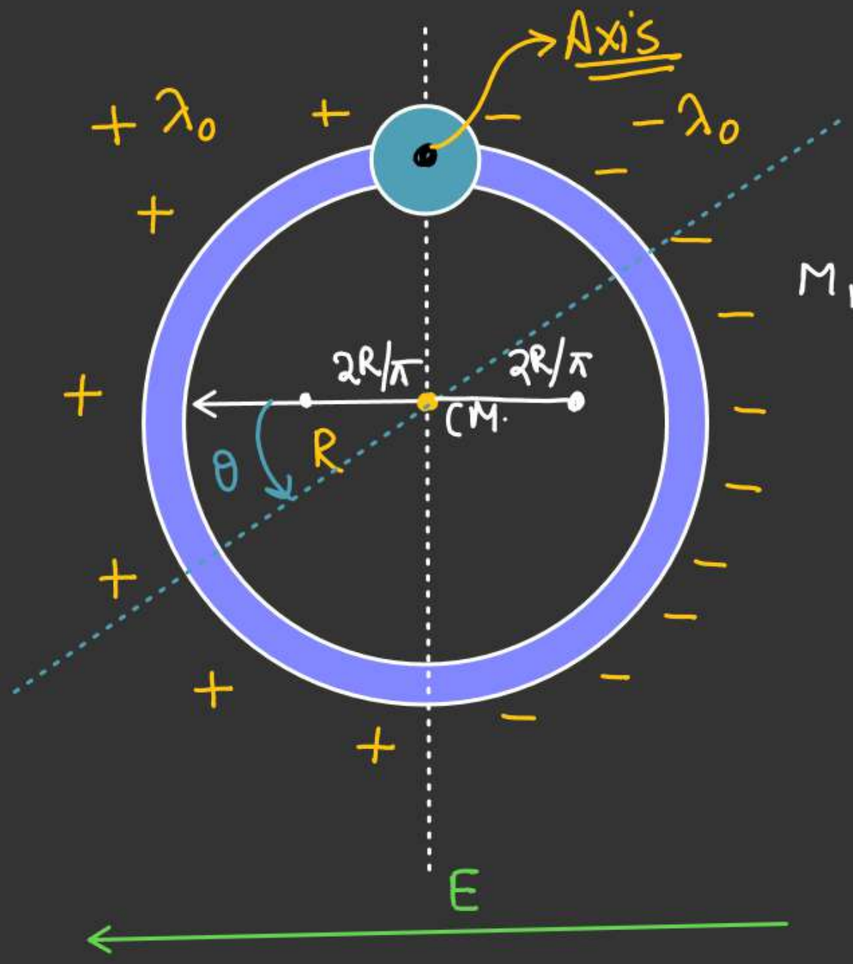

θ → Small angle

$T_{osc} = ?$

$$p = Q \cdot \frac{4R}{\pi} = \lambda_0 \cdot \pi R \cdot \frac{4R}{\pi} = 4\lambda_0 R^2$$

$$I_{axis} = I_{cm} + MR^2$$

$$\Rightarrow MR^2 + MR^2$$

$$\Rightarrow 2MR^2$$

$$T_{osc} = 2\pi \sqrt{\frac{I_{axis}}{PE}}$$

$$= 2\pi \sqrt{\frac{2MR^2}{4\lambda_0 R^2 \cdot E_0}} = 2\pi \sqrt{\frac{M}{2\lambda_0 E_0}}$$

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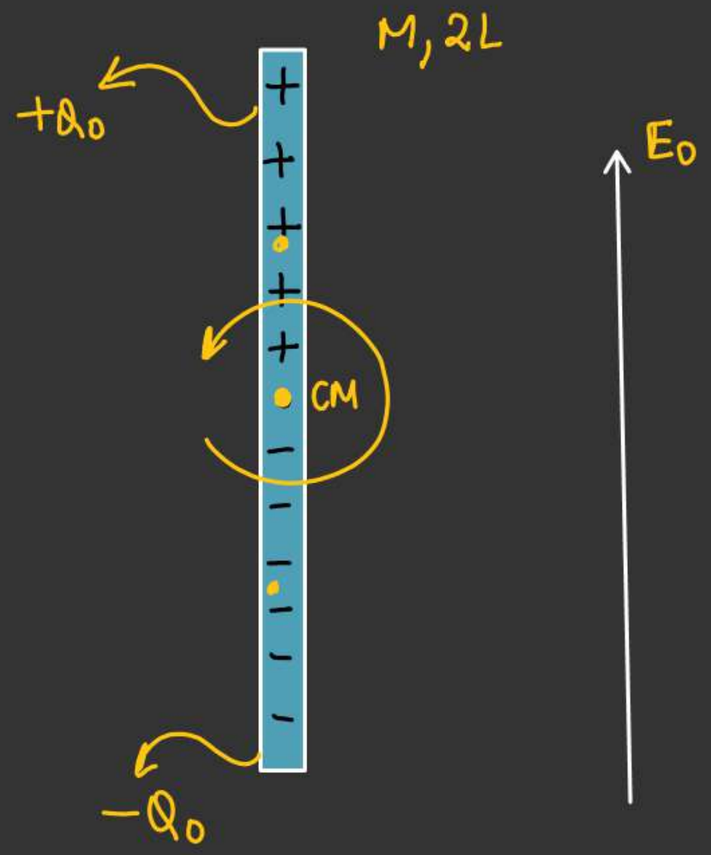
* free to rotate



$$T_{osc} = 2\pi\sqrt{\frac{I_{axis}}{pE}} = 2\pi\sqrt{\frac{ML^2}{3 \cdot Q_0 L E}} = 2\pi\sqrt{\frac{ML}{3Q_0E}}$$

$$p = Q_0 \cdot L \qquad I_{axis} = \frac{M(2L)^2}{12} = \frac{ML^2}{3}$$

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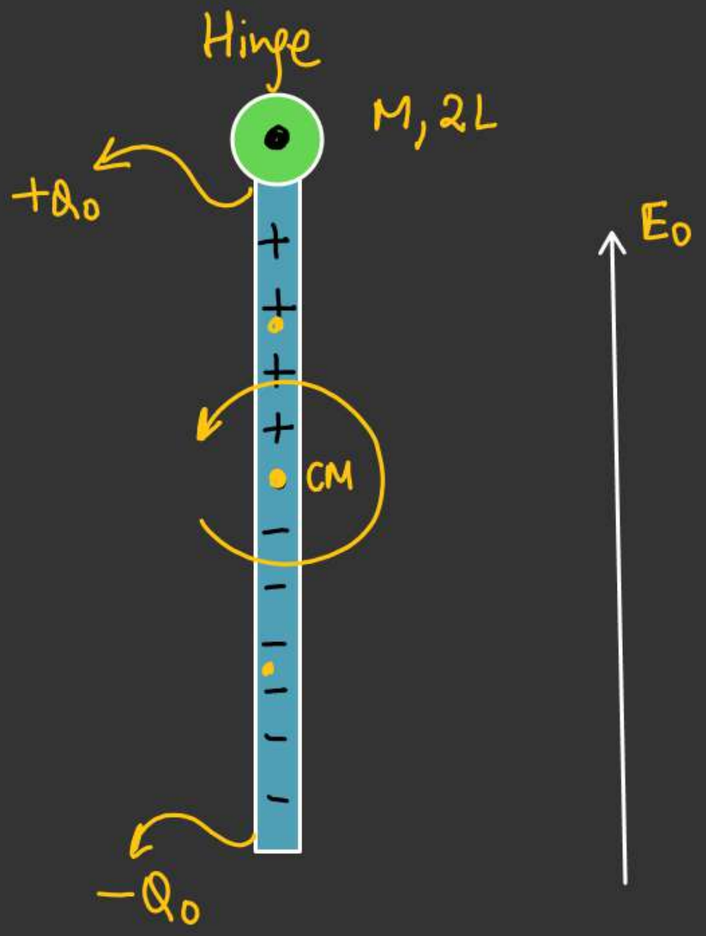

$$T_{osc} = 2\pi\sqrt{\frac{I_{axis}}{pE}} = 2\pi\sqrt{\frac{4ML^2}{3 \cdot Q_0 L E}} = 2\pi\sqrt{\frac{4ML}{3Q_0E}}$$

$$p = Q_0 \cdot L \qquad I_{axis} = \frac{M(2L)^2}{3} = 4\frac{ML^2}{3}$$

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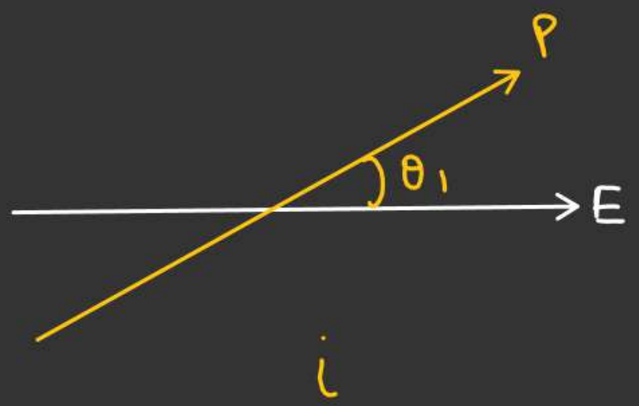
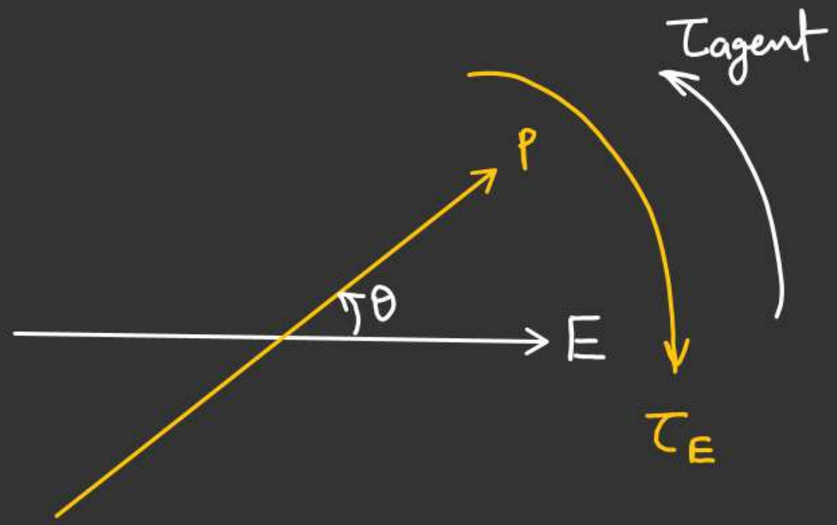
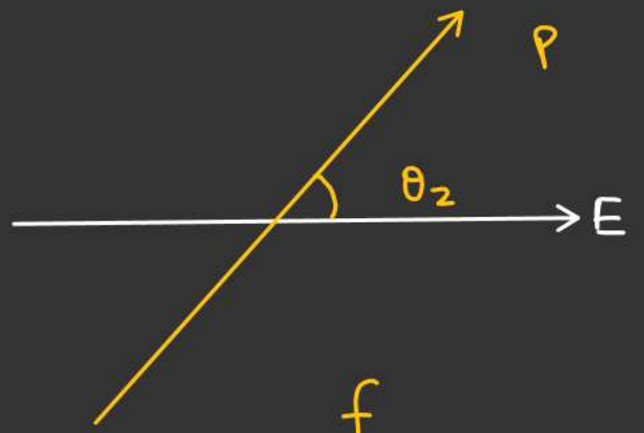
* Potential Energy of dipole in electric field :-



i





f

$$\vec{\tau}_E = \vec{p} \times \vec{E}$$

$$= pE \sin\theta$$

$$\vec{\tau}_{agent} = pE \sin\theta$$

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$$dW_E = -dU \qquad W_e = -\Delta U = U_i - U_f$$

$$dW_{agent} = dU \qquad W_{agent} = U_f - U_i \quad \text{(Very slowly)}$$

$\theta \uparrow$

$$dW_{agent} = \vec{\tau}_{agent} \cdot \vec{d\theta} = \tau_{agent} \cdot d\theta = pE\sin\theta \cdot d\theta$$

$$W_{agent} = U_f - U_i = pE \int_{\theta_1}^{\theta_2} \sin\theta \cdot d\theta = U_{\theta_2} - U_{\theta_1}$$

$$U_{\theta_2} - U_{\theta_1} = pE(\cos\theta_1 - \cos\theta_2)$$

$$\boxed{U_\theta = -pE\cos\theta = -\vec{p}\cdot\vec{E}}$$

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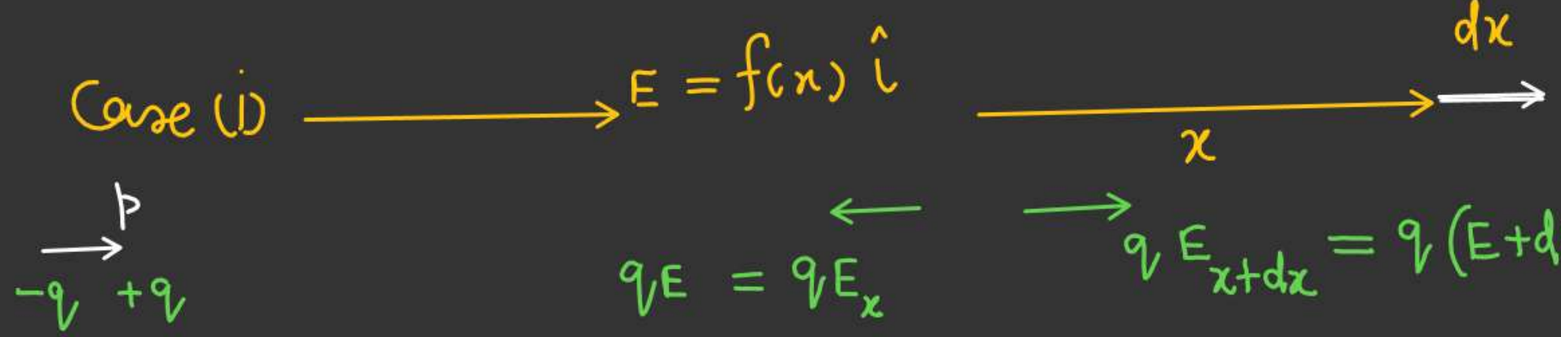
* Force on dipole in electric field :-

* Uniform E $\quad F = 0$

* Non Uniform E

Case (i) $\longrightarrow E = f(x)\,\hat{i}$



$p = q \cdot dx$

$F = q(E+dE) - qE = q \cdot dE$

$$F = \underbrace{q \cdot dx}_{} \cdot \frac{dE}{dx} = p \cdot \frac{dE}{dx}$$

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*

$$U = -\vec{p} \cdot \vec{E} = -pE$$

$$F = \left| -\frac{dU}{dx} \right| = p\frac{dE}{dx}$$

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$$* \quad \vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

$$\vec{P} = P_1 \hat{i} + P_2 \hat{j} + P_3 \hat{k}$$

$$U = -\vec{P} \cdot \vec{E} = -P_1 E_x - P_2 E_y - P_3 E_z$$

$$\left(\begin{array}{l} F_x = -\dfrac{\partial U}{\partial x} \\ \\ F_y = -\dfrac{\partial U}{\partial y} \\ \\ F_z = -\dfrac{\partial U}{\partial z} \end{array} \right)$$

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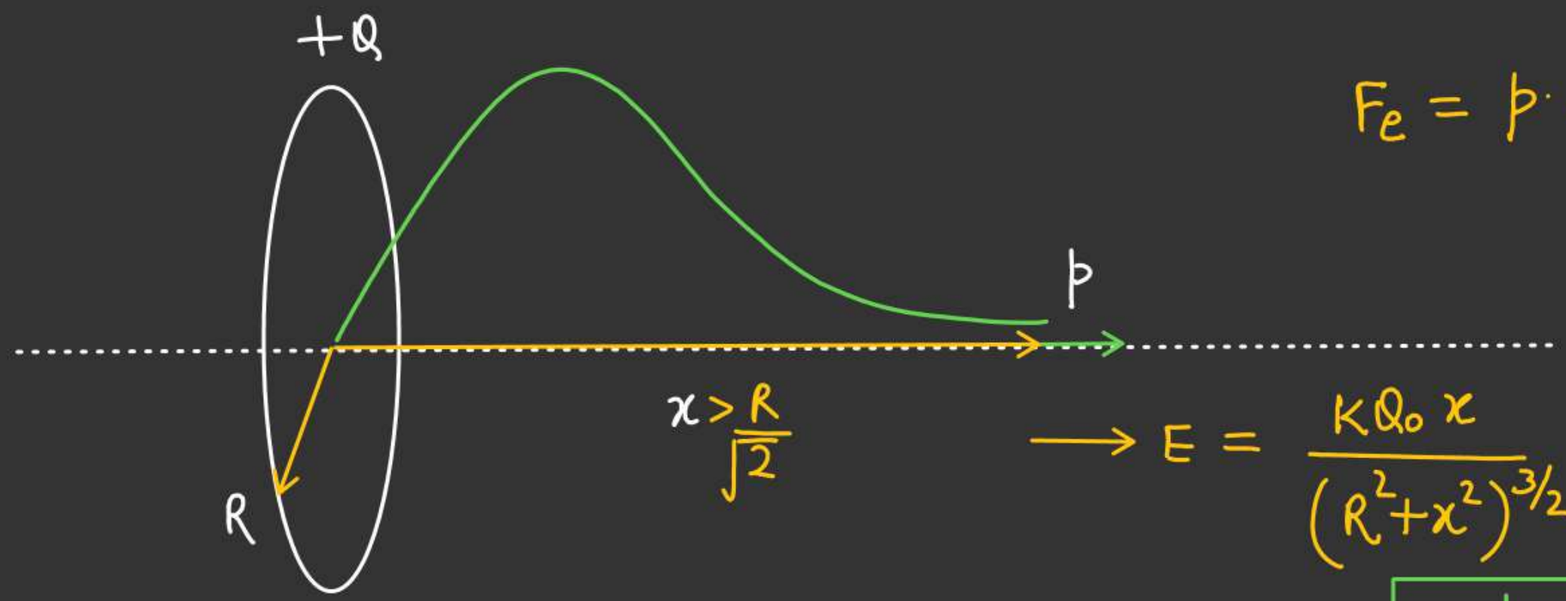

$$F_e = p \cdot \frac{dE}{dx}$$

$$\longrightarrow E = \frac{KQ_0 x}{(R^2+x^2)^{3/2}}$$

$E \downarrow$ $\quad F_e \longrightarrow$ attractive

$x > \frac{R}{\sqrt{2}}$

$qE_x > qE_{x+dx}$

Force on Ring =

Force on dipole =

$x < \frac{R}{\sqrt{2}}$ $\quad E \uparrow$ $\quad F_e \rightarrow$ repulsive

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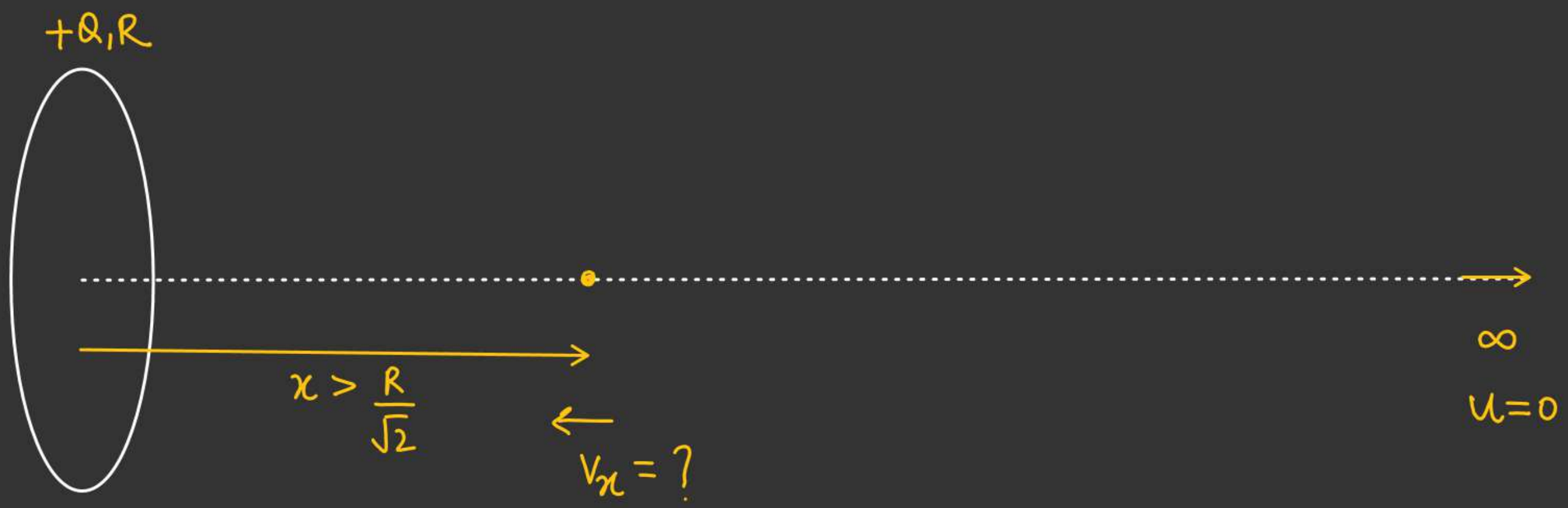

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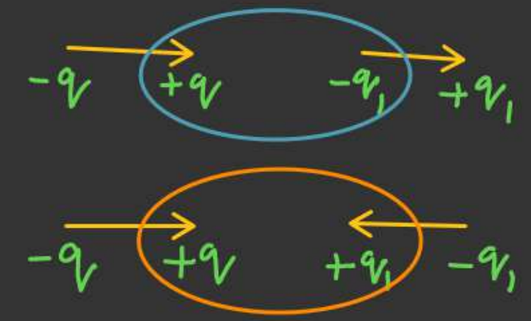
* Force b/w the dipoles :-

Tiny dipoles $F \neq 0$ $\tau = 0$



Attraction

repulsion

**

$$\longrightarrow E_r = \frac{K \cdot 2P_1}{r^3}$$

$$U = -\vec{P_2} \cdot \vec{E_r} = -P_2 E_r$$

$$F = -\frac{dU}{dr} = P_2 \cdot \frac{dE_r}{dr} = 2KP_1P_2 \frac{d}{dr}\left(\frac{1}{r^3}\right) = -\frac{6KP_1P_2}{r^4}$$ (Attractive)

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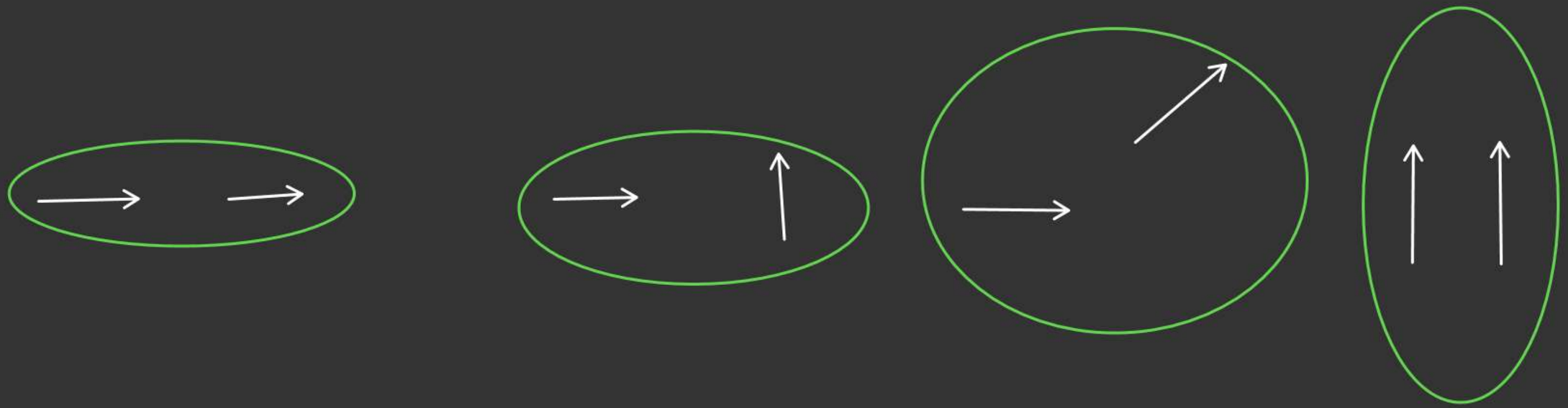
H.W.

Dipole - Dipole Interaction at INSP YT



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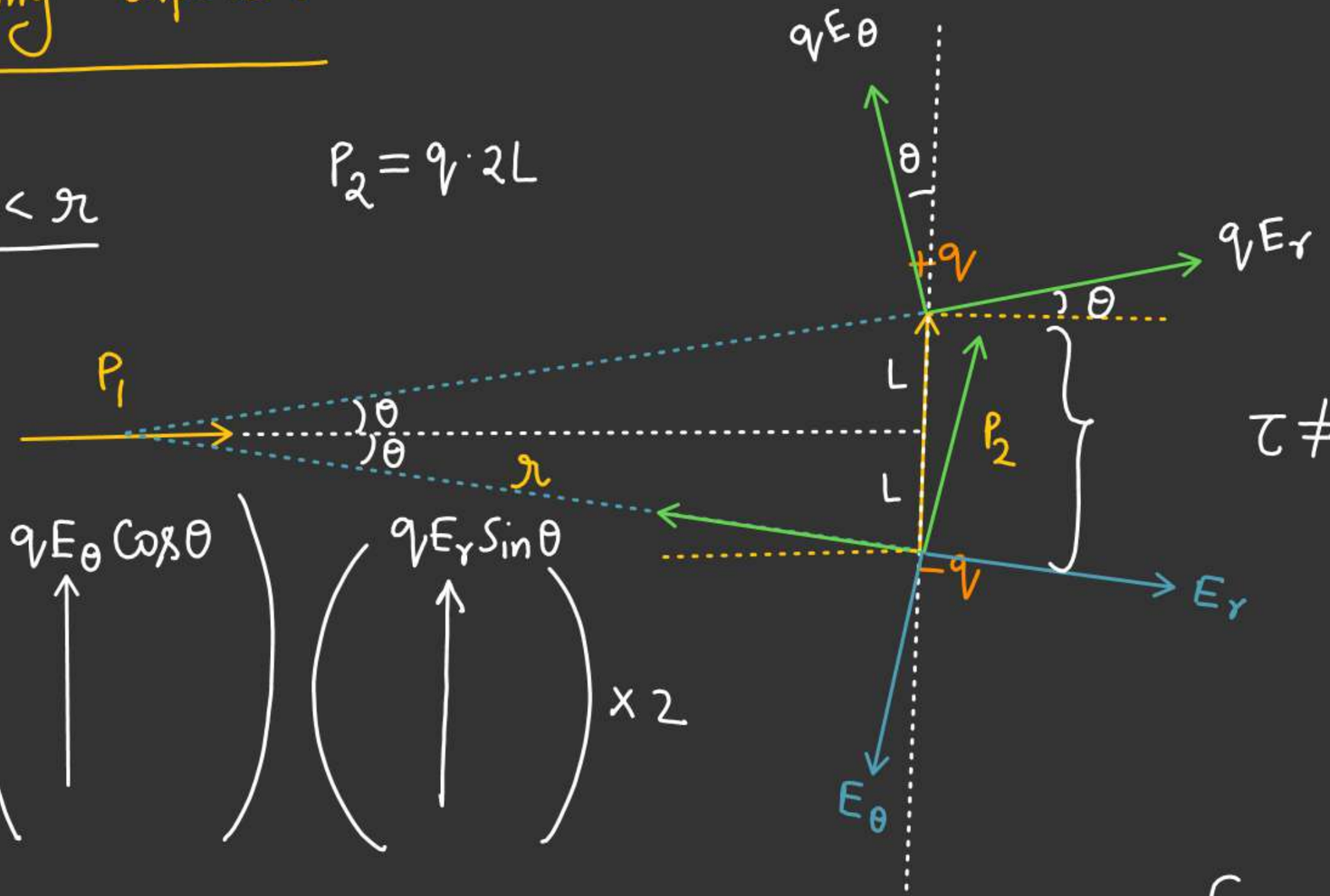
Tiny dipoles :-

$L << r$

$P_2 = q \cdot 2L$



$$2\left(qE_\theta \cos\theta\right)\left(qE_r \sin\theta\right) \times 2$$

$$E_r = \frac{K \cdot 2P_1 \cos\theta}{r^3}$$

$$E_\theta = \frac{K \cdot P_1 \sin\theta}{r^3}$$

$\tau \neq 0$ $F \neq 0$

$\downarrow$

$F_x = 0$

$F_y \neq 0$

$$F_y = 2q\left[E_\theta \cdot \cos\theta + E_r \sin\theta\right]$$

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Tiny dipoles :-

$$P_2 = q \cdot 2L$$

$$E_r = \frac{K \cdot 2P_1 \cos\theta}{r^3}$$

$$E_\theta = \frac{K \cdot P_1 \sin\theta}{r^3}$$

$$F_y = 2q\left[E_\theta \cdot \cos\theta + E_r \sin\theta\right]$$

$$= 2q \cdot \left[\frac{KP_1 \sin\theta \cdot \cos\theta}{r^3} + \frac{K \cdot 2P_1 \cos\theta \cdot \sin\theta}{r^3}\right]$$

$$= 2 \cdot q \cdot \frac{KP_1}{r^3} \cdot 3 \cdot \sin\theta \cdot \cos\theta$$

$$\Rightarrow 2q \cdot \frac{KP_1 \cdot 3}{r^3} \cdot \frac{L}{r} \cdot \Rightarrow \frac{3KP_1P_2}{r^4}$$

$$\sin\theta \approx \frac{L}{r}$$

$$\cos\theta \approx 1$$

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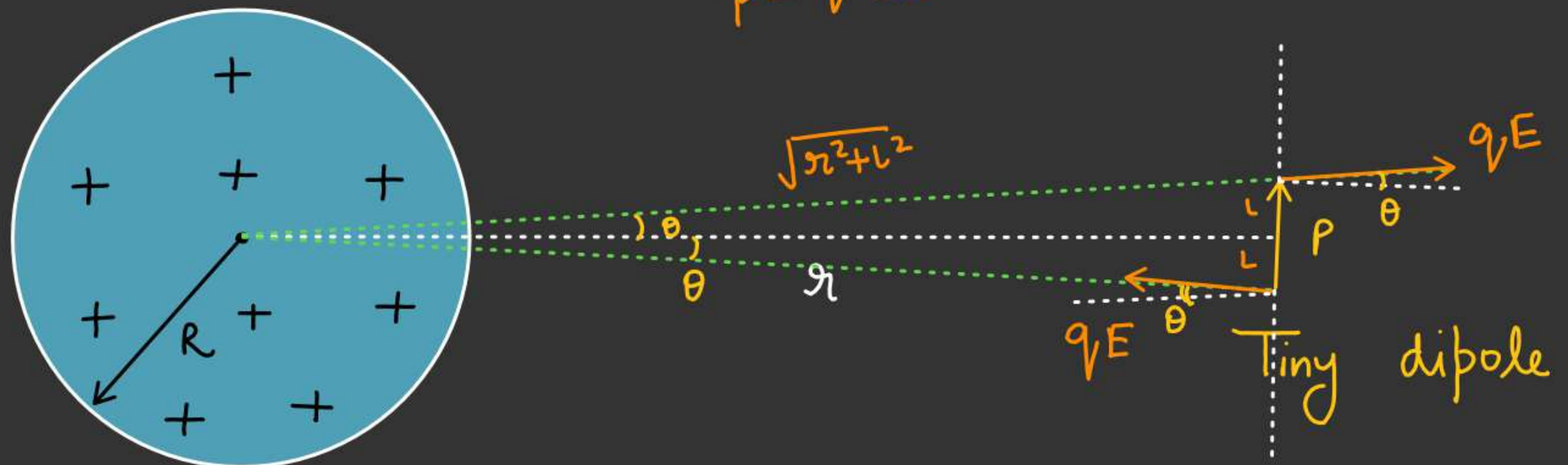
* Force b/w sphere and dipole

$p = q \cdot 2L$

M1)



$\tau \neq 0 \quad F \neq 0$

$F_x = 0$

$F_y \neq 0$

$$\text{Force} = 2qE\sin\theta$$

$$= 2q\left(\frac{KQ}{r^2+L^2}\right)\cdot\frac{L}{(r^2+L^2)^{1/2}} \Rightarrow \frac{KQp}{(r^2+L^2)^{3/2}}$$

** $r >> L$

$$F = \frac{KQp}{r^3}$$

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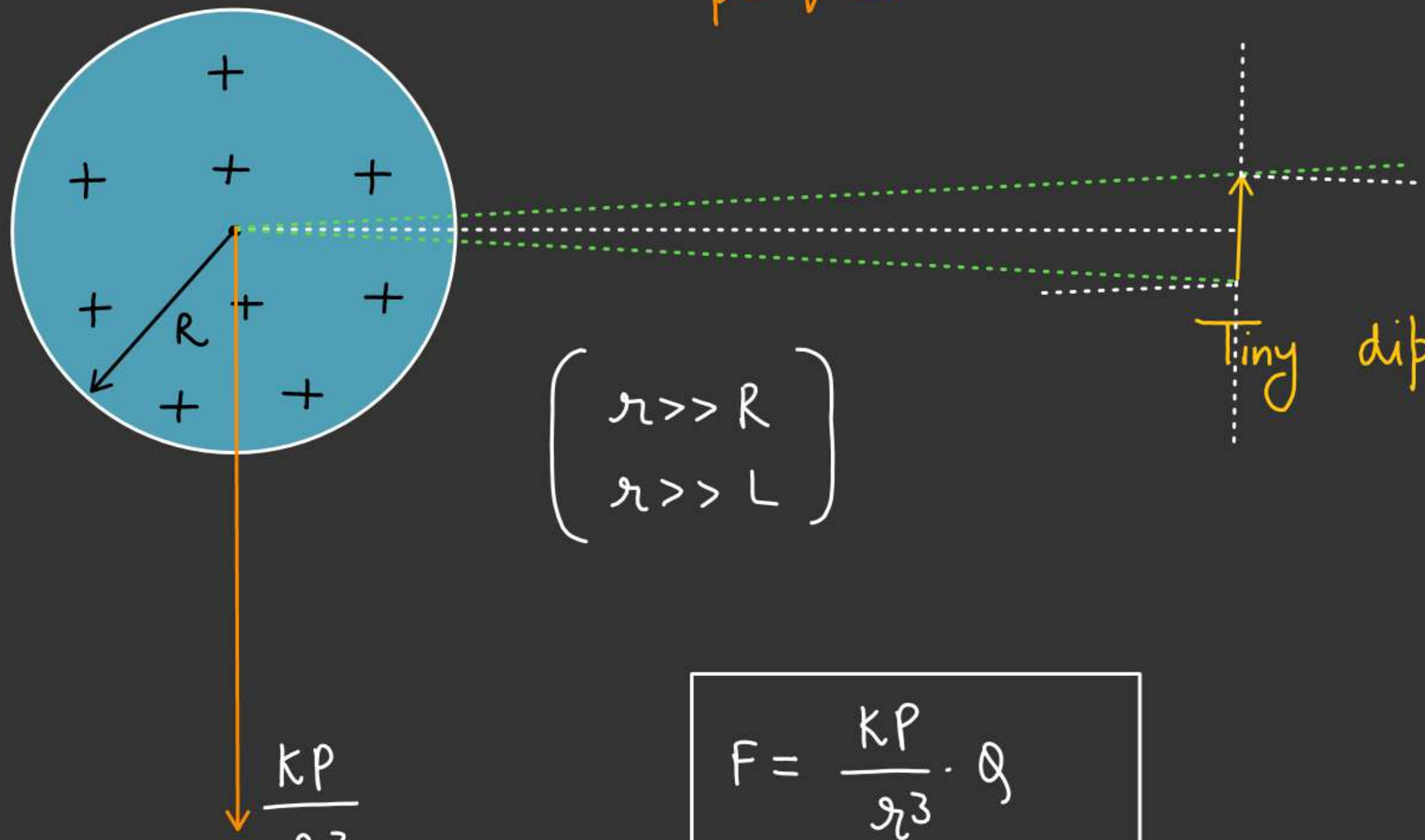
* Force b/w sphere and dipole

$p = q \cdot 2L$

M2)



$\tau \neq 0 \quad F \neq 0$

$F_x = 0$

$F_y \neq 0$

$$\left(\begin{array}{l} r >> R \\ r >> L \end{array} \right)$$

$\frac{KP}{r^3}$

$$F = \frac{KP}{r^3} \cdot Q$$

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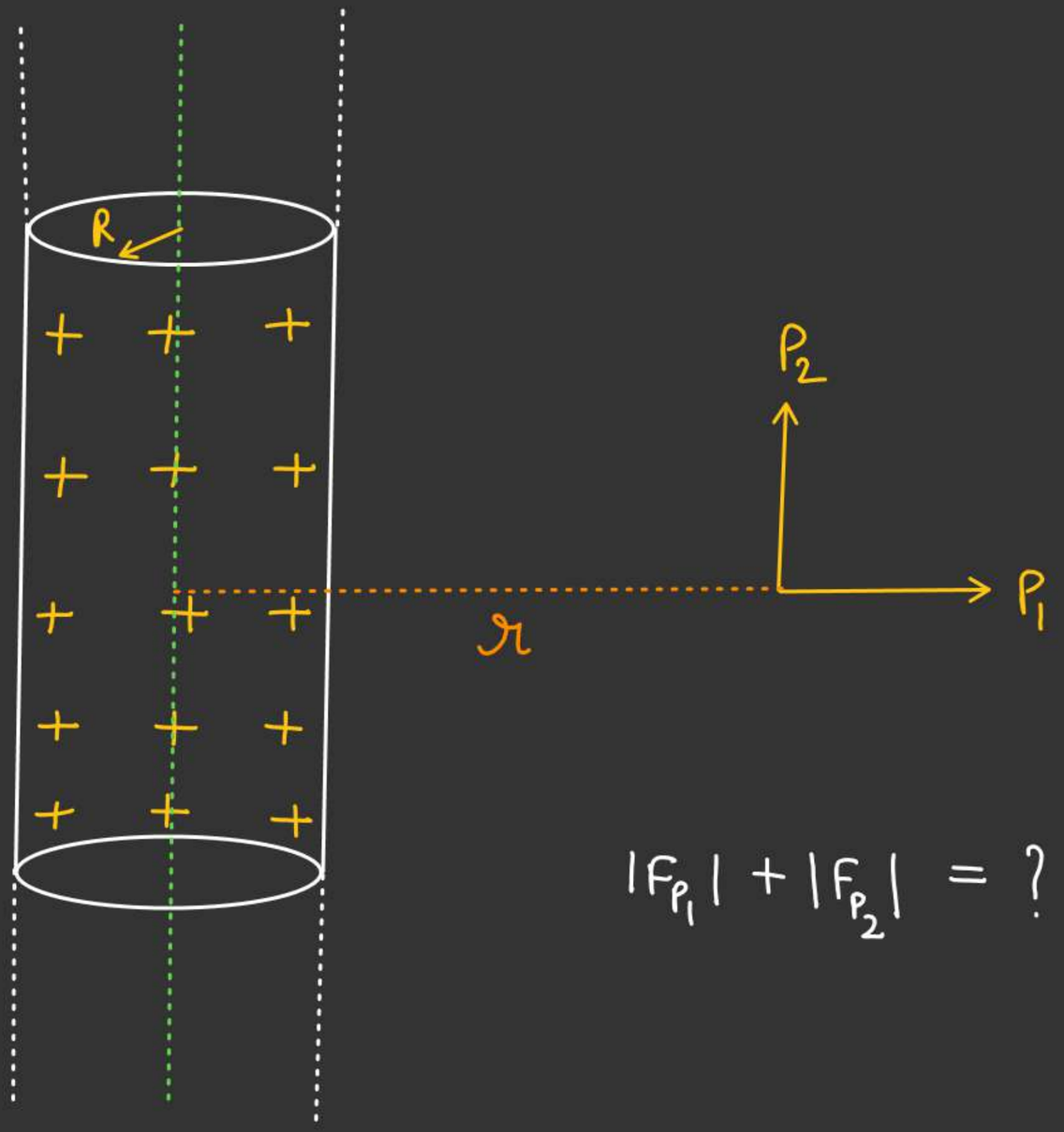
R
+
+
+

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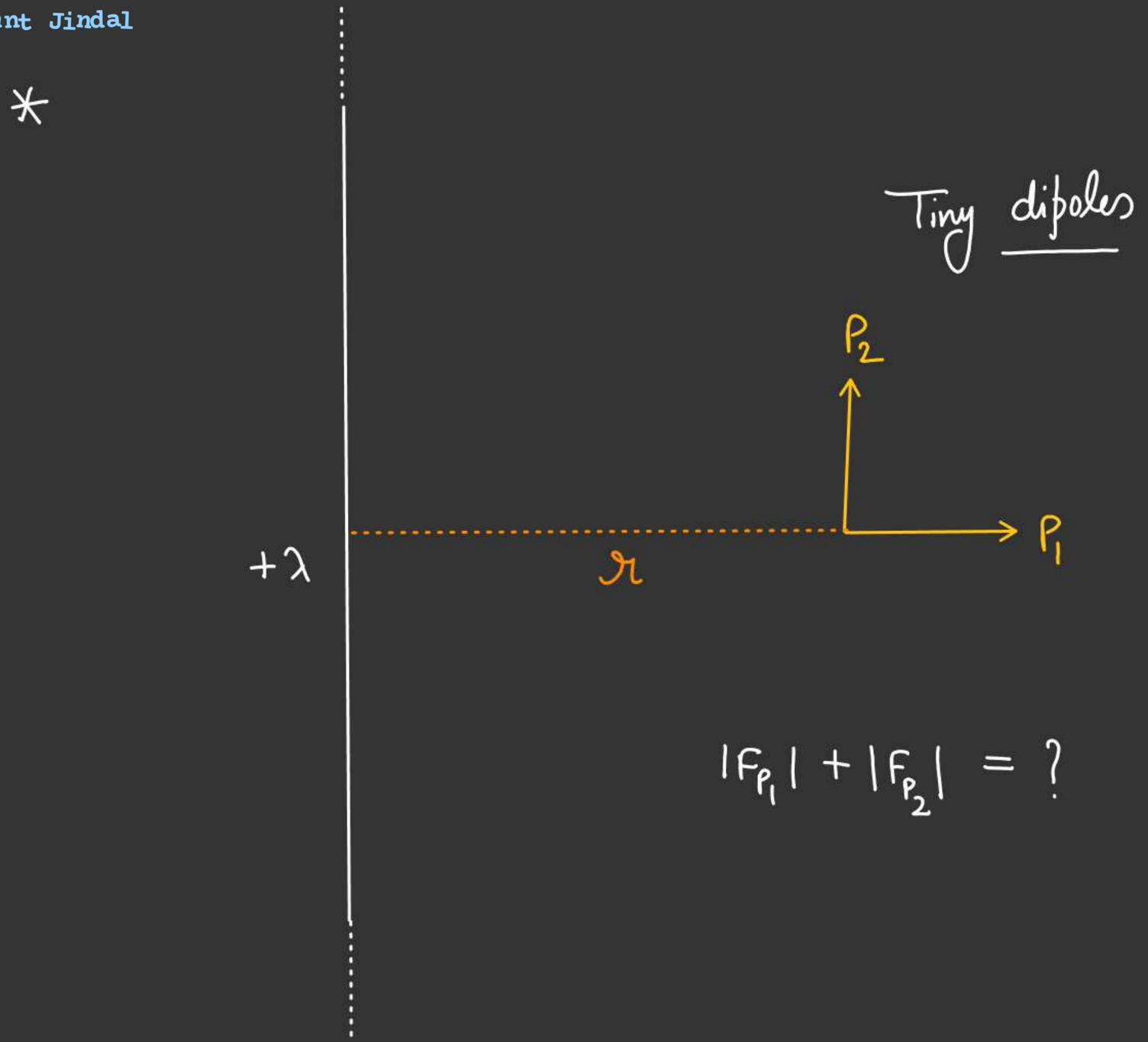

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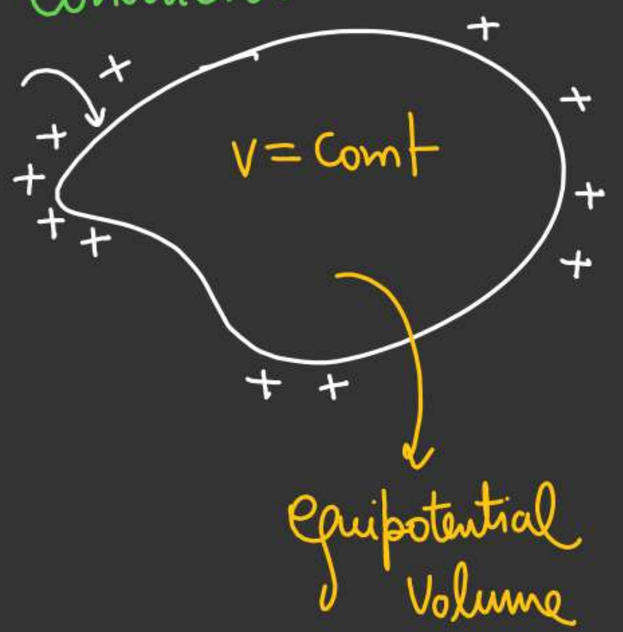
Equipotential Surfaces & Volumes :-

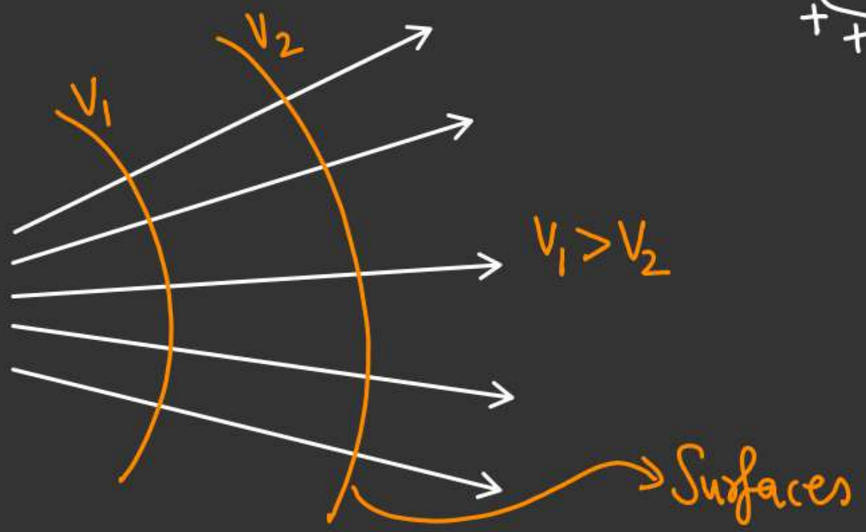
$$E = -\frac{\partial V}{\partial r} \quad , \quad V_f - V_i = -\int \vec{E} \cdot d\vec{r}$$

$V = \text{Const} \longrightarrow$ i) $E = 0$ → Inside Conductor



(ii) $\vec{E} \perp d\vec{r}$ → Path

$$V_1 - V_2 = \int_1^2 E_x \cdot dx$$



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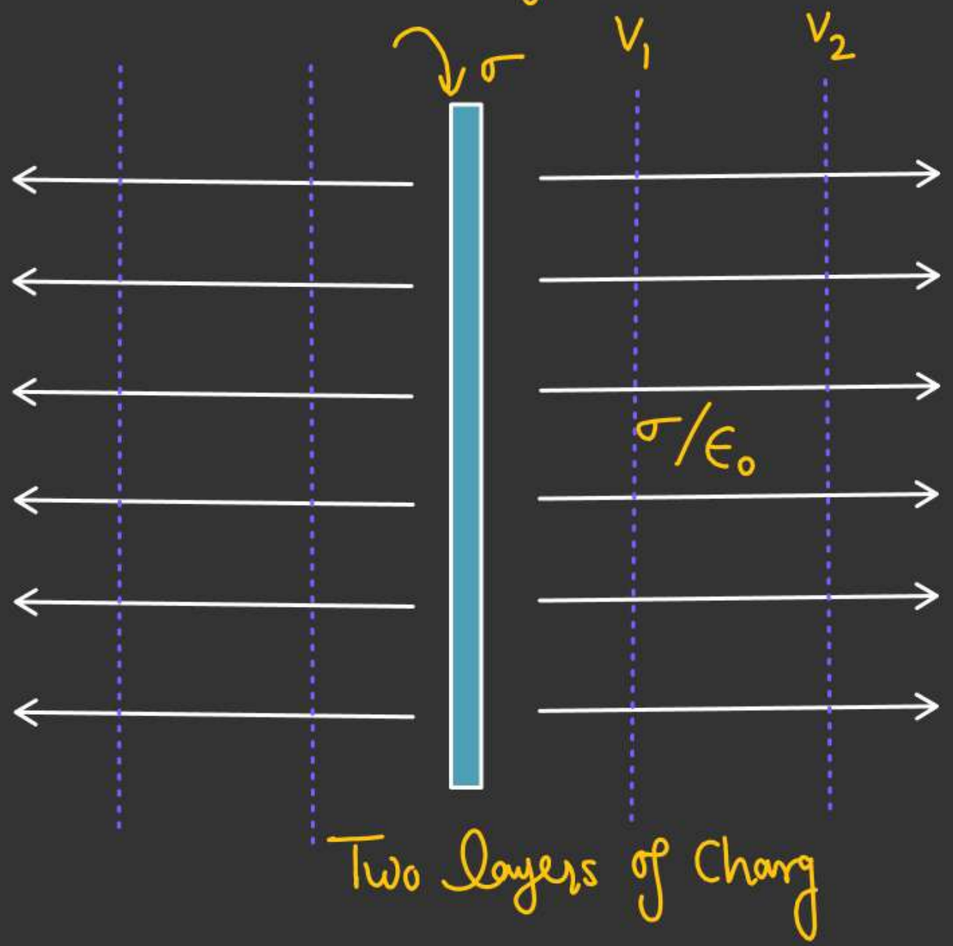
* Infinite Plate & sheet :

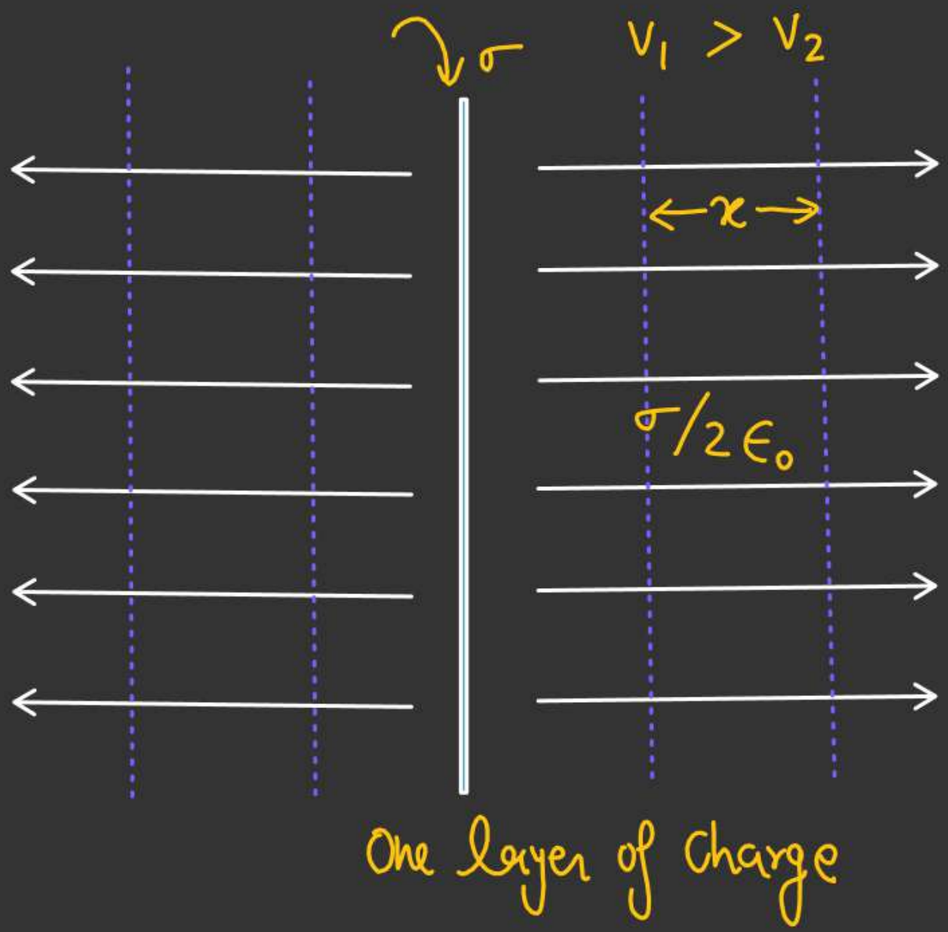
Conducting

$V_1 - V_2 = \frac{\sigma}{\epsilon_0} \cdot x$



Two layers of Charg

$V_1 - V_2 = \frac{\sigma}{2\epsilon_0} \cdot x$



One layer of Charge

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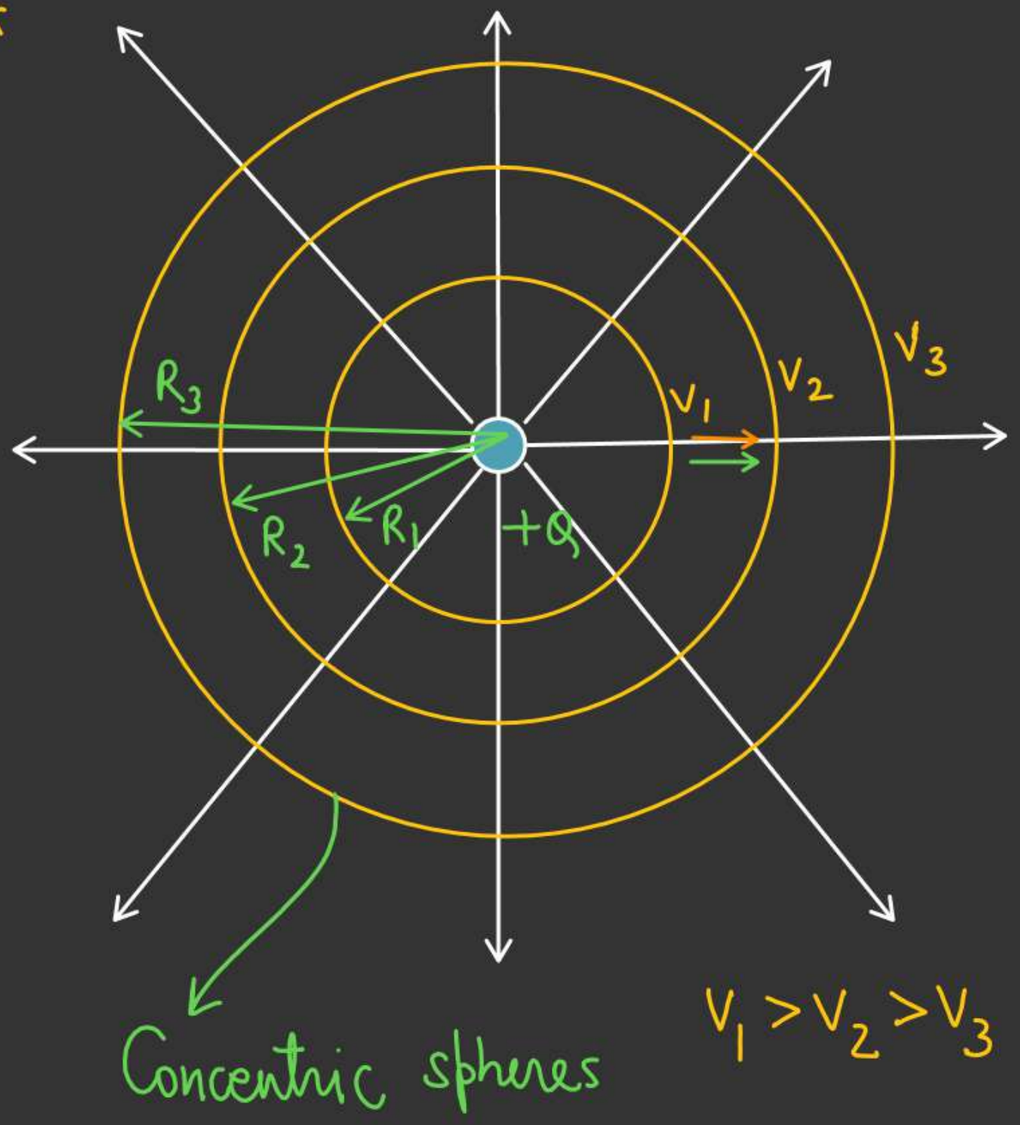

$$V_f - V_i = -\int \vec{E} \cdot d\vec{r}$$

$$V_2 - V_1 = -\int E_r \cdot dr = -\int_{R_1}^{R_2} \frac{KQ}{r^2} \cdot dr$$

$$V_1 - V_2 = \int_{R_1}^{R_2} \frac{KQ}{r^2} \cdot dr$$

$$V_2 - V_3 = \int_{R_2}^{R_3} \frac{KQ}{r^2} \cdot dr$$

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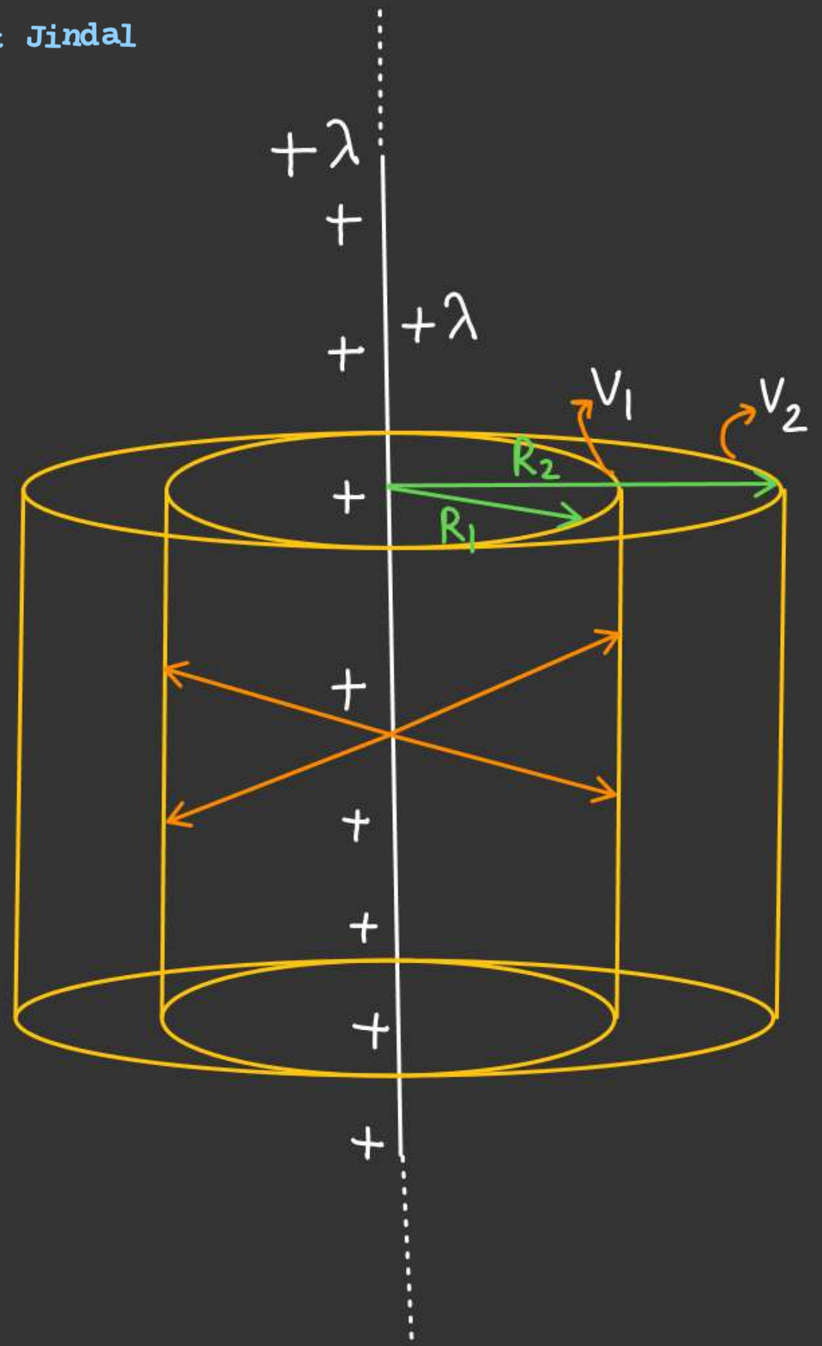

$$V_1 - V_2 = \int_{R_1}^{R_2} \frac{\lambda}{2\pi\epsilon_0 r} \cdot dr$$

$$\boxed{V_1 - V_2 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right)}$$

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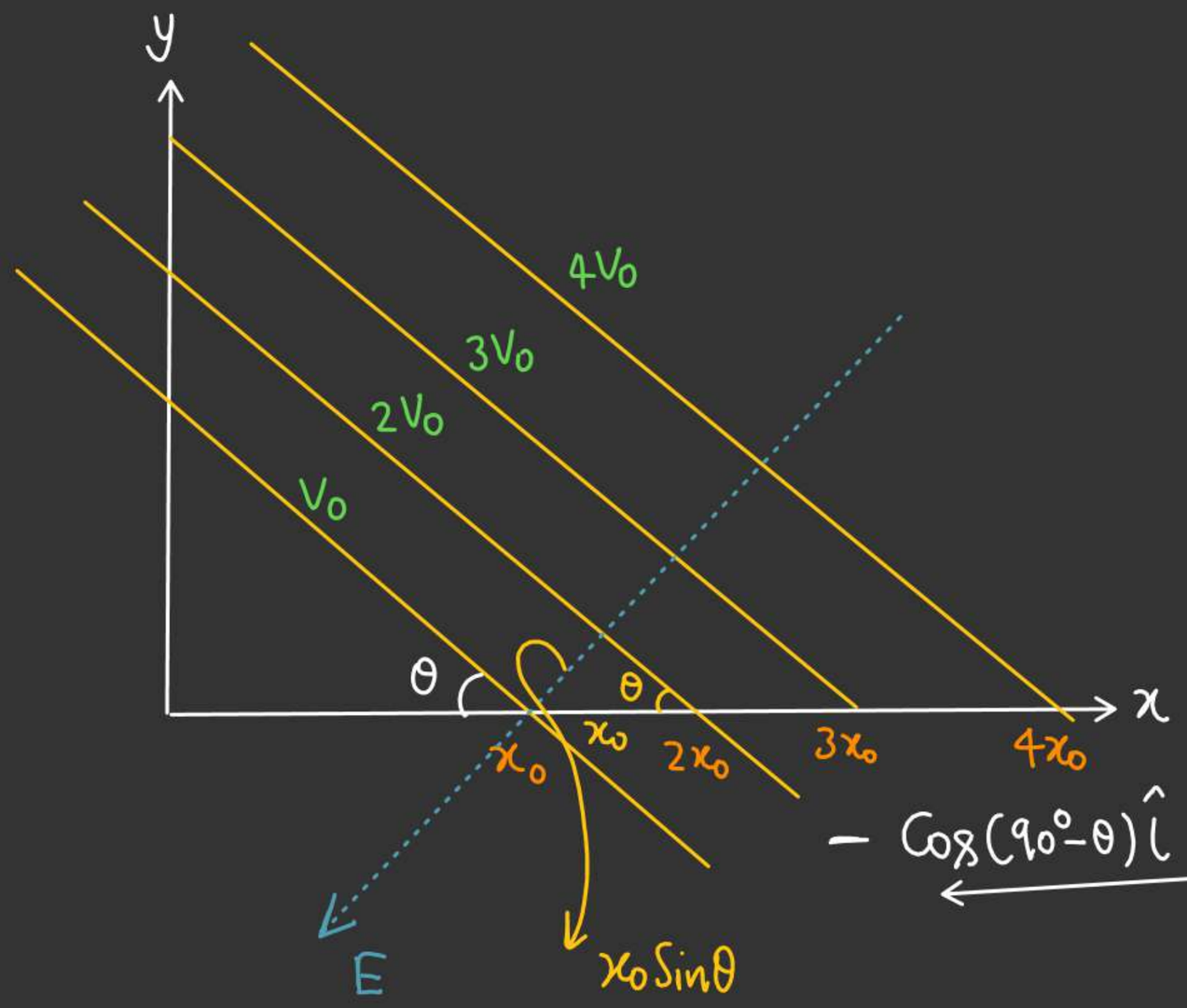

$$E = \frac{\Delta V}{d} = \frac{2V_0 - V_0}{x_0 \sin\theta}$$

$$\boxed{E \Rightarrow \frac{V_0}{X_0 \sin\theta}}$$

$$-\cos(90^\circ - \theta)\hat{i}$$

$$\cos\theta(-\hat{j})$$

$$\vec{E} = -E\sin\theta\,\hat{i} - E\cos\theta\,\hat{j}$$

$$\boxed{\vec{E} \Rightarrow -\frac{V_0}{X_0}\left(\hat{i} - \cot\theta\,\hat{j}\right)}$$

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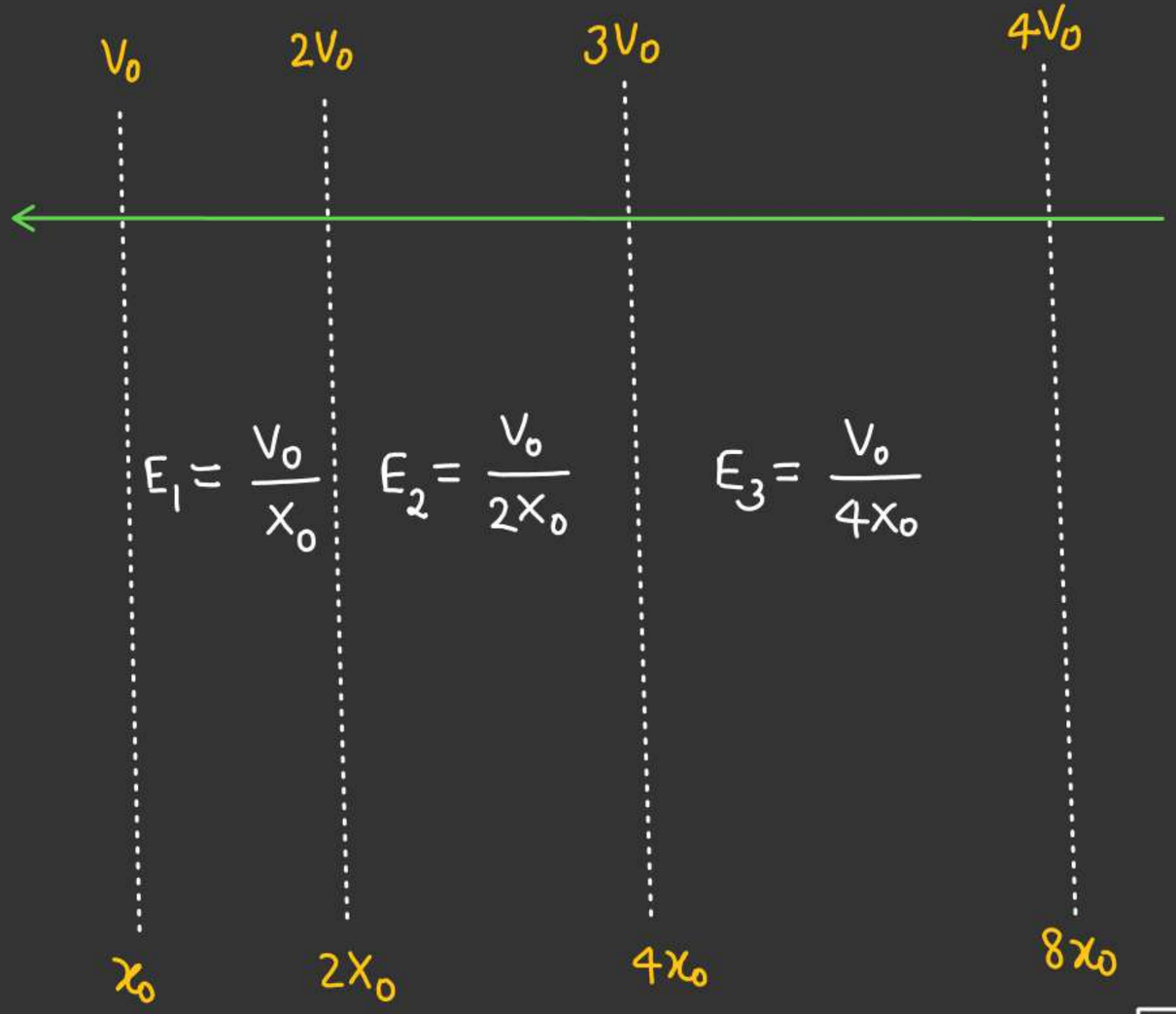
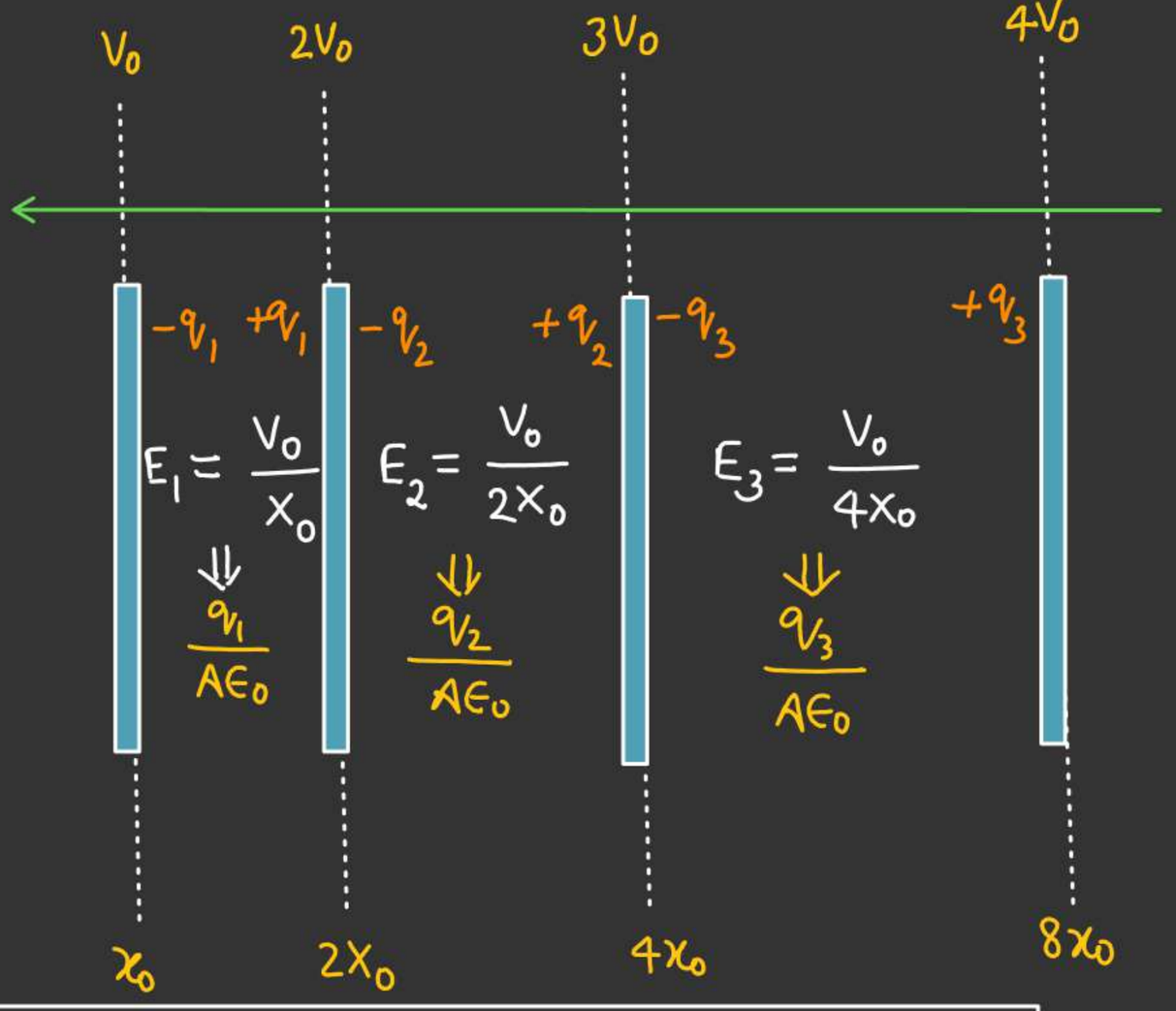


For system of Plates



$$q_1 = \frac{A\epsilon_0 V_0}{x_0}, \quad q_2 = \frac{A\epsilon_0 V_0}{2x_0}, \quad q_3 = \frac{A\epsilon_0 V_0}{4x_0}$$

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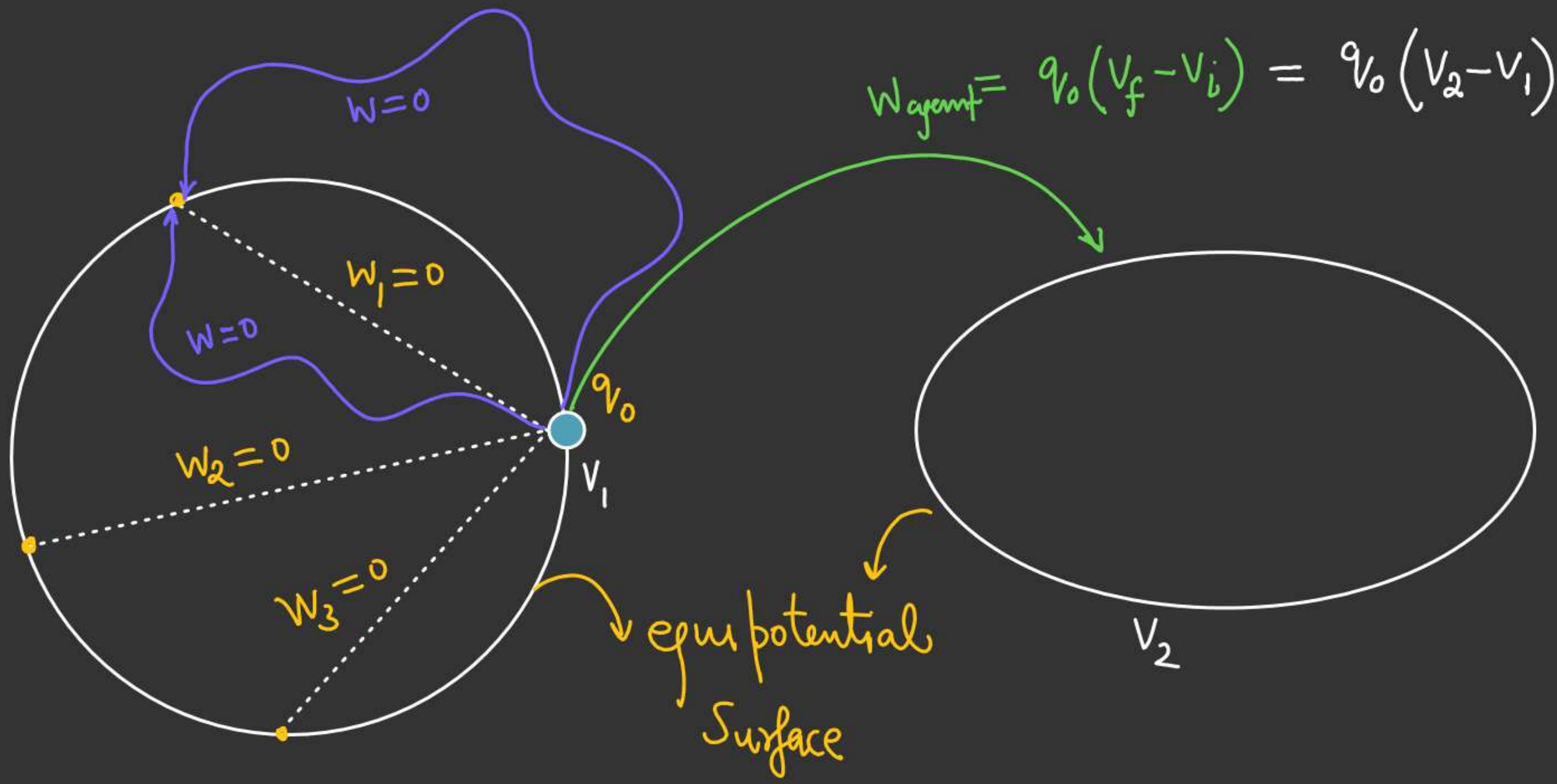

$$W_{agent} = q_0(V_f - V_i) = q_0(V_2 - V_1)$$

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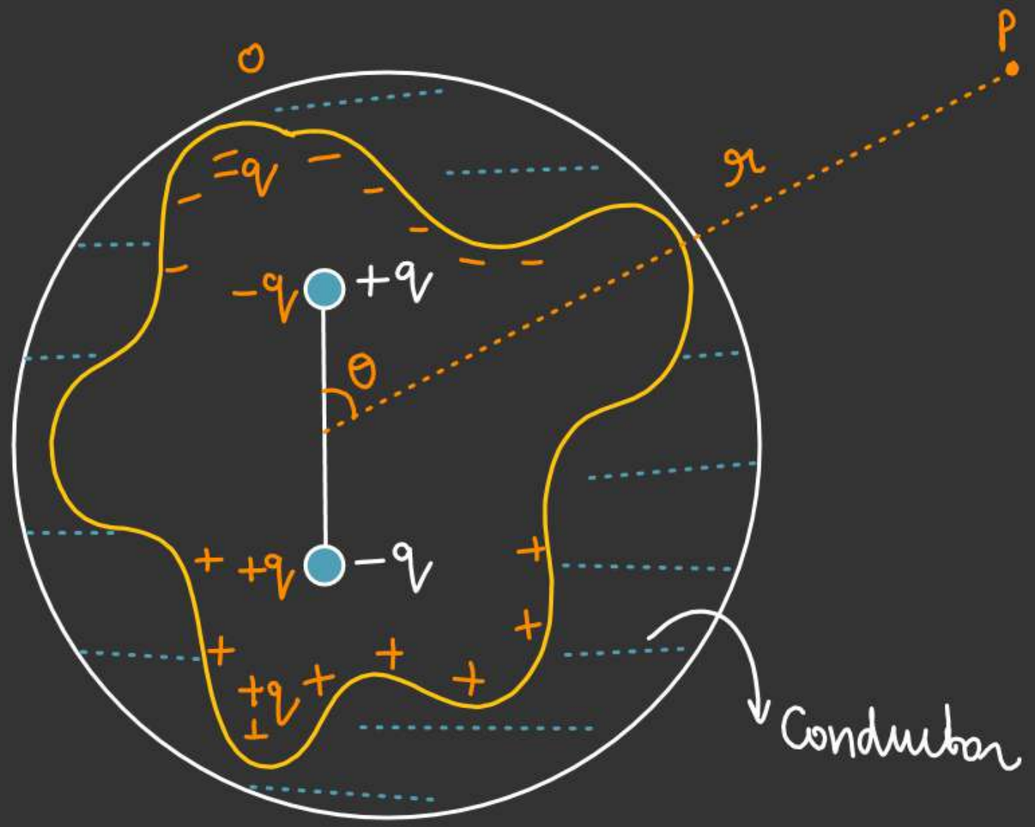

$V_p = 0$ ✓

$E_p = 0$ ✓

$q_{out}\big|_{shell} = 0$ ✓

$q_{surface}\big|_{cavity} = 0$ ✓

✓ $q_{surface}\big|_{cavity} \rightarrow$ non Uniform distribution

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